An Introduction to Mathematical Analysis

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Alpha Science

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Preface

In an elementary course on Calculus, one comes across concepts such as limit, continuity, differentiation and integration from a computational and intuitive point of view. In this book, we present a logical development of these concepts and the standard elementary functions to which they are applied. The development is based on an axiomatic definition of the real number system. However, certain properties of the natural numbers and integers are taken for granted, because it would have been too burdensome to avoid doing so.

Although the target readership consists primarily of senior undergraduate and beginning graduate students, we hope that other young mathematicians will also find an occasional topic that catches the eye. The material has been arranged keeping in view a variety of needs. In particular, it provides not only an introduction to the subject but also enrichment material, usually indicated by a star (*), that would be appropriate for a second reading. The latter includes a discussion of some matters that do not ordinarily find a place in books on Mathematical Analysis. Besides starred items, there are also a few other matters that can be postponed during a first reading, such as the justification of the computation $\sum_{k=0}^{6} \frac{1}{k!} = \frac{1957}{720} > 2.718$ and the arithmetical procedures involved.

The initial chapters proceed at a leisurely pace so that the reader with limited experience of the axiomatic approach has enough opportunity to get used to it. The pace increases in later chapters. Basic ideas of set theory, including bijective mappings and cartesian products are explained with essential details to ensure that the book is self contained. The reader is encouraged to visualise mathematical situations "geometrically" as this will help get a deeper understanding of the topics covered. Ability to do so even with ε-δ is presumed and we therefore present very few figures. The book contains nearly 450 problems, many with hints. Several of them are hard to find in other texts, even in a related form. It is expected that the reader will attempt to work out these problems and refer to the hints only as a last resort.

A working knowledge of complex numbers is helpful in understanding the discussion of trigonometric and exponential functions. Section 1-10 summarises the construction of the complex field and its basic properties that will be used.

With a focus on clarity rather than brevity, this text gives clear motivation, definitions, examples and transparent proofs of the theorems. It will be ideal as a classroom text or a self study resource for students.

Some forays into topics that go beyond the most basic level are preceded by an appropriate caveat. Those who disagree with us on the philosophical need to bring trigonometric and exponential functions within the purview of axioms can avoid Section 5-3 on special functions.

The reader with previous experience of the subject who is looking for something different in this book may wish to glance at Section 1-1, Problem 1-4. P15, Appendix to Chapter 1, Theorem 2-2.1, Props. 6-2.6 and 7, Problem 7-4. P13, Example 7-5.8(b), Problems 7-5. P13 and 14, Cor. 8-4.3, Prop. 8-4.4, Appendix 2 to Chapter 8 and proofs of the Mean Value Theorems for Integrals 10-3.1 and 2.

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Chapter 1
The Real Number System

1-1 Introduction

It is a familiar experience in Calculus, that one adds an "arbitrary constant" to an indefinite integral. Thus for example, the most general function with derivative $2x$ is $x^2 + C$. If there were to be any nonconstant function f with derivative 0 at every point, then $x^2 + f(x)$ would also have derivative $2x$ and therefore $x^2 + C$ would not be the most general such function. The justification for the constant is not that every constant function has derivative 0 at each point, but the converse: no nonconstant function can have this property. In other words, a function with derivative 0 at every point must necessarily be a constant. This converse, and the fact that it is basic to something we do every time we solve an integration problem, are often not emphasised in Calculus.

It is also familiar from Calculus that the most general function with derivative $1/x$ for $x > 0$ is $\ln|x| + C$. The same is true for $x < 0$. Now consider positive and negative x together and let g be the function defined as

$$g(x) = \begin{cases} \ln|x| + 1 & \text{if } x > 0 \\ \ln|x| & \text{if } x < 0. \end{cases}$$

This function has derivative $1/x$ for all values of x under consideration, and yet it is *not* of the form $\ln|x| + C$. Thus $\ln|x| + C$ is not the most general function having derivative $1/x$ when positive and negative values of x are considered simultaneously, i.e. when the domain consists of all nonzero real numbers. This example illustrates how the procedure of adding an "arbitrary constant" to obtain the most general indefinite integral is not valid when the domain is not an interval. This too is not emphasised in Calculus.

It is possible to go further and question why the procedure is valid even in an interval.

One of the purposes of Analysis is to clarify such matters. There are other matters that need clarification but one does not dwell upon in a Calculus course. For instance, what is the guarantee that the series by which the number e is defined really has a sum (is convergent)? If one chooses instead to define it as a limit as n tends to infinity of $(1+\frac{1}{n})^n$, then the existence of this limit can be questioned.

It turns out that satisfactory answers to such questions become possible through a precise definition of what we understand by "numbers". We begin with several pages of preliminary work on what consequences we expect the eventual definition to have. Clarification of matters relating to Calculus will occupy us through much of this book.

What makes numbers useful for commercial or scientific purposes is that they can be added, multiplied, compared and so on. They are collectively interrelated through addition, multiplication etc., and our concern is these interrelationships among numbers taken collectively. The precise nature of an individual number like, say 5, is not a matter of concern to a banker, scientist or a mathematician, although it may be a legitimate question for philosophy. To take them collectively is to regard them as a *set*. Since the elements of the set are interrelated, we refer to it as a *system*. We shall be dealing extensively with the set of all numbers and also with parts of it, called subsets. Therefore we go on to present preliminaries about sets and the way they will be used.

1-2 Preliminaries on Sets

The following symbols will be employed in this book when convenient:

$\forall$ means "for all" or "for every"

$\exists$ means "there exists"

$\ni$ means "such that"

$\Rightarrow$ means "implies"

$\Leftrightarrow$ means "if, and only if,".

The concept of *set* is basic to every area of modern mathematics. Although a set is a collection of objects called its "elements" or "members", it is not possible to regard every collection of objects as a set. Doing so would allow for a set having all possible sets, including itself, as an element, and this is known to lead to a contradiction. The concept of set is therefore left undefined. Two of the most common ways to describe a set are (a) simply to list its elements and (b) to specify a property shared by all its elements and by none others. Thus $\{x_1, x_2, \dots, x_n\}$ is the set whose elements are $x_1, x_2, \dots, x_n$; and $\{x\}$ is the set whose sole element is x. If X is the set of all elements x such that some property $P(x)$ is true, we shall write

$$X = \{x : P(x)\}.$$

In defining a set X in this manner, it is necessary for technical reasons to take care that all the objects x under consideration belong to one single set. For the most part, we shall be working with the set of real numbers, which will *not* be described in either of the above two ways.

The objects belonging to a set may themselves be sets. It is cutomary to refer to such a set as a *family* or *collection* or *class* of sets and a subset thereof as a *subfamily* or *subcollection* or a *subclass*.

The symbol $\varnothing$ denotes the empty set, having no elements.

We write $x \in X$ if x is an element of the set X and $x \notin X$ when x is not an element of X. If Y is a *subset* of X, that is, if $x \in Y \Rightarrow x \in X$, we write $Y \subseteq X$. If $Y \subseteq X$ and $X \subseteq Y$, then $X = Y$. When $Y \subseteq X$ and $Y \neq X$, we say that Y is *proper subset* of X. It is trivial that $\varnothing \subseteq X$ for every set X.

The standard symbols for the most important sets of numbers are as follows:

$\mathbb{N}$ the set of all natural numbers (positive integers)

$\mathbb{Z}$ the set of all integers

$\mathbb{Q}$ the set of all rational numbers

$\mathbb{R}$ the set of all real numbers (to be defined in Section 1-6)

$\mathbb{C}$ the set of all complex numbers (to be defined in Section 1-10).

We note that $\{k \in \mathbb{N} : 1 \le k \le n\}$ means the same thing as $\{1, 2, \ldots, n\}$.

Let X and Y be two sets. Then we can construct the following new sets from them:

$$X \cup Y = \{x : x \in X \text{ or } x \in Y\}$$
$$X \cap Y = \{x : x \in X \text{ and } x \in Y\}.$$

The sets $X \cup Y$ and $X \cap Y$ are the *union* and *intersection* respectively of X and Y.

Let $\mathcal{F}$ be a family or collection of sets. Its *union* consists of all elements that belong to some set in the family and its *intersection* consists of all elements that belong to every set in the family:

$$\bigcup \mathcal{F} = \{x : \exists\, F \in \mathcal{F} \ni x \in F\} \quad \text{and} \quad \bigcap \mathcal{F} = \{x : x \in F \ \forall\, F \in \mathcal{F}\}.$$

The alternative notations $\bigcup_{F \in \mathcal{F}} F$ and $\bigcap_{F \in \mathcal{F}} F$ may be used when appropriate.

The symbol

$$f : X \to Y$$

means that f is a *function* (or *mapping* or *map*) from the set X into the set Y; that is, f assigns to each $x \in X$ an element $f(x) \in Y$. It is convenient to say that the function "maps x into $f(x)$" or that "$f(x)$ is the *image* of x". The elements assigned to members of X by f are often called values of f. If $A \subseteq X$ and $B \subseteq Y$, the *image* of A and *inverse image* of B are, respectively,

$$f(A) = \{f(x) : x \in A\}$$

and

$$f^{-1}(B) = \{x : f(x) \in B\}.$$

Note that $f^{-1}(B)$ may be empty even when $B \neq \varnothing$. The assertion that $C = f(A)$ is sometimes conveniently rephrased as "f maps (the subset) A *onto* C".

The *domain* of f is X and the *range* is $f(X)$; the *range space* is Y. If $f(X) = Y$, the function f is said to map X *onto* Y (or the function is said to be *surjective*). We write $f^{-1}(y)$ instead of $f^{-1}(\{y\})$ for every $y \in Y$. If $f^{-1}(y)$ consists of at most one element for each $y \in Y$, then f is said to be *one-to-one* (or *injective*). If f is one-to-one, then f^{-1} is a function with domain $f(X)$ and range X, such that $f^{-1}(y) = x$ if, and only if, $f(x) = y$. It is called the *inverse of f*. A function is said to be *invertible* if it has an inverse; thus being invertible is the same as being injective. The inverse of f^{-1} is f. Two functions f and g with domains X and Y respectively are inverses of each other if, and only if,

$$g(f(x)) = x \quad \forall\, x \in X \qquad \text{and also} \qquad f(g(y)) = y \quad \forall\, y \in Y.$$

A function which is both injective and surjective is said to be *bijective*. One also speaks of a *bijection* or *one-to-one correspondence*. In the case when f is bijective, f^{-1} is a function with domain Y and range X.

A set X is said to be *finite* if there is a bijection between it and a set of the form $\{k \in \mathbb{N} : 1 \leq k \leq n\}$, for some $n \in \mathbb{N}$. It is *infinite* if it is not finite, and is *denumerable* if there exists a bijection between it and the set $\mathbb{N}$. A denumerable set is necessarily infinite and any infinite subset of a denumerable set is itself denumerable. For example, even integers form a denumerable set. Moreover, it follows that any infinite set from which there is an injection into $\mathbb{N}$ is denumerable.

Finite and denumerable sets together constitute what we shall call *countable* sets.

It is sometimes necessary to consider a function f only on a subset S of its domain X. Technically, that makes it a different function and it is called the *restriction* of f to S. Introducing a new symbol to denote a restriction can clutter the notation and we shall avoid it as far as possible.

Sometimes a family of sets $\mathcal{F}$ occurs as the range of a function with some domain Λ. The set associated by the function with an element $\alpha \in \Lambda$ may be denoted by X_α instead of introducing a symbol for the function. Then the range $\mathcal{F}$ may be described as $\{X_\alpha : \alpha \in \Lambda\}$. If $\{X_\alpha\}$ is a collection of sets, where α runs through some indexing set Λ, we write

$$\bigcup_{\alpha\in\Lambda} X_\alpha \text{ and } \bigcap_{\alpha\in\Lambda} X_\alpha$$

for the union $\bigcup\mathcal{F}$ and intersection $\bigcap\mathcal{F}$, respectively, of the sets X_α :

$$\bigcup_{\alpha\in\Lambda} X_\alpha = \{x : x \in X_\alpha \text{ for at least one } \alpha \in \Lambda\}$$

$$\bigcap_{\alpha\in\Lambda} X_\alpha = \{x : x \in X_\alpha \text{ for every } \alpha \in \Lambda\}.$$

If $\Lambda = \mathbb{N}$, the set of all natural numbers, the customary notations are

$$\bigcup_{n=1}^{\infty} X_n \text{ and } \bigcap_{n=1}^{\infty} X_n .$$

If $Y \subseteq X$, the *complement* of Y in X is the set of those elements in X that are not in Y, that is

$$X \setminus Y = \{x : x \in X, x \notin Y\}.$$

The complement of Y is denoted by Y^c whenever it is clear from the context which larger set X is intended.

If $\{X_\alpha\}$ is a collection of subsets of X, then the following *De Morgan's laws* hold:

$$\left(\bigcup_{\alpha\in\Lambda} X_\alpha\right)^c = \bigcap_{\alpha\in\Lambda} (X_\alpha)^c \text{ and } \left(\bigcap_{\alpha\in\Lambda} X_\alpha\right)^c = \bigcup_{\alpha\in\Lambda} (X_\alpha)^c .$$

As the reader knows, a point in the plane is represented by two numbers, wherein the first and second numbers must be distinguished even if they are equal. This amounts to saying that they actually represent a function from the two element set $\{1,2\}$ into $\mathbb{R}$. In general, if A and B are sets, not necessarily distinct, then a map from the set $\{1,2\}$ into $A \cup B$ such that 1 is mapped into A and 2 is mapped into B is called an *ordered pair*. The ordered pair that maps 1 into $a \in A$ and 2 into $b \in B$ is denoted by (a,b). The reader is cautioned that the same notation is used for an open interval from a to b but no confusion will arise. The set of all such ordered pairs is called the *Cartesian product* of the sets and is denoted by $A \times B$.

We note the following relations between unions, intersections, images and inverse images.

$$f(\bigcap_{\alpha\in\Lambda} X_\alpha) \subseteq \bigcap_{\alpha\in\Lambda} f(X_\alpha).$$

Also, if $\{Y_\alpha : \alpha \in \Lambda\}$ is a family of subsets of Y, then

$$f^{-1}(\bigcup_{\alpha\in\Lambda} Y_\alpha) = \bigcup_{\alpha\in\Lambda} f^{-1}(Y_\alpha)$$

and

$$f^{-1}(\bigcap_{\alpha\in\Lambda} Y_\alpha) = \bigcap_{\alpha\in\Lambda} f^{-1}(Y_\alpha).$$

If Y_1 and Y_2 are subsets of Y, then

$$f^{-1}(Y_1 \setminus Y_2) = f^{-1}(Y_1) \setminus f^{-1}(Y_2).$$

Finally, if $f: X \to Y$ and $g: Y \to Z$, the *composite* function (or *composition*) $g \circ f : X \to Z$ is defined by

$$(g \circ f)(x) = g(f(x)).$$

1-3 Field Axioms

We know from elementary experience that integers $\mathbb{Z}$ are interrelated among themselves and so are rational numbers $\mathbb{Q}$. In other words, they form two systems. Without further ado, we take them as known. However, the systems $\mathbb{Z}$ and $\mathbb{Q}$ are not adequate for the purposes of Euclidean Geometry (because of the Pythagoras Theorem) and certainly not for Calculus.

The interrelationship between the elements of the system is through addition, multiplication and comparison, and we therefore begin by stating their basic properties which we expect to include in our eventual definition of the number system we want to work with. The list is long and we shall first state the properties that relate to addition alone. However, these properties are shared by other systems as well, such as matrices. Therefore we shall initially be talking about a general system until we finish stating *all* the properties.

Both addition and multiplication are carried out on two numbers at a time and result in a number. They are instances of what are called "binary operations" in whatever set of numbers F we may have at hand, whether of integers, rational numbers or any other. More precisely, they are mappings from the cartesian product $F \times F$ into F.

The basic properties we expect addition to have are:

(A1) **Closure Law.** If $x \in F$ and $y \in F$, then $x + y \in F$.

(A2) **Commutative Law.** If $x \in F$ and $y \in F$, then $x + y = y + x$.

(A3) **Associative Law.** If $x \in F,\ y \in F,\ z \in F$, then $(x + y) + z = x + (y + z)$.

(A4) **Identity Law.** There exists $0 \in F$ such that, if $x \in F,$ then $0 + x = x$.

(A5) **Inverse Law.** If $x \in F$, then there is an element $-x \in F$ such that

$$(-x) + x = 0.$$

The Closure Law merely repeats a property of any binary operation. It could be omitted on the grounds that addition is already understood to be a binary operation. Since the Associative Law (A3) make no sense without Closure, we are automatically using it when we use the former.

Examples of systems in which the above properties hold are $\mathbb{Z}$, $\mathbb{Q}$ and matrices with entries from $\mathbb{Q}$. However, the system $\mathbb{N}$ of natural numbers satisfies only (A1)–(A3).

The numbering $(\text{A1}), (\text{A2}), \ldots$ is not standard but will be used for reference within this book. However, outside the context of this book, it is better to use the names given in boldface, because they are commonly understood by the mathematical public. The same is true of Laws which will be stated in the sequel.

The reader can certainly think of many other properties of addition. They have not been included above, because they can be derived. The next Proposition provides an example of such derivations.

1-3.1. Proposition. *Suppose* (A1)—(A5) *hold and* $x \in F,\ y \in F,\ z \in F$.

(a) *If* $x + y = x + z$, *then* $y = z$.

(b) *If* $y + x = x$, *then* $y = 0$.

(c) *If* $y + x = 0$, *then* $y = -x$.

(d) $-(-x) = x$.

(e) $0 = -0$.

Proof. (a)
$$\begin{aligned} y &= 0 + y && \text{(A4)}\\ &= (-x + x) + y && \text{(A5)}\\ &= -x + (x + y) && \text{(A3)}\\ &= -x + (x + z) && \text{hypothesis}\\ &= (-x + x) + z && \text{(A3)}\\ &= 0 + z && \text{(A5)}\\ &= z && \text{(A4).} \end{aligned}$$

(b) As above, $y = -x + (x + y)$. Now by (A2), $-x + (x + y) = -x + (y + x)$, which equals $-x + x$ by hypothesis, and $-x + x = 0$ by (A5). Thus $y = 0$.

(c) Once again, $y = -x + (x + y)$. By using (A2), the hypothesis, (A2) again and then (A4), we get $-x + (x + y) = -x + (y + x) = -x + 0 = 0 + (-x) = -x$.

(d) By (A5) and (A2), $x + (-x) = 0$. Applying part (c) with x in place of y and $-x$ in place of x, we get $x = -(-x)$.

(e) $0 + 0 = 0$ by (A4). Therefore $0 = -0$ by (c). ☺

For an alternative approach to the above proofs, the interested reader is referred to [8].

Remarks. (A4) only guarantees that there is *at least one* special number whose sum with any number is the latter number. It does not guarantee that there is only one such special number. However, Part (b) of Prop.1-3.1 guarantees this. Similarly, (A5) only guarantees that, for a given number, there is *at least one* related number whose sum with that number is 0. It is Part (c) of Prop.1-3.1 that guarantees that there is only one such related number. It is commonly called the **negative of** the given number, or also as its **additive inverse**.

By using (A2) in conjunction with Prop.1-3.1, we can easily establish also the following under the same hypotheses:

(a) If $y + x = z + x$, then $y = z$.
(b) If $x + y = x$, then $y = 0$.
(c) If $x + y = 0$, then $y = -x$.

We shall consider this to be part of Prop.1-3.1.

The basic properties expected for multiplication are:

(M1) **Closure Law.** If $x \in F$ and $y \in F$, then $xy \in F$.

(M2) **Commutative Law.** If $x \in F$ and $y \in F$, then $xy = yx$.

(M3) **Associative Law.** If $x \in F,\ y \in F,\ z \in F$, then $(xy)z = x(yz)$.

(M4) **Identity Law.** There exists $1 \in F$ such that $1 \neq 0$ and,

if $x \in F$ then $1x = x$.

(M5) **Inverse Law.** If $x \in F$ and $x \neq 0$, then there is an element $x^{-1} \in F$ such that $x^{-1}x = 1$.

Note that these properties resemble the ones for addition, but that (M4) says that $1 \neq 0$, while (M5) includes the restriction $x \neq 0$, which is not there in (A5). So a Proposition similar to Prop.1-3.1 should be possible, but duly modified to take care of the restriction $x \neq 0$. What we have denoted by "x^{-1}" could instead have been denoted by "$1/x$".

Just as with (A1), the Law (M1) will have been used indirectly whenever (M3) is used. Therefore no explicit references to (M1) will be made.

The above properties hold in $\mathbb{Q}$ but not in $\mathbb{Z}$ or $\mathbb{N}$. With the exception of (M5), they hold in $\mathbb{Z}$ as well. The Laws (M1)–(M4) hold in $\mathbb{N}$, the stipulation that "$1 \neq 0$" being superfluous because zero is not in $\mathbb{N}$ anyhow. The reader who has had a brush with matrices will note that nonsingular matrices of a given dimension satisfy (M1) and (M3)–(M5), the stipulations that "$1 \neq 0$" and "$x \neq 0$" both being superfluous.

1-3.2. Proposition. *Suppose* (M1)—(M5) *hold and* $x \in F$, $y \in F$, $z \in F$.
(a) *If* $x \neq 0$ *and* $xy = xz$, *then* $y = z$.
(b) *If* $x \neq 0$ *and* $yx = x$, *then* $y = 1$.
(c) *If* $yx = 1$ *and* $x \neq 0$, *then* $y = x^{-1}$.
(d) *If* $x^{-1} \neq 0$, *then* $(x^{-1})^{-1} = x$.

The proof is similar to that of the previous Proposition and is left to the reader. It may appear that, in (c), one should be able to *argue* $x \neq 0$ and should not have to assume it. But the argument requires more than (M1)—(M5). See proofs of Prop.1-3.3(a) and Prop.1-3.4(a) below.

Part (c) of the above Proposition shows that, for a given nonzero $x \in F$, there cannot be two different elements $x^{-1} \in F$ of the kind guaranteed by (M5). The one and only element that serves the purpose is called the **reciprocal of** or also **multiplicative inverse of** x.

The reader is undoubtedly conversant with the properties we have proved so far and also the ones we shall prove in the next few Propositions. At this stage, the point is not to present any new properties but to show that the familiar properties can be deduced from the ones we list as Laws. Even familiar properties are to be used in subsequent work only on the grounds that we have deduced them, and not because of habits drummed into us since school days. In order to emphasise this, we have initially given constant references to previous Propositions.

The expected property that would link addition and multiplication is:
(D) **Distributive Law.** If $x \in F$, $y \in F$, $z \in F$, then $x(y + z) = xy + xz$.

1-3.3. Proposition. *Suppose* (A1)—(A5), (M1)—(M5) *and* (D) *all hold and* $x \in F$, $y \in F$. *Then*
(a) $0x = 0$.
(b) *If* $x \neq 0$ *and* $y \neq 0$ *then* $xy \neq 0$.
(c) $(-x)y = x(-y) = -(xy)$.

(d) $(-x)(-y) = xy$.

***Proof.**(a) $0x = (0 + 0)x$ (A4)

$= 0x + 0x$ (D)

It now follows by applying Prop.1-3.1(b) that $0x = 0$.

(b) Suppose, if possible, that

$$x \neq 0 \text{ and } y \neq 0 \text{ but } xy = 0. \tag{1}$$

Then by (M5), y has a multiplicative inverse y^{-1} and

$0 = 0\, y^{-1}$ (part (a))

$= (xy)y^{-1}$ (last part of (1)). (2)

Using (M3), (M5), (M2), (M4) respectively, we have $(xy)y^{-1} = x(yy^{-1}) = x1 = 1x = x$. Hence by (2), $x = 0$, contradicting the first part of (1). Therefore it is not possible for xy to be 0 when neither x nor y is 0.

(c) Using (D), (A2), (A4), (M2) and part (a), we have $xy + x(-y) = x(y + (-y)) = x(-y + y) = x0 = 0$. It now follows by applying Prop.1-3.1(c) that $x(-y) = -(xy)$. Similarly, $y(-x) = -(yx)$. By (M2), we now get $(-x)y = -(xy)$.

(d) $(-x)(-y) = -(x(-y))$ part (c)

$= -(-(xy))$ again part (c).

It now follows by applying Prop.1-3.1(d) that $(-x)(-y) = xy$. ☺

1-3.4. Proposition. *Suppose* (A1)—(A5), (M1)—(M5) *and* (D) *hold and* $x \in F$, $y \in F$.

(a) *If* $xy = 1$, *then* $x \neq 0$, $y \neq 0$ *and* $y = x^{-1}$.

(b) *If* $x \neq 0$, *then* $x^{-1} \neq 0$ *and* $(x^{-1})^{-1} = x$.

(c) *If* $x \neq 0$, *then* $-x \neq 0$ *and* $(-x)^{-1} = -x^{-1}$.

***Proof.** (a) Suppose $xy = 1$. If x were to be 0, then $xy = 0y = 0$ by Prop.1-3.3(a). Therefore $1 = xy = 0$. This contradicts (M4). Therefore $x \neq 0$. It now follows that $y = x^{-1}$ by (M2) and Prop.1-3.2(c). If y were to be 0, then by (M2) we would have $yx = 1$. Proceeding as above with x and y interchanged, we would arrive at the same contradiction. Hence $y \neq 0$.

(b) If $x \neq 0$ and $x^{-1} = 0$, then by (M5) and Prop.1-3.3(a), $1 = x^{-1}x = 0x = 0$, contradicting (M4). Therefore $x \neq 0 \Rightarrow x^{-1} \neq 0$. The rest follows by Prop.1-3.2(d).

(c) If $x \neq 0$, it follows by (M5) that x^{-1} exists and hence by (A5) that $-x$ and $-x^{-1}$ both exist. Besides, by Prop.1-3.3(d), (M2) and (M5),

$$(-x)(-x^{-1}) = x(x^{-1}) = (x^{-1})x = 1.$$

It now follows by (a) that $-x \neq 0$ and $-x^{-1} = (-x)^{-1}$. ☺

It is an immediate consequence of Prop.1-3.4(a) and (M2) that

$$\text{if } xy = 1, \text{ then } x \neq 0, y \neq 0 \text{ and } x = y^{-1}.$$

When we use such a consequence of a Proposition or Law obtained by applying (M2) and/or (A2), we shall quote only the original Proposition or Law without explicitly pointing out that (A2) and/or (M2) is also being used. Thus the consequence of (A5) and (A2) that $x + (-x) = 0$ will be used by quoting only (A5).

In the preceding three Propositions, we have worked exclusively on the assumption that (A1)—(A5), (M1)—(M5) and (D) hold. The question of whether the elements of the set F are some kind of "numbers" did not enter into our considerations. In fact, we only need the set F to have two binary operations for which the properties (A1)—(A5), (M1)—(M5) and (D) hold. There are many such situations, apart from the ones we shall be concerned with. It is convenient to have a name for such situations.

1-3.5. Definition. *A* **field** *is a set F with two binary operations, called addition and multiplication, such that the Laws* (A1)—(A5), (M1)—(M5) *and* (D) *hold.*

The properties (A1)—(A5), (M1)—(M5) and (D) are together referred to as "field axioms". Since $\mathbb{Q}$ has them all, it is a field. But $\mathbb{Z}$ is not because it does not have property (M5).

We close this Section with the derivation of a few more elementary properties starting from field axioms.

1-3.6. Proposition. *Suppose F is a field. If* $x, y, a, b \in F$ *and* $x \neq 0, y \neq 0$, *then*
(a) $-(a + b) = (-a) + (-b)$;
(b) $(xy)^{-1} = x^{-1}y^{-1}$;
(c) $ax^{-1} + by^{-1} = (ay + bx)(xy)^{-1}$;
(d) $(ax^{-1})(by^{-1}) = (ab)(xy)^{-1}$.

***Proof.** (a) It follows by (A3) that

$$(b + a) + ((-a) + (-b)) = b + (a + ((-a) + (-b))). \tag{1}$$

Now,

$$\begin{aligned}(a + b) + ((-a) + (-b)) &= (b + a) + ((-a) + (-b)) = b + (a + ((-a) + (-b))) \\ &= b + ((a + (-a)) + (-b)) \\ &= b + (0 + (-b)) = b + (-b) = 0,\end{aligned}$$

where we have used (A2), (1), (A3) again, (A5), (A4), (A5) in that order. It now follows from Prop.1-3.1(c), $(-a)+(-b) = -(a+b)$. Of course, it is taken as understood that we have also used (A2) while applying Prop.1-3.1(c).

(b) $(xy)(x^{-1}y^{-1}) = (yx)(x^{-1}y^{-1}) = y(x(x^{-1}y^{-1})) = y((xx^{-1})y^{-1}) = y(1y^{-1}) = yy^{-1} = 1$, where we have used (M2), (M3), (M3) again, (M5), (M4), (M5) in that order. It now follows from Prop.1-3.4(a) that $x^{-1}y^{-1} = (xy)^{-1}$.

(c) $ax^{-1} = (a1)x^{-1} = (a(yy^{-1}))x^{-1} = ((ay)y^{-1})x^{-1} = (ay)(y^{-1}x^{-1}) = (ay)(x^{-1}y^{-1}) = ay(xy)^{-1}$, where we have used (M4), (M5), (M3), (M3) again, (M2) and part (b) above. By a similar argument, $by^{-1} = bx(xy)^{-1}$. Upon adding these two equalities and using (D), we get the required equality.

(d) Repeated use of (M2) and (M3) leads to $(ax^{-1})(by^{-1}) = (ab)(x^{-1}y^{-1})$. Now one may use (b). ☺

1-3.7. Definition. *If x and y are elements of a field and $y \neq 0$, then the* **quotient of** x **by** y *is the element xy^{-1} of the field and is denoted by x/y or also by ${}^{x}/_{y}$ or by $\frac{x}{y}$.*

It follows from this definition that $x^{-1} = 1x^{-1} = 1/x$. Parts (b), (c) and (d) of Prop.1-3.6 can be expressed in terms of quotients as saying that, when $x \neq 0$, $y \neq 0$,

$$1/(xy) = (1/x)(1/y), \quad a/x + b/y = (ay + bx)/(xy) \quad \text{and} \quad (a/x)(b/y) = (ab)/(xy).$$

Prop.1-3.4 can also be rephrased in terms of quotients.

1-3.8. Definition. *If x and y are elements of a field, the* **difference** *$x - y$ is defined to mean $x + (-y)$.*

It can be shown that

$$(1)\ x(y - z) = xy - xz \quad \text{and} \quad (2)\ a/x - b/y = (ay - bx)/(xy),$$

where it is assumed of course that $x \neq 0$, $y \neq 0$. Details are left to the reader in 1-3.P2 below. We shall use the results (1) and (2) without giving any reference or explanation.

Rearranging terms in a sum such as 5 + 8 + 25 is another familiar practice from elementary Mathematics. By not writing such brackets as (5 + 8) + 25 or 5 + (8 + 25), we are taking (A3) for granted. By rearranging as 8 + (5 + 25), which makes it slightly easier to add up, we are also taking for granted (A2). The point here is that, since (A3) and (A2) hold in any field, we can omit brackets and also do such rearranging in any field. Thus $x + z + y$ makes sense in any field and $x + z + y = x + y + z = z + y + x$ and so on. This simple fact will henceforth be used without explanation. See Prop.1-3.9 and 1-3.P5 below. The same will be done with products as well.

In any field we use the usual notation x^2 to mean the product xx, and 2 to mean $1+1$. We record for later use the following simple facts as consequences of the Laws stated so far.

1-3.9. Proposition. *If F is a field and* $x \in F$, $z \in F$, *then*

$$z^2 - x^2 = (z+x)(z-x), \quad (z+x)^2 = z^2 + 2zx + x^2 \quad \textit{and} \quad z - x = -(x-z).$$

Proof. This is left to the reader in 1-3.P5 below. ☺

Problem Set 1-3

1-3.P1. Rephrase Prop.1-3.4 in terms of quotients.

1-3.P2. Write out proofs of the following, pointing out at each step the Law, definition or proposition used: In any field,
(a) $x(y-z) = xy - xz$;
(b) $a/x - b/y = (ay - bx)/(xy)$ when $x \neq 0$, $y \neq 0$.

1-3.P3. In a field, if $x \neq 0 \neq y$, show that $x/y \neq 0 \neq y/x$ and $1/(x/y) = y/x$.

1-3.P4. Prove that, in a field, if $x \neq 0$ then (a) $-x \neq 0$ and (b) $1/(-x) = -(1/x)$.

1-3.P5. Show that, in any field,
(a) $2x = x + x$ (b) $(x+y)^2 = x^2 + 2xy + y^2$ (c) $(x+y)(x-y) = x^2 - y^2$
(d) $x - y = -(y-x)$.
[Note: This is supposed to be proved in *any field*. Therefore the traditional formulas from school level algebra are no justification for (a), (b) and (c). These have to be derived from field axioms and Propositions already derived from them. (M5) will not be needed.]

1-3.P6. In any field, show that (a) $a/x = b/y \Leftrightarrow ay = bx$ and (b) $xa/xy = a/y$, assuming (of course) that $x \neq 0$, $y \neq 0$.

1-3.P7. Using only (A1)—(A5) and their consequences, show that $x \neq 0 \Rightarrow -x \neq 0$.

1-3.P8. [Needed in 1-4.P13&14] (a) Prove that in any field, if $ax_1^2 + bx_1 + c = 0$ and $a \neq 0$, then $x_2 = -(b/a + x_1)$ satisfies $ax_2^2 + bx_2 + c = 0$.

(b) Prove that in any field, if $ax_1^2 + bx_1 + c = 0 = ax_2^2 + bx_2 + c$, $x_1 \neq x_2$, $a \neq 0$, then $x_1 + x_2 = -b/a$ and $x_1 x_2 = c/a$. [Caution: Square roots are not guaranteed to exist; so the quadratic formula cannot be presumed to be applicable.]
(c) If $a \neq 0$, show that there cannot exist three distinct elements x such that $ax^2 + bx + c = 0$.
(d) Let $a \neq 0$. If there exists one x_1 such that $ax_1^2 + bx_1 + c = 0$ and no other, show that $b^2 - 4ac = 0$.

1-4 Order Axioms

The Laws stated so far tell us nothing about some numbers being bigger than others or about some numbers being positive and others being negative. The basic Laws which we expect regarding the "less than" relation < in a field F are as follows:

(O1) **Trichotomy Law:** If $a \in F,\ b \in F,$ then one of the following three must be true:

$$a < b, \quad a = b, \quad b < a;$$

besides, no two of the three are true for the same pair of elements a and b.

(O2) **Transitive Law of Order:** If $a \in F,\ b \in F,\ c \in F$ then

$$a < b,\ b < c \Rightarrow a < c.$$

(O3) **Addition Law of Order:** If $a \in F,\ b \in F,\ x \in F$ then

$$a < b \Rightarrow a + x < b + x.$$

(O4) **Multiplication Law of Order:** If $a \in F,\ b \in F$ then

$$0 < a,\ 0 < b \Rightarrow 0 < ab.$$

There are many fields in which there is no such relation as < for which these Laws hold, but we shall not be concerned with them. When there is such a relation, we call it the "order relation" in that field. Note that such a relation is available in $\mathbb{Q}$.

1-4.1. Definition. *A field in which the Laws* (O1)—(O4) *hold is called an* **ordered field**.

Thus $\mathbb{Q}$ is an example of an ordered field.

As usual, $a > b$ is interpreted to mean $b < a$. Also, "$a \geq b$" is interpreted to mean "$a > b$ or $a = b$", and "$a \leq b$" is interpreted to mean "$a < b$ or $a = b$".

We remind the reader that the point here is not to present any new properties but to show that the familiar properties can be deduced from the ones we have listed as Laws. When field axioms or their consequences are used, no explicit references will be given.

1-4.2. Proposition. *Suppose F is an ordered field and $a, b, c \in F$. Then*
(a) $a > 0 \Leftrightarrow -a < 0$. *Also*, $a < 0 \Leftrightarrow -a > 0$.
(b) $a > 0, b < 0 \Rightarrow ab < 0$.
(c) $a > 0, b > c \Rightarrow ab > ac$.
(d) $a < 0, b > c \Rightarrow ab < ac$.
(e) $a \neq 0 \Rightarrow a^2 > 0$.
(f) $a > 0 \Rightarrow a^{-1} > 0$; *and* $a < 0 \Rightarrow a^{-1} < 0$.
(g) $a > b > 0 \Rightarrow b^{-1} > a^{-1} > 0$.
(h) $a > b \Leftrightarrow -b > -a$.
(i) $a < 0, b < 0 \Rightarrow ab > 0$.

Proof. (a) $a > 0 \Rightarrow -a + a > -a + 0 \Rightarrow 0 > -a$, where we have used (O3),. This proves that $a > 0 \Rightarrow -a < 0$. The converse [that $-a < 0 \Rightarrow a > 0$] is proved in a similar manner. The second assertion of (a) follows by applying to $-a$ what has already been proved and noting that $-(-a) = a$.

(b) Let $a > 0$ and $b < 0$. Then by (a), we have $a > 0$ and $-b > 0$. Therefore $a(-b) > 0$ by (O4) i.e. $-ab > 0$. It follows by (a) that $-(-ab) < 0$, so that $ab < 0$.

(c) Let $a > 0$ and $b > c$. Then by (O3), $b - c > 0$. Using (O4), we now get $a(b - c) > 0$. Therefore $ab - ac > 0$, and hence $ab > ac$ by (O3).

(d) Let $a < 0$ and $b > c$. Then by (O3), $b - c > 0$. Using (b), we get $a(b - c) < 0$. Thus $ab - ac < 0$. Hence by (O3), we have $ab < ac$.

(e) By (O1), either $a > 0$ or $a < 0$. If $a > 0$, then $a^2 > 0$ by (O4). If $a < 0$, then $-a > 0$ by (a), and therefore $(-a)(-a) > 0$ by (O4). But $(-a)(-a) = a^2$.

(f) Let $a > 0$. Then $a \neq 0$ by (O1) and hence $a^{-1} \neq 0$. Using (O1) again, either $a^{-1} > 0$ or $a^{-1} < 0$. Suppose, if possible, that $a^{-1} < 0$. Then $1 = a^{-1}a < 0$ by (b) above. But since $1 = 1^2$, therefore $1 > 0$ by (e). Hence by (O1), it is false that $1 < 0$. Thus it is not possible that $a^{-1} < 0$. It follows that $a^{-1} > 0$. Therefore $a > 0 \Rightarrow a^{-1} > 0$.

The second part [which says that $a < 0 \Rightarrow a^{-1} < 0$] can be proved from what has been just proved by using it together with (a), and the facts that $-(-x) = x$ and $(-x)^{-1} = -x^{-1}$. Details are left to the reader.

(g) Suppose $a > b > 0$. Then $ab > 0$ by (O4). It follows by (f) that $(ab)^{-1} > 0$. So $a^{-1}b^{-1} > 0$. It now follows by (c) that

$$a(a^{-1}b^{-1}) > b(a^{-1}b^{-1}) > 0(a^{-1}b^{-1}),$$

i.e. $b^{-1} > a^{-1} > 0$.

(h) Let $a > b$. Upon applying (O3) with $x = (-a) + (-b)$, it follows that

$$a + [(-a) + (-b)] > b + [(-a) + (-b)],$$

i.e. $-b > -a$. The converse follows from the fact that $-(-x) = x$ for every $x \in F$.

(i) This is an easy consequence of part (a) and (O4). ☺

1-4.3. Proposition. *Let F be an ordered field and $x \in F$. If $x = -x$, then $x = 0$.*

Proof. From the hypothesis that $x = -x$, it follows that $x + x = 0$. Suppose, if possible, that $x \neq 0$. Then either $x > 0$ or $x < 0$, by (O1). In case $x > 0$, we obtain from (O3) that $x + x > x$, so that $x + x > 0$ by (O2). But then $x + x \neq 0$ by (O1). This contradicts our earlier conclusion that $x + x = 0$. In case $x < 0$, we deduce via (O3) that $x + x < x$, so that $x + x < 0$ by (O2). But then $x + x \neq 0$ by (O1) and we again have a contradiction. Therefore $x = 0$. ☺

With reference to any *ordered* field, we refer to an element x as being **positive** if $x > 0$ and **negative** if $x < 0$. The element 0 is *neither positive nor negative*. The notation "$x \geq y$" has the usual meaning. An element x is called **nonnegative** if $x \geq 0$ and **nonpositive** if $x \leq 0$. It is easy to see that x is nonnegative if, and only if $-x$ is nonpositive and that x is nonpositive if, and only if $-x$ is nonnegative.

As already mentioned in Section 1-3, while working with any field whatsoever, we use the usual notation x^2 to mean the product xx. Also, we call any element z for which $z^2 = x$ as a *square root* of x. Unless specified otherwise, it is understood that a square root z (if any) is in the same field. There is no presumption that an element of a field must have a square root. On the basis of the Laws (A1)—(A5), (M1)—(M5) and (D), we know that $0^2 = 0$ and $1^2 = 1$, and hence that 0 and 1 have square roots. The aforementioned Laws do not provide a basis to conclude that any other elements must have square roots.

1-4.4. Proposition. *In an ordered field, if $x > y \geq 0$ then $x^2 > y^2$. Also, if $x \geq 0$, $y \geq 0$ and $x^2 \leq y^2$, then $x \leq y$.*

Proof. If $y = 0$, then Prop.1-4.2(e) shows that $x^2 > y^2$. If $y \neq 0$, then $x > y > 0$, and it follows by (O4) that $xx > xy$ and also that $xy > yy$. Hence $x^2 > y^2$ by (O2). For the second part, let $x \geq 0$, $y \geq 0$ and $x^2 \leq y^2$. If it were not true that $x \leq y$, then by (O1) we would have $x > y$ and it would follow from what has already been proved that $x^2 > y^2$, a contradiction. ☺

1-4.5. Proposition. *An element of a field can have at most two square roots. The element* 0 *has only one square root* (*which is* 0, *of course*). *If an element of an ordered field has a square root and is not* 0, *then it has one positive square root and one negative square root.*

Proof. If y and z are both square roots of the same element x, then $y^2 - z^2 = x - x = 0$. Therefore $(y+z)(y-z) = 0$. It follows that either $y = -z$ or $y = z$. Thus $-z$ is the only square root that x can have besides z. Therefore there are at most two square roots. If $x \neq 0$, then $z \neq 0$. If the field is ordered, then it follows from Prop.1-4.3 that $z \neq -z$, so that these are two different square roots of x. By (O1) and Prop.1-4.2(a), one among z and $-z$ is positive and the other is negative.

If $z^2 = 0$, then $z = 0$. So 0 has only one square root, whether the field is ordered or not. ☺

Towards the end of Section 1-3, we mentioned that "2" means $1+1$ in *any field*. We now note that $2 > 0$ in any ordered field. This is so because $1 = 1^2 > 0$ by Prop.1-4.2(e) and $1+1 > 1+0 = 1$ by (O3); since $1 > 0$, it follows by (O2) that $1+1 > 0$. The fact that $2 > 1$ and hence that $2 > 0$ will be used without reference and will therefore not be formulated as a Proposition.

Now we formally define *absolute value*, a concept already encountered in Calculus. The definition presumes (O1). From now on, we shall often use (O1) without saying so explicitly.

1-4.6. Definition. *The* **absolute value** $|x|$ *of an element x of an ordered field is*

$$|x| = \begin{cases} x \textit{ if } x \textit{ is positive} \\ 0 \textit{ if } x \textit{ is } 0 \\ -x \textit{ if } x \textit{ is negative.} \end{cases}$$

It is a consequence of this Definition and the fact that $0 = -0$ that

$$|x| = \begin{cases} x \text{ if } x \geq 0 \\ -x \text{ if } x \leq 0. \end{cases}$$

We shall not state this consequence as a Proposition, but shall use it freely without explanation.

1-4.7. Proposition. *For any x and y in an ordered field F,*

(a) $|x| \geq 0$; *and* $|x| > 0$ *except when* $x = 0$;

(b) $|x|^2 = x^2$;

(c) $|xy| = |x||y|$;

(d) $|-x| = |x|$;

(e) $x \le |x|$ *and* $-x \le |x|$; *hence* $-|x| \le x \le |x|$;
(f) $|x+y| \le |x| + |y|$; [This is called the **triangle inequality**.]
(g) $|x-z| \le |x-y| + |y-z|$; [Also called the **triangle inequality**.]
(h) $\big||x| - |y|\big| \le |x-y|$.

Proof. (a) When $x = 0$, we have $|x| = 0$ by definition of absolute value. When $x > 0$, we have $|x| = x$ by definition of absolute value and hence $|x| > 0$ in this case. When $x < 0$, we have $|x| = -x$ by definition and $-x > 0$ by Prop.1-4.2(a). [By (O1), there are no other possibilities to consider.]

(b) If $x \ge 0$, then $|x| = x$ and hence $|x|^2 = x^2$. If $x < 0$, then $|x|^2 = (-x)^2 = x^2$.

(c) If $x = 0$, then $|x||y| = |0||y| = 0|y| = 0$, while $|xy| = |0y| = |0| = 0$. So $|xy| = |x||y|$. A similar argument works when $y = 0$. So, consider the situation when $x \ne 0 \ne y$. Then $|x||y| > 0$ by (a) and (O4). By (b), we have $|xy|^2 = (xy)^2 = x^2y^2 = |x|^2|y|^2 = (|x||y|)^2$. Now $xy \ne 0$ and hence $|xy| > 0$ by (a). This means that $|xy|$ and $|x||y|$ are both positive square roots of $(|x||y|)^2$. It follows from Prop.1-4.5 (last part) that $|xy| = |x||y|$.

(d) If $x = 0$, then $-x = -0 = 0$. Hence $-x = x$ and $|-x| = |x|$. If $x > 0$, then $|x| = x$. Also, $-x < 0$ by Prop.1-4.2(a). Therefore $|-x| = -(-x) = x$. Thus $|x|$ and $|-x|$ are both equal to x when $x > 0$. A similar argument shows that $|x|$ and $|-x|$ are both equal to $-x$ when $x < 0$.

(e) This is clear when $x = 0$. Suppose $x > 0$. Then, as shown in the proof of (d), $-x < 0 < x = |x|$ and hence $-x < |x|$ by (O2). Therefore we have $x = |x|$ and $-x < |x|$. It follows that $x \le |x|$ and $-x \le |x|$ in the case that $x > 0$. For the case $x < 0$, we can either give a similar argument or proceed as follows: By Prop.1-4.2(a), $-x > 0$. Hence by what has just been proved, $-x \le |-x|$ and $-(-x) \le |x|$. But $|-x| = |x|$ by (d). Therefore $x \le |x|$ and $-x \le |x|$ also in the case that $x < 0$. From the latter one of these two inequalities, we further get $-|x| \le x$ by applying Prop.1-4.2(h).

(f) By (e), $x \le |x|$ and $y \le |y|$. Using (O3) twice and (O2) once, it follows that $x + y \le |x| + |y|$. Similarly we have, $(-x) + (-y) \le |x| + |y|$. Therefore

$$-(x+y) \le |x| + |y| \text{ and } x + y \le |x| + |y|.$$

By definition of absolute value, $|x + y|$ is equal to either $-(x + y)$ or $x + y$. So, $|x + y| \le |x| + |y|$.

(g) $|x-z| = |(x-y)+(y-z)| \le |x-y| + |y-z|$ by part (f).

(h) By (f),

$$|x| = |(x-y)+y| \le |x-y| + |y|. \tag{1}$$

By (f) and (d),

$$|y| = |(y-x)+x| \le |y-x| + |x| = |-(y-x)| + |x| = |x-y| + |x|. \tag{2}$$

From (1), (O3), it follows that $|x|-|y| \le |x-y|$. From (2), (O3), it follows that $|y|-|x| \le |x-y|$. But $|y|-|x| = -(|x|-|y|)$. Therefore we have

$$|x|-|y| \le |x-y| \quad \text{and} \quad -(|x|-|y|) \le |x-y|.$$

By definition of absolute value, $||x|-|y||$ is equal to either $-(|x|-|y|)$ or $|x|-|y|$. Therefore $||x|-|y|| \le |x-y|$. ☺

1-4.8. Proposition. *In an ordered field, if* $a > 0$, *then*

$$|x| < a \Leftrightarrow -a < x < a,$$

and

$$|x| \le a \Leftrightarrow -a \le x \le a.$$

Proof. Suppose $|x| < a$. By Prop.1-4.7(e), $x \le |x|$. Therefore by (O2), $x < a$. Similarly, $-x < a$. Adding $x - a$ to both sides and using (O3), we get $-a < x$. This proves that $|x| < a \Rightarrow -a < x < a$. To prove the converse, assume that $-a < x < a$. By using (O3), we get $-x < a$ and $x < a$. But $|x|$ is either x or $-x$. Therefore $|x| < a$. So, the converse is also true.

The second part follows from the first part and the observation that, if $|x| = a$, then

$$-a < x = a \qquad \text{if } x > 0 \quad \text{and}$$
$$-a = x < a \qquad \text{if } x < 0. \text{ ☺}$$

If $a \in F$ and $b \in F$, then $\mathbf{min}\,\{\boldsymbol{a},\boldsymbol{b}\}$ will mean a if $a \le b$ and will mean b if $b \le a$. The following consequences are easy to derive and will be used on several occasions: If $a > 0$ and $b > 0$, then $\min\{a,b\} > 0$; if $\delta < \min\{a,b\}$, then $\delta < a$ as well as $\delta < b$. We leave it to the reader [see 1-4.P12] to show that $\min\{a,b\} = \frac{1}{2}[(a+b)-|a-b|]$.

Similarly, $\mathbf{max}\,\{\boldsymbol{a},\boldsymbol{b}\}$ will mean b if $a \le b$ and will mean a if $b \le a$. The corresponding properties of $\max\{a,b\}$ are obvious.

Problem Set 1-4

1-4.P1. Prove that in any ordered field,

(a) $a < 0 \Rightarrow a^{-1} < 0$ (b) $a < b < 0 \Rightarrow b^{-1} < a^{-1} < 0$

(c) $a \le b \Leftrightarrow -b \le -a$ (d) $x < y < 0 \Rightarrow x^2 > y^2$

(e) if $x \le 0$, $y \le 0$ and $x^2 \le y^2$, then $x \ge y$.

1-4.P2. In an ordered field, if $a > 0$, then prove that

$$|x-s| < a \Leftrightarrow s-a < x < s+a$$

and

$$|x-s| \le a \Leftrightarrow s-a \le x \le s+a.$$

1-4.P3. In an ordered field, show that if $a < b$, then $a < \frac{a+b}{2} < b$ [The presumption that 2 is not 0 is justified, because we have proved that $2 > 0$, which guarantees by (O1) that $2 \neq 0$.]

1-4.P4. In an ordered field, show that $|2ab| \le a^2 + b^2$.

1-4.P5. Let a be a positive element of an ordered field, $a \neq 1$. Prove $a + 1/a > 2$. [The result of the preceding Problem is relevant, but it must be borne in mind that there is no guarantee that a is a square just because it is positive!] Hence show that $a + 1/a \ge 2$ whenever $a > 0$.

1-4.P6. Let a, b and c be positive elements of an ordered field such that $abc = 1$. If $a \neq 1$ or $b \neq 1$ or $c \neq 1$, then prove that $a + b + c > 3$ (which is defined to mean $2 + 1$, of course).

1-4.P7. Check whether it is true in any ordered field that $-1 < x < 2 \Rightarrow x^2 - x - 2 < 0$.

1-4.P8. If x and y are elements of an ordered field, show that

$$x < y \Leftrightarrow y = x + \varepsilon, \text{ where } \varepsilon > 0.$$

1-4.P9. If x and y are elements of an ordered field, show that

$$\begin{aligned} x \le y &\Leftrightarrow x \le y + \varepsilon \quad \text{for all } \varepsilon > 0 \\ &\Leftrightarrow x < y + \varepsilon \quad \text{for all } \varepsilon > 0. \end{aligned}$$

1-4.P10. Write out the proof of Prop.1-4.2(i).

1-4.P11. Suppose $p < s < q$. If $a \le x_1 < q$, $a \le x_2 < q$ and $|x_1 - x_2| < s - p$, show that

$$\begin{array}{llll} \text{either} & a \le x_1 < s & \text{as well as} & a \le x_2 < s \\ \text{or} & p < x_1 < q & \text{as well as} & p < x_2 < q. \end{array}$$

1-4.P12. If a and b are elements of an ordered field, show that

$$\min\{a, b\} = \tfrac{1}{2}[(a+b) - |a-b|] \quad \text{and}$$
$$\max\{a, b\} = \tfrac{1}{2}[(a+b) + |a-b|].$$

1-4.P13. In an ordered field, if $ax_1^2 + bx_1 + c = 0 = ax_2^2 + bx_2 + c$, $x_1 \neq x_2$, $a \neq 0$, $c/a < 0$, then show that x_1 and x_2 cannot be both positive or both negative.

1-4.P14. In an ordered field, suppose $ax_1^2 + bx_1 + c = 0 = ax_2^2 + bx_2 + c$, $x_1 < x_2$ and $a > 0$. Show that $x_1 < x < x_2$ if, and only if, $ax^2 + bx + c < 0$.

1-4.P15. Let a and b be elements of an ordered field. Let

$f(a,b)$ denote the sum $|a+b|+|a|+|b|$

and $g(a,b)$ denote the sum $\big|a+b+|a|-|b|\big|+\big|a-b-|a+b|\big|$.

Comment on the following: (i) Is $f(a,b)=f(b,a)$? (ii) Is $f(a,b)=f(-a,-b)$? (iii) Is $g(a,b)=g(b,a)$? (iv) Is $g(a,b)=g(-a,-b)$ (v) Is $f(a,b)=g(a,b)$?

1-4.P16. [Needed in 1-8.P12] If $a \geq b+1 \geq 1$ in an ordered field, show that $a(a-1)-b(b-1) \geq 2b$.

1-5 Bounded Subsets

In order to understand how our definition of the number system will lead to satisfactory answers to the matters raised in Section 1-1, it will be essential for the reader to acquire facility with the topic discussed in the present Section and the next.

1-5.1. Definition. *For any subset S of an ordered field F, an element $a \in F$ is called an* **upper bound of** *S if every element of S is either less than a or equal to a.* [One also speaks of an upper bound *for S*.]

Another way to formulate this would be: *An element a in an ordered field F is an upper bound of a subset $S \subseteq F$ if $x \leq a\ \forall\ x \in S$.* Recall that the sequence of symbols "$\forall\ x \in$" means "for all x belonging to".

By saying "$x \leq a\ \forall\ x \in S$", we do not commit ourselves to the possibility of x being equal to a ever occurring. It is correct to say it even when we know that in actual fact no element $x \in S$ is equal to a; in other words, when it would also be correct to say "$x < a\ \forall\ x \in S$" if we wanted to.

We pause to note that the apparently "negative" statement that *no* element of S is greater than a is equivalent to the "positive" statement that *every* element of S is less than or equal to a.

It is also important to note that some subsets do not have an upper bound at all. For example, F itself has no upper bound. In other words, no $a \in F$ can have the property that *every* element $x \in F$ is less than or equal to a, because $x = a+1 \in F$ but is greater than a. If this example seems trivial, consider the subset $T = \{x \in F : x > 0\}$. Any nonpositive element $a \in F$ is less than every element of T, and is therefore not an upper bound; a positive element a also cannot be an upper bound because $a+1 \in T$ while $a+1 > a$. So, T does not have any upper bound at all. We need a name for such a situation.

1-5.2. Definition. *A subset of an ordered field that has an upper bound is said to be* **bounded above**. *Otherwise, it is said to be* **unbounded above**.

Since it is true regarding the empty subset $\varnothing$ and *any* element a that no element of $\varnothing$ is greater than a, the subset $\varnothing$ is bounded above. Moreover, every element of the field is an upper bound for it.

At this stage, the reader can probably anticipate what the analogous definitions of lower bound, a set being bounded below and related propositions are. Nevertheless, they are presented below and we hope that the reader will find them entirely in accordance with expectations.

1-5.3. Definition. *For any subset S of an ordered field F, an element $a \in F$ is called a* **lower bound of** *S if no element of S is less than a.* [One also speaks of a lower bound *for S.*]

In mathematical symbols, we have two "positive" versions

$$\forall\, x \in F,\ x \in S \Rightarrow x \geq a \qquad \text{or} \qquad x \geq a\ \forall\, x \in S.$$

Another way to formulate Def.1-5.3 would be: *An element a in an ordered field F is a lower bound of a subset $S \subseteq F$ if $x \geq a\ \forall\, x \in S$.*

Some subsets do not have a lower bound at all. For example, F itself and $T = \{x \in F : x < 0\}$.

1-5.4. Definition. *A subset of an ordered field that has a lower bound is said to be* **bounded below**. *Otherwise, it is said to be* **unbounded below**.

Since it is true regarding the empty subset $\varnothing$ and *any* element a that no element of $\varnothing$ is less than a, the subset $\varnothing$ is bounded below. Moreover, every element of the field is a lower bound for it.

1-5.5. Proposition. *If a is an upper bound of a subset S of an ordered field and T $= \{x : -x \in S\}$, then $-a$ is a lower bound of T. Similarly, if a is a lower bound of a subset S of an ordered field and $T = \{x : -x \in S\}$, then $-a$ is an upper bound of T.*

Proof. Consider any $t \in T$. By the definition of T, it follows that $-t \in S$. Now a is an upper bound of S. Therefore $-t \leq a$, and hence by Prop.1-4.2(h), $-(-t) \geq -a$. Thus $t \geq -a\ \forall\ t \in T$. By definition of lower bound, $-a$ is a lower bound of T.

The proof of the second part is similar and is left as an exercise in 1-5.P4. ☺

The statement "$t \leq a\ \forall\ t \in T$" asserts that the inequality that $t \leq a$ is true of every element t of the set T without exception. For this to be false, all we need is one instance of an element t belonging to T for which $t > a$ (Trichotomy). There could of course be many such. When such an element exists in T, we express it in mathematical symbols by writing

$$t > a \text{ for some } t \in T \quad \text{or} \quad \exists\ t \in T \text{ such that } t > a \quad \text{or} \quad \exists\ t \in T \ni t > a.$$

The latter two are both read as "There exists t in T such that $t > a$". The symbol "∃" stands for "there exists" and "∋" stands for "such that". Stated in plain English, "Some element of T, which we shall call t, is greater than a". Note that this amounts to declaring what the symbol t stands for and therefore any previous meaning of t is cancelled. In order to prevent confusion, it is better to avoid reusing a symbol as far as possible.

In terms of these symbols, the "negative" assertion that a is not an upper bound of T is equivalent to the "positive" statement: $\exists\ t \in T \ni t > a$. Similarly, a is not a lower bound of T if, and only if, $\exists\ t \in T \ni t < a$ (i.e. some element of T is less than a).

It should be no surprise that, when $S = \{s \in F : -s \in T\}$, and $a \in F$ is not an upper bound of S, the element $-a$ is not a lower bound of T. This is the same as saying that, if $\exists\ s \in S \ni s > a$, then $\exists\ t \in T \ni t < -a$. We shall prove this in order to illustrate how to handle situations involving these kinds of statements: Suppose $s \in S$ has the property that $s > a$. Then $-s < -a$. Since $s \in S$, therefore $-s \in T$. Let $t = -s$. Then $t \in T$ and $t < -a$. Thus $\exists\ t \in T \ni t < -a$. This means that $-a$ is not a lower bound of T.

The next result is extremely simple and will therefore be used in later Sections without giving any reference.

1-5.6. Proposition. *In an ordered field, any element greater than an upper bound of a subset S must itself be an upper bound of S.*

Proof. Let a be an upper bound of S and $b > a$. Then $s \leq a$ whenever $s \in S$ and by the Transitive Law of Order, $s < b$ whenever $s \in S$. Thus b is an upper bound of S. ☺

Remarks. The element 1 of the set $S = \{0, 1\}$ is the greatest and therefore is an upper bound. Moreover, if $b \in F$ is also an upper bound of S, then by definition, b is greater than or equal to *every* element of S, in particular greater than or equal to 1. Thus, 1 is not only an upper bound, but is also less than or equal to every upper bound; in other words, 1 is the *least* upper bound.

There can be a least one among all upper bounds even when the set contains no greatest element, as we now illustrate.

Let $S = \{x \in F : x < 1\}$. This set has 1 as an upper bound. Consider any $b \in F$ such that $b < 1$. Then $b < \frac{b+1}{2} < 1$, so that $\frac{b+1}{2}$ is an element of S but is greater than b. This shows that no element less than 1 can be an upper bound. Consequently, by the Trichotomy Law, every upper bound must be greater than or equal to 1. Thus, 1 is not only an upper bound, but is also less than or equal to every upper bound; in other words, 1 is the *least* upper bound, although without being an element of S.

Finally, consider the example of the set $S = \{x \in F : x < 1/(1+x^4)\}$. This set has 0 as an element of it, because $0 < 1/(1+0^4)$. Also, 1 is an upper bound, because $1/(1+x^4) \leq 1\ \forall\ x \in F$. Since it does have some upper bound, the set is bounded above. But if we were to ask for a least upper bound for the set, there would be no ready answer. One may even wonder whether a least upper bound exists at all. We shall return to this question in the next Section.

1-5.7. Definition. *A subset S of an ordered field F is said to be* **bounded** *if there exists some $M \geq 0$ such that $|x| \leq M$ for every $x \in S$. Any such element $M \in F$ is called a* **bound** *for the set S.*

If $a > 0$, the set $\{x \in F : -a \leq x \leq a\}$ is bounded, the element a being a bound for it. The set $\{x \in F : -a \leq x \leq a + 1\}$ is also bounded with bound $a + 1$ and $\{x \in F : -a - 1 \leq x \leq a\}$ has bound $a + 1$. The sets $\{x \in F : -a \leq x\}$ and $\{x \in F : x \leq a\}$ are not bounded.

1-5.8. Proposition. *A subset S of an ordered field F is bounded if, and only if, it is bounded above as well as below.*

Proof. If M is a bound for S, then M is an upper bound and $-M$ is a lower bound. So, every bounded set is bounded above as well as below. Conversely, suppose $S \subseteq F$ has a lower bound α and an upper bound β. Then $\alpha \leq x \leq \beta\ \forall\ x \in S$. Therefore every $x \in S$ satisfies both the inequalities $x \leq \beta \leq |\beta| \leq \max\{|\alpha|, |\beta|\}$ and $-x \leq -\alpha \leq |\alpha| \leq \max\{|\alpha|, |\beta|\}$. Hence $x \in S \Rightarrow |x| \leq \max\{|\alpha|, |\beta|\}$, so that $\max\{|\alpha|, |\beta|\}$ is a bound for S. ☺

Problem Set 1-5

1-5.P1. Name any two upper bounds for each of the following subsets of an ordered field F (several answers possible for each):

(a) $\{-2, -1, -1/2\}$ (b) $\{-2, -1/2, 0\}$ (c) $\{x \in F : -1/2 < x < 1\}$

(d) $\{x \in F : -2 \leq x < 1\}$ (e) $\{x \in F : -2 \leq x \leq 1\}$

(f) $\{x \in F : -2 \leq x \leq -1/2\} \cup \{x \in F : 0 \leq x \leq 1\}$

(g) $\{x \in F : -2 \leq x \leq 1/2\} \cap \{x \in F : 0 \leq x \leq 1\}$

1-5.P2. Suppose S and T are subsets of an ordered field F. Prove the following:
(a) If $S \subseteq T$, then any lower bound of T is also a lower bound of S.
(b) If a and b are lower bounds of S and T respectively and $a \leq b$, then a is a lower bound of $S \cup T$ and b is a lower bound of $S \cap T$.
(c) If a and b are lower bounds of S and T respectively and $U = \{s + t : s \in S \text{ and } t \in T\}$, then $a + b$ is a lower bound of U.

1-5.P3. Suppose S is a subset of an ordered field F, $a \in F$ and

$$\forall\, s \in S, \forall\, \varepsilon \in F, \quad \varepsilon > 0 \Rightarrow s < a + \varepsilon.$$

Show that a is an upper bound of S.

1-5.P4. Prove the second part of Prop.1-5.5.

1-5.P5. Suppose T is a subset of an ordered field F and $S = \{s \in F : -s \in T\}$. If $a \in F$ is not a lower bound of S, show that the element $-a$ is not an upper bound of T.

1-5.P6. Suppose S and T are subsets of an ordered field F, a is a lower bound of S and b an upper bound of T. Let $U = \{s - t : s \in S \text{ and } t \in T\}$. Show that $a - b$ is a lower bound of U.

1-5.P7. If a is not an upper bound of a subset S of an ordered field F and $b < a$, show that b is also not an upper bound of S.

1-5.P8. Suppose a is not an upper bound of a subset S of an ordered field F. Prove that there exists $b \in F$ such that $b > a$ and b is not an upper bound of S.

1-5.P9. Let $S = \{1\}$. Then 0 and 1 are both lower bounds of S.
(a) Is there a greater element than 0 that is also a lower bound of the same set?
(b) Is there a greater element than 1 that is also a lower bound of the same set?

1-5.P10. Given an element of an ordered field that is an upper bound of a subset (of the field), must there exist a lesser element that is also an upper bound of that subset?

1-5.P11. Fill in the blanks below using mathematical symbols, so that the resulting question is the same as in 1-5.P10. Denote the field by F, the subset by S and the upper bound by a.
Given that _____ an ordered field, __ $\subseteq$ __ and _____ has the property that __________, is it true that ________________________?

1-5.P12. Describe the set $\{x \in F : x^2 < 2^2\}$ without mentioning any squares. Find a lower bound for it such that no greater element of F is a lower bound.

1-5.P13. Describe the set $\{x \in F : x(1-x) > 0\}$ in the form $\{x \in F : \alpha < x < \beta\}$. [You may use (O4) and Prop.1-4.2(b) if you wish.] Find a lower bound for it such that no greater element of F is a lower bound.

1-5.P14. Formulate what is to be proved in 1-4.P8 using the symbols $\exists$ and $\ni$.

1-5.P15. Suppose S and T are subsets of an ordered field F. Prove:
(a) If $S \subseteq T$, then any upper bound of T is also an upper bound of S.
(b) If a and b are upper bounds of S and T respectively and $a \leq b$, then b is an upper bound of $S \cup T$ and a is an upper bound of $S \cap T$.
(c) If a and b are upper bounds of S and T respectively and

$$U = \{s + t : s \in S \text{ and } t \in T\},$$

then $a + b$ is an upper bound of U.

1-5.P16. Prove that, given any element in an ordered field that is not a lower bound of a subset (of the field), there exists a smaller element that is also not a lower bound of that subset.

1-6 Completeness Axiom

Towards the end of Section 1-5, we noted that the sets $\{0,1\}$ and $\{x \in F : x < 1\}$ both have 1 as the least upper bound. While 1 belongs to the first of our two sets, it does not belong to the second. We also noted that the set $\{x \in F : x < 1/(1 + x^4)\}$ has 1 as an upper bound but that a least upper bound was not so easy to find. We now lay down the formal, though perhaps obvious, definition of the term "least upper bound".

1-6.1. Definition. *For a subset S of an ordered field, a* **least upper bound** *is an element of the ordered field which is an upper bound of S and is less than or equal to every upper bound of S. It is also called a* **supremum** *of S.*

If a set is not bounded above (i.e. has no upper bound at all) then it certainly cannot have a least upper bound. There is one obvious case of a set being bounded above and yet not having any least upper bound: the empty set $\varnothing$. As noted in the preceding Section, this particular set has *every* element of the ordered field as an upper bound and therefore there can be no least one among them.

1-6.2. Definition. *For a subset S of an ordered field, a* **greatest lower bound** *is an element of the ordered field which is a lower bound of S and is greater than or equal to every lower bound of S. It is also called an* **infimum** *of S.*

If a set is not bounded below (i.e. has no lower bound at all) then it certainly cannot have a greatest lower bound. The empty set $\varnothing$ has *every* element of the ordered field as a lower bound and therefore there can be no greatest one among them.

1-6.3. Proposition. *If a subset S of an ordered field F has a sup (or inf), then it is unique. In other words, two different elements of F cannot both be suprema (or infima) of the same set.*

Proof. Suppose a and b are both suprema of S. Being suprema, both are upper bounds and both are least among upper bounds. It follows that $a \le b$ and $b \le a$. By the Trichotomy Law, it follows that $a = b$. The argument for infima is similar. ☺

Since we have established that there cannot be two different suprema of a set, from now on, we may speak of *the* supremum of a set when there is a supremum at all. Similarly for *the* infimum.

1-6.4. Notation. The supremum (if any) of a set S is denoted by **sup *S*** or by **LUB *S***. Similarly, the infimum (if any) of a set S is denoted by **inf *S*** or by **GLB *S***.

We note that the notation **lub *S*** or **l.u.b. *S*** may also be used. The same is true of **glb *S*** and **g.l.b. *S***.

As noted in Section 1-5, it is a consequence of the Trichotomy Law that the assertion that a is not an upper bound of T is equivalent to: $\exists\, t \in T \ni t > a$.

1-6.5. Proposition. *In an ordered field, an upper bound a of a subset S is the supremum of S if, and only if,*

$$b < a \Rightarrow \exists\, s \in S \ni s > b.$$

Proof. Suppose $a = \sup S$. Then by definition of supremum, it is an upper bound. Consider any $b < a$. Again by definition of supremum, b is not an upper bound and hence $\exists\, s \in S \ni s > b$.

For the converse, suppose the upper bound a of S has the property that $b < a \Rightarrow \exists\, s \in S \ni s > b$. Then no element b that is less than a can be an upper bound, so that by Trichotomy, an upper bound must necessarily be greater than or equal to a. Since a is itself an upper bound, it now follows by the definition of supremum that $a = \sup S$. ☺

The condition in the preceding Proposition can be rephrased as:

$$\forall\, \varepsilon > 0, \exists\, s \in S \ni s > a - \varepsilon.$$

1-6.6. Proposition. *In an ordered field, a lower bound a of a subset S is the infimum of S if, and only if,*

$$b > a \Rightarrow \exists\, s \in S \ni s < b.$$

Proof. Left to the reader [see 1-6.P3]. ☺

1-6.7. Definition. *An ordered field which satisfies the following additional property is said to be* **complete** (*or* **order complete**):

Every nonempty subset which is bounded above has a supremum.

The property is referred to as the **Completeness Axiom**, *or also as the* **Order Completeness Axiom**.

In particular, the subset $\{x \in F : x < 1/(1+x^4)\}$, for which we did not claim to know any supremum, does have a supremum if F is order complete. Similarly, the subset $\{x \in F : 0 < x \text{ and } x^2 < 2\}$ has a supremum. As the reader may have guessed, the supremum of the latter subset is what we would ordinarily call √2. In other words, an element $a \in F$ for which $a > 0$ and $a^2 = 2$. However, it can happen in an ordered field that there is no such element (we justify this claim later), while there must always exist such an element if the field is complete (we justify this also later).

We are now ready to describe the number system we want to work with:

1-6.8. Definition. *The* **real number system** $\mathbb{R}$ *is a complete ordered field containing the rational number system* $\mathbb{Q}$.

Actually, any ordered field always contains $\mathbb{Q}$ but a proof of this fact will divert attention from more essential matters; therefore we prefer to assume $\mathbb{Q} \subseteq \mathbb{R}$ by definition.

$\mathbb{R}$ is also called the **real number field** or simply **real field**.

We demonstrate the power of the Order Completeness property in the next result. Recall that Prop.1-4.5 states that an element of an ordered field has at most two square roots. Now, an element of an ordered field need not have a square root at all [see Prop.1-9.3 below]. This however is not the case if one is dealing with an order complete field, as we shall presently show.

In the statement of the Proposition below, the word "unique" is to be understood as "one and not more". This is how the word is used in Mathematics; the phrases "one and only one" and "exactly one" are also used instead of "unique".

1-6.9. Proposition. *Every positive element a of a complete ordered field F has a unique positive square root, i.e. a positive element b such that* $b^2 = a$.

Proof. Let $S = \{x \in F : 0 < x \text{ and } x^2 < a\}$. In case $a > 1$, we have $1 \in S$ because $1 > 0$ and $1^2 = 1 < a$. In case $a \le 1$, we have $\frac{1}{2}a \in S$ because not only is $\frac{1}{2}a > 0$ but also $\frac{1}{2}a \le \frac{1}{2}$, so that $(\frac{1}{2}a)^2 < \frac{1}{2}a < a$. Thus S is nonempty whether or not $a > 1$.

Next, we show that:

$$\text{Any positive number } c \text{ for which } c^2 > a \text{ is an upper bound of } S. \tag{1}$$

Let $s \in S$. Then s and c are both positive and $s^2 < a < c^2$. It follows by Prop.1-4.4 that $s < c$. This proves (1). In particular, $a+1$ is an upper bound of S, considering that $a+1 > 0$ and $(a+1)^2 = a^2 + 2a + 1 > 2a > a$.

It follows by the Completeness Axiom that S has a supremum, which we shall call b. Since S consists of positive numbers, its supremum b must also be positive. We shall argue that $b^2 = a$.

Let

$$h = \frac{a - b^2}{a + b}. \tag{2}$$

It is an easy computation to show that $b + h = \frac{a(b+1)}{a+b}$, which implies that

$$b + h \text{ is positive,} \tag{3}$$

and

$$a - (b+h)^2 = \frac{a(a-1)(a-b^2)}{(a+b)^2}. \tag{4}$$

Note that a, $a-1$ and $(a+b)^2$ are all positive. Now suppose, if possible, that $b^2 < a$. Then it follows from (2) and (4) that both h and $a-(b+h)^2$ are positive. This has the consequence that $b+h > b$ and $(b+h)^2 < a$. But this means $b+h$ is an element of S that is greater than b, contradicting the fact that b is an upper bound of S. Therefore the possibility that $b^2 < a$ is ruled out. So, suppose that $b^2 > a$. Then it follows from (2) and (4) that both h and $a-(b+h)^2$ are negative. This has the consequence that $b+h < b$ and $(b+h)^2 > a$. Together with (1)and (3), this means $b+h$ is an upper bound of S that is less than b, contradicting the fact b is the least one of the upper bounds of S. Therefore the possibility that $b^2 > a$ is also ruled out. Hence $b^2 = a$.

There cannot be another positive square root, because this is so in any ordered field [see Prop.1-4.5]. ☺

1-6.10. Remark. In view of Props.1-4.5 & 1-6.9, the standard quadratic formula can be derived by the usual computation. Moreover, elementary facts about quadratic equations, such as conditions for the roots to have opposite signs, can be justified. Therefore they will be used freely in the sequel.

Problem Set 1-6

In this Problem Set, the symbol F denotes a complete ordered field.

1-6.P1. Name the least upper bound (i.e. supremum) of each of the subsets in 1-5.P1.

1-6.P2. For each of the subsets S in 1-5.P1, let $T = \{x \in F : -x \in S\}$. What is $\inf T$ in each case?

1-6.P3. Prove Prop.1-6.6.

1-6.P4. If $x \in F$, prove that $x > 0 \Leftrightarrow x = y^2$ for some $y \in F$.

1-6.P5. Let $\varnothing \neq S \subseteq F$ and $a = \sup S$. For the subset $T = \{x \in F : -x \in S\}$, show that $\inf T = -a$.

1-6.P6. Suppose S and T are subsets of F. If a and b are suprema of S and T respectively and

$$U = \{s + t : s \in S \text{ and } t \in T\},$$

then show that $a + b$ is the supremum of U.

1-6.P7. If $S \subseteq F$, $\sup S$ exists and $k \geq 0$, show that $\sup \{ks : s \in S\} = k \cdot \sup S$.

1-6.P8. If $S \subseteq F$, $\sup S$ exists and $k < 0$, show that $\inf \{ks : s \in S\} = k \cdot \sup S$.

1-6.P9. If $S \subseteq F$, $\inf S$ exists and $k < 0$, show that $\sup \{ks : s \in S\} = k \cdot \inf S$.

1-6.P10. If $\varnothing \neq S \subseteq T \subseteq F$ and T is bounded above, show that $\sup S \leq \sup T$. If $\varnothing \neq S \subseteq T \subseteq F$ and T is bounded below, show that $\inf T \leq \inf S$.

1-6.P11. If $S \subseteq F$, is bounded above and contains a positive number, then

$$\sup S = \sup \{s \in S : s > 0\}.$$

1-6.P12. If S and T are subsets of F with suprema a and b respectively, and each subset contains at least one positive number, then

$$\sup \{st : s \in S, t \in T, s > 0, t > 0\} = ab.$$

1-6.P13. Let $\varnothing \neq S \subseteq F$ and $\inf S > 0$. For the subset $T = \{x \in F : 1/x \in S\}$, show that $\sup T = 1/\inf S$ and $\inf T = 1/\sup S$.

1-6.P14. Suppose S and T are nonempty subsets of F, both of which are bounded above. If $\sup S \le \sup T$, prove that $\sup(S \cup T) = \sup T$. If $S \cap T$ is nonempty, prove that $\sup(S \cap T) \le \sup S$.

1-7 Intervals

Functions dealt with in Calculus generally have real numbers as values and certain kinds of subsets of $\mathbb{R}$ as domains. Examples of such subsets are

$\{x \in \mathbb{R} : 3 \le x \le 5\}$ for $f(x) = \sqrt{\{(3-x)(x-5)\}}$,

$\{x \in \mathbb{R} : 0 < x \le 1\}$ for $f(x) = x^{-1/4}(1-x)^{1/2}$

and

$\{x \in \mathbb{R} : x \ne 0\}$ for $f(x) = 1/x$.

In the first example here, the domain can be described in terms of the double inequality

$$3 \le x \quad \textit{and} \quad x \le 5,$$

and in the second one, it can be described in terms of the double inequality

$$0 < x \quad \textit{and} \quad x \le 1.$$

In the third example, the domain cannot be described in this manner. However, it can be described in terms of inequalities as

$$x < 0 \quad \textit{or} \quad x > 0.$$

The first two are examples of what is called a *bounded interval*. It could be described as the set of all real numbers lying between two numbers, inclusive or exclusive of one or both. [We note that the word "between" will ordinarily be understood to mean "exclusive", unless something different is specified.]

1-7.1. Definition. *If a and b are real numbers, where $a \le b$, then*

$$[a, b] \textit{ denotes } \{x \in \mathbb{R} : a \le x \le b\},$$

and if $a < b$, then

$$[a, b) \textit{ denotes } \{x \in \mathbb{R} : a \le x < b\}$$

$$(a, b] \textit{ denotes } \{x \in \mathbb{R} : a < x \le b\}$$

and

$$(a, b) \textit{ denotes } \{x \in \mathbb{R} : a < x < b\}.$$

By a **bounded interval** *we mean a set of one of these four kinds. The numbers a and b are called the* **left endpoint** *and* **right endpoint** *respectively of the bounded interval.* [When we use the notation $[a, b]$ for an interval, it is presumed that $a \le b$; similarly, when we write $[a, b)$, etc., it is understood that $a < b$.]

Remark. If $a = b$, then $[a, b]$ consists of the single element a. The other three kinds of bounded intervals are never empty.

Bounded intervals are bounded sets (above as well as below) in the sense of Def.1-5.7. There are other kinds of intervals too, which we now define. It will be seen immediately that they are unbounded sets:

1-7.2. Definition. *An* **unbounded interval** *is a subset of* $\mathbb{R}$ *of one of the forms*

$\{x \in \mathbb{R} : a \le x\}$, *which is denoted by* $[a, \infty)$,

$\{x \in \mathbb{R} : a < x\}$, *which is denoted by* (a, ∞),

$\{x \in \mathbb{R} : x \le b\}$, *which is denoted by* $(-\infty, b]$,

$\{x \in \mathbb{R} : x < b\}$, *which is denoted by* $(-\infty, b)$,

and $\mathbb{R}$, *sometimes denoted by* $(-\infty, \infty)$.

The intervals $[a, \infty)$ *and* (a, ∞) *have* **left endpoint** *a but have no right endpoint. Similarly, the intervals* $(-\infty, b]$ *and* $(-\infty, b)$ *have* **right endpoint** *b but have no left endpoint.*

It should be noted that the symbol ∞ has been introduced at this stage not as a mathematical object but only as an abbreviation for "no number belongs in this spot".

The word "interval" includes bounded and unbounded intervals alike. Since we have defined nine different kinds of intervals, four bounded and five unbounded, it is natural to ask what they have in common so as to deserve a common name. This will be answered in Prop. 1-7.5 below, but first we attend to some simple but frequently used properties of intervals. No reference need be given when Prop.1-7.3 or 1-7.5 is used.

1-7.3. Proposition. *A number lying between two numbers belonging to an interval itself belongs to that interval. In other words, for any interval I,*

$$p \in I,\ q \in I,\ p < x < q \Rightarrow x \in I. \tag{A}$$

Proof. We shall show that, if $a < b$, then $I = [a, b)$ has property (A). For other kinds of intervals, the argument is similar. Let $p \in [a, b)$, $q \in [a, b)$ and $p < x < q$. Then $a \le p < x < q < b$. It follows that $a \le x < b$ [actually more: $a < x < b$, but we do not need this]. Therefore $x \in [a, b)$. ☺

1-7.4. Proposition. [Needed in Sections 2-2, 2-3, 2-4, 6-1, 7-1] *If c belongs to an interval I but is not an endpoint of it, then for any* $\delta_1 > 0$, *there exists* $\delta > 0$ *such that* $0 < \delta < \delta_1$ *and* $[c - \delta, c + \delta] \subseteq I$.

If c belongs to an interval I but is not the right endpoint of *it, then for any* $\delta_1 > 0$, *there exists* $\delta > 0$ *such that* $0 < \delta < \delta_1$ *and* $[c, c + \delta] \subseteq I$.

If c belongs to an interval I but is not the left endpoint of it, then for any $\delta_1 > 0$, *there exists* $\delta > 0$ *such that* $0 < \delta < \delta_1$ *and* $[c-\delta, c] \subseteq I$.

***Proof.** Suppose $c \in I$ is not an endpoint of I. Then I contains some $x_1 < c$ and also some $x_2 > c$, so that the numbers $c - x_1$ and $x_2 - c$ are both positive. Let $\delta = \min\{\frac{1}{2}\delta_1, c-x_1, x_2-c\}$. It is left to the reader to show that $[c-\delta, c+\delta] \subseteq I$ by using Prop.1-7.3.

Now suppose $c \in I$ is not the right endpoint of I. Then I contains some $x_2 > c$ so that $x_2 - c$ is positive. Let $\delta = \min\{\frac{1}{2}\delta_1, x_2-c\}$. The reader may now show that $[c, c+\delta] \subseteq I$ by using Prop.1-7.3.

The last part is proved by an exactly analogous argument. ☺

The reader may recall that a major item in the agenda laid out for this book in Section 1-1 was a proof that a function with derivative 0 at each point of an interval has to be a constant. Although we have not yet formally defined a derivative, the "ammunition" required for the proof is almost ready, now that we have given a precise definition of what we mean by the number system (a complete ordered field) and defined what an interval is. The reader who wishes to see right away how the Completeness Axiom makes it possible to prove the aforementioned result is welcome to proceed directly to the Appendix to this Chapter. The result proved there does not explicitly mention a derivative, but its hypothesis is an obvious consequence of the derivative being 0 at each point. The Appendix does not draw upon any material discussed in the rest of this Chapter or later.

1-7.5. Proposition. [Converse of Prop.1-7.3; needed in Section 2-4] *A nonempty subset S of* $\mathbb{R}$ *is an interval if every number lying between two numbers belonging to S itself belongs to S, i.e.*

$$p \in S,\ q \in S,\ p < x < q \Rightarrow x \in S. \tag{A}$$

***Proof.** Consider the case when S has a lower as well as an upper bound and let $a = \inf S$, $b = \sup S$. Then for every $s \in S$, we have $a \le s \le b$, so that $S \subseteq [a, b]$. Let $x \in (a, b)$, i.e. $\inf S < x < \sup S$. Then x is neither a lower nor an upper bound and hence there exist $p, q \in S$ such that $p < x < q$ and hence by (A), $x \in S$. This shows that $(a, b) \subseteq S$. But $S \subseteq [a, b]$, as noted above. It follows that S equals one among $[a, b]$, $[a, b)$, $(a, b]$, (a, b), depending on whether a and/or b belongs to S or not.

The case when there is only an upper bound proceeds along the same lines with a replaced by $-\infty$, except that the reason why x is not a lower bound is easier.

Similarly when there is only a lower bound. Finally, the case when there is neither a lower nor an upper bound is even easier. ☺

Recall that a set whose elements are again sets is often called a *class* or *family* of sets. Thus a family of intervals is the same as a set of intervals.

***1-7.6. Proposition.** *The intersection of a family of intervals is either empty or an interval. In case it is nonempty, the union of the family is an interval.*

Proof. Let $\mathcal{F}$ be a family of intervals. Assume the intersection $\bigcap\mathcal{F}$ to be nonempty. Let $p \in \bigcap\mathcal{F}$, $q \in \bigcap\mathcal{F}$ and $p < x < q$. Then for every interval A of the family $\mathcal{F}$, we have $p \in A$, $q \in A$ and $p < x < q$. By Prop.1-7.3, $x \in A$. Since this holds $\forall\ A \in \mathcal{F}$, we have $x \in \bigcap\mathcal{F}$. By Prop.1-7.5, $\bigcap\mathcal{F}$ is an interval.

To see why $\bigcup\mathcal{F}$ must be an interval when $\bigcap\mathcal{F}$ is nonempty, consider any x such that $p < x < q$, where $p \in \bigcup\mathcal{F}$ and $q \in \bigcup\mathcal{F}$. Then there exist intervals $A, B \in \mathcal{F}$ such that $p \in A$, $q \in B$. Since $\bigcap\mathcal{F}$ is nonempty, there exists some $x_0 \in \bigcap\mathcal{F}$. If $x = x_0$, then certainly $x \in \bigcup\mathcal{F}$. In case $x < x_0$, we have $p < x < x_0$, where $p \in A$ and $x_0 \in A$, so that by Prop.1-7.3, $x \in A$ and hence $x \in \bigcup\mathcal{F}$. A similar argument applies when $x > x_0$. Thus $x \in \bigcup\mathcal{F}$ in all cases and $\bigcup\mathcal{F}$ must be an interval by Prop.1-7.5. ☺

Problem Set 1-7

1-7.P1. Suppose $p < s < q$. For any $x_1, x_2 \in [a, q)$ such that $|x_1 - x_2| < s - p$, show that either x_1 and x_2 both belong to $[a, s)$ or they both belong to (p, q).

1-7.P2. Suppose $a \le p < s < q$. Show that $[a, s) \cup (p, q) = [a, q)$. [When this has to be used in a proof, no explanation will be given; it is to be regarded as "obvious".]

1-7.P3. Suppose I is an interval with left endpoint a. If I contains numbers besides a, i.e. $I \ne [a, a]$, show that for any $\varepsilon > 0$, there exists $x \in I$ such that $a < x < a + \varepsilon$.

1-7.P4. If $A = [0,1) \cup [2,3]$, show that when $x \in A$ and $a \in A$ with $|x - a| < 1$, the numbers x and a must either both belong to $[0,1)$ or both belong to $[2,3]$.

1-7.P5. For intervals $[a, b]$ and $[p, q]$, show that $[a, b] \subseteq [p, q]$ if, and only if, $a \ge p$ as well as $b \le q$. Is it true that $(a, b) \subseteq (p, q)$ if, and only if, $a \ge p$ as well as $b \le q$?

1-8 System of Integers in $\mathbb{R}$

Since $\mathbb{R}$ contains $\mathbb{Q}$, it not only contains the subset $\mathbb{Z}$ of $\mathbb{Q}$ but also the subset consisting of the positive elements 1, 2, 3, … of $\mathbb{Z}$, which are called "natural numbers" or "positive integers". In this Section, we consider how $\mathbb{N}$ and $\mathbb{Z}$ "sit inside" $\mathbb{R}$.

The first result of this Section expresses what is known as the **archimedean property** of $\mathbb{R}$. It says essentially that no element of the ordered field $\mathbb{R}$ can be greater than every positive integer, something that lies at the root of the fact that $\lim_{n\to\infty} \frac{1}{n} = 0$. The property is shared by the field $\mathbb{Q}$ of rational numbers and by several other ordered fields. Although there do exist ordered fields that are not archimedean, they will not concern us.

1-8.1. Theorem. Archimedean property of $\mathbb{R}$. *Given $x \in \mathbb{R}$ and $y \in \mathbb{R}$, where $x > 0$, there exists some $n \in \mathbb{N}$ such that $y < nx$.*

Proof. Since $x > 0$, the inequality $y < nx$ is equivalent to $(y/x) < n$. If no such n exists, then y/x is an upper bound for $\mathbb{N}$. Therefore $\mathbb{N}$ is a nonempty subset of $\mathbb{R}$ having an upper bound. By the Completeness Axiom, $\mathbb{N}$ must have a least upper bound; call it a. Then $a-1$ is not an upper bound and hence there exists $n \in \mathbb{N}$ such that $a-1 < n$. But this implies that $a < n+1$ although $n+1 \in \mathbb{N}$ and a is an upper bound of $\mathbb{N}$. This is a contradiction. ☺

We have just used the property that $n \in \mathbb{N} \Rightarrow n+1 \in \mathbb{N}$. Any subset of $\mathbb{R}$ having this property and containing 1 is said to be **inductive**. Any subset of $\mathbb{N}$ that is inductive must be the whole of $\mathbb{N}$. Therefore $\mathbb{N}$ is a subset of any inductive subset S of $\mathbb{R}$, as can be seen by considering $S \cap \mathbb{N}$. The fact that any inductive subset of $\mathbb{N}$ must be the whole of $\mathbb{N}$ is the basis of any induction proof.

The following three well known properties of $\mathbb{N}$ and $\mathbb{Z}$ will also be used at some stage and are recorded here for later reference:

1-8.2 Remark. A natural number cannot be less than 1. In particular, if 1 belongs to a subset of $\mathbb{N}$, then 1 is the least element in that subset.

1-8.3. Remark. If $m \in \mathbb{Z}$, $n \in \mathbb{Z}$, and $n \neq m$, then $|n-m| \in \mathbb{N}$, so that $|n-m| \geq 1$. In particular, if $m \in \mathbb{Z}$, $n \in \mathbb{Z}$ and $n < m$, then $n \leq m-1$. Also, if $n < m < n+1$, where $n \in \mathbb{Z}$, then $m \notin \mathbb{Z}$.

1-8.4. Remark. Every integer is either even or odd but not both.

1-8.5. Proposition. *For any real number x, there exists a unique integer n such that $n \le x < n + 1$.*

Proof. Consider the set

$$S = \{k \in \mathbb{Z} : k \le x\}. \tag{1}$$

By the archimedean property [see Theorem 1-8.1], there exist integers k_1 and k_2 such that $x \le k_1$ and $-x \le k_2$. Therefore $-k_2 \le x \le k_1$, so that $-k_2 \in S$ and k_1 is an upper bound for S. Let $n = \sup S$.

Since $n-1$ is not an upper bound of S, there exists some $s \in S$ such that $n-1 < s \le n$. We shall show that $s = n$. Suppose, if possible, that $s < n$. Then there exists $s_1 \in S$ such that

$$n-1 < s < s_1 \le n. \tag{2}$$

It follows from (2) that $s \ne s_1$ and $|s - s_1| < 1$, contradicting Remark 1-8.3. Therefore $s = n$. Since $s \in S$, it now follows from (1) that

$$n \le x \quad \text{as well as} \quad n \in \mathbb{Z}.$$

Hence $n+1 \in \mathbb{Z}$. Moreover, $n+1 > n = \sup S$ and therefore $n+1 \notin S$. Together with (1), this implies that $x < n+1$, thus completing the proof that

$$n \in \mathbb{Z} \quad \text{and} \quad n \le x < n+1.$$

It remains to prove the uniqueness of such an integer n. Suppose

$$n \in \mathbb{Z},\ m \in \mathbb{Z},\ n \le x < n+1 \text{ and } m \le x < m+1.$$

Then $n \le x < m+1$ and $m \le x < n+1$. Therefore $-1 < n-m < 1$, or $|n-m| < 1$. It follows from Remark 1-8.3 that $n = m$. ☺

The unique integer n such that $n \le x < n + 1$ is called the **integer part** of x and is denoted by $[x]$, but usually only after stating that the symbol is being used in this manner. The name "greatest integer in" x is also commonly used.

For real numbers a and b with $b > 0$, the integer part $\left[\frac{a}{b}\right]$ is the unique integer k such that $kb \le a < (k+1)b$, as this is equivalent to $k \le \frac{a}{b} < k+1$. Thus the justification of the statement that $\left[\frac{7}{3}\right] = 2$ is that $2\times 3 < 7$ but $(2+1)\times 3 > 7$. The reader may recall from elementary arithmetic that in representing $\frac{7}{3}$ as the sum of 2 and a fraction less than 1, one first obtains the integer 2 by noting that $2\times 3 < 7$ but $(2+1)\times 3 > 7$. This amounts to obtaining the value of $\left[\frac{7}{3}\right]$ as 2. In other words, one has already learnt to work with the integer part function in elementary arithmetic!

1-8.6. Proposition. Well Ordering Principle. *Any nonempty subset of* $\mathbb{N}$ *contains a least number.*

Proof. Let $\varnothing \neq S \subseteq \mathbb{N}$. Suppose, if possible, that S contains no least number. Then by Remark 1-8.2, it follows that $1 < s$ for each $s \in S$. In other words, $1 \in T = \{n \in \mathbb{N} : n < s \ \forall\ s \in S\}$. We note that $T \cap S = \varnothing$. We shall proceed by proving that T is inductive.

Consider any $k \in T$. Then $s > k$ for each $s \in S$. But by Remark 1-8.3, $s > k \Rightarrow s \geq k+1$. So, $s \geq k+1$ whenever $s \in S$. It follows from this that $k+1 \notin S$, because otherwise $k+1$ would be the least element of S, contrary to our supposition. Therefore $s > k+1$ whenever $s \in S$, which shows that $k+1 \in T$. Thus T is inductive, and it follows that $T = \mathbb{N}$. As already noted, $T \cap S = \varnothing$, which now means $\mathbb{N} \cap S = \varnothing$. This is impossible when S is a nonempty subset of $\mathbb{N}$. Therefore the supposition that S contains no least number is false. ☺

1-8.7. Proposition. [Needed in Theorems 1-9.4 and 1-11.7] *A nonempty subset of* $\mathbb{Z}$ *that has a lower bound contains a least number. Analogously, a nonempty subset of* $\mathbb{Z}$ *that has an upper bound contains a greatest number.*

Proof. Let $\varnothing \neq S \subseteq \mathbb{Z}$ and let $b \in \mathbb{R}$ have the property that every integer in S is greater than or equal to b. Let $T = \{s - [b] + 1 : s \in S\}$, which is nonempty. Since $s \in S \Rightarrow s \geq b \geq [b]$, it follows that $T \subseteq \mathbb{N}$. By the Well Ordering Principle [Prop.1-8.6], T contains a least number, say $s_1 - [b] + 1$, with $s_1 \in S$. It follows that s_1 is the least number in S.

For the second part, we consider the set consisting of the negatives of the numbers of the given set and apply the first part. ☺

The next result is sometimes useful in an induction proof when the induction is on an integer variable that ranges between two specific values [see 1-11.P2(b)].

***1-8.8. Proposition.** *Let* $m, n \in \mathbb{N}$, $m \leq n$ *and* $\mathbb{Z}_{m,n} = \{k \in \mathbb{Z} : m \leq k \leq n\}$. *Then any set* S *such that*

$$\text{(a) } S \subseteq \mathbb{Z}_{m,n} \quad \text{(b) } m \in S \quad \textit{and} \quad \text{(c) } k \in S,\ k < n \Rightarrow k+1 \in S$$

is equal to $\mathbb{Z}_{m,n}$.

Proof. Left to the reader in 1-8.P11. ☺

Problem Set 1-8

1-8.P1. If S is a subset of $\mathbb{N}$ such that (a) $1 \in S$ and (b) whenever $\{j \in \mathbb{N} : 1 \leq j \leq k\} \subseteq S$, we have $k+1 \in S$, show that $S = \mathbb{N}$.

1-8.P2. If S is a subset of $\mathbb{N}$ and m is an element of $\mathbb{N}$ such that (a) $m \in S$ and (b) $k+1 \in S$ whenever $k \in S$, prove that $S \supseteq \{n \in \mathbb{N} : n \geq m\}$. [This justifies "induction from m onwards".]

1-8.P3. Let $2 \leq n \in \mathbb{N}$ and $\mathbb{N}_n = \{j \in \mathbb{N} : 1 \leq j \leq n\}$. If $n_1 \in \mathbb{N}_n$, $n_2 \in \mathbb{N}_n$ and $n_1 \neq n_2$, show that there exists a bijection $\phi : \mathbb{N}_n \to \mathbb{N}_n$ such that $\phi(n_1) = 1$ and $\phi(n_2) = 2$. [This allows us to rearrange the sum or product of a finite sequence with two particular terms of our choice as the first two terms.]

1-8.P4. If a map $\phi : \mathbb{N} \to \mathbb{N}$ satisfies $\phi(n) < \phi(n+1)$ for every $n \in \mathbb{N}$, show that (a) $\phi(n) \geq n$ for every $n \in \mathbb{N}$, and that (b) $\phi(n) < \phi(m)$ whenever $n < m$.

1-8.P5. Let a and b be real numbers, $b > 0$. Show that there exists a unique pair of real numbers q and r such that (i) $a = bq + r$ (ii) $0 \leq r < b$ (iii) q is an integer.

1-8.P6. Show that $1 < x < 3 \Rightarrow |x+2| < 5$ and that $1 < x < 3 \Rightarrow |x^2 + 2x + 4| < 19$.

1-8.P7. For any $k \in \mathbb{Z}$, show that $k - 3[\frac{k-1}{3}]$ is either 1 or 2 or 3. Here [ ] denotes the integer part function.

1-8.P8. For every odd integer n, show that there exists a unique integer k such that either $n = 4k+1$ or $n = 4k+3$.

1-8.P9. [Needed in Example after Prop.4-2.3] By the result of 1-8.P7, with [ ] denoting the integer part function, the following describes a map ϕ from $\mathbb{Z}$ into $\mathbb{Z}$.

$$\phi(k) = \begin{cases} (4k-1)/3 & \text{if } k - 3[\frac{k-1}{3}] = 1 \\ (4k+1)/3 & \text{if } k - 3[\frac{k-1}{3}] = 2 \\ 2k/3 & \text{if } k - 3[\frac{k-1}{3}] = 3. \end{cases}$$

(a) Find the nine values $\phi(k)$ for $1 \leq k \leq 9$. Try to describe in words, without mentioning the greatest integer function, how one can go on writing the subsequent values $\phi(k)$.

(b) Show that $k > 0 \Rightarrow \phi(k) > 0$. (Thus ϕ maps $\mathbb{N}$ into $\mathbb{N}$.)

(c) Show that the restriction $\phi : \mathbb{N} \to \mathbb{N}$ is a bijection.

1-8.P10. If $x \in \mathbb{R}$, $y \in \mathbb{R}$ and $x - y$ is an integer, show that $x - [x] = y - [y]$.

***1-8.P11.** Prove Prop.1-8.8.

1-8.P12. [Used in the proof of Theorem 3-3.8] (a) If N, M, n, m are natural numbers such that $N(N-1) + 2M = n(n-1) + 2m$ and $N > n$, show that $m > n$.

(b) Let $X = \{(n,m) \in \mathbb{N}\times\mathbb{N}: n \geq m\}$ and $\phi: X\to\mathbb{N}$ be the map for which $\phi(n,m) = n(n-1)+2m$. Show that ϕ is injective.

1-9 System of Rational Numbers in $\mathbb{R}$

In this Section, we examine how $\mathbb{Q}$ "sits inside" $\mathbb{R}$.

1-9.1. Definition. *A number in $\mathbb{R}$ that is not rational is called* **irrational**.

1-9.2. Remark. The Order Completeness Axiom is not valid in $\mathbb{Q}$. If it were, then 2 would have to have a rational square root by Prop.1-6.9, and it can be shown that this is not so [see Prop.1-9.3 below]. Thus the number $\sqrt{2}$, which must exist in $\mathbb{R}$ by Prop.1-6.9, must be irrational. We emphasise that, although $\mathbb{Q}$ and $\mathbb{R}$ are both ordered fields, the major distinction between them is that $\mathbb{R}$ has the order completeness property.

We now show that the ordered field $\mathbb{Q}$ is not complete by proving that 2 does not have a square root in $\mathbb{Q}$. The reader is undoubtedly aware of the usual contradiction argument, which starts with the representation of the square root of 2 as p/q, where p and q have no common "factors". As we have not discussed the latter concept, our proof starts with a representation p/q that is described differently but amounts to the same thing.

1-9.3. Proposition. *No number in $\mathbb{Q}$ has square equal to* 2. *In particular, the ordered field $\mathbb{Q}$ is not complete.*

Proof. Suppose, if possible, that $x \in \mathbb{Q}$ and that $x^2 = 2$. There must exist integers p and q such that $x = p/q = (-p)/(-q)$. Either p or $-p$ is positive and therefore there exists a positive integer p such that $x = p/q$ for some $q \in \mathbb{Z}$. By the Well Ordering Principle [Prop.1-8.6], there must exist a smallest such positive integer p. Then $2 = x^2 = p^2/q^2$, so that $p^2 = 2q^2$, which means p^2 is even. If p were not even then by the first part of Remark 1-8.4, it would be odd, say $2k-1$; consequently, $p^2 = 4k^2-4k+1$ would also be odd and therefore not even, in view of the second part of Remark 1-8.4. So p must be even, say $p = 2p_1$. Then $4p_1^2 = p^2 = 2q^2$, which leads to $q^2 = 2p_1^2$. This implies that q^2 and hence also q is even, say $q = 2q_1$. But then $x = p/q = (2p_1)/(2q_1) = p_1/q_1$, where p_1 is a smaller positive integer than p, a contradiction. Therefore no such x can exist. ☺

1-9.4. Theorem. *Given $x \in \mathbb{R}$ and $y \in \mathbb{R}$, $x < y$, there exists some $r \in \mathbb{Q}$ such that $x < r < y$.* Briefly put, *the rationals are "dense" in the reals.*

Proof. By applying the archimedean property of $\mathbb{R}$ [see Theorem 1-8.1] to the numbers $1/(y-x)$ and $1 > 0$, we get a natural number n such that $n > 1/(y-x) > 0$, or

$$-\frac{1}{n} > x - y. \tag{1}$$

Again applying the archimedean property, but to x and $\frac{1}{n}$ this time, we get a natural number k_1 such that $k_1/n > x$. Applying it yet again, this time to $-x$ and $\frac{1}{n}$, we get a natural number k_2 such that $(-k_2)/n < x$. Thus there exist integers m such that $\frac{m}{n} > x$ (e.g. k_1) and all of them are greater than $-k_2$. The set of such integers is therefore nonempty and bounded below. By Prop.1-8.7 (a version of the Well Ordering Principle), there must be a smallest such integer. Denote it by m. Then

$$\frac{m-1}{n} \le x < \frac{m}{n}.$$

By (1), $$\frac{m-1}{n} = \frac{m}{n} - \frac{1}{n} > \frac{m}{n} + (x-y).$$

Therefore $$\frac{m}{n} + (x-y) < x < \frac{m}{n}$$

and hence $$x < \frac{m}{n} < y.$$

Thus $r = \frac{m}{n}$ has the required properties. ☺

1-9.5. Example. Find the supremum and infimum of $S = \{\frac{1}{n} : n \in \mathbb{N}\}$. [Note that the archimedean property is needed in the solution.]

Since $1 \le n$ for every $n \in \mathbb{N}$ [by Remark 1-8.2], therefore 1 is the greatest element of S and hence also its supremum.

Also, $0 < \frac{1}{n}$ for every $n \in \mathbb{N}$, and therefore 0 is a lower bound of S. Furthermore, whenever $p > 0$, the archimedean property of $\mathbb{R}$ [see Prop.1-8.1] applied to $\frac{1}{p}$ and the positive number 1 yields some $n \in \mathbb{N}$ such that $n > \frac{1}{p} > 0$. Now $s = \frac{1}{n} \in S$ and $s < p$. Thus $p > 0 \Rightarrow \exists\ s \in S \ni s < p$. By Prop.1-6.6, $\inf S = 0$.

1-9.6. Proposition. *Any real number is the supremum of the set of all rational numbers less than it. In other words,* $x = \sup\{r \in \mathbb{Q} : r < x\}\ \forall\ x \in \mathbb{R}$. *Similarly, any real number is the infimum of the set of all rational numbers greater than it. That is to say,* $x = \inf\{r \in \mathbb{Q} : r > x\}\ \forall\ x \in \mathbb{R}$.

Proof. Let $x \in \mathbb{R}$. Then the set $S = \{r \in \mathbb{Q} : r < x\}$ has x as an upper bound. Furthermore, for any $y < x$, Prop.1-9.4 yields a number $r \in \mathbb{Q}$ such that $y < r < x$, i.e. such that $r \in S$ and $r > y$. By Prop.1-6.5, $x = \sup S$.

The proof that $x = \inf\{r \in \mathbb{Q} : r > x\}$ is exactly similar. ☺

We note also that $x = \sup\{r \in \mathbb{Q} : r \le x\} = \inf\{r \in \mathbb{Q} : r \ge x\}$.

Problem Set 1-9

1-9.P1. Find the supremum and infimum of each of the following sets. [Results of 1-6.P7–1-6.P11, and Example 1-9.5 may be used as and when appropriate.]
(a) $\{(-1)^n/n : n \in \mathbb{N}\}$ (b) $\{(n+1)/n : n \in \mathbb{N}\}$ (c) $\{(n+2)/(n+6) : n \in \mathbb{N}\}$
(d) $\{(1-\frac{1}{n})/n : n \in \mathbb{N}\}$ (e) $\{5(1-5/7n)/7n : n \in \mathbb{N}\}$ (f) $\{\frac{1}{m}+\frac{1}{n} : m, n \in \mathbb{N}\}$

1-9.P2. If $a > 0$, show that $\{r \in \mathbb{Q} : 0 < r \text{ and } r^2 < a\}$ is nonempty and bounded above. Then show that its supremum b satisfies $b^2 = a$.

1-9.P3. Show that, for any real numbers x, y and rational number $t < x+y$, there exist rational numbers $r < x$ and $s < y$ such that $t = r+s$. Use this to describe the set $C = \{t \in \mathbb{Q} : t < x+y\}$ in terms of the sets $A = \{r \in \mathbb{Q} : r < x\}$ and $B = \{s \in \mathbb{Q} : s < y\}$.

1-9.P4. Show that, for any positive real numbers x, y and positive rational number $t < xy$, there exist positive rational numbers $r < x$ and $s < y$ such that $t = rs$. Use this to describe the set $C = \{t \in \mathbb{Q} : 0 < t < xy\}$ in terms of the sets $A = \{r \in \mathbb{Q} : 0 < r < x\}$ and $B = \{s \in \mathbb{Q} : 0 < s < y\}$.

***1-9.P5.** Show that the only map $\phi : \mathbb{R} \to \mathbb{R}$ such that

$$\phi(x+y) = \phi(x) + \phi(y) \text{ and } \phi(xy) = \phi(x)\,\phi(y) \qquad (H)$$

but $\phi(x) \neq 0$ for some $x \in \mathbb{R}$ is the "identity map" given by: $\phi(x) = x$ for every $x \in \mathbb{R}$.

1-9.P6. [Needed in Examples 2-1.4(b),(c),(d)] Prove the following complement to Theorem 1-9.4:

Given $x \in \mathbb{R}$ and $y \in \mathbb{R}$, $x < y$, there exists some $s \notin \mathbb{Q}$ such that $x < s < y$.

(In other words, the irrationals are dense in $\mathbb{R}$.)

1-9.P7. (a) Without using the fact that only even integers can have even squares, prove that $\sqrt{2}$ is irrational as follows: Let q be the smallest positive integer for which there exists some positive integer p such that $p^2/q^2 = 2$. Show that $2q > p > q$, so that $0 < p-q < q$ and $2q-p > 0$. Then show that $p/q = (2q-p)/(p-q)$, which is a contradiction.

(b) Prove that $\sqrt{3}$ and $\sqrt{5}$ are irrational.

1-10 System $\mathbb{C}$ of Complex Numbers

We now define a sum and a product in the cartesian product $\mathbb{R}\times\mathbb{R}$, denoted by $\mathbb{R}^2$, in such a manner that (A1)—(A5), (M1)—(M5) and the Distributive Law (D) are valid.

The sum $(a,b)+(c,d)$ of the elements (a,b) and (c,d) of $\mathbb{R}^2$ is defined to be

$$(a,b)+(c,d)=(a+c,b+d). \tag{1}$$

It is straightforward to verify that (A1)—(A5) hold, i.e. $(a,b)+(c,d) \in \mathbb{R}^2$, $(a,b)+(c,d)=(c,d)+(a,b)$ and so forth.

The product $(a,b)(c,d)$ of the elements (a,b) and (c,d) of $\mathbb{R}^2$ is defined to be

$$(a,b)(c,d)=(ac-bd,ad+bc). \tag{2}$$

Since the product of two elements of $\mathbb{R}^2$ is defined to be an element of $\mathbb{R}^2$, the Closure Law (M1) holds. Since

$$(c,d)(a,b)=(ca-db,cb+da)=(ac-bd,ad+bc)=(a,b)(c,d),$$

the Commutative Law (M2) holds. Also,

$$\begin{aligned}\big((a,b)(c,d)\big)(e,f) &= (ac-bd,ad+bc)(e,f)\\ &= ((ac-bd)e-(ad+bc)f, (ac-bd)f+(ad+bc)e)\\ &= (ace-bde-adf-bcf, acf-bdf+ade+bce)\\ &= (a(ce-df)-b(cf+de), a(cf+de)+b(ce-df))\\ &= (a,b)(ce-df, cf+de)=(a,b)\big((c,d)(e,f)\big).\end{aligned}$$

This shows that the Associative Law (M3) holds. For the Identity Law, the role of 1 is played by $(1,0)$, because $(a,b)(1,0)=(a,b)$. For the Inverse Law (M5), consider any $(a,b)\neq(0,0)$. Then $a^2+b^2\neq 0$; let $(a,b)^{-1}$ stand for $(a/(a^2+b^2),-b/(a^2+b^2))$. Then

$$\begin{aligned}(a,b)(a,b)^{-1} &= (a^2/(a^2+b^2)+b^2/(a^2+b^2), -ab/(a^2+b^2)+ba/(a^2+b^2))\\ &= (1,0).\end{aligned}$$

Finally, we verify that the Distributive Law (D) holds.

$$\begin{aligned}(a,b)\big((c,d)+(e,f)\big) &= (a,b)(c+e, d+f)\\ &= (a(c+e)-b(d+f), a(d+f)+b(c+e))\\ &= (ac-bd,ad+bc)+(ae-bf, af+be)\\ &= (a,b)(c,d)+(a,b)(e,f).\end{aligned}$$

Thus, the set $\mathbb{R}^2$ becomes a field, called the **complex field**, and it is denoted by $\mathbb{C}$. An element of the complex field $\mathbb{C}$ is called a **complex number.** Whatever we have proved for a field, i.e. on the basis of (A1)—(A5), (M1)–(M5) and (D), but without using (O1)–(O4), is valid in $\mathbb{C}$.

For any complex number $x = (a, b)$, the real number a is called its **real part** and is denoted by $\boldsymbol{\Re x}$ or $\boldsymbol{\Re(x)}$; the real number b is called its **imaginary part** and is denoted by $\boldsymbol{\Im x}$ or $\boldsymbol{\Im(x)}$. Therefore any complex number x is $(\Re x, \Im x)$. It is an immediate consequence of the definitions of the sum and product of complex numbers that

$$\Re(x+y) = \Re x + \Re y, \ \Im\,(x+y) = \Im x + \Im y$$

and

$$\Re(xy) = (\Re x)(\Re y) - (\Im x)(\Im y), \quad \Im(xy) = (\Re x)(\Im y) + (\Im x)(\Re y).$$

The subset $\mathbb{C}_0 = \{(a,b) \in \mathbb{C} : b = 0\}$ of $\mathbb{C}$, i.e. the set of complex numbers with imaginary part 0, can be identified with $\mathbb{R}$ via the bijection ϕ which maps any complex number $(a, 0)$ into the element a of $\mathbb{R}$. This is because for any complex numbers $x = (a, 0)$ and $y = (b, 0)$ in $\mathbb{C}_0$,

$$\phi(x+y) = \phi((a,0)+(b,0)) = \phi((a+b,0)) = a+b = \phi(x)+\phi(y)$$

and

$$\phi(xy) = \phi((a,0)(b,0)) = \phi((ab-0, a0+0b)) = \phi((ab,0)) = ab = \phi(x)\phi(y).$$

Thus the sum [resp. product] of the complex numbers $(a, 0)$ and $(b, 0)$ in accordance with (1) [resp. (2)] is effectively the same thing as the sum [resp. product] of the real numbers $\phi((a,0))$ and $\phi((b,0))$ in $\mathbb{R}$. It is therefore not necessary to distinguish between the real number $\phi(x) \in \mathbb{R}$ and the complex number $x \in \mathbb{C}_0$. This means we regard $(a,0) \in \mathbb{C}_0$ as being the same as $a \in \mathbb{R}$. With this understanding, $\mathbb{R}$ is a subset of $\mathbb{C}$. Therefore every real number is a complex number, but of a special kind in which the imaginary part is 0.

The complex number $(0, 1)$ is denoted by i. Any complex number $z = (a, b)$ can be written as $(a,b) = (a,0)+(0,b) = (a,0)+(0,1)(b,0) = (a,0)+i(b,0)$. Since we regard $(a, 0)$ and $(b, 0)$ as a and b respectively, we have $(a,b) = a + ib$. Another way to write this is that $z = \Re z + i\Im z$.

The **conjugate of** a complex number $z = \Re z + i\Im z$ is the complex number $\Re z - i\Im z$ and is denoted by $\bar{z}$. The following properties are easy to verify by computation:

(a) $\overline{z+w} = \bar{z} + \bar{w}$ (b) $\overline{zw} = \bar{z}\,\bar{w}$ (c) $z + \bar{z} = 2\Re z$ (d) $z - \bar{z} = 2i\Im z$
(e) $z\bar{z} = (\Re z)^2 + (\Im z)^2 \geq 0$, and $z\bar{z} = 0 \Leftrightarrow z = 0$ (f) $\Im z = 0 \Leftrightarrow z = \bar{z}$.

By virtue of (e), we may define the **absolute value of** a complex number z as $|z| = \sqrt{(z\bar{z})}$.

When z is real, i.e. $\Im z = 0$, its absolute value according to Def.1-4.6 is the same as the absolute value as we have just defined it for a complex number. This is because of Prop.1-4.7(b).

1-10.1. Proposition. *Let x and y be complex numbers (possibly real). Then*
(a) $|-x| = |x| \geq 0$; *also* $|x| = 0$ *if, and only if,* $x = 0$;
(b) $|\bar{x}| = |x|$, $|\Re x| \leq |x|$, $|\Im x| \leq |x|$ *and* $x + \bar{x} \leq 2|x| = 2\sqrt{(x\bar{x})}$;
(c) $|x \cdot y| = |x||y|$;
(d) $|x + y| \leq |x| + |y|$; **(triangle inequality)**
(e) $|x - z| \leq |x - y| + |y - z|$;
(f) $\big||x| - |y|\big| \leq |x - y|$.

Proof. (a) and (b) are trivial.

(c) We have $|xy|^2 = (\overline{xy})(xy) = (\bar{x}\bar{y})(xy) = (\bar{x}x)(\bar{y}y) = |x|^2|y|^2$. Since by Prop.1-4.5, the nonnegative square root of a nonnegative number is unique, it follows that $|xy| = |x||y|$.

(d) The triangle inequality follows upon applying the last assertion in (b) to $x\bar{y}$, adding $x\bar{x} + y\bar{y}$ to both sides, noting that $|x\bar{y}| = |x||y|$ by (c) and the first assertion in (b), and then taking square roots.

(e) and (f) follow from the triangle inequality exactly as in Prop.1-4.7. ☺

A subset S of $\mathbb{C}$ is said to be **bounded** if there exists a real number $M \geq 0$ such that $|z| \leq M$ for every $z \in S$. Any such M is called a **bound** for S. The set $\{z \in \mathbb{C} : |z| \leq M\}$ is of course bounded with M as a bound; the set $\{z \in \mathbb{C} : \Re z \leq 1, \Im z \leq 1\}$ has $(1^2 + 1^2)^{1/2} = \sqrt{2}$ as a bound. If S is a subset of $\mathbb{R}$, then its being bounded in the sense just defined is the same thing as its being bounded as a subset of the ordered field as defined in Def.1-5.7. This is because $|z|$ is the same whether we think of it as an element of $\mathbb{C}$ or as an element of $\mathbb{R}$.

A pertinent question would be whether (O1)—(O4) hold in $\mathbb{C}$ if we define "<" is some suitable manner. It is easy to see that the answer is No. If this could be done, then by Prop.1-4.2, we would have $1 = 1^2 > 0$ and $-1 = i^2 > 0$, which contradicts part (a) of the same result. Therefore $\mathbb{C}$ cannot be made into an ordered field. Consequently, there is no such thing as a subset of $\mathbb{C}$ being bounded "above" or "below", as these concepts make sense only in an ordered field.

Problem Set 1-10

1-10.P1. For any $z \in \mathbb{C}$, let $a = \left(\frac{|z| + \Re z}{2}\right)^{1/2}$ and $b = \left(\frac{|z| - \Re z}{2}\right)^{1/2}$. Show that $(a+ib)^2 = z \Leftrightarrow \Im z \geq 0$ and $(a-ib)^2 = z \Leftrightarrow \Im z \leq 0$. [It follows that every nonzero complex number has at least two square roots; but since an element of a field can have at most two square roots by Prop.1-4.5, it follows that a nonzero complex number has precisely two square roots.] Find the two square roots of i and of $-i$.

1-10.P2. Let $\omega = (-1 + i\sqrt{3})/2$. Show that $\omega^2 = \bar{\omega}$ and that $\omega^3 = 1$ (so that $\bar{\omega}^3 = 1$ as well). Use this to show that, if $w \in \mathbb{C}$ is a cube root of $z \in \mathbb{C}$, then ωw and $\omega^2 w$ are also cube roots of z. [Note that, if $z \neq 0$, then the three cube roots are different from each other. This shows that if at all a nonzero complex number has a cube root then it has at least three of them.] If $x \in \mathbb{C}$ and $y \in \mathbb{C}$ are distinct cube roots of the same complex number, then either $x = \omega y$ or $x = \omega^2 y$, so that a complex number cannot have more than three different cube roots.

1-10.P3. Show that every nonzero complex number has four distinct fourth roots. Find the fourth roots of 1 and of -1.

***1-10.P4.** Let x,y,z be three complex numbers, not all three equal (i.e. $y \neq z$ OR $z \neq x$ OR $x \neq y$) and such that $x^3 + y^3 + z^3 - 3xyz = 0$ and $x + y + z \neq 0$. Show that (1) $y-z, z-x, x-y$ are all distinct (2) $y \neq z$ AND $z \neq x$ AND $x \neq y$ and (3) $(y-z)^3 = (z-x)^3 = (x-y)^3$.

***1-10.P5.** Let z be a complex number such that $z^7 = 1$ but none of $z, z^2, z^3, z^4, z^5, z^6$ is 1. Find the set of all positive integers k such that $1 + z + z^2 + \cdots + z^k = 0$.

***1-10.P6.** Using the answer to 1-10.P3, find the seven eighth roots of $(2-5i)^8$ other than $2-5i$.

***1-10.P7.** If a and b are complex numbers such that $a(a^2 - 3b^2) = 11$ and $b(3a^2 - b^2) = 2$, and if $a^2 + b^2$ is real, prove that $a^2 + b^2 = 5$.

***1-10.P8.** If z is a nonreal complex number such that $\frac{z^3+1}{z^3-1}$ is a real number greater than 1, then show that $\frac{z}{|z|} + \frac{z^2}{|z|^2} = -1$.

1-11 Finite Sums and Products

For $n \in \mathbb{N}$, we denote the set $\{j \in \mathbb{N} : 1 \leq j \leq n\}$ by $\mathbb{N}_n$. A map $a:\mathbb{N}_n \to S$ is known as a **sequence of *n* terms**, or as an *n-tuple* in S, depending on the context. When we do not wish to specify the natural number n, we speak of simply a *finite sequence*. For any $k \in \mathbb{N}_n$, the element $a(k)$ of S will be called the **k^{th} term** of the finite sequence and will be denoted by a_k whenever convenient. Often it is convenient to denote the sequence by $\{a_1, a_2, \dots, a_n\}$ or simply $a_1, a_2, \dots, a_n$ or by $\{a_j\}_{j=1}^n$. When one has a function b with domain $\{k \in \mathbb{Z} : 0 \le k \le n-1\}$, i.e. what may be denoted by $\{b_0, b_1, \dots, b_{n-1}\}$ or by $\{b_j\}_{j=0}^{n-1}$, the associated sequence a defined on $\mathbb{N}_n$ by taking $a(j) = b(j-1)$ is also called a sequence (of n terms). This amounts to renaming the terms $b_0, b_1, \dots, b_{n-1}$ as $a_1, a_2, \dots, a_n$ respectively. Whatever we do with sequences with domain $\mathbb{N}_n$ carries over to sequences with domain $\{k \in \mathbb{Z} : 0 \le k \le n-1\}$ with obvious modifications, which do not need to be stated separately.

When $S = \mathbb{R}$ [resp. $\mathbb{C}$], the finite sequence is described as being "real" [resp. "complex"], which is just a way of saying that its terms are real numbers [resp. complex numbers]. In the rest of this Section, all finite sequences will be understood to be one of these kinds and $\mathbb{K}$ will denote $\mathbb{R}$ or $\mathbb{C}$. For a finite sequence $a : \mathbb{N}_n \to \mathbb{K}$, we inductively define the sum

$$a_1 + a_2 + \cdots + a_n \text{ or } \sum_{j=1}^n a_j$$

as follows.

1-11.1. Definition. *For all sequences* $a : \mathbb{N}_1 \to \mathbb{K}$ (1 term only), *the sum is* a_1. *The sum of any sequence of* $n + 1$ *terms is defined in terms of the sum of a sequence of n terms as*

$$(a_1 + a_2 + \cdots + a_n) + a_{n+1} .$$

We shall use the notation Σa_j *or* Σa_k *or* $\sum_{j=1}^n a_j$ *to mean* $a_1 + a_2 + \cdots + a_n$. *If m* $<$ *n and* $b : \mathbb{N}_m \to \mathbb{K}$ *is defined as*

$$b_k = a_k \text{ for } k \in \mathbb{N}_m ,$$

then the symbol $\sum_{j=1}^m a_j$ *will mean the same thing as* Σb_j. *Similarly, if* $c : \mathbb{N}_{n-m} \to \mathbb{K}$ *is defined as*

$$c_j = a_{j+m} \text{ for } j \in \mathbb{N}_{n-m} ,$$

then the symbol $\sum_{j=m+1}^n a_j$ *will mean the same thing as* Σc_j .

It is easy to prove by induction that for any finite complex sequence $x : \mathbb{N}_n \to \mathbb{C}$, we have

$$\sum_{k=1}^n x_k = \Big(\sum_{k=1}^n \Re x_k, \sum_{k=1}^n \Im x_k\Big).$$

Another way to state the same thing is that $\Re\big(\sum_{k=1}^n x_k\big) = \sum_{k=1}^n \Re x_k$ as well as $\Im\big(\sum_{k=1}^n x_k\big) = \sum_{k=1}^n \Im x_k$.

We now present what is sometimes called the **Generalised Associative Property**.

1-11.2. Proposition. *Let $a:\mathbb{N}_{n+1}\to\mathbb{K}$ be a finite sequence and m be a natural number less than $n+1$. Then*

$$\Big(\sum_{j=1}^{m} a_j\Big) + \Big(\sum_{j=m+1}^{n+1} a_j\Big) = \sum_{j=1}^{n+1} a_j .$$

Proof. (Induction on n) When $n = 1$, both sides of the equality are equal to $a_1 + a_2$. Assume that the equality holds for some $k \in \mathbb{N}$ (induction hypothesis). For $m < k+2$, apply Def.1-11.1 to the second sum in $\big(\sum_{j=1}^{k} a_j\big) + \big(\sum_{j=m+1}^{k+2} a_j\big)$ and then use (A3), the induction hypothesis and the definition again. ☺

It should be noted that the preceding proof did not use (A2) at all. The next result is the **Generalised Commutative Property** and uses not only (A2) but also Prop.1-11.2.

1-11.3. Proposition. *Let $a:\mathbb{N}_{n+1}\to\mathbb{K}$ be a finite sequence and $\psi:\mathbb{N}_{n+1}\to\mathbb{N}_{n+1}$ be a bijection.*

Then
$$\sum_{j=1}^{n+1} a_{\psi(j)} = \sum_{j=1}^{n+1} a_j .$$

***Proof.** For $n = 1$, the right side of the equality is equal to $a_1 + a_2$, while the left side can only be either $a_1 + a_2$ or $a_2 + a_1$. So, the equality must hold in this case.

Assume the equality to be true for some $k \in \mathbb{N}$. Let $\psi:\mathbb{N}_{k+2}\to\mathbb{N}_{k+2}$ be a bijection. We need only prove that

$$\sum_{j=1}^{k+2} a_{\psi(j)} = \sum_{j=1}^{k+2} a_j . \tag{1}$$

If $1 < \psi^{-1}(k+2) < k+2$, then denoting $\psi^{-1}(k+2)$ by m, we have

$$\begin{aligned}
\sum_{j=1}^{k+2} a_{\psi(j)} &= \Big[\sum_{j=1}^{m} a_{\psi(j)}\Big] + \Big[\sum_{j=1}^{k+2-m} a_{\psi(j+m)}\Big] && \text{Prop.1-11.2}\\
&= \Big[\Big(\sum_{j=1}^{m-1} a_{\psi(j)}\Big) + a_{\psi(m)}\Big] + \sum_{j=1}^{k+2-m} a_{\psi(j+m)} && \text{Def.1-11.1}\\
&= \sum_{j=1}^{m-1} a_{\psi(j)} + \Big[\Big(\sum_{j=1}^{k+2-m} a_{\psi(j+m)}\Big) + a_{\psi(m)}\Big] && \text{(A2) and (A3).}
\end{aligned}$$

Now, define a map $\varphi:\mathbb{N}_{k+1}\to\mathbb{N}_{k+1}$ as

$$\varphi(j) = \psi(j) \text{ for } j < m \quad\text{and}\quad \varphi(j) = \psi(j+1) \text{ for } j \geq m.$$

Then φ is a bijection. Also, it follows from the above equality that

$$\sum_{j=1}^{k+2} a_{\psi(j)} = \Big(\sum_{j=1}^{m-1} a_{\varphi(j)}\Big) + \Big[\Big(\sum_{j=1}^{k+2-m} a_{\varphi(j+m-1)}\Big) + a_{k+2}\Big]$$

$$= \Big[\Big(\sum_{j=1}^{m-1} a_{\varphi(j)}\Big) + \Big(\sum_{j=1}^{k+2-m} a_{\varphi(j+m-1)}\Big)\Big] + a_{k+2} \qquad \text{(A3)}$$

$$= \Big(\sum_{j=1}^{k+1} a_{\varphi(j)}\Big) + a_{k+2} \qquad \text{Prop.1-11.2}$$

$$= \Big(\sum_{j=1}^{k+1} a_j\Big) + a_{k+2} \qquad \text{induction hypothesis}$$

$$= \sum_{j=1}^{k+2} a_j \qquad \text{Def.1-11.1.}$$

This proves (1) when $1 < \psi^{-1}(k+2) < k+2$. When $\psi^{-1}(k+2) = 1$, which means $m = 1$, the same argument goes through, except that there is no need for the sum $\sum_{j=1}^{m-1} a_{\psi(j)}$.

When $\psi^{-1}(k+2) = k+2$, the restriction of ψ to $\mathbb{N}_{k+1}$ is a bijection from $\mathbb{N}_{k+1}$ to itself. Denoting this bijection by φ, we have

$$\sum_{j=1}^{k+2} a_{\psi(j)} = \sum_{j=1}^{k+1} a_{\psi(j)} + a_{\psi(k+2)} \qquad \text{Def.1-11.1}$$

$$= \sum_{j=1}^{k+1} a_{\varphi(j)} + a_{k+2} \qquad \text{definition of } \varphi$$

$$= \sum_{j=1}^{k+1} a_j + a_{k+2} \qquad \text{induction hypothesis}$$

$$= \sum_{j=1}^{k+2} a_j \qquad \text{Def.1-11.1.}$$

Thus (1) holds in this case as well. ☺

Remark. The definition of $\sum_{j=1}^{n+1} a_j$ allows us to say only that $\sum_{j=1}^{n+1} a_j = \sum_{j=1}^{n} a_j + a_{n+1}$, but not that $\sum_{j=1}^{n+1} a_j = a_1 + \sum_{j=2}^{n+1} a_j$. The truth of the latter equality is obvious when $n = 1$ and follows for $n > 1$ by induction:

$$\sum_{j=1}^{k+1} a_j = \sum_{j=1}^{k} a_j + a_{k+1} = (a_1 + \sum_{j=2}^{k} a_j) + a_{k+1} = a_1 + (\sum_{j=2}^{k} a_j + a_{k+1}) = a_1 + \sum_{j=2}^{k+1} a_j.$$

An informal statement of the next result would be that

$$(a_1 + b_1) + (a_2 + b_2) + \cdots + (a_n + b_n) = (a_1 + a_2 + \cdots + a_n) + (b_1 + b_2 + \cdots + b_n).$$

It relates three sums of n terms each. There are n parentheses on the left side and if we remove them, it will become a sum of $2n$ terms and therefore, strictly speaking, will no longer be covered by the preceding Proposition without further application of the Commutative and Associative Laws. The $2n$ terms will be $c_1 = a_1,\ c_2 = b_1,\ c_3 = a_2, \ldots, c_{2n} = b_n$. Such removal of parentheses will become necessary with infinite series at a later stage and we provide for it now.

1-11.4. Proposition. [Referenced in Prop.4-1.4, Theorems 5-3.9 and 11-4.11] *If* $a:\mathbb{N}_n\to\mathbb{K}$, $b:\mathbb{N}_n\to\mathbb{K}$, $c:\mathbb{N}_n\to\mathbb{K}$ *are finite sequences and* $c_j = a_j + b_j$ *for all* $j\in\mathbb{N}_n$, *then* $\Sigma c_j = \Sigma a_j + \Sigma b_j$.

***Proof.** Induction on n. If $n = 1$, the assertion is that $c_1 = a_1 + b_1$, which is true by hypothesis. Assuming it to be true for some k, we have

$$\sum_{j=1}^{k+1} c_j = \sum_{j=1}^{k} c_j + c_{k+1} = \Big(\sum_{j=1}^{k} a_j + \sum_{j=1}^{k} b_j\Big) + (a_{k+1} + b_{k+1}),$$

which can then be shown via (A2), (A3) and Def.1-11.1 to be equal to $\sum_{j=1}^{k+1} a_j + \sum_{j=1}^{k+1} b_j$. ☺

1-11.5. Proposition. [Referenced in Theorem 5-3.9] *If* $a:\mathbb{N}_n\to\mathbb{K}$, $b:\mathbb{N}_n\to\mathbb{K}$ *are finite sequences and* $c:\mathbb{N}_{2n}\to\mathbb{K}$ *is the finite sequence such that*

$$c_j = \begin{cases} a_i & \text{if } j = 2i-1 \quad \text{with } 1 \le i \le n \\ b_i & \text{if } j = 2i \quad \text{with } 1 \le i \le n, \end{cases}$$

then $\sum_{j=1}^{2n} c_j = \sum_{j=1}^{n} a_j + \sum_{j=1}^{n} b_j$.

***Proof.** First we check that c is indeed defined on $\mathbb{N}_{2n}$. This is so because $1 \le 2i \le 2n \Leftrightarrow 1 \le i \le n$ while $1 \le 2i-1 \le 2n \Leftrightarrow 1 \le i \le n+\frac{1}{2} \Rightarrow 1 \le i \le n$.

Before proceeding, we note that it is a consequence of the definition of c_j that $c_1 = a_1$, $c_2 = b_1$ and $c_{2n-1} = a_n$, $c_{2n} = b_n$.

When $n = 1$, the equality to be proved asserts that $c_1 + c_2 = a_1 + b_1$, which is true because $c_1 = a_1$, $c_2 = b_1$. Assuming it to be true for some k, for the next positive integer $k + 1$, we have

$$\sum_{j=1}^{2k+2} c_j = \sum_{j=1}^{2k+1} c_j + c_{2k+2} = \Big(\sum_{j=1}^{2k} c_j + c_{2k+1}\Big) + c_{2k+2},$$

which can be shown to be equal to $\sum_{j=1}^{k+1} a_j + \sum_{j=1}^{k+1} b_j$ by the induction hypothesis, (A2) and (A3). ☺

1-11.6. Proposition. Generalised Distributive Law. *Suppose* $a:\mathbb{N}_n\to\mathbb{K}$ *is a finite sequence and* $x\in\mathbb{C}$. *Then* $\Sigma(xa_k) = x\Sigma a_k$.

Proof. This is an easy induction argument. ☺

One can define a **product** $a_1a_2\cdots a_n$ or $\prod_{j=1}^{n} a_j$ of a sequence of n real or complex terms in a similar way and prove the Generalised Associative and Commutative Properties for it, as also the analogue of Prop.1-11.4. Details are left to the reader. For any $x\in\mathbb{C}$ and $n\in\mathbb{N}$, one defines x^n as the product of the sequence $a(j) = x$, $1 \le j \le n$. Also, $x^n = 1$ if $n = 0$ (even for $x = 0$). The usual "laws of indices for a nonnegative integral index" then follow by induction. For $x \neq 0$, one defines x^{-n} as the multiplicative inverse (reciprocal) of x^n. [Note that the meaning of x^{-n} when $n = 1$ according to this definition is the same as what we have denoted by x^{-1} all along.] The "laws of indices" can then be proved in the usual manner for all integral indices, positive, negative as well as 0.

It is routine to start from the definition that $n!$ is the product of the sequence $a(j) = j$, $1 \leq j \leq n$, when $n \in \mathbb{N}$ while $0! = 1$, and prove that

$$\frac{n!}{j!(n-j)!} + \frac{n!}{(j-1)!(n-j+1)!} = \frac{(n+1)!}{j!(n-j+1)!}$$

for $1 \leq j \leq n$. In the usual notation $C(n,j) = n!/j!(n-j)!$, this is the well known identity $C(n,j) + C(n,j-1) = C(n+1,j)$. Undoubtedly, the reader has seen an induction argument that uses this identity to prove the **binomial theorem**, which asserts that $(x+y)^n$ is the sum of the sequence

$$a(j) = C(n,j-1)x^{n-j+1}y^{j-1}, \quad 1 \leq j \leq n+1.$$

It is easily checked that the argument is valid whenever $x \in \mathbb{C}$, $y \in \mathbb{C}$ and $n \in \mathbb{N}$ by using the Remark after Prop.1-11.3, and Props.1-11.4, 1-11.6. We shall therefore take the binomial theorem as proved.

In a similar spirit, the summation formula for a geometric progression and the factorisation formula

$$(x^{n+1} - y^{n+1}) = (x-y)\sum_{j=1}^{n+1} x^{n-j+1}y^{j-1}$$

that follows from it will also be taken as proved. It may be noted that, in case $y = 0$, the power y^0 is to be interpreted as 1, not 0.

We note that it is easily proved by induction that, for any finite sequence $a : \mathbb{N}_n \to \mathbb{R}$, there exists a least value, i.e. a real number α such that (i) $\alpha \leq a_j$ for all $j \in \mathbb{N}_n$ and (ii) $\alpha = a_j$ for some $j \in \mathbb{N}_n$. It will be denoted by $\min\{a_j : j \in \mathbb{N}_n\}$ or $\min\{a_1, \ldots, a_n\}$. Similarly, the greatest value will be denoted by $\max\{a_j : j \in \mathbb{N}_n\}$ or $\max\{a_1, \ldots, a_n\}$.

The remaining result in this Section is devoted to showing that every nonnegative integer can be expressed in terms of the powers of a given "base" $b > 1$ with nonnegative integer coefficients less than b. This is akin to the commonly used *decimal* representation.

1-11.7. Theorem. *Suppose $n, b \in \mathbb{N}$ and $b > 1$. Then for any nonnegative integer a which is less than b^n, there exists a unique finite sequence $a_0, \ldots, a_{n-1}$ of integers such that*

$$\text{(A)} \quad a = a_0 + a_1 b + \cdots + a_{n-1} b^{n-1},$$
$$\text{(B)} \quad 0 \le a_j \le b-1 \textit{ for } 0 \le j \le n-1.$$

Proof. First we prove the existence of such a sequence by induction on n. When $n = 1$, we have $a < b$ and the sequence (single term because $n = 1$) given by $a_0 = a$ serves the purpose.

Assume (induction hypothesis) that existence is guaranteed for $n = k \in \mathbb{N}$ and let $0 \le a < b^{k+1}$. Then nonnegative integers p such that $pb^k \le a < b^{k+1}$ exist (e.g. 0) and are all smaller than b. Let a_k be the largest among them. Then $a_k b^k \le a$, $0 \le a_k \le b-1$ and also $(a_k + 1)b^k > a$. Hence $0 \le a - a_k b^k < b^k$, and therefore by the induction hypothesis, $a - a_k b^k = a_0 + a_1 b + \cdots + a_{k-1} b^{k-1}$, where $0 \le a_j \le b-1$ for $0 \le j \le k-1$. Since $0 \le a_k \le b-1$ as already noted, this completes the proof of existence. We proceed to prove uniqueness.

The case $n = 1$ is obvious and we assume $n > 1$. Consider the possibility that there are two distinct sequences of the kind in question. Subtraction leads to $0 = \alpha_0 + \alpha_1 b + \cdots + \alpha_{n-1} b^{n-1}$, where each α_j is an integer satisfying $|\alpha_j| \le b-1$ and at least one is nonzero. Now α_0 cannot be the only one that is nonzero and there is a largest j such that $\alpha_j \ne 0$ by Prop.1-8.7. Call it k. Then $k > 0$ and $\alpha_0 + \alpha_1 b + \cdots + \alpha_{k-1} b^{k-1} = -\alpha_k b^k$, while at the same time

$$|\alpha_0 + \alpha_1 b + \cdots + \alpha_{k-1} b^{k-1}| \le (b-1)(1 + b + \cdots + b^{k-1}) = b^k - 1 < b^k \le |-\alpha_k b^k|,$$

which is a contradiction. ☺

The sum $a_0 + a_1 b + \cdots + a_{n-1} b^{n-1}$ with $a_{n-1} \ne 0$ is called the expansion or representation of a in terms of the **base** b, and the nonnegative integers $a_0, \ldots, a_{n-1}$ (each less than the base b) are called the **digits** in the expansion. When the base b is ten, a digit may be called *decimal digit* for emphasis. The standard way to write the expansion is of course to write only the digits and in reverse order, the powers of the base being taken as understood: $a_{n-1} a_{n-2} \cdots a_0$. With this notation, the base b itself is represented as "10", because $b = 0 + 1 \cdot b$. The traditional processes of column addition, multiplication, etc. for integers expanded in terms of a base in accordance with Theorem 1-11.7 are easily derived and will be taken for granted.

Needless to say, we shall normally work with the number ten as the base b. The possible digits $0, 1, \ldots, b-1$ are then the usual 0, 1, … , 9. For practical work, one needs the basic addition and multiplication facts about digits, such as $8+9 = 8+(2+7) = (8+2)+7 = b+7 = 17$ and $7\times 2 = 7+7 = 7+(3+4) = (7+3)+4 = b+4 = 14$, and so on. Such facts and the multiplication table are consequences of our axioms and we can now say such things as

$$1+1+\frac{1}{2}+\frac{1}{6}+\frac{1}{24}+\frac{1}{120}+\frac{1}{720}=\frac{1957}{720}=\frac{1440+517}{720}=2+\frac{517}{720}.$$

However, representations of positive numbers less than 1 are yet to be discussed, which will be done after infinite series.

Problem Set 1-11

1-11.P1. Let $n \in \mathbb{N}$, and suppose $a:\{j \in \mathbb{N} : 1 \le j \le n\} \to \mathbb{R}$ and $b:\{j \in \mathbb{N} : 1 \le j \le n\} \to \mathbb{R}$ are two finite sequences such that $a_j \le b_j$ for $1 \le j \le n$. Show (by induction on n) that $\Sigma a_j \le \Sigma b_j$. With the additional hypothesis that $a_j < b_j$ for some j, show that $\Sigma a_j < \Sigma b_j$. State the corresponding result for products.

1-11.P2. (a) Prove that $(n+1)! > 2^n$ whenever $n \in \mathbb{N}$ and $n \ge 2$.
*(b) For $n \in \mathbb{N}$ and $k \in \mathbb{N}$, where $2 \le k \le n$, show that $n!/(n-k)! < n^k$. [Prop.1-8.8 may help.]
*(c) For $n \in \mathbb{N}$, where $n > 1$, show that $(1+1/n)^n < \sum_{j=1}^{n+1} 1/(j-1)!$
(d) Show that $\sum_{j=1}^{n+1} 1/(j-1)! < 1+\sum_{j=1}^{n} 1/2^{j-1}$ when $n > 2$.

1-11.P3. Show that a sum of n terms, each of which is equal to x, is nx. In other words, let $a_k = x$ for every $k \in \mathbb{N}_n$ and show that $\Sigma a_k = nx$.

1-11.P4. For a positive integer n and positive x and y, show that $x^n > y^n$ if, and only if, $x > y$.

1-11.P5. [To be used in Prop.2-4.6] Let $2 \le n \in \mathbb{N}$, and $a_1, \ldots, a_n$ be a finite sequence of positive real numbers with product equal to 1. If they are not all equal to one another, show that $\Sigma a_k > n$.

1-11.P6. Show that $(-1)^n = 1$ if n is even and -1 if n is odd.

1-11.P7. Show that in any ordered field, $\left|\sum_{j=1}^{n} a_j\right| \le \sum_{j=1}^{n} |a_j|$.

1-11.P8. Show that in $\mathbb{C}$, $\left|\sum_{k=1}^{n} z_k\right| \le \sum_{k=1}^{n} |z_k|$.

***1-11.P9.** [This problem is about divisibility criteria.] An integer m is called a **divisor** of an integer n if $n = mk$ for some $k \in \mathbb{Z}$. Suppose the expansion of $a \in \mathbb{N}$ in terms of the base b (which means that $b > 1$) is $a = a_0 + \cdots + a_{n-1}b^{n-1}$.

(a) For any divisor β of $b-1$, show that β is a divisor of a if, and only if, β is a divisor of Σa_k.
(b) For any divisor β of $b+1$, show that β is a divisor of a if, and only if, β is a divisor of $\Sigma a_k(-1)^k$.

1-11.P10. (**Bernoulli's Inequality**) For any real $x \geq -1$ and any $n \in \mathbb{N}$, show that

$$(1+x)^n \geq 1+nx.$$

1-11.P11. [Used in Prop.3-1.6] For any real $x \geq 0$ and any $n \in \mathbb{N}$, show (by induction) that

$$(1+x)^n \geq 1+nx+\tfrac{1}{2}n(n-1)x^2.$$

For any real $x > 0$ and any positive integer $n > 2$, show that

$$(1+x)^n > 1+nx+\tfrac{1}{2}n(n-1)x^2.$$

*Appendix 1 (Completeness and Everywhere Zero Derivative)

The purpose of this Appendix is to demonstrate two things. One of them is how the completeness of $\mathbb{R}$ will make it possible to prove (after defining derivatives) that a function with derivative 0 at each point of an interval must be a constant. The other is why nothing short of completeness will serve the purpose. The first of these is a consequence of the following

Proposition. *Let* $\varepsilon > 0$ *and* $f:[a,b]\to\mathbb{R}$ *be a function such that*, *for every* $x \in [a,b]$, *there exists* $\delta > 0$ *such that* $|f(y)-f(x)| \leq \varepsilon|y-x|$ *whenever* $|y-x| < \delta$. *Then* $|f(\alpha)-f(\beta)| \leq \varepsilon|\alpha-\beta|$ *for all* $\alpha, \beta \in [a,b]$.

Proof. We present only an informal description of the argument. Consider any α, $\beta \in [a,b]$, where $\alpha < \beta$. The case when $\alpha > \beta$ is similar and need not be considered separately. The inequality to be proved will be called "amenability" of the interval $[\alpha,\beta]$. It is easy to check that if two intervals with a common endpoint are both amenable, then so is their union (call this fact "contiguity"). Define S to be the set of all $x \in [\alpha,\beta]$ such that $[\alpha,x]$ is amenable. Then S has an upper bound β and is nonempty by hypothesis. Let c be its supremum. Then $\alpha < c \leq \beta$, but we shall show that $c = \beta$.

Consider the possibility that $c < \beta$. Then by hypothesis and contiguity, there exists some interval $(c-\delta_1, c+\delta_1)$ such that every subinterval of it that includes c as well as its own endpoints is amenable. Since $c-\delta_1$ is not an upper bound of S, there is some $s_1 > c-\delta_1$ such that $s_1 \in S$. But then $[\alpha,s_1]$ and $[s_1, c+\tfrac{1}{2}\delta_1]$ are both amenable, so that $[\alpha, c+\frac{1}{2}\delta_1]$ is amenable by contiguity. But this implies $c+\frac{1}{2}\delta_1 \in S$, which is a contradiction because $c+\frac{1}{2}\delta_1 > c = \sup S$. Thus $c = \beta$, as claimed.

Again by hypothesis, there exists some interval $(\beta-\delta_2, \beta]$ such that every subinterval of it that includes β as well as its own endpoints is amenable. Using contiguity and the fact that $\beta = c = \sup S$, we can proceed as in the preceding paragraph to show that $[\alpha, \beta]$ must be amenable. ☺

Remark. After the derivative is defined (as we shall do later), it will be clear that a function f having derivative 0 at each point of an interval $[a, b]$ has the property that, for every $\varepsilon > 0$ and for every $x \in [a, b]$, there exists $\delta > 0$ such that

$$|f(y)-f(x)| \le \frac{\varepsilon}{(b-a)}|y-x| \quad \text{whenever} \quad |y-x| < \delta.$$

It will then follow by the above Proposition that, for every $\varepsilon > 0$,

$$|f(\alpha)-f(\beta)| \le \frac{\varepsilon}{(b-a)}|\alpha-\beta| \le \varepsilon \quad \text{for all } \alpha, \beta \in [a, b].$$

This has the consequence [see 1-4.P9] that $f(\alpha) = f(\beta)$ for all $\alpha, \beta \in [a, b]$, i.e. f is a constant.

Before proceeding, we note that the function described below is modelled on $f: \mathbb{Q} \to \mathbb{Q}$, where $f(x) = 0$ when x lies to the left of $\sqrt{2}$ and $f(x) = 1$ when x lies to the right of $\sqrt{2}$.

To see why nothing short of completeness will ensure that a function having derivative 0 at each point of an interval must necessarily be a constant, we consider an ordered field F which is not complete. That is to say, F has a nonempty subset S which has an upper bound in F but has no least upper bound. We describe below how to construct a function $f: F \to F$ which is not constant, but has the property that for every $x \in F$,

$$\text{there exists } \delta > 0 \text{ such that } f(x+h)-f(x) = 0 \text{ whenever } |h| < \delta. \tag{A}$$

Once the derivative is defined, this property will be seen to have the consequence that f has derivative 0 at each $x \in F$. The function f is:

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is not an upper bound of } S \\ 1 & \text{if } x \text{ is an upper bound of } S. \end{cases}$$

Since S is not empty, all numbers less than some number in S are not upper bounds of S. Therefore f does take the value 0. Since S is bounded above, f also takes the value 1. Thus f is not constant. If x is not an upper bound, then some $s \in S$ is greater than x. Let $\delta = s - x$, so that $x + \delta = s$. Then the inequality $|h| < \delta$ has the consequence that $-\delta < h < \delta$, and hence that $x - \delta < x + h < x + \delta = s$, i.e. $x + h$

is not an upper bound of S. Hence $f(x+h) = 0 = f(x)$, so that $f(x + h) - f(x) = 0$. Therefore (A) holds when x is not an upper bound of S. Next, suppose x is an upper bound. Since there is no least upper bound, there exists an upper bound less than x. Let t denote an upper bound such that $t < x$ and define δ to be $x - t$, so that $x - \delta = t$. Then the inequality $|h| < \delta$ has the consequence that $\delta > h > -\delta$, and hence that $x + \delta > x + h > x - \delta = t$, i.e. $x + h$ is also an upper bound of S. Hence $f(x+h) = 1 = f(x)$, so that $f(x+h) - f(x) = 0$. Therefore (A) holds also when x is an upper bound of S.

This shows that if we work with a number system in which comparison of numbers is possible (ordered field), there will always exist nonconstant functions with derivative 0 everywhere, unless the Completeness Axiom holds. For Integral Calculus, completeness is thus a mathematical compulsion and not a matter of choice. However, it can be useful to go beyond a complete ordered field; the interested reader could consult [2], [4] or the Internet source [5].

Chapter 2
Continuous Functions

2-1 Elementary Properties of Continuous Functions

In this Chapter, we discuss continuous functions whose domain is a subset of $\mathbb{R}$, or what could be called "functions of a single real variable". We begin with the definition of continuity for such functions and prove some of their "elementary" properties that do not rely on the completeness of $\mathbb{R}$. Some more elementary properties will be proved in the next Section.

2-1.1. Definition. *A function* $f: A \to \mathbb{R}$ *is said to be* **continuous at** $a \in A$ *if, for every* $\varepsilon > 0$, *there exists some* $\delta > 0$ *such that* $|f(x) - f(a)| < \varepsilon$ *whenever* $x \in A$ *and* $|x - a| < \delta$. *In symbols, this condition states that*

$$\forall\ \varepsilon > 0\ \exists\ \delta > 0 \ni \forall\ x \in A,\ |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon.$$

Remark. If one positive number δ satisfies this condition, then any positive $\delta_1 < \delta$ also satisfies it. This is obvious because whenever $x \in A$ and $|x - a| < \delta_1$, it is also true that $x \in A$ and $|x - a| < \delta$. Therefore such a number δ is far from being unique.

Sometimes continuity is defined via limits, which turns out in most situations to be equivalent to the way we have defined it above, but not all. The exceptions will be discussed elsewhere. Def.2-1.1 is often called the "ε-δ definition" of continuity.

While working with the ε-δ definition, we can omit writing "$x \in A$", because the subsequent reference to $f(x)$ makes it clear that x is intended to be in the domain of f.

Many authors prefer to begin by defining continuity only when the domain is an interval, making a distinction between "left", "right" and "two sided" continuity at the outset. For readers who are rooted in that approach, we make the following comment: In case the domain A is an interval and $a \in A$ is its left endpoint, the proviso that $x \in A$ automatically ensures that we are considering only those values of x that are greater than a. Therefore continuity at a means "right continuity" without our having to say so separately. Similarly for "left continuity".

One learns in elementary Calculus that the function f given by $f(x) = x^2$ is continuous at any $a \in \mathbb{R}$. However, if one were to justify this directly on the basis of the preceding ε-δ definition, one would have to show for a general $\varepsilon > 0$ what the number δ would be. If $a = 2$, then one appropriate number δ would be $\min\{1, \frac{\varepsilon}{5}\}$. In other words, it is true that

$$|x^2 - 4| < \varepsilon \text{ whenever } x \in \mathbb{R} \text{ and } |x-2| < \delta = \min\{1, \tfrac{\varepsilon}{5}\},$$

as can be seen by the following steps of reasoning:

$$|x-2| < \min\{1, \tfrac{\varepsilon}{5}\} \Rightarrow |x-2| < 1 \Rightarrow -1 < x-2 < 1$$
$$\Rightarrow 3 < x+2 < 5 \Rightarrow 3 < |x+2| < 5;$$

and therefore

$$|x-2| < \min\{1, \tfrac{\varepsilon}{5}\} \Rightarrow |x-2| < \tfrac{\varepsilon}{5}$$
$$\Rightarrow |x^2-4| < \tfrac{\varepsilon}{5}|x+2| < \tfrac{\varepsilon}{5} \cdot 5 = \varepsilon.$$

The reader is invited to show in a similar manner that

$$|x^2 - 4| < \varepsilon \text{ whenever } x \in \mathbb{R} \text{ and } |x-2| < \delta = \min\{\tfrac{1}{2}, \tfrac{2\varepsilon}{9}\}.$$

An example that presents no challenge at all is the function f such that $f(x) = x$. For any $\varepsilon > 0$, the positive number $\delta = \varepsilon$ satisfies the requirement that $|f(x) - f(a)| < \varepsilon$ whenever $x \in A$ and $|x-a| < \delta$, because what this actually says is that $|x-a| < \varepsilon$ whenever $x \in A$ and $|x-a| < \delta$. And this is certainly true when the number δ is none other than ε itself.

As yet another example, consider the function f with domain $A = [0,1) \cup [2,3]$ and given by $f(x) = x$ when $x \in [0,1)$ and $f(x) = x-1$ when $x \in [2,3]$. This domain A has the property [see 1-7.P4] that, if x, $a \in A$ and $|x-a| < 1$, then either x and a both belong to $[0,1)$ or they both belong to $[2,3]$. Therefore $|x-a| < 1 \Rightarrow |f(x)-f(a)| = |x-a|$. Hence for any $\varepsilon > 0$, the positive number $\delta = \min\{1, \varepsilon\}$ has the property that $|f(x)-f(a)| < \varepsilon$ whenever $|x-a| < \delta$. This means f is continuous at any $a \in A$.

Since continuity has not been defined here in terms of limits, any argument to establish continuity of, say $f(x) = x^{10} + 4x^3 + 1$, would ultimately have to draw upon the ε-δ definition. This may be done directly as was done above for x^2 or indirectly through the forthcoming Theorem 2-1.3. But it would be impractical to fall back upon the definition.

For functions $f:A\to\mathbb{R}$ and $g:A\to\mathbb{R}$ with any domain A, we denote the function which maps every $x\in A$ to $f(x)+g(x)$ by $f+g$ and the function which maps every $x\in A$ to the product $f(x)g(x)$ by fg. These are called the *sum* and *product* respectively of f and g. If g is a constant function (i.e. $g(x_1)=g(x_2)\ \forall\ x_1, x_2\in A$), then fg is also denoted by αf, where α is the constant value of g, and it is called a *constant multiple* of f. As usual, $(-1)f$ will be denoted by $-f$. The function which maps every $x\in A$ to $|f(x)|$ will be denoted by $|f|$ and will be called the *absolute value* of f. All these have domain A. If there exists some $x\in A$ for which $g(x)\neq 0$, then the function which maps such $x\in A$ to $f(x)/g(x)$ is called the *quotient* of f by g and is denoted by f/g. Its domain can be a proper subset of A. Finally, the function which maps every $x\in A$ to $\min\{f(x),g(x)\}$ [resp. $\max\{f(x),g(x)\}$] is denoted by $\mathbf{min\{f,g\}}$ [resp. $\mathbf{max\{f,g\}}$]; each one has domain A. Except for max and min, what we have just said applies equally well to functions with values in $\mathbb{C}$ or $\mathbb{Q}$.

2-1.2. Proposition. *Let $f:A\to\mathbb{R}$ be continuous at $a\in A$. Then there exist $M>0$ and $\delta_1>0$ such that $|f(x)|<M$ whenever $|x-a|<\delta_1$. If $f(a)\neq 0$, then there exist $K>0$ and $\delta_2>0$ such that $|f(x)|>K$ whenever $|x-a|<\delta_2$.* [*It is understood that $x\in A$.*]

Proof. Since $1>0$, Def.2-1.1 is applicable with $\varepsilon=1$. Therefore there exists $\delta_1>0$ such that $|f(x)-f(a)|<1$ whenever $|x-a|<\delta_1$. But

$$|f(x)-f(a)|<1\Rightarrow |f(x)|=|f(x)-f(a)+f(a)|\le|f(x)-f(a)|+|f(a)|<1+|f(a)|.$$

Therefore $M=1+|f(a)|$ and δ_1 have the required property.

Now suppose $f(a)\neq 0$. Then $\frac{1}{2}|f(a)|>0$ and Def.2-1.1 is applicable with $\varepsilon=\frac{1}{2}|f(a)|$. Therefore there exists $\delta_2>0$ such that $|f(x)-f(a)|<\frac{1}{2}|f(a)|$ whenever $|x-a|<\delta_2$. But

$$\begin{aligned}|f(x)-f(a)|<\tfrac{1}{2}|f(a)| \quad\Rightarrow\quad |f(x)|&=|f(x)-f(a)+f(a)|\\ &\ge|f(a)|-|f(x)-f(a)| \qquad \text{by Prop.1-4.7(h)}\\ &>|f(a)|-\tfrac{1}{2}|f(a)|=\tfrac{1}{2}|f(a)|>0.\end{aligned}$$

Therefore $K=\frac{1}{2}|f(a)|$ and δ_2 have the required property. ☺

2-1.3. Theorem. *Suppose $A\subseteq\mathbb{R}$ and $f:A\to\mathbb{R}$, $g:A\to\mathbb{R}$ are continuous at $a\in A$. Then the following are also continuous at a:*

(a) $f+g$;

(b) *fg and αf, where α is any real number*;

(c) *f/g provided $g(x)$ is never 0*;

(d) $|f|$, $\min\{f,g\}$ *and* $\max\{f,g\}$.

Proof. (a) Let $\varepsilon > 0$. Then $\frac{\varepsilon}{2} > 0$ and (by Def.2-1.1) there exist $\delta_1, \delta_2 > 0$ such that

$$|f(x)-f(a)| < \tfrac{\varepsilon}{2} \text{ whenever } |x-a| < \delta_1$$

and

$$|g(x)-g(a)| < \tfrac{\varepsilon}{2} \text{ whenever } |x-a| < \delta_2.$$

Let δ be the lesser among δ_1 and δ_2. Then $\delta > 0$. If $|x-a| < \delta$, then $|x-a| < \delta_1$ as well as $|x-a| < \delta_2$ and hence $|f(x)-f(a)| < \frac{\varepsilon}{2}$ as well as $|g(x)-g(a)| < \frac{\varepsilon}{2}$. Therefore

$$\begin{aligned}|(f+g)(x)-(f+g)(a)| &= |[f(x)-f(a)]+[g(x)-g(a)]| \\ &\leq |f(x)-f(a)|+|g(x)-g(a)| < \tfrac{\varepsilon}{2}+\tfrac{\varepsilon}{2} = \varepsilon.\end{aligned}$$

Thus there exists $\delta > 0$ such that $|(f+g)(x)-(f+g)(a)| < \varepsilon$ whenever $|x-a| < \delta$. Since the existence of such a positive δ has been shown for every positive ε, the continuity of $f+g$ at a has been proved.

(b) Let $\varepsilon > 0$. By Prop.2-1.2, there exist $M > 0$ and $\delta_1 > 0$ such that $|f(x)| < M$ whenever $|x-a| < \delta_1$. Also, there exists $\delta_2, \delta_3 > 0$ such that $|g(x)-g(a)| < \varepsilon/2M$ whenever $|x-a| < \delta_2$, and $|f(x)-f(a)| < {}^{\varepsilon}\!/2(|g(a)|+1)$ whenever $|x-a| < \delta_3$. Let δ be the least among δ_1, δ_2 and δ_3. Then $\delta > 0$. Consider any $x \in A$ for which $|x-a| < \delta$. For such a number x, we have $|f(x)| < M$, $|g(x)-g(a)| < \varepsilon/2M$ and also $|f(x)-f(a)| < {}^{\varepsilon}\!/2(|g(a)|+1)$. Therefore

$$\begin{aligned}|(fg)(x)-(fg)(a)| &= |f(x)\cdot(g(x)-g(a))+g(a)\cdot(f(x)-f(a))| \\ &\leq |f(x)|\cdot|g(x)-g(a)|+|g(a)|\cdot|f(x)-f(a)| \\ &\leq M\cdot(\varepsilon/2M)+|g(a)|\cdot\left[{}^{\varepsilon}\!/2(|g(a)|+1)\right] < \tfrac{\varepsilon}{2}+\tfrac{\varepsilon}{2} = \varepsilon.\end{aligned}$$

Thus there exists $\delta > 0$ such that $|(fg)(x)-(fg)(a)| < \varepsilon$ whenever $|x-a| < \delta$. Since the existence of such a positive δ has been shown for every positive ε, the continuity of fg at a has been proved.

If $\alpha = 0$, then $(\alpha f)(x) = 0$ for each $x \in A$, so that αf is certainly continuous. For $\alpha \neq 0$, continuity of αf follows either by applying what has just been proved with g chosen so as to have $g(x) = \alpha\ \forall\ x \in A$ [which is continuous at a], or by selecting a δ such that $|f(x)-f(a)| < \varepsilon/|\alpha|$ whenever $|x-a| < \delta$.

(c) Let $\varepsilon > 0$. By Prop.2-1.2, there exist $K > 0$ and $\delta_1 > 0$ such that $|g(x)| > K$ whenever $|x-a| < \delta_1$. Furthermore, there exists $\delta_2 > 0$ such that $|f(x)-f(a)| < \varepsilon K/2$ whenever $|x-a| < \delta_2$, and there exists $\delta_3 > 0$ such that $|g(x)-g(a)| < \varepsilon K|g(a)|/2(|f(a)|+1)$ whenever $|x-a| < \delta_3$. Let δ be the least among δ_1, δ_2 and δ_3.

Then $\delta > 0$. Consider any $x \in A$ for which $|x-a| < \delta$. For such a number x, we have

$$1/|g(x)| < 1/K, \ |f(x)-f(a)| < \varepsilon K/2$$

and also $$|g(x)-g(a)| < \varepsilon K|g(a)|/2[|f(a)|+1].$$

Therefore

$$\begin{aligned}
|(f/g)(x)-(f/g)(a)| &= \frac{|f(x)g(a)-f(a)g(x)|}{|g(x)g(a)|} \\
&= \frac{|f(x)g(a)-f(a)g(a)+f(a)g(a)-f(a)g(x)|}{|g(x)g(a)|} \\
&\le \frac{|g(a)|\cdot|f(x)-f(a)| + |f(a)|\cdot|g(x)-g(a)|}{|g(x)g(a)|} \\
&\le \frac{|f(x)-f(a)|}{|g(x)|} + \frac{|f(a)|\cdot|g(x)-g(a)|}{|g(x)g(a)|} \\
&< \frac{\varepsilon K/2}{K} + \frac{|f(a)|\cdot(\varepsilon K|g(a)|)}{K|g(a)|\cdot 2[|f(a)|+1]} \\
&< \frac{\varepsilon}{2}+\frac{\varepsilon}{2} = \varepsilon.
\end{aligned}$$

Thus there exists $\delta > 0$ such that $|(f/g)(x)-(f/g)(a)| < \varepsilon$ whenever $|x-a| < \delta$. Since the existence of such a positive δ has been shown for every positive ε, the continuity of f/g at a has been proved.

(d) Continuity of $|f|$ is a simple consequence of the inequality $||f(x)|-|f(a)|| \le |f(x)-f(a)|$. The rest follows from this, the equalities

$$\min\{f,g\} = \tfrac{1}{2}[(f+g)-|f-g|], \qquad \max\{f,g\} = \tfrac{1}{2}[(f+g)+|f-g|]$$

[from 1-4.P12], and from parts (a) and (b). ☺

It is an immediate consequence of the above result that a function which has a "polynomial" expression [formal definition later], such as the one given by the expression $x^{10}+4x^3+1$, or a quotient of such functions (usually called a *rational function*) is continuous at each a in its domain. However, this does not cover the square root and other root functions. The existence of these functions and their continuity will follow from a result to be proved in Section 2-2.

2-1.4. Examples. (a) Consider the function f with domain $[0,2]$, as below:

$$f(x) = \begin{cases} x & \text{if } x \in [0,1) \\ x+1 & \text{if } x \in [1,2]. \end{cases}$$

We show that f is *not* continuous at 1. What this means by Def.2-1.1 is that it is *not true* that the kind of δ referred to therein is always possible for every $\varepsilon > 0$. In other words, that kind of δ does not exist at least for some positive number ε (although it may well exist for some others). In order to show this, we produce a positive ε for which no positive number δ can have the property that $|x-1| < \delta \Rightarrow |f(x)-f(1)| < \varepsilon$. For $\varepsilon = 1$, no positive number δ can have the property that $|x-1| < \delta \Rightarrow |f(x)-f(1)| < 1$, because $x = 1 - \min\{1, \frac{\delta}{2}\}$ belongs to $[0,1)$ and $|x-1| = \min\{1, \frac{\delta}{2}\} < \delta$, but $|f(x)-f(1)| = |x-2| = 2-x = 1+1-x = 1+|x-1| > 1$.

(b) Let f be the function defined on $\mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$$

This function is not continuous at any $c \in \mathbb{R}$. Another way to say this is that f is *discontinuous* at every $c \in \mathbb{R}$. Indeed, if c is a rational number, every interval $(c-\delta, c+\delta)$ contains an irrational number x [see 1-9.P6], so that $|f(x)-f(c)| = |0-1| = 1$. Therefore, when $\varepsilon = 1$, no positive number δ can have the property that $|x-c| < \delta \Rightarrow |f(x)-f(c)| < \varepsilon$. On the other hand, if c is irrational then every interval $(c-\delta, c+\delta)$ contains a rational number y [by Theorem 1-9.4], so that $|f(y)-f(c)| = |1-0| = 1$. Once again, when $\varepsilon = 1$, no positive number δ can have the property that $|x-c| < \delta \Rightarrow |f(x)-f(c)| < \varepsilon$.

(c) Let f be the function defined on $\mathbb{R}$ by

$$f(x) = \begin{cases} x & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$$

This function is discontinuous at every nonzero $c \in \mathbb{R}$ and is continuous at $c = 0$. This is because we have

$$|f(x)-f(0)| = \begin{cases} |x-0| = |x| & \text{if } x \text{ is rational} \\ |0-0| = 0 & \text{if } x \text{ is irrational.} \end{cases}$$

In either case, $|f(x)-f(c)| < \varepsilon$ whenever $|x-0| = |x| < \delta$, where we choose $\delta = \varepsilon$. Now suppose $c \neq 0$. If c is a rational number, every interval $(c-\delta, c+\delta)$ contains an irrational number x [see 1-9.P6], so that $|f(x)-f(c)| = |0-c| = |c| > |\frac{c}{2}|$. Therefore, when $\varepsilon = |\frac{c}{2}|$, no positive number δ can have the property that $|x-c| < \delta \Rightarrow |f(x)-f(c)| < \varepsilon$. Before considering irrational c, observe that

$$x \in (c - |\tfrac{c}{2}|, c + |\tfrac{c}{2}|) \Rightarrow |x - c| < |\tfrac{c}{2}| \Rightarrow |x| \geq |c| - |x - c| > |\tfrac{c}{2}|.$$

Now, for any irrational c and any $\delta > 0$, the interval

$$(c - \min\{\delta, |\tfrac{c}{2}|\},\ c + \min\{\delta, |\tfrac{c}{2}|\})$$

is a subset of both the intervals $(c-\delta, c+\delta)$ and $(c - |\tfrac{c}{2}|, c + |\tfrac{c}{2}|)$. Therefore any number x belonging to it satisfies $|x - c| < \delta$ as well as $|x| > |\tfrac{c}{2}|$. However, it contains a rational number x [by Theorem 1-9.4]. It follows on the one hand that $|x - c| < \delta$ and on the other hand that $|f(x) - f(c)| = |x - 0| = |x| > |\tfrac{c}{2}|$. Once again, when $\varepsilon = |\tfrac{c}{2}|$, no positive number δ can have the property that $|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon$.

(d) Since this example is somewhat difficult, the reader may prefer to skip it for now and return to it later when needed.

Define $f: \mathbb{R} \to \mathbb{R}$ as follows.

$$f(x) = \begin{cases} \dfrac{1}{n} & \text{where } n \text{ is least in } \mathbb{N} \text{ such that } x = \dfrac{m}{n} \text{ with } m \in \mathbb{Z} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$$

[Note that such a least n exists by the Well Ordering Principle Prop.1-8.6.] This function f is continuous at every irrational number and discontinuous at every rational number c. To see why, we consider rational and irrational c separately.

Let $c \in \mathbb{R}$ be rational, so that $f(c) = 1/n$, where n is the least integer in $\mathbb{N}$ such that $c = m/n$ and $m \in \mathbb{Z}$. Choose $\varepsilon = 1/2n$. For any $\delta > 0$, the interval $(c - \delta, c + \delta)$ contains an irrational number x [see 1-9.P6], so that $|f(x) - f(c)| = |0 - 1/n| = 1/n > \varepsilon$. Therefore, when $\varepsilon = 1/2n$, no positive number δ can have the property that $|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon$.

On the other hand, if $c \in \mathbb{R}$ is an irrational number and ε any positive number whatsoever, there exists (by the archimedean property) some natural number $n_0 \in \mathbb{N}$ such that $1/n_0 < \varepsilon$. Now consider the interval $(c - 1/2n_0^2, c + 1/2n_0^2)$. For any p and q in this interval, $|p - q| < 1/n_0^2$. It follows that this interval can contain at most one rational number of the form $r = m/n$ with $n \leq n_0$, because when $n_1 \leq n_0$, $n_2 \leq n_0$ and $m_1/n_1 \neq m_2/n_2$, we have $m_1 n_2 - m_2 n_1 \neq 0$ and hence

$$\left|\frac{m_1}{n_1} - \frac{m_2}{n_2}\right| = \left|\frac{m_1 n_2 - m_2 n_1}{n_1 n_2}\right| \geq \frac{1}{n_1 n_2} \geq \frac{1}{n_0^2}.$$

If there is any such r in the interval $(c - 1/2n_0^2, c + 1/2n_0^2)$, let $\delta = |c - r|$. Otherwise, let $\delta = 1/2n_0^2$. In both cases, no number x in the interval $(c - \delta, c + \delta)$

can be of the form $x = m/n$ with $n \le n_0$. Thus every number in this interval is either irrational or is a rational number of the form m/n with $n > n_0$. Therefore

$$\begin{aligned} |x-c| < \delta &\Rightarrow \text{ either } f(x) = 0 \text{ or } f(x) = 1/n \text{ with } n > n_0 \\ &\Rightarrow \text{ either } |f(x)-f(c)| = 0 \text{ or } |f(x)-f(c)| = 1/n \text{ with } n > n_0 \\ &\Rightarrow |f(x)-f(c)| < 1/n_0 < \varepsilon. \end{aligned}$$

Problem Set 2-1

2-1.P1. Prove the continuity of $f(x) = x^2$ at 3 from "first principles" by showing that, for any $\varepsilon > 0$, the positive number $\delta = \min\{1, \frac{\varepsilon}{7}\}$ has the property that $|x-3| < \delta \Rightarrow |x^2-9| < \varepsilon$.

2-1.P2. Prove the continuity of $f(x) = x^3$ at 2 from "first principles" by showing that, for any $\varepsilon > 0$, the positive number $\delta = \min\{1, \frac{\varepsilon}{19}\}$ has the property that $|x-2| < \delta \Rightarrow |x^3-8| < \varepsilon$.

2-1.P3. Prove directly from the definition [Def.2-1.1] that the function f given by

$$f(x) = \begin{cases} x^2 & \text{if } x \in [0,1) \\ x+1 & \text{if } x \in [1,2]. \end{cases}$$

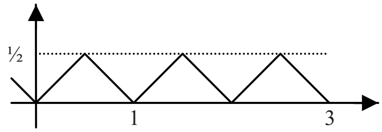
is not continuous at 1.

2-1.P4. If a is not an integer, show that the integer part function [as defined just after Prop.1-8.5] is continuous at a.

2-1.P5. Define a function f with domain $\mathbb{R}$ as

$$f(x) = \min\{x-[x], 1-(x-[x])\},$$

where the brackets [ ] denote the integer part function. This function can also be described as initially defined on $[0,1]$ to be $f(x) = x$ on $[0,\frac{1}{2}]$ and $1-x$ on $(\frac{1}{2},1]$, and then extended to all of $\mathbb{R}$ "by periodicity" [see accompanying graph]. By 2-1.P4 and Prop.2-1.3(d), f is continuous at any a that is not an integer. Now show that for any integer n,



$$n-\tfrac{1}{2} \le x \le n+\tfrac{1}{2} \Rightarrow f(x) = |x-n|.$$

Use this to prove that f is continuous at a even when a is an integer, so that f is continuous at each $a \in \mathbb{R}$. Also, show that

(i) $0 \le f(x) \le \frac{1}{2}\ \forall\ x \in \mathbb{R}$;

(ii) $f(n+\frac{1}{2}) = \frac{1}{2}$ and $f(n) = 0 \ \forall \ n \in \mathbb{Z}$;

(iii) If $n \neq x \neq n + \frac{1}{2} \ \forall \ n \in \mathbb{Z}$, then $0 < f(x) < \frac{1}{2}$.

What are the corresponding properties of ϕ, where $\phi(x) = 1 - 2f(x)$?

2-1.P6. Let $A \subseteq \mathbb{R}$ and $f: \mathbb{R} \to \mathbb{R}$ be a function such that $a \in A \Rightarrow f(a) = 0$. Suppose $x \in \mathbb{R}$ has the property that: for all $\delta > 0$, there exists some $a \in A$ such that $|a - x| < \delta$ and that f is continuous at x. Show that $f(x) = 0$.

2-1.P7. If a function $f: \mathbb{R} \to \mathbb{R}$ satisfies $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$ and if f is continuous at any one number $b \in \mathbb{R}$, show that f is continuous at every $a \in \mathbb{R}$.

2-1.P8. Suppose $f: \mathbb{R} \to \mathbb{R}$ satisfies $f(x) = x^2$ if $x \in \mathbb{Q}$ and $f(x) = x^3$ if $x \notin \mathbb{Q}$. Show that f is continuous at 0 and 1 only.

2-2 Inverses and Compositions

It is cumbersome to keep repeating the phrase "at a for each a in". When a function is continuous at a for each a in a subset of the domain, we speak of the function being **continuous on** the subset or **continuous everywhere on** the subset. Also, a function is called simply "continuous" if it is continuous on its entire domain.

2-2.1. Theorem. *Let I be an interval and the function $f: I \to \mathbb{R}$ satisfy the condition*

$$x_1 < x_2 \Rightarrow f(x_1) < f(x_2). \tag{1}$$

Then f has an inverse g with domain $B = \{f(x) : x \in I\}$ and satisfying

$$y_1 < y_2 \Rightarrow g(y_1) < g(y_2). \tag{2}$$

Moreover, g is continuous everywhere on its domain (*even if f is not*!).

If instead of (1), *f satisfies*

$$x_1 < x_2 \Rightarrow f(x_1) > f(x_2),$$

then the same conclusion holds with (2) *altered correspondingly*.

Proof. It follows from (1) that f is injective (i.e. one-to-one). So f must have an inverse g whose domain is the range of f, which is what B is. To establish (2), consider $y_1, y_2 \in B$ with $y_1 < y_2$. Then $y_1 = f(x_1)$ and $y_2 = f(x_2)$ for some $x_1, x_2 \in I$, and $g(y_1) = x_1$, $g(y_2) = x_2$. We must show $x_1 < x_2$. This is true because, by (1), $x_1 \geq x_2 \Rightarrow f(x_1) \geq f(x_2) \Rightarrow y_1 \geq y_2$, contradicting the hypothesis that $y_1 < y_2$.

It remains to show that g is continuous. To this end, consider any $b \in B$ and any $\varepsilon > 0$. Then $b = f(a)$ for some $a \in I$, so that $g(b) = a$. We must show that there exists some $\delta > 0$ such that $|y - b| < \delta \Rightarrow |g(y) - g(b)| < \varepsilon$. The case when I

contains only the single number a needs no argument. So we assume that there exist numbers in I besides a.

First consider the case when a is not an endpoint of I. By Prop.1-7.4, I contains a number x_1 between $a-\varepsilon$ and a, and also a number x_2 between a and $a+\varepsilon$. [See figure further below.] Since $a = g(b)$, we have

$$g(b)-\varepsilon < x_1 < a < x_2 < g(b)+\varepsilon \text{ and } x_1, x_2 \in I. \tag{3}$$

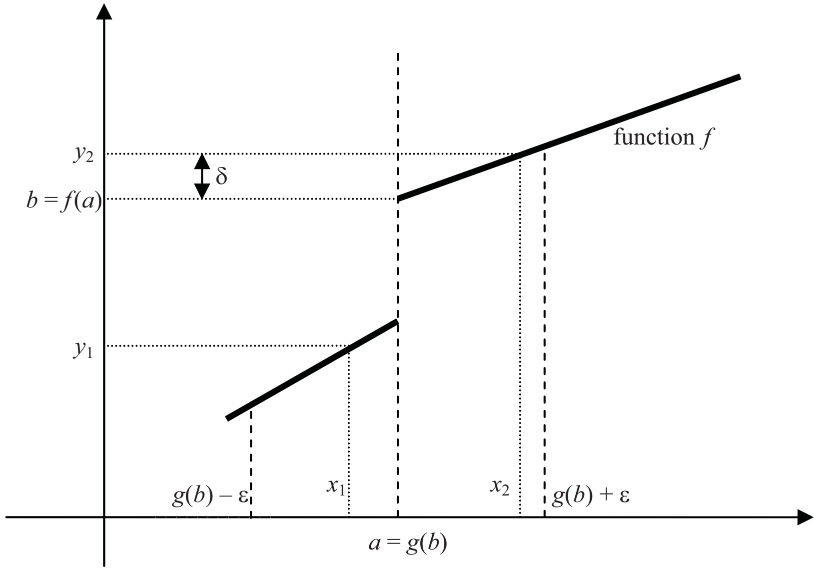
Let $y_1 = f(x_1)$ and $y_2 = f(x_2)$, so that

$$x_1 = g(y_1) \text{ and } x_2 = g(y_2). \tag{4}$$

By (1) and (3), we have $f(x_1) < f(a) < f(x_2)$, which means the same thing as $y_1 < b < y_2$. Let $\delta = \min\{b-y_1, y_2-b\}$, so that $\delta > 0$ and, for any $y \in B$,

$$\begin{aligned}
-\delta < y-b < \delta \Rightarrow y_1-b < y-b < y_2-b &\Rightarrow y_1 < y < y_2 \\
&\Rightarrow g(y_1) < g(y) < g(y_2) && \text{by (2)} \\
&\Rightarrow x_1 < g(y) < x_2 && \text{by (4)} \\
&\Rightarrow g(b)-\varepsilon < g(y) < g(b)+\varepsilon && \text{by (3).}
\end{aligned}$$

Thus $y \in B, |y-b| < \delta \Rightarrow |g(y)-g(b)| < \varepsilon$. Since the existence of such a positive δ has been shown for every positive ε, the continuity of g has been proved.



Now consider the case when a is an endpoint. We need handle only the case when a is the left endpoint. First we note that

$$y \in B \Rightarrow b \le y, \tag{5}$$

because by (2), $b > y \Rightarrow g(b) > g(y) \Rightarrow a > g(y) \in I$, which is impossible when a is the left endpoint of I. Let $x_2 \in I$ satisfy $a < x_2 < a+\varepsilon$. [Existence follows from Prop.1-7.4.] Then

$$g(b)-\varepsilon < g(b) = a < x_2 < a+\varepsilon = g(b)+\varepsilon. \tag{6}$$

Let $y_2 = f(x_2)$, so that

$$x_2 = g(y_2). \tag{7}$$

Since $a < x_2$, we have $b = f(a) < f(x_2) = y_2$. Let $\delta = y_2 - b$, so that $\delta > 0$. One can now proceed in a manner very similar to the case above when a was not an endpoint [details left as Problem 2-2.P8].

The last part can be proved either by modifying the above argument suitably or by considering the function $-f$ and noting that when h is continuous at some b lying in its domain, the associated function $g(y) = h(-y)$ has $-b$ in its domain and is continuous there. ☺

2-2.2. Theorem. *Suppose S and T are subsets of $\mathbb{R}$, and $f{:}S\to\mathbb{R}$, $g{:}T\to\mathbb{R}$ are functions for which T contains the range of f. Let $g\circ f{:}S\to\mathbb{R}$ be the composition defined by* $(g\circ f)(x) = g(f(x))$. *If f is continuous at $a \in S$ and g is continuous at* $f(a)$, *then $g\circ f$ is continuous at a.*

Proof. Consider any $\varepsilon > 0$. Since g is continuous at $f(a)$, there exists $\eta > 0$ such that

$$|g(y)-g(f(a))| < \varepsilon \text{ whenever } |y-f(a)| < \eta \text{ and } y \in T. \tag{1}$$

Since $\eta > 0$ and f is continuous at a, there exists $\delta > 0$ such that

$$|f(x)-f(a)| < \eta \text{ whenever } |x-a| < \delta \text{ and } x \in S. \tag{2}$$

Now if $x \in S$ and $|x-a| < \delta$, it follows from (2) that $|f(x)-f(a)| < \eta$. Since T contains the range of f, we have $f(x), f(a) \in T$. Therefore (1) is applicable with $y = f(x)$, and hence $|g(f(x))-g(f(a))| < \varepsilon$, i.e. $|(g\circ f)(x)-(g\circ f)(a)| < \varepsilon$. What this shows is that

$$|(g\circ f)(x)-(g\circ f)(a)| < \varepsilon \text{ whenever } |x-a| < \delta \text{ and } x \in S.$$

Since such a positive δ has been shown to exist for every positive ε, the continuity of $g\circ f$ has been proved. ☺

Problem Set 2-2

2-2.P1. Consider the functions f with domain $[0,2]$ and g with domain $[0,1)\cup[2,3]$ as below:

$$f(x) = \begin{cases} x & \text{if } x \in [0,1) \\ x+1 & \text{if } x \in [1,2], \end{cases}$$

$$g(y) = \begin{cases} y & \text{if } y \in [0,1) \\ y-1 & \text{if } y \in [2,3]. \end{cases}$$

Show that f and g are inverses of each other, that g satisfies

$$y_1 < y_2 \Rightarrow g(y_1) < g(y_2),$$

and that g is continuous everywhere on its domain. Note that f is not continuous everywhere on its domain, as shown in Section 2-1.

2-2.P2. As in 2-1.P5, define a function f with domain $\mathbb{R}$ as

$$f(x) = \min\{x - [x], 1 - (x - [x])\},$$

where the brackets [ ] denote the integer part function. Show that the function $g{:}(0,1]\to\mathbb{R}$ defined by $g(x) = f\left(\frac{1}{x}\right)$ is continuous but that the function $h{:}[0,1]\to\mathbb{R}$ defined by $h(0) = 0$, $h(x) = g(x)$ for $0 < x \le 1$ is not continuous at 0.

2-2.P3. Show that there is an infinite subset $S \subseteq \mathbb{R}$ with a continuous function $f{:}S\to\mathbb{R}$ such that $f(x)^5 + f(x) = x - 1$ for all $x \in S$. [It will be seen later that $S = \mathbb{R}$ has this property.]

2-2.P4. Let $a < b < c$, the functions $f{:}[a,b]\to\mathbb{R}$ and $g{:}[b,c]\to\mathbb{R}$ both be continuous on their respective domains and suppose $f(b) = g(b)$. Define the function $\phi{:}[a,c]\to\mathbb{R}$ as $\phi(x) = f(x)$ when $x \in [a,b]$ while $\phi(x) = g(x)$ when $x \in [b,c]$. Show that ϕ is continuous on $[a,c]$.

2-2.P5. Suppose $a < \alpha < b < \beta$ and $f{:}[a,b)\to\mathbb{R}$ and $g{:}[\alpha,\beta]\to\mathbb{R}$ are both continuous on their respective domains. If $f(x) = g(x)$ whenever $\alpha \le x < b$, show that the function $\phi{:}[a,\beta]\to\mathbb{R}$ defined to be $f(x)$ when $x \in [a,b)$ and $g(x)$ when $x \in [\alpha,\beta]$ is continuous.

2-2.P6. Give an example of disjoint subsets S and T of $\mathbb{R}$ with continuous functions f and g on them such that the function $\phi{:}S\cup T\to\mathbb{R}$ defined to be $f(x)$ when $x \in S$ and $g(x)$ when $x \in T$ is not continuous at any $a \in S\cup T$.

2-2.P7. Suppose that, for each $n \in \mathbb{N}$, $f_n{:}[-n,n]\to\mathbb{R}$ is a continuous function, and that $f_m(x) = f_n(x)$ whenever $m > n$ and $x \in [-n,n]$. Show that there exists a continuous function $f{:}\mathbb{R}\to\mathbb{R}$ such that $f(x) = f_n(x)$ for all $x \in [-n,n]$.

2-2.P8. Complete the argument of Theorem 2-2.1 for the case when a is the left endpoint of the domain I.

2-3 Boundedness Property

Problems asking for the maximum or minimum value of the volume of a solid or the area of a plane region are standard fare in Calculus. They usually reduce to finding the maximum or minimum value of some function with an interval as its domain. Although such problems are designed so as to have a valid answer, there do exist functions having neither a maximum nor a minimum, such as $\phi(x) = (x-1)/x(x-2)$, $x \in (0,2)$. This function has values larger than any number one may choose to name; for instance $\phi(1/300000) > 100000$. It also has values less than any number one may choose to name, as for instance $\phi(2-1/300000) < -100000$. So it can have neither a maximum nor a minimum value.

A somewhat subtle situation is illustrated by the function

$$\psi(x) = x/(1+x), x \in [0,\infty).$$

All values of this function are obviously less than 1 and yet it has no maximum value. This follows from the easily proven inequality $\psi(x) < \psi(x+1)$. Although range has 1 as an upper bound, it contains no greatest number; thus the supremum of the range does not belong to it.

2-3.1. Definition. *A function f is said to be* **bounded above on** *a nonempty subset T of its domain if there exists a number M such that $f(x) \le M$ whenever $x \in T$. Similarly, f is said to be* **bounded below on** *a nonempty subset T of its domain if there exists a number m such that $f(x) \ge m$ whenever $x \in T$. If a function is bounded above as well as below on T, it is simply said to be* **bounded on** *T. When no subset T is specified, it is understood to be the entire domain.*

Remark. A function f being bounded above [or below] on T means the same thing as the subset $\{f(x) : x \in T\}$ of its range being bounded above in the sense of Def.1-5.2 [or below in the sense of Def.1-5.4]. An upper [or lower] bound α of the subset $\{f(x) : x \in T\}$ is also called an **upper [lower] bound of f on T**. Sometimes it is convenient to speak of f being **bounded above [below] by α**.

If a number M satisfies the "strict" inequality $f(x) < M$ whenever $x \in T$ (where we have replaced $\le$ by $<$), then it certainly satisfies $f(x) \le M$ whenever $x \in T$. The converse is not true, but the number $M+1$ satisfies the strict inequality: $f(x) < M+1$ whenever $x \in T$. Thus the statements

$$\exists\, M \ni f(x) \le M \text{ whenever } x \in T$$

and

$$\exists\, M \ni f(x) < M \text{ whenever } x \in T$$

are equivalent, although the number M whose existence is being asserted may not be the same in the two statements.

In terms of the terminology introduced in Def.2-3.1, what we have said about the function $\phi(x) = (x-1)/x(x-2)$, $x \in (0, 2)$ is that it is neither bounded above nor bounded below. And what we have said about $\psi(x) = x/(1+x)$, $x \in [0,\infty)$ is that it is bounded above but has no greatest value. Note that both functions are continuous everywhere on their respective domains. We show in Theorems 2-3.2 and 2-3.3 below that such things cannot happen if the domain of the function is a bounded interval whose endpoints belong to it.

Before going on to the two Theorems, we record two simple observations.

(I) If f is bounded on each of two subsets T_1 and T_2 of its domain, then it is bounded on the union $T_1 \cup T_2$. This is because when there exist M_1 and M_2 such that

$$f(x) \le M_1 \text{ whenever } x \in T_1 \quad \text{and} \quad f(x) \le M_2 \text{ whenever } x \in T_2,$$

the greater of the numbers M_1 and M_2, call it M, has the property that

$$f(x) \le M \text{ whenever } x \in T_1 \cup T_2.$$

(II) If $f: A \to \mathbb{R}$ is continuous at $s \in A$, then, by Prop.2-1.2, there exists $\delta > 0$ such that f is bounded on $(s-\delta, s+\delta) \cap A$. When the domain A is an interval $[a, b]$ and $s = a$, it follows by Prop.1-7.4 that there exists a positive δ_0 such that $[a, a+\delta_0) \subset [a, a+\delta_0] \subseteq A$ and f is bounded on $[a, a+\delta_0)$. A corresponding statement holds regarding b. It also follows by Prop.1-7.4 that, if s is not an endpoint of $[a, b]$, then there exists a positive δ_1 such that $(s-\delta_1, s+\delta_1) \subset [s-\delta_1, s+\delta_1] \subseteq A$ and f is bounded on $(s-\delta_1, s+\delta_1)$.

The two observations above will be used in the forthcoming proof without further explanation.

2-3.2. Boundedness Theorem. *If $f: [a,b] \to \mathbb{R}$ is continuous everywhere on $[a, b]$, then f is bounded on $[a, b]$.*

Proof. Define $S = \{s \in [a, b] : f \text{ is bounded on } [a, s)\}$. By continuity at a, there exists $\delta_0 > 0$

$$f \text{ is bounded on } [a, a+\delta_0) \subseteq [a, b].$$

Then $a + \delta_0 \in S$. Therefore $S \ne \varnothing$ and is bounded above by b. We set $c = \sup S$, whereupon we have

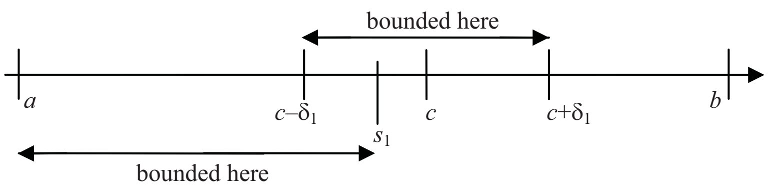
$$a < a+\delta_0 \le c \le b.$$

Suppose, if possible, that $c < b$. Then $c \in (a,b)$ and it therefore follows from the continuity of f at c that there exists $\delta_1 > 0$ such that

$$f \text{ is bounded on } (c-\delta_1, c+\delta_1) \subseteq [a,b]. \tag{1}$$

In particular, $a \le c-\delta_1$. Now $c-\delta_1$ is not an upper bound for S. Therefore there exists $s_1 \in S$ such that $c-\delta_1 < s_1 \le c$. Then

$$f \text{ is bounded on } [a,s_1). \tag{2}$$



Since $a \le c-\delta_1 < s_1 \le c < c+\delta_1$, therefore

$$[a,s_1) \cup (c-\delta_1, c+\delta_1) = [a, c+\delta_1). \tag{3}$$

It follows from (1), (2) and (3) that f is bounded on $[a, c+\delta_1)$. Thus $c+\delta_1 \in S$ even though $c+\delta_1 > c = \sup S$. This contradiction shows that $c = b$.

By continuity at b, there exists $\delta_2 > 0$ such that

$$f \text{ is bounded on } (b-\delta_2, b] \subseteq [a,b]. \tag{4}$$

Since $b = c = \sup S$, we know that $b-\delta_2$ is not an upper bound for S. Therefore there exists $s_2 \in S$ such that $b-\delta_2 < s_2 \le b$ and f is bounded on $[a,s_2)$. Therefore by (4), f is bounded on $[a,s_2) \cup (b-\delta_2, b]$. But $a \le b-\delta_2 < s_2 \le b$ and therefore $[a,s_2) \cup (b-\delta_2, b] = [a,b]$. Thus f is bounded on $[a,b]$. ☺

Recall the examples of unbounded functions ϕ and ψ given just before Def.2-3.1. Of these, ϕ had a bounded domain whose endpoints were not in it and ψ had an unbounded domain. According to Theorem 2-3.2, when the domain is a bounded interval and both endpoints belong to it, a function continuous everywhere on that domain has to be bounded.

As illustrated by the function called ψ above, it is possible for a function to be bounded above and still not have a greatest value. The next result rules this out for a continuous function on a bounded interval that includes both its endpoints.

2-3.3. Extreme Value Theorem. *If $f:[a,b]\to\mathbb{R}$ is continuous everywhere on $[a,b]$, then there exists $c\in[a,b]$ such that $f(x)\le f(c)$ whenever $x\in[a,b]$. Also, there exists $c'\in[a,b]$ such that $f(x)\ge f(c')$ whenever $x\in[a,b]$.*

Proof. By Theorem 2-3.2, the range of f is bounded and therefore has a supremum. Let M be the supremum. First we prove that M belongs to the range of f.

Suppose, if possible, that M does not belong to the range of f. Then $f(x) < M$ for all $x \in [a,b]$. Consider the function g with domain $[a,b]$ such that $g(x) = 1/(M-f(x))$. By Prop.2-1.3, g is continuous everywhere on $[a,b]$ and hence by Theorem 2-3.2, it must be bounded above. Let N be a positive upper bound of (the range of) g. Then $M-1/N < M$ and is therefore not an upper bound of f. Hence there exists some $x_1 \in [a,b]$ such that $M-1/N < f(x_1) < M$, from where it follows that $0 < M-f(x_1) < 1/N$. But this implies that $g(x_1) = 1/(M-f(x_1)) > N$, which is not possible, because N is an upper bound of g. The contradiction shows that M belongs to the range of f.

Since M belongs to the range of f, we have $M = f(c)$ for some $c \in [a, b]$, so that $f(x) \le M = f(c)$ whenever $x \in [a, b]$.

The existence of c' such that $f(x) \ge f(c')$ whenever $x \in [a, b]$ can either be proved analogously or by applying to the function $-f$ what has just been proved. ☺

The greatest value of a function f is of course also known as its *maximum value*. The number c in the domain such that $f(c)$ is the maximum value is called *a maximum*, or also *a point of maximum* in order to distinguish it from the maximum value. Similarly with "minimum". The name "extremum" includes points of maximum and minimum both. Similarly, the name "extreme value" includes maximum and minimum values both.

We shall need a name for describing intervals that contain any endpoints they may have: An interval is said to be **closed** if its endpoints, *if any*, belong to the interval. [No problem if there are no endpoints at all, i.e. $\mathbb{R}$.] Thus $[a, b]$, $[a, \infty)$, $(-\infty, b]$ and $\mathbb{R}$ are closed intervals, but (a, b), $(a, b]$, $[a, b)$, (a, ∞) and $(-\infty, b)$ are not. An interval is said to be **open** if its endpoints, *if any*, do not belong to the interval. Thus (a, b), (a, ∞), $(-\infty, b)$ and $\mathbb{R}$ are open intervals, but $[a, b]$, $(a, b]$, $[a, b)$, $[a, \infty)$ and $(-\infty, b]$ are not. Note that the interval $\mathbb{R}$ is closed and open at the same time. [It may be absurd for a door or a window to be closed and open at the same time, but $\mathbb{R}$ is neither a door nor a window!]

In this terminology, the Boundedness Theorem states that a continuous function whose domain is a closed bounded interval is bounded. The Extreme Value Theorem states that such a function must also have a maximum value and a minimum value.

Problem Set 2-3

2-3.P1. (a) Show that a function $f: X \to \mathbb{R}$ is bounded (i.e. above as well as below) on a subset T of its domain X if, and only if, there exists some $K > 0$ such that $|f(x)| \le K$ for all $x \in T$.
(b) If the functions $f: X \to \mathbb{R}$ and $g: X \to \mathbb{R}$ are bounded above on $T \subseteq X$, show that the function $h: X \to \mathbb{R}$ defined by $h(x) = f(x) + g(x)$ is also bounded above on T.
(c) If the functions $f: X \to \mathbb{R}$ and $g: X \to \mathbb{R}$ are bounded on $T \subseteq X$, show that the function $h: X \to \mathbb{R}$ defined by $h(x) = f(x)g(x)$ is also bounded on T.
(d) Show that, if the functions $f: X \to \mathbb{R}$ and $g: X \to \mathbb{R}$ are bounded above on $T \subseteq X$, the function $h: X \to \mathbb{R}$ defined by $h(x) = f(x)g(x)$ need not be bounded above on T.

2-3.P2. Let the function f be the one that was shown in 2-1.P5 to be continuous on $\mathbb{R}$. Now, show that it is bounded on $\mathbb{R}$. Also, show that the function $g: (0,1] \to \mathbb{R}$ defined by $g(x) = [f(\frac{1}{x}) - \frac{1}{4}]/x$ is continuous on $(0,1]$ but is bounded neither above nor below.

2-3.P3. Call a function $f: S \to \mathbb{R}$ "locally bounded" if, for every $x \in S$, there exists some $\delta > 0$ and some $M > 0$ such that $|f(y)| \le M$ for every $y \in S \cap (x - \delta, x + \delta)$.
(a) Prove that sums, products, constant multiples and absolute values of locally bounded functions are locally bounded. Give an example to show that a quotient of locally bounded functions need not be locally bounded.
(b) Prove that if a function $f: [a,b] \to \mathbb{R}$ is locally bounded, then it is bounded. Does it have to take "extreme values" as in the Extreme Value Theorem for continuous functions?

2-3.P4. Suppose $f: [a,b] \to \mathbb{R}$ is continuous and $f(x) > 0$ for all $x \in [a,b]$. Show that there exists $\alpha > 0$ such that $f(x) \ge \alpha$ for all $x \in [a,b]$.

2-3.P5. Suppose $f: \mathbb{R} \to \mathbb{R}$ is continuous and $a < b$. Show that the function g: $[a,b] \to \mathbb{R}$ defined by

$$g(x) = f(x) + 1/(1 + [x]^2),$$

where [ ] denotes the integer part function, is bounded.

2-3.P6. Show that the function $f: (-\infty, 0) \to \mathbb{R}$ defined by

$$f(x) = 1/(1+x^2) + 1/[x]^2$$

is bounded above.

2-4 Intermediate Value Property

An informal but suggestive description of a continuous function is that it is one whose graph can be drawn without lifting the pencil. Looking at continuous functions $f:[a,b]\to\mathbb{R}$ in this manner, it seems reasonable that one cannot get from the initial point $(a, f(a))$ of the graph to the final point $(b, f(b))$ unless one stays within some bounds. In other words, the function should be bounded. That this guess is actually correct is confirmed by the Boundedness Theorem.

It seems equally reasonable that if the initial and final points lie on opposite sides of the x-axis, then one must necessarily cross the x-axis in going from the initial point to the final point; in other words, if $f(a)$ and $f(b)$ have opposite signs, then $f(c) = 0$ for some c between a and b. The main purpose of this Section is to confirm this.

For the reader who may wonder why it is necessary to confirm something so obvious as this, we offer some explanation. First of all, a graph is only a convenient depiction of a function and is not to be confused with the function itself. Besides, drawing a graph requires the pencil to move and thus have a direction of movement, except when it has to halt temporarily to change direction. Since halting to change direction can only be done at some points but not at all points, it would seem a reasonable guess that a continuous function should have a derivative somewhere or other. However, this seemingly reasonable guess is actually false, as we shall show in Section 8-4.

The hypothesis that $f(a)$ and $f(b)$ have opposite signs means that either $f(a) < 0 < f(b)$ or $f(a) > 0 > f(b)$. It is sufficient to consider any one of these two cases.

We abbreviate the phrase

$$f(x) < 0 \text{ whenever } x \in S \qquad \text{as} \qquad f < 0 \text{ on } S.$$

Other similar abbreviations like "f is continuous on S", and so forth, will be used freely from now on.

2-4.1. Intermediate Value Theorem. *Let $f:[a,b]\to\mathbb{R}$ be continuous on $[a,b]$ and $f(a) < 0 < f(b)$. Then there exists $c \in (a, b)$ such that $f(c) = 0$.*

Proof. Define $S = \{s \in [a,b] : f(s) \leq 0\}$. Then $a \in S$ and therefore $S \neq \varnothing$; moreover, it is bounded above by b. Let $c = \sup S$. Since $-f(a) > 0$, the continuity of f at a leads to the existence of some $\delta_0 > 0$ such that $|f(x)-f(a)| < -f(a)$ whenever $|x-a| < \delta_0$ and $x \in [a,b]$. This implies that $f(x)-f(a) < -f(a)$ whenever $x \in (a-\delta_0, a+\delta_0)\cap[a,b]$, i.e. $f(x) < 0$ for all $x \in [a, a+\delta_0)\cap[a,b]$, so that $[a, a+\delta_0)\cap[a,b] \subseteq S$. Now this intersection contains numbers greater than a, such as $a+\min\{\frac{\delta_0}{2}, b-a\}$. Therefore

$$c \geq a + \min\{\frac{\delta_0}{2}, b-a\} > a, \quad \text{so that} \quad a < c \leq b.$$

Suppose, if possible, that $f(c) < 0$. Then $c \neq b$, because $f(b) > 0$. So $a < c < b$. Therefore $c \in (a,b)$. By the continuity of f at c, it further follows [by Prop.1-7.4] that there exists $\delta_1 > 0$ such that

$$f < 0 \text{ on } [c-\delta_1, c+\delta_1] \subseteq [a, b]. \tag{1}$$

In particular, $f(c+\delta_1) < 0$. Thus $c+\delta_1 \in S$ even though $c + \delta_1 > c = \sup S$. This contradiction shows that $f(c) \geq 0$.

Next suppose, if possible, that $f(c) > 0$. Remark: This time we do not know at the outset that $c < b$; so our next statement has to be slightly different from (1). By continuity at c, there exists $\delta_2 > 0$ such that

$$f > 0 \text{ on } (c-\delta_2, c]\cap[a,b]. \tag{2}$$

Now $c-\delta_2$ is not an upper bound for S. So there exists $s_2 \in S$ such that $c-\delta_2 < s_2 \leq c$. By definition of S, we must have $f(s_2) \leq 0$ even though $s_2 \in (c-\delta_2, c]\cap[a,b]$. This contradicts (2) and shows that $f(c) \leq 0$.

As it was already shown that $f(c) \geq 0$, it now follows that $f(c) = 0$. ☺

2-4.2. Corollary. *The range of a continuous function whose domain is a bounded closed interval is again a bounded closed interval.*

Proof. Let $f:[a, b]\to\mathbb{R}$ be continuous and T be its range. First we prove that T is an interval by showing that a number lying between two numbers of T itself belongs to T [Prop.1-7.5]. For this purpose, suppose $p < u < q$, where $p, q \in T$. Since p, q belong to the range of f, there exist $p', q' \in [a, b]$ such that $f(p') = p$ and $f(q') = q$. Since $p \neq q$, therefore $p' \neq q'$ and hence we must have either $p' < q'$ or $p' > q'$. We need consider only one case, say $p' < q'$. Let $g:[p', q']\to\mathbb{R}$ be the function defined as $g(x) = f(x)-u$. Then g is continuous, $g(p') = f(p')-u = p - u < 0$ and $g(q') = f(q')-u = q-u > 0$. By the Intermediate Value Theorem, it follows that there exists some $c \in [p', q']$ for which $g(c) = 0$, i.e. $f(c)-u = 0$. Thus $u \in T$.

Having shown that T is an interval, we shall argue that it is bounded and closed. By the Extreme Value Theorem [Theorem 2-3.3], the range T contains a greatest as well as a least number. Therefore the interval is bounded and closed. ☺

Example. If $f:[0,1]\to[0,1]$ is continuous, then $f(s) = s$ for some $s \in [0,1]$. Since f maps into $[0,1]$, we have $f(0) \geq 0$ and $f(1) \leq 1$. If $f(0) = 0$, then $s = 0$ satisfies the required equality. If $f(1) = 1$, then $s = 1$ satisfies the required equality. So we need consider only the case when $f(0) > 0$ and $f(1) < 1$. Let $g:[0,1]\to[0,1]$ be given by $g(x) = f(x)-x$. Then $g(0) = f(0)-0 > 0$ and $g(1) = f(1)-1 < 0$. The required conclusion follows from here by the Intermediate Value Theorem.

In proving the Intermediate Value Theorem, the result that every positive real number has a unique positive n^{th} root has not been used either directly or indirectly. The latter can now be proved as a consequence of the former, as we now do. We note however that a proof in the same spirit as that of Prop.1-6.9 is also possible [see Rudin [8]].

2-4.3. Corollary. *If x is a positive real number and n a positive integer, there exists a unique positive real number y such that $y^n = x$.*

Proof. Given a positive real number x, the number $x+1$ must be greater than 1 and therefore $(x+1)^n > x+1 > x$. This means that the continuous function $f:[0,x+1]\to\mathbb{R}$ defined by $f(u) = u^n-x$ satisfies $f(0) < 0 < f(x+1)$. By the Intermediate Value Theorem, there exists $y \in (0,x+1)$ such that $f(y) = 0$, i.e. $y^n-x = 0$. [The uniqueness, as may be recalled, can be easily proved in any ordered field, whether complete or not.] ☺

We remark that 0 is the unique number such that $0^n = 0$. Therefore the above Cor.2-4.3 could have been formulated as saying that, if x is a nonnegative real number and n a positive integer, there exists a unique nonnegative real number y such that $y^n = x$.

2-4.4. Definition. *If $n \in \mathbb{N}$, the* ***n^{th}* root** *of a nonnegative real number a is the unique nonnegative real number b such that $b^n = a$, and is denoted by* $\boldsymbol{a^{1/n}}$.

2-4.5. Definition. *For any finite sequence a_k of numbers, $1 \leq k \leq n$, the* **arithmetic mean** *is $(\Sigma a_k)/n$. If each $a_k \geq 0$, the* **geometric mean** *is $(\Pi a_k)^{1/n}$. If each $a_k > 0$, the* **harmonic mean** *is $n/(\Sigma a_k^{-1})$.*

2-4.6. Proposition. (AM-GM Inequality) *If a_k $(1 \leq k \leq n)$ is a finite sequence of positive real numbers with arithmetic mean A, geometric mean G and harmonic mean H, then $A \geq G \geq H$. Also,*

$$A = G \quad\Leftrightarrow\quad G = H \quad\Leftrightarrow\quad a_k = a_j \text{ whenever } k \neq j.$$

Proof. There is nothing to prove when $n = 1$. Suppose $n \geq 2$. Since each $a_k > 0$, therefore $G > 0$. Set $b_k = a_k/G$ for $1 \leq k \leq n$. Then $\Pi b_k = 1$. By 1-11.P5, $\Sigma b_k \geq n$ and equality holds if, and only if, the numbers b_k are all equal to one another. But the numbers b_k are all equal to one another if, and only if, the a_k are. By the Generalised Distributive Law (Prop.1-11. 6), it follows that $(\Sigma a_k)/G \geq n$; besides, equality holds if, and only if, the numbers a_k are all equal to one another. The statement about H follows by applying what has already proved to the numbers a_k^{-1}, which are also positive. ☺

We are almost ready to define rational powers a^r, where $r \in \mathbb{Q}$. If $r = p/q$, where p and q are integers, $q > 0$, we take a^r to mean $(a^{1/q})^p$. However, the same rational number r can be represented as p/q in different ways and we must ensure that the particular representation makes no difference to what a^r is supposed to be. For example, must $(a^{1/4})^3$ be the same number as $(a^{1/8})^6$? More precisely, it is necessary to check that, when $p/q = m/n$ with m, n also integers and $n > 0$, the numbers $(a^{1/q})^p$ and $(a^{1/n})^m$ turn out to be the same. We leave this and the matter of laws of indices for rational indices to the reader (Problem 2-4.P7). Henceforth, we shall take rational indices as defined and their "laws" as established.

It is clear from Def.2-4.4 that the function $f:[0,\infty)\to[0,\infty)$ defined by $f(x) = x^{1/n}$ is the inverse of the n^{th} power function. It is easy to verify that, when n is odd, such a function is defined on the whole of $\mathbb{R}$. From Theorem 2-2.1, it follows immediately that functions given by expressions such as $x^{1/13}$ are continuous everywhere on their domains. It then follows by Theorem 2-1.3(b) that their powers, such as $x^{5/13}$ are continuous and further by Theorem 2-1.3(b) and (c) that expressions like $2x^{3/7} - 3x^{-5/9}$ also describe functions continuous everywhere on their domains.

Problem Set 2-4

2-4.P1. If $f:[0, 1]\to\mathbb{R}$ is continuous and $f(0) = f(1)$, show that some $s \in [\frac{1}{2}, 1]$ satisfies $f(s) = f(s - \frac{1}{2})$.

2-4.P2. Show that the function $h:[0, 1]\to\mathbb{R}$ of 2-2.P2 has the intermediate value property on $[0, 1]$ although it is not continuous on $[0, 1]$.

2-4.P3. Show that the equation $x^3 + x^2 = 1$ has a solution between 0 and 1. [In other words, there exists $x \in (0, 1)$ such that $x^3 + x^2 = 1$.]

2-4.P4. Show that there is a continuous function $f:\mathbb{R}\to\mathbb{R}$ such that $f(x)^5 + f(x) = x - 1$ for every $x \in \mathbb{R}$.

2-4.P5. Suppose $f:[0,\infty)\to[0,\infty)$ is a continuous function such that $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$. Show that there exists a continuous function $g:[0,\infty)\to[0,\infty)$ such that $g(x)\cdot f(g(x)) = x$ for every $x \in [0,\infty)$.

***2-4.P6.** Let $f:[0,1]\to\mathbb{R}$ be continuous and $f(0) = f(1) = 0 < f(\frac{1}{2})$.
(a) Show that some $K > 0$ has the property that: the equation $f(x) = kx$ has a solution in $(0,1)$ whenever $0 < k \le K$ [in other words, $k \in (0,K] \Rightarrow \exists\, x \in (0,1) \ni f(x) = kx$.]
(b) Depending on the function f, can there exist such K for which the solution x is unique for all $k \in (0,K]$?
(c) Must there always be a largest such K?

2-4.P7. (a) Let $a > 0$ and let p, q, m and n be integers, $q > 0$, $n > 0$. If $p/q = m/n$, show that $(a^{1/q})^p = (a^{1/n})^m$, using the fact that "laws of indices" are valid for integer powers. [Their validity was pointed out just after Prop.1-11.6.] In view of part (a), one can define a^r for any rational $r = p/q$, $q > 0$, as $(a^{1/q})^p$, independently of which representation of r in the form p/q is taken. The remaining parts of this Problem refer to this definition.
(b) For $a > 0$, show that $(a^p)^{1/q} = (a^{1/q})^p$, i.e. that $(a^p)^{1/q} = a^{p/q}$. [Another way to say this is $b^q = a^p \Leftrightarrow b = a^{p/q}$, where $b > 0$. Note that $(x^{1/q})^q = x$ by definition of $x^{1/q}$.]
(c) If r and s are rational numbers, show that $(a^r)^s = a^{rs}$. [Note that we know this for integers r and s as noted in this Section, and now by part (b), also when one among r and s is an integer and the other the reciprocal of a positive integer.]
(d) Prove that $a^r a^s = a^{r+s}$ when r and s are rational numbers.
(e) Prove that $(ab)^r = a^r b^r$ when $a > 0$, $b > 0$ and r is rational.

2-4.P8. If $x > y > 0$ and r is a positive rational number, show that $x^r > y^r$. What if $r < 0$?

2-4.P9. Bernoulli's Inequality. If r is a rational number such that $0 < r \le 1$ and x is any positive real number, show that $(1+x)^r \le 1 + rx$. Hence show that if r is a rational number such that $r \ge 1$ and x is any positive real number, then $(1+x)^r \ge 1 + rx$.

2-4.P10. If $a > 1$ and r is a positive rational number, show that $a^{-r} > 1 - ra$.

***2-4.P11.** If $a > 1$ prove that $a^x - a^y < a^{x+1}(x - y)$ for any rational numbers x, y with $x > y$.

***2-4.P12.** Let $a > 1$ and $f:\mathbb{Q}\to\mathbb{R}$ be defined by $f(x) = a^x$. Show that f is continuous on $\mathbb{Q}$.

2-5 Uniform Continuity

The definition of continuity in 2-1.1 begins by saying "A function $f: A \to \mathbb{R}$ is said to be continuous *at* $a \in A$ if ... ". The requirement that for every ε a certain kind of δ should exist is intended to apply to the particular a mentioned at the beginning. Therefore the δ is specific not only to the ε mentioned before it but also to the number a in the domain. Solutions of Problems 2-1.P1 and 2-1.P2 illustrate how the number a comes into the picture while determining δ based on a given ε. One could ask whether there exists such a positive δ that is specific only to the positive ε but not to a, i.e. one which works for *all* numbers a in the domain. The description of such a δ has to include the phrase "whenever $a \in A$" somewhere. More precisely, we have

2-5.1. Definition. *A function $f: A \to \mathbb{R}$ is said to be* **uniformly continuous on a subset** *S of its domain if, for every $\varepsilon > 0$, there exists some $\delta > 0$ such that $|f(x) - f(a)| < \varepsilon$ whenever $a \in S$, $x \in S$ and $|x - a| < \delta$. In symbols, this condition states that*

$$\forall\ \varepsilon > 0\ \exists\ \delta > 0 \ni \forall\ a, x \in S,\ |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon.$$

When the subset S is the entire domain A, we speak of the function being simply "uniformly continuous" without specifying a subset, or for emphasis, "uniformly continuous on its domain".

In the definition of continuity at a, the number a had a special status and was not covered under some such phrase as "whenever ...". But in the definition of uniform continuity, the numbers x and a have the same status. In order to keep this in focus, it is sometimes convenient to rephrase the condition of uniform continuity in the following manner:

$$\forall\ \varepsilon > 0\ \exists\ \delta > 0 \ni \forall\ x_1, x_2 \in S,\ |x_1 - x_2| < \delta \Rightarrow |f(x_1) - f(x_2)| < \varepsilon.$$

A function that is uniformly continuous on its domain is certainly continuous at each a belonging to the domain. However there is a distinction between the two and we highlight it in the Examples below:

2-5.2. Examples. (a) The function $f: (0, \infty) \to \mathbb{R}$ given by $f(x) = x^2$ is continuous at each $a \in (0, \infty)$ but not uniformly continuous on $(0, \infty)$. Given $a \in (0, \infty)$ and $\varepsilon > 0$, consider $\delta = \min\{1, \varepsilon/(2a + 1)\}$. Then $\delta > 0$ and

$$|x - a| < \delta \Rightarrow |x - a| < 1 \Rightarrow x < a + 1 \Rightarrow 0 \le x + a < 2a + 1 \Rightarrow |x + a| < 2a + 1,$$

and also

$$|x-a|<\delta \Rightarrow |x-a|<\frac{\varepsilon}{2a+1} \Rightarrow |x^2-a^2| = |x-a|\,|x+a| < \frac{\varepsilon}{2a+1}\cdot|x+a|.$$

From the two inequalities derived, it follows that

$$|x-a|<\delta \;\Rightarrow\; |x^2-a^2| < \frac{\varepsilon}{2a+1}\cdot(2a+1) = \varepsilon.$$

Since the number δ has been described with explicit reference to a, it is specific to a. However, that alone does not rule out the possibility that some δ with a different description could be independent of a. We now proceed to rule out such a possibility.

Suppose some positive number δ is claimed to have the property that

$$|x^2-a^2| < \varepsilon \quad \text{whenever} \quad a, x \in (0,\infty) \text{ and } |x-a|<\delta.$$

Let $a = \frac{\varepsilon}{\delta}$ and $x = a + \frac{\delta}{2}$. Then $a, x \in (0,\infty)$ and $|x-a| = \frac{\delta}{2} < \delta$. However, $|x^2-a^2|$ $= a\delta + \frac{\delta^2}{4} > a\delta = \varepsilon$. This contradicts the property that δ is claimed to have. Therefore there can be no δ which "works" for all $a \in (0,\infty)$. Thus f is not uniformly continuous, though it is continuous at every $a \in (0,\infty)$.

(b) In the preceding paragraph, it was essential that the domain available was all of $(0,\infty)$. Had the domain been, say $(0,10]$, then there would have been no way to show that the numbers $a = \frac{\varepsilon}{\delta}$ and $x = a + \frac{\delta}{2}$ are in this domain. And in fact, the function $f:(0,10]\to\mathbb{R}$ given by the same formula $f(x) = x^2$ is uniformly continuous. To see why, let $\delta = \frac{\varepsilon}{20}$. Note that $a, x \in (0,10] \Rightarrow |x+a| \le 20$. Taking this into account, it follows that

$$\begin{aligned}|x-a|<\delta &\Rightarrow |x-a| < \frac{\varepsilon}{20}\\ &\Rightarrow |x^2-a^2| = |x-a|\,|x+a| < \left(\frac{\varepsilon}{20}\right)20 = \varepsilon \text{ whenever } a, x \in (0,10].\end{aligned}$$

This shows that the function $f:(0,10]\to\mathbb{R}$ given by $f(x) = x^2$ is uniformly continuous. In other words, the function which was shown to be *not* uniformly continuous on $(0,\infty)$ is nonetheless uniformly continuous on the subset $(0,10]$ of its domain.

(c) By a similar argument, it can be shown that the function given by $f(x) = x^3$ is uniformly continuous on $(0,12]$. One just has to take $\delta = \frac{\varepsilon}{432}$, the reason for the denominator 432 being that, as long as the numbers a and x belong to $(0,12]$, we have $|x^2+xa+a^2| \le 3(12^2) = 432$.

(d) In Example (b) above, where f is not uniformly continuous, the domain is unbounded. Here is an instance of lack of uniform continuity even when the domain is bounded: Let $f:(0,1)\to\mathbb{R}$ be given by $f(x)=\frac{1}{x}$. By Theorem 2-1.3(c), f is continuous on its domain, which is a bounded subset of $\mathbb{R}$. Nevertheless, we can show it is not uniformly continuous there. For $\varepsilon = 1$, there can exist no positive number δ with the property that

$$\left|\frac{1}{x}-\frac{1}{a}\right| < \varepsilon = 1 \quad \text{whenever} \quad a, x \in (0,1) \text{ and } |x-a| < \delta.$$

Suppose some positive number δ is claimed to have this property. Let δ' be the smaller of the two positive numbers $\frac{1}{2}$ and δ. Then $\frac{1}{\delta'} > 2$, and also $\delta' \in (0,1)$ and $\frac{\delta'}{2} \in (0,1)$. Choose $x = \delta'$ and $a = \frac{\delta'}{2}$. We find that $|x-a| = \frac{\delta'}{2} < \delta' \le \delta$ although $\left|\frac{1}{x}-\frac{1}{a}\right| = \left|\frac{1}{\delta'}-\frac{2}{\delta'}\right| = \frac{1}{\delta'} > 2 > \varepsilon$. This shows that f is not uniformly continuous on $(0,1)$.

For a constant function f or for a function g given by $g(x)=x$ on any domain, uniform continuity can be demonstrated effortlessly: for a constant, take δ to be any positive number, because $|f(x)-f(a)| = 0 < \varepsilon$, no matter what x and a are; and for g, take $\delta = \varepsilon$, because $|g(x)-g(a)| = |x-a|$.

The Examples (a), (b) and (c) in the preceding paragraphs illustrate how the nature of the domain can make a dramatic difference in the matter of uniform continuity. A general result regarding this is given below. [Its proof may be easier to understand if the reader solves Problem 2-5.P3 first.]

2-5.3. Uniform Continuity Theorem: *Let $f:[a,b]\to\mathbb{R}$ be continuous on $[a,b]$. Then f is uniformly continuous on $[a,b]$.* In other words, *a continuous function whose domain is a closed bounded interval is uniformly continuous.*

Proof. Consider any $\varepsilon > 0$. We have to prove that for some $\delta > 0$ and for all $x_1 \in [a,b]$ and $x_2 \in [a,b]$,

$$|f(x_1)-f(x_2)| < \varepsilon \quad \text{whenever} \quad |x_1-x_2| < \delta. \tag{A}$$

Define S to be the set of all $s \in [a,b]$ such that, for some $\delta > 0$ and for all x_1, $x_2 \in [a,s)$, the statement (A) holds. By continuity at a, there exists $\delta_0 > 0$ such that $a+\delta_0 < b$ and

$$|f(x)-f(a)| < \frac{\varepsilon}{2} \quad \text{whenever} \quad |x-a| < \delta_0.$$

For this δ_0, it follows by the triangle inequality that for all $x_1, x_2 \in [a, a+\delta_0)$,

$$|f(x_1)-f(x_2)| \le |f(x_1)-f(a)| + |f(x_2)-f(a)| < \varepsilon,$$

and hence that

$$|f(x_1)-f(x_2)| < \varepsilon \quad \text{whenever} \quad x_1, x_2 \in [a, a+\delta_0) \text{ and } |x_1-x_2| < \delta_0.$$

Thus, for $\delta = \delta_0 > 0$ and for all $x_1, x_2 \in [a, a+\delta_0)$, (A) holds. In other words, $a + \delta_0 \in S$. Therefore $S \neq \varnothing$ and is bounded above by b. Now let $c = \sup S$. Then

$$c \geq a + \delta_0 > a, \text{ so that } a < c \leq b.$$

Suppose, if possible, that $c < b$. Then $c \in (a, b)$ and it therefore follows from the continuity of f at c that there exists $\delta_1 > 0$ such that

$$(c-\delta_1, c+\delta_1) \subseteq [a, b] \qquad \text{and} \qquad |f(x)-f(c)| < \tfrac{\varepsilon}{2} \text{ whenever } |x-c| < \delta_1.$$

In particular, $a \leq c - \delta_1$. Furthermore, it follows by the triangle inequality (again) that for all $x_1, x_2 \in (c-\delta_1, c+\delta_1)$,

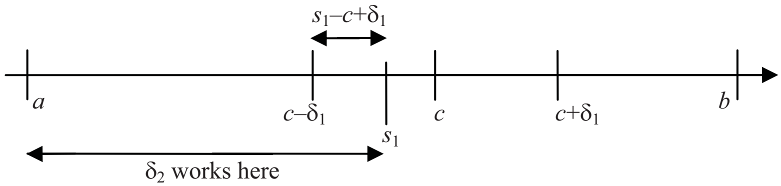
$$|f(x_1)-f(x_2)| \leq |f(x_1)-f(c)| + |f(x_2)-f(c)| < \varepsilon.$$

In other words,

$$|f(x_1)-f(x_2)| < \varepsilon \quad \text{whenever} \quad x_1, x_2 \in (c-\delta_1, c+\delta_1). \tag{1}$$

Now $c - \delta_1$ is not an upper bound for S. So there exists $s_1 \in S$ such that $c - \delta_1 < s_1$. Since $s_1 \in S$, therefore for some $\delta_2 > 0$,

$$|f(x_1)-f(x_2)| < \varepsilon \quad \text{whenever} \quad x_1, x_2 \in [a, s_1) \text{ and } |x_1 - x_2| < \delta_2. \tag{2}$$



Since $a \leq c - \delta_1 < s_1 \leq c < c + \delta_1$, we have

$$[a, s_1) \cup (c-\delta_1, c+\delta_1) = [a, c+\delta_1) \tag{3}$$

and $\delta_3 = \min\{s_1 - c + \delta_1, \delta_2\} > 0$. Consider any $x_1, x_2 \in [a, c+\delta_1)$, where $|x_1 - x_2| < \delta_3$. Then

$$|x_1 - x_2| < \delta_2 \qquad \text{and} \qquad |x_1 - x_2| < s_1 - c + \delta_1. \tag{4}$$

Suppose $x_1 \notin [a, s_1)$. Then by (3), $x_1 \in (c-\delta_1, c+\delta_1)$. Also, $x_1 \geq s_1$ and, by (4), $x_2 - x_1 > -s_1 + c - \delta_1$. Therefore $x_2 = (x_2 - x_1) + x_1 > c - \delta_1$, so that both x_1 and x_2 belong to $(c-\delta_1, c+\delta_1)$. It therefore follows from (1) that $|f(x_1)-f(x_2)| < \varepsilon$. The same argument applies in case $x_2 \notin [a, s_1)$. On the other hand, if both x_1 and x_2 belong to $[a, s_1)$, then the inequality $|f(x_1)-f(x_2)| < \varepsilon$ follows from (4) and (2). This shows that

$$|f(x_1)-f(x_2)| < \varepsilon \quad \text{whenever} \quad x_1, x_2 \in [a, c+\delta_1) \text{ and } |x_1 - x_2| < \delta_3. \tag{5}$$

Therefore $c+\delta_1 \in S$ even though $c+\delta_1 > c = \sup S$. This contradiction shows that $c = b$.

By repeating the argument leading to (5), but with the modification of replacing $(c-\delta_1, c+\delta_1)$ by $(c-\delta_1, c]$ and $[a, c+\delta_1)$ by $[a, c]$, one can show that, for some $\delta_4 > 0$,

$$|f(x_1)-f(x_2)| < \varepsilon \quad \text{whenever} \quad x_1, x_2 \in [a, c] \text{ and } |x_1 - x_2| < \delta_4.$$

Since this holds true for any positive ε whatsoever, f is uniformly continuous on $[a, c]$. As it has already been shown that $c = b$, this proves the uniform continuity of f on $[a, b]$. ☺

A natural question is whether there is an analogue of Theorem 2-1.3 for uniformly continuous functions. One can prove by arguments analogous to the ones used for proving the aforementioned result that, if f and g are uniformly continuous (on some subset S of their common domain), and α is a real number, then $f+g$ and αf are uniformly continuous. However fg need not be uniformly continuous. This is demonstrated by the fact that when $f(x) = g(x) = x$, both are uniformly continuous on $[0, \infty)$, but their product $(fg)(x) = x^2$ is not. Other matters of this kind are pursued in 2-5.P1.

Problem Set 2-5

2-5.P1. (a) Suppose f and g are uniformly continuous on S and α is a real number. Show that $f+g$ and αf are also uniformly continuous. If f and g are also bounded, show that fg is uniformly continuous and also (of course) bounded.
(b) Give an example of a uniformly continuous bounded function $f:(0,1)\to\mathbb{R}$ such that f is nonzero everywhere on (0,1) but $1/f$ is neither bounded nor uniformly continuous. For a uniformly continuous function $f:(0,1)\to\mathbb{R}$ that is nonzero everywhere, prove that if $1/f$ is bounded, then it is uniformly continuous.

2-5.P2. (a) Let $M > 0$. Show that, if $|x_1| \le M$ and $|x_2| \le M$, then $5M^4 \ge |\Sigma x_1^k x_2^{4-k}|$, the summation being over $0 \le k \le 4$. Hence show that $|x_1 - x_2| < \varepsilon/5M^4 \Rightarrow |x_1^5 - x_2^5| < \varepsilon$. For which function does this prove uniform continuity on $[-M, M]$?
(b) What restriction, if any, is necessary on the real numbers x_1 and x_2 in order that $|x_1x_2| \le x_1^2 + x_2^2 + x_1^2x_2^2$? Use this inequality to prove the uniform continuity of the function $f(x) = \frac{x}{1+x^2}$ on an appropriate subset of $\mathbb{R}$.

2-5.P3. Show that a function uniformly continuous on subsets S_1 and S_2 of its domain need not be continuous on $S_1 \cup S_2$. But if there exists $\delta_0 > 0$ such that

$$x_1 \in S_1,\ x_1 \notin S_2,\ x_2 \in S_2,\ x_2 \notin S_1 \ \Rightarrow\ |x_1 - x_2| \geq \delta_0,$$

show that the function must be uniformly continuous on $S_1 \cup S_2$.

2-5.P4. Suppose a function $f: A \to \mathbb{R}$ is uniformly continuous and the subset A of $\mathbb{R}$ is bounded. Show that f is bounded.

Chapter 3
Real and Complex Sequences

3-1 Sequences and Their Limits

The earliest succession of numbers one meets in life is 1, 2, 3, 4, … . Then one comes across 2, 4, 6, 8, … and later on the decimals 0.3, 0.33, 0.333, …, which have a predictable pattern, and also 1, 1.4, 1.41, 1.414, 1.4142, …, which do not. Having a pattern that can be recognised is not an essential ingredient of an endless succession of numbers, or what is formally called an infinite sequence or just "sequence". For more on this question, see 3-1.P5.

While sequences may provide fodder for number games and television quiz shows, they also have a rather more serious role in Mathematics. The practical usefulness of knowing a distance to be, say $100\sqrt{2}$ metres, stems from the fact that $\sqrt{2}$ can be approximated as 1.41. However, if the distance were needed with centimetre accuracy i.e. $10000\sqrt{2}$ cm, then 1.41 would not serve the purpose; one would need to take 1.4142 as an approximant. For other purposes, one could conceivably need an even higher degree of accuracy. What we would therefore like is a succession of decimals or other familiar kinds of numbers, such as rational numbers, that provide approximations of higher and higher degrees of accuracy. An informal description of the fact that, in practice, one can deal with $\sqrt{2}$ through a decimal approximation is that this number can be "accessed" through decimal numbers. One of the important concerns of Mathematics is producing successions of numbers to "access" a variety of numbers. However, it is not possible to insist from day one that the approximants must be of a familiar kind or that the accuracy improve steadily at each step. Therefore, we shall not restrict the numbers in a sequence to be of any particular kind. Moreover, we shall allow the improvement in accuracy to be something like "two steps forward and one step backward", or worse. This matter is further clarified in 3-1.P6.

In Section 1-11, we discussed *finite sequences*. What we are about to define is an *infinite* sequence, generally referred to as simply *sequence*. From this point onwards, a sequence will be understood in the sense of the definition below, unless it is specifically called a finite sequence.

3-1.1. Definition. *A* **sequence** *in a set S is a function defined on the set* $\mathbb{N}$ *of all natural numbers with values in S.*

For a sequence $x:\mathbb{N}\to S$, the value $x(n)$ is often written in subscript notation as x_n and the sequence itself as $\{x_n\}$. This notation is particularly convenient when the sequence has been described by an expression, such as $x_n = \frac{1}{n}$, in which case the sequence can be denoted by $\{\frac{1}{n}\}$. Besides this and the usual function notation $x(n)$, we shall also use the notation $x_1, x_2, x_3, \ldots$, whenever it seems helpful to do so. When there is no risk of confusion, the sequence $\{x_n\}$ may also be denoted by just x_n.

The value x_n will informally be called the n^{th} term of the sequence and n will be called its *index*. A sequence may take values in any set S, but in our early examples, the set S will be usually $\mathbb{R}$ or $\mathbb{C}$. A sequence in $\mathbb{R}$ [resp. $\mathbb{C}$] is called a real [resp. complex] sequence. It should be noted that every real sequence is a complex sequence as well, and therefore any general statement about complex sequences is automatically true for real sequences too.

Remarks. (i) The terms of a sequence need not be distinct.

(ii) It is sometimes convenient to consider a sequence as being defined on the set of nonnegative integers. In this case the sequence can be denoted as $x_0, x_1, \ldots$ or also as x_n, $n = 0, 1, \ldots$.

(iii) As with all functions, a sequence is to be distinguished from its range. For example, the sequence $f(n) = (-1)^n$, $n = 1, 2, \ldots$ alternates between -1 and 1, whereas the set of its values, i.e. its range is $\{(-1)^n : n \in \mathbb{N}\} = \{-1, 1\}$.

Examples. (a) If c is a constant, the sequence $\{x_n\}$, where $x_n = c$ for $n = 1, 2, \ldots$ is called a **constant sequence**.

(b) The sequence of reciprocals of natural numbers is the sequence $1, \frac{1}{2}, \frac{1}{3}, \ldots$.

(c) If $z \in \mathbb{C}$, then the sequence $1, z, z^2, \ldots$ is the function $f(n) = z^n$, $n = 0, 1, 2, \ldots$.

(d) It is easy to see that there exists a sequence $x:\mathbb{N}\to\mathbb{Z}$ for which $x(k+2) = 2x(k+1) - 2x(k)$ for every $k \in \mathbb{N}$. Readers who have come across "recurrence relations" in computer science may recall that the general expression for $x(k)$ is $\alpha(1+i)^k + \beta(1-i)^k$, where α and β are complex numbers that can be determined from the first two values $x(1)$ and $x(2)$. This illustrates one of the benefits of having the system $\mathbb{C}$ at our disposal.

3-1.2. Definition. *A complex sequence* $\{x_n\}$ *is said to be* **bounded** *if its range is a bounded subset of* $\mathbb{C}$. *A real sequence is said to be* **bounded above** [*resp*. **below**] *if its range is bounded above* [*resp. below*].

Remark. For a complex sequence $\{x_n\}$ to be bounded, there must exist some $M > 0$ such that $|x_n| \le M$ for all $n \in \mathbb{N}$. For a real sequence, the condition that $|x_n| \le M$ can be reformulated as $-M \le x_n \le M$. Therefore a real sequence is bounded if, and only if, it is bounded above as well as below. If it is bounded only above or only below, then it is unbounded. By upper bound, least upper bound, lower bound and greatest lower bound of a real sequence we mean those of the range of the sequence.

Examples. (a) The sequence $\{n^2\}$ is bounded below but not above. Since it is not bounded above, it is considered as an unbounded sequence. The sequence $\{-n\}$ is bounded above but not below.

(b) The real sequence $x_n = n/(1+n^2)$ is bounded. A lower bound (for its range) is 0 and an upper bound is $\frac{1}{2}$ [the latter because $2n \le 1+n^2$].

(c) For $z \in \mathbb{C}$, the complex sequence $\{z^n\}$ is bounded when $|z| \le 1$ and unbounded otherwise. This is because $|z^n| = |z|^n \le 1$ when $|z| \le 1$, while $|z^n| = |z|^n = (1+(|z|-1))^n \ge 1+n(|z|-1)$ when $|z| > 1$, by Bernoulli's Inequality [see 1-11.P10]. It now follows from the archimedean property of $\mathbb{R}$ that $\{z^n\}$ is unbounded when $|z| > 1$.

3-1.3. Definition. *Let* $\{x_n\}$ *be a sequence in* $\mathbb{C}$. *A number* $x \in \mathbb{C}$ *is said to be a* **limit of** $\{x_n\}$ *if*, *for every* $\varepsilon > 0$, *there exists a natural number* n_0 *such that*

$$|x_n - x| < \varepsilon \quad \textit{whenever} \quad n \ge n_0.$$

In this case, *we also say that* $\{x_n\}$ **converges to** x, *and write it in symbols as* $\boldsymbol{x_n \to x}$. *If there is no such* x, *we say that the sequence* **diverges**. *A sequence is said to be* **convergent** *if it converges to some limit or other*, *and* **divergent** *otherwise.*

Remark. If $x \in \mathbb{C}$ and $x' \in \mathbb{C}$ are such that $x_n \to x$ and $x_n \to x'$, then $x = x'$. Indeed, for $\varepsilon > 0$, there exist natural numbers n_1 and n_2 such that $|x_n - x| < \frac{\varepsilon}{2}$ whenever $n \ge n_1$ and $|x_n - x'| < \frac{\varepsilon}{2}$ whenever $n \ge n_2$. Consequently, $|x - x'| \le |x - x_n| + |x_n - x'| < \varepsilon$ provided that $n \ge \max\{n_1, n_2\}$. It follows that $|x - x'| \le 0$ [see 1-4.P9] and hence that $x = x'$. This means that a sequence can have at most one limit and we can speak of *the* (one and only) limit of a sequence, provided that the sequence under reference has a limit in the first place. We can now unambiguously denote the limit by $\lim x_n$ or by $\lim\limits_{n\to\infty} x_n$.

A statement about the limit of a sequence is to be interpreted as including the assertion that the limit exists, unless the context indicates otherwise. In particular, the equality $\lim_{n\to\infty} x_n = x$ ordinarily means that the limit *exists and* is x.

It will be shown in the next Section that if a real sequence $\{x_n\}$ has a limit x, then the limit is real. Therefore, the defining condition for $x_n \to x$ in Def.3-1.3 can be restated as: for every $\varepsilon > 0$, there exists a natural number n_0 such that

$$-\varepsilon < x_n - x < \varepsilon \quad \text{whenever} \quad n \geq n_0.$$

Examples. (a) If $x_n = (-1)^n/n$, the sequence $\{x_n\}$ is bounded, has an infinite range and converges to 0. Indeed, the range is $\{(-1)^n/n : n \in \mathbb{N}\}$, which may be described informally as $\{-1, \frac{1}{2}, -\frac{1}{3}, \ldots\}$. Since $-1 \leq (-1)^n/n \leq 1$ for all $n \in \mathbb{N}$, the range has -1 as a lower bound and 1 as an upper bound. [The least upper bound is $\frac{1}{2}$.] Also, it can be shown that $\lim_{n\to\infty} x_n = 0$ as follows: Let $\varepsilon > 0$. Choose a positive integer $n_0 > 1/\varepsilon$ (possible by the archimedean property of $\mathbb{R}$). Then for $n \geq n_0$, we have $|x_n - 0| = 1/n \leq 1/n_0 < \varepsilon$. In future we shall sometimes use the archimedean property without explicit mention.

(b) If $x_n = (-1)^n(1 - \frac{1}{n})$, the sequence $\{x_n\}$ is bounded and has infinite range, but has no limit. Upper and lower bounds (for the range) are 1 and -1. The sequence can be shown to be an injective function as follows. To begin with, for even n, we have $x_n = (1 - \frac{1}{n}) > 0$ and, for odd n, we have $x_n = -(1 - \frac{1}{n}) \leq 0$. If $x_m = x_n$, then m and n must be both even or both odd. In either case, we must also have $(1 - \frac{1}{m}) = (1 - \frac{1}{n})$, from which it follows that $m = n$. Now we argue that $\{x_n\}$ has no limit. Suppose, if possible, that $\{x_n\}$ converges to x. Since $\frac{1}{2} > 0$, there exists some natural number n_0 such that $n \geq n_0 \Rightarrow |x_n - x| < \frac{1}{2}$. Consider any even positive integer $n \geq n_0$. Then $n + 1$ is odd and greater than n_0. Therefore $|x_n - x| < \frac{1}{2}$ and $|x_{n+1} - x| < \frac{1}{2}$ and hence

$$|x_n - x_{n+1}| = |(x_n - x) - (x_{n+1} - x)| \leq |x_n - x| + |x_{n+1} - x| < \tfrac{1}{2} + \tfrac{1}{2} = 1. \qquad (1)$$

But since n is even, we have $x_n = (1 - \frac{1}{n})$ and $x_{n+1} = -(1 - \frac{1}{n+1})$ and therefore

$$|x_n - x_{n+1}| = |(1 - \tfrac{1}{n}) + (1 - \tfrac{1}{n+1})| > 2 - \tfrac{2}{n},$$

which contradicts (1).

(c) If $x_n = \frac{n^2}{n^2+1}$, the sequence $\{x_n\}$ is bounded, has infinite range and converges to 1. Indeed, $|x_n| = \frac{n^2}{n^2+1} < 1$ for $n = 1, 2, \ldots$ and so the sequence is bounded. The range is $\{\frac{1}{2}, \frac{4}{5}, \frac{9}{10}, \ldots\}$, and the reader can easily supply the precise proof that $x_m = x_n \Rightarrow m = n$. Also, $|x_n - 1| = \left|\frac{n^2}{n^2+1} - 1\right| = \frac{1}{n^2+1} \le \frac{1}{n^2} \le \frac{1}{n} < \varepsilon$ whenever $n \ge n_0$, where n_0 is any natural number greater than or equal to $1/\varepsilon$.

(d) If $x_n = \frac{n^3}{n^2+1}$, the sequence $\{x_n\}$ has infinite range and is unbounded. Besides, it has no limit. Again, we invite the reader to supply the easy proof that $x_m = x_n \Rightarrow m = n$. This will show that the range, namely $\{\frac{1}{2}, \frac{8}{5}, \frac{27}{10}, \ldots\}$, is infinite. For unboundedness, we observe that $x_n = n/(1 + 1/n^2) \ge n/2$ for all $n \in \mathbb{N}$. This alone is sufficient to show that the sequence cannot have a limit. Suppose, if possible, that $x_n \to x$. Since $1 > 0$, there exists some $n_0 \in \mathbb{N}$ such that $|x_n - x| < 1$ whenever $n \ge n_0$. Choose $n > \max\{2(|x| + 1), n_0\}$. Then on the one hand, $n > n_0$ and therefore $|x_n - x| < 1$, so that $x_n = |x_n| < |x| + 1$, while on the other hand, $x_n \ge n/2 > |x| + 1$. This contradiction shows that no number x can have the property that $x_n \to x$. Thus the sequence has no limit.

(e) Now consider the sequence $\{x_n\}$ with $x_n = i^n$, where i is the usual complex number with $i^2 = -1$. The sequence has finite range consisting only of the numbers i, -1, $-i$ and 1. In particular, $|x_n| = 1$ for all $n \in \mathbb{N}$, which means the sequence is bounded. The fact that $|x_n - x_{n+1}| = \sqrt{2}$ can be used for proving that the sequence does not converge.

In discussing Example (d), we had pointed out that unboundedness alone was sufficient to show that the sequence could not have a limit. This is true for any sequence, as we now prove.

3-1.4. Proposition. *Let $\{x_n\}$ be a sequence in $\mathbb{R}$ or $\mathbb{C}$. If $\{x_n\}$ has a limit, then it is also bounded.*

Proof. Suppose $x_n \to x$. For $\varepsilon = 1$, there exists n_0 such that $n \ge n_0 \Rightarrow |x_n - x| < 1$, so that $|x_n| < |x| + 1$ whenever $n \ge n_0$. Choose M to be

$$\max\{|x_1|, |x_2|, \ldots, |x_{n_0-1}|, |x| + 1\}.$$

Then $|x_n| \le M$ for all $n \in \mathbb{N}$. ☺

The specific sequences discussed below play a role in our considerations later.

3-1.5. Proposition. *If $a > 0$, then* $\lim_{n\to\infty} a^{1/n} = 1$.

Proof. If $a = 1$, there is nothing to prove, because $|a^{1/n} - 1| = 0$ for all n.

If $a > 1$, we write $d_n = a^{1/n} - 1 > 0$. Then $a = (1 + d_n)^n \ge 1 + nd_n$ by Bernoulli's Inequality. Therefore $d_n \le (a-1)/n$. Consequently, $|a^{1/n} - 1| = d_n \le (a-1)/n$. The right side is less than ε provided $n \geq n_0$, where n_0 is an integer greater than $(a-1)/\varepsilon$.

If $0 < a < 1$, then $a^{1/n} = 1/(1+h_n)$ for some $h_n > 0$. So,

$$a = \frac{1}{(1+h_n)^n} \leq \frac{1}{1+nh_n} < \frac{1}{nh_n} \qquad \text{for all } n \in \mathbb{N},$$

from which it follows that $0 < h_n < 1/na$ for $n \in \mathbb{N}$. Thus

$$|a^{1/n} - 1| = |-h_n/(1+h_n)| = h_n/(1+h_n) < h_n < 1/na \text{ for all } n \in \mathbb{N}.$$

The right side of the above inequality is less than ε provided $n \geq n_0$, where n_0 is any natural number greater than $1/a\varepsilon$. ☺

3-1.6. Proposition. $\lim_{n\to\infty} n^{1/n} = 1$.

Proof. When $n > 1$, we have $n^{1/n} > 1$ and we may therefore write $n^{1/n} = 1 + h_n$ for some $h_n > 0$. With this notation,

$$n = (1+h_n)^n \geq 1 + nh_n + (\tfrac{1}{2})\,n(n-1)\,h_n^2 \geq 1 + (\tfrac{1}{2})\,n(n-1)\,h_n^2,$$

by 1-11.P11. It follows that

$$h_n^2 \leq \frac{2(n-1)}{n(n-1)} \leq \frac{2}{n} \qquad \text{for } n > 1.$$

The right hand side is less than ε^2 provided $n \geq n_0$, where n_0 is any natural number greater than $2/\varepsilon^2$. Let $n_1 = \max\{2, n_0\}$. Then it follows that for $n \geq n_1$, we have

$$0 < n^{1/n} - 1 = h_n \leq (2/n_1)^{1/2} < \varepsilon.$$

Thus $\lim_{n\to\infty} n^{1/n} = 1$. ☺

3-1.7. Proposition. *If* $|z| < 1$, *then* $\lim_{n\to\infty} z^n = 0$. [*Note*: *z may be complex*.]

Proof. If $|z| = 0$, there is nothing to prove. So, suppose $0 < |z| < 1$. Then $|z| = 1/y$, where $y > 1$ and we may therefore write $y = 1 + b$ with $b > 0$. Then $y^n = (1+b)^n \geq 1 + nb$. So, $|z^n - 0| = |z|^n = 1/y^n \leq \dfrac{1}{1+nb} < \dfrac{1}{nb}$. The right side of the last inequality is less than ε provided $n \geq n_0$, where n_0 is any natural number greater than $\dfrac{1}{b\varepsilon}$. ☺

3-1.8. Proposition. *If* $\{z_n\}$ *is any complex sequence and* $z \in \mathbb{C}$, *then the following are equivalent.*

(α) $\lim_{n\to\infty} z_n = z$;

(β) $\lim_{n\to\infty} |z_n - z| = 0$;

(γ) $\lim_{n\to\infty} (z_n - z) = 0$.

Proof. This is immediate from the fact that

$$|z_n - z| = \big||z_n - z| - 0\big| = |(z_n - z) - 0|. ☺$$

Example. If $\{z_n\}$ is any complex sequence, define the sequence $\{\zeta_n\}$ of its arithmetic means by $\zeta_n = (z_1 + z_2 + \cdots + z_n)/n$, for $n \in \mathbb{N}$. If $\lim_{n\to\infty} z_n = l$, then $\lim_{n\to\infty} \zeta_n = l$.

Let $u_n = z_n - l$. Since $\lim_{n\to\infty} z_n = l$, it follows from Prop.3-1.8 that $\lim_{n\to\infty} u_n = 0$. Also,

$$(z_1 + z_2 + \cdots + z_n)/n = l + (u_1 + u_2 + \cdots + u_n)/n \qquad \text{for all } n.$$

It is therefore enough to show that $\lim_{n\to\infty} (u_1 + u_2 + \cdots + u_n)/n = 0$ when $\lim_{n\to\infty} u_n = 0$. For this purpose, consider any $\varepsilon > 0$. Since $\lim_{n\to\infty} u_n = 0$, there exists n_0 such that $n \geq n_0$ implies $|u_n| < \frac{\varepsilon}{2}$. By Prop.3-1.4, there also exists some $k > 0$ such that $|u_n| < k$ for all n. By Prop.1-11.2, we now have

$$\begin{aligned}
\left|\frac{u_1 + u_2 + \cdots + u_n}{n}\right| &= \left|\frac{u_1 + u_2 + \cdots + u_{n_0-1} + u_{n_0} + \cdots + u_n}{n}\right| \\
&\leq \frac{|u_1| + \cdots + |u_{n_0-1}|}{n} + \frac{|u_{n_0}| + \cdots + |u_n|}{n} \\
&\leq \frac{(n_0 - 1)k}{n} + \frac{n - n_0 + 1}{n} \cdot \frac{\varepsilon}{2} \\
&\leq \frac{(n_0 - 1)k}{n} + \frac{\varepsilon}{2} \qquad \text{for all } n \geq n_0.
\end{aligned}$$

Choose $n_1 > 2(n_0 - 1)(k/\varepsilon)$, so that $(n_0 - 1)(k/n) < \frac{\varepsilon}{2}$ whenever $n > n_1$. Then

$$n > \max\{n_0, n_1\} \;\Rightarrow\; \left|\frac{u_1 + u_2 + \cdots + u_n}{n}\right| < \varepsilon,$$

which shows (as required) that $\lim_{n\to\infty} (u_1 + u_2 + \cdots + u_n)/n = 0$.

We note that the converse is not true. The sequence $\{z_n\}$ for which $z_n = (-1)^n$ has no limit, but the sequence $\{\zeta_n\}$ of its arithmetic means has a limit, namely 0, because $\zeta_n = 0$ if n is even and $-\frac{1}{n}$ if n is odd.

Problem Set 3-1

3-1.P1. Show directly from the definition (first principles) that the following sequences have limit 0:

(a) $\{n^{-1/2}\}$ (b) $\{1/2n^{1/2}\}$ (c) $\{1/[n^{1/2} + (n+1)^{1/2}]\}$ (d) $\{\sqrt{(n+1)} - \sqrt{n}\}$

3-1.P2. Show that the sequence $\{\sum_{k=1}^{n} z^{k-1}\}$ is bounded if the complex number z satisfies $|z| \leq 1$ and $z \neq 1$. Show also that it is unbounded for all other complex z.

3-1.P3. (a) Which of the following says precisely that the sequence $\{x_n\}$ is *unbounded*?

(A) $\exists\ M > 0$ such that $|x_n| > M$ whenever $n \in \mathbb{N}$;
(B) $\exists\ M > 0$ and $\exists\ N \in \mathbb{N}$ such that $|x_n| > M$ whenever $n \geq N$;
(C) $\forall\ M > 0\ \exists\ n \in \mathbb{N}$ such that $|x_n| > M$;
(D) $\forall\ M > 0\ \exists\ N \in \mathbb{N}$ such that $|x_n| > M$ whenever $n \geq N$.

(b) Which of the following says precisely that the sequence $\{x_n\}$ does *not* have the number x as its limit?

(A) $\exists\ \varepsilon > 0$ and $\exists\ N \in \mathbb{N}$ such that $|x_n - x| > \varepsilon$ whenever $n \geq N$;
(B) $\exists\ \varepsilon > 0$ such that $\forall\ N \in \mathbb{N}\ \exists\ n \geq N$ such that $|x_n - x| > \varepsilon$;
(C) $\forall\ \varepsilon > 0$ and $\forall\ N \in \mathbb{N}\ \exists\ n \geq N$ such that $|x_n - x| > \varepsilon$;
(D) $\forall\ \varepsilon > 0\ \exists\ N \in \mathbb{N}$ such that $|x_n - x| > \varepsilon$ whenever $n \geq N$.

(c) Show that the sequence $\{x_n\}$ with $x_n = n(1 + (-1)^n)$, and with first eight terms

$$0, 4, 0, 8, 0, 12, 0, 16,$$

is an unbounded sequence, by proving that it satisfies the appropriate condition from part (a).

(d) Show that the sequence $\{x_n\}$ with $x_n = (1/10)(1 + (-1)^n)$ does not converge to 0, by proving the appropriate condition from part (b).

3-1.P4. Which two of the following (if any) describe the same sequence? The symbol $[\ ]$ denotes the integer part function.

(a) $x_n = \frac{1}{2}(1 + (-1)^n)$ (b) $x_n = \left[\frac{n+1}{2}\right] - \left[\frac{n-1}{2}\right]$

(c) $x_n = (n + 1) - 2\left[\frac{n+1}{2}\right]$.

3-1.P5. (a) The succession of three numbers $1, 8, 27$ has a "recognisable" pattern. Now calculate the first four terms x_1, x_2, x_3, x_4 of the sequence $\{x_n\}$ given by $x_n = 6n^2 - 11n + 6$.

(b) Guess the fifth number that succeeds the four numbers $3, 0, 3, 0$. Then compute the first five terms of the sequence $y_n = (24 - 34n + 15n^2 - 2n^3)$.

(c) Let A, B, C and D be any four numbers, not necessarily distinct. Suppose

$$x_n = A\frac{(n-2)(n-3)(n-4)}{(1-2)(1-3)(1-4)} + B\frac{(n-1)(n-3)(n-4)}{(2-1)(2-3)(2-4)} + C\frac{(n-1)(n-2)(n-4)}{(3-1)(3-2)(3-4)} + D\frac{(n-1)(n-2)(n-3)}{(4-1)(4-2)(4-3)}.$$

Show that the first four terms of this sequence are A, B, C, D. This formula provides a "pattern" for any succession of four numbers, no matter how random one may imagine them to be. The formula can of course be easily extended to more than four numbers [Lagrange Interpolation Formula]. The sequence

$1, 1, 6, 9, 9, 3, \ldots$ may seem random, but these are first few digits in the decimal expansion of $\sqrt[7]{3}$.

3-1.P6. The sequence $x_n = \frac{1}{n}(1 + (-1)^n)$ converges to the limit $x = 0$ and may be thought of as approximating 0. Is x_4 a better approximation to x than x_3? Does the accuracy of approximation improve steadily with each next term?

3-1.P7. Show that $n \geq 4(8^{99}) \Rightarrow (\frac{1}{3})^n < (\frac{1}{8})^{100}$.

3-2 Elementary Results on Convergence and Boundedness

The hypothesis in the next Proposition requiring both the limits to be real will be seen to be unnecessary after Prop.3-2.8 is proved later.

3-2.1. Proposition. *If* $\{x_n\}$, $\{y_n\}$ *and* $\{z_n\}$ *are sequences of real numbers such that*

$$x_n \leq y_n \leq z_n \quad \text{for all } n \in \mathbb{N},$$

and $\lim_{n\to\infty} x_n = \lim_{n\to\infty} z_n$, *both real, then* $\lim_{n\to\infty} y_n$ *exists and* $\lim_{n\to\infty} x_n = \lim_{n\to\infty} y_n = \lim_{n\to\infty} z_n$.

Proof. Indeed, if $\lim_{n\to\infty} x_n = \lim_{n\to\infty} z_n = w$, then for any $\varepsilon > 0$ there exist positive integers n_1 and n_2 such that

$$n \geq n_1 \Rightarrow -\varepsilon < x_n - w < \varepsilon \quad \text{and} \quad n \geq n_2 \Rightarrow -\varepsilon < z_n - w < \varepsilon.$$

Let n_0 be max $\{n_1, n_2\}$. Then

$$n \geq n_0 \Rightarrow -\varepsilon < x_n - w < \varepsilon \quad \text{as well as} \quad -\varepsilon < z_n - w < \varepsilon.$$

Since the hypothesis implies

$$x_n - w \leq y_n - w \leq z_n - w \quad \text{for all } n \in \mathbb{N},$$

it follows that $$-\varepsilon < y_n - w < \varepsilon \quad \text{for all } n \geq n_0.$$

Thus $\lim_{n\to\infty} y_n$ exists and equals w. ☺

Examples. (a) Let f be the function shown to be bounded in 2-3.P2. One bound for the function is 1, because $-1 \leq f(x) \leq 1$ for all real x. In particular $-1 \leq f(n) \leq 1$ for all positive integers n and hence $-1/n \leq f(n)/n \leq 1/n$. Since $\lim_{n\to\infty} (-1/n) = \lim_{n\to\infty} (1/n) = 0$, it follows by Prop.3-2.1 that $\lim_{n\to\infty} f(n)/n = 0$.

(b) When trigonometric functions are defined over the real number field in a later Chapter, it will be shown that $-1 \leq \sin x \leq 1$ for all real x. Then $\lim_{n\to\infty} (\sin n)/n = 0$ exactly as in the preceding Example.

(c) It is a straightforward application of the archimedean property that $1/n \to 0$. Since $n^2 > n$ for $n > 1$, one can deduce that $1/n^2 \to 0$ by invoking Prop.3-2.1. The fact that $1/n^2 \to 0$ has the consequence that $(1 + 1/n^2) \to 1$ by Prop.3-1.8.

Now consider $x_n = \sqrt{(1+1/n^2)}$. Since $x_n > 1$, we have $1 < x_n < x_n^2 = 1+1/n^2$. By Prop.3-2.1, it follows that $x_n \to 1$. Thus $\lim_{n\to\infty} \sqrt{(1+1/n^2)} = 1$. Similarly, from the fact that $1/n \to 0$, one can show that $\lim_{n\to\infty} \sqrt{(1+1/n)} = 1$.

An important way of obtaining new sequences from given (or known) ones is by employing algebraic operations, as we do for functions on any domain with values in $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{Q}$ [see the paragraph before Prop.2-1.2]. To put it explicitly, $\{x_n\} + \{y_n\} = \{x_n + y_n\}$, $\{x_n\}\{y_n\} = \{x_n y_n\}$ and $\alpha\{x_n\}$ *is the sequence* $\{\alpha x_n\}$ (where $\alpha \in \mathbb{C}$).

3-2.2. Examples. (a) If $\{x_n\}$ is given by $x_n = 1$ and $\{y_n\}$ is given by $y_n = (-1)^n$, then the sum $\{x_n\} + \{y_n\} = \{x_n + y_n\}$ is given by $x_n + y_n = 1 + (-1)^n$. The first four terms are 2, 0, 2, 0. The sequence of course alternates between the two values 2 and 0.

(b) If $x_n = n$ and $y_n = 1/(1 + n^2)$ for all $n \in \mathbb{N}$, then the product sequence $\{x_n\}\{y_n\} = \{x_n y_n\}$ is given by $x_n y_n = n/(1+n^2)$ for $n \in \mathbb{N}$. Its first three terms are 1/2, 2/5 and 3/10.

3-2.3. Theorem. *Let* $\{x_n\}$ *and* $\{y_n\}$ *be complex sequences. Suppose* $\lim_{n\to\infty} x_n = x$ *and* $\lim_{n\to\infty} y_n = y$. *Then*
(a) $\lim_{n\to\infty} (x_n \pm y_n) = (x \pm y)$;
(b) $\lim_{n\to\infty} \alpha x_n = \alpha x$;
(c) $\lim_{n\to\infty} x_n y_n = xy$;
(d) $\lim_{n\to\infty} {}^{x_n}\!/\!_{y_n} = {}^{x}\!/\!_{y}$ *provided that* $y_n \neq 0$ *for all n and also* $y \neq 0$.

Proof. (a) Let $\varepsilon > 0$ be given. There exist n_1 and n_2 such that

$$n \geq n_1 \Rightarrow |x_n - x| < \tfrac{\varepsilon}{2} \qquad \text{and} \qquad n \geq n_2 \Rightarrow |y_n - y| < \tfrac{\varepsilon}{2}.$$

Let $n_0 = \max\,\{n_1, n_2\}$. Then for $n \geq n_0$, we have

$$|(x_n \pm y_n) - (x \pm y)| \leq |x_n - x| + |y_n - y| < \varepsilon.$$

(b) This is easy and is left to the reader.

(c) Observe that

$$|x_n y_n - xy| = |(x_n - x)y_n + x(y_n - y)| \leq |y_n||x_n - x| + |x||y_n - y|.$$

By Prop.3-1.4, there exists M_1 such that $|y_n| \leq M_1$ for all n. Let $M = \max\,\{M_1, |x|\} + 1$. Then $M > 0$ and we have the estimate (i.e. inequality)

$$|x_n y_n - xy| \leq M(|x_n - x| + |y_n - y|).$$

For $\varepsilon > 0$, choose n_1 and n_2 such that $|x_n - x| < \frac{\varepsilon}{2M}$ for $n \geq n_1$ and $|y_n - y| < \frac{\varepsilon}{2M}$ for $n \geq n_2$. Let $n_0 = \max\ \{n_1, n_2\}$. Then $n \geq n_0$ implies that $|x_n y_n - xy| < \varepsilon$. Thus $\lim_{n\to\infty} x_n y_n = xy$.

(d) We show that, if $\{y_n\}$ is a sequence of nonzero complex numbers that converges to y ($y \neq 0$), then the sequence $\left\{\frac{1}{y_n}\right\}$ converges to $\frac{1}{y}$. Since y_n converges to y, there exists a natural number n_1 such that $n \geq n_1$ implies that $|y_n - y| < \frac{1}{2}|y|$. Therefore $|y_n| \geq |y| - |y_n - y| > |y| - \frac{1}{2}|y| = \frac{1}{2}|y|$ for $n \geq n_1$. Now if $\varepsilon > 0$ is given, there exists a natural number n_2 such that $n \geq n_2$ implies that $|y_n - y| < \frac{1}{2}\varepsilon|y|^2$. Let $n_0 = \max\ \{n_1, n_2\}$. For $n \geq n_0$, we have

$$\left|\frac{1}{y_n} - \frac{1}{y}\right| = \left|\frac{y - y_n}{y y_n}\right| = \frac{1}{|y y_n|}|y_n - y| \leq \frac{2}{|y|^2}|y_n - y| < \varepsilon.$$

Thus $\lim_{n\to\infty} \frac{1}{y_n} = \frac{1}{y}$. The proof of (d) is now completed by using (c). ☺

Examples. (a) $\lim_{n\to\infty} n(\sqrt{(n^2 + 1)} - n) = \frac{1}{2}$. This is because

$$n(\sqrt{(n^2 + 1)} - n) = \frac{n}{\sqrt{(n^2 + 1)} + n} = \frac{1}{\sqrt{(1 + 1/n^2)} + 1},$$

and, from Example (c) discussed after Prop.3-2.1 and Theorem 3-2.3(a), $\lim_{n\to\infty} \sqrt{(1 + 1/n^2)} + 1 = 2$. By Theorem 3-2.3(d), the limit in question is $\frac{1}{2}$.

(b) Recall from Example (c) following 3-2.1 that $\lim_{n\to\infty} \sqrt{(1 + \frac{1}{n})} = 1$. It follows from 3-2.3(d) that $\lim_{n\to\infty} 1/\sqrt{(1 + \frac{1}{n})} = 1$. Now $1/\sqrt{(1 + \frac{1}{n})} = n/\sqrt{(n^2 + n)}$. It follows that the sequence with n^{th} term $n/\sqrt{(n^2 + n)}$ has limit 1.

3-2.4. Theorem. *Let $\{z_n\}$ and $\{w_n\}$ be bounded complex sequences. Then the sequences $\{z_n\} + \{w_n\}$, $\{z_n\}\{w_n\}$, and $\alpha\{z_n\}$ (where α is a complex number) are also bounded.*

Proof. Left as a Problem [see 3-2.P2]. ☺

3-2.5. Proposition. *If the sequence $\{s_n\}$ is defined by $s_n = 1 + z + z^2 + \cdots + z^{n-1}$, where $|z| < 1$, then $\lim_{n\to\infty} s_n = 1/(1 - z)$.*

Proof. It is known that $s_n = (1 - z^n)/(1 - z)$. Therefore $|s_n - 1/(1 - z)| = |z|^n/|1 - z|$. Since $|z| < 1$, it follows by Prop.3-1.7 that $\lim_{n\to\infty} |z|^n = 0$. It then follows from Theorem 3-2.3(b) with $\alpha = 1/|1 - z|$ that $\lim_{n\to\infty} (|z|^n/|1 - z|) = 0$. Hence

$$\lim_{n\to\infty} |s_n - 1/(1 - z)| = 0,$$

from where it follows that $\lim_{n\to\infty} s_n = 1/(1 - z)$ by Prop.3-1.8. ☺

3-2.6. Proposition. *Let* $\{z_n\}$ *be a sequence of complex numbers. Suppose* $\lim_{n\to\infty} z_n = z$. *Then* $\lim_{n\to\infty} |z_n| = |z|$. *The converse is false.*

Proof. It follows from the triangle inequality that

$$\big||z_n|-|z|\big| \le |z_n - z|.$$

Since $\lim_{n\to\infty} z_n = z$, it follows that, for any $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that $n \ge n_0$ implies that $|z_n - z| < \varepsilon$. Therefore $n \ge n_0$ implies that $\big||z_n|-|z|\big| < \varepsilon$.

For the converse, consider the sequence $\{(-1)^n\}$. The sequence of absolute values, namely $1, 1, 1, \ldots$ converges to 1. However, the sequence itself does not converge; for if there exists z such that $|(-1)^n - z| < \varepsilon$ for $n \ge n_0$, then $|-1-z| < \varepsilon$ as well as $|1 - z| < \varepsilon$, as there are even as well as odd natural numbers greater than n_0. Since these inequalities must be true for any $\varepsilon > 0$ whatsoever, we get [see 1-4.P9] $z = 1$ as well as $z = -1$, which is absurd. ☺

3-2.7. Theorem. *Let* $\{z_n\}$ *be a complex sequence and* $z \in \mathbb{C}$. *Then the following are equivalent*:

(α) $\lim_{n\to\infty} z_n = z$;

(β) $\lim_{n\to\infty} \bar{z}_n = \bar{z}$;

(γ) $\lim_{n\to\infty} \Re z_n = \Re z$ *and also* $\lim_{n\to\infty} \Im z_n = \Im z$.

Proof. (α) is equivalent to (β) because $|z_n - z| = |\bar{z}_n - \bar{z}|$.

We shall prove that (α) is equivalent to (γ). First of all,

$$|\Re z_n - \Re z| \le |z_n - z| \text{ and } |\Im z_n - \Im z| \le |z_n - z|.$$

If (α) is true, then for any $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that $n \ge n_0$ implies that $|z_n - z| < \varepsilon$. Therefore $n \ge n_0$ implies that $|\Re z_n - \Re z| < \varepsilon$ and also $|\Im z_n - \Im z| < \varepsilon$. Thus (γ) is also true.

For the converse, suppose (γ) is true, i.e. $\lim_{n\to\infty} \Re z_n = \Re z$ and $\lim_{n\to\infty} \Im z_n = \Im z$. Then for any $\varepsilon > 0$, there exist n_1 and n_2 such that

$$|\Re z_n - \Re z| < \varepsilon/\sqrt{2} \quad \text{whenever} \quad n \ge n_1$$

and

$$|\Im z_n - \Im z| < \varepsilon/\sqrt{2} \quad \text{whenever} \quad n \ge n_2.$$

Let $n_0 = \max\{n_1, n_2\}$. For $n \ge n_0$, we then have

$$|z_n - z|^2 = |\Re z_n - \Re z|^2 + |\Im z_n - \Im z|^2 < \varepsilon^2/2 + \varepsilon^2/2 = \varepsilon^2,$$

so that $|z_n - z| < \varepsilon$. Thus (α) is true. ☺

3-2.8. Corollary. *If a real sequence converges, then its limit is real.*

Proof. Let $\{x_n\}$ be a real sequence having limit $z \in \mathbb{C}$. We must show that $\Im z = 0$. By Prop.3-2.7, $\lim\limits_{n\to\infty} \Im x_n = \Im z$. But $\Im x_n = 0$ for all $n \in \mathbb{N}$, because $\{x_n\}$ is a real sequence. Therefore $\lim\limits_{n\to\infty} \Im x_n = 0$. Thus $\Im z = 0$. ☺

3-2.9. Proposition. *If $\{x_n\}$, $\{y_n\}$ are sequences of real numbers for which there exists some n_0 such that*

$$x_n \le y_n \quad \textit{whenever} \quad n \ge n_0,$$

and if both have limits, then $\lim\limits_{n\to\infty} x_n \le \lim\limits_{n\to\infty} y_n$.

Proof. Denote $\lim\limits_{n\to\infty} x_n$ by x and $\lim\limits_{n\to\infty} y_n$ by y. Consider any $\varepsilon > 0$. Then there exists n_1 such that $|x_n - x| < \frac{\varepsilon}{2}$ whenever $n \ge n_1$, and also n_2 such that $|y_n - y| < \frac{\varepsilon}{2}$ whenever $n \ge n_2$. Equivalently,

$$-\tfrac{\varepsilon}{2} < x_n - x < \tfrac{\varepsilon}{2} \text{ whenever } n \ge n_1 \quad \text{and} \quad -\tfrac{\varepsilon}{2} < y_n - y < \tfrac{\varepsilon}{2} \text{ whenever } n \ge n_2.$$

Now, $-\frac{\varepsilon}{2} < x_n - x < \frac{\varepsilon}{2} \Rightarrow x < x_n + \frac{\varepsilon}{2}$ while $-\frac{\varepsilon}{2} < y_n - y < \frac{\varepsilon}{2} \Rightarrow y_n < y + \frac{\varepsilon}{2}$. Denote $\max\{n_0, n_1, n_2\}$ by n. Then we have $x < x_n + \frac{\varepsilon}{2} \le y_n + \frac{\varepsilon}{2} < (y + \frac{\varepsilon}{2}) + \frac{\varepsilon}{2} = y + \varepsilon$. Since the inequality $x < y + \varepsilon$ has been proved to be true for any positive ε whatsoever, it follows that $x \le y$ [see 1-4.P9]. ☺

3-2.10. Corollary. *If $\{x_n\}$ is a convergent sequence of real numbers and a is a real number such that $x_n \le a$ [resp. $x_n \ge a$] for every $n \in \mathbb{N}$, then $\lim\limits_{n\to\infty} x_n \le a$ [resp. $\lim\limits_{n\to\infty} x_n \ge a$].*

Proof. The sequence $\{y_n\}$ with every term equal to a converges to a. The required conclusion therefore follows from the preceding Proposition. ☺

3-2.11. Example. For any real number p and nonnegative integer n, define $\binom{p}{n}$ inductively by setting

$$\binom{p}{0} = 1 \qquad \text{and} \qquad \binom{p}{n+1} = \binom{p}{n}\frac{p-n}{n+1}.$$

As is easily verified,

$$\binom{p}{n} = \frac{p(p-1)\cdots(p-n+1)}{n!} \quad \text{for every } p \in \mathbb{R} \text{ and every } n \in \mathbb{N} \text{ (but not } n = 0\text{)}.$$

In case p is an integer and $p \ge n$, then $\binom{p}{n}$ is the same as the well known "binomial coefficient", which is often denoted by $C(p, n)$. Now let a denote any complex number, and consider the sequence $\{x_n\}$ for which

$$x_n = \binom{p}{n} a^n \quad \text{for every } n \in \mathbb{N}.$$

Then $x_{n+1} = x_n\left(\dfrac{p-n}{n+1}a\right)$ and hence

$$|x_{n+1}| \le |x_n|\left|\frac{p-n}{n+1}a\right|. \tag{1}$$

We shall prove that, if $|a| < 1$, then $\lim\limits_{n\to\infty} x_n = 0$, i.e.

$$\lim_{n\to\infty}\binom{p}{n}a^n = 0 \qquad \text{for every } p \in \mathbb{R}.$$

Let b be a real number such that $|a| < b < 1$. Since

$$\frac{|p|+n}{n+1} = 1 + \frac{|p|-1}{n+1} \quad \text{and} \quad \lim_{n\to\infty}\frac{|p|-1}{n+1} = 0,$$

there exists $N \in \mathbb{N}$ such that [e.g. take $N > {}^{(|p|b-1)}\!/_{(1-b)}$]

$$\begin{aligned} n \ge N \;&\Rightarrow\; \frac{|p|+n}{n+1} < \frac{1}{b} \\ &\Rightarrow\; 0 \le \frac{|p-n|}{n+1} < \frac{1}{b} \\ &\Rightarrow\; |x_{n+1}| \le |x_n|\frac{1}{b}|a| \quad \text{by (1).} \end{aligned}$$

It follows from this by an induction argument that

$$n \ge N \;\Rightarrow\; 0 \le |x_n| \le |x_N|\left(\frac{|a|}{b}\right)^{n-N}.$$

Since $|a| < b$, it follows from Props.3-1.7 and Prop.3-2.1 that $\lim\limits_{n\to\infty} x_n = 0$.

Problem Set 3-2

3-2.P1. Suppose $\{z_n\}$ is a complex sequence and z a complex number such that $\{z_n^2\}$ converges to z^2. Show that $\{z_n\}$ need not converge to z. If $z + z_n \ne 0$ for all n, and if the sequence $\{1/(z+z_n)\}$ is bounded, show that $\{z_n\}$ does converge to z.

3-2.P2. Prove Theorem 3-2.4. What about a quotient of bounded sequences?

3-2.P3. If a convergent sequence $\{x_n\}$ satisfies $x_n < a$ for every $n \in \mathbb{N}$, where a is some real number, does it follow that $\lim\limits_{n\to\infty} x_n < a$? If a convergent sequence $\{x_n\}$ satisfies the condition that

$$\exists\, n_0 \ni \forall\, n \in \mathbb{N}, \quad n \ge n_0 \Rightarrow x_n \le a,$$

show that $\lim\limits_{n\to\infty} x_n \le a$.

3-2.P4. Let $\{x_n\}$ be any real sequence. Show that

$$\lim_{n\to\infty} \min \tfrac{1}{n}\{x_n - [x_n],\, 1 - (x_n - [x_n])\} = 0,$$

where the brackets [ ] denote the integer part function.

3-2.P5. Let $\{z_n\}$ be the sequence defined as

$$z_n = \frac{(1+(-1)^n)}{n} + \frac{i(1-(-1)^n)}{n}.$$

Show that $\lim_{n\to\infty} z_n = 0$.

3-2.P6. Let $\{x_n\}$ be a sequence such that $x_n \neq -1$ for every $n \in \mathbb{N}$ and let $y_n = x_n/(1+x_n)$ for every $n \in \mathbb{N}$. Show that $\lim_{n\to\infty} y_n = 0 \Leftrightarrow \lim_{n\to\infty} x_n = 0$.

3-2.P7. If a sequence $\{x_n\}$ in $\mathbb{R}$ satisfies $x_n > 0$ for every n and $\lim_{n\to\infty} (x_{n+1}/x_n) = \frac{2}{3}$, show that $\lim_{n\to\infty} x_n = 0$.

3-3 Monotone Sequences

We have seen sequences whose terms sometimes increase from one to the next but sometimes decrease. Two of the simplest examples would be (i) $x_n = (-1)^n$, or $-1, 1, -1, 1, \ldots$ and (ii) $x_n = (-1)^n/n$, or $-1, 1/2, -1/3, 1/4, \ldots$. What we intend to study in this Section is the situation when such "jumping back and forth" does not occur. An example that may come to mind immediately is 1, 2, 3, …, which one would call "increasing". The sequence 1, 1, 2, 2, 3, 3, … may seem a bit different but will also be regarded as increasing. In both these examples, each succeeding term is greater than or equal to the one before it: $x_{n+1} \geq x_n$ for all $n \in \mathbb{N}$.

3-3.1. Definition. *A sequence $\{x_n\}$ of real numbers is said to be* **increasing** *if $x_{n+1} \geq x_n$ for all $n \in \mathbb{N}$. It is said to be* **decreasing** *if the inequality is reversed. A sequence which is either increasing or decreasing is called* **monotone**.

Examples. (a) The sequence $\{2^n\}$ is increasing

(b) The sequences $\{\frac{1}{n}\}$ and $\{-n\}$ are decreasing.

(c) A constant sequence is both increasing and decreasing. Conversely, any sequence which is both increasing and decreasing must be a constant sequence.

(d) All sequences in the preceding Examples are monotone, because each one is either increasing or decreasing. The sequence $\{(-1)^n\}$ is not monotone, because it is neither increasing nor decreasing. Note that this sequence is bounded and has no limit. Our next result will be that a monotone sequence cannot be bounded and also have no limit.

3-3.2. Remarks. (i) For an increasing sequence $\{x_n\}$, it is easy to prove by induction that $x_1 \le x_n$ for all n. Therefore the first term is a lower bound for the (range of the) sequence. In particular, an increasing sequence is always bounded below. It is therefore bounded if, and only if, it is bounded above. Similarly for a decreasing sequence. Thus the words "above" and "below" can be omitted in the statement of the Theorem below, if one wishes to.

(ii) It can be easily proved by induction for an increasing sequence $\{x_n\}$ that $m > n \Rightarrow x_m \ge x_n$, and for a decreasing sequence that $m > n \Rightarrow x_m \le x_n$. One proof is by induction on $m - n$. These simple properties will be used in the proof below.

3-3.3. Theorem. (a) *If $\{x_n\}$ is increasing and is bounded above, then it converges to its least upper bound.*
(b) *If $\{x_n\}$ is decreasing and is bounded below, then it converges to its greatest lower bound.*

Proof. (a) Let B be the least upper bound of (the range of) the sequence $\{x_n\}$. Let $\varepsilon > 0$ be given. Since $B - \varepsilon$ is not an upper bound, there exists at least one term, x_{n_0} say, such that $x_{n_0} > B - \varepsilon$. Since the sequence $\{x_n\}$ is increasing, it follows [see Remark 3-3.2 (ii)] that $x_n \ge x_{n_0} > B - \varepsilon$ for $n \ge n_0$. As B is an upper bound of the sequence, we have $x_n \le B$ for every $n \in \mathbb{N}$. Thus

$$B - \varepsilon < x_n \le B < B + \varepsilon \quad \text{whenever} \quad n \ge n_0.$$

Therefore $|x_n - B| < \varepsilon$ whenever $n \ge n_0$. Since such an integer n_0 has been shown to exist for any given $\varepsilon > 0$, therefore $\lim\limits_{n\to\infty} x_n = B$.

(b) If $\{x_n\}$ is decreasing and bounded below, $\{-x_n\}$ is increasing and bounded above. The proof can be completed by using (a). ☺

3-3.4. Example. Consider the sequences $\left\{\left(1 \pm \frac{1}{n}\right)^n\right\}$. We first show that both are increasing. In the AM-GM Inequality [see Prop. 2-4.6],

$$(a_1 + a_2 + \cdots + a_n)/_n \ge (a_1 a_2 \cdots a_n)^{1/n},$$

(obviously valid even if one or more of the a_j are zero), set $a_1 = a_2 = \cdots = a_{n-1} = 1 \pm \dfrac{1}{n-1}$ and $a_n = 1$ to obtain

$$\frac{(n-1)\left(1 \pm \dfrac{1}{n-1}\right) + 1}{n} \ge \left(1 \pm \frac{1}{n-1}\right)^{(n-1)/n} \text{ for } n = 2,\ 3, \ldots,$$

that is,

$$\left(1 \pm \frac{1}{n-1}\right)^{n-1} \le \left(1 \pm \frac{1}{n}\right)^n \quad \text{for } n = 2, 3, \ldots,$$

which means the sequences are increasing. We next show that they bounded above.

The simple equality $(1+\frac{1}{n})(1-\frac{1}{n+1}) = (\frac{n+1}{n})(\frac{n}{n+1}) = 1$ shows that $(1+\frac{1}{n})^{n+1}(1-\frac{1}{n+1})^{n+1} = 1$. Since we have just proved that $\{(1-\frac{1}{n+1})^{n+1}\}$ is increasing, it now follows that the sequence $\{(1+\frac{1}{n})^{n+1}\}$ is decreasing. Hence $(1+\frac{1}{n})^{n+1} \le (1+\frac{1}{1})^{1+1} = 4$. However, $(1-\frac{1}{n})^n \le (1+\frac{1}{n})^n \le (1+\frac{1}{n})^{n+1}$. Consequently, the sequences $\{(1-\frac{1}{n})^n\}$ and $\{(1+\frac{1}{n})^n\}$ have an upper bound 4.

From Theorem 3-3.3, it follows that each of them converges to a limit. After we formally define the natural logarithm function in the context of a complete ordered field, it will turn out of course that the limit is e, the "base of natural logarithms", which will have been shown to lie between 2.718 and 2.719 [see Prop.7-4.12]. For the interested reader, we mention that it is possible to obtain the smaller upper bound 3 right away if desired. Indeed, $n \ge 5$ implies $(1+\frac{1}{n})^n \le (1+\frac{1}{n})^{n+1} \le (1+\frac{1}{5})^{5+1} = \frac{46656}{15625} < 3$, and the same holds for the first four terms since the sequence is increasing.

More examples. (A) If $x_1 = \sqrt{2}$ and $x_{n+1} = \sqrt{(2+\sqrt{x_n})}$, $n \in \mathbb{N}$, then $\{x_n\}$ is increasing and bounded above. Observe that $x_{n+1}^2 - x_n^2 = \sqrt{x_n} - \sqrt{x_{n-1}}$. Since $x_2 = \sqrt{(2+\sqrt{x_1})} > \sqrt{2} = x_1$, it follows by induction that $x_{n+1} \ge x_n$ for each n. Moreover we can show by induction that $x_n \le 2$ for each n, because $x_1 = \sqrt{2} \le 2$, while $x_m \le 2 \Rightarrow x_{m+1} = \sqrt{(2+\sqrt{x_m})} \le \sqrt{(2+\sqrt{2})} \le 2$. It follows from Theorem 3-3.3 that the sequence has a limit. Since each term is greater than $\sqrt{2}$, the limit must be at least $\sqrt{2}$. If one tries to compute the limit x by the usual elementary methods, one obtains from $x = \sqrt{(2+\sqrt{x})}$ that $x^4 - 4x^2 - x + 4 = 0$. The left hand side equals $(x-1)(x^3+x^2-3x-4)$. Since $x > 1$, it must be a root of the cubic factor. The adventurous reader is invited to find a root that lies between $\sqrt{2}$ and 2. In case an approximation to the root is to be obtained, it may be noted that the given sequence already provides one, although it is not obvious up to how many decimal places of accuracy. In any case, the sequence provides a way of approximating an irrational root of a cubic without ever computing a cube root, although square roots have to be computed.

(B) Let $a > 0$ and $\{x_n\}$ be the sequence defined inductively by

$$x_{n+1} = \tfrac{1}{2}(x_n + ax_n^{-1}) \text{ and } x_1 \text{ is any number} > \sqrt{a}.$$

We shall show that $\lim x_n = \sqrt{a}$.

Obviously each $x_n > 0$. Moreover, $2x_n x_{n+1} = x_n^2 + a$, that is, $x_n^2 - 2x_n x_{n+1} + a = 0$. This can be written as $(x_n - x_{n+1})^2 = x_{n+1}^2 - a$, so that $x_{n+1}^2 - a \geq 0$. Thus $x_{n+1} \geq \sqrt{a}$ for all n. We next show that the sequence $\{x_n\}$ is decreasing. Indeed,

$$x_n - x_{n+1} = x_n - \tfrac{1}{2}(x_n + a x_n^{-1}) = (1/2x_n)(x_n^2 - a) \geq 0 \text{ for all } n.$$

Thus the sequence is decreasing and is bounded below (by $\sqrt{a}$). Therefore it converges to some limit, say x. From the defining equation $x_{n+1} = \tfrac{1}{2}(x_n + a x_n^{-1})$ it follows that $x = \tfrac{1}{2}(x + a x^{-1})$, which gives $x^2 = a$. Since we know that $x \geq \sqrt{a}$, we have $x = \sqrt{a}$.

When $a = 13$ and $x_1 = 4$, the second and third terms are 3.625 and 3.6056…; this third term agrees with $\sqrt{13}$ up to the third decimal place.

(C) Let $x > y > 0$. Define two sequences $\{x_n\}$ and $\{y_n\}$ as follows:

$$x_1 = (x+y)/2 \text{ and } y_1 = \sqrt{(xy)}; \qquad x_{n+1} = \tfrac{1}{2}(x_n + y_n) \text{ and } y_{n+1} = \sqrt{(x_n y_n)}.$$

Then $\lim x_n = \lim y_n$.

Since x_1 is the arithmetic mean and y_1 the geometric mean of x and y, therefore $0 < y < y_1 < x_1 < x$. If we assume (as induction hypothesis) that $y_n < x_n$, then it follows in view of the facts $x_{n+1} = \tfrac{1}{2}(x_n + y_n)$ and $y_{n+1} = \sqrt{(x_n y_n)}$ that $y_n < y_{n+1} < x_{n+1} < x_n$. Therefore, by induction, this triple inequality holds for all n. Thus $\{x_n\}$ and $\{y_n\}$ are respectively decreasing and increasing, both being bounded (below by y_1 and above by x_1). Therefore both converge to limits, say l and m respectively. Then $l = \tfrac{1}{2}(l + m)$ and $m = \sqrt{lm}$. The first of these implies that $l = m$. So does the second, but upon using the fact that $m \neq 0$, which is true because $m > y_1 > 0$.

(D) Let $k > 0$ and $x_1 > 0$. Define $x_{n+1} = k/(1 + x_n)$ for all n. We shall show that $\lim x_n$ exists and equals the positive root of the equation $x^2 + x - k = 0$.

Observe that

$$x_n > 0 \text{ and } k - x_{n+1} = k - k/(1 + x_n) = (kx_n)/(1 + x_n) > 0 \text{ for all } n.$$

Thus $0 < x_n < k$ for $n = 2, 3, \ldots$. Again,

$$\begin{aligned} x_n - x_{n-2} &= \frac{k}{1 + x_{n-1}} - \frac{k}{1 + x_{n-3}} = \frac{k(x_{n-3} - x_{n-1})}{(1 + x_{n-1})(1 + x_{n-3})} \\ &= -\frac{k(x_{n-1} - x_{n-3})}{(1 + x_{n-1})(1 + x_{n-3})}. \end{aligned}$$

Hence if $x_{n-1} - x_{n-3} \geq 0$, then $x_n - x_{n-2} \leq 0$. It follows upon using an induction argument that, if $x_1 \leq x_3$ [resp. $x_1 \geq x_3$] then the sequence $\{x_{2n+1}\}$ of odd terms is increasing [resp. decreasing] while the sequence $\{x_{2n}\}$ of even terms is decreasing [resp. increasing]. Since each of the sequences is bounded below by 0 and above by max $\{x_1, k\}$, they both converge. Suppose lim $x_{2n} = l$ and lim $x_{2n+1} = m$. Then $l = k/(1+m)$ and $m = k/(1+l)$, that is, $l + lm = k$ and $m + lm = k$. So, $l = m$ and $l^2 + l - k = 0$. Since one of the roots of this quadratic equation is negative, it must be discarded, as the terms of our sequence are all positive.

3-3.5. Definition. *A sequence of intervals* $[a_n, b_n]$, $n = 1, 2, \dots$ *is called a* **nested sequence of intervals** *if each one contains the next*: $[a_n, b_n] \supseteq [a_{n+1}, b_{n+1}]$, $n = 1, 2, \dots$.

For an interval with two endpoints, the absolute value of the difference between the endpoints is called its **length**.

3-3.6. Nested Interval Theorem. *The intersection of a nested sequence of intervals is nonempty. If the sequence of the lengths of the intervals converges to* 0, *then the intersection of the intervals contains only one number.*

Proof. For a nested sequence of intervals $[a_n, b_n]$, $n = 1, 2, \dots$, we have [see 1-7.P5]

$$a_n \le a_{n+1} \quad \text{and also} \quad b_n \ge b_{n+1} \quad \text{for every } n \in \mathbb{N},$$

which means that the left endpoints form an increasing sequence and the right endpoints a decreasing sequence. By Theorem 3-3.3, the sequence $\{b_n\}$ converges to it greatest lower bound b and $\{a_n\}$ converges to its least upper bound a. Now, consider any $m, n \in \mathbb{N}$. If $m \le n$, then $a_m \le a_n \le b_n$, while if $m \ge n$, we have $a_m \le b_m \le b_n$. Therefore $a_m \le b_n$, whatever the integers m and n may be. Thus *every* a_n is a lower bound for the sequence $\{b_n\}$. Hence the limit b of the sequence $\{b_n\}$, being its greatest lower bound, must be greater than or equal to a_n for every n: $b \ge a_n$ for every n. In other words, b is an upper bound for the sequence $\{a_n\}$. Therefore the limit a of the sequence $\{a_n\}$, being its least upper bound, must be less than or equal to b. Thus we have $a_n \le a \le b \le b_n$ for all n, i.e. $[a, b] \subseteq [a_n, b_n]$ for all n. Therefore the intersection of all the intervals $[a_n, b_n]$ contains $[a, b]$, which is nonempty, as it contains a (or b, which could be equal to a).

Actually the intersection is precisely $[a, b]$, as we now argue. Let $p \notin [a, b]$. Then either $p < a$ or $p > b$. Suppose $p < a$. Since $a = \sup \{a_n\}$, therefore p is not an upper bound of $\{a_n\}$ and hence there must exist some number a_n such that $a_n > p$. But this means that $p \notin [a_n, b_n]$ and p cannot be in the intersection of the intervals. The case when $p > b$ is similar.

Since $a_n \le a \le b \le b_n$, we have $0 \le b - a \le b_n - a_n$, the length of the interval $[a_n, b_n]$. If the sequence $b_n - a_n$ converges to 0, it follows from the preceding inequality that $b-a=0$. Therefore the intersection $[a,b]$ consists of the single number a. ☺

The preceding result can be used for proving various theorems, including the three main results of Chapter 2, namely Theorems 2-3.2, 2-4.1 and 2-5.3. We shall now use it for proving something in a different direction.

First we remark that, if $a<b$, then for any numbers x_1 and x_2 such that $a<x_1<x_2<b$, we have not only

$$[a,x_1]\cup[x_1,x_2]\cup[x_2,b]=[a,b],$$

but also

$$[a,x_1]\cap[x_1,x_2]\cap[x_2,b]=\varnothing.$$

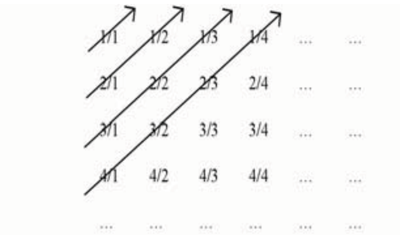
Therefore every number in $[a,b]$ must be in one of the three intervals, possibly in two of them, but never in all three. We shall refer to any such triplet of intervals as a "trifurcation" of $[a,b]$.

3-3.7. Theorem. *Any interval $[a,b]$ with $a<b$ is uncountable. Hence any subset of $\mathbb{R}$ containing such an interval is uncountable.*

Proof. Suppose, if possible, that there is a bijection ϕ from $\mathbb{N}$ to $[a,b]$. Let I_1 be the interval $[a,b]$. Now $\phi(1)$ cannot be in all the three intervals of a trifurcation of $I_1=[a,b]$. Let I_2 be any one of the three intervals that does *not* contain $\phi(1)$. Similarly, let I_3 be any one of the three intervals of a trifurcation of I_2 that does not contain $\phi(2)$. Being a subset of I_2, it cannot contain $\phi(1)$. Proceeding in this manner, we get a nested sequence of intervals $I_1\supseteq I_2\supseteq\cdots\supseteq I_n\supseteq I_{n+1}\supseteq\cdots$ such that I_n does not contain $\phi(k)$ for $1\le k\le n-1$. Therefore the intersection of all the intervals I_n does not contain $\phi(k)$ for any k at all. By the Nested Interval Theorem 3-3.6, the intersection contains some number, x say. Since ϕ is supposed to be a bijection (in particular, a surjection), there must be some m for which $\phi(m)=x$. Thus $\phi(m)$ belongs to the intersection of all the intervals in our nested sequence, although the intersection has been shown not to contain $\phi(k)$ for any k at all. This contradiction shows that a bijection from $\mathbb{N}$ to $[a,b]$ is not possible. ☺

For the set $\mathbb{Q}$ of rational numbers, the situation is the contrary: $\mathbb{Q}$ is countable. The proof is however much more elementary than that of the uncountability of $\mathbb{R}$, as it does not use the completeness property of the latter, nor any consequences of it.

The most direct way to see the plausibility of $\mathbb{Q}$ being countable is to arrange all positive rational numbers in a two dimensional infinite array [see below], the first row having all numerators 1 and denominators ranging over $1,2,\ldots,$ the second having all numerators 2 and denominators ranging over $1, 2, \dots,$ and so on.



The arrows shown in the figure suggest how one might arrange all positive rational numbers in a sequence, omitting all duplications like 2/2, 4/2, 2/4 and so on. The rational number 1/5 will be the 11th one in the sequence. The step of omitting duplications can be looked upon this way. Each positive rational number has a unique representation as m/n with least positive $n \in \mathbb{N}$ among all such representations. This provides an injective map from the set of positive rational numbers into $\mathbb{N}\times\mathbb{N}$. Like any injective map, it is a bijection onto its own range. Now the range of this map is an infinite subset of $\mathbb{N}\times\mathbb{N}$. The aforementioned scheme with arrows suggests how to set up a bijection between $\mathbb{N}$ and $\mathbb{N}\times\mathbb{N}$, making the latter denumerable. The range of the injective map is thus an infinite subset of a denumerable set and therefore denumerable. Consequently, so is its domain, which is the set of positive rational numbers. In this version of the argument, the purpose served by the array and arrows is only to provide a bijection between $\mathbb{N}$ and $\mathbb{N}\times\mathbb{N}$.

3-3.8. Theorem. *The sets* $\mathbb{N}\times\mathbb{N}$ *and* $\mathbb{Q}$ *are denumerable.*

Proof. Let $\phi:\mathbb{N}\times\mathbb{N}\to\mathbb{N}$ be the map for which $\phi(n,m) = (n+m-1)(n+m-2)+2m$. Since $n+m-1 \geq m$, it is an immediate consequence of 1-8.P12(b) that ϕ is injective. It follows from this that $\mathbb{N}\times\mathbb{N}$ must be denumerable.

Let $\mathbb{Q}^+$ be the set of positive rational numbers and $\psi:\mathbb{Q}^+\to\mathbb{N}\times\mathbb{N}$ be the following map. If $x \in \mathbb{Q}^+$, then there exist $p, q \in \mathbb{N}$ such that $x = p/q$. Among all such $q \in \mathbb{N}$, let n be the least (Well Ordering Principle). There exists a unique $m \in \mathbb{N}$ such that $x = m/n$. Let $\psi(x) = (m, n)$. Then ψ is a bijection between $\mathbb{Q}^+$ and an infinite subset of $\mathbb{N}\times\mathbb{N}$. But the latter set is denumerable and therefore any infinite subset is also denumerable. It follows that $\mathbb{Q}^+$ is denumerable, i.e. there exists a bijection ψ_1: $\mathbb{Q}^+\to\mathbb{N}$. From ψ_1, we construct the map ψ_2: $\mathbb{Q}\to\mathbb{N}$ as follows.

$$\psi_2(x) = \begin{cases} 2\psi_1(x) & \text{if } x > 0 \\ 1 & \text{if } x = 0 \\ 2\psi_1(-x) + 1 & \text{if } x < 0. \end{cases}$$

This map is a bijection between $\mathbb{Q}$ and $\mathbb{N}$, showing that $\mathbb{Q}$ is denumerable. ☺

Problem Set 3-3

3-3.P1. Let $\{x_n\}$ be a real sequence such that $x_{n+1} = 1/4 + x_n^2/2$.
(a) Show by induction that, if $0 \le x_1 \le x_2$, then the sequence is increasing, and that if $x_1 \ge x_2$, the sequence is decreasing.
(b) Show that the limit, if any, must be $1 \pm 1/\sqrt{2}$.
(c) If $x_1 = 1$, show that the sequence is decreasing with lower bound 0 and that it therefore has a limit. Show that the limit must be $1 - 1/\sqrt{2}$.
(d) The hypothesis that $0 \le x_1$ is not satisfied when $x_1 = -1$. Is the sequence increasing or decreasing; what about a limit?

3-3.P2. Show that the sequence $\{x_n\}$ with $x_n = (n+2)^{1/(n+2)}$ is decreasing. [We have shown in Prop.3-1.6 that its limit is 1.] Show also that, if $x_n = (n+1)^{1/(n+1)}$, then $\{x_n\}$ is not decreasing.

3-3.P3. Show that the sequence $\{(1-\frac{1}{n})^n\}$ is increasing. It will follow for $n \ge 2$ that $(1-\frac{1}{n})^n \ge (1-\frac{1}{2})^2 = \frac{1}{4}$. Use this and the fact that $(1+\frac{1}{n})(1-\frac{1}{n}) \le 1$ to show that $(1+\frac{1}{n})^n \le 4$. [This is an alternative argument for the boundedness of $\{(1+\frac{1}{n})^n\}$. We shall see later that the limits of the two sequences are reciprocals.] Sharpen this argument to obtain the estimate $(1+\frac{1}{n})^n \le 4$.

3-3.P4. If $\{x_n\}$ is an increasing sequence of real numbers, show that the sequence of real numbers $\{y_n\}$ defined by $y_n = (\sum_{k=1}^{n} x_k)/n$ is also increasing.

3-3.P5. Let $x_{n+1} = \sqrt{(k+x_n)}$, where $k > 0$, $x_1 > 0$, and let α be the (only) positive root of the quadratic equation $x^2 = x + k$ [see Remark 1-6.10]. If $x_1 > \alpha$, show that $\{x_n\}$ is a decreasing sequence and find its limit. Determine what happens if $x_1 = \alpha$ and if $0 \le x_1 < \alpha$.

3-3.P6. Let $0 < k < \frac{1}{4}$, so that the quadratic equation $x^2 - x + k = 0$ has distinct positive roots a and b. Let $0 < a < b$. Suppose the sequence $\{x_n\}$ satisfies $a < x_1 < b$ and $x_{n+1} = x_n^2 + k$. Show that $\{x_n\}$ is a decreasing sequence and find its limit.

3-3.P7. (a) Let $0 < a \le b$ and $\{x_n\}$ be a sequence such that $x_1 = a + b$ and $x_{n+1} = a + b - ab/x_n$. Show that $a \le x_{n+1} \le x_n \le b$ for all n and that $\lim x_n = b$. In case $a < b$, show that $x_n = (a^{n+1} - b^{n+1})/(a^n - b^n)$.
(b) What is $\lim (p^{n+1} - t^{n+1})/(p^n - t^n)$ when p and t are both positive and unequal?

3-4 Subsequences, lim sup and lim inf

The sequence $\{x_n\}$ given by $x_n = (1+(-1)^n)/2$, or what may be described as $0,1,0,1,\ldots$, has no limit. But if we select only the terms in odd position ("odd" terms, for short), the resulting sequence $0,0,0,\ldots$ certainly converges. The selection can be described as $y_k = x_{2k-1}$. Similarly, if in the sequence $1,-1,1/2,2,-2,2/3,3,-3,3/4,\ldots$, we select every third term starting from 1/2, i.e. $y_k = x_{3k}$, the resulting sequence $y_k = k/(k+1)$ converges. A sequence obtained by selecting terms from a sequence is called a *subsequence* of the latter. The understanding is that the selected terms must appear in the same order as in the original. Note that our first selection was via the function $\phi:\mathbb{N}\to\mathbb{N}$ given by $\phi(k) = 2k-1$; in our second selection, we had $\phi(k) = 3k$. The way in which ϕ was used for making the selection in each case was that $y_k = x_{\phi(k)}$. What ensures that the selected terms appear in the same order as in the original is that ϕ has the property that $j > k \Rightarrow \phi(j) > \phi(k)$.

3-4.1. Definition. *If $\{x_n\}$ is a sequence in any set and the sequence $\phi:\mathbb{N}\to\mathbb{N}$ has the property that $j > k \Rightarrow \phi(j) > \phi(k)$, then the sequence $\{y_k\}$ for which $y_k = x_{\phi(k)}$ is called a* **subsequence of** *$\{x_n\}$. A* **subsequential limit of** *a complex sequence is the limit of any subsequence of it.*

By way of illustration, consider the sequence given by $x_n = n^{(-1)^n}$, or 1, 2, 1/3, 4, 1/5, 6, 1/7, …. If we take $\phi(k) = 2k-1$, the resulting subsequence is $y_k = 1/(2k-1)$. This sequence has limit 0 and therefore 0 is a subsequential limit of $\{x_n\}$. By choosing ϕ differently, we can construct other subsequences, such as

$$y_k = x_{3k} : 1/3, 6, 1/9, 12, 1/15, \ldots \quad \text{or} \quad z_k = x_{5k} : 1/5, 10, 1/15, 20, 1/25, \ldots.$$

They will not necessarily have a limit. The reader is invited to try constructing a subsequence of $n^{(-1)^n}$ which converges to a real number different from 0.

The sequence $(-1)^n$ has subsequences with limit –1 and also others with limit 1. These two numbers are subsequential limits and it is easy to see that there are no others.

The function ϕ used for generating a subsequence need not be given by a simple formula as in our illustrations. When a sequence is denoted by x_n, it is often convenient to denote the expression $\phi(k)$ by n_k or by $n(k)$, and we shall do so henceforth. With this notation, the subsequence is written as $\{x_{n_k}\}$ or $\{x_{n(k)}\}$. Note that, as a consequence of the property $j > k \Rightarrow n_j > n_k$, we have $n_k \geq k$ for all k [see 1-8.P4].

3-4.2. Proposition. *If the sequence* $\{x_n\}$ *has a limit* x, *then any subsequence of it also has limit* x.

Proof. Let $\{x_{n_k}\}$ be a subsequence and ε be any number greater than 0. Since $\lim_{n\to\infty} x_n = x$, there exists a natural number n_0 such that $|x_n - x| < \varepsilon$ whenever $n \geq n_0$. In view of the inequality $n_k \geq k$, it is true for any $k \geq n_0$ that we have $n_k \geq k \geq n_0$, and hence that $|x_{n_k} - x| < \varepsilon$. Thus, for any positive ε, there exists a natural number n_0 such that $|x_{n_k} - x| < \varepsilon$ whenever $n_k \geq n_0$. This means that the subsequence $\{x_{n_k}\}$ has limit x. ☺

We shall soon show that a bounded sequence in $\mathbb{R}$ always has at least one subsequential limit. In other words, that it has at least one subsequence which has a limit. Before doing so, we introduce two related concepts. For any bounded sequence $\{x_n\}$, consider the subsequences

$$x_1, x_2, x_3, \ldots, \qquad x_2, x_3, x_4, \ldots, \qquad x_3, x_4, x_5, \ldots,$$

and so on. This a sequence of subsequences consisting of the first term onwards, second term onwards, ..., n^{th} term onwards, and so forth. Each is bounded because $\{x_n\}$ is assumed bounded. Therefore each has a finite supremum. Moreover, the range of each subsequence contains that of the next and therefore the sup of each subsequence is greater than or equal to that of the next [see 1-6.P10]. Denote their sups by $B_1, B_2, \ldots$. Then $\{B_k\}$ is a decreasing sequence, which must be bounded below by any lower bound of the original sequence $\{x_n\}$. Therefore by Theorem 3-3.3, $\{B_k\}$ converges to its inf. We are going to prove that this limit is not only a subsequential limit of $\{x_n\}$, but also that it is the largest among them.

3-4.3. Definition. *For a bounded real sequence* $\{x_n\}$ *the limit of the* (*decreasing*) *sequence*

$$B_k = \sup\{x_k, x_{k+1}, x_{k+2}, \ldots\},$$

which is also $\inf\{B_k : k \in \mathbb{N}\}$, *is called the* **limit superior of** $\{x_n\}$. *It is denoted by* **lim sup** $\boldsymbol{x_n}$ *or also by* $\boldsymbol{\limsup_{n\to\infty} x_n}$. *Similarly, the limit of the* (*increasing*) *sequence*

$$A_k = \inf\{x_k, x_{k+1}, x_{k+2}, \ldots\},$$

which is also $\sup\{A_k : k \in \mathbb{N}\}$, *is called the* **limit inferior of** $\{x_n\}$. *It is denoted by* **lim inf** $\boldsymbol{x_n}$ *or also by* $\boldsymbol{\liminf_{n\to\infty} x_n}$.

Remark. Since A_k and B_k are the infimum and supremum of the same set $\{x_k, x_{k+1}, x_{k+2}, \ldots\}$, we have $A_k \leq B_k$ and therefore by Prop.3-2.9, $\lim A_k \leq \lim B_k$, which shows that

$$\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n \text{ for any } \textit{bounded} \text{ real sequence } \{x_n\}.$$

Also, if $\{x_n\}$ and $\{y_n\}$ are bounded real sequences such that $x_n \le y_n$ for every n, then $\liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} y_n$ and $\limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} y_n$.

Examples. (a) For the sequence $\{x_n\}$, wherein $x_n = (1+(-1)^n)/2$, i.e. $0,1,0,1,\dots$, we have $\sup\{x_n, x_{n+1}, x_{n+2}, \dots\} = 1$ for every n and therefore $\limsup_{n\to\infty} x_n = 1$. Similarly, $\liminf_{n\to\infty} x_n = 0$.

(b) When $x_n = n - 3\left[\frac{n}{3}\right]$, which is to say, $1,2,0,1,2,0,1,2,0,\dots$, we have $\sup\{x_n, x_{n+1}, x_{n+2}, \dots\} = 2$ for every n and therefore $\limsup_{n\to\infty} x_n = 2$. Similarly, $\liminf_{n\to\infty} x_n = 0$.

(c) When $x_n = (-1)^n/n$, we have $\sup\{x_n, x_{n+1}, x_{n+2}, \dots\} = 1/(n+1)$ if n is odd and $1/n$ if n is even. The first six terms of the sequence denoted by B_k in Def.3-4.3 are 1/2, 1/2, 1/4, 1/4, 1/6, 1/6. One can see that $\limsup_{n\to\infty} x_n = 0$. As for the limit inferior, $\inf\{x_n, x_{n+1}, x_{n+2}, \dots\} = -1/n$ if n is odd and $-1/(n+1)$ if n is even. The first six terms of the sequence denoted by A_k in Def.3-4.3 are -1, $-1/3$, $-1/3$, $-1/5$, $-1/5$, $-1/7$. Thus the limit inferior is also 0.

3-4.4. Proposition. *Let the sequence* $\{x_n\}$ *be bounded. Then the number* $A = \liminf\{x_n\}$ *is the largest one with the property that*:

$$\textit{For any } \varepsilon > 0, \textit{ there exists } n_0 \textit{ such that } x_n > A - \varepsilon \textit{ whenever } n \ge n_0. \tag{1}$$

Similarly, the number $B = \limsup\{x_n\}$ *is the smallest one with the property that*:

$$\textit{For any } \varepsilon > 0, \textit{ there exists } n_0 \textit{ such that } x_n < B + \varepsilon \textit{ whenever } n \ge n_0.$$

Proof. Let $A_k = \inf\{x_k, x_{k+1}, x_{k+2}, \dots\}$. Then $A = \sup A_k$ by definition. Now consider any $\varepsilon > 0$. Since $A - \varepsilon$ is not an upper bound of $\{A_k\}$, therefore there exists some n_0 such that $A - \varepsilon < A_{n_0} = \inf\{x_{n_0}, x_{n_0+1}, x_{n_0+2}, \dots\}$. Therefore $x_n > A - \varepsilon$ whenever $n \ge n_0$.

Suppose A' has property (1). Then for any $\varepsilon > 0$, there exists n_0 such that $A' - \varepsilon \le \inf\{x_{n_0}, x_{n_0+1}, x_{n_0+2}, \dots\} \le \sup A_k = A$. Since this has been proved to be true for any $\varepsilon > 0$, it follows that $A' \le A$.

The proof of the corresponding statement regarding the limit superior is similar. ☺

3-4.5. Proposition. *The limit inferior and limit superior of a bounded real sequence are equal if, and only if, the sequence has a limit, in which case, the limit is equal to the limits superior and inferior.*

Proof. Suppose $\{x_n\}$ is a bounded sequence. Let $A = \liminf_{n\to\infty} x_n$ and $B = \limsup_{n\to\infty} x_n$. First assume $A = B$ and consider any $\varepsilon > 0$. By Prop.3-4.4, there exist n_0 and n_1 such that $x_n > A - \varepsilon$ whenever $n \ge n_0$ and also $x_n < B + \varepsilon$ whenever $n \ge n_1$. For $n \ge$

max $\{n_0, n_1\}$, we have $A - \varepsilon < x_n < B + \varepsilon$. Since $A = B$, this implies that $\lim_{n\to\infty} x_n$ exists and equals A and (of course) B.

Next, suppose that $\lim_{n\to\infty} x_n = x$. In order to show that $A = B = x$, consider any $\varepsilon > 0$. By definition of limit, there exists n_0 such that $x - \varepsilon < x_n < x + \varepsilon$ whenever $n \geq n_0$. In particular, the number x has the following property: for any $\varepsilon > 0$, there exists n_0 such that $x - \varepsilon < x_n$ whenever $n \geq n_0$. Now, according to Prop.3-4.4, A is the largest number having this property. It follows that $x \leq A$. By a similar argument, it also follows that $B \leq x$. But $A \leq B$ [see Remark following Def. 3-4.3]. Therefore $A = B = x$. ☺

Examples. (a) Let $\{x_n\}$ be a sequence of nonzero complex numbers such that $\lim_{n\to\infty} |x_{n+1}/x_n| = \beta$. Then we can show that $\lim_{n\to\infty} |x_n|^{1/n}$ exists and equals β. Let $\varepsilon > 0$ be given. There exists a positive integer n_0 such that

$$\beta - \varepsilon < |x_{n+1}/x_n| < \beta + \varepsilon \text{ for } n \geq n_0. \tag{1}$$

First suppose that $\beta > 0$; then we may assume that $\varepsilon < \beta$ by replacing ε with min $\{\varepsilon, \beta/2\}$. Then $0 < \beta - \varepsilon$. Putting $n = n_0 + 1, n_0 + 2, \ldots, m - 1$ in (1) and multiplying the resulting inequalities, we get

$$(\beta - \varepsilon)^{m-1-n_0} < |x_m/x_{n_0+1}| < (\beta + \varepsilon)^{m-1-n_0} \quad \text{for } m \geq n_0+2$$

or
$$(\beta - \varepsilon)^{m-1-n_0}|x_{n_0+1}| < |x_m| < (\beta + \varepsilon)^{m-1-n_0}|x_{n_0+1}| \quad \text{for } m \geq n_0+2,$$

which can be expressed as

$$(\beta - \varepsilon)^{1-(n_0+1)/m}|x_{n_0+1}|^{1/m} < |x_m|^{1/m} < (\beta + \varepsilon)^{1-(n_0+1)/m}|x_{n_0+1}|^{1/m} \quad \text{for } m \geq n_0+2. \tag{2}$$

Now, $\lim_{m\to\infty} (\beta - \varepsilon)^{1-(n_0+1)/m}|x_{n_0+1}|^{1/m} = \beta - \varepsilon$ and $\lim_{m\to\infty} (\beta + \varepsilon)^{1-(n_0+1)/m}|x_{n_0+1}|^{1/m} = \beta + \varepsilon$ by Prop.3-1.5 and Theorem 3-2.3. Therefore these two sequences are bounded and hence it follows from (2) that $\{|x_m|^{1/m}\}$ is also bounded. From Prop.3-4.5 and the Remark following Def.3-4.3, it further follows that

$$\beta - \varepsilon \leq \liminf_{m\to\infty} |x_m|^{1/m} \leq \limsup_{m\to\infty} |x_m|^{1/m} \leq \beta + \varepsilon.$$

Since this is true for an arbitrary positive ε, it follows that $\liminf_{m\to\infty} |x_m|^{1/m} = \limsup_{m\to\infty} |x_m|^{1/m} = \beta$. By Prop.3-4.5, $\lim_{m\to\infty} |x_m|^{1/m}$ exists and equals β. This establishes our claim when $\beta > 0$. In the case when $\beta = 0$, we modify the above argument by replacing $\beta - \varepsilon$ by 0 everywhere.

(b) This example is a concrete illustration of the preceding one. Let $\{x_n\}$ be given by

$$x_n = n^n/(n+1)(n+2)\cdots(n+n).$$

Then $$\frac{x_{n+1}}{x_n} = \frac{(n+1)^{n+1}}{(n+2)\cdots(2n)(2n+1)(2n+2)} \cdot \frac{(n+1)(n+2)\cdots(n+n)}{n^n}$$

$$= \frac{(1+\frac{1}{n})^n(n+1)^2}{(2n+1)(2n+2)} = \frac{(1+\frac{1}{n})^n(1+\frac{1}{n})^2}{(2+\frac{1}{n})(2+\frac{2}{n})}.$$

As shown in Example 3-3.4, the sequence $\{(1+\frac{1}{n})^n\}$ has a limit, which we shall denote by e. Then we find from the preceding equality that $|x_{n+1}/x_n|$ has limit $e/4$. By the result of the preceding Example, we conclude that $\lim_{n\to\infty}|x_n|^{1/n}$ exists and equals $e/4$.

3-4.6. Proposition. *If* $\{x_n\}$ *is a bounded sequence, then* $\limsup x_n$ *is the largest subsequential limit and* $\liminf x_n$ *the smallest.*

Proof. We first show that there is a subsequence having $B = \limsup x_n$ as its limit. In order to construct such a subsequence, we begin by showing the following:

For every $m \in \mathbb{N}$ and every $\varepsilon > 0$, there exists $n > m$ for which $B-\varepsilon < x_n < B+\varepsilon$. (1)

Consider any $\varepsilon > 0$ and any $m \in \mathbb{N}$. By Prop.3-4.4, on the one hand B has the property that there exists some n_0 such that $x_n < B+\varepsilon$ whenever $n \geq n_0$. On the other hand, $B-\varepsilon$ does not have such a property and therefore there exists some $\eta > 0$ and $n > \max\{n_0, m\}$ such that $x_n \geq (B-\varepsilon)+\eta > B-\varepsilon$. The integer n must then satisfy $n > m$ as well as $B-\varepsilon < x_n < B+\varepsilon$. This establishes (1).

The existence of a subsequence converging to B is now proved as follows.

Let $m = 1$ and $\varepsilon = 1/2$ in (1). Then there exists $n_1 > 1$ such that $B - 1/2 < x_{n_1} < B + 1/2$. Next let $m = n_1$ and $\varepsilon = 1/2^2$ in (1). Then there exists $n_2 > n_1$ such that $B - 1/2^2 < x_{n_2} < B + 1/2^2$. Continuing in this manner, we can inductively define a sequence n_k of positive integers such that $n_{k+1} > n_k$ and $B - 1/2^k < x_{n_k} < B + 1/2^k$, for each k. [If such n_j are defined for $1 \leq j \leq k$, then take $m = n_k$ and $\varepsilon = 1/2^k$ in (1) to get n_{k+1}.] Since $1/2^k \to 0$ by Prop.3-1.7, it follows that the subsequence $\{x_{n_k}\}$ converges to B. Thus $\limsup x_n$ is a subsequential limit. We go on to show that it is the largest.

Let $\{x_{n_k}\}$ be any convergent subsequence and ε be any positive number. By Prop.3-4.4, there exists some n_0 such that $x_n < B+\varepsilon$ whenever $n \geq n_0$. For $k \geq n_0$, we have $n_k \geq k \geq n_0$, so that $x_{n_k} < B+\varepsilon$. By Prop.3-2.9, $\lim x_{n_k} \leq B+\varepsilon$. Since this holds for any positive ε whatsoever, we have $\lim x_{n_k} \leq B$, as claimed.

The analogous statement regarding the limit inferior has a similar proof and is left to the reader [see 3-4.P7]. ☺

3-4.7. Bolzano-Weierstrass Theorem. *A bounded complex sequence has a subsequence that converges.*

Proof. A bounded real sequence has a limit superior and, by Prop.3-4.6, there is a subsequence converging to the limit superior. Therefore there is some subsequence that converges. The case of a complex sequence is left to the reader [see 3-4.P8]. ☺

For a proof of the Bolzano-Weierstrass Theorem without using lim sup and lim inf, the interested reader is referred to [7] and [8].

Problem Set 3-4

3-4.P1. (a) For the sequence $x_n = (1+(-1)^n)/n$, what are the first four terms of the subsequence $x_{\phi(k)} = x_{5k}$?

(b) Let $x_n = n - 3\left[\frac{n}{3}\right]$, where $[\ ]$ denotes the integer part function. Write down the first five terms of the sequence obtained by composing $\{x_n\}$ with $\phi(k) = 3k+1$, i.e. the fourth, seventh, tenth, thirteenth and the sixteenth terms of $\{x_n\}$ in order.

(c) Does the sequence $x_n = \left[\frac{n}{2}\right] + ((1+(-1)^{n+1})/2$ have a subsequence with first six terms

(i) $1, 3, 3, 4, 8, 9$?

(ii) $10, 11, 11, 11, 12, 12$?

(d) Give an example of a sequence having precisely three subsequential limits, but in which no term repeats.

3-4.P2. Let $\{s_k\}$ and $\{t_k\}$ be the subsequences of a sequence $\{x_n\}$ such that $s_k = x_{2k}$ and $t_k = x_{2k-1}$. If $\lim_{k\to\infty} s_k = \lim_{k\to\infty} t_k = a$, show that $\lim_{n\to\infty} x_n = a$. [If the subsequences of even terms and of odd terms have a common limit, then the sequence itself has that limit.]

3-4.P3. (a) Let $\{z_n\}$ be a sequence and z a number such that the sequence does *not* converge to z. Show that there exists a subsequence $\{z_{n(k)}\}$ and a number $\eta > 0$ such that

$$|z_{n(k)} - z| > \eta \text{ for every } k \in \mathbb{N}.$$

(b) Let $\{z_n\}$ be a sequence and z a number such that every subsequence of $\{z_n\}$ itself has a subsequence converging to z. Show that $\{z_n\}$ converges to z.

3-4.P4. (a) For bounded sequences $\{x_n\}$ and $\{y_n\}$, show that $\liminf_{n\to\infty}(x_n+y_n) \geq \liminf_{n\to\infty} x_n + \liminf_{n\to\infty} y_n$, and give an example of bounded sequences $\{x_n\}$ and $\{y_n\}$ for which $\liminf_{n\to\infty}(x_n+y_n) > \liminf_{n\to\infty} x_n + \liminf_{n\to\infty} y_n$.

(b) For bounded sequences $\{x_n\}$ and $\{y_n\}$, show that $\limsup_{m\to\infty}(x_n+y_n) \leq \limsup_{m\to\infty} x_n + \limsup_{m\to\infty} y_n$, and give an example of bounded sequences $\{x_n\}$ and $\{y_n\}$ for which $\limsup_{m\to\infty}(x_n+y_n) < \limsup_{m\to\infty} x_n + \limsup_{m\to\infty} y_n$.

3-4.P5. For a bounded sequence $\{x_n\}$, if $\liminf_{n\to\infty} x_{2n} = 1$ and $\liminf_{n\to\infty} x_{2n-1} = 2$, show that $\liminf_{n\to\infty} x_n = 1$.

3-4.P6. (a) Does there exist a bounded sequence $\{x_n\}$ such that $\liminf_{n\to\infty} x_{2n} = 6$, $\liminf_{n\to\infty} x_{2n-1} = 7$ and $\liminf_{n\to\infty} x_{3n} = 8$?
(b) Does there exist a bounded sequence $\{x_n\}$ such that $\liminf_{n\to\infty} x_{2n} = 6$, $\liminf_{n\to\infty} x_{2n-1} = 7$ and $\liminf_{n\to\infty} x_{3n} = 5$?

3-4.P7. If $\{x_n\}$ is a bounded sequence, show that $\liminf x_n$ is the smallest subsequential limit.

3-4.P8. Prove the Bolzano-Weierstrass Theorem for a complex sequence.

3-5 Cauchy Criterion

Convergence of any complex sequence $\{x_n\}$ to a limit x means, roughly speaking, that the terms of the sequence get as close to x as one may wish and stay that close, provided one goes far enough along the sequence. It seems reasonable that when all terms stay close to x, they stay close to each other as well. In more precise terminology, this says that

$$|x_n - x| < \varepsilon \text{ and } |x_m - x| < \varepsilon \Rightarrow |x_n - x_m| < 2\varepsilon,$$

where ε is a measure of closeness between the terms of the sequence and x. This statement is easily seen to be true by using the triangle inequality. The assertion that $|x_n - x_m| < 2\varepsilon$ makes no reference to the limit, but is a consequence of the fact that it exists. We proceed with this background in mind.

3-5.1. Definition. *A sequence $\{x_n\}$ is called a* **Cauchy sequence** *if, for every $\varepsilon > 0$, there exists a natural number n_0 such that*

$$|x_n - x_m| < \varepsilon \quad \textit{whenever} \quad m \geq n_0 \textit{ and } n \geq n_0.$$

3-5.2. Theorem. Cauchy criterion of convergence. *A sequence $\{x_n\}$ is convergent if, and only if, it is a Cauchy sequence.*

Proof. First suppose that $\{x_n\}$ is convergent. With a view to showing that it is Cauchy, consider any $\varepsilon > 0$. There must exist some natural number n_0 such that $|x_n - x| < \frac{\varepsilon}{2}$ whenever $n \geq n_0$. Let m and n be any natural numbers such that $m \geq n_0$ and $n \geq n_0$. It follows by the triangle inequality that $|x_n - x_m| \leq |x_n - x| + |x_m - x| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$. Thus $\{x_n\}$ is a Cauchy sequence.

For the converse, suppose $\{x_n\}$ is any Cauchy sequence. Applying the Cauchy property with $\varepsilon = 1$, we find that there exists some n_0 such that $|x_n - x_m| <$

1 whenever $m \geq n_0$ and $n \geq n_0$. In particular, $|x_{n_0} - x_m| < 1$ whenever $m \geq n_0$. By the triangle inequality, $|x_m| \leq |x_{n_0} - x_m| + |x_{n_0}| \leq 1 + |x_{n_0}|$ whenever $m \geq n_0$. Let

$$M = \max\{|x_1|, |x_2|, \ldots, |x_{n_0}|, 1 + |x_{n_0}|\}.$$

Then we have $|x_m| \leq M$ not only for $1 \leq m \leq n_0$, but also for $m > n_0$. Thus $\{x_n\}$ is bounded. Now by the Bolzano-Weierstrass Theorem [see 3-4.7], there exists a subsequence $\{x_{n(k)}\}$ with some limit x. We shall show that x is also the limit of $\{x_n\}$. To this end, consider any $\varepsilon > 0$. Since $\{x_n\}$ is a Cauchy sequence, there exists a natural number k_1 such that

$$|x_n - x_m| < \frac{\varepsilon}{2} \text{ whenever } m \geq k_1 \text{ and } n \geq k_1. \tag{1}$$

Since the subsequence $\{x_{n(k)}\}$ converges to x, there exists a natural number k_2 such that

$$|x_{n(k)} - x| < \frac{\varepsilon}{2} \text{ whenever } k \geq k_2. \tag{2}$$

Now let m be any positive integer such that $m \geq n_1 = \max\{k_1, k_2\}$. Then $m \geq k_1$ and also $n(m) \geq m \geq k_1$. Therefore by (1), $|x_m - x_{n(m)}| < \frac{\varepsilon}{2}$. Moreover $n(m) \geq m \geq k_2$ and therefore by (2), $|x_{n(m)} - x| < \frac{\varepsilon}{2}$. From these two inequalities and the triangle inequality, it follows that

$$|x_m - x| \leq |x_m - x_{n(m)}| + |x_{n(m)} - x| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

This inequality has been shown to hold whenever $m \geq n_1 = \max\{k_1, k_2\}$. Therefore by the definition of convergence, the sequence $\{x_n\}$ converges to x. ☺

Remark. The Cauchy criterion of convergence and Theorem 3-3.3 both enable us to infer the existence of a limit without even having any information about what that limit may be, e.g. $\{(1 + \frac{1}{n})^n\}$ as discussed in Example 3-3.4.

3-5.3. Examples. (a) Consider the sequence $\{x_n\}$ defined by $x_1 = 1$, $x_2 = 2$ and $x_n = \frac{1}{2}(x_{n-2} + x_{n-1})$ for $n \geq 3$. We shall show that $\{x_n\}$ is a Cauchy sequence in $\mathbb{R}$. Observe first of all that $|x_n - x_{n+1}| = |\frac{1}{2}(x_{n-1} + x_n) - x_n| = \frac{1}{2}|x_{n-1} - x_n|$. It follows by induction that

$$|x_n - x_{n+1}| = (\tfrac{1}{2})^{n-1}|x_1 - x_2| = (\tfrac{1}{2})^{n-1} \text{ for every } n \in \mathbb{N}.$$

Thus, if $n > m$, we may apply the triangle inequality to obtain

$$\begin{aligned}|x_m - x_n| &\leq |x_m - x_{m+1}| + |x_{m+1} - x_{m+2}| + \cdots + |x_{n-1} - x_n| \\ &= (\tfrac{1}{2})^{m-1} + (\tfrac{1}{2})^m + \cdots + (\tfrac{1}{2})^{n-2} \\ &\leq (\tfrac{1}{2})^{m-1}\left(1 + \tfrac{1}{2} + \cdots + (\tfrac{1}{2})^{n-m-1}\right) \\ &= (\tfrac{1}{2})^{m-1}\left((1 - (\tfrac{1}{2})^{n-m})/(1 - \tfrac{1}{2})\right)\end{aligned}$$

$$\leq (\tfrac{1}{2})^{m-1}\big(1/(1-\tfrac{1}{2})\big) = (\tfrac{1}{2})^{m-1}\big(1/(\tfrac{1}{2})\big) = (\tfrac{1}{2})^{m-2}.$$

Now consider any $\varepsilon > 0$. Choose $n_0 \in \mathbb{N}$ such that $2^{n_0} > \frac{4}{\varepsilon}$; this is possible because $2^{n_0} = (1+1)^{n_0} \geq 1 + n_0$ by 1-11.P10 and we may choose $n_0 > \frac{4}{\varepsilon} - 1$. Then for $n > m \geq n_0$, it follows that $|x_m - x_n| < \varepsilon$. By the Cauchy criterion, the sequence has a limit. Thus the limit has been shown to exist without making any effort to find what it is. [But see 3-5.P1.]

The use of the triangle inequality followed by the formula for the sum $\sum_{k=1}^{n} a^{k-1}$ is a common feature of a very large number of applications of the Cauchy criterion.

(b) Let $\{x_n\}$ be the sequence such that $x_n = \sum_{k=1}^{n} (-1)^{k-1}/(k-1)!$ For $k \geq 3$, we have $(k-1)! \geq 2^{k-2}$ [see 1-11.P2(a)]. Therefore, for $n > m \geq 3$, we have

$$|x_n - x_m| \leq \sum_{k=m+1}^{n} 1/(k-1)! \leq \sum_{k=m+1}^{n} 1/2^{k-2} < 1/2^{m-2}.$$

Now proceed as in (a). It should be noted that, if the factor $(-1)^{k-1}$ were to be removed from each term, which amounts to replacing each term by its absolute value, the resulting sequence would equally well be convergent.

(c) In this example, we illustrate the use of the Cauchy criterion for showing that a sequence does *not* converge. Let $\{x_n\}$ be the sequence given by $x_n = 1 + 1/2 + 1/3 + \cdots + 1/n$. Then we have $x_n - x_m = 1/(m+1) + 1/(m+2) + \cdots + 1/n$ for $n > m$. In particular, when $n = 2m$, we have $x_{2m} - x_m = 1/(m+1) + \cdots + 1/2m \geq 1/2m + \cdots + 1/2m = m/2m = 1/2$. This shows that, for $\varepsilon = 1/2$, there can be no positive integer n_0 such that $|x_m - x_n| < \varepsilon$ whenever m and n are both greater than or equal to n_0, as we would then have $|x_{2n_0} - x_{n_0}| < \varepsilon$. Thus $\{x_n\}$ is not a Cauchy sequence and therefore cannot converge. Since it is an increasing sequence, according to Theorem 3-3.3, it would have to converge if it were bounded above. Therefore it is not bounded above.

(d) Now an example of the use of the Cauchy criterion to prove convergence. Let $x_n = 1 + 1/2^2 + 1/3^2 + \cdots + 1/n^2$. For $k > 1$, we have $k^2 \geq k(k-1) > 0$ and hence $0 < 1/k^2 \leq 1/k(k-1) = 1/(k-1) - 1/k$. Therefore, for $n > m$, we have

$$x_n - x_m = \sum_{k=m+1}^{n} 1/k^2 \leq \sum_{k=m+1}^{n} [1/(k-1) - 1/k].$$

But $\sum_{k=m+1}^{n} [1/(k-1) - 1/k] = 1/m - 1/n$, as can be shown by induction on n. It follows that $0 < x_n - x_m < 1/m$. For a given $\varepsilon > 0$, choose $n_0 > 1/\varepsilon$. Then $n > m > n_0 \Rightarrow |x_m - x_n| < \varepsilon$.

The inequality $0 < x_n - x_m < 1/m$ also shows that $x_n < x_1 + 1 = 2$ and that $\{x_n\}$ is increasing. One can therefore conclude that the sequence is convergent by using

Theorem 3-3.3 instead of the Cauchy criterion. We go on to present an example of a sequence that is not monotone and there is no simple way to obtain an expression for the n^{th} term.

(e) Let $a > 0$ and the sequence $\{x_n\}$ be defined by $x_{n+1} = \frac{1}{2}(x_n + ax_n^{-2})$, with x_1 any positive number. Then $x_{n+1} - x_n = \frac{1}{2}(-x_n + ax_n^{-2}) = \frac{1}{2}(-x_n^3 + a)x_n^{-2}$, which shows that

$$x_{n+1} \lesseqgtr x_n \text{ according as } a \lesseqgtr x_n^3.$$

Now if $a = 1$ and $x_1 = \frac{1}{2}$, we get $x_2 = \frac{9}{4}$, which has cube greater than 1. Therefore by the preceding observation, $x_3 < x_2$. But we also have $x_2 > x_1$. Therefore the sequence is not monotone. We proceed to show that it is always a Cauchy sequence.

$$\begin{aligned} x_{n+1} - x_n &= \tfrac{1}{2}(x_n - x_{n-1}) + \tfrac{a}{2}(x_n^{-2} - x_{n-1}^{-2}) \\ &= \tfrac{1}{2}(x_n - x_{n-1}) - \tfrac{a}{2}(x_n^2 - x_{n-1}^2)x_{n-1}^{-2}x_n^{-2} \\ &= \tfrac{1}{2}(x_n - x_{n-1})\left[1 - a(x_n + x_{n-1})x_{n-1}^{-2}x_n^{-2}\right]. \end{aligned} \tag{1}$$

We shall show that

$$\left|1 - a(x_n + x_{n-1})x_{n-1}^{-2}x_n^{-2}\right| \le \tfrac{5}{4}. \tag{2}$$

Since a, x_n and x_{n-1} are positive, we already have

$$1 - a(x_n + x_{n-1})x_{n-1}^{-2}x_n^{-2} < 1 < \tfrac{5}{4}.$$

We need only show that $1 - a(x_n + x_{n-1})x_{n-1}^{-2}x_n^{-2} \ge -\frac{5}{4}$, i.e. $a(x_n + x_{n-1})x_{n-1}^{-2}x_n^{-2} \le \frac{9}{4}$. This is a consequence of the routinely verified fact that

$$4a(x + y) - 9x^2y^2 = -\tfrac{1}{4}(3x^2 - ax^{-1})^2 \quad \text{when} \quad y = \tfrac{1}{2}(x + ax^{-2}).$$

This establishes (2). It now follows from (1) that

$$|x_{n+1} - x_n| \le \tfrac{5}{8}|x_n - x_{n-1}|.$$

By induction, we have $|x_{n+1} - x_n| \le \left(\frac{5}{8}\right)^{n-1}|x_2 - x_1|$. Hence

$$\begin{aligned} n > m \Rightarrow |x_n - x_m| &\le \sum_{k=m}^{n-1}\left(\tfrac{5}{8}\right)^{k-1}|x_2 - x_1| \\ &\le |x_2 - x_1|\left(\tfrac{5}{8}\right)^{m-1}\Big/\left(1 - \tfrac{5}{8}\right) \le \tfrac{8}{3}|x_2 - x_1|\left(\tfrac{5}{8}\right)^{m-1}. \end{aligned}$$

As in the foregoing Example (a), it follows that $\{x_n\}$ is a Cauchy sequence. Once we know that the sequence does have a limit x, it is easy to obtain it as the solution of the equation

$$x = \tfrac{1}{2}(x + ax^{-2}),$$

which leads to $x = a^{1/3}$. It should be noted that this computation of the limit is valid only if we know that the limit exists in the first place.

Problem Set 3-5

3-5.P1. Show by induction that for any sequence $\{x_n\}$ such that $x_n = \frac{1}{2}(x_{n-2} + x_{n-1})$ when $n \geq 3$, we have

$$x_n = (x_2/3)(2 + (-\tfrac{1}{2})^{n-2}) + (x_1/3)(1 - (-\tfrac{1}{2})^{n-2}) \quad \text{when } n \geq 3.$$

Hence find the limit in terms of x_1 and x_2 without using the Cauchy criterion.

3-5.P2. Let $\{x_n\}$ be a Cauchy sequence and $\{x_{n(k)}\}$ be a subsequence having limit x. Show that $\{x_n\}$ has limit x.

3-5.P3. Give an example of a sequence $\{x_n\}$ for which the subsequences $\{x_{2n}\}$ and $\{x_{2n+1}\}$ are both Cauchy sequences, but $\{x_n\}$ is not a Cauchy sequence.

3-5.P4. Suppose α is a real number such that $0 < \alpha < 1$ and $\{z_k\}$ is a sequence in $\mathbb{C}$ such that $|z_{n+2} - z_{n+1}| \leq \alpha|z_{n+1} - z_n|$ for all $n \in \mathbb{N}$. Show that $\{z_k\}$ is a Cauchy sequence.

3-5.P5. Give an example of a sequence $\{z_n\}$ such that $|z_{n+2} - z_{n+1}| < |z_{n+1} - z_n|$ for all $n \in \mathbb{N}$ but is not a Cauchy sequence.

3-5.P6. Suppose $\{z_k\}$ is a sequence in $\mathbb{C}$ such that $|z_{k+2} - z_{k+1}| \leq \frac{3}{k}|z_{k+1} - z_k|$ for all $k \in \mathbb{N}$. Show that $\{z_k\}$ is a Cauchy sequence.

Chapter 4
Real and Complex Series

4-1 Convergence of Infinite Series

One comes across infinite series long before one has heard of them, for instance 0.3333… or 1.414213562…. These representations of numbers are *infinite series* in the sense that

$$0.3333\ldots = 3/10 + 3/10^2 + 3/10^3 + 3/10^4 + \cdots$$

and $$1.414213\ldots = 1 + 4/10 + 1/10^2 + 4/10^3 + 2/10^4 + 1/10^5 + 3/10^6 + \cdots.$$

When we terminate the decimal somewhere as a matter of practical necessity, we drop all the terms in the sum on the right side from a certain stage onwards; the resulting approximation is an example of what is called a *partial sum* of an infinite series.

4-1.1. Definition. *Given a sequence* $\{z_k\}$ *of complex numbers, the associated sequence* $s_n = \sum_{k=1}^{n} z_k$, $n \in \mathbb{N}$, *is called the sequence of* **partial sums of the series** $\sum_{k=1}^{\infty} z_k$. *We say that the* **series** $\sum_{k=1}^{\infty} z_k$ **converges** *to a complex number z if the sequence* $\sum_{k=1}^{n} z_k$ *of its partial sums converges to z. We refer to z as the* **sum of the series** $\sum_{k=1}^{\infty} z_k$ *and write this in symbols as* $\sum_{k=1}^{\infty} z_k = z$. *If the sequence of partial sums diverges, we say that the* **series** $\sum_{k=1}^{\infty} z_k$ **diverges**.

Remarks. (i) The symbol $\sum_{k=1}^{\infty} z_k$ often stands for the sum to which its sequence of partial sums converges. One has to judge from the context whether this is intended.

(ii) While speaking of an infinite series $\sum_{k=1}^{\infty} z_k$, the words "terms" refers to z_k and not the terms s_n of the sequence of partial sums. It is convenient to refer to z_n as the "n^{th} term" of the series and s_n as the "n^{th} partial sum".

(iii) Any given sequence $\{x_n\}$ is the sequence of partial sums of some series. Indeed, set $z_1 = x_1$ and $z_k = x_k - x_{k-1}$ for $k \geq 2$. Then the sequence of partial sums of the series $\sum_{k=1}^{\infty} z_k$ is precisely the sequence $\{x_n\}$ with which we started.

(iv) As usual, the name "complex numbers" includes real numbers and our definition by no means rules out the possibility that all terms of a series are real; when that is the case, the partial sums, and hence also the sum of the series (if any), will be real.

4-1.2. Examples. (a) Let $z_k = 3/10^k$. Then $\sum_{k=1}^{\infty} z_k$ is the series $3/10 + 3/10^2 + \cdots$, or what is denoted by .333… in elementary arithmetic. Its sequence of partial sums is given by $s_n = \frac{3}{10}(1-(\frac{1}{10})^n)/(1-\frac{1}{10})$, which has limit $\frac{1}{3}$ (of course).

(b) According to Prop.3-2.5, the limit of the sequence $\{x_n\}$ defined by $x_n = 1+z+z^2+\cdots+z^{n-1}$, where $|z| < 1$, is $1/(1-z)$. Since x_n is, by its very definition, the n^{th} partial sum of the series $\sum_{k=1}^{\infty} z^{k-1}$, it follows that the sum of this series (called geometric series with common ratio z) is $\frac{1}{1-z}$ when $|z| < 1$.

(c) According to Example 3-5.3(b), the sequence $\{x_n\}$ given by

$$x_n = \sum_{k=1}^{n} \frac{(-1)^{k-1}}{(k-1)!}$$

has a limit. Now x_n is, by its very definition, the n^{th} partial sum of the series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{(k-1)!}$$

and therefore this series is convergent.

(d) According to Example 3-5.3(c), the sequence $\{x_n\}$ given by

$$x_n = 1 + 1/2 + 1/3 + \cdots + 1/n$$

is divergent. Since x_n is the n^{th} partial sum of the series $\sum_{k=1}^{\infty}\left(\frac{1}{k}\right)$, it follows that this series is divergent. As noted there, the sequence $\{x_n\}$ is unbounded above.

(e) Consider the series $\sum_{k=1}^{\infty} k^{-1/2}$. Since $k^{-1/2} \geq \frac{1}{k}$, the partial sums of this series are greater than or equal to the corresponding partial sums of the series $\sum_{k=1}^{\infty}\left(\frac{1}{k}\right)$. Since the latter are unbounded above (as just observed in (d)), so are these. It follows by Prop.3-1.4 that these partial sums do not converge and consequently the series $\sum_{k=1}^{\infty} k^{-1/2}$ is divergent.

(f) It may be recalled that the inequality $(k-1)! \geq 2^{k-2}$ for $k \geq 3$ played a crucial role in Example 3-5.3(b). For the similar series $\sum_{k=1}^{\infty} \frac{2^{k-1}}{(k-1)!}$, one needs the different inequality $(k-1)! \geq 4^{k-3}$ for $k \geq 3$, because it shows that $\frac{2^{k-1}}{(k-1)!} \leq 1/2^{k-5}$ for $k \geq 3$. To implement the same idea with powers of some number other than 2, we would have to find yet another inequality. Methods will be developed later for more efficient ways of checking convergence. [See Section 4-3.]

4-1.3. Proposition. *If a series converges, then the* n^{th} *term has limit* 0. *In other words, if* $\sum_{k=1}^{\infty} z_k$ *converges, then* $\lim_{n\to\infty} z_n = 0$.

Proof. Since $\sum_{k=1}^{n} z_k = \sum_{k=1}^{n-1} z_k + z_n$, we have $z_n = s_n - s_{n-1}$ for $n > 1$. Since $\lim_{n\to\infty} s_n = \lim_{n\to\infty} s_{n-1}$, we can infer that $\lim_{n\to\infty} z_n = 0$. ☺

The converse of the preceding Proposition is false, as seen from Example 4-1.2(d).

4-1.4. Proposition. *Suppose* $\sum_{k=1}^{\infty} z_k$ *and* $\sum_{k=1}^{\infty} w_k$ *are convergent series with sum* z *and* w *respectively. Then* $\sum_{k=1}^{\infty} (z_k + w_k)$ *is a convergent series with sum* $z + w$. *If* α *is any complex number, then* $\sum_{k=1}^{\infty} (\alpha z_k)$ *is convergent with sum* αz. *A series* $\sum_{k=1}^{\infty} z_k$ *is convergent if, and only if, the series whose terms are respectively the real and imaginary parts of* z_k *are both convergent, in which case,* $\sum_{k=1}^{\infty} z_k = \sum_{k=1}^{\infty} \Re(z_k) + i \sum_{k=1}^{\infty} \Im(z_k)$.

Proof. Let the n^{th} partial sums of $\sum_{k=1}^{\infty} z_k$ and $\sum_{k=1}^{\infty} w_k$ be Z_n and W_n respectively. By the definition of the sum of a series $\lim_{n\to\infty} Z_n = z$ and $\lim_{n\to\infty} W_n = w$. Now the n^{th} partial sum of $\sum_{k=1}^{\infty} (z_k + w_k)$ is $Z_n + W_n$ [see Prop.1-11.4]. It follows by the corresponding result on sequences, Prop.3-2.3(a), that $\lim_{n\to\infty} (Z_n + W_n) = z + w$. Therefore $\sum_{k=1}^{\infty} (z_k + w_k) = z + w$. The rest is left to the reader in 4-1.P6. ☺

As an application of the preceding Proposition, we note that the series whose n^{th} term is the sum $\frac{3}{10^k} + \frac{2^{k-1}}{(k-1)!}$ is convergent in view of Examples 4-1.2(a) and (f).

4-1.5. Proposition. *A series of nonnegative terms is convergent if, and only if, the sequence of its partial sums is bounded.*

Proof. Let the n^{th} partial sum of the series $\sum_{k=1}^{\infty} a_k$, where every term a_k is nonnegative, be s_n. Then $s_n = s_{n-1} + a_n \geq s_{n-1}$ for $n > 1$. Therefore $\{s_n\}$ is an increasing sequence. By Theorem 3-3.3(a) and Prop.3-1.4, $\{s_n\}$ is convergent if, and only if, it is bounded. But the convergence of the sequence $\{s_n\}$ is equivalent to the convergence of the series $\sum_{k=1}^{\infty} a_k$ by definition. ☺

4-1.6. Theorem. Comparison Test. *Suppose the series* $\sum_{k=1}^{\infty} b_k$ *is convergent and* $\sum_{k=1}^{\infty} a_k$ *is a series such that* $0 \leq a_k \leq b_k$ *for every* $k \in \mathbb{N}$. *Then the series* $\sum_{k=1}^{\infty} a_k$ *is also convergent.*

Proof. Let $\{s_n\}$ be the sequence of partial sums of $\sum_{k=1}^{\infty} a_k$ and $\{t_n\}$ be that of $\sum_{k=1}^{\infty} b_k$. Then $0 \leq s_n = \sum_{k=1}^{n} a_k \leq \sum_{k=1}^{n} b_k = t_n$ and $\{t_n\}$ is bounded by Prop.4-1.5. Therefore $\{s_n\}$ is bounded and hence the series $\sum_{k=1}^{\infty} a_k$ is convergent by Prop.4-1.5. ☺

Examples. (a) In Example 4-1.2(f), the series $\sum_{k=1}^{\infty} \left(2^{k-1}/(k-1)!\right)$ was shown to be convergent. Since $3^{k-1}/2^{k-1}(k-1)! \leq 2^{k-1}/(k-1)!$, it follows by the Comparison Test (Theorem 4-1.6) that $\sum_{k=1}^{\infty} \left(3^{k-1}/2^{k-1}(k-1)!\right)$ is convergent.

(b) For the series $\sum_{k=1}^{\infty} \left(1/k - 1/(k+1)\right)$, the partial sum s_n can be easily computed as

$$s_n = (1-1/2) + (1/2-1/3) + \cdots + (1/n - 1/(n+1)) = 1 - 1/(n+1).$$

[This can be confirmed independently by an induction argument, so that one does not have to place faith in the cancellations that are shielded from public view by the three dots!] It follows that the series converges to the sum 1. Now, we have $1/k - 1/(k+1) = 1/k(k+1) \geq 1/(k+1)^2$. It follows by the Comparison Test (Theorem 4-1.6) that $\sum_{k=1}^{\infty} 1/(k+1)^2$ also converges. Since $1/(k+1)^3 \leq 1/(k+1)^2$, it further follows that the series $\sum_{k=1}^{\infty} 1/(k+1)^3$ also converges.

The next result concerns real series with first term nonnegative and the subsequent ones alternating in sign: $\sum_{k=1}^{\infty} (-1)^{k+1} a_k$ with each a_k a nonnegative (real) number. A corresponding result can be derived for the case when the first term is nonpositive, but will not be stated explicitly.

4-1.7. Leibnitz' Alternating Series Theorem (or **Test**). *If the terms of the series* $\sum_{k=1}^{\infty} (-1)^{k+1} a_k$ *satisfy the two conditions*

$$(1) \lim_{k\to\infty} a_k = 0 \qquad (2)\ 0 \leq a_{k+1} \leq a_k \text{ for all } k \in \mathbb{N},$$

then the series converges. Moreover, if s is the sum of the series and s_n its n^{th} partial sum, then

$$\text{(a) } 0 \leq s - s_{2n} \leq a_{2n+1} \text{ for all } n \in \mathbb{N} \quad \text{and} \quad \text{(b) } 0 \leq s_{2n-1} - s \leq a_{2n} \text{ for all } n \in \mathbb{N}.$$

Proof. We begin by showing that, for every $n \in \mathbb{N}$, we have

$$(3)\ s_{2n} \leq s_{2n+2}, \qquad (4)\ s_{2n} \leq s_{2n-1} \qquad \text{and} \qquad (5)\ s_{2n+1} \leq s_{2n-1}.$$

By (2), $a_{2n+1} - a_{2n+2} \geq 0$. Therefore

$$s_{2n+2} = s_{2n} + (-1)^{2n+2} a_{2n+1} + (-1)^{2n+3} a_{2n+2} = s_{2n} + a_{2n+1} - a_{2n+2} \geq s_{2n}.$$

This proves (3). By (2), $-a_{2n} \leq 0$. Therefore

$$s_{2n} = s_{2n-1} + (-1)^{2n+1} a_{2n} = s_{2n-1} - a_{2n} \leq s_{2n-1}.$$

This proves (4). Again by (2), $-a_{2n} + a_{2n+1} \leq 0$. Therefore

$$s_{2n+1} = s_{2n-1} + (-1)^{2n+1} a_{2n} + (-1)^{2n+2} a_{2n+1} = s_{2n-1} - a_{2n} + a_{2n+1} \leq s_{2n-1}.$$

This proves (5).

From (3) and (5), it follows that for every $n \in \mathbb{N}$, we have

$$s_{2n} \geq s_2 \tag{6}$$

and

$$s_{2n-1} \leq s_1. \tag{7}$$

Combining (7) with (4), we find further that

$$s_{2n} \leq s_1. \tag{8}$$

Similarly, combining (6) with (4), we find that

$$s_{2n-1} \geq s_2. \tag{9}$$

By (3) and (8), $\{s_{2n}\}$ is an increasing sequence bounded above; by (5) and (9), $\{s_{2n-1}\}$ is a decreasing sequence bounded below. By Theorem 3-3.3, both sequences converge. From (1), we have $\lim_{n\to\infty}(s_{2n}-s_{2n-1}) = \lim_{n\to\infty}(-a_{2n}) = 0$. Therefore $\lim_{n\to\infty} s_{2n} = \lim_{n\to\infty} s_{2n-1} = s$, say. It follows [see 3-4.P2] that $\lim_{n\to\infty} s_n = s$. Thus the series $\sum_{k=1}^{\infty}(-1)^{k+1}a_k$ is convergent.

Since s is the limit of the decreasing sequence $\{s_{2n-1}\}$ as well as of the increasing sequence $\{s_{2n}\}$, we have

$$s_{2n+1} \geq s \geq s_{2n}. \tag{10}$$

But $s_{2n+1} = s_{2n} + a_{2n+1}$ and $s_{2n} = s_{2n-1} - a_{2n}$. Therefore by (10), $s_{2n}+a_{2n+1} \geq s \geq s_{2n-1}-a_{2n}$. Using (10) once again, it follows from here that $0 \leq s - s_{2n} \leq a_{2n+1}$ and that $0 \leq s_{2n-1} - s \leq a_{2n}$. ☺

Remark. The inequality in (a) of the Theorem just proved can be restated as $s_{2n} \leq s \leq s_{2n+1}$ and the one in (b) as $s_{2n} \leq s \leq s_{2n-1}$. We shall often use this version.

4-1.8. Examples. The series $\sum_{k=1}^{\infty}(-1)^{k+1}k^{-1/2}$ and $\sum_{k=1}^{\infty}(-1)^{k+1}k^{-1}$ both satisfy the hypotheses of the foregoing Theorem and therefore converge. Of course, we can have any negative rational power of k instead of the powers $-1/2$ and -1 that we have specifically mentioned. After defining what irrational powers mean, it will turn out that even with a negative irrational power of k, such a series converges. According to Examples 4-1.2(d) and (e), the series $\sum_{k=1}^{\infty}k^{-1/2}$ and $\sum_{k=1}^{\infty}k^{-1}$, which are obtained by replacing every term by its absolute value in the series $\sum_{k=1}^{\infty}(-1)^{k+1}k^{-1/2}$ and $\sum_{k=1}^{\infty}(-1)^{k+1}k^{-1}$ respectively, are both divergent series.

Let s be the sum of $\sum_{k=1}^{\infty}(-1)^{k+1}k^{-1}$ and s_n be its n^{th} partial sum. Then $s_2 = 1-1/2 = 1/2$ and $s_3 = 1-1/2+1/3 = 5/6$. Since $s_2 \leq s \leq s_3$ by Theorem 4-1.7, therefore $1/2 \leq s \leq 5/6$ for this series.

We now turn our attention to "decimal" type representations of numbers between 0 and 1, but with an arbitrary integer $b > 1$ as the base, not necessarily 10.

4-1.9. Theorem. *Suppose* $x \in \mathbb{R}$, $b \in \mathbb{N}$ *where* $b > 1$ and $0 \leq x < 1$. *Then there exists an infinite sequence* $\{a_k : k \in \mathbb{N}\}$ *of integers such that*

(A) $x = a_1 b^{-1} + \cdots + a_k b^{-k} + \cdots = \sum_{k=1}^{\infty} a_k b^{-k}$,

(B) $0 \leq a_k \leq b-1$ *for all* $k \in \mathbb{N}$,

(C) *for every n, there exists $k > n$ such that $a_k < b-1$.*

Proof. When $x = 0$, there is nothing to prove and so we assume $x > 0$. Using the integer part function [ ], define the sequence $\{a_k : k \in \mathbb{N}\}$ inductively as

$$a_1 = [bx] \quad \text{and} \quad a_{n+1} = [b^{n+1}(x - \sum_{k=1}^{n} a_k b^{-k})].$$

We shall prove that

$$0 \le x - \sum_{k=1}^{n} a_k b^{-k} < b^{-n} \text{ for all } n \in \mathbb{N} \tag{1}$$

and

$$0 \le a_n < b \text{ for all } n \in \mathbb{N}. \tag{2}$$

These imply (A) and (B) respectively.

By definition of the integer part function, $a_1 \le bx < a_1 + 1$. Therefore $0 \le bx - a_1 < 1$ and hence $0 \le x - a_1 b^{-1} < b^{-1}$. Now assume that $0 \le x - \sum_{k=1}^{n} a_k b^{-k} < b^{-n}$ for some $n \in \mathbb{N}$. Since

$$a_{n+1} \le b^{n+1}(x - \sum_{k=1}^{n} a_k b^{-k}) < a_{n+1} + 1,$$

we have

$$0 \le b^{n+1}(x - \sum_{k=1}^{n} a_k b^{-k} - a_{n+1} b^{-n-1}) < 1,$$

so that

$$0 \le b^{n+1}(x - \sum_{k=1}^{n+1} a_k b^{-k}) < 1, \quad \text{or} \quad 0 \le x - \sum_{k=1}^{n+1} a_k b^{-k} < b^{-n-1}.$$

This completes the induction proof of (1), from which (A) follows immediately.

Since $0 < x < 1$, therefore $0 < bx < b$, so that $0 \le [bx] < b$, i.e. $0 \le a_1 < b$. Therefore (2) holds when $n = 1$. Multiplying (1) by b^{n+1}, we get $0 \le b^{n+1}(x - \sum_{k=1}^{n} a_k b^{-k}) < b$ and hence $0 \le a_{n+1} < b$. Therefore $0 \le a_n < b$ for $n \ge 2$, showing that the inequality in (2) holds for all positive integers n. This completes the proof of (2) and hence also (B).

To prove (C), suppose, if possible, that $n \in \mathbb{N}$ has the property that $a_k = b-1$ whenever $k > n$. Then

$$x = \sum_{k=1}^{n} a_k b^{-k} + \sum_{k=n+1}^{\infty} a_k b^{-k} = \sum_{k=1}^{n} a_k b^{-k} + \sum_{k=n+1}^{\infty} (b-1) b^{-k} = \sum_{k=1}^{n} a_k b^{-k} + b^{-n}.$$

Therefore $b^{n+1}(x - \sum_{k=1}^{n} a_k b^{-k}) = b$, an integer. Since a_{n+1} is the integer part of $b^{n+1}(x - \sum_{k=1}^{n} a_k b^{-k})$, it follows that $a_{n+1} = b$, contradicting (B) and thus proving (C). ☺

Remark. We do not claim that the representation of x as $\sum_{k=1}^{\infty} a_k b^{-k}$ is unique when only (A) and (B) of the preceding Theorem hold. It is a familiar situation when b is the usual base ten that 0.1999… and 0.2000… both represent $\frac{1}{5}$. Actually, this is the only kind of exception to uniqueness that can occur but we shall not need this fact. However, uniqueness does obtain when (C) also holds. This will be taken up later in Cor.8-4.3.

The formal argument that has just been presented should not obscure the fact that practical computations of decimal representation of fractions follow precisely the prescription given in the proof for obtaining the digits a_k. According to the prescription, the first two digits in the decimal representation of $\frac{517}{720}$ are to be computed as follows:

$a_1 = [\frac{10\times 517}{720}] = 7$ because $7\times 720 = 5040 < 10\times 517$ but $8\times 720 > 10\times 517$. So, the first digit after the decimal point is $a_1 = 7$.
$\frac{517}{720} - \frac{7}{10} = \frac{517-504}{720} = \frac{13}{720}$ and $[10^2 \times \frac{13}{720}] = [\frac{1300}{720}] = 1$ because $1\times 72 < 130 < 2\times 72$. So the next digit is $a_2 = 1$. Thus $\frac{517}{720} = 0.71\ldots$.

One can see that the very same computations are carried out in the traditional long division

```
720 ) 517 ( 0.71
      000
      5170
      5040
        1300
         720
         580
```

but they are arranged so compactly as to camouflage the connection with the prescription of Theorem 4-1.9.

The next digit a_3 would be obtained from

$$\frac{517}{720} - \frac{7}{10} - \frac{1}{100} = \frac{13}{720} - \frac{1}{100} = \frac{1300-720}{72000} = \frac{58}{7200}$$

as
$$a_3 = [10^3 \times \frac{580}{72000}] = [\frac{580}{72}],$$

which is 8, because $8\times 72 < 580 < 9\times 72$. Therefore, the long division should continue with 8 as the next digit of the quotient. Thus $\frac{517}{720} = 0.718\ldots > 0.718$.

Problem Set 4-1

4-1.P1. Show by the Comparison Test that the series $\sum_{k=1}^{\infty} 1/k^{3+(-1)^k} = 1 + 1/2^4 + 1/3^2 + 1/4^4 + 1/5^2 + \cdots$ converges. The convergence of the series $\sum_{k=1}^{\infty} 1/k^2$ may be taken for granted without proof.

4-1.P2. Show that the series $\sum_{k=1}^{\infty} (-1)^{k+1} k^{-2}$ is convergent and find the second and third partial sums s_2, s_3. Hence show that $3/4 \le \sum_{k=1}^{\infty} (-1)^{k+1} k^{-2} \le 31/36$.

4-1.P3. Show that the series $\sum_{k=1}^{\infty}(-1)^{k+1}(\sqrt{k+1}-\sqrt{k})$ converges and that its sum lies between $2\sqrt{2}-1-\sqrt{3}$ and $\sqrt{2}-1$.

4-1.P4. Show that the series $\sum_{k=}^{\infty}(-1)^{k-1}\dfrac{2^{2k-2}}{(2k-2)!}$ converges to s, where $s \leq -\frac{1}{3}$.

4-1.P5. Show that $\sum_{k=1}^{\infty}(-1)^{k+1}\left[k/(1+k^2)\right]$ converges to a sum s, where $\frac{1}{10} \leq s \leq \frac{2}{5}$.

4-1.P6. Prove the rest of Prop.4-1.4.

4-1.P7. Prove that $\sum_{k=1}^{\infty}(1+\frac{1}{k})^{-k^2}$ is convergent.

4-2 Cauchy Criterion and Absolute Convergence

So far, we have seen two general results which guarantee convergence of series, namely the Comparison Test and the Leibnitz Test. The former is applicable only to series of nonnegative terms and the latter only to series of terms with alternating signs. We now turn our attention to series that may not fall in either of these categories. The general results that we discuss will provide for convergence not only of the series concerned but also of the series obtained by replacing each term by its absolute value. Besides, they will be applicable to complex series as well. But first we need a theoretical tool.

4-2.1. Theorem. Cauchy criterion of convergence for series. *The series* $\sum_{k=1}^{\infty} z_k$ *is convergent if, and only if, for every* $\varepsilon > 0$, *there exists a natural number* n_0 *such that*

$$\left|\sum_{k=m+1}^{n} z_k\right| < \varepsilon \quad \textit{whenever} \quad n > m \geq n_0.$$

Proof. By definition, $\sum_{k=1}^{\infty} z_k$ converges if, and only if the sequence $\{s_n\}$ of its partial sums converges. By the Cauchy criterion for sequences, Theorem 3-5.2, this sequence converges if, and only if, for every $\varepsilon > 0$, there exists a natural number n_0 such that

$$|s_n - s_m| < \varepsilon \quad \text{whenever} \quad m \geq n_0 \text{ and } n \geq n_0.$$

This is always true if $n = m$. When $n \neq m$, one of them must be greater and the above condition is equivalent to

$$|s_n - s_m| < \varepsilon \quad \text{whenever} \quad n > m \geq n_0.$$

But by Prop.1-11.2, $s_n - s_m = \sum_{k=m+1}^{n} z_k$ when $n > m$. ☺

4-2.2. Definition. *A series* $\sum_{k=1}^{\infty} z_k$ *is said to be* **absolutely convergent** *if* $\sum_{k=1}^{\infty}|z_k|$ *is convergent. A series* $\sum_{k=1}^{\infty} z_k$ *is said to be* **conditionally convergent** *if it is convergent while* $\sum_{k=1}^{\infty}|z_k|$ *is divergent.*

As is clear from Example 4-1.8, the two series

$$\sum_{k=1}^{\infty}(-1)^{k+1}k^{-1/2} \quad \text{and} \quad \sum_{k=1}^{\infty}(-1)^{k+1}k^{-1}$$

are conditionally convergent; but the series $\sum_{k=1}^{\infty}(-1)^{k-1}/(k-1)!$ is absolutely convergent because $\sum_{k=1}^{\infty}1/(k-1)!$ is convergent, as observed in Example 3-5.3(b).

When z is a complex number with $|z| < 1$, the series $\sum_{k=1}^{\infty}|z^{k-1}|$ is convergent, as seen in Example 4-1.2(b). This says that the series $\sum_{k=1}^{\infty}z^{k-1}$ is absolutely convergent when $|z| < 1$.

Note that the definition of absolute convergence of $\sum_{k=1}^{\infty}z_k$ requires only that the related series $\sum_{k=1}^{\infty}|z_k|$ converge, without requiring that the original series $\sum_{k=1}^{\infty}z_k$ itself converge. However, the convergence of the original series is taken care of by the following result:

4-2.3. Proposition. *An absolutely convergent series is convergent.*

Proof. Suppose the series $\sum_{k=1}^{\infty}z_k$ is absolutely convergent. By definition, this means that the series $\sum_{k=1}^{\infty}|z_k|$ is convergent. By the Cauchy criterion [see Theorem 4-2.1], for every $\varepsilon > 0$, there exists a natural number n_0 such that

$$\Big|\sum_{k=m+1}^{n}|z_k|\Big| < \varepsilon \quad \text{whenever} \quad n > m \geq n_0.$$

But by the triangle inequality, $\Big|\sum_{k=m+1}^{n}z_k\Big| \leq \sum_{k=m+1}^{n}|z_k| = \Big|\sum_{k=m+1}^{n}|z_k|\Big|$. Therefore, for every $\varepsilon > 0$, there exists a natural number n_0 such that

$$\Big|\sum_{k=m+1}^{n}z_k\Big| < \varepsilon \quad \text{whenever} \quad n > m \geq n_0.$$

By the Cauchy criterion, the series $\sum_{k=1}^{\infty}z_k$ is convergent. ☺

In Example 4-1.2(f), it was shown that the series $\sum_{k=1}^{\infty}2^{k-1}/(k-1)!$ is convergent. From the Comparison Test [see Theorem 4-1.6], it follows that the series $\sum_{k=1}^{\infty}|z|^{k-1}/(k-1)!$ is convergent when $|z| < 2$, which means that $\sum_{k=1}^{\infty}z^{k-1}/(k-1)!$ is absolutely convergent when $|z| < 2$. Therefore this series is also convergent by Prop.4-2.3.

If the terms in a finite summation $\sum_{k=1}^{n}z_k$ are "rearranged", then the value of the sum remains the same. This is just the Generalised Commutative Property Prop.1-11.3; recall that the precise formulation is that, if $\phi:\mathbb{N}_n\to\mathbb{N}_n$ is a bijection, then $\sum_{k=1}^{n}z_k = \sum_{k=1}^{n}z_{\phi(k)}$. It is natural to ask whether $\sum_{k=1}^{\infty}z_k = \sum_{k=1}^{\infty}z_{\phi(k)}$ whenever $\phi:\mathbb{N}\to\mathbb{N}$ is a bijection. We shall show that the answer in general is No; that is to say, there is at least one series for which the equality in question does not hold. However, in

Theorem 4-2.4 below, we shall show that if $\sum_{k=1}^{\infty} z_k$ is *absolutely convergent*, then the answer is Yes.

Example. The series $\sum_{k=1}^{\infty} (-1)^{k+1} k^{-1}$ converges to a sum s by the Leibnitz Alternating Series Test [see Theorem 4-1.7]. Suppose we rearrange this series as

$$1 + 1/3 - 1/2 + 1/5 + 1/7 - 1/4 + 1/9 + 1/11 - 1/6 + \cdots .$$

The rearrangement process may be described in words in the following manner: Begin by writing the first two positive terms and then the first negative term; then the next two positive terms and the second negative term. Continue with the scheme that after every negative term, write the two positive terms of smallest index not written so far and then the negative term of smallest index not written so far. A proper mathematical description of the bijection ϕ that is involved would be as in 1.8.P9. Note that $\phi(3m+1) = 4m+1$, $\phi(3m+2) = 4m+3$ and $\phi(3m+3) = 2m+2$.

We intend to show that the rearranged series cannot have the same sum as the original one. Let the partial sums of the original be s_n and of the rearrangement be t_m. The Leibnitz Alternating Series Test tells us that

$$s \le s_5 = 1 - 1/2 + 1/3 - 1/4 + 1/5 = 47/60 < 5/6.$$

Now,

$$\begin{aligned} t_{3m+3} - t_{3m} &= (-1)^{\phi(3m+1)+1}/\phi(3m+1) \\ &\qquad + (-1)^{\phi(3m+2)+1}/\phi(3m+2) \\ &\qquad\qquad + (-1)^{\phi(3m+3)+1}/\phi(3m+3) \\ &= (-1)^{4m+2}/(4m+1) + (-1)^{4m+4}/(4m+3) + (-1)^{2m+3}/(2m+2) \\ &= 1/(4m+1) + 1/(4m+3) - 1/(2m+2) \\ &= 1/(4m+1) + 1/(4m+3) - 2/(4m+4) \\ &= [1/(4m+1) - 1/(4m+4)] + [1/(4m+3) - 1/(4m+4)] > 0. \end{aligned}$$

Therefore every t_{3m} with $m > 1$ is greater than $t_3 = 1 + 1/3 - 1/2 = 5/6$. It follows that the sequence $\{t_m\}$ cannot converge to a limit less than 5/6, while $s < 5/6$. So $\{t_m\}$ cannot converge to s.

4-2.4. Theorem. *Let* $\sum_{k=1}^{\infty} z_k$ *be absolutely convergent and* $\phi : \mathbb{N} \to \mathbb{N}$ *be a bijection. Then the series* $\sum_{k=1}^{\infty} z_{\phi(k)}$, *called a* **rearrangement** *of* $\sum_{k=1}^{\infty} z_k$, *is convergent and has the same sum.*

Proof. We prove this in three steps.

First suppose that all terms are nonnegative; then absolute convergence is the same as convergence. Let s_m be the m^{th} partial sum of $\sum_{k=1}^{\infty} z_k$. We claim that for each partial sum t_n of the rearranged series, there exists some m such that $t_n \le s_m$. Let $m = \max\{\phi(j) : 1 \le j \le n\}$. Then $1 \le j \le n \Rightarrow \phi(j) \le m$, so that ϕ is an injection from $\mathbb{N}_n$ into $\mathbb{N}_m$. This means that all the terms of the finite sum $\sum_{j=1}^{n} z_{\phi(j)}$ appear in the finite sum $\sum_{k=1}^{m} z_k$. Since the latter has nonnegative terms, it follows as claimed, that $t_n = \sum_{j=1}^{n} z_{\phi(j)} \le \sum_{k=1}^{m} z_k = s_m$. Now $\{s_m\}$ is an increasing sequence, because the terms z_k are nonnegative. Therefore $s_m \le \sum_{k=1}^{\infty} z_k$ for all $m \in \mathbb{N}$ and hence the sequence $\{t_n\}$ of partial sums is increasing and is bounded above by $\sum_{k=1}^{\infty} z_k$. It follows that it converges to a limit less than or equal to this bound. Thus $\sum_{k=1}^{\infty} z_{\phi(k)} \le \sum_{k=1}^{\infty} z_k$. The reverse inequality follows from the fact that $\sum_{k=1}^{\infty} z_k$ is itself a rearrangement of $\sum_{k=1}^{\infty} z_{\phi(k)}$ via ϕ^{-1}. Therefore $\sum_{k=1}^{\infty} z_k = \sum_{k=1}^{\infty} z_{\phi(k)}$.

Next, suppose that the terms are real, but not necessarily nonnegative. Since $\sum_{k=1}^{\infty} |z_k|$ and $\sum_{k=1}^{\infty} z_k$ are both convergent [see Prop.4-2.3], the series $\sum_{k=1}^{\infty} (|z_k| + z_k)$ and $\sum_{k=1}^{\infty} (|z_k| - z_k)$ are both convergent by Prop.4-1.4. Denote their sums by t and u respectively. Since these two series have nonnegative terms, it follows from what has been proved in the preceding paragraph that they converge to the same sums when rearranged: $\sum_{k=1}^{\infty} (|z_{\phi(k)}| + z_{\phi(k)}) = t$ and $\sum_{k=1}^{\infty} (|z_{\phi(k)}| - z_{\phi(k)}) = u$. From the equality

$$z_k = \tfrac{1}{2}[(|z_k| + z_k) - (|z_k| - z_k)]$$

and the corresponding equality for $z_{\phi(k)}$, it follows that

$$\begin{aligned}
\sum_{k=1}^{\infty} z_k &= \tfrac{1}{2}(t - u) \quad \text{by Prop.4-1.4} \\
&= \tfrac{1}{2}\Big[\sum_{k=1}^{\infty} (|z_{\phi(k)}| + z_{\phi(k)}) - \sum_{k=1}^{\infty} (|z_{\phi(k)}| - z_{\phi(k)})\Big] \\
&= \sum_{k=1}^{\infty} \tfrac{1}{2}[(|z_{\phi(k)}| + z_{\phi(k)}) - (|z_{\phi(k)}| - z_{\phi(k)})] \quad \text{by Prop.4-1.4} \\
&= \sum_{k=1}^{\infty} z_{\phi(k)}.
\end{aligned}$$

Finally, we consider series with complex terms z_k. Since the real series $\sum_{k=1}^{\infty} |z_k|$ is convergent, while $|\Re z| \le |z|$ and $|\Im z| \le |z|$ for every complex z, therefore by the Comparison Test, the series $\sum_{k=1}^{\infty} \Re z_k$ and $\sum_{k=1}^{\infty} \Im z_k$ are absolutely convergent. The rest now follows by what has just been proved for the real case and Theorem 3-2.7. [Details are left to the reader; see 4-2.P4.] ☺

Problem Set 4-2

4-2.P1. If the series $\sum_{k=1}^{\infty} z_k$ is absolutely convergent, show that the following are also absolutely convergent:

(i) $\sum_{k=1}^{\infty} z_k^3$ (ii) $\sum_{k=1}^{\infty} z_k \big/ (\frac{1}{2} + |z_k|^{1/9})$ (iii) $\sum_{k=1}^{\infty} iz_k$

4-2.P2. (a) Let $\sum_{k=1}^{\infty} z_k$ be an absolutely convergent series and $\phi:\mathbb{N}\to\mathbb{N}$ be an injection. Show that the series $\sum_{k=1}^{\infty} z_{\phi(k)}$ is absolutely convergent.
(b) Give an example of a convergent series $\sum_{k=1}^{\infty} z_k$ and an injection $\phi:\mathbb{N}\to\mathbb{N}$ such that the series $\sum_{k=1}^{\infty} z_{\phi(k)}$ is divergent.

4-2.P3. (a) If $\sum_{k=1}^{\infty} z_k$ is absolutely convergent and $z_k \neq \pm i$ for every k, show that the series $\sum_{k=1}^{\infty} z_k^2/(1+z_k^2)$ is also absolutely convergent.
(b) Give an example of a conditionally convergent series $\sum_{k=1}^{\infty} z_k$ such that $\sum_{k=1}^{\infty} z_k^2$ is divergent.

4-2.P4. Complete the proof of Theorem 4-2.4 (complex case).

4-3 Ratio and Root Tests

The need for more efficient methods for checking convergence of series was illustrated in Example 4-1.2(f). Three of the simplest ones will be discussed in the present Section. It should be noted that they apply only to series of positive terms and therefore can be used for checking absolute convergence of some series.

We begin with the obvious remark that, if the series obtained by deleting a finite number of terms converges, then the original converges. In other words, if the series $\sum_{k=N+1}^{\infty} z_k$ converges for some $N \geq 1$, then $\sum_{k=1}^{\infty} z_k$ converges.

4-3.1. Theorem. Limit Comparison Test. *Suppose there is some positive integer N such that $a_k \geq 0$ and $b_k > 0$ for every $k > N$ and suppose $\lim_{k\to\infty} (a_k/b_k) = c$.*
(a) *If $c > 0$, then $\sum_{k=1}^{\infty} a_k$ converges $\Leftrightarrow$ $\sum_{k=1}^{\infty} b_k$ converges.*
(b) *If $c = 0$ and $\sum_{k=1}^{\infty} b_k$ converges, then $\sum_{k=1}^{\infty} a_k$ converges.*

Proof. (a) By definition of limit, since $c/2 > 0$, there exists some $n_0 > N$ such that

$$\left|a_k/b_k - c\right| < c/2 \quad \text{whenever} \quad k \geq n_0.$$

Therefore $-c/2 < a_k/b_k - c < c/2$, or $c/2 < a_k/b_k < 3c/2$ whenever $k \geq n_0$. Since $a_k \geq 0$ and $b_k > 0$ whenever $k > N$, it follows that

$$0 < b_k < (2/c)\, a_k \quad \text{and also that} \quad 0 < a_k < (3c/2)\, b_k \quad \text{whenever} \quad k \geq n_0. \qquad (1)$$

If $\sum_{k=1}^{\infty} a_k$ converges, then $\sum_{k=1}^{\infty} (2/c)\, a_k$ converges by Prop.4-1.4. Therefore by the first inequality in (1) and the Comparison Test of Theorem 4-1.6, the series $\sum_{k=1}^{\infty} b_k$ also converges. The converse follows by a similar argument using the second inequality in (1).

(b) By definition of limit, since $1 > 0$, there exists some $n_0 > N$ such that

$$|a_k/b_k| < 1 \quad \text{whenever} \quad k \geq n_0.$$

Therefore $0 \leq a_k < b_k$ whenever $k \geq n_0$. By the Comparison Test, convergence of $\sum_{k=1}^{\infty} b_k$ implies that of $\sum_{k=1}^{\infty} a_k$. ☺

Like the Comparison Test, this Test also allows us to infer convergence of one series from that of another. But this means that the convergence of the latter must have been established already. It is a convention that for this purpose, the convergence of the geometric series $\sum_{k=1}^{\infty} ar^{k-1}$ with $|r| < 1$ and of the series $\sum_{k=1}^{\infty}(1/k^2)$ is taken as established. After the convergence of $\sum_{k=1}^{\infty}(1/k^p)$ with $p > 1$ is proved, this series can also be used for comparison purposes.

4-3.2. Examples. (a) The series $\sum_{k=1}^{\infty}[100k/(k^3+1)]$ is convergent because $\lim_{k\to\infty}(a_k/b_k) = 100$, where $a_k = 100k/(k^3+1)$ and $b_k = 1/k^2$, and $100 > 0$. Here we have used the fact that $\sum_{k=1}^{\infty}(1/k^2)$ is known to be convergent.

(b) In proving the convergence of $\sum_{k=1}^{\infty}(1/k^2)$ in Section 4-1, the inequality $1/k - 1/(k+1) = 1/k(k+1) \geq 1/(k+1)^2$ and the fact that it was possible for us to obtain an explicit expression for the n^{th} partial sum of $\sum_{k=1}^{\infty}(1/k - 1/(k+1))$ played a crucial role. Doing this for $\sum_{k=1}^{\infty}(1/k^p)$ for (rational) $p \neq 2$ is not so easy. Although convergence of this series for $p > 2$ follows from that of $\sum_{k=1}^{\infty}(1/k^2)$ by an application of the Comparison Test, the situation for $1 < p < 2$ requires careful handling. Note that the divergence of $\sum_{k=1}^{\infty}(1/k)$, which was established earlier on, now implies the divergence of $\sum_{k=1}^{\infty}(1/k^p)$ for $p < 1$ by the same Test; all one has to do is to use the inequality $0 < k^p \leq k$ for $p < 1$.

With a view to proving the convergence of $\sum_{k=1}^{\infty}(1/k^p)$ for $1 < p$, we begin by proving the inequality

$$1/k^p \leq \frac{1}{p-1}\left[1/(k-1)^{p-1} - 1/k^{p-1}\right] \text{ for } k > 1 \text{ and } p > 1. \tag{1}$$

We use Bernoulli's Inequality [see 2-4.P9]: $(1+x)^p \geq 1+px$ with $x = 1/(k-1)$. Since $1+x = k/(k-1)$, we get $[k/(k-1)]^p \geq 1+p/(k-1)$. Multiplying by the positive number $(k-1)/k^p$, we get

$$1/(k-1)^{p-1} \geq (k-1)/k^p + p/k^p = 1/k^{p-1} + (p-1)/k^p.$$

Upon adding $-1/k^{p-1}$ to both sides and then dividing by the positive number $p-1$, we get (1).

It follows from (1) that

$$\sum_{k=2}^{n}(1/k^p) \leq \frac{1}{p-1}\sum_{k=2}^{n}\left[1/(k-1)^{p-1} - 1/k^{p-1}\right] = \frac{1}{p-1}\left[1 - 1/n^{p-1}\right].$$

Therefore the increasing sequence of partial sums of $\sum_{k=1}^{\infty}(1/k^p)$ is bounded above by $1+\frac{1}{p-1}=\frac{p}{p-1}$. Thus the series is convergent with sum not greater than $\frac{p}{p-1}$.

After we define x^p for irrational p, it will turn out that $x^p > x^q$ whenever $x > 1$ and $p > q$ regardless of the rationality or otherwise of p and q. It will then follow by the Comparison Test that $\sum_{k=1}^{\infty}(1/k^p)$ converges when $p > 1$ even if p is irrational; all we shall need to do is to select a rational number q such that $p > q > 1$.

(c) The series $\sum_{k=1}^{\infty}\sqrt{k}/(1+k^2)$ is convergent, because $\lim_{k\to\infty}(a_k/b_k) = 1$ when we choose $b_k = 1/k^{3/2}$ and $a_k = \sqrt{k}/(1+k^2)$, while $\sum_{k=1}^{\infty} b_k$ is convergent (as just shown in the preceding Example).

(d) The series $1+1/3+1/5+\cdots+1/(2k-1)+\cdots = \sum_{k=1}^{\infty}1/(2k-1)$ is divergent because $\lim_{k\to\infty}(a_k/b_k) = 1/2$ when $b_k = 1/k$, $a_k = 1/(2k-1)$, and $\sum_{k=1}^{\infty} b_k$ is divergent.

4-3.3. Theorem. Ratio Test (d'Alembert). *Let $a_k > 0$ for every k. Then the series* $\sum_{k=1}^{\infty} a_k$

(a) *converges if* $\limsup_{k\to\infty}(a_{k+1}/a_k) < 1$,

(b) *diverges if* $\limsup_{k\to\infty}(a_k/a_{k+1}) < 1$, *and*

(c) *may either converge or diverge in the event that*

neither $\limsup_{k\to\infty}(a_{k+1}/a_k) < 1$ *nor* $\limsup_{k\to\infty}(a_k/a_{k+1}) < 1$.

Proof. (a) Let ρ be a number between $\limsup_{k\to\infty}(a_{k+1}/a_k)$ and 1. By Prop.3-4.4, there exists $N\in\mathbb{N}$ such that $a_{k+1}/a_k < \rho$ whenever $k \ge N$ and therefore $a_{k+1} < \rho a_k$ whenever $k \ge N$. It follows that $a_{N+2} < \rho a_{N+1} < \rho(\rho a_N) = \rho^2 a_N$, hence $a_{N+3} < \rho a_{N+2} < \rho(\rho^2 a_N) = \rho^3 a_N$, and so on. Generally one can show by induction that $a_{N+p} < \rho^p a_N$ whenever p is any positive integer. Another way to state this is that $a_k < \rho^{k-N} a_N$ whenever $k > N$. Now $\sum_{k=N+1}^{\infty}\rho^{k-N}a_N$ converges because $0 < \rho < 1$. By the Comparison Test, the series $\sum_{k=N+1}^{\infty} a_k$ converges and hence $\sum_{k=1}^{\infty} a_k$ converges.

(b) Let ρ be a number between $\limsup_{k\to\infty}(a_k/a_{k+1})$ and 1. By Prop.3-4.4, there exists $N\in\mathbb{N}$ such that $a_k/a_{k+1} < \rho$ whenever $k \ge N$ and therefore $a_k < \rho a_{k+1}$ whenever $k \ge N$. Since $0 < \rho < 1$, this implies that $a_{k+1} > a_k/\rho > a_k$ whenever $k \ge N$. It follows by induction that $a_k > a_N$ for $k > N$. Therefore a_k cannot converge to 0. By Prop.4-1.3, the series $\sum_{k=1}^{\infty} a_k$ cannot converge.

(c) To prove this part, it is sufficient to produce two series, one convergent and one divergent, both having $\lim_{k\to\infty}(a_{k+1}/a_k) = \lim_{k\to\infty}(a_k/a_{k+1}) = 1$. The series $\sum_{k=1}^{\infty} a_k$ with $a_k = 1/k^2$ converges while $\lim_{k\to\infty}(a_{k+1}/a_k) = \lim_{k\to\infty} k^2/(k+1)^2 = 1 = \lim_{k\to\infty}(a_k/a_{k+1})$. On the other hand, the series $\sum_{k=1}^{\infty} a_k$ with $a_k = 1/k$ diverges while $\lim_{k\to\infty}(a_{k+1}/a_k) = \lim_{k\to\infty} k/(k+1) = 1 = \lim_{k\to\infty}(a_k/a_{k+1})$. ☺

4-3.4. Examples. (a) Consider the series $\sum_{k=1}^{\infty} 2^{k-1}/(k-1)!$ of Example 4-1.2(f). If a_k denotes the k^{th} term of this series, then $a_{k+1}/a_k = 2/k$ and therefore $\lim_{k\to\infty}(a_{k+1}/a_k) = 0$. By part (a) of the Ratio Test, it follows that the series converges. Note that we have not used the inequality $(k-1)! \geq 4^{k-3}$ for $k \geq 3$, as we did in the earlier discussion. It was pointed out then that this inequality would be inadequate if the number 2 in the numerator of a_k were to be replaced by some other number. In contrast, the Ratio Test is just as easily applicable for obtaining *absolute convergence* if 2 is replaced even by a complex number! Indeed, the ratio $|a_{k+1}|/|a_k|$ becomes $|z|/k$ if 2 is replaced by z, and this has limit 0 for complex z as well.

(b) For the series $\sum_{k=1}^{\infty} a_k$ with $a_k = (2k)!/(k!)^2$, we find after some computation that $a_{k+1}/a_k = (4k+2)/(k+1)$, which has limit $4 > 1$. Therefore by part (b) of the Ratio Test, the series diverges.

(c) The series $\sum_{k=1}^{\infty} a_k = \frac{1}{2} + (\frac{1}{2})^2 + 3(\frac{1}{2})^3 + (\frac{1}{2})^4 + 5(\frac{1}{2})^5 + \cdots$ has the property that the even terms of the sequence $\{a_{k+1}/a_k\}$ form the unbounded sequence 3/2, 5/2, 7/2, ..., and the odd terms of the sequence $\{a_k/a_{k+1}\}$ form the unbounded sequence 2, 6, 10, ... Therefore the Ratio Test is not applicable. However, the series is the sum of $\sum_{k=1}^{\infty} (\frac{1}{2})^{2k}$ and $\sum_{k=1}^{\infty} (2k-1)(\frac{1}{2})^{2k-1}$; the former is a convergent geometric series and the latter can be seen to be convergent by comparison [note that $2k-1 \leq 2^k$] with $\sum_{k=1}^{\infty} (2^k)(\frac{1}{2})^{2k-1}$, which is the same as the convergent geometric series $\sum_{k=1}^{\infty} (\frac{1}{2})^{k-1}$. We go on to present another convergence test, which easily shows the aforementioned series $\sum_{k=1}^{\infty} a_k$ to be convergent.

4-3.5. Theorem. Root Test (Cauchy). *Let $a_k > 0$ for every k greater than some n_0, so that $a_k^{1/k}$ exists for all $k > n_0$. Then the series $\sum_{k=1}^{\infty} a_k$*

(a) *converges if* $\limsup_{k\to\infty} a_k^{1/k} < 1$,

(b) *diverges if either* $\{a_k^{1/k}\}$ *is unbounded above or* $\limsup_{k\to\infty} a_k^{1/k} > 1$, *and*

(c) *may either converge or diverge if* $\limsup_{k\to\infty} a_k^{1/k} = 1$.

Proof. (a) Let ρ be a number between $\limsup_{k\to\infty} a_k^{1/k}$ and 1. By Prop.3-4.4, there exists $N \in \mathbb{N}$ such that $a_k^{1/k} < \rho$ whenever $k \geq N$, and therefore $a_k < \rho^k$ whenever $k \geq N$. Now $\sum_{k=N+1}^{\infty} \rho^k$ converges because $0 < \rho < 1$. The Comparison Test now shows that the series $\sum_{k=N+1}^{\infty} a_k$ converges and hence $\sum_{k=1}^{\infty} a_k$ converges.

(b) First suppose $\{a_k^{1/k}\}$ is bounded and $\limsup_{k\to\infty} a_k^{1/k} > 1$. Let ρ be a number between $\limsup_{k\to\infty} a_k^{1/k}$ and 1. Then $\rho < \limsup_{k\to\infty} a_k^{1/k}$ and, by Prop.3-4.4, cannot have the property that every $a_k^{1/k}$ is less than ρ beyond a certain index $k = N$. This means that for every $N \in \mathbb{N}$, there exists $k > N$ such that $a_k^{1/k} \geq \rho$, i.e. $a_k \geq \rho^k$. Since $\rho > 1$, this implies that $\{a_k\}$ cannot converge to 0. Thus the series $\sum_{k=1}^{\infty} a_k$ must diverge. Now suppose $\{a_k^{1/k}\}$ is unbounded. If $\{a_k\}$ were to converge to 0, then it would be bounded and hence so would $\{a_k^{1/k}\}$; therefore $\{a_k\}$ cannot converge to 0 and, once again, the series $\sum_{k=1}^{\infty} a_k$ must diverge.

(c) To prove this part, we must produce two series, one convergent and one divergent, both having $\limsup_{k\to\infty} a_k^{1/k} = 1$. The series $\sum_{k=1}^{\infty} a_k$ with $a_k = 1/k^2$ converges while $\lim_{k\to\infty} a_k^{1/k} = \lim_{k\to\infty}(1/k^2)^{1/k} = 1$ by Props.3-1.6 and 3-2.3. On the other hand, the series $\sum_{k=1}^{\infty} a_k$ with $a_k = 1/k$ diverges while $\lim_{k\to\infty} a_k^{1/k} = \lim_{k\to\infty}(1/k)^{1/k} = 1$, once again by Props.3-1.6 and 3-2.3. ☺

Example. Consider the series $\sum_{k=1}^{\infty} a_k = \frac{1}{2} + (\frac{1}{2})^2 + 3(\frac{1}{2})^3 + (\frac{1}{2})^4 + 5(\frac{1}{2})^5 + \cdots$ again. For the sequence $\{a_k^{1/k}\}$, the subsequences of even terms and of odd terms both have limit $\frac{1}{2}$ and therefore the sequence itself has limit $\frac{1}{2}$. It follows by the Root Test that the series converges. Thus it can happen that a series is found to be convergent by the Root Test but not by the Ratio Test. The reverse cannot happen, as the next result shows.

4-3.6. Proposition. *Let $\{a_k\}$ be a sequence of positive terms. If $\{a_{k+1}/a_k\}$ is bounded, then so is $\{a_k^{1/k}\}$ and, furthermore,* $\limsup_{k\to\infty} a_k^{1/k} \leq \limsup_{k\to\infty} (a_{k+1}/a_k)$.

Proof. Let $c = \limsup_{k\to\infty} (a_{k+1}/a_k)$. Choose any real number $b > c$. By Prop.3-4.4, there exists $N \in \mathbb{N}$ such that $a_{k+1}/a_k < b$ and hence $a_{k+1} < ba_k$ whenever $k \geq N$. Therefore for any $k > N$, we have (by induction on $k-N$) $a_k < b^{k-N}a_N = (a_N b^{-N})b^k$, which implies that $a_k^{1/k} < (a_N b^{-N})^{1/k} b$. Now, $\lim_{k\to\infty} (a_N b^{-N})^{1/k} = 1$ by Prop.3-1.5, and therefore by Prop.3-1.4, this sequence is bounded. Therefore $\{a_k^{1/k}\}$ is not only bounded but also $\limsup_{k\to\infty} a_k^{1/k} \leq b$. Since this is true for every $b > c$, we have $\limsup_{k\to\infty} a_k^{1/k} \leq c$. ☺

Problem Set 4-3

4-3.P1. Determine the convergence or divergence of the following series:

(i) $\sum_{k=1}^{\infty} \left(2k^2/(3k^2+1)\right)^k$ (ii) $\sum_{k=1}^{\infty} \left(2^k (k!)^2/(2k)!\right)$ (iii) $\sum_{k=1}^{\infty} \left(3^k (k!)^2/(2k)!\right)$

(iv) $\sum_{k=1}^{\infty} \left(4^k (k!)^2/(2k)!\right)$ (v) $\sum_{k=1}^{\infty} \left(5^k (k!)^2/(2k)!\right)$.

4-3.P2. If $\sum_{k=1}^{\infty} a_k$ is a convergent series of nonnegative terms, show that $\sum_{k=1}^{\infty} k^{-p}\sqrt{a_k}$ converges if $p > \frac{1}{2}$.

4-3.P3. For each real x, determine the values of p for which the series $\sum_{k=1}^{\infty} x^{2k}/(2k)^p$ converges.

4-3.P4. Determine for which positive x the following series converge:
(i) $\sum_{k=1}^{\infty} k^{1/2}x^k(k^2+1)^{-1/2}$ (ii) $\sum_{k=1}^{\infty} (k+1)^k k^{-(k+1)} x^k$.

4-3.P5. Determine the values of p for which $\sum_{k=1}^{\infty} k^p/k!$ converges.

4-3.P6. (a) Suppose $\sum_{k=1}^{\infty} a_k$ is a series of positive terms and there exist $\lambda \in \mathbb{R}$ and $N \in \mathbb{N}$ such that

$$\lambda > 1 \qquad \text{and} \qquad k\left(1-\frac{a_{k+1}}{a_k}\right) > \lambda \quad \text{for all } k \geq N.$$

Show that the series converges.
(b) Suppose $\sum_{k=1}^{\infty} a_k$ is a series of positive terms and there exists $N \in \mathbb{N}$ such that

$$k\left(1-\frac{a_{k+1}}{a_k}\right) \leq 1 \quad \text{for all } k \geq N.$$

Show that the series diverges. [Parts (a) and (b) together constitute what is known as **Raabe's Test**.]
(c) Suppose $\sum_{k=1}^{\infty} a_k$ is a series of positive terms and $\liminf k\left(1-\frac{a_{k+1}}{a_k}\right) > 1$. Show that the series converges. If $\limsup k\left(1-\frac{a_{k+1}}{a_k}\right) < 1$, show that it diverges.
(d) If $a_k = \frac{1}{k}$, show that the series $\sum_{k=1}^{\infty} a_k$ satisfies the divergence condition of (b) but not the one in (c).

*4-4 Abel's and Dirichlet's Tests

The convergence tests presented so far are applicable only when the terms of the series are either nonnegative or alternate in sign. In practice, one rarely encounters any other types of real series, but we nevertheless present two other convergence tests, which are not subject to this limitation. They are both based on the following computational trick:

If $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$ are finite sequences of (complex) numbers and $S_k = b_1 + \cdots + b_k$, then $b_k = S_k - S_{k-1}$ for $k = 2, \ldots, n$ and therefore

$$\begin{aligned} a_1b_1 + \cdots + a_nb_n &= a_1S_1 + a_2(S_2-S_1) + a_3(S_3-S_2) + \cdots + a_n(S_n - S_{n-1}) \\ &= (a_1-a_2)S_1 + (a_2-a_3)S_2 + (a_3-a_4)S_3 + \cdots + (a_{n-1}-a_n)S_{n-1} + a_nS_n. \end{aligned}$$

4-4.1. Proposition. Abel summation formula. *Let $\{a_k\}_{k=1}^{n}$ and $\{b_k\}_{k=1}^{n}$ be finite sequences of complex numbers, where $n \geq 2$, and let $S_j = \sum_{k=1}^{j} b_k$ for $1 \leq j \leq n$. Then*

$$\sum_{k=1}^{n} a_kb_k = \sum_{k=1}^{n-1} (a_k - a_{k+1})S_k + a_nS_n.$$

Proof. If $n = 2$, the equality states that $a_1 b_1 + a_2 b_2 = (a_1 - a_2)S_1 + a_2 S_2$. This is true because the right side is equal to $(a_1 - a_2) b_1 + a_2(b_1 + b_2) = a_1 b_1 - a_2 b_1 + a_2 b_1 + a_2 b_2$, which is equal to the left side.

Assume it true for some n. If n is replaced by $n+1$, then

$$\begin{aligned}\sum_{k=1}^{n+1} a_k b_k &= \sum_{k=1}^{n} a_k b_k + a_{n+1} b_{n+1} = \sum_{k=1}^{n-1} (a_k - a_{k+1}) S_k + a_n S_n + a_{n+1} b_{n+1} \\ &= \sum_{k=1}^{n-1} (a_k - a_{k+1}) S_k + a_n S_n + a_{n+1} (S_{n+1} - S_n) \\ &= \sum_{k=1}^{n-1} (a_k - a_{k+1}) S_k + a_n S_n + a_{n+1} S_{n+1} - a_{n+1} S_n \\ &= \sum_{k=1}^{n-1} (a_k - a_{k+1}) S_k + (a_n - a_{n+1}) S_n + a_{n+1} S_{n+1} \\ &= \sum_{k=1}^{n} (a_k - a_{k+1}) S_k + a_{n+1} S_{n+1} . \end{aligned}$$ ☺

4-4.2. Theorem. Dirichlet's Test. *Suppose* $\sum_{k=1}^{\infty} b_k$ *is a complex series with bounded partial sums and* $\{a_k\}$ *a monotone* (*real*) *sequence with limit* 0. *Then* $\sum_{k=1}^{\infty} a_k b_k$ *converges.*

Proof. Denote the n^{th} partial sum of $\sum_{k=1}^{\infty} b_k$ by S_n. By the Abel summation formula,

$$\sum_{k=1}^{n} a_k b_k = \sum_{k=1}^{n-1} (a_k - a_{k+1}) S_k + a_n S_n . \tag{1}$$

According to the hypothesis, there exists some $M > 0$ such that $|S_n| \le M$ for every n. Therefore in view of the hypothesis that $\{a_k\}$ is monotone, we have

$$\sum_{k=1}^{n-1} |(a_k - a_{k+1}) S_k| \le M \sum_{k=1}^{n-1} |a_k - a_{k+1}| = M|a_1 - a_n|. \tag{2}$$

The hypothesis that $\{a_k\}$ has limit 0 and that the partial sums S_n are bounded implies on the one hand that $\lim_{n\to\infty} a_n S_n = 0$ and on the other hand that the right side of (2) is bounded, so that the series

$$\sum_{k=1}^{\infty} (a_k - a_{k+1}) S_k$$

is (absolutely) convergent by Prop.4-1.5. These two facts, together with (1), show that $\sum_{k=1}^{\infty} a_k b_k$ converges. ☺

Remark. The first part of the Leibnitz Alternating Series Test (4-1.7) follows from this result upon taking $b_k = (-1)^{k+1}$. Note that the sequence of partial sums of $\sum_{k=1}^{\infty} (-1)^{k+1}$ is $1, 0, 1, 0, \ldots$. To obtain a series with bounded partial sums, all one has to do is to start with a bounded sequence $\{c_k\}$ such as $1, 0, 1, 0, \ldots$ or $1, 2, 0, 1, 2, 0, \ldots$ and set $b_1 = c_1$, $b_{k+1} = c_{k+1} - c_k$, so that $\sum_{k=1}^{n} b_k = c_n$. It may be noted that the second of the two sequences just mentioned could have been described in terms of the integer part function [ ] as $c_k = k - 3\left[\frac{k}{3}\right]$. This leads to the series with bounded partial sums $\sum_{k=1}^{n} b_k$, where $b_1 = 1$ and $b_{k+1} = 1 - 3\left[\frac{k+1}{3}\right] + 3\left[\frac{k}{3}\right]$. Since the sequence $\{k^{-1/2}\}$ is decreasing with limit 0, it follows by Dirichlet's Test that the series

$$\sum_{k=1}^{\infty}\left(1 - 3\left[\frac{k}{3}\right] + 3\left[\frac{k-1}{3}\right]\right)k^{-1/2}$$

is convergent. Its first six terms are $1 + 1/\sqrt{2} - 2/\sqrt{3} + 1/\sqrt{4} + 1/\sqrt{5} - 2/\sqrt{6}$.

After defining trigonometric functions in Section 5-3, it will follow that $\sin x$ is a bounded function (of course) and hence that the series

$$\sum_{k=1}^{\infty} (\sin(k+1) - \sin k)\,k^{-1/2}$$

converges.

4-4.3. Theorem. Abel's Test. *Suppose* $\sum_{k=1}^{\infty} b_k$ *is a convergent complex series and* $\{a_k\}$ *a convergent monotone (real) sequence. Then* $\sum_{k=1}^{\infty} a_k b_k$ *converges.*

Proof. Denote the n^{th} partial sum of $\sum_{k=1}^{\infty} b_k$ by S_n. By the Abel summation formula,

$$\sum_{k=1}^{n} a_k b_k = \sum_{k=1}^{n-1} (a_k - a_{k+1})S_k + a_n S_n. \tag{1}$$

Since $\sum_{k=1}^{\infty} b_k$ is convergent by hypothesis, there exists some $M > 0$ such that $|S_n| \le M$ for every n. Therefore in view of the hypothesis that $\{a_k\}$ is monotone, we have

$$\sum_{k=1}^{n-1} |(a_k - a_{k+1})S_k| \le M\sum_{k=1}^{n-1} |a_k - a_{k+1}| = M|a_1 - a_n|. \tag{2}$$

The hypothesis that $\sum_{k=1}^{\infty} b_k$ is convergent and that $\{a_k\}$ has a limit implies on the one hand that $\lim_{n\to\infty} a_n S_n$ exists and on the other hand that the right side of (2) is bounded, so that the series

$$\sum_{k=1}^{\infty} (a_k - a_{k+1})S_k$$

is (absolutely) convergent by Prop.4-1.5. These two facts, together with (1), show that $\sum_{k=1}^{\infty} a_k b_k$ converges. ☺

Example. The series $\sum_{k=1}^{\infty} \frac{1}{k}\left(1 + \frac{1}{k}\right)^k (-1)^{k+1}$ can be handled by Abel's Test as follows: The series with k^{th} term $\frac{1}{k}(-1)^{k+1}$ is convergent by the Leibnitz Alternating Series Test and the sequence with k^{th} term $\left(1 + \frac{1}{k}\right)^k$ is monotone and convergent as seen in Example 3-3.4.

Problem Set 4-4

4-4.P1. If z is a complex number such that $|z| = 1 \ne z$, show that the series $\sum_{k=1}^{\infty} (z^{k-1})/\sqrt{k}$ converges.

4-4.P2. It was shown in 2-1.P5 that the function f defined on $\mathbb{R}$ by

$$f(x) = \min\{x-[x], 1-(x-[x])\},$$

where the brackets [ ] denote the integer part function, is bounded. Use this to prove that for any real number a, the series $\sum_{k=1}^{\infty}\big(f(ka)-f((k-1)a)\big)k^{-1/2}$ is convergent. [When $a = 0.3$, the first fifteen values of $\big(f(ka)-f((k-1)a\big)$ are

0.3, 0.1, –0.3, 0.1, 0.3, –0.3, –0.1, 0.3, –0.1, –0.3, 0.3, 0.1, –0.3, 0.1, 0.3.

Thus the series is not alternating and the Leibnitz Test cannot be applied.]

4-4.P3. The formula $\sum_{k=1}^{n-1}\sin kb = -\frac{1}{2}\Big(\cos\,(n-\frac{1}{2})b-\cos\frac{b}{2}\Big)\Big/\sin\frac{b}{2}$ when $\sin\frac{b}{2}\neq 0$ will be proved later [see 7-4.P9]. Use it now to show that, if a_n is a decreasing sequence with limit 0, then $\sum_{k=1}^{\infty}a_k\sin kx$ converges.

4-4.P4. The formula $\sum_{k=1}^{n-1}\cos kb = \frac{1}{2}\Big(\sin\,(n-\frac{1}{2})b-\sin\frac{b}{2}\Big)\Big/\sin\frac{b}{2}$ when $\sin\frac{b}{2}\neq 0$ will be proved later [end of Section 7-4]. Use it now to show that, if a_n is a decreasing sequence with limit 0, then $\sum_{k=1}^{\infty}a_k\cos kx$ converges provided that $\sin\frac{b}{2}\neq 0$.

4-5 Infinite Limits of Real Sequences

A careful examination of the proof of part (b) of the Ratio Test [see Theorem 4-3.3] reveals that we could have started with the hypothesis that $\liminf_{k\to\infty}(a_{k+1}/a_k) > 1$. However such a hypothesis would restrict us to the case when the sequence with kth term a_{k+1}/a_k is bounded, considering that we defined limits superior and inferior only for bounded sequences. In order to avoid the restriction while starting with the hypothesis $\liminf_{k\to\infty}(a_{k+1}/a_k) > 1$, we need to extend some concepts regarding limits. Carrying out this extension will also provide some advantage when dealing with power series in Section 5-2.

In the process, we shall introduce another use of the symbols ∞ and $-\infty$, in a manner that practically gives them the status of mathematical objects. We refrain from setting forth the formal description of the resulting "extended real number system", as the reader will undoubtedly learn to use such a system to whatever extent required by actually working with it.

4-5.1. Definition. *A sequence $\{x_n\}$ in $\mathbb{R}$ is said to* **diverge to** ∞ *if, for every $M > 0$, there exists a natural number n_0 such that*

$$x_n > M \quad \textit{whenever} \quad n \geq n_0.$$

We also write $\lim_{n\to\infty}x_n = \infty$, or $x_n\to\infty$, and speak of the sequence as having ∞ as its limit. Similarly, a sequence $\{x_n\}$ in $\mathbb{R}$ is said to **diverge to** $-\infty$ if $\{-x_n\}$ diverges to

∞ and we speak of the sequence as having $-\infty$ as its limit. [This terminology and notation are similar to what is used when a sequence *converges* to a real number, but it should be borne in mind that, in the present situation, the sequence *diverges*.]

A sequence $\{x_n\}$ diverges to $-\infty$ if, and only if, for every $M > 0$, there exists a natural number n_0 such that

$$x_n < -M \quad \text{whenever} \quad n \geq n_0.$$

It is immediate from the definition that a sequence which diverges to ∞ must be unbounded above but bounded below; also every subsequence must diverge to ∞. Analogously for a sequence which diverges to $-\infty$ (details are left as 4-5.P2 for the reader to complete).

Examples. (a) The sequence $\{n\}$ diverges to ∞, because for any given $M > 0$, the integer $n_0 = [M]+1$, where [ ] denotes the integer part function, has the property that $n > M$ whenever $n \geq n_0$. It follows that the sequence $\{-n\}$ diverges to $-\infty$.

(b) The sequence $\{(-1)^n n\}$ diverges but does not diverge either to ∞ or to $-\infty$. It does not diverge to ∞ because the positive number $M = 7$ (for instance) has the property that, whatever natural number n_0 one may try, an odd integer n greater than $\max\{n_0, 7\}$ satisfies $(-1)^n n < M$. Choosing even instead of odd shows that the sequence $\{-(-1)^n n\}$ does not diverge to ∞. Thus $\{(-1)^n n\}$ does not diverge to $-\infty$ either. Note that this sequence is unbounded.

(c) The sequence $\{8+n+(-1)^n n\}$ also does not diverge either to ∞ or to $-\infty$. To see why it does not diverge to ∞, consider the positive number $M = 8$. Whatever natural number n_0 one may try, an odd integer n greater than n_0 satisfies $8+n+(-1)^n n = 8$. We could have argued the same thing with any $M > 8$, because an odd integer $n > n_0$ will satisfy $8+n+(-1)^n n = 8 < M$. (However, we cannot argue our case with $M < 8$.) To see that $\{8+n+(-1)^n n\}$ also does not diverge to $-\infty$ is even easier: $-(8+n+(-1)^n n) \leq -8 < 0$ and one can therefore argue our case with any positive M whatsoever. The sequence of this Example too is unbounded.

The preceding two Examples show that an unbounded sequence need not necessarily diverge to ∞ or to $-\infty$. With regard to monotone sequences however, the situation is different, as we now show.

4-5.2. Proposition. *If an increasing sequence is unbounded, then it diverges to* ∞. *If a decreasing sequence is unbounded, then it diverges to* $-\infty$.

Proof. Let the increasing sequence $\{x_n\}$ be unbounded. Being an increasing sequence, it is bounded below by its first term x_1. Therefore it has to be unbounded above. So, for any $M > 0$, there exists some $n_0 \in \mathbb{N}$ such that $x_{n_0} > M$. Since the sequence is increasing, it follows that $n \geq n_0 \Rightarrow x_n \geq x_{n_0} > M$. Thus the sequence diverges to ∞. If $\{x_n\}$ is a decreasing unbounded sequence, then $\{-x_n\}$ is an increasing unbounded sequence and hence diverges to ∞, as already proved. ☺

It follows from the preceding result that a monotone sequence always has a limit; moreover, the limit is $\pm\infty$ if, and only if, the monotone sequence is unbounded.

Before defining limits superior and inferior for unbounded sequences, we make the following observation about a sequence $\{x_n\}$ that is bounded above: $\{x_k, x_{k+1}, \dots\}$ is a nonempty subset of $\mathbb{R}$ with an upper bound and therefore $B_k = \sup\ \{x_k, x_{k+1}, \dots\}$ is a real number; moreover, $B_{k+1} \leq B_k$, so that $\{B_k\}$ is a decreasing sequence and therefore has a limit, which can be $-\infty$.

4-5.3. Definition. *Let $\{x_n\}$ be an unbounded real sequence. Then the* **limit superior** *of the sequence is given by*

$$\limsup_{n\to\infty} x_n = \begin{cases} \infty \text{ if the sequence is unbounded above} \\ \lim\limits_{k\to\infty} \left(\sup\ \{x_k, x_{k+1}, \dots\}\right) \text{ if the sequence is bounded above.} \end{cases}$$

The **limit inferior** *is defined analogously*:

$$\liminf_{n\to\infty} x_n = \begin{cases} -\infty \text{ if the sequence is unbounded below} \\ \lim\limits_{k\to\infty} \left(\inf\ \{x_k, x_{k+1}, \dots\}\right) \text{ if the sequence is bounded below.} \end{cases}$$

Examples. (a) For the sequence $\{x_n\}$ given by $x_n = -n$, which may be described also as $-1, -2, -3, \dots$, the sequence $\{B_k\}$ is the same as $\{x_n\}$, which diverges to $-\infty$. So, $\limsup\limits_{n\to\infty} x_n = -\infty$.

(b) For the sequence $-1, 0, -2, 0, -3, 0, -4, 0, \dots$, the sequence $\{B_k\}$ is $0, 0, 0, \dots$. Therefore the limit superior is 0.

4-5.4. Lemma. *If $\{x_n\}$ is bounded above, then*

$$\limsup_{n\to\infty} x_n = \lim_{k\to\infty} \left(\sup\ \{x_k, x_{k+1}, \dots\}\right),$$

irrespective of whether it is bounded below. When $\{x_n\}$ is bounded below,

$$\liminf_{n\to\infty} x_n = \lim_{k\to\infty} \left(\inf\ \{x_k, x_{k+1}, \dots\}\right),$$

irrespective of whether it is bounded above.

Proof. This follows immediately upon comparing the definition of $\limsup$ and $\liminf$ for unbounded sequences [Def.4-5.3] with the one for bounded sequences [Def.3-4.3]. ☺

Remarks. (i) Every real sequence has a limit superior and a limit inferior, although one or both may turn out to be ∞ or $-\infty$.

(ii) The limit of a decreasing sequence cannot be ∞. It follows that, for a bounded sequence $\{x_n\}$, $\limsup_{n\to\infty} x_n$ cannot be ∞. Thus $\limsup_{n\to\infty} x_n = \infty$ if, and only if, $\{x_n\}$ is unbounded above. Similarly, $\liminf_{n\to\infty} x_n = -\infty$ if, and only if, $\{x_n\}$ is unbounded below.

(iii) For bounded sequences, we had argued [see Remark following Def.3-4.3] that the limit inferior cannot exceed the limit superior. For the corresponding statement for unbounded sequences to make sense, we need to make the convention that $-\infty < x < \infty$ for all real x. [The reader is reminded that ∞ and $-\infty$ are not elements of $\mathbb{R}$ and therefore inequalities involving them are not governed by the Order Axioms for $\mathbb{R}$ given in Chapter 1.] It is then "automatic" that $\liminf_{n\to\infty} x_n \le \limsup_{n\to\infty} x_n$ for a sequence that is either unbounded below or unbounded above. Any other kind of sequence is bounded. Therefore the inequality is valid for *all* real sequences without exception.

(iv) Another consequence of the convention laid down in (iii) above is that $\liminf_{n\to\infty} x_n > 1$ if, and only if, $\liminf_{n\to\infty} x_n$ is either a real number greater than 1 or ∞. For a sequence bounded below, $\liminf_{n\to\infty} x_n = \lim_{k\to\infty}\left(\inf\{x_k, x_{k+1}, \dots\}\right)$; therefore $\liminf_{n\to\infty} x_n > 1$ implies that $\inf\{x_k, x_{k+1}, \dots\} > 1$ for all "sufficiently large" k. Since the ratios a_{k+1}/a_k in d'Alembert's Ratio Test (Theorem 4-3.3) are bounded below by 0, $\liminf_{k\to\infty}(a_{k+1}/a_k) > 1$ would imply that $a_{k+1} > a_k$ for all "sufficiently large" k. This makes it possible to prove part (b) of that Theorem with the hypothesis $\liminf_{k\to\infty}(a_{k+1}/a_k) > 1$ instead of $\limsup_{k\to\infty}(a_k/a_{k+1}) < 1$.

(v) Also, it is easy to show that when $x_n \le y_n$ for every n, and the sequences $\{x_n\}$ and $\{y_n\}$ both have limits (which could be ∞ or $-\infty$), we have $\lim_{n\to\infty} x_n \le \lim_{n\to\infty} y_n$.

Props.3-4.5 and 3-4.6 can be extended to include unbounded sequences but we leave the details to the reader.

4-5.5. Proposition. *The limit inferior and limit superior of a real sequence are equal if, and only if, the sequence has a limit, in which case, the limit is equal to the limits superior and inferior.*

Proof. See 4-5.P9. ☺

4-5.6. Proposition. *If* $\{x_n\}$ *is a real sequence, then* $\limsup_{n\to\infty} x_n$ *is the largest subsequential limit and* $\liminf_{n\to\infty} x_n$ *the smallest.*

Proof. See 4-5.P10. ☺

Problem Set 4-5

4-5.P1. Which of the following sequences diverge to ∞? Which ones to $-\infty$?
(i) $\{n!\}$ (ii) $\{n(n-10)\}$ (iii) $\{-k^{1/10}\}$ (iv) $\{k(1+(-1)^k)\}$
(v) $\{[\sqrt{k}]\}$, where [ ] denotes the integer part function
(vi) $\{1+(-1)^k(k+1/k)+k\}$.

4-5.P2. If a real sequence $\{a_n\}$ diverges to $-\infty$, show that (a) the sequence is unbounded below but is bounded above and that (b) every subsequence diverges to $-\infty$.

4-5.P3. Let $\{a_k\}$ and $\{b_k\}$ be real sequences such that the former diverges to ∞ and the latter converges to $c \in \mathbb{R}$. Show that the sequence $\{a_k + b_k\}$ diverges to ∞. If $c > 0$, show that the sequence $\{a_k b_k\}$ also diverges to ∞. What can be said about the convergence or divergence of $\{a_k b_k\}$ if $c < 0$ and when $c = 0$?

4-5.P4. Show that the sequence $\{6+1/n+n+(-1)^n n\}$ does not diverge to ∞ by selecting an appropriate positive number M and showing that, whatever positive integer n_0 may be, the inequality $6+1/n+n+(-1)^n n \leq M$ holds for some $n > n_0$.

4-5.P5. Find all subsequential limits of (i) $x_n = n[1+(-1)^n]$ (ii) $x_n = n[6+(-1)^n]$.

4-5.P6. For a sequence $\{x_n\}$ of nonzero terms, if $\lim_{n\to\infty} x_n = \infty$, show that $\lim_{n\to\infty} 1/x_n = 0$. For a sequence $\{x_n\}$ of positive terms, if $\lim_{n\to\infty} 1/x_n = 0$, show that $\lim_{n\to\infty} x_n = \infty$.

4-5.P7. Let $\{x_n\}$ be a sequence of nonzero terms such that $0 < \limsup_{n\to\infty} 1/x_n < \infty$. If $\liminf_{n\to\infty} x_n > 0$, show that $\liminf_{n\to\infty} x_n = 1/(\limsup_{n\to\infty} 1/x_n)$.

4-5.P8. Give an example of a sequence $\{x_n\}$ of nonzero terms such that $\{1/x_n\}$ is bounded above and $\limsup_{n\to\infty} 1/x_n > 0$, but $\liminf_{n\to\infty} x_n \neq 1/(\limsup_{n\to\infty} 1/x_n)$.

4-5.P9. Prove Prop.4-5.5.

4-5.P10. Prove Prop.4-5.6.

4-6 Multiplication of Series

One of the simplest results about infinite series is that the series $\sum_{k=0}^{\infty}(a_k+b_k)$ converges to the sum of the separate sums $A=\sum_{k=0}^{\infty}a_k$ and $B=\sum_{k=0}^{\infty}b_k$. It is natural to think of the first mentioned series as the sum of the other two. Regarding products however, one would not expect that $\sum_{k=0}^{\infty}(a_kb_k)$ has anything to do with the product of the separate sums, since $a_0b_0+a_1b_1 \neq (a_0+a_1)(b_0+b_1)$. The limit of the sequence $(\sum_{k=0}^{n}a_k)(\sum_{k=0}^{n}b_k)$ is, of course, equal to AB. If we could write $(\sum_{k=0}^{n}a_k)(\sum_{k=0}^{n}b_k)$ as a partial sum of a suitable series [cf. 4-6.P1], we could call it the "product" of the series and its sum will be the product AB.

However, there is another way to arrive at a series which could reasonably be called the "product" of the series. If one attempts to compute the product $(\sum_{k=0}^{\infty}a_kz^k)(\sum_{k=0}^{\infty}b_kz^k)$ on the *assumption* that the terms obtained can be *freely rearranged* and grouped, one gets

$$\begin{aligned}(\sum_{k=0}^{\infty}a_kz^k)(\sum_{k=0}^{\infty}b_kz^k) &= (a_0+a_1z+a_2z^2+\cdots)(b_0+b_1z+b_2z^2+\cdots)\\ &= a_0b_0+(a_0b_1+a_1b_0)z+(a_0b_2+a_1b_1+a_2b_1)z^2+\cdots\\ &= \sum_{k=0}^{\infty}(\sum_{j=0}^{k}a_jb_{k-j})z^k.\end{aligned}$$

Upon substituting $z=1$ in this, one is led to expect that if the assumption about freely rearranging terms can be justified, then $\sum_{k=0}^{\infty}(\sum_{j=0}^{k}a_jb_{k-j})$ should be equal to the product $(\sum_{k=0}^{\infty}a_k)(\sum_{k=0}^{\infty}b_k)$. In fact, it turns out that the assumption can be justified if the series $\sum_{k=0}^{\infty}a_k$ and $\sum_{k=0}^{\infty}b_k$ both converge absolutely. What is more, it turns out that $\sum_{k=0}^{\infty}(\sum_{j=0}^{k}a_jb_{k-j})$ converges to the product $(\sum_{k=0}^{\infty}a_k)(\sum_{k=0}^{\infty}b_k)$ if *one* among the two series *converges absolutely*, although we do not claim that the terms can be freely rearranged in this case! It is this form of the result, due to Mertens, that we shall prove below. But before doing so, we present an example to show that, without absolute convergence, the series $\sum_{k=0}^{\infty}(\sum_{j=0}^{k}a_jb_{k-j})$ need not even converge.

Example. Let $a_k=b_k=(-1)^k(k+1)^{-1/2}$. By the Leibnitz Test, the series $\sum_{k=0}^{\infty}a_k$ converges; and since $(k+1)^{-1/2}>(k+1)^{-1}$ while $\sum_{k=0}^{\infty}(k+1)^{-1}$ diverges, it follows that the series does not converge absolutely. Now,

$$\begin{aligned}\sum_{j=0}^{k}a_jb_{k-j} &= \sum_{j=0}^{k}[(-1)^j(j+1)^{-1/2}(-1)^{k-j}((k-j)+1)^{-1/2}]\\ &= (-1)^k\sum_{j=0}^{k}[(j+1)^{-1/2}(k-j+1)^{-1/2}].\end{aligned}$$

Since $(j+1)(k-j+1)=(\frac{k}{2}+1)^2-(\frac{k}{2}-j)^2\leq(\frac{k}{2}+1)^2$, we have

$$[(j+1)^{-1/2}(k-j+1)^{-1/2}] \geq \frac{2}{k+2}$$

and hence

$$\sum_{j=0}^{k}[\,(j+1)^{-1/2}((k-j+1)^{-1/2}] \geq \,(k+1)\frac{2}{k+2}\,.$$

This shows that the terms of the series $\sum_{k=0}^{\infty}(\sum_{j=0}^{k} a_j b_{k-j})$ do not converge to 0 and hence that the series does not converge.

4-6.1. Definition. *The* **Cauchy product** *of the series* $\sum_{k=0}^{\infty} a_k$ *with the series* $\sum_{k=0}^{\infty} b_k$ *is the series* $\sum_{k=0}^{\infty} c_k$, *where* $c_k = \sum_{j=0}^{k} a_j b_{k-j}$.

4-6.2. Mertens' Theorem. *Suppose* $\sum_{k=0}^{\infty} a_k = A$, $\sum_{k=0}^{\infty} b_k = B$ and $c_k = \sum_{j=0}^{k} a_j b_{k-j}$. *If either* $\sum_{k=0}^{\infty} a_k$ *or* $\sum_{k=0}^{\infty} b_k$ *converges absolutely, then their Cauchy product* $\sum_{k=0}^{\infty} c_k$ *converges and has sum AB.*

Proof. Let A_n, B_n, C_n be the partial sums $\sum_{k=0}^{n} a_k$, $\sum_{k=0}^{n} b_k$, $\sum_{k=0}^{n} c_k$ respectively. Then

$$\lim_{n\to\infty} A_n = A \quad \text{and} \quad \lim_{n\to\infty} B_n = B \tag{1}$$

and we need to prove that

$$\lim_{n\to\infty} C_n = AB. \tag{A}$$

We begin by showing that, for every integer $n \in \mathbb{N}\cup\{0\}$,

$$C_n = A_n B + \sum_{k=0}^{n} a_k (B_{n-k} - B). \tag{2}$$

For $n = 0$, this asserts that $c_0 = a_0 B + a_0(b_0 - B)$, which is true. Assuming it to be true for some integer $n \geq 0$, we have

$$\begin{aligned} C_{n+1} = C_n + c_{n+1} &= A_n B + \sum_{k=0}^{n} a_k (B_{n-k} - B) + \sum_{k=0}^{n+1} a_k b_{n+1-k} \\ &= A_n B + \sum_{k=0}^{n} a_k (B_{n-k} - B) + \sum_{k=0}^{n} a_k b_{n+1-k} + a_{n+1} b_0 \\ &= A_n B + \sum_{k=0}^{n} a_k (B_{n-k} - B + b_{n+1-k}) + a_{n+1}(b_0 - B) + a_{n+1} B \\ &= (A_n + a_{n+1})B + \sum_{k=0}^{n} a_k (B_{n+1-k} - B) + a_{n+1}(b_{n+1-(n+1)} - B) \\ &= A_{n+1} B + \sum_{k=0}^{n+1} a_k (B_{n+1-k} - B). \end{aligned}$$

This completes the proof of (2) by induction.

By (1), we have $\lim_{n\to\infty} A_n B = AB$. From this and (2), it follows that (A) will hold if

$$\lim_{n\to\infty} \sum_{k=0}^{n} a_k (B_{n-k} - B) = 0. \tag{B}$$

Therefore we need establish only (B).

By hypothesis, one among $\sum_{k=0}^{\infty}|a_k|$ and $\sum_{k=0}^{\infty}|b_k|$ converges and we may, without loss of generality, take it to be the former. If its sum is 0 there is nothing to prove, because in this case, every $|a_k| = 0$ and hence every $a_k = 0$ and every $c_k = 0$, so that $A = 0 = C$, which implies that $C = AB$. So, we assume $0 < \sum_{k=0}^{\infty}|a_k| = \alpha$, say.

Now consider any $\varepsilon > 0$. We have to show that there exists $N \in \mathbb{N}$ such that

$$n \geq N \;\Rightarrow\; \left|\sum_{k=0}^{n} a_k(B_{n-k} - B)\right| < \varepsilon.$$

By (1), there exists N_1 such that

$$m \geq N_1 \Rightarrow |B_m - B| < \varepsilon/2\alpha. \tag{3}$$

Since any convergent sequence is bounded, there exists $M > 0$ such that

$$|B_m - B| < M \quad \text{for every } m \in \mathbb{N}. \tag{4}$$

Since $a_k \to 0$, therefore there exists N_2 such that

$$k \geq N_2 \;\Rightarrow\; |a_k| < \varepsilon/2N_1M. \tag{5}$$

Now let $N = N_1 + N_2$ and consider any $n \geq N$. Then firstly, $n \geq N_1 + 1$, and secondly, for $k \leq n - N_1$ we have $n - k \geq N_1$. Therefore by (3),

$$\left|\sum_{k=0}^{n-N_1} a_k(B_{n-k} - B)\right| < (\varepsilon/2\alpha)\sum_{k=0}^{n-N_1}|a_k| \leq \tfrac{\varepsilon}{2}. \tag{6}$$

Furthermore, for $k \geq n - N_1 + 1$ we have $k \geq N - N_1 + 1 = N_2 + 1$. Therefore by (4) and (5),

$$\begin{aligned}\left|\sum_{k=n-N_1+1}^{n} a_k(B_{n-k} - B)\right| &\leq M\sum_{k=n-N_1+1}^{n}|a_k| < M\sum_{k=n-N_1+1}^{n}(\varepsilon/2N_1M)\\ &= (\varepsilon/2N_1)(n-(n-N_1)) = \tfrac{\varepsilon}{2}.\end{aligned} \tag{7}$$

It follows from (6), (7) and Prop.1-11.2 that we have $\left|\sum_{k=0}^{n} a_k(B_{n-k} - B)\right| < \varepsilon$ whenever $n \geq N$. This establishes (B) and, as already noted, thereby completes the proof. ☺

Problem Set 4-6

4-6.P1. Suppose $\sum_{k=0}^{\infty} a_k = A$, $\sum_{k=0}^{\infty} b_k = B$, $c_0 = a_0b_0$ and $c_k = a_kb_k + b_k\sum_{j=0}^{k-1}a_j + a_k\sum_{j=0}^{k-1}b_j$ for $k > 0$. Show that $\sum_{k=0}^{\infty} c_k$ converges to AB.

4-6.P2. Let $a_k = (-1)^k/(k+1)$, so that $\sum_{k=0}^{\infty} a_k$ is conditionally convergent. Using the Leibnitz Test, show that its Cauchy product with itself is convergent nevertheless.

Chapter 5
Sequences and Series of Functions

5-1 Convergence of a Sequence of Functions

Recall the convergence result $\lim_{n\to\infty} z^n = 0$ when $|z| < 1$. This can be stated in terms of the sequence of functions $\{f_n\}$ defined by $f_n(z) = z^n$ on the domain $|z| < 1$ in the following manner: For each z in the domain, the sequence of values $\{f_n(z)\}$ converges to the limit 0. Here we may take z to be real or complex. For convenience, we shall take z to be real. In terms of the definition of limit, the full statement would be that:

$$\text{for every } \varepsilon > 0, \text{ there exists } n_0 \text{ such that } n \geq n_0 \Rightarrow |z^n| < \varepsilon.$$

When this was proved in Prop.3-1.7, the existence of n_0 for nonzero z was obtained by showing that any integer greater than $1/b\varepsilon$, where

$$b = \frac{1}{|z|} - 1 = \frac{1-|z|}{|z|},$$

serves the purpose. In other words, $n_0 \geq \frac{1}{\varepsilon}\frac{|z|}{(1-|z|)}$. Note how z is explicitly involved in the description of n_0. This has the consequence that this integer n_0 which works for one z need not work for another. Is there another way to select an integer n_0 which works simultaneously for *every* number z satisfying $0 < |z| < 1$? The answer is negative, as we now argue. If such an integer n_0 were to exist for $\varepsilon = \frac{1}{3}$ say, then the number $z = (\frac{1}{2})^{1/n_0}$, which satisfies $0 < |z| < 1$, would also have to satisfy $|z^{n_0}| < \frac{1}{3}$. But $|z^{n_0}| = \frac{1}{2} > \frac{1}{3}$. Therefore when $\varepsilon = \frac{1}{3}$, no integer n_0 can have the property that $n \geq n_0 \Rightarrow |z^n| < \varepsilon$ simultaneously for *every* number z satisfying $0 < |z| < 1$. There is a name for the phenomenon when such a common integer indeed exists, and also for the more general phenomenon when a common integer may or may not exist. We give examples after stating the definitions formally.

In the remainder of this Chapter, functions will be assumed to be real or complex valued, unless specified otherwise.

5-1.1. Definition. *A sequence of functions* $\{f_n\}$ *with domain* S *is said to* **converge uniformly to** *a function* f (*with domain* S) *if, for every* $\varepsilon > 0$, *there exists* n_0 *such that*

$$|f_n(s) - f(s)| < \varepsilon \qquad \textit{for every } s \in S \textit{ and every } n \geq n_0.$$

While working with functions whose domain is intended to be a subset of $\mathbb{R}$, we shall denote a general element of the domain by x; if however the domain is a subset of $\mathbb{C}$ that may or may not be a subset of $\mathbb{R}$, we shall use the letter z.

Examples. (a) Let $0 < a < 1$ and $\{f_n\}$ be the sequence of functions on $S = [0, a]$ defined by $f_n(x) = x^n$. Then $\{f_n\}$ converges uniformly to the constant function $f(x) = 0$. Consider any $\varepsilon > 0$. Since $0 < a < 1$, therefore by Prop.3-1.7, we have $\lim\limits_{n\to\infty} a^n = 0$. This means there exists n_0 such that $n \geq n_0 \Rightarrow |a^n| < \varepsilon$. Now,

$$x \in S \Rightarrow 0 \leq x^n \leq a^n \Rightarrow |x^n| \leq |a^n|.$$

Therefore $n \geq n_0 \Rightarrow |x^n| < \varepsilon$. Thus $\{f_n\}$ converges uniformly to the constant function $f(x) = 0$.

(b) Let $\{f_n\}$ be the sequence of functions on $S = (\frac{1}{2}, 1)$ defined by $f_n(x) = x^n$. Then $\{f_n\}$ does *not* converge to any function uniformly. If it were to converge uniformly to some function f, then it would be a consequence that, for every $x \in S$, the real sequence $\{f_n(x)\}$, i.e. $\{x^n\}$, converges to $f(x)$. By Prop.3-1.7, the function f would have to have the value 0 everywhere on S. Therefore, for every $\varepsilon > 0$, there would exist n_0 such that

$$|x^n - 0| = |x^n| < \varepsilon \quad \text{for all } x \in S \text{ and all } n \geq n_0.$$

As argued above, no such n_0 exists for $\varepsilon = \frac{1}{3}$, because when $x = (\frac{1}{2})^{1/n_0}$, we have $x \in S$ but $|x^{n_0}| = \frac{1}{2} > \frac{1}{3}$.

As noted in the second Example above, it is a consequence of uniform convergence that, for every $x \in S$, the real (or complex) sequence $\{f_n(x)\}$ converges to $f(x)$. This situation has a name to distinguish it from the more stringent one of uniform convergence. We formulate it below.

5-1.2. Definition. *A sequence of functions $\{f_n\}$ with domain S is said to* **converge pointwise to** *a function f (with domain S) if, for every $s \in S$ and every $\varepsilon > 0$, there exists n_0 such that*

$$|f_n(s) - f(s| < \varepsilon \qquad \textit{for every } n \geq n_0.$$

Remark. There cannot be two different functions f to which a sequence $\{f_n\}$ converges, whether pointwise or uniformly. Therefore one can unambiguously refer to f as the limit of the sequence $\{f_n\}$ and write $f = \lim\limits_{n\to\infty} f_n$. Whether the convergence is pointwise or uniform has to be understood from the context.

The sequence $\{f_n\}$, where $f_n(x) = x^n$ on the domain $[0, 1]$, converges pointwise to the *discontinuous* function f, which has the value 0 when $0 \leq x < 1$ and has the value 1 when $x = 1$. The convergence is not uniform and this is no coincidence, as the following result shows.

5-1.3. Theorem. *Let X be a subset of $\mathbb{R}$ or $\mathbb{C}$ and let $\{f_n\}$ be a sequence of continuous functions on X converging uniformly to f. Then f is continuous.* Briefly put: *the uniform limit of a sequence of continuous functions is continuous.*

Proof. Let a be any point of X whatsoever and consider any $\varepsilon > 0$. Since $\{f_n\}$ converges to f uniformly, there exists some $n_0 \in \mathbb{N}$ such that

$$|f_n(z) - f(z)| < \tfrac{\varepsilon}{3} \qquad \text{for every } z \in X \text{ and every } n \geq n_0. \tag{1}$$

By continuity of f_{n_0}, there exists some $\delta > 0$ such that

$$|z - a| < \delta \Rightarrow |f_{n_0}(z) - f_{n_0}(a)| < \tfrac{\varepsilon}{3}. \tag{2}$$

Now consider any $z \in X$ such that $|z - a| < \delta$. By (2), we have

$$|f_{n_0}(z) - f_{n_0}(a)| < \tfrac{\varepsilon}{3} \tag{3}$$

and by (1), we have

$$|f_{n_0}(z) - f(z)| < \tfrac{\varepsilon}{3} \quad \text{and} \quad |f_{n_0}(a) - f(a)| < \tfrac{\varepsilon}{3}. \tag{4}$$

It follows from (3) and (4) that

$$\begin{aligned} |f(z) - f(a)| &= |\big(f(z) - f_{n_0}(z)\big) + \big(f_{n_0}(z) - f_{n_0}(a)\big) + \big(f_{n_0}(a) - f(a)\big)| \\ &\leq |f(z) - f_{n_0}(z)| + |f_{n_0}(z) - f_{n_0}(a)| + |f_{n_0}(a) - f(a)| \\ &< \tfrac{\varepsilon}{3} + \tfrac{\varepsilon}{3} + \tfrac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Thus $|z - a| < \delta \Rightarrow |f(z) - f(a)| < \varepsilon$. Since the existence of such a positive δ has been proved for any $\varepsilon > 0$ and any point a of X, the continuity of f at any point of X is established. ☺

A point worth noting about the preceding proof is that it actually shows that, if each f_n is continuous at one particular $a \in X$, then the uniform limit f is continuous there.

The above Theorem can be used towards an alternative proof that the sequence in Example (b) above is not uniformly convergent. The sequence $\{f_n\}$ consists of the restrictions to the interval $(\frac{1}{2}, 1)$ of the functions g_n on $(\frac{1}{2}, 1]$ described by the same formula. Since $\{g_n(1)\}$ is a convergent sequence in $\mathbb{R}$, it follows [see 5-1.P2] that, if $\{f_n\}$ were to be a uniformly convergent sequence, then so would $\{g_n\}$. But the functions g_n are continuous and the limit, being 0 on $(\frac{1}{2}, 1)$ and 1 at 1, is discontinuous. By the above Theorem, $\{g_n\}$ therefore cannot be uniformly convergent and hence nor can $\{f_n\}$. When the limit is a continuous function, other methods have to be used to in order to show that convergence is not uniform. See Examples 5-1.6 later in this Section.

The true significance of Theorem 5-1.3 is that it allows us to deduce the continuity of a function which is accessible to us only as the uniform limit of a sequence of continuous functions and is not known by means of an explicit formula. Consider any bounded continuous function h with domain $\mathbb{R}$, such as

$$h(x) = x/(1+x^2) \quad \text{or} \quad h(x) = \min\{x-[x], 1-(x-[x])\},$$

where [ ] denotes the greatest integer function [see 2-1.P5]. Let $f_n(x) = \sum_{k=0}^{n} a^k h(b^k x)$, where $|a| < 1$ and b is any real number. For example, if

$$h(x) = x/(1+x^2),$$

the function f_2 would be

$$f_2(x) = x/(1+x^2) + abx/(1+b^2x^2) + a^2b^2x/(1+b^4x^2).$$

It is a consequence of the Cauchy criterion of uniform convergence below [see Theorem 5-1.5] that this sequence converges uniformly. We leave it to the reader to ponder over what a general formula for $f_n(x)$ and for $\lim_{n\to\infty} f_n(x)$ could look like! The point here is that once the uniform convergence is proved, the continuity of the limit function will follow from Theorem 5-1.3 even if we do not know a formula for it that is easy to manipulate.

5-1.4. Definition. *A sequence $\{f_n\}$ of functions with domain S is said to be a* **uniformly Cauchy sequence** *if, for every $\varepsilon > 0$, there exists a natural number n_0 such that*

$$|f_n(s) - f_m(s)| < \varepsilon \quad \text{for all } s \in S, \text{ all } m \geq n_0 \text{ and all } n \geq n_0.$$

5-1.5. Theorem. Cauchy criterion of uniform convergence. *A sequence $\{f_n\}$ of functions with domain S is uniformly convergent if, and only if, it is a uniformly Cauchy sequence.*

Proof. First suppose that $\{f_n\}$ is uniformly convergent. With a view to showing that it is uniformly Cauchy, consider any $\varepsilon > 0$. Then $\frac{\varepsilon}{2} > 0$ and, by the definition of uniform convergence, there must exist some natural number n_0 such that

$$|f_n(s) - f(s)| < \tfrac{\varepsilon}{2} \ \ \forall\, s \in S \text{ whenever } n \geq n_0.$$

Let m and n be any natural numbers such that $m \geq n_0$ and $n \geq n_0$. Then

$$|f_n(s) - f(s)| < \tfrac{\varepsilon}{2} \text{ and also } |f_m(s) - f(s)| < \tfrac{\varepsilon}{2} \ \ \forall\, s \in S.$$

It follows by the triangle inequality that

$$|f_n(s) - f_m(s)| \le |f_n(s) - f(s)| + |f_m(s) - f(s)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \ \forall\ s \in S.$$

This shows that $\{f_n\}$ is a uniformly Cauchy sequence.

For the converse, suppose $\{f_n\}$ is any uniformly Cauchy sequence. Then the real (or complex) sequence of values $\{f_n(s)\}$ at any $s \in S$ is a Cauchy sequence. Therefore it converges to some limit [by Theorem 3-5.2]. Define the function f on the domain S by $f(s) = \lim_{n\to\infty} f_n(s)$. We shall show that the sequence $\{f_n\}$ converges to f uniformly. Consider any $\varepsilon > 0$. Let $n_0 \in \mathbb{N}$ be such that

$$|f_n(s) - f_m(s)| < \frac{\varepsilon}{2} \ \forall\ s \in S \quad \text{whenever} \quad m \ge n_0 \text{ and } n \ge n_0.$$

Since this holds whenever $m \ge n_0$, it follows upon taking the limit as $m\to\infty$ [and using results from Section 3-2] that

$$|f_n(s) - f(s)| < \varepsilon \ \forall\ s \in S \quad \text{whenever} \quad n \ge n_0.$$

Thus the sequence $\{f_n\}$ converges to f uniformly. ☺

Returning to the question of uniform convergence of $f_n(x) = \sum_{k=0}^{n} a^k h(b^k x)$, where $|a| < 1$ and b is any real number, we note that

$$|f_n(x) - f_m(x)| = \left|\sum_{k=m+1}^{n} a^k h(b^k x)\right| \le \sum_{k=m+1}^{n} |a^k|\, M,$$

where M is an upper bound for $|h|$. Since $|a| < 1$, the geometric series $\Sigma|a|^k$ converges and the right side of the preceding inequality can be made smaller than any $\varepsilon > 0$ by taking both m and n greater than or equal to some n_0. This makes the sequence $\{f_n\}$ uniformly Cauchy, and hence uniformly convergent by the preceding Theorem.

5-1.6. Examples. (a) We examine the sequence $\{f_n\}$ given by

$$f_n(x) = \frac{nx}{1 + n^2x^2}$$

for uniform convergence on $\mathbb{R}$. Since $f_n(0) = 0$ for every n, we have $\lim_{n\to\infty} f_n(0) = 0$. For nonzero x, we have

$$f_n(x) = \frac{1}{\dfrac{1}{nx} + nx},$$

so that $\lim_{n\to\infty} f_n(x) = 0$ even if $x \ne 0$. Therefore the sequence converges pointwise to the identically zero function. To check whether it converges uniformly, we examine the function

$$|f_n(x) - 0| = |f_n(x)| = \frac{|nx|}{1 + n^2x^2} .$$

Select $x = \frac{1}{n}$. We find that $|f_n(x)| = \frac{1}{2}$, which shows that, for $\varepsilon = \frac{1}{2}$, the inequality $|f_n(x)| < \varepsilon$ cannot hold for all $x \in \mathbb{R}$. Therefore the convergence is not uniform. Note that the argument applies even if the domain of each f_n is taken to be some interval containing 0 (among other numbers) instead of $\mathbb{R}$.

(b) Let $f_n(x) = nx^n$ for $0 < x < 1$. We shall show that $\{f_n\}$ converges pointwise to the limit function $f(x) = 0$ for every $x \in (0, 1)$, but not uniformly. We shall start with the following inequality [see 1-11.P11 or apply the binomial theorem]:

When $n \in \mathbb{N}$ and $0 < h \in \mathbb{R}$, we have

$$(1+h)^n \geq 1+nh+\tfrac{1}{2}n(n-1)h^2.$$

For $x \in (0, 1)$, let $h = \frac{1}{x} - 1$. Then $h > 0$ and

$$0 < nx^n = n(1+h)^{-n} \leq n(1+nh+\tfrac{1}{2}n(n-1)h^2)^{-1} = (\frac{1}{n}+h+\tfrac{1}{2}(n-1)h^2)^{-1}$$
$$\leq (\tfrac{1}{2}(n-1)h^2)^{-1} \quad \text{for } n > 1.$$

But this has limit 0 and hence $\lim\limits_{n\to\infty} nx^n = 0$. To check for uniform convergence, we examine the function $|f_n(x) - f(x)| = |f_n(x)| = nx^n$. For $x = n^{-1/n} \in (0, 1)$, its value is 1. Therefore, when $\varepsilon < 1$, it is not possible to have $|f_n(x) - f(x)| < \varepsilon$ for every $x \in (0, 1)$.

An alternative to starting with the inequality quoted above is this: Let $1 < a < \frac{1}{x}$. Since $n^{1/n} \to 1$, there exists N such that

$$n \geq N \Rightarrow n^{1/n} < a \Rightarrow 0 < n^{1/n}x < ax \Rightarrow 0 < nx^n < (ax)^n$$

and $\lim\limits_{n\to\infty} (ax)^n = 0$ since $0 < ax < 1$.

(c) Let $f_n(x) = 1/(nx+1)$, $x \in [0, b]$. This sequence converges pointwise to the function f for which $f(x) = 1$ if $x = 0$ and $f(x) = 0$ if $x > 0$. Since each f_n is continuous but the limit f is not, it follows by Theorem 5-1.3 that the convergence is not uniform.

If we were to consider the restrictions to $(0, b]$, the limit function would be 0 everywhere and hence continuous; therefore Theorem 5-1.3 would not be applicable. However, we could still prove the convergence to be *not uniform* by proceeding as follows: Suppose the sequence of restrictions were to converge uniformly. Then, for any $\varepsilon > 0$, there would exist $n_1 \in \mathbb{N}$ such that

$$n \geq n_1, x \in (0, b] \Rightarrow |f_n(x) - f(x)| < \varepsilon.$$

Since $\lim\limits_{n\to\infty} f_n(0) = 1$, there exists $n_2 \in \mathbb{N}$ such that

$$n \geq n_2 \Rightarrow |f_n(0) - f(0)| < \varepsilon.$$

The integer $n_3 = \max\ \{n_1, n_2\}$ would then have the property that

$$n \geq n_3,\ x \in [0, b] \Rightarrow |f_n(x) - f(x)| < \varepsilon.$$

Thus the sequence $\{f_n\}$ of (unrestricted) functions with domain $[0, b]$ would converge uniformly; but, as already argued on the basis of Theorem 5-1.3, this does not happen. So, the sequence of restrictions to $(0, b]$ also does not converge uniformly. This conclusion could also have been obtained by using the result of 5-1.P2.

5-1.7. Proposition. *If a sequence $\{f_n\}$ of bounded functions with domain S converges uniformly to f, then f is also bounded.* Briefly put: *the uniform limit of a sequence of bounded functions is bounded.*

Proof. By definition of uniform convergence [applied with $\varepsilon = 1$], there exists n_0 such that

$$|f_n(x) - f(x)| < 1 \qquad \text{whenever } x \in S \text{ and } n \geq n_0.$$

In particular,

$$|f_{n_0}(x) - f(x)| < 1 \quad \text{whenever } x \in S. \tag{1}$$

Now f_{n_0} is bounded and hence there exists some $M > 0$ such that

$$|f_{n_0}(x)| \leq M \quad \text{whenever } x \in S. \tag{2}$$

It follows from (1) and (2) that

$$|f(x)| \leq 1 + M \quad \text{whenever } x \in S.$$

Thus f is bounded. ☺

A sequence of real valued functions $\{f_n\}$ is called *decreasing* if, for each x in their domain, the real sequence $\{f_n(x)\}$ is decreasing.

5-1.8. Dini's Theorem. *Suppose a decreasing sequence $\{f_n\}$ of continuous functions on a closed bounded interval X has a pointwise limit f which is continuous. Then the convergence is uniform.*

Proof. It is sufficient to show that the decreasing sequence $\{f_n - f\}$ of nonnegative continuous functions, which converges to 0, does so uniformly. Suppose, if possible, that the convergence is not uniform. Then there exists some $\varepsilon > 0$ such that, for every $k \in \mathbb{N}$, there exist $n_k \in \mathbb{N}$ and $x_k \in X$ such that $n_k \geq k$ and

$$(f_{n_k} - f)(x_k) > \varepsilon. \tag{1}$$

By the Bolzano-Weierstrass Theorem 3-4.7, $\{x_k\}$ has a subsequence $\{x_{k(j)}\}$ that converges to some limit x and, since X is closed, we must have $x \in X$. Denote $x_{k(j)}$ by y_j and $(f_{n_{k(j)}} - f)$ by g_j. Then it follows from (1) that

$$g_j(y_j) > \varepsilon \text{ whenever } j \in \mathbb{N}. \tag{2}$$

Also, $\lim_{j\to\infty} g_j(x) = 0$. Therefore there exists $J \in \mathbb{N}$ such that.

$$0 \le g_J(x) < \tfrac{\varepsilon}{2}. \tag{3}$$

By continuity of g_J at x, there exists $\delta > 0$ such that

$$|y - x| < \delta \Rightarrow |g_J(y) - g_J(x)| < \tfrac{\varepsilon}{2}. \tag{4}$$

Since $\lim_{j\to\infty} y_j = x$, there exists some $J' \in \mathbb{N}$ such that

$$j \ge J' \Rightarrow |y_j - x| < \delta. \tag{5}$$

Now, let $j \ge \max\{J, J'\}$. Then $|y_j - x| < \delta$ by (5) and hence

$$\begin{aligned} 0 \le g_j(y_j) \le g_J(y_j) &= g_J(y_j) - g_J(x) + g_J(x) \\ &< \tfrac{\varepsilon}{2} + g_J(x) \quad \text{by (4)} \\ &< \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon \quad \text{by (3).} \end{aligned}$$

But this contradicts (2). ☺

Problem Set 5-1

5-1.P1. For the following values of ε, show that no integer n_0 can exist for which $|z^n| < \varepsilon$ whenever $|z| < 1$ and $n \ge n_0$: (a) $\varepsilon = \frac{1}{4}$ (b) $\varepsilon = \frac{2}{3}$.

5-1.P2. Suppose $S = A \cup B$, $\{f_n\}$ is a sequence of functions on S and the restrictions of f_n to A and B form uniformly convergent sequences. Show that $\{f_n\}$ is uniformly convergent.

5-1.P3. (a) Discuss the uniform convergence of $\{f_n\}$, where $f_n(x) = \dfrac{nx}{n+x}$ on $[0,\infty)$.

(b) Show that the sequence $\{f_n\}$, where $f_n(x) = nx/(1+nx)$, $0 \le x \le 1$, is not uniformly convergent.

5-1.P4. Show that the sequence $\{f_n\}$, where $f_n(x) = \dfrac{x}{1+nx^2}$, $x \in \mathbb{R}$, converges uniformly on any closed interval.

5-1.P5. Show that the sequence $\{f_n\}$, where $f_n(x)$ is given by

$$f_n(x) = n^2x \quad \text{when} \quad 0 \le x \le \tfrac{1}{n} \qquad \text{and} \qquad 0 \quad \text{when} \quad \tfrac{1}{n} < x \le 1,$$

is not uniformly convergent on $[0, 1]$.

5-1.P6. Show that the sequence $\{f_n\}$, where $f_n(x)$ is given by

$$f_n(x) = n^2x \quad \text{when} \quad 0 \le x \le \tfrac{1}{n}, \quad -n^2x + 2n \quad \text{when} \quad \tfrac{1}{n} < x \le \tfrac{2}{n} \quad \text{and} \quad 0 \quad \text{when} \quad \tfrac{2}{n} < x \le 1,$$

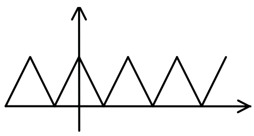
is not uniformly convergent on $[0, 1]$. Why does Dini's Theorem not apply?

5-1.P7. (a) Give an example of a sequence $\{f_n\}$ of continuous functions on $[0, 1]$ converging pointwise to a continuous function f and a sequence $\{x_n\}$ in $[0, 1]$ converging to a limit x such that $f_n(x_n)$ does not converge to $f(x)$.
(b) If $\{f_n\}$ is a sequence of continuous functions on $[0, 1]$, converging uniformly to f and $\{x_n\}$ is a sequence in the domain converging to x, show that $\lim_{n\to\infty} f_n(x_n) = f(x)$.

5-1.P8. Show that the sequence $\{f_n\}$, where $f_n(x) = n^2x(1-x^2)^n$ on the domain $0 \le x \le 1$, is not uniformly convergent.

5-1.P9. Find $\lim_{m\to\infty}\left[\lim_{n\to\infty}\left(\tfrac{1}{4} + \tfrac{3}{4}\cos(\pi m!x)\right)^n\right]$, assuming π is irrational [proved in App 1 to Ch 8]. The properties of $\cos(\pi\theta)$ that are needed here are consequences of what has been discussed later in this Chapter, but may be used freely now.

5-1.P10. Let ϕ be a continuous function on $\mathbb{R}$ that takes the value 1 at every integer and has absolute value less than 1 everywhere else. An example of such a function is $\cos^2 \pi x$ (to be defined formally later in this Chapter) or, alternatively, the one illustrated in the figure, i.e. $\phi(x) = 1 - 2\cdot\min\{x - [x], 1 - (x - [x])\}$, where the brackets [ ] denote the integer part function; see 2-1.P5. Let $g(x) = \lim_{n\to\infty}(\phi(5!x))^n$. Show that $g(x) = 1$ if x is a rational number $p/5!$ and $g(x) = 0$ otherwise. Is g the uniform limit of $(\phi(5!x))^n$?



5-1.P11. Let a be a positive number such that the function $f:(0, \infty)\to\mathbb{R}$ satisfies $|uf(u)| \le a$ everywhere. [Elementary examples would be $f(u) = 1/(1+u)$, $1/(1+u^2)$, and so on; Calculus methods for showing that $f(u) = (\sin u^2)/u^2$, u^pe^{-u} $(p > -1)$ have the property will be justified later.]
(a) Show that the sequence $f_n(x) = n^{7/8}xf(nx)$ converges uniformly to 0 on $(0, \infty)$.
(b) What other numbers can be taken in place of 7/8 in part (a)?
(c) Write expressions for $f_n(x)$ when $f(x)$ is given by the examples mentioned above.

(d) If $g_n(x) = nxf(nx)$ converges to 0 uniformly on $(0,\infty)$, then show that f is zero everywhere.
(e) Show that $f_n(x) = x\sqrt{n}\cdot(n^8x^8 - 17n^2x^2 - 200)/(n^{10}x^{10} + n^4x^4 + 7)$ approaches 0 uniformly on $[0,\infty)$.

5-2 Series of Functions and Power Series

Just as a sequence $\{z_k\}$ of numbers gives rise [see Def.4-1.1] to a series Σz_k, a sequence of functions $\{f_k\}$ on a common domain S also gives rise to a series Σf_k of functions. The concepts of uniform, pointwise and absolute convergence carry over to series of functions as follows.

5-2.1. Definition. *Given a sequence* $\{f_k\}$ *of functions on a domain* S, *the associated sequence* $\sum_{k=0}^{n} f_k$, $n \in \mathbb{N}$, *is called the sequence of* **partial sums of the series** $\sum_{k=0}^{\infty} f_k$. *We say that the* **series** $\sum_{k=0}^{\infty} f_k$ **converges uniformly** *to a limit function* f *on* S *if the sequence of its partial sums converges uniformly to* f *in the sense of* Def.5-1.1. *Similarly*, *we say that the* **series** $\sum_{k=0}^{\infty} f_k$ **converges pointwise** *to a limit function* f *on* S *if the sequence* $\sum_{k=0}^{n} f_k$ *of its partial sums converges pointwise to* f *in the sense of* Def.5-1.2. *Whether pointwise or uniform*, *we refer to* f *as the* **sum of** *the series* $\sum_{k=0}^{\infty} f_k$ *and write this in symbols as* $\sum_{k=0}^{\infty} f_k = f$. *If the series* $\sum_{k=0}^{\infty} |f_k|$ *converges*, *we say that* $\sum_{k=0}^{\infty} f_k$ **converges absolutely**. *If it converges uniformly*, *we say that* $\sum_{k=0}^{\infty} f_k$ **converges uniformly absolutely**.

[For an instance of a series which converges uniformly and absolutely but not uniformly absolutely, see 5-2.P9.]

5-2.2. Theorem. Cauchy criterion for uniform convergence of series *The series* $\sum_{k=0}^{\infty} f_k$ *of functions on a domain* S *is uniformly* [resp. *uniformly absolutely*] *convergent if*, *and only if*, *for every* $\varepsilon > 0$, *there exists a natural number* n_0 *such that*

$$\Big|\sum_{k=m+1}^{n} f_k(s)\Big| \text{ [resp. } \sum_{k=m+1}^{n} |f_k(s)|] < \varepsilon \quad \textit{whenever} \quad n > m \geq n_0 \textit{ and } s \in S.$$

Proof. Let $S_n(s)$ denote the n^{th} partial sum $\sum_{k=0}^{n} f_k(s)$ [resp. $\sum_{k=0}^{n} |f_k(s)|$]. By Def.5-2.1, the series $\sum_{k=0}^{\infty} f_k$ [resp. $\sum_{k=0}^{\infty} |f_k|$] converges uniformly if, and only if the sequence $\{S_n\}$ converges uniformly. By the Cauchy criterion for uniform convergence of sequences, Theorem 5-1.5, this sequence converges uniformly if, and only if, it is a uniformly Cauchy sequence, i.e. if, and only if, for every $\varepsilon > 0$, there exists a natural number n_0 such that

$$|S_n(s) - S_m(s)| < \varepsilon \quad \text{whenever} \quad m \geq n_0,\, n \geq n_0 \text{ and } s \in S.$$

This is always true if $n = m$. When $n \neq m$, one of them must be greater and the above condition is equivalent to

$$|S_n(s) - S_m(s)| < \varepsilon \quad \text{whenever} \quad n > m \geq n_0 \text{ and } s \in S.$$

But by Prop. 1-11.2, $S_n(s) - S_m(s) = \sum_{k=m+1}^{n} f_k(s)$ [resp. $\sum_{k=m+1}^{n} |f_k(s)|$] when $n > m$. ☺

A more convenient result for purposes of application is the following:

5-2.3. Theorem. Weierstrass *M*-test. *Let $\{f_k\}$ be a sequence of functions on a domain S. If there is a sequence $\{M_k\}$ of nonnegative numbers such that*

$$\sum_{k=0}^{\infty} M_k \textit{ is convergent} \tag{1}$$

and $$\exists\, N \in \mathbb{N} \textit{ such that } |f_k(s)| \leq M_k \textit{ for every } s \in S \textit{ and } k \geq N, \tag{2}$$

then $\sum_{k=0}^{\infty} f_k$ is uniformly absolutely convergent.

Proof. Consider any $\varepsilon > 0$. In view of (1), there is a positive integer n_0 (depending upon ε) such that

$$\sum_{k=m+1}^{n} M_k < \varepsilon \quad \text{whenever} \quad n > m \geq n_0.$$

Let $S_n(s)$ denote the n^{th} partial sum of $\sum_{k=0}^{\infty} |f_k(s)|$. If $n_1 = \max\,\{N, n_0\}$, then it follows from (2) that, whenever $n > m \geq n_1$ and $s \in S$, we have

$$|S_n(s) - S_m(s)| = \sum_{k=m+1}^{n} |f_k(s)| \leq \sum_{k=m+1}^{n} M_k < \varepsilon.$$

By the Cauchy criterion, $\sum_{k=0}^{\infty} f_k$ is uniformly absolutely convergent. ☺

If the series obtained by restricting the functions f_k in $\sum_{k=0}^{\infty} f_k$ to a subset T of the domain is uniformly convergent, we speak of the series $\sum_{k=0}^{\infty} f_k$ as being uniformly convergent **on *T***. If condition (2) in the Weierstrass *M*-test is altered by replacing "$\forall\ s \in S$" by "$\forall\ s \in T$", we get uniform and absolute convergence on the subset T.

5-2.4. Examples. (a) The series

$$\sum_{k=1}^{\infty} 1/(k^2 + z^2)$$

converges uniformly absolutely on $\{z \in \mathbb{C} : 1 < |z| < 2\}$. Indeed, for $k \geq 3$ and $1 < |z| < 2$, we have

$$|k^2 + z^2| \geq k^2 - |z|^2 > k^2 - 4 \geq \tfrac{1}{2}k^2,$$

whence $$1/|k^2 + z^2| < 2/k^2 \text{ for } k \geq 3.$$

Since $\sum_{k=1}^{\infty} 1/k^2$ is convergent [see Example (b) after Theorem 4-1.6], it follows from the Weierstrass M-test that

$$\sum_{k=3}^{\infty} 1/(k^2+z^2) \text{ and hence } \sum_{k=1}^{\infty} 1/(k^2+z^2)$$

is uniformly absolutely convergent on $\{z \in \mathbb{C} : 1 < |z| < 2\}$.

(b) The series $\sum_{k=1}^{\infty} x/(kx+1)((k-1)x+1)$ does not converge uniformly on $[0, b]$ for any $b > 0$, although it converges pointwise. Since

$$\frac{x}{(kx+1)((k-1)x+1)} = \frac{1}{((k-1)x+1)} - \frac{1}{(kx+1)},$$

therefore the n^{th} partial sum of the series is

$$1 - 1/(nx+1),$$

which has pointwise limit 0 if $x = 0$ and 1 if $x > 0$. The convergence is not uniform in view of Example 5-1.6(c).

(c) The series $2x/(1+x^2) + 4x^3/(1+x^4) + 8x^7/(1+x^8) + \cdots$ converges uniformly on $[-\frac{1}{2}, \frac{1}{2}]$. The series is

$$\sum_{k=1}^{\infty} f_k \text{ with } f_k(x) = 2^k x^{(2^k-1)}/(1+x^{2^k}).$$

When $x \in [-\frac{1}{2}, \frac{1}{2}]$, we have

$$|f_k(x)| \le 2^k |x|^{(2^k-1)} \le 2^k (\tfrac{1}{2})^{(2^k-1)} \le (\tfrac{1}{2})^{(2^k-1-k)} \le (\tfrac{1}{2})^{(k-1)},$$

where the final step uses the inequality $2^k \ge 2k$. If $M_k = (\frac{1}{2})^{(k-1)}$, the geometric series $\sum_{k=1}^{\infty} M_k$ converges. By the Weierstrass M-test, $\sum_{k=1}^{\infty} f_k$ converges uniformly on $[-\frac{1}{2}, \frac{1}{2}]$.

5-2.5. Definition. *A series*

$$\sum_{k=0}^{\infty} a_k (z-z_0)^k,$$

in which each term is a function f_k of the form $f_k(z) = a_k(z-z_0)^k$, where $z_0 \in \mathbb{C}$ (or $\mathbb{R}$), each $a_k \in \mathbb{C}$ (or $\mathbb{R}$), and the domain is $\mathbb{C}$ (or $\mathbb{R}$) is called a **power series centred at** z_0. *The set of $z \in \mathbb{C}$ (or $\mathbb{R}$) for which the series converges is called its* **domain of convergence**. *The number a_k is called the* **k^{th} coefficient** *of the power series. When z_0 and the coefficients a_k are all real and the domain of each f_k is restricted to be $\mathbb{R}$, we speak of a* **real power series**.

Remark. Sometimes it is convenient to refer to a power series as a *complex* power series if we wish to emphasise that it may or may not be a real power series. Since every partial sum is a_0 when $z = z_0$, a power series always converges when $z = z_0$; in other words, the domain of convergence always has the number z_0 in it.

We shall mostly be dealing with the special case of power series centred at 0, because our results have obvious analogues for the general case. Explicit mention of the centre will therefore be unnecessary.

5-2.6. Examples. (a) It has been noted before that the geometric series $\sum_{k=0}^{\infty} z^k$ converges to the sum $1/(1-z)$ when $|z| < 1$. This is a power series with all coefficients $a_k = 1$. When we intend to consider only real z here, we can call it a real power series, although the expression for it would remain the same. Some authors prefer to replace the letter z by x in order to indicate the restriction that z be real.

The domain of convergence of the real power series $\sum_{k=0}^{\infty} z^k$ is the open interval $(-1, 1)$, as the series certainly does not converge when $z = \pm 1$. The domain of convergence of the complex power series $\sum_{k=0}^{\infty} z^k$ is $\{z \in \mathbb{C} : |z| < 1\}$, which is to say that, when z is any *complex* number with $|z| \geq 1$, the series does not converge; this is because the k^{th} term z^k does not have limit 0.

(b) Consider the series $\sum_{k=0}^{\infty} (z^k / k!)$. Its coefficients are $a_k = \dfrac{1}{k!}$ and when $z \neq 0$, we have

$$|z^{k+1}/(k+1)!| \Big/ |z^k / k!| = |z| \Big/ (k+1),$$

which converges to 0, no matter what the nonzero complex number z may be. Therefore we conclude on the basis of the Ratio Test [see Theorem 4-3.3] that the series converges absolutely for every complex number z. Thus the domain of convergence is $\mathbb{C}$.

(c) The domain of convergence of the power series

$$\sum_{k=0}^{\infty} (-1)^k [z^k / (2k)!]$$

which has $a_k = (-1)^k [1/(2k)!]$ is $\mathbb{C}$, as can be seen by using the Ratio Test, because

$$|z^{k+1}/(2k+2)!| \Big/ |z^k / (2k)!| = |z| \Big/ 2(k+1)(2k+1), \quad (z \neq 0),$$

which converges to 0, no matter what the nonzero complex number z may be. It follows that we may replace z by z^2 and therefore the power series

$$\sum_{k=0}^{\infty} (-1)^k [z^{2k} / (2k)!]$$

also converges for every complex z.

(d) Another power series with domain of convergence $\mathbb{C}$ is

$$\sum_{k=0}^{\infty} (-1)^k [z^{2k+1} / (2k+1)!]$$

The related series $\sum_{k=0}^{\infty}(-1)^k[z^k/(2k+1)!]$ converges absolutely for all complex z, because

$$|z^{k+1}/(2k+3)!| \big/ |z^k/(2k+1)!| = |z| \big/ 2(k+1)(2k+3), \quad (z \neq 0),$$

which converges to 0, no matter what the nonzero complex number z may be. Replacing z by z^2, we find that the power series

$$\sum_{k=0}^{\infty}(-1)^k[z^{2k}/(2k+1)!] \text{ and hence also } \sum_{k=0}^{\infty}(-1)^k[z^{2k+1}/(2k+1)!]$$

converges for every complex z.

(e) In contrast to the preceding three Examples, the power series $\sum_{k=0}^{\infty}(k!\,z^k)$ has $\{0\}$ as its domain of convergence. This is because $|(k!\,z)^k|$ does not have limit 0 when $z \neq 0$. In fact, with $K = [1/|z|]$, the integer part of $1/|z|$, we have $k \geq K \Rightarrow (k+1)|z| \geq (K+1)|z| > 1$ and it therefore follows by induction on k that

$$k \geq K \Rightarrow |(k+1)!\,z^{k+1}| > |K!\,z^K|.$$

(Note: The Ratio Test does not help here because the terms need not be positive.)

In the Examples above, each power series was handled by some strategy appropriate to the situation at hand. We now present general result in this direction; it will be easier to state it if we introduce the convention that 1/0 means ∞ and that $1/\infty$ means 0. This convention is intended strictly for the result to be presented and is not to be interpreted to mean that division by 0 or by ∞ comes within the purview of an ordered field, as studied in Chapter 1. Any reference to the sequence $\{|a_k|^{1/k}\}$ tacitly assumes that the first term, wherein $k = 0$, is deleted.

5-2.7. Theorem. *Given the power series* $\sum_{k=0}^{\infty} a_k z^k$, *let*

$$R = 1/\limsup_{k\to\infty} |a_k|^{1/k}.$$

(a) *If* $|z| < R$, *the series converges absolutely.*

(b) *If* $0 < \rho < R$, *the series converges uniformly absolutely on the set*

$$\{z \in \mathbb{C} : |z| \leq \rho\}.$$

(c) *If* $|z| > R$, *the terms of the series are unbounded, so that the series diverges.*

Proof. (a) If $|z| < R$, there exists r such that $|z| < r < R$. Then $1/r > 1/R$ so that $1/R$ is not ∞. It follows from Remark (ii) after Lemma 4-5.4 that the sequence $\{|a_k|^{1/k}\}$ is bounded above. Since it is already bounded below by 0, it is a bounded sequence and by Prop.3-4.4, it follows that there exists some $k_0 \in \mathbb{N}$ such that

$$k \geq k_0 \Rightarrow |a_k|^{1/k} < 1/r.$$

Hence

$$k \geq k_0 \Rightarrow |a_k z^k| < |{}^z\!/_r|^k. \tag{1}$$

Since $|z| < r$, the geometric series $\sum_{k=0}^{\infty} |{}^z\!/_r|^k$ converges. Therefore from the inequality (1) and the Comparison Test [see Theorem 4-1.6], it follows that $\sum_{k=0}^{\infty} a_k z^k$ converges absolutely.

(b) To prove uniform absolute convergence on $\{z \in \mathbb{C} : |z| \leq \rho\}$ when $0 < \rho < R$, choose r' such that $\rho < r' < R$. As above, there exists some $k_0 \in \mathbb{N}$ such that

$$k \geq k_0 \Rightarrow |a_k|^{1/k} < {}^1\!/_{r'}.$$

Hence

$$k \geq k_0 \Rightarrow |a_k z^k| < |{}^z\!/_{r'}|^k < ({}^\rho\!/_{r'})^k. \tag{2}$$

Since $\rho < r'$, the geometric series $\sum_{k=0}^{\infty} ({}^\rho\!/_{r'})^k$ converges. Therefore from the inequality (2) and the Weierstrass M-Test [see Theorem 5-2.3], it follows that $\sum_{k=0}^{\infty} a_k z^k$ converges uniformly absolutely on $\{z \in \mathbb{C} : |z| \leq \rho\}$.

(c) Suppose $|z| > R$. Choose r such that $|z| > r > R$. Then ${}^1\!/_r < {}^1\!/_R$. In case the sequence $\{|a_k|^{1/k}\}$ is unbounded, there exist integers k greater than any given integer such that

$${}^1\!/_r < |a_k|^{1/k}. \tag{3}$$

In the contrary case, the same is again true, because ${}^1\!/_R = \limsup\limits_{k\to\infty} |a_k|^{1/k}$ is a subsequential limit by Prop.3-4.6. It follows from (3) that

$$|a_k z^k| > |{}^z\!/_r|^k.$$

Since $|{}^z\!/_r| > 1$, it further follows that the sequence $\{|a_k z^k|\}$ is unbounded. ☺

It is clear that when R is a real number, no other real number nor ∞ can have the combination of properties (a) and (c). Similarly, when R is ∞, no real number can have the combination of properties (a) and (c). In this sense R is unique.

5-2.8. Definition. *The number* $R = 1/\limsup\limits_{k\to\infty} |a_k|^{1/k}$ *is called the* **radius of convergence** *of the power series* $\sum_{k=0}^{\infty} a_k z^k$. *The subset* $\{z \in \mathbb{C} : |z| < R\}$ *of* $\mathbb{C}$ *is called its* **disc of convergence**. *When considering real power series* (*i.e. each* a_k *real and* z *restricted to be real*), *the interval* $(-R, R)$ *is called the* **interval of convergence**.

The above result makes no mention of convergence or otherwise of the power series when $|z| = R$. The reason is that nothing general can be said about this situation, as we now illustrate.

5-2.9. Examples. (a) The radius of convergence of $\sum_{k=0}^{\infty} z^k$ is 1; this follows not only from the fact that each $|a_k|^{1/k} = 1$ ($k > 0$), but also from the fact that this power series converges when $|z| < 1$ and diverges when $|z| > 1$. As noted in Example 5-2.6(a), this power series diverges whenever $|z| = 1$. So, we have an instance of a power series that diverges for *every z* with absolute value equal to the radius of convergence.

(b) The radius of convergence of the power series $\sum_{k=1}^{\infty} z^k/k^2$ is 1, because

$$\limsup_{k\to\infty} |1/k^2|^{1/k} = 1,$$

considering that $\lim_{k\to\infty} k^{1/k} = 1$ by Prop.3-1.6. Alternatively, we note that

$$(z^{k+1}/(k+1)^2)/(z^k/k^2) = z\left[k/(k+1)\right]^2,$$

and apply the Ratio Test to conclude that the series converges when $|z| < 1$ and diverges when $|z| > 1$. When $|z| = 1$, we have $|z^k/k^2| = 1/k^2$. Since $\sum_{k=1}^{\infty} 1/k^2$ converges, the Comparison Test implies that $\sum_{k=1}^{\infty} z^k/k^2$ converges absolutely when $|z| = 1$. Thus we have an instance of a power series that converges for *every z* with absolute value equal to the radius of convergence.

(c) Consider the power series $\sum_{k=1}^{\infty} z^{k+1}k^{-1}$. As in the preceding two Examples, it can be checked that its radius of convergence is 1. When $z = 1$, the series becomes $\sum_{k=1}^{\infty} 1/k$, which is divergent [see (d) of Examples 4-1.2]. However, when $z = -1$, it becomes $\sum_{k=1}^{\infty} (-1)^{k+1}k^{-1}$, which is convergent [see Examples 4-1.8].

The fact that we can often find the disc of convergence of a power series opens up the possibility of defining a function on that disc as being the sum of the series. The series in Examples (b), (c), (d) of 5-2.6 above have $\mathbb{C}$ as the disc of convergence. Studying the functions defined by the three series is the agenda for the next Section.

Problem Set 5-2

5-2.P1. Prove that the series $\sum_{k=1}^{\infty} (k^2 + k^4x^4)^{-1}$ is uniformly convergent on $\mathbb{R}$.

5-2.P2. Prove that the series $\sum_{k=1}^{\infty} x/k(k+1)$ is uniformly convergent on $[0,5]$ but not on $\mathbb{R}$.

5-2.P3. Show that $\sum_{k=1}^{\infty} x/(1+k^2x^2)$ converges uniformly on $[a,b]$ when $0 < a < b$.

5-2.P4. (a) If $|a_k| \le |b_k|$ for every k, show that the radius of convergence of $\Sigma\, a_k z^k$ is no smaller than that of $\Sigma\, b_k z^k$.

(b) If $|b_k| \geq 7$ for infinitely many k, show that the radius of convergence of $\Sigma\, b_k z^k$ is no greater than 1.

(c) If $|b_k| \geq |a_k|$ for infinitely many k, does it follow that the radius of convergence of $\Sigma\, b_k z^k$ is no greater than that of $\Sigma\, a_k z^k$?

5-2.P5. Let $I:\mathbb{R}\to\mathbb{R}$ be defined to be 0 when $x \leq 0$ and to be 1 when $x > 0$, and let Σc_j be any absolutely convergent series, where the numbers c_j may be real or complex. Suppose $\{x_j\}$ is a sequence of distinct real numbers and define $f:\mathbb{R}\to\mathbb{R}$ as $f(x) = \Sigma c_j I(x - x_j)$. Prove each of the following:

(a) The series defining $f(x)$ is uniformly convergent and the function f is continuous at any point which is distinct from every x_j.

(b) If $c_k \neq 0$, the function f is discontinuous at x_k. In the particular case when the sequence runs through every rational number and every c_k is nonzero, the function f is continuous at every irrational point and discontinuous at every rational point. [Another such function was seen in Example 2-1.4(d).]

5-2.P6. Define a sequence $\{g_k\}$ of functions on $[0,1]$ in terms of the function f of 2-1.P5 in the following manner: $g_k(x) = f(1/x)$ if $1/(k+1) < x \leq 1/k$ and $g_k(x) = 0$ otherwise. Show that

(a) g_k converges to the zero function but not uniformly;

(b) $\sum\limits_{k=1}^{\infty} g_k$ converges absolutely but not uniformly.

5-2.P7. Show that the series with k^{th} term $f(x/k^2)$, where f is as in 2-1.P5, is not uniformly convergent on $(0,\infty)$ but is uniformly convergent on any bounded closed interval $[0, A]$, with $A > 0$.

5-2.P8. If $\lim\limits_{k\to\infty} |a_k/a_{k+1}|$ exists, show that the power series $\sum\limits_{k=0}^{\infty} a_k z^k$ has radius of convergence equal to the limit. Give an example where the limit does not exist.

5-2.P9. Show that the series $\sum\limits_{k=1}^{\infty} \frac{(-x)^k}{k}$ converges

(a) uniformly on $(0, 1)$,

(b) absolutely for each $x \in (0, 1)$,

(c) but not uniformly absolutely on $(0, 1)$.

5-3 Special Functions

Although polynomials have the advantage that they are easy to compute with, it turns out that they can generally approximate a continuous function only on a bounded interval. The reason is not far to seek: The absolute value of any nonconstant polynomial $f(x)$ will be "large" when $|x|$ is "large". Such a function cannot reflect certain natural phenomena with a repetitive pattern that may have seasonal or daily cycles. It is therefore necessary to have a broad class of functions with repetitive behaviour, called *periodicity*, a phenomenon which the reader has undoubtedly come across in trigonometry. However, the elementary description of trigonometric functions in terms of circles and angles does not fit in with our approach in terms of the real and complex number systems as defined in Chapter 1. In the present Section, we describe them exclusively in the context of these number systems, totally independent of circles and angles. For an approach that avoids $\mathbb{C}$, see Deshpande [3].

5-3.1. Proposition. *The series* $\sum_{k=0}^{\infty}(z^k/k!)$ *converges absolutely for all complex z.*

Proof. We use the Ratio Test [see Theorem 4-3.3]:

$$|z^{k+1}/(k+1)!| \big/ |z^k/k!| = |z| \big/ (k+1),$$

which converges to 0, no matter what the nonzero complex number z may be. ☺

5-3.2. Definition. *The function defined on the domain* $\mathbb{C}$ *by*

$$\exp z = \sum_{k=0}^{\infty}(z^k/k!)$$

is called the **exponential function**. *Alternative notations are* $\boldsymbol{e^z}$ *and* $\mathbf{exp}\,(\boldsymbol{z})$, *depending on convenience.*

5-3.3. Remarks. (i) In view of Prop.5-3.1, the series that defines the exponential function is absolutely convergent.

(ii) Another consequence of Prop.5-3.1 is that $\lim_{k\to\infty}(z^k/k!) = 0$ for any complex z.

5-3.4. Theorem. $\exp(z+w) = (\exp z)(\exp w)$ *for all* $z, w \in \mathbb{C}$.

Proof. By Remark 5-3.3(i) and Mertens' Theorem 4-6.2, (exp z)(exp w) equals the Cauchy product of the series for exp z and exp w, which is to say,

$$(\exp z)(\exp w) = \sum_{k=0}^{\infty}\sum_{j=0}^{k}\left[(z^j/j!)(w^{k-j}/(k-j)!)\right].$$

But $\sum_{j=0}^{k}\left[(z^j/j!)(w^{k-j}/(k-j)!)\right] = \sum_{j=0}^{k}[z^j w^{k-j}/j!(k-j)!]$

$$= [\sum_{j=0}^{k} C(k,j)\, z^j w^{k-j}]/k!$$

$$= (z+w)^k/k! \text{ by the binomial theorem.}$$

Therefore $(\exp z)(\exp w) = \sum_{k=0}^{\infty}(z+w)^k/k! = \exp(z+w)$ by Def. 5-3.2. ☺

5-3.5. Corollary. $\exp 0 = 1$ *and* $\exp z$ *is never* 0. *Besides*, $\exp(-z) = 1/\exp z$ *for all* $z \in \mathbb{C}$.

Proof. It is an immediate consequence of the definition of exp z as the sum of a series that $\exp 0 = 1$. By Theorem 5-3.4, $(\exp z)(\exp(-z)) = \exp 0 = 1$. It follows that $\exp z$ is never 0 and that $\exp(-z) = 1/\exp z$. ☺

5-3.6. Definition. *For any* $z \in \mathbb{C}$,

$$\mathbf{cos}\, z = \tfrac{1}{2}[\exp(iz) + \exp(-iz)] \quad \text{and} \quad \mathbf{sin}\, z = \tfrac{1}{2i}[\exp(iz) - \exp(-iz)].$$

These are known as the **cosine** *and* **sine** *function respectively*.

Remark. It follows from this Definition that

$$\exp(iz) + \exp(-iz) = 2\cos z \quad \text{and} \quad \exp(iz) - \exp(-iz) = 2i \sin z.$$

5-3.7. Theorem. *For any complex z and w*,
(a) $\sin(-z) = -\sin z$ *and* $\sin 0 = 0$;
(b) $\cos(-z) = \cos z$ *and* $\cos 0 = 1$;
(c) $\cos(z + w) = \cos z \cos w - \sin z \sin w$;
(d) $\sin(z - w) = \sin z \cos w - \cos z \sin w$;
(e) $\cos^2 z + \sin^2 z = 1$;
(f) $\sin 2z = 2 \sin z \cos z$;
(g) $\cos 2z = 2\cos^2 z - 1 = 1 - 2\sin^2 z = \cos^2 z - \sin^2 z$.

Proof. (a) and (b) follow immediately from Def.5-3.6.

(c) For any complex numbers u, v, s, t we have

$$(u + v)(s + t) + (u - v)(s - t) = 2us + 2vt.$$

Substituting $u = \exp(iz)$, $v = \exp(-iz)$, $s = \exp(iw)$, $t = \exp(-iw)$, and taking into account the equalities noted in the above Remark, we get

$$\begin{aligned} 4\cos z \cos w - 4\sin z \sin w &= 2\exp(iz)\exp(iw) + 2\exp(-iz)\exp(-iw) \\ &= 2[\exp(i(z + w)) + \exp(-i(z + w))] \text{ by Theorem 5-3.4} \\ &= 4\cos(z + w) \text{ by Def.5-3.6.} \end{aligned}$$

(d) Making the same substitutions as above in the identity

$$(u - v)(s + t) - (u + v)(s - t) = -2vs + 2ut,$$

we get

$$\begin{aligned} 4i \sin z \cos w - 4i \cos z \sin w &= -2\exp(-iz)\exp(iw) + 2\exp(iz)\exp(-iw) \\ &= 2[\exp(i(z - w)) - \exp(-i(z - w))] \text{ by Theorem 5-3.4} \\ &= 4i \sin(z - w) \text{ by Def.5-3.6.} \end{aligned}$$

(e) If we substitute $u = \exp(iz)$ in the identity

$$\left(u+\tfrac{1}{u}\right)^2-\left(u-\tfrac{1}{u}\right)^2=4$$

and use the fact that $\frac{1}{u}=\exp(-iz)$, we get $(2\cos z)^2-(2i\sin z)^2=4$, which leads to $\cos^2 z+\sin^2 z=1$.

(f) and (g) now follow from parts (a) to (e). ☺

5-3.8. Theorem. *If z is real, then* $\exp z$ *is real,* $\cos z=\Re(\exp(iz))$ *and* $\sin z=\Im(\exp(iz))$. *In particular,* cos *and* sin *are real valued on* $\mathbb{R}$.

Proof. Since the partial sums $S_n(z)=\sum_{k=0}^{n}(z^k/k!)$ of the series that defines $\exp z$ satisfy $S_n(\bar{z})=\overline{S_n(z)}$, we have

$$\exp(\bar{z})=\overline{\exp z}\quad\text{for all } z\in\mathbb{C}. \tag{1}$$

In particular, $z\in\mathbb{R}\Rightarrow z=\bar{z}\Rightarrow\overline{\exp z}=\exp z\Rightarrow\exp z\in\mathbb{R}$.

Since $-(iz)=(-iz)=\bar{i}z=\overline{i\bar{z}}$, therefore $-(iz)=\overline{iz}$ if $z\in\mathbb{R}$. It follows for $z\in\mathbb{R}$ that $\exp(-(iz))=\exp(\overline{iz})=\overline{\exp(iz)}$ by (1). Consequently from Def.5-3.6, $\cos z=\Re(\exp(iz))$ and $\sin z=\Im(\exp(iz))$ when $z\in\mathbb{R}$. ☺

The analogue of (1) holds for cos and sin as well. This will follow by a similar argument using the series we are about to establish.

5-3.9. Theorem. *The series*

$$\sum_{k=0}^{\infty}(-1)^k[z^{2k}/(2k)!]\quad\textit{and}\quad\sum_{k=0}^{\infty}(-1)^k[z^{2k+1}/(2k+1)!]$$

converge absolutely for all complex z. Their sums are $\cos z$ *and* $\sin z$ *respectively.*

Proof. Absolute convergence of both series for all nonzero complex z follows by the Ratio Test. For $z=0$, the convergence is trivial. Denote their sums by $C(z)$ and $S(z)$ respectively.

Let E_{2n+1} denote the partial sum $\sum_{k=0}^{2n+1}[(iz)^k/k!]$ of the series that defines $\exp iz$, and similarly, let $C_n=\sum_{k=0}^{n}(-1)^k[z^{2k}/(2k)!]$ and $S_n=\sum_{k=0}^{n}(-1)^k[z^{2k+1}/(2k+1)!]$. Since $(-1)^k=i^{2k}$ and $i(-1)^k=i^{2k+1}$, we have

$$C_n=\sum_{k=0}^{n}[(iz)^{2k}/(2k)!]\quad\text{and}\quad iS_n=\sum_{k=0}^{n}[(iz)^{2k+1}/(2k+1)!].$$

It follows that

$$\begin{aligned}C_n+iS_n&=\sum_{k=0}^{n}[(iz)^{2k}/(2k)!+(iz)^{2k+1}/(2k+1)!]\text{ by Prop.1-11.4}\\&=\sum_{k=0}^{2n+1}[(iz)^k/k!]\qquad\text{by Prop.1-11.5}\\&=E_{2n+1}.\end{aligned}$$

[In applying Prop.1-11.5, we have taken

$$a_k=(iz)^{2(k-1)}/(2(k-1))!\ ,\quad b_k=(iz)^{2k-1}/(2k-1)!\ ,$$

from which it follows that the finite sequence $\{c_k\}$ required there is given by $c_k = (iz)^{k-1}/(k-1)!$.] From the above equality, we get $\exp iz = C(z) + iS(z)$.

It is immediate from the definition that $S(-z) = -S(z)$ and $C(-z) = C(z)$. Hence $\exp(iz) = C(z) + iS(z)$ and $\exp(-iz) = C(-z) + iS(-z) = C(z) - iS(z)$, so that

$$C(z) = \tfrac{1}{2}[\exp(iz) + \exp(-iz)] \text{ and } S(z) = \tfrac{1}{2i}[\exp(iz) - \exp(-iz)].$$

From here and Def.5-3.6, it follows for $z \in \mathbb{R}$, that $C(z) = \cos z$ and $S(z) = \sin z$. ☺

5-3.10. Proposition. *For any* $z \in \mathbb{C}$ *such that* $|z| < 2$, *we have*

$$|\exp z - 1| \leq 2|z|/(2 - |z|).$$

Proof. We shall use the summation formula $\sum_{k=0}^{\infty} r^k = 1/(1-r)$ for $|r| < 1$, and the inequality $(k+1)! \geq 2^k$ for $k \geq 0$ (easily proved by induction).

$$|\exp z - 1| = |\sum_{k=1}^{\infty}(z^k/k!)| \leq |z\sum_{k=1}^{\infty}(z^{k-1}/k!)| = |z\sum_{k=0}^{\infty}(z^k/(k+1)!)|$$
$$\leq |z|\sum_{k=0}^{\infty}(|z^k|/2^k) = |z|\,[1/(1 - \tfrac{|z|}{2})] = 2|z|/(2 - |z|). \text{ ☺}$$

Suppose $A \subseteq \mathbb{C}$. Continuity of a function $f\colon A \to \mathbb{C}$ at a point $a \in A$ means that, for every $\varepsilon > 0$, there exists $\delta > 0$ such that every $z \in A$ satisfying $|z - a| < \delta$ also satisfies $|f(z) - f(a)| < \varepsilon$. In case $a \in \mathbb{R} \cap A$ and $f\colon A \to \mathbb{C}$ is continuous at a, then it is easily seen that the restriction of f to $\mathbb{R} \cap A$ is also continuous at a. If $f\colon A \to \mathbb{C}$, $g\colon A \to \mathbb{C}$ are continuous at $a \in A$, one can prove the analogue of Theorem 2-1.3 following the same ideas; however, min $\{f, g\}$ and max $\{f, g\}$ are now meaningful only when the ranges happen to be subsets of $\mathbb{R}$.

5-3.11. Theorem. *The functions* exp, cos *and* sin *are continuous at each point of* $\mathbb{C}$.

Proof. First we show that exp is continuous at 0. By Prop.5-3.10, when $|z| < 1$, we have $|\exp z - 1| \leq 2|z|/(2 - |z|) \leq 2|z|$. For any $\varepsilon > 0$, take $\delta = \min\{1, \frac{\varepsilon}{2}\}$. Then

$$|z - 0| < \delta \Rightarrow |\exp z - 1| \leq 2\delta < \varepsilon.$$

Thus exp is continuous at 0.

Now consider any $a \in \mathbb{C}$. By Theorem 5-3.4, we have

$$\begin{aligned}\exp z - \exp a &= (\exp a)(\exp(z - a)) - \exp a \\ &= (\exp a)(\exp(z - a) - 1). \qquad (1)\end{aligned}$$

Since exp is continuous at 0 and $\exp a \neq 0$ [see Cor.5-3.5], for any $\varepsilon > 0$, there exists $\delta > 0$ such that any $z \in \mathbb{C}$ satisfying $|z-a| < \delta$ also satisfies $|\exp(z-a)-1| < \varepsilon/|\exp a|$, and hence by (1), also satisfies $|\exp z - \exp a| < \varepsilon$.

Since $|iz - ia| = |z-a|$, it follows that the functions which map $z \in \mathbb{C}$ into $\exp(iz)$ and $\exp(-iz)$ are also continuous. Therefore by Def.5-3.6, cos and sin are also continuous. ☺

5-3.12. Proposition. *There is a least positive number* α *for which* $\cos\alpha = 0$. *Furthermore,* $1 < \alpha < 2$ *and* cos *is positive on* $[0, \alpha)$.

Proof. Since $\exp 0 = 1$, it follows from Def.5-3.6 that $\cos 0 = 1$. We shall show that $\cos 2 < 0$. From the series for $\cos z$ established in Theorem 5-3.9, we have $\cos 2 = 1 - \sum_{k=1}^{\infty}(-1)^{k+1}[2^{2k}/(2k)!]$. For the series here, we have $2^{2(k+1)}/(2k+2)! \leq 2^{2k}/(2k)!$ because $2^2 \leq (2k+2)(2k+1)$ when $k \geq 1$; moreover, $\lim_{k\to\infty}[2^{2k}/(2k)!] = 0$ by Remark 5-3.3(ii). It follows by Leibnitz' Test 4-1.7(a), that the sum of the series is greater than or equal to the sum of the first two terms, which is $2^2/2! - 2^4/4! = 2 - 16/24 = 4/3$. Therefore $\cos 2 \leq 1 - 4/3 = -1/3 < 0$. [For an alternative proof that cos has a negative value somewhere or other, see 5-3.P1.]

By Theorem 5-3.11, the restriction of cos to $[0,2]$ is a continuous function and it is real valued by Theorem 5-3.8. Since it equals 1 at 0 and is negative at 2, it follows by the Intermediate Value Theorem that it takes the value 0 at some number between 0 and 2. Let α be the infimum of all such numbers. By continuity of cos, we have $\cos\alpha = 0$. Also, $\alpha < 2$ and

$$0 \leq x < \alpha \Rightarrow \cos x \neq 0. \tag{1}$$

Since $\cos 0 = 1$, we have $\alpha \neq 0$ so that $0 < \alpha < 2$. We shall show that cos is positive on the open interval $(0, \alpha)$.

Suppose, if possible, that there exists some x such that $0 < x < \alpha$ and $\cos x < 0$. By the Intermediate Value Theorem, there must exist some y between 0 and x such that $\cos y = 0$. Therefore $0 < y < x < \alpha < 2$ and $\cos y = 0$. Since α is the infimum of all numbers in $(0,2)$ where cos is 0, such a number y cannot exist. It follows that there cannot be any number x such that $0 < x < \alpha$ and $\cos x < 0$. Together with (1), this shows that cos is positive on $[0, \alpha)$.

It remains to prove that $\alpha > 1$. This will follow if we show that cos is positive on $[0,1]$. When $0 \leq x \leq 1$, the series $\cos x = \sum_{k=0}^{\infty}(-1)^k[x^{2k}/(2k)!]$ is found to satisfy the conditions of Leibnitz' Test because $\lim_{k\to\infty}[x^{2k}/(2k)!] = 0$ and

$$[x^{2(k+1)}/(2(k+1))!]/[x^{2k}/(2k)!] = x^2/(2k+2)(2k+1) \leq 1$$

when $0 \le x \le 1$ and $k \ge 0$. It follows as above that $\cos x \ge 1 - x^2/2! \ge 1 - 1^2/2! = 1/2$. Since $\cos 0 = 1$, it follows that cos is positive on $[0, 1]$. Thus $\alpha > 1$. ☺

5-3.13. Definition. *The number* π *is twice the least positive number* α *for which* $\cos\alpha = 0$.

The number π that has just been introduced is no different from the one known from elementary mathematics as the ratio of the circumference of a circle to its diameter; this turns out to be easy to prove after defining "circle" and "length of a curve" in the framework of our axioms.

It follows from the above Definition and Prop. 5-3.12 that (i) $\cos\frac{\pi}{2} = 0$ (ii) $0 \le x < \frac{\pi}{2} \Rightarrow \cos x > 0$ and (iii) $1 < \frac{\pi}{2} < 2$.

We now come to what is called the "periodic" behaviour of sin and cos, which was promised in the introductory paragraph above.

5-3.14. Theorem. (a) $-\frac{\pi}{2} < x < \frac{\pi}{2} \Rightarrow \cos x > 0$ and $0 < x < \pi \Rightarrow \sin x > 0$.

(b) $\sin\frac{\pi}{2} = 1$;

(c) $\cos\pi = -1$, $\sin\pi = 0$, $\cos 2\pi = 1$, $\sin 2\pi = 0$;

(d) $\cos(z + 2\pi) = \cos z$ *and* $\sin(z + 2\pi) = \sin z$ *for every* $z \in \mathbb{C}$.

Proof. (a) The first part follows from the fact that $0 \le x < \frac{\pi}{2} \Rightarrow \cos x > 0$ and that $\cos(-x) = \cos x$.

The series $\sin x = \sum_{k=0}^{\infty}(-1)^k[x^{2k+1}/(2k+1)!]$ satisfies the conditions of Leibnitz' Test when $0 < x < 2$, because $\lim_{k\to\infty}[x^{2k+1}/(2k+1)!] = 0$ and

$$[x^{2k+3}/(2k+3)!]/[x^{2k+1}/(2k+1)!] = x^2/(2k+3)(2k+2) \le 1$$

when $0 < x < 2$ and $k \ge 0$. It follows by (a) of Leibnitz' Test that $\sin x \ge x - x^3/3! = x(1 - x^2/6)$, which is positive when $0 < x < 2$. The same holds when $0 < x < \frac{\pi}{2}$ because $\frac{\pi}{2} < 2$. Therefore $\sin x$ and $\cos x$ are both positive when $0 < x < \frac{\pi}{2}$. It follows that $0 < x < \pi \Rightarrow \sin\frac{x}{2}\cos\frac{x}{2} > 0$. But by Theorem 5-3.7(f), we have $\sin x = 2\sin\frac{x}{2}\cos\frac{x}{2}$. Therefore $0 < x < \pi \Rightarrow \sin x > 0$.

(b) Since $\cos\frac{\pi}{2} = 0$, it follows from the identity $\cos^2 z + \sin^2 z = 1$ [proved in Theorem 5-3.7(e) above] that $\sin\frac{\pi}{2} = 1$ or -1. But $1 < \frac{\pi}{2} < 2$ and hence by (a), $\sin\frac{\pi}{2} > 0$. So $\sin\frac{\pi}{2} = 1$.

(c) By Theorem 5-3.7(g), $\cos(2z) = 2\cos^2 z - 1$ and hence $\cos\pi = 2\cos^2\frac{\pi}{2} - 1 = -1$. Since $\cos^2\pi + \sin^2\pi = 1$, we have $\sin\pi = 0$. So, $\cos 2\pi = 2\cos^2\pi - 1 = 2(-1)^2 - 1 = 1$ and $\sin 2\pi = 0$.

(d) This follows from Theorem 5-3.7(c),(d) and part (c) of the present Theorem. ☺

The following consequences of Theorem 5-3.7(a)–(d) are easy to derive by substituting $z = (u+v)/2$ and $w = (u-v)/2$:

$$\cos u - \cos v = -2\sin[(u+v)/2]\sin[(u-v)/2]$$
$$\sin u - \sin v = 2\cos[(u+v)/2]\sin[(u-v)/2].$$

One can ask whether a positive number smaller than 2π can have the property stated with reference to *all* complex numbers z in Theorem 5-3.14(d). Our next result says that a smaller positive number *cannot* have such a property with reference to *even a single* complex z.

5-3.15. Theorem. *If* $0 < \alpha < 2\pi$ *and* z *is any complex number, then either* $\cos(z+\alpha) \neq \cos z$ *or* $\sin(z+\alpha) \neq \sin z$.

Proof. We have

$$\cos(z+\alpha) - \cos z = -2\sin[(2z+\alpha)/2]\sin[\alpha/2]$$

and

$$\sin(z+\alpha) - \sin z = 2\cos[(2z+\alpha)/2]\sin[\alpha/2].$$

If both left sides were to be 0 while $\sin[\alpha/2] \neq 0$, then we would have $\sin[(2z+\alpha)/2] = 0 = \cos[(2z+\alpha)/2]$, which is not possible in view of the identity $\cos^2 u + \sin^2 u = 1$. Therefore $\sin[\alpha/2]$ would have to be 0, which is also not possible when $0 < \alpha < 2\pi$ in view of Theorem 5-3.14(a). Therefore the left sides in the equalities displayed above cannot both be zero. ☺

It is a fact that for any complex α and z with $\Im\alpha \neq 0$, we have either $\cos(z+\alpha) \neq \cos z$ or $\sin(z+\alpha) \neq \sin z$. But we do not take up this matter here.

The preceding two results together say that $p = 2\pi$ is the smallest positive number such that $f(x) = f(x+p)$ for every $x \in \mathbb{R}$, where f may be either sin or cos.

5-3.16. Definition. *A function* $f:\mathbb{R}\to\mathbb{R}$ *is called* **periodic** *if there is some real number* $p \neq 0$ *such that* $f(x) = f(x+p)$ *for all* $x \in \mathbb{R}$. *The smallest such positive number, if any, is called the* **period** *of the function.*

A constant function is periodic (any p will do) but has no period; the functions cos and sin have period 2π. It is easy to verify that $\cos nx$ and $\sin nx$ have period $2\pi/n$, where $n \in \mathbb{N}$. A broad class of periodic functions that can be built up from these two is described in the following

5-3.17. Definition. *A function $f:\mathbb{R}\to\mathbb{R}$ is called a* **trigonometric polynomial** *if there exist finite sequences $\{a_0, a_1, \ldots, a_n\}$ and $\{b_1, b_2, \ldots, b_n\}$ of real numbers such that*

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{n}(a_k\cos kx + b_k\sin kx) \quad \textit{for every } x \in \mathbb{R}.$$

It is clear that sums and constant multiples of trigonometric polynomials are again trigonometric polynomials. From the formulas noted just after the proof of Theorem 5-3.14 and two others of the same kind, it follows that the same is true of a product of two trigonometric polynomials. Details are left to the reader [see 5-3.P4], but the fact is of significance in connection with the use of such polynomials for approximation.

Problem Set 5-3

5-3.P1. (a) The function $f:\mathbb{R}\to\mathbb{R}$ given by $f(x) = 1$ for each $x \in \mathbb{R}$ satisfies the identity $f(2x) = 2f(x)^2 - 1$ for each $x \in \mathbb{R}$. Give another example of a function with domain $\mathbb{R}$ satisfying the same identity.

(b) Suppose the function $f:\mathbb{R}\to\mathbb{R}$ satisfies the identity $f(2x) = 2f(x)^2 - 1$ for each $x \geq 0$. If $f(x) \geq 0$ for each $x \geq 0$, show that $f(x) \geq 1$ for each $x \geq 0$.

5-3.P2. Show directly from the series defining $\exp z$ that $|\exp\frac{1}{2}i - 1| \leq 2/3$.

5-3.P3. Show that the function $f:\mathbb{R}\to\mathbb{R}$ defined as $f(x) = 0$ or 1 according as $x \in \mathbb{Q}$ or $x \notin \mathbb{Q}$ is periodic but has no period.

5-3.P4. Show by induction that, if f and g are trigonometric polynomials, then so is fg.

5-3.P5. Show that the series $\sum_{k=1}^{\infty}\sin(1/k)$ diverges.

5-3.P6. Prove the following on the basis of results derived so far:

(a) $\pi < x < 2\pi \Rightarrow \sin x < 0$.

(b) $0 < x < 2\pi$ and $\sin x = 0 \;\Rightarrow\; x = \pi$.

(c) $0 \leq x < y \leq 2\pi$ and $\cos x = \cos y \;\Rightarrow\; y = 2\pi - x$.

(d) $0 \leq x < y \leq \pi \;\Rightarrow\; \cos x > \cos y$.

5-3.P7. Prove the following on the basis of results derived so far:

(a) The terms of the alternating series for $\cos x$ decrease if $|x| < \sqrt{2}$. Hence $\cos 1 > \frac{1}{2}$.

(b) $\cos 3x = 4\cos^3 x - 3\cos x$.

(c) $\cos\frac{\pi}{3} = \frac{1}{2}$.
(d) $\pi > 3$.

5-3.P8. Prove the following on the basis of results derived so far:
(a) $\frac{7}{8} < \cos\frac{1}{2} < \frac{337}{384}$.
(b) $\cos\frac{3}{2} > \frac{3}{64}$.

***5-3.P9.** Prove that for any real number a, the series

$$\sum_{k=1}^{\infty} [\sin(ka) - \sin((k-1)a)]\, k^{-1/2}$$

is convergent.

5-3.P10. (a) If $t \in [0, 2\pi)$ and $\cos t = 1$, show that $t = 0$.
(b) If $t \in (0, 2\pi)$, show that $\cos t \neq 1$.
(c) If $t \in \mathbb{R}$ and $\cos t = 1$, show that $t = 2\pi k$ for some integer k.
(d) If $t, u \in \mathbb{R}$, $\cos t = \cos u$ and $\sin t = \sin u$, show that $u - t = 2\pi k$ for some integer k.

***5-3.P11.** ("Parametrisation of the unit circle") If x_1 and x_2 are real numbers such that $x_1^2 + x_2^2 = 1$, show that there exists a unique real number $t \in [0, 2\pi)$ such that $\cos t = x_1$ and $\sin t = x_2$. Also show that if $u \in \mathbb{R}$ satisfies $\cos u = x_1$ and $\sin u = x_2$, then $u - t = 2\pi k$ for some integer k.

5-3.P12. Define a sequence $\{g_k\}$ of functions on [0,1] in the following manner: $g_k(x) = \sin^2(\frac{\pi}{x})$ if $1/(k+1) < x \leq 1/k$ and $g_k(x) = 0$ otherwise. Show that
(a) g_k converges to the zero function but not uniformly;
(b) $\sum_{k=1}^{\infty} g_k$ converges absolutely but not uniformly.

5-3.P13. Show that the series with k^{th} term $\sin(x/k^2)$ is not uniformly convergent on $(0, \infty)$ but is uniformly convergent on any bounded closed interval $[0, A]$, with $A > 0$.

Chapter 6
Uniform Approximation

*6-1 Heine-Borel Theorem

The Theorem named in the title is not specifically about uniform approximation but will be used for obtaining a fundamental result regarding the latter.

6-1.1. Definition. *A* **cover of** *a set K is a family $\mathscr{U}$ of sets whose union $\bigcup\mathscr{U}$ contains the set K. A subfamily of a cover which is also a cover of the set is called a* **subcover**.

Remarks. (i) Starting from a family $\mathscr{U}$ of sets, it is easy to determine which sets it is a cover of. The answer is that it is a cover of all those sets that are contained in the union $\bigcup\mathscr{U}$, i.e. of all subsets of this union. The issue is usually the reverse: Starting from a set, is it possible to have a cover of some particular kind? [The terminology "$\mathscr{U}$ covers K" is also used.]

(ii) The statement that $\mathscr{U}$ covers the set K can be expressed as: Every element of K belongs to some set of the family $\mathscr{U}$. In formal terms, it can also be expressed as:

$$\forall\ x \in K\ \exists\ U \in \mathscr{U} \ni x \in U.$$

6-1.2. Examples. (a) The union of the subsets $\{a,b\}$ and $\{b,c\}$ of $X = \{a,b,c,d\}$ contains the subset $K = \{a,c\}$; therefore the family $\mathscr{U} = \{\{a,b\},\{b,c\}\}$ is a cover of K. Of course, the union also contains several other subsets, e.g. $\{a,b\}$ and $\{a,b,c\}$, and $\mathscr{U}$ is therefore a cover of these subsets well. However the family is not a cover of any set containing d, because the union does not contain d.

(b) For the set $\{a,b,c\}$, the family $\mathscr{U} = \{\{a,b\},\{b,c\}\}$ is a cover consisting of two-element subsets of $\{a,b,c\}$. There are of course other covers consisting of two-element subsets [see 6-1.P1].

(c) Consider the subset $[1,\infty)$ of $\mathbb{R}$. The infinite family $\mathscr{U} = \{[1,n] : n \in \mathbb{N}\}$ of subsets of $\mathbb{R}$ is a cover of $K = [1,\infty)$. Note that, if we were to delete one set, say $[1,7]$, from the family, the resulting subfamily will have the same union and will therefore cover $[1,\infty)$. The subfamily will thus be a subcover. In fact, we could delete a finite number of sets from $\mathscr{U}$ and the subfamily formed by the remaining sets would still be a subcover.

(d) Consider the same subset of $\mathbb{R}$ as in (c) above and also the same cover $\mathcal{U}$. We have already noted that the subfamily formed by deleting a finite number of sets of $\mathcal{U}$ is also a cover of $K = [1,\infty)$. Here we give an example of a subfamily, which is formed by deleting an infinite number of sets of $\mathcal{U}$ and is a cover of $[1,\infty)$. Delete all sets $[1,n]$ for which n is even. The resulting subfamily is $\{[1,n] : n \in \mathbb{N}$ and n odd$\}$. This is also a cover of $[1,\infty)$ and is thus a subcover. Of course, we could have deleted all $[1,n]$ with odd n; the resulting subfamily $\{[1,n] : n \in \mathbb{N}$ and n even$\}$ is also a subcover.

(e) In the examples described in (c) and (d) above, the subcover was infinite. However, if we were to take a *finite* subfamily of $\mathcal{U} = \{[1,n] : n \in \mathbb{N}\}$, then the union would be $[1,N]$, where N is the greatest among all the finitely many n for which $[1,n]$ belongs to the subfamily. Therefore a finite subfamily cannot cover $[1,\infty)$. Thus $[1,\infty)$ cannot be covered by any finite subfamily of $\mathcal{U}$. In other words, the cover $\mathcal{U}$ contains no finite subcover.

(f) The family $\mathcal{U} = \{(\frac{1}{n},2) : n \in \mathbb{N}\}$ is a cover of $(0,1]$. The union of any finite subfamily is $(\frac{1}{N},2)$, where N is the greatest among all the finitely many n for which $(\frac{1}{n},2)$ belongs to the subfamily. Therefore a finite subfamily cannot be a cover of $(0,1]$. Thus the cover $\mathcal{U}$ contains no finite subcover.

6-1.3. Definition. *A subinterval of an interval I is called a* **relatively open subinterval** *of I if it is the intersection of I with an open interval.*

Examples. (a) Let $I = [0,5]$. Then the subinterval $[0,1)$ is the intersection of I with the open interval $(-1,1)$ and is therefore a relatively open subinterval of I. Note that it is not an open interval. Similarly, the interval $(3,5]$, though not open, is a relatively open subinterval of I, because $(3,5] = I \cap (3,6)$. The closed interval I itself is a relatively open subinterval, because $I = I \cap (-1,6)$. The closed interval $[4,5]$ is not a relatively open subinterval, because whichever way it may be expressed as an intersection of I with an interval J, the number 4 has to be in the latter, while no smaller number can. Therefore 4 must belong to it and also be its left endpoint. So, J cannot be open.

(b) Let $I = (0,5)$. Since this is an open interval and the intersection of any two open intervals must be an open interval, it follows that a relatively open subinterval of I [or of any open interval] must be an open interval. So, in this case, a relatively open subinterval is the same thing as an open interval contained in I.

6-1.4. Remarks. (i) Suppose $\phi: I \to \mathbb{R}$ is continuous and $\phi(c) > 0$, where $c \in I$. By selecting $\varepsilon = \phi(c)$, we find that there exists $\delta > 0$ such that $x \in I \cap (c-\delta, c+\delta)$

implies $|\phi(x)-\phi(c)| < \phi(c)$, which further implies $\phi(x) > 0$. This can be rephrased as saying that there is a relatively open subinterval U of I, namely, $U = I \cap (c-\delta, c+\delta)$ such that $x \in U$ implies $\phi(x) > 0$. This will be used in the proof of Theorem 6-3.6 below.

(ii) Suppose I is a closed and bounded interval $[a,b]$, where $a < b$. Consider a relatively open subinterval $U = I \cap J$, where J is an open interval. Recall from Prop.1-7.4 that when an open interval J contains a number c, it also contains $[c-\delta, c+\delta]$ for some $\delta > 0$. If $a \in U = I \cap J$, then both I and J contain numbers greater than a, therefore so does the relatively open subinterval U. Thus $[a, a+\delta] \subseteq U$ for some $\delta > 0$. Similarly, if $b \in U$, then $[b-\delta, b] \subseteq U$ for some $\delta > 0$. If $c \in U$ and $a < c < b$, then $c \in (a,b) \cap J$, a nonempty intersection of open intervals and hence an open interval. It follows that $[c-\delta, c+\delta] \subseteq U$ for some $\delta > 0$. These simple facts about what happens when a relatively open subinterval U of $[a,b]$ contains a, contains b or contains a number in (a,b), will be used in the forthcoming proof without reference.

6-1.5. Heine-Borel Theorem. *Every cover of a bounded closed interval $[a,b]$ consisting of relatively open subintervals contains a finite subcover.*

Proof. We need consider only the case when $a < b$. Let $\mathscr{U}$ be a cover of $[a,b]$ consisting of relatively open subintervals. Define $S = \{s \in [a,b] : \mathscr{U}$ contains a finite subcover of $[a,s]\}$ Since $[a,b]$ is covered by $\mathscr{U}$, its element a must belong to some $U_0 \in \mathscr{U}$. The set U_0, being an element of $\mathscr{U}$, is a relatively open subinterval of $[a,b]$. Therefore there exists $\delta_0 > 0$ such that $[a, a+\delta_0] \subseteq U_0$. It follows that the finite subfamily $\{U_0\}$ of $\mathscr{U}$ covers $[a, a+\delta_0]$ and hence that $a+\delta_0 \in S$. Thus S is nonempty and is bounded above by b. Let $c = \sup S$. Then

$$c \geq a+\delta_0 > a \text{ so that } a < c \leq b.$$

Suppose, if possible, that $c < b$. Since $[a,b]$ is covered by $\mathscr{U}$, its element c must belong to some $U_1 \in \mathscr{U}$. The set U_1, being an element of $\mathscr{U}$, is a relatively open subinterval of $[a,b]$. Therefore

$$\text{there exists } \delta_1 > 0 \text{ such that } [c-\delta_1, c+\delta_1] \subseteq U_1. \tag{1}$$

Now, $c-\delta_1$ is not an upper bound of S. So, $\exists\, s_1 \in S \ni c-\delta_1 < s_1$. By definition of S, this implies that

$$[a, s_1] \text{ is covered by a finite subfamily } \mathscr{V} \text{ of } \mathscr{U}. \tag{2}$$

By (1) and (2), the union $[a,s_1] \cup [c-\delta_1, c+\delta_1]$ is covered by the finite subfamily $\mathscr{V} \cup \{U_1\}$ of $\mathscr{U}$ and hence

$$\mathscr{U} \text{ contains a finite subcover of } [a,s_1] \cup [c-\delta_1, c+\delta_1]. \tag{3}$$

But $c-\delta_1 < s_1 \le c$. Therefore $[a,s_1]\cup[c-\delta_1, c+\delta_1] = [a, c+\delta_1]$. Hence by (3), $\mathscr{U}$ contains a finite subcover of $[a, c+\delta_1]$. Thus $c+\delta_1 \in S$ even though $c+\delta_1 > c =$ sup S. This contradiction shows that $c = b$.

Since $[a,b]$ is covered by $\mathscr{U}$, its element b must belong to some $U_2 \in \mathscr{U}$. The set U_2, being an element of $\mathscr{U}$, is a relatively open subinterval of $[a,b]$. Since we now know that $c = b$, it follows that

$$\text{there exists } \delta_2 > 0 \text{ such that } [c-\delta_2, c] \subseteq U_2. \tag{4}$$

Considering that $c-\delta_2$ is not an upper bound for S, there must exist $s_2 \in S \ni c-\delta_2 < s_2 \le c$. It follows from the definition of S that $\mathscr{U}$ contains a finite subcover of $[a,s_2]$ and hence, in view of (4), that $\mathscr{U}$ contains a finite subcover of the union $[a,s_2]\cup[c-\delta_2, c]$. But the inequality satisfied by s_2 shows that this union equals $[a,c]$. Thus $\mathscr{U}$ contains a finite subcover of $[a,c]$.

Since it was already shown that $c = b$, it now follows that $\mathscr{U}$ contains a finite subcover of $[a,b]$. ☺

Problem Set 6-1

6-1.P1. Describe three covers of the set $\{a,b,c\}$, each consisting of two-element subsets of it. (One was described in Example 6-1.2(b).)

6-1.P2. Describe a cover of the set $\{a,b,c\}$, which consists of proper subsets of it, each one of them containing a.

6-1.P3. (a) An example of a cover of $[0,3]$ consisting of closed intervals of length 1 is $\{[0,1],[1,2],[2,3]\}$. Give an example of a cover of $[0,3]$ consisting of closed intervals of length $\frac{3}{2}$.
(b) Give any one example of a cover of $[0,2)$ consisting of open intervals of length $\frac{3}{2}$.

6-1.P4. (a) The set $\{a,b,c,d\}$ is covered by the family $\{\{a,b\},\{b,c\},\{c,d\}\}$. Describe a proper subfamily which is a subcover, i.e. covers the same set.
(b) The set $\{a,b,c,d,e\}$ is covered by $\{\{a,b,c\},\{b,c,d\},\{c,d,e\}\}$. Describe a proper subcover.

6-1.P5. The family $\{(-\frac{1}{n}, 2-\frac{1}{n}) : n \in \mathbb{N}\}$ is a cover of $(-\frac{3}{4}, 2)$ consisting of open intervals. Determine whether there is a finite subfamily that covers $(0,2)$.

6-1.P6. Which of the following are relatively open subintervals of $[6,9]$?
(a) $(6,9]$ (b) $(7,9)$ (c) $(7,9]$ (d) $(7,10)$ (e) $(7,8)$ (f) $[7,8]$

6-1.P7. For each of the following subsets of $\mathbb{R}$, give an example of a cover whose elements are intervals of the kind indicated and which contains no finite subcover: (a) $[0,\infty)$; open intervals (b) $(0,2]$; open intervals (c) $[0,2]$; all intervals open with one exception.

6-1.P8. Give an example of a sequence of bounded closed intervals such that the union is bounded but not closed.

6-2 Polynomials and Step Functions

We have mentioned polynomials only "unofficially" so far. In this Section, we define them formally and also introduce another variety of functions on an interval. We exclude from consideration the case when the interval on which the function is defined consists of a single point.

6-2.1. Definition. *A complex valued function f whose domain is a subset S of $\mathbb{C}$ is called a* **polynomial** *if there exists a finite sequence $a_0, \dots, a_n$ of complex numbers such that*

$$f(z) = a_0 + a_1 z + \cdots + a_n z^n \quad \textit{for all } z \in S.$$

6-2.2. Remarks. (i) For most of the polynomial functions considered in the sequel, the domain will be an interval of $\mathbb{R}$. In order to avoid losing sight of this fact, we shall replace the letter z in the definition by the letter x.

(ii) The Definition does not presume that the finite sequence is unique. It would not be appropriate to call the terms of the sequence as "the" coefficients before establishing uniqueness under suitable conditions. We propose to do so in Section 7-2. Until then, we avoid any reference to the "degree" of a polynomial. Note however that a_0 is easily seen to be unique at least when $0 \in S$, because it equals $f(0)$. In particular, if $f(0) = 0$, then $a_0 = 0$.

(iii) What we have called a polynomial resembles, but is distinct from, what is known by the same name in the study of "polynomial rings" in Algebra.

(iv) Polynomials are among the easiest functions to compute with. However, there are functions that are not polynomials, the most elementary one to think of being perhaps $\sqrt{x}$. The mere fact that we are not aware of any representation of the square root function as a polynomial does not, by itself, guarantee that no such representation exists. The possibility that $\sqrt{x}$ may be a polynomial in disguise needs to be ruled out. The simplest way is by noting that it has no derivative at 0, whereas polynomials have a derivative everywhere. A proof that uses only the concepts we have discussed so far—we have not discussed derivatives yet—would be along these lines: If $(a_0 + a_1x + \cdots + a_nx^n)^2 = x$ for $0 \le x \le A$, then $a_0 = 0$, and therefore $(a_1 + \cdots + a_nx^{n-1})^2 = 1/x$ for $0 < x \le A$. But the left side here cannot exceed $(|a_1| + |a_2|A + \cdots + |a_n|A^{n-1})^2$, while the right side is unbounded above.

(v) One thing that makes polynomials easy to compute with is that if every a_j is rational, then $f(x)$ turns out to be rational when x is rational. This lends polynomials a kind of earthy practicality somewhat like rational numbers have. For example, even if it is theoretically determined that a certain length needs to be $\sqrt{2}$, a carpenter can only work with a rational approximation to it. This is feasible because a rational approximation always exists, in view of Theorem 1-9.4. In the realm of functions, it is polynomials that have such earthiness about them and it makes sense to ask what other kinds of functions can be approximated by polynomials. Here approximation is meant to be uniform, the domain being some closed bounded interval; in other words, we are asking what kinds of functions on a closed bounded interval are uniform limits of sequences of polynomials. Since polynomials are continuous, it follows by Theorem 5-1.3 that only continuous functions have a fighting chance of being such limits. According to a famous theorem of Weierstrass, called Weierstrass Approximation Theorem, all continuous real valued functions on a closed bounded interval are uniform limits of polynomials! So, trying to approximate a real valued continuous function uniformly by a sequence of polynomials is not a wild goose chase. How we go about computing a concrete polynomial approximant depends on what we know about the function. The Weierstrass Approximation Theorem will be discussed fully in Section 6-3.

We go on to describe another class of functions.

6-2.3. Definition. *If $[a,b]$ is a closed and bounded interval, any finite sequence P of points $x_0, x_1, \ldots, x_n$ such that*

$$a = x_0 < x_1 < \cdots < x_n = b$$

is called a **partition** *of $[a,b]$. A complex valued function f with $[a,b]$ as its domain is called a* **step function** *if there exists a partition $x_0, x_1, \ldots, x_n$ of $[a,b]$ such that f is constant on each of the subintervals (x_{j-1}, x_j), $1 \le j \le n$.*

The succession of inequalities $x_0 < x_1 < \cdots < x_n$ can often be expressed more conveniently by saying that the numbers $x_0, \ldots, x_n$ are in "increasing order".

Example. On the interval $[0,4]$, the function f defined as

$$f(0) = 10, \quad f(x) = 11 \text{ when } x \in (0,3), \quad f(3) = -12,$$
$$f(x) = 13 \text{ when } x \in (3,4), \quad \text{and} \quad f(4) = 51$$

is a step function. The partition consists of $0,3,4$ and the range of f is $\{10,11,-12,13,51\}$.

The same function can also be described with a different partition as

$$f(0) = 10, \quad f(x) = 11 \text{ when } x \in (0,2), \quad f(2) = 11,$$
$$f(x) = 11 \text{ when } x \in (2,3), \quad f(3) = -12,$$
$$f(x) = 13 \text{ when } x \in (3,4), \quad \text{and} \quad f(4) = 51.$$

The partition is now $0,2,3,4$ while the range is of course the same as before.

The range of a step function is finite and can have at most $2n+1$ numbers in it.

The elements of any finite set of real numbers can be arranged in increasing order. Therefore any finite set of numbers belonging to $[a,b]$ that includes the numbers a and b forms a partition of $[a,b]$ upon being arranged in increasing order.

When $a < b < c$, and

$$a = x_0 < x_1 < \cdots < x_n = b$$
$$b = y_0 < y_1 < \cdots < y_m = c$$

are partitions of $[a,b]$ and $[b,c]$ respectively, the sequence

$$a = x_0 < x_1 < \cdots < x_n < y_1 < \cdots < y_m = c$$

is a partition of $[a,c]$. It follows that, if f is a step function on $[a,b]$ and g is a step function on $[b,c]$, then a function that agrees with f on (a,b) and with g on (b,c) while having domain $[a,c]$ must be a step function.

Now suppose

$$a = x_0 < x_1 < \cdots < x_n = b$$

is a partition of $[a,b]$ and f_j is a step function on (x_{j-1}, x_j), $1 \le j \le n$. Then any function f on $[a,b]$ such that $f(x) = f_j(x)$ whenever $x \in (x_{j-1}, x_j)$, $1 \le j \le n$, is a step function; this follows by an induction argument that is best left to the imagination!

6-2.4. Definition. *Let f be a complex valued function with an interval I as its domain and let c be either a point belonging to I or the left endpoint of I, but not the right endpoint. A number L is called a* **right limit of f at** *c if for every $\varepsilon > 0$, there exists $\delta > 0$ such that $c + \delta \in I$ and*

$$|f(x) - L| < \varepsilon \text{ whenever } c < x < c + \delta.$$

The definition of **left limit** when c is not the left endpoint of I is similar. A number which is either a left limit or a right limit is called a **one sided limit** at the point.

Examples. (a) For the step function in the Example above, a left limit at 3 is 11 and at 4 is 13. There is no left limit at 0, because 0 is the left endpoint of the interval on which the function is defined. Of course, *the only* left limit at 3 is 11. Similarly, *the only* right limit at 1 is 11 and at 3 is 13.

(b) If a function with an interval as domain is continuous at a point other than the right [resp. left] endpoint, then the right [resp. left] limit exists and equals the value of the function there. When it is continuous at a point which is neither the left nor the right endpoint, both one sided limits exist and are equal to the value at the point.

(c) The function $f(x) = \frac{1}{x},\ 0 < x < 2,$ has no right limit at 0 but has left limit $\frac{1}{2}$ at 2.

(d) Let $g(x) = \min\{x - [x], 1 - (x - [x])\}$. Define f on $(0,1)$ as $f(x) = g(\frac{1}{x})$. For any $n \in \mathbb{N}$, if $x = 1/n$, we have $f(x) = 0$ and if $x = 1/(n + \frac{1}{2})$, we have $f(x) = \frac{1}{2}$. Therefore f has no right limit at 0. But unlike in the preceding Example, this f is bounded. [If the function g seems outlandish, replace it by $\sin x$.]

It is left to the reader to show that a one sided limit, if it exists, is unique. Therefore we speak of *the* left limit or of *the* right limit. We shall denote them by $f(c-)$ and $f(c+)$ respectively, although other notations are also in use.

A part of what we noted about continuous functions in Example (b) is that, if the domain is an interval and c is a point of the domain other than the right endpoint, the right limit $f(c+)$ exists; also if c is a point of the domain other than the left endpoint, the left limit $f(c-)$ exists. This much is true of any step function. It is also true of another class of functions which we now introduce formally:

6-2.5. Definition. *A function* $f: I \to \mathbb{R}$, *where* I *is an interval*, *is said to be* **increasing** *if*

$$x_1 < x_2 \;\Rightarrow\; f(x_1) \le f(x_2)$$

and is said to be **decreasing** *if*

$$x_1 < x_2 \;\Rightarrow\; f(x_1) \ge f(x_2).$$

A function is **monotone** *if it is either increasing or decreasing.*

A monotone function on a bounded closed interval can be shown to have a left limit everywhere except at the left endpoint and a right limit everywhere except at the right endpoint. Indeed, if f is increasing on $(u, v]$, the nonempty set $\{f(x) : u < x < v\}$ is bounded above by $f(v)$ and therefore has a sup, say L. For any $\varepsilon > 0$, there exists $x_1 \in (u, v)$ such that $L - \varepsilon < f(x_1) \leq L$. Since f is increasing, it follows that

$$x_1 < x < v \Rightarrow L - \varepsilon < f(x) \leq L \Rightarrow |f(x) - L| < \varepsilon.$$

Now take $\delta = v - x_1$ to see why $f(v-) = L$.

If a function having an interval domain has a left limit at each point except the left endpoint (if any) of the interval and also has a right limit at very point except the right endpoint (if any) will be described as **having a one sided limit at each point**.

Suppose A is a set of functions with domain X and f is also a function with domain X, but not necessarily belonging to A. We say that f **can be uniformly approximated by functions of** A if

for every $\varepsilon > 0$, there exists $\phi \in A$ such that $|f(x) - \phi(x)| < \varepsilon$ for every $x \in X$.

This terminology makes it possible to state the next two results with almost no technical symbols. We leave it to the reader to check that an equivalent form of the above condition is that there exists a sequence of functions in A that converges uniformly to f.

***6-2.6. Proposition.** *If a complex valued function f on a closed and bounded interval $[a, b]$ can be uniformly approximated by functions having a one sided limit at each point, then f also has a one sided limit at each point.*

Proof. Let $\{s_n\}$ be a sequence of functions having a one sided limit at each point and converging uniformly to f. Let $c \in [a, b)$; we shall show that the right limit $f(c+)$ exists.

Since each s_n has a right limit at c, we have the sequence $s_n(c+)$ in $\mathbb{C}$. We shall show that this is a Cauchy sequence. Consider any $\varepsilon > 0$. Since s_n converges to f uniformly, there exists some n_0 such that

$$n \geq n_0 \Rightarrow |f(x) - s_n(x)| < \tfrac{\varepsilon}{4} \text{ for all } x \in [a, b].$$

Choose any n and m both greater than or equal to n_0. Then

$$|s_n(x) - s_m(x)| < \tfrac{\varepsilon}{2} \text{ for all } x \in [a, b].$$

By definition of right limit, there exists $\delta > 0$ such that $c + \delta \leq b$ and

$$c < x < c + \delta \Rightarrow |s_n(x) - s_n(c+)| < \tfrac{\varepsilon}{4} \text{ as well as } |s_m(x) - s_m(c+)| < \tfrac{\varepsilon}{4}.$$

Upon considering any x such that $c < x < c + \delta$, it follows that $|s_n(c+) - s_m(c+)| < \frac{\varepsilon}{4} + \frac{\varepsilon}{4} + \frac{\varepsilon}{2} = \varepsilon$. Therefore $s_n(c+)$ is a Cauchy sequence, as claimed.

Denote the limit of the sequence $s_n(c+)$ by L. We shall show that $f(c+)$ exists and equals L. Consider any $\varepsilon > 0$. Choose N such that

$$|s_N(c+) - L| < \frac{\varepsilon}{3} \text{ as well as } |f(x) - s_N(x)| < \frac{\varepsilon}{3} \text{ for all } x \in [a, b].$$

Let $\delta > 0$ be such that $c + \delta \leq b$ and

$$c < x < c + \delta \Rightarrow |s_N(x) - s_N(c+)| < \frac{\varepsilon}{3}.$$

Then

$$\begin{aligned} c < x < c + \delta \Rightarrow |f(x) - L| &\leq |f(x) - s_N(x)| + |s_N(x) - s_N(c+)| + |s_N(c+) - L| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Thus $f(c+)$ exists and equals L.

By a similar argument, the left limit $f(c-)$ exists at any $c \in (a, b]$. Thus f has a one sided limit at each point. ☺

It follows from this Proposition that a uniform limit of a sequence of step functions on a closed bounded interval must have a one sided limit at each point. We go on to prove the converse.

***6-2.7. Proposition.** *A complex valued function f with domain a closed bounded interval $[a, b]$ and having a one sided limit at each point can be uniformly approximated by step functions. Moreover, if f is real valued, then it can be uniformly approximated by real valued step functions.*

Proof. Let ε be any positive number. We shall show that there exists a step function s such that $|f(x) - s(x)| < \varepsilon$ for all $x \in [a, b]$. It will be clear from the way s is constructed that it is real valued whenever f is.

For each $c \in [a, b)$, there exists some $\delta_1 > 0$ such that $c + \delta_1 \leq b$ and

$$c < x < c + \delta_1 \Rightarrow |f(x) - f(c+)| < \varepsilon.$$

On the domain $[c, c + \delta_1)$, let g_1 be the function for which $g_1(c) = f(c)$ and $g_1(x) = f(c+)$ when $c < x < c + \delta_1$. Then g_1 satisfies

$$|f(x) - g_1(x)| < \varepsilon \text{ for all } x \text{ such that } c \leq x < c + \delta_1.$$

Similarly, for each $c \in (a, b]$, there exists some $\delta_2 > 0$ such that $c - \delta_2 \geq a$ and the function g_2, which is defined on the domain $(c - \delta_2, c]$ as $g_2(c) = f(c)$ and $g_2(x) = f(c-)$ when $c - \delta_2 < x < c$, satisfies

$$|f(x) - g_2(x)| < \varepsilon \text{ for all } x \text{ such that } c - \delta_2 < x \leq c.$$

It follows that each $c \in [a,b]$ belongs to some relatively open subinterval U of $[a,b]$, on which there is a step function g_U satisfying $|f(x)-g_U(x)| < \varepsilon$ for all $x \in U$. [In case $a < c < b$, the interval U is $(c-\delta_2, c+\delta_1)$ and function g_U is constant on $(c-\delta_2, c)$ and on $(c, c+\delta_1)$; in case $c = a$, the interval U is $[c, c+\delta_1)$ and g_U is constant on $(c, c+\delta_1)$.] These relatively open subintervals therefore form a cover of $[a,b]$, and by the Heine-Borel Theorem 6-1.5, it must contain a finite subcover. Denote the finite subcover by $\mathscr{U}$.

The subintervals containing a and b must have these as one endpoint. Therefore a and b are among the finitely many endpoints of all the various subintervals in $\mathscr{U}$. The finitely many endpoints form a partition of $[a,b]$, which we shall denote by

$$a = x_0 < x_1 < \cdots < x_n = b.$$

Now define a function s on $[a,b]$ in the following manner. If x is one of the endpoints, then $s(x) = f(x)$. For each j $(1 \le j \le n)$, choose any ξ_j belonging to (x_{j-1}, x_j); it must belong to some $U \in \mathscr{U}$. The restriction of g_U to (x_{j-1}, x_j) is obviously a step function; call it g_j. Since there are no endpoints between x_{j-1} and x_j, the left endpoint of U, being less than ξ_j, must be less than or equal to x_{j-1}; similarly its right endpoint must be greater than or equal to x_j. Thus U must contain the entire interval (x_{j-1}, x_j). Now the function g_U satisfies $|f(x)-g_U(x)| < \varepsilon$ for all $x \in U$ and hence $|f(x)-g_j(x)| < \varepsilon$ for all $x \in (x_{j-1}, x_j)$. Define $s(x) = g_j(x)$ for all $x \in (x_{j-1}, x_j)$ and $s(x_j) = f(x_j)$, $1 \le j \le n$. In the light of the observation immediately preceding Def.6-2.4, s is a step function with domain $[a,b]$ and satisfies $|f(x)-s(x)| < \varepsilon$ for all $x \in [a,b]$. ☺

It is a consequence of what has just been proved that on a closed bounded interval, a function having a one sided limit at each point is bounded. For a proof of this by other means, see 6-2.P3.

***6-2.8. Proposition.** *On a closed bounded interval, a continuous function is a uniform limit of step functions. So also an increasing or a decreasing function.*

Proof. Since continuous functions as well as increasing or decreasing functions on a closed bounded interval have a one sided limit at each point, both kinds of functions are uniform limits of step functions by Prop.6-2.7. ☺

Problem Set 6-2

6-2.P1. Show that a function cannot have two different left limits at a point.

6-2.P2. Give an example of a step function f on $[0, 1]$ such that $|f(x) - x^2| \leq 1/4$ for every $x \in [0, 1]$.

6-2.P3. (a) Show that a function f having a one sided limit at each point is "locally bounded at each point of the domain" in the sense that for each c in the domain, there exists some $\delta > 0$ and some $M > 0$ such that $|f(x)| \leq M$ for those $x \in (c - \delta, c + \delta)$ that are in the domain.
(b) Show that a locally bounded function on a closed bounded interval as domain is bounded.

***6-2.P4.** Show that the function f of 5-2.P5 has a one sided limit at each point.

6-2.P5. Let $f:[8, 11] \to \mathbb{R}$ be the step function having values $0, 1, 2, 1$ on the respective subintervals $[8, 9], (9, 10), \{10\}, (10, 11]$ of its domain.
(a) For each $t \in [0, 5]$, describe $\{x \in [8, 11] : f(x) \leq t\}$.
(b) For each $t \in [0, 5]$, find the total length of the subintervals having union equal to $\{x \in [8, 11] : f(x) \leq t\}$.

6-2.P6. Let the symbols m, n, k denote nonnegative integers. Denote by P_n the class of all polynomials f such that $f(x) = a_0 + a_1 x + \cdots + a_n x^n$ on $\mathbb{R}$, for some finite sequence of real numbers $a_0, \ldots, a_n$. It is not required that a_n be nonzero. Prove each of the following:

(a) If $f \in P_n$, and $g(x) = xf(x)$, then $g \in P_{n+1}$. Also $P_n \subset P_{n+1}$.

(b) If $f, g \in P_n$, then $f + g \in P_n$. Also, if $f \in P_n$ and $\alpha \in \mathbb{R}$, then $\alpha f \in P_n$.

(c) If $f \in P_n$ and $g(x) = x^m f(x)$, then $g \in P_{m+n}$.

(d) If $\alpha \in \mathbb{R}$ and f, g are polynomials, then $f + g$, fg and αf are also polynomials.

(e) If $c \in \mathbb{R}$, then $x^{n+1} - c^{n+1} = (x - c)g(x)$, where $g \in P_n$.

(f) If $c \in \mathbb{R}$ and $f \in P_{n+1}$, then $f(x) - f(c) = (x - c)g(x)$, where $g \in P_n$.

(g) If $f \in P_n$ but $f \notin P_0$, then the equation $f(x) = 0$ has at most n roots.

(h) If $c \in \mathbb{R}, f \in P_{n+1}, f(c) = 0$ but f is not zero everywhere, then there exist k and m such that $m > 0$ and

$$f(x) = (x - c)^m g(x), \quad \text{where } \; g \in P_k,\ g(c) \neq 0 \; \text{ and } \; m + k = n + 1.$$

*6-3 Stone-Weierstrass Theorem

The Weierstrass Approximation Theorem promised in Section 6-2 has a proof that was discovered by Stone, and is valid in a more general setting than what we indicated there. This is the proof that will be presented here, but in a form that can also be read in the original context of intervals rather than the more general setting. The original theorem of Weierstrass asserts that every continuous real valued function f on a closed bounded interval can be uniformly approximated by polynomials. To put it explicitly, for every $\varepsilon > 0$, there exists a polynomial g such that $|f(x) - g(x)| < \varepsilon$ for every x in the closed bounded interval. Stone's approach involves exploiting a certain property of the set of all polynomials that other sets of functions also have. Consequently, the result is applicable to these other sets of functions as well. One of the features of the relevant property of the set of polynomials is spelt out here:

6-3.1. Definition. *A nonempty set A of real valued functions on a domain S is called an* **algebra of functions** *if, whenever* $f, g \in A$ *and* $\alpha \in \mathbb{R}$, *we have*

$$\text{(Alg1)}\ \ f+g \in A \quad \text{(Alg2)}\ \ fg \in A \quad \text{and}\ \ \text{(Alg3)}\ \ \alpha f \in A.$$

6-3.2. Examples. (a) The set of all polynomials $f(x) = a_0 + a_1x + \cdots + a_nx^n$ on an interval, say $[0, 1]$, where each a_j is real, is an algebra of functions. This is because the sum, product and constant multiples of polynomials are polynomials, as can be easily verified on the basis of Def.6-2.1.

(b) The set of all polynomials $f(x) = a_0 + a_3x^3 + \cdots + a_nx^n$ on an interval, say $[0, 1]$, where each a_j is real, is an algebra of functions. This is because the sum, product and constant multiples of polynomials of this particular kind (with $a_1 = a_2 = 0$) are polynomials of the same kind, as can also be verified on the basis of Def.6-2.1. If we further restrict a_0 also to be 0, so that all terms are of "degree 3 or higher", we still have an algebra, but it will have the property that all functions in it have value 0 (i.e. vanish) at 0. Thus there is at least one point in the domain $[0, 1]$ where every function in the algebra vanishes. In the Stone-Weierstrass Theorem below, such an algebra is excluded by the hypothesis.

Another kind of algebra that is also excluded is the following. Let A consist of all polynomials function f on $[0, 1]$ which are of the form

$$f(x) = a_0 + x(1-x)P(x),$$

where P is a polynomial function. It is easy to verify these functions also form an algebra; but all of them have the additional property that $f(0) = f(1)$. Thus the algebra A *does not* "separate the pair of points 0 and 1". The exclusion of such algebras will be expressed by saying that the algebra considered should "separate points", meaning that no such pair of points must exist in the domain. The algebra of polynomials separates points; it also fulfils the requirement that there be no point where every function in the algebra vanishes. This is the property of the set of polynomials that Stone's approach exploits.

(c) The set of real valued step functions on an interval form an algebra of functions. It can be checked by a slightly cumbersome argument that a sum and product of step functions are step functions. Constant multiples are easy to check.

(d) If in (a), we permit either complex values of x or complex values of the numbers a_j, the requirements (Alg1)–(Alg3) will still be fulfilled; however, the functions will cease to be real valued. In some situations, it would be appropriate to regard this as an algebra by modifying the definition to allow complex valued functions. However, doing so does not suit the purpose in this Section. It may be noted in passing that complex α could also be allowed, in which case, one would need to distinguish between real and complex algebras; Def.6-3.1 admits only "real algebras".

(e) The set of continuous real valued functions on the domain $\mathbb{R}$ form an algebra and so do the continuous bounded real valued functions. However, the uniformly continuous functions do not form an algebra, because (Alg2) breaks down. Indeed, $f(x) = x$ is uniformly continuous on $\mathbb{R}$ but $f(x) = x^2$ is not. The remaining requirements of Def.6-3.1 are fulfilled by this set of functions.

6-3.3. Definition. *An algebra of functions on a set X is said to* **separate the points $x_1, x_2 \in X$,** *where* $x_1 \neq x_2$, *if there exists* $f \in A$ *such that* $f(x_1) \neq f(x_2)$. *It is said to* **separate points** *if it separates* $x_1, x_2 \in X$ *whenever* $x_1 \neq x_2$.

In the Examples above, when we recognised certain sets of functions as being algebras, the basis for doing so was our knowledge of a fairly explicit form of the functions in the set. The next result provides, among other things, a different kind of basis for recognising an algebra. In the proof of the Stone-Weierstrass Theorem, recognising an algebra on such a basis will constitute an essential step.

6-3.4. Proposition. *Let A be an algebra of bounded functions on a set X and B be the set of functions f on X that can be uniformly approximated by functions of A, i.e.*

for every $\varepsilon > 0$, *there exists* $\phi \in A$ *such that* $|f(x) - \phi(x)| < \varepsilon$ *whenever* $x \in X$. (I)

Then

(a) *$A \subseteq B$ and B is an algebra of bounded functions*;

(b) *if f is a function on X such that* (I) *holds with* $\phi \in B$ (*instead of* $\phi \in A$), *then* $f \in B$;

(c) *any pair of points that is not separated by A is also not separated by B*;

(d) *if all functions of* A *vanish at some point, then so do all functions of* B.

Proof. (a) It is clear that $A \subseteq B$. Consider any $f \in B$. We argue that f is bounded. Let ϕ satisfy (I) with some ε. Then $|f(x)| < |\phi(x)| + \varepsilon$. Being an element of A, the function ϕ is bounded. Therefore f is bounded.

Now consider any $f, g \in B$. For any $\varepsilon > 0$, there exist $\phi, \psi \in A$ such that

$$|f(x) - \phi(x)| < \tfrac{\varepsilon}{2} \text{ and } |g(x) - \psi(x)| < \tfrac{\varepsilon}{2} \quad \text{whenever } x \in X.$$

It follows that $\phi + \psi \in A$ and that $|(f+g)(x) - (\phi+\psi)(x)| < \varepsilon$ for all $x \in X$. Therefore $f + g \in B$.

We shall now show that $fg \in B$. Let $M = \sup\ \{|f(x)| : x \in X\}$ and $N = \sup\ \{|g(x)| : x \in X\}$. If either M or N is 0, then $fg = 0$ and hence $fg \in B$. So, assume $M \neq 0 \neq N$. Then there exist $\phi, \psi \in A$ such that

$$|f(x) - \phi(x)| < \min\ \{{}^{\varepsilon}\!/_{2N}, \varepsilon\}, \quad |g(x) - \psi(x)| < \tfrac{1}{2}\frac{\varepsilon}{M+\varepsilon} \text{ whenever } x \in X.$$

It follows for every $x \in X$ that $|\phi(x)| < |f(x)| + \varepsilon \leq M + \varepsilon$ and hence that

$$\begin{aligned} |f(x)g(x) - \phi(x)\psi(x)| &\leq |f(x)g(x) - \phi(x)g(x) + \phi(x)g(x) - \phi(x)\psi(x)| \\ &\leq |g(x)| \cdot |f(x) - \phi(x)| + |\phi(x)| \cdot |g(x) - \psi(x)| \\ &< N({}^{\varepsilon}\!/_{2N}) + (M+\varepsilon)\tfrac{1}{2}\frac{\varepsilon}{M+\varepsilon} = \varepsilon. \end{aligned}$$

Since $\phi\psi \in A$, this shows that $fg \in B$.

If $f \in B$, $0 \neq \alpha \in \mathbb{R}$ and $\varepsilon > 0$, then the function $\phi \in A$ for which $|f(x) - \phi(x)| < {}^{\varepsilon}\!/_{|\alpha|}$ for all $x \in X$ has the property that $|\alpha f(x) - \alpha\phi(x)| < \varepsilon$ for all $x \in X$. Since $\alpha\phi \in A$, this shows that $\alpha f \in B$ when $\alpha \neq 0$. When $\alpha = 0$, we have $\alpha f = 0 \in B$.

(b) Suppose f has property (I) but with $\phi \in B$. For a given $\varepsilon > 0$, take some $\phi \in B$ such that $|f(x) - \phi(x)| < \frac{\varepsilon}{2}$ for all $x \in X$. Since $\phi \in B$, there exists $\psi \in A$ such that $|\phi(x) - \psi(x)| < \frac{\varepsilon}{2}$ for all $x \in X$. It follows that $|f(x) - \psi(x)| < \varepsilon$ for all $x \in X$. Since $\psi \in A$, this shows that $f \in B$.

(c) and (d) These facts will not be used towards any major result in the sequel and are left to the reader to prove [see 6-3.P9]. ☺

For any set A of bounded functions on a set X, the set B of bounded functions on X satisfying condition (I) above, i.e. can be uniformly approximated by functions of A, is called the **uniform closure of** A. Thus the uniform closure of an algbra of functions is also an algebra.

We are about to use Dini's Theorem 5-1.8 on $X = [0,1]$, according to which, *if a decreasing sequence* $\{f_n\}$ *of continuous functions on* $[0,1]$ *has a pointwise limit* f *which is continuous, then the convergence is uniform.*

6-3.5. Proposition. *There exists a sequence* $\{P_n\}$ *of polynomials with domain* $[0,1]$ *which converges uniformly to the square root function* $f(x) = \sqrt{x}$ *and satisfies* $P_n(0) = 0$ *for every* n.

Proof. Let $\{P_n\}$ be the sequence of functions on $[0,1]$ defined inductively by

$$P_1(x) = 0, \quad P_{n+1}(x) = P_n(x) + \tfrac{1}{2}(x - P_n(x)^2).$$

It follows by a straightforward induction argument that each P_n is a polynomial and that $P_n(0) = 0$ for every n. Assume $\sqrt{x} \geq P_{n+1}(x) \geq P_n(x) \geq 0$ as induction hypothesis. Then $x - P_{n+1}(x)^2 \geq 0$ and hence $P_{n+2}(x) \geq P_{n+1}(x) \geq 0$. As $1 \geq \sqrt{x}$, we also have $1 - P_{n+1}(x) \geq 1 - \sqrt{x} \geq 0$, which leads successively to

$$(1 - P_{n+1}(x))^2 \geq (1 - \sqrt{x})^2,$$
$$1 - 2P_{n+1}(x) + P_{n+1}(x)^2 \geq 1 - 2\sqrt{x} + x,$$
$$2\sqrt{x} \geq x + 2P_{n+1}(x) - P_{n+1}(x)^2,$$
$$\sqrt{x} \geq P_{n+1}(x) + \tfrac{1}{2}(x - P_{n+1}(x)^2) = P_{n+2}(x).$$

For $n = 1$, the first inequality in the induction hypothesis says that $\sqrt{x} \geq \frac{1}{2}x$, which is equivalent to $2 \geq \sqrt{x}$ and therefore true for $x \in [0,1]$. The other two inequalities therein say that $\frac{1}{2}x \geq 0 \geq 0$, which is also true. The induction is now complete.

So, $\{P_n\}$ is a nonnegative valued increasing sequence of continuous functions and must therefore have a pointwise limit f. Now f must satisfy $f(x) = f(x) + \frac{1}{2}(x - f(x)^2)$, and hence $f(x) = \sqrt{x}$. This means $\sqrt{x} - P_n(x)$ gives a decreasing sequence of continuous functions with pointwise limit 0 everywhere. The limit function is continuous and the convergence is uniform by Dini's Theorem 5-1.8. It is immediate that the convergence of $\{P_n\}$ is also uniform. ☺

A noteworthy consequence of what we have just proved is that a uniform limit of polynomials on $[0,1]$ need not be a polynomial.

We now come to the major result of this Section. A statement of the result without mathematical symbols would be as follows: The uniform closure of a point separating algebra of continuous functions on a closed bounded interval that do not all vanish at the same point of the domain is the algebra of *all* continuous functions.

6-3.6. Stone-Weierstrass Theorem. *Let* f *be a continuous real valued function on a closed bounded interval* X *and let* A *be a point separating algebra of continuous real valued functions on* X *such that there is no point where all functions of* A *vanish. Then* f *can be uniformly approximated by functions in* A, *meaning that*

$$\text{for every } \varepsilon > 0 \text{ there exists } \phi \in A \text{ such that } |f(x) - \phi(x)| < \varepsilon \text{ whenever } x \in X. \qquad \text{(I)}$$

Proof. Let B denote the set of real valued functions f on X for which (I) holds. We have to show that every continuous real valued function on X belongs to B. Since A consists of continuous functions, it follows by Prop.5-1.3 that every function in B is continuous. Since X is a closed bounded interval, it further follows by the Boundedness Theorem 2-3.2 that every function in B is also bounded. Therefore by Prop.6-3.4(a), (b), B is an algebra and it is sufficient to show that any continuous real valued function f on X has property (I) with $\phi \in B$ (instead of $\phi \in A$). We show this in five steps. With this in view, consider any continuous real valued function f on X.

Step 1. *If $x_1, x_2 \in X$ and $c_1, c_2 \in \mathbb{R}$, where $x_1 \neq x_2$, then there exists $F \in A$ such that $F(x_1) = c_1$ and $F(x_2) = c_2$.*

First we show that there exists $h \in A$ such that $0 \neq h(x_1) \neq h(x_2) \neq 0$.

Since A separates points, there exists $f_0 \in A$ taking different values at x_1 and x_2, one of which must be nonzero, $f_0(x_1)$ say. Then

$$0 \neq f_0(x_1) \neq f_0(x_2). \tag{1}$$

If $f_0(x_2) \neq 0$, then $h = f_0$ serves our purpose. Consider the possibility that $f_0(x_2) = 0$. Since there is no point where all the functions of A vanish, some $g \in A$ must satisfy $g(x_2) \neq 0$. By (1), there exists some real number λ such that

$$\lambda f_0(x_1) \neq -g(x_1) \quad \text{and} \quad \lambda[f_0(x_1) - f_0(x_2)] \neq -[g(x_1) - g(x_2)]. \tag{2}$$

Then $h = \lambda f_0 + g$ serves our purpose, because

$$h(x_1) = \lambda f_0(x_1) + g(x_1) \neq 0 \qquad \text{by the first part of (2),}$$

$$h(x_2) = \lambda f_0(x_2) + g(x_2) = g(x_2) \neq 0,$$

and $$h(x_1) = \lambda f_0(x_1) + g(x_1) \neq \lambda f_0(x_2) + g(x_2) = h(x_2) \quad \text{by the second part of (2).}$$

Now $h(x_1)h(x_2)[h(x_1) - h(x_2)] \neq 0$. This implies [via an elementary computation] the existence of real numbers α and β such that

$$\alpha h(x_1) + \beta h(x_1)^2 = c_1 \text{ and } \alpha h(x_2) + \beta h(x_2)^2 = c_2.$$

The function $F = \alpha h + \beta h^2$ then satisfies $F(x_1) = c_1$ and $F(x_2) = c_2$, and also belongs to A.

Step 2. *If $g \in B$, then $|g| \in B$.*

As already noted, g is bounded. Let $M = \sup\{|g(x)| : x \in X\}$. In case $M = 0$, we have $g = 0$ and there is nothing to prove. So, assume $M > 0$ and consider any $\varepsilon > 0$. By Prop. 6-3.5, there exists a polynomial P on $[0, 1]$ such that $|P(t) - \sqrt{t}| < \varepsilon/M$ for each $t \in [0, 1]$ and $P(0) = 0$. It follows that

$$|t| \leq 1 \;\Rightarrow\; |P(t^2) - \sqrt{t^2}| < \varepsilon/M \;\Rightarrow\; |P(t^2) - |t|| < \varepsilon/M$$

and hence that

$$|t| \le M \;\Rightarrow\; |P(t^2/M^2) - |t|/M| < \varepsilon/M \;\Rightarrow\; |MP(t^2/M^2) - |t|| < \varepsilon.$$

Therefore $|MP(g(x)^2/M^2) - |g(x)|| < \varepsilon$ for every $x \in X$. Since B is an algebra and $P(0) = 0$, the function given by $MP(g(x)^2/M^2)$ belongs to B. Thus $|g|$ satisfies (I) with $\phi \in B$ given by $\phi(x) = MP(g(x)^2/M^2)$. Hence by Prop.6-3.4(b), we have $|g| \in B$.

Step 3. *If* $f_1, \ldots, f_n \in B$, *then* $\min\{f_1, \ldots, f_n\} \in B$ *and* $\max\{f_1, \ldots, f_n\} \in B$.

Since we have [see 1-4.P12]

$$\min\{f_1, f_2\} = \tfrac{1}{2}(f_1 + f_2 - |f_1 - f_2|) \text{ and } \max\{f_1, f_2\} = \tfrac{1}{2}(f_1 + f_2 + |f_1 - f_2|),$$

it follows from Step 2 that $\min\{f_1, f_2\} \in B$ and $\max\{f_1, f_2\} \in B$. An induction argument now completes the Step.

Step 4. *For any* $\xi \in X$ *and* $\varepsilon > 0$, *there exists* $g_\xi \in B$ *such that*

$$g_\xi(\xi) = f(\xi) \text{ and } g_\xi(x) > f(x) - \varepsilon \text{ for all } x \in X. \tag{3}$$

By Step 1, for each $y \in X$, there exists $F_y \in A$ such that

$$F_y(\xi) = f(\xi). \tag{4}$$

and $F_y(y) = f(y) > f(y) - \varepsilon$. Since F_y and f are both continuous, by Remark 6-1.4(i) there exists relatively open subinterval $U_y \subseteq X$ such that $y \in U_y$ and

$$F_y(x) > f(x) - \varepsilon \quad \text{for all } x \in U_y. \tag{5}$$

Since $y \in U_y$, the family $\{U_y : y \in X\}$ is a cover of X and, by the Heine-Borel Theorem, must contain a finite subcover, say $\{U_{y_1}, U_{y_2}, \ldots, U_{y_n}\}$:

$$X = U_{y_1} \cup U_{y_2} \cup \cdots \cup U_{y_n}. \tag{6}$$

Let $g_\xi = \max\{F_{y_1}, F_{y_2}, \ldots, F_{y_n}\}$. By Step 3, we have $g_\xi \in B$. Besides, it follows from (4) that $g_\xi(\xi) = f(\xi)$ and from (5) and (6) that $g_\xi(x) > f(x) - \varepsilon$ for all $x \in X$.

Step 5. *For any* $\varepsilon > 0$, *there exists* $g \in B$ *such that*

$$|f(x) - g(x)| < \varepsilon \quad \textit{for all } x \in X.$$

For each $\xi \in X$, consider the function g_ξ of Step 4. Since $g_\xi(\xi) = f(\xi) < f(\xi) + \varepsilon$, and both the functions f and g_ξ are continuous, there exists a relatively open subinterval $V_\xi \subseteq X$ such that $\xi \in V_\xi$ and

$$g_\xi(x) < f(x) + \varepsilon \quad \text{for every } x \in V_\xi. \tag{7}$$

Since $\xi \in V_\xi$, the family $\{V_y : y \in X\}$ is a cover of X and, by the Heine-Borel Theorem, must contain a finite subcover, say $\{V_{y_1}, V_{y_2}, \ldots, V_{y_n}\}$:

$$X = V_{y_1} \cup V_{y_2} \cup \cdots \cup V_{y_n}. \tag{8}$$

Let $g = \min\{g_{\xi_1}, g_{\xi_2}, \ldots, g_{\xi_n}\}$. By Step 3, we have $g \in B$. Besides, it follows from (3) that $g(\xi) = f(\xi)$ and from (3), (7) and (8) that $f(x) - \varepsilon < g(x) < f(x) + \varepsilon$, i.e. $|f(x) - g(x)| < \varepsilon$ for every $x \in X$. ☺

The Stone-Weierstrass Theorem has been used in the study of "neural networks"; see *The Stone-Weierstrass theorem and its applications in neural networks* by N.E.Cotter (1990) in *IEEE Transactions on Neural Networks*, 1, 290-295.

Problem Set 6-3

6-3.P1. For each of the following sets of functions on $[0, 2]$, determine whether it is an algebra. If not, state which requirement of the definition of algebra is not fulfilled.

(a) Rational valued step functions.

(b) Monotone functions.

(c) $f(x) = ax + b$ for $x \in [0, 2]$, where a, b are real.

(d) Restriction of a polynomial on each of $[0, 1)$ and $[1, 2]$, though not necessarily the same polynomial on the two subintervals.

6-3.P2. What is the uniform closure of the algebra of real valued step functions on a closed bounded interval?

6-3.P3. Determine whether the set of functions f defined on $\mathbb{C}$ such that there is at least one expression $f(z) = a_0 + a_1 z + \cdots + a_n z^n$, where each $a_j \in \mathbb{C}$, is an algebra.

6-3.P4. Show that the uniformly continuous real valued functions on $(0, 1)$ form an algebra.

6-3.P5. (a) Fix some function $g: [0, \infty) \to \mathbb{R}$. Then form the set Φ of all functions that can be expressed as a finite product $g(c_1 x) g(c_2 x) \cdots g(c_j x)$ for some choice of a finite sequence $\{c_1, \ldots, c_j\}$ of positive numbers, not necessarily integers. Now let A consist of all functions f on $[0, 1]$ that are finite sums

$$f(x) = \sum_{k=1}^{n} a_k \phi_k(x), \text{ where each } a_k \text{ is real and each } \phi_k \in \Phi,$$

and A_1 consist of all functions f on $[0, 1]$ that are finite sums

$$f(x) = a_0 + \sum_{k=1}^{n} a_k \phi_k(x), \text{ where each } a_k \text{ is real and each } \phi_k \in \Phi.$$

Are A and A_1 algebras?

(b) In part (a), modify Φ to consist of finite products of the same kind but with $j \geq 3$ (or 17 if desired). Are A and A_1 algebras? Describe a general function in A when $g(x) = x$ and also when $g(x) = \sqrt{x}$.
(c) Define a function $g:[0,\infty)\to\mathbb{R}$ such that the set A_1 of part (b) consists of all functions of the form $f(x) = a_0 + a_1x^5 + a_2x^{20/3} + a_3x^{25/3} + \cdots$ to finitely many terms.

6-3.P6. For any $\varepsilon > 0$, show that there exists some $n \in \mathbb{N}$ and a polynomial $P(x) = a_3x^3 + a_4x^4 + \cdots + a_nx^n$ such that $|x - P(x)| < \varepsilon$ for each $x \in [0,1]$.

6-3.P7. Let A be an algebra of continuous real valued functions on a closed bounded interval X. Suppose f is a continuous real valued function that does not separate any pair of points that A does not separate, and vanishes at any point where all the functions of A vanish. Show that, for any $\varepsilon > 0$, there exists $g \in A$ such that $|f(x) - g(x)| < \varepsilon$ for each $x \in X$.

6-3.P8. Let f be a continuous real valued function on $[0,2\pi]$ such that $f(0) = f(2\pi)$. Show that, for any $\varepsilon > 0$, there exists a trigonometric polynomial g such that $|f(x) - g(x)| < \varepsilon$ for each $x \in [0,2\pi]$.

6-3.P9. (a) Prove parts (c) and (d) of Prop.6-3.4. Show that the uniform closure of an algebra of continuous real valued functions on a closed bounded interval consists of those continuous functions on the space that (1) do not separate any pair of points unless the algebra separates them and (2) vanish at any point where all functions of the algebra may vanish.
(b) Obtain the following result as a corollary of part (a): Given any $\varepsilon > 0$, there exists a finite sum $P(x) = \sum_{k=5}^{n} a_kx^k(1-x)^k$ such that $|x^{1/3} + (1-x)^{1/3} - 1 - P(x)| < \varepsilon\ \forall\ x \in [0,1]$.

6-3.P10. Fix some function $g:[0,\infty)\to\mathbb{R}$ and let Φ be the set of all functions that can be expressed as a finite product $g(n_1x)g(n_2x)\cdots g(n_jx)$ for some choice of a finite sequence $n_1,\ldots,n_j$ of positive integers. Now let A consist of all functions f on $[0,\alpha]$ that are finite sums

$$f(x) = \sum_{k=1}^{n} a_k\phi_k(x), \text{ where each } a_k \text{ is real and each } \phi_k \in \Phi,$$

and A_1 consist of all functions f on $[0,\alpha]$ that are finite sums

$$f(x) = a_0 + \sum_{k=1}^{n} a_k\phi_k(x), \text{ where each } a_k \text{ is real and each } \phi_k \in \Phi.$$

(a) Show that A and A_1 are algebras. In case g vanishes at 0, show that all functions in Φ do the same, and hence all the functions in A, but not the ones in A_1. If g is continuous and its restriction to $[0,\alpha]$ separates points, what are the uniform closures of A and of A_1?

(b) If $g(x) = \cos x$ and $\alpha = 2\pi$, show that A_1 consists of the restrictions to $[0, 2\pi]$ of certain kinds of trigonometric polynomials. What is the uniform closure of A_1?

Chapter 7
Differentiation

7-1 Preliminaries on Limit Points and Limits

It is a familiar definition that the derivative of a function f at a point x of its domain A is the limit of the so called "difference quotient"

$$\frac{f(x+h)-f(x)}{h} \tag{1}$$

as h approaches 0. For the quotient in (1) to be meaningful, one must have

$$|h|>0,\ x\in A \text{ and } x+h\in A.$$

Informally speaking, the limiting "process" is only possible if A always contains some number $x+h$ with "small" but nonzero $|h|$, which is to say that A always contains a number that belongs to any interval of the form $(x-\delta,x+\delta)$ but is different from x. It will often be convenient to rephrase this by saying that any interval $(x-\delta,x+\delta)$ contains a number of A different from x. We now make this formal:

7-1.1. Definition. *A* **limit point of** *a nonempty subset $A\subseteq\mathbb{R}$ is a real number x* [*which may or may not belong to the subset*] *such that, for every $\delta>0$, the interval $(x-\delta,x+\delta)$ contains at least one number that belongs to set A but is different from x*; in alternative formulation: *for every $\delta>0$, there exists $t\in A$ for which* $0<|t-x|<\delta$.

An **isolated point of** *a nonempty subset $A\subseteq\mathbb{R}$ is a real number which belongs to A but is not a limit point of it.*

In this terminology, what we have said above regarding the difference quotient (1) is that the point x must not only belong to the domain of f but also be a limit point of it in order for us to consider the limit. For a fixed x, the quotient represents a function of h with domain

$$A' = \{h\in\mathbb{R} : |h|>0 \text{ and } x+h\in A\}.$$

It is a straightforward verification that x is a limit point of A if, and only if, 0 is a limit point of A'.

7-1.2. Examples. (a) For the subset $A=(0,1]\subseteq\mathbb{R}$, every number in $[0,1]$ is a limit point. Let us see why $\frac{1}{3}$, for example, is a limit point. Consider any $\delta>0$. If $\delta<\frac{1}{3}$, then $(\frac{1}{3}-\delta,\frac{1}{3}+\delta)$ contains the numbers of $(\frac{1}{3}-\delta,\frac{1}{3})$, all of which are not only in A but also different from $\frac{1}{3}$. If instead $\delta \geq \frac{1}{3}$, then $(\frac{1}{3}-\delta,\frac{1}{3}+\delta)$ contains the numbers of $(0,\frac{1}{3})$, all of which are not only in A but also different from $\frac{1}{3}$.

The number 0 is also a limit point: If $0 < \delta < 1$, then $(-\delta,\delta)$ contains the numbers of $(0,\delta)$; these are all in A and different from 0. If $\delta \geq 1$, then $(-\delta,\delta)$ contains the numbers of $(0,1)$ and these are all in A and different from 0. Similar considerations show that 1 is a limit point of $(0,1]$.

The general situation is clarified in Prop.7-1.3 below.

(b) The set $A = \{0,1\}$ has no limit points. The number 0 is not a limit point, because the interval $(-\frac{1}{2},\frac{1}{2})$ contains no number of A that is different from 0. Similarly, number 1 is not a limit point, because the interval $(\frac{1}{2},\frac{3}{2})$ contains no number of A that is different from 1. If $0 \neq x \neq 1$, then x is still not a limit point. To see this, consider $\delta = \min\{|x-0|,|x-1|\} > 0$. Then

$$u \in (x-\delta,x+\delta) \Rightarrow |u-x| < \delta \Rightarrow |u-x| < |x-0| \text{ and } |u-x| < |x-1|$$
$$\Rightarrow 0 \neq u \neq 1.$$

Thus $(x-\delta,x+\delta)$ contains no number of A.

The elements 0 and 1 of A are isolated points of it. It is easy to extend these ideas to show that a finite set has no limit points, so that all its elements are isolated points.

(c) Let $A = \{\frac{1}{k} : k \in \mathbb{N}\}$. It is left to the reader [see 7-1.P4] to verify that 0 is a limit point and that no other number is. Thus all points of A are isolated points.

(d) For the set $\{x \in \mathbb{R} : x \neq 0\}$, the number 0 is a limit point.

7-1.3. Proposition. *Let a and b be real numbers, $a < b$. The set of limit points of any of the four possible intervals $[a,b]$, $[a,b)$, $(a,b]$, (a,b) with endpoints a and b is $[a,b]$.*

Proof. Consider any x such that $a \leq x < b$. Then $x \in [a,b]$ but is not its right endpoint. It follows by Prop.1-7.4 that, for every $\delta > 0$, there exists δ' such that $0 < \delta' < \delta$ and $[x, x+\delta'] \subseteq [a,b]$. Now $(x-\delta,x+\delta)$ contains the number $x+\frac{1}{2}\delta'$, which belongs to (a,b) but is different from x. This shows that x is a limit point of any interval with endpoints a and b as long as $a \leq x < b$. A similar argument applies when $a < x \leq b$. Thus every number belonging to $[a,b]$ is a limit point of any interval with endpoints a and b.

Now suppose $x \notin [a,b]$. In case $x > b$, consider $\delta = x-b > 0$. Then $(x-\delta,x+\delta)$ contains no number of $[a,b]$ and *a fortiori* contains no number of any other interval with endpoints a and b. Therefore $x > b$ implies that x is not a limit point of any of the intervals in question. A similar argument shows that the same is true when $x < a$. ☺

7-1.4. Definition. *Let x be a limit point of the domain A of a function f with values in $\mathbb{R}$ or $\mathbb{C}$. A number λ is called a* **limit of** f **at** x *if, and only if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that*

$$|f(x+h)-\lambda| < \varepsilon \quad \textit{whenever} \quad 0 < |h| < \delta \;\; \textit{and} \;\; x+h \in A.$$

In symbols,

$$\forall\, \varepsilon > 0\; \exists\, \delta > 0 \ni \forall\, h \in \mathbb{R},\;\; 0 < |h| < \delta \textit{ and } x+h \in A \Rightarrow |f(x+h)-\lambda| < \varepsilon.$$

There cannot be two distinct numbers λ with this property, the reasons being analogous to why a sequence cannot have two different limits. Details are left to the reader [see 7-1.P2] but we note here that the argument uses the fact that x is a limit point of A. Thus the limit, if it exists, is unique. Therefore we shall henceforth refer to it as *the* limit of f at x, and shall denote it by $\lim\limits_{t \to x} f(t)$.

It is sometimes more convenient to express the above Definition without explicit reference to h by introducing $t = x + h$:

$$\forall\, \varepsilon > 0\; \exists\, \delta > 0 \ni \forall\, t \in A,\, 0 < |t-x| < \delta \Rightarrow |f(t)-\lambda| < \varepsilon.$$

The similarity between this and Def.2-1.1 will be essential to the forthcoming Prop.7-1.6; it is also the reason for the similarity between the proofs of Props.2-1.2 and Prop.7-1.7 and also the proofs of Theorem 2-1.3 and Theorem 7-1.9.

If the phrase "$0 < |h| < \delta$" in the above Definition is replaced by "$0 < h < \delta$", the meaning changes and a number λ having the changed version of the property is called a **right limit of** f at x. We shall denote it by $f(x+)$. However, one considers right limits at x only if, for any $\delta > 0$, there exists h such that $0 < h < \delta$ and $x + h \in A$, because without this restriction on x, it is not possible to show that two distinct λ cannot have the property. In case A is an interval and the number x is the right endpoint of it, the restriction does not hold, and therefore there is no such thing as a right limit at the right endpoint of an interval. By contrast, if x is the left endpoint instead, any h such that $x + h \in A$ must necessarily be positive, so that the limit at x means the same thing as the right limit. Similar remarks apply to **left limits** $f(x–)$, which are defined by replacing the phrase "$0 < |h| < \delta$" by "$0 > h > –\delta$". See 7-1.P5.

It is left to the reader to check that when A is an interval, the right and left limit as described here are the same as in Def.6-2.4. Examples to illustrate the concept were given there and the reader may find it useful to go through them before proceeding to the material below.

7-1.5. Remark. Suppose f is a function with domain A having a limit point x, and $\lim_{t \to x} f(t) = \lambda$. Let f_B denote the restriction of f to a subset B of A, and suppose x is a limit point of the subset B as well. Then it is clear that $\lim_{t \to x} f_B(t) = \lambda$. Indeed, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < |h| < \delta \text{ and } x + h \in A \Rightarrow |f(x+h) - \lambda| < \varepsilon.$$

Since $B \subseteq A$, it is certainly true that

$$0 < |h| < \delta \text{ and } x + h \in B \Rightarrow |f(x+h) - \lambda| < \varepsilon$$

and hence that

$$0 < |h| < \delta \text{ and } x + h \in B \Rightarrow |f_B(x+h) - \lambda| < \varepsilon.$$

Since x is a limit point of B, this means that $\lim_{t \to x} f_B(t) = \lambda$. In practice, it is too cumbersome to introduce the notation for the subset B and the restriction f_B. These will be taken as understood and no explicit reference to the content of this Remark will be made as long as we avoid certain pitfalls.

On this basis, we can say "$\lim_{t \to x} t^2 = x^2$" with the understanding that t^2 is to be taken as the function f for which $f(t) = t^2$ with some domain that is appropriate to the context and has x as a limit point.

7-1.6. Proposition. *Suppose that x is a limit point of the domain A of a real or complex valued function f and that $x \in A$. Then f is continuous at x if, and only if, $\lim_{t \to x} f(t)$ exists and equals $f(x)$. In the event that x is an isolated point of A, every real or complex valued function f with domain A is continuous at x.*

Proof. The first part is an immediate consequence of the definitions of continuity and of limit. Suppose x is an isolated point of A. Then $x \in A$ and there exists $\delta > 0$ such that $(x - \delta, x + \delta)$ contains no number of A that is different from x. Therefore $|h| < \delta, x + h \in A \Rightarrow x + h \in (x - \delta, x + \delta), x + h \in A \Rightarrow x + h = x \Rightarrow |f(x+h) - f(x)| = 0$. Consequently, for any $\varepsilon > 0$,

$$|f(x+h) - f(x)| < \varepsilon \quad \text{whenever} \quad |h| < \delta \quad \text{and} \quad x + h \in A.$$

Thus f is continuous at x. ☺

One often refers to Def.7-1.4 as the "ε-δ definition" of a limit. While working with the ε-δ definition, we can usually omit writing "$x + h \in A$", because the subsequent reference to $f(x + h)$ makes it clear that $x + h$ is intended to be in the domain of f.

7-1.7. Proposition. *Let x be a limit point of A and f be a real or complex valued function with domain A such that $\lim_{t\to x} f(t)$ exists. Then there exist some $M > 0$ and $\delta_1 > 0$ such that $|f(t)| < M$ whenever $0 < |t-x| < \delta_1$. If $\lim_{t\to x} f(t) \neq 0$, then there exist some $K > 0$ and $\delta_2 > 0$ such that $|f(t)| > K$ whenever $0 < |t-x| < \delta_2$.*

Proof. Denote $\lim_{t\to x} f(t)$ by λ. Since $1 > 0$, Def.7-1.4 is applicable with $\varepsilon = 1$. Therefore there exists $\delta_1 > 0$ such that $|f(t)-\lambda| < 1$ whenever $0 < |t-x| < \delta_1$. But

$$|f(t)-\lambda| < 1 \Rightarrow |f(t)| = |f(t)-\lambda+\lambda| \le |f(t)-\lambda| + |\lambda| < 1+|\lambda|.$$

Therefore $M = 1+|\lambda|$ and δ_1 have the required property.

Now suppose $\lambda \neq 0$. Then $|\lambda|/2 > 0$ and Def.7-1.4 is applicable with $\varepsilon = |\lambda|/2$. Therefore there exists $\delta_2 > 0$ such that $|f(t)-\lambda| < |\lambda|/2$ whenever $0 < |t-x| < \delta_2$. But

$$\begin{aligned}|f(t)-\lambda| < |\lambda|/2 \;\Rightarrow\; |f(t)| &= |f(t)-\lambda+\lambda| \\ &\ge |\lambda| - |f(t)-\lambda| \quad \text{by Prop.1-4.7(h)} \\ &> |\lambda| - |\lambda|/2 = |\lambda|/2 > 0.\end{aligned}$$

Therefore $K = |\lambda|/2$ and δ_2 have the required property. ☺

7-1.8. Lemma. *Let x be a limit point of the domain A of a real valued function g and $\lim_{t\to x} g(t) \neq 0$. Then x is a limit point of $B = \{t \in A : g(t) \neq 0\}$.*

Proof. Since $\lim_{t\to x} g(t) \neq 0$, it follows by Prop.7-1.7 that there exist $K > 0$ and $\delta_1 > 0$ such that $|g(t)| > K$ whenever $0 < |t-x| < \delta_1$. In particular, $t \in B$ when $t \in A$ and $0 < |t-x| < \delta_1$. Since x is a limit point of A, for every $\delta > 0$, [by the alternative formulation in Def.7-1.1] there exists $t \in A$ such that $0 < |t-x| < \min\{\delta, \delta_1\}$, so that $t \in B$ and $0 < |t-x| < \delta$. This shows that x is a limit point of B. ☺

In the result below, the existence of the limits called λ and μ is assumed; but the existence of every other limit involved is proved and not assumed; nonetheless, we avoid the awkward, though more accurate, phrasing "exists and equals" regarding each one of them.

7-1.9. Theorem. *Let x be a limit point of A. Suppose f and g are real or complex valued functions with domain A such that $\lim_{t\to x} f(t) = \lambda$ and $\lim_{t\to x} g(t) = \mu$. Then*

(a) $\lim_{t\to x} (f+g)(t) = \lambda + \mu$;

(b) $\lim_{t\to x} (fg)(t) = \lambda\mu$;

(c) $\lim_{t\to x} (\alpha f)(t) = \alpha\lambda$, *where α is any complex number*;

(d) *if* $\mu \neq 0$, *then* $B = \{t \in A : g(t) \neq 0\}$ *has x as a limit point*; *and when f/g is restricted to any subset of B having x as a limit point,*

$$\lim_{t\to x}(f/g)(t) = \lambda/\mu;$$

(e) $\lim_{t\to x}(\Re f) = \Re\lambda$, $\lim_{t\to x}(\Im f) = \Im\lambda$; *if f is real valued, then λ is real*;
(f) $\lim_{t\to x}|f(t)| = |\lambda|$;
(g) *if f and g are real valued,*

$$\lim_{t\to x}[\min\{f,g\}(t)] = \min\{\lambda,\mu\} \quad \textit{and} \quad \lim_{t\to x}[\max\{f,g\}(t)] = \max\{\lambda,\mu\};$$

(h) *if f is real valued, x is a limit point of a subset B of A and if* $f(t) \le K$ *for all* $t \in B$, *then* $\lambda \le K$.

Proof. (a) Let $\varepsilon > 0$. Then $\frac{\varepsilon}{2} > 0$ and (by Def.7-1.4) there exist $\delta_1, \delta_2 > 0$ such that

$$|f(t)-\lambda| < \tfrac{\varepsilon}{2} \quad \text{whenever} \quad 0 < |t-x| < \delta_1$$

and

$$|g(t)-\mu| < \tfrac{\varepsilon}{2} \quad \text{whenever} \quad 0 < |t-x| < \delta_2.$$

Let δ be the lesser among δ_1 and δ_2. Then $\delta > 0$. If $0 < |t-x| < \delta$, then $0 < |t-x| < \delta_1$ as well as $0 < |t-x| < \delta_2$, so that we have both $|f(t)-\lambda| < \frac{\varepsilon}{2}$ and $|g(t)-\mu| < \frac{\varepsilon}{2}$. Therefore

$$\begin{aligned}|(f+g)(t)-(\lambda+\mu)| &= |[f(t)-\lambda]-[g(t)-\mu]| \\ &\le |f(t)-\lambda| + |g(t)-\mu| < \tfrac{\varepsilon}{2}+\tfrac{\varepsilon}{2} = \varepsilon.\end{aligned}$$

Thus there exists $\delta > 0$ such that

$$|(f+g)(t) - (\lambda+\mu)| < \varepsilon \quad \text{whenever} \quad 0 < |t-x| < \delta.$$

Since the existence of such a positive δ has been shown for every positive ε, it follows that $\lim_{t\to x}(f+g)(t) = \lambda+\mu$.

(b) Let $\varepsilon > 0$. By Prop.7-1.7, there exist $M > 0$ and $\delta_1 > 0$ such that

$$|f(t)| < M \quad \text{whenever} \quad 0 < |t-x| < \delta_1.$$

Also, there exists $\delta_2 > 0$ such that

$$|g(t)-\mu| < \varepsilon/2M \quad \text{whenever} \quad 0 < |t-x| < \delta_2,$$

and there exists $\delta_3 > 0$ such that

$$|f(t)-\lambda| < \varepsilon/2(|\mu|+1) \text{ whenever } 0 < |t-x| < \delta_3.$$

Let δ be the least among δ_1, δ_2 and δ_3. Then $\delta > 0$. Consider any $t \in A$ such that $0 < |t-x| < \delta$. For such a number t, we have

$$|f(t)| < M,\ |g(t)-\mu| < \varepsilon/2M \text{ and also } |f(t)-\lambda| < \varepsilon/2(|\mu|+1).$$

Therefore

$$\begin{aligned}|(fg)(t)-\lambda\mu| &= |f(t)\cdot(g(t)-\mu) + \mu\cdot(f(t)-\lambda)| \\ &\le |f(t)|\cdot|g(t)-\mu| + |\mu|\cdot|f(t)-\lambda| \\ &\le M\cdot(\varepsilon/2M) + |\mu|\cdot\left[\varepsilon/2(|\mu|+1)\right] < \tfrac{\varepsilon}{2}+\tfrac{\varepsilon}{2} = \varepsilon.\end{aligned}$$

Thus there exists $\delta > 0$ such that

$$|(fg)(t)-\lambda\mu|<\varepsilon \quad \text{whenever} \quad 0<|t-x|<\delta.$$

Since the existence of such a positive δ has been shown for every positive ε, it follows that $\lim\limits_{t\to x}(fg)(t)=\lambda\mu$.

(c) If $\alpha = 0$, then $(\alpha f)(t) = 0$ for all $t \in A$, so that $\lim\limits_{t\to x}(\alpha f)(t)$ and $\alpha\lambda$ are both 0. For $\alpha \neq 0$, the result follows either by applying what has just been proved with g chosen so as to have $g(t) = \alpha$ for all $t \in A$, or by selecting a δ such that $|f(t)-\lambda| < \frac{\varepsilon}{|\alpha|}$ whenever $0<|t-x|<\delta$.

(d) By Lemma 7-1.8, x is a limit point of B. Now let B_1 denote any subset of B having x as a limit point. Since $\mu \neq 0$, by Prop.7-1.7, there exist $K > 0$ and $\delta_1 > 0$ such that $|g(t)| > K$ whenever $0 < |t-x| < \delta_1$. Now, consider any $\varepsilon > 0$. There exists $\delta_2 > 0$ such that

$$|f(t)-\lambda|<\tfrac{1}{2}\varepsilon K \quad \text{whenever} \quad 0<|t-x|<\delta_2$$

and there also exists $\delta_3 > 0$ such that

$$|g(t)-\mu|<{}^{\varepsilon K|\mu|}\!/_{2(|\lambda|+1)} \quad \text{whenever} \quad 0<|t-x|<\delta_3.$$

Let δ be the least among δ_1, δ_2 and δ_3. Then $\delta > 0$. Consider any $t \in B_1$ such that $0 < |t-x| < \delta$. For such a number t, we have

$$1/|g(t)| < 1/K, \quad |f(t)-\lambda|<\tfrac{1}{2}\varepsilon K \quad \text{and} \quad |g(t)-\mu|<{}^{\varepsilon K|\mu|}\!/_{2(|\lambda|+1)}.$$

Therefore

$$\begin{aligned}
|(f/g)(t)-\lambda/\mu| &= \frac{|f(t)\mu-\lambda g(t)|}{|g(t)\mu|} \\
&= \frac{|f(t)\mu-\lambda\mu+\lambda\mu-\lambda g(t)|}{|g(t)\mu|} \\
&\le \frac{|\mu|\cdot|f(t)-\lambda|+|\lambda|\cdot|g(t)-\mu|}{|g(t)\mu|} \\
&\le \frac{|f(t)-\lambda|}{|g(t)|}+\frac{|\lambda|\cdot|g(t)-\mu|}{|g(t)\mu|} \\
&\le \frac{\varepsilon K/2}{K}+\frac{|\lambda|\cdot\varepsilon K|\mu|}{K|\mu|\cdot 2(|\lambda|+1)} \\
&< \frac{\varepsilon}{2}+\frac{\varepsilon}{2}=\varepsilon.
\end{aligned}$$

Thus there exists $\delta > 0$ such that $|(f/g)(t) - \lambda/\mu| < \varepsilon$ whenever $0 < |t-x| < \delta$. and $t \in B_1$. Since the existence of such a positive δ has been shown for every positive ε, it follows that $\lim_{t\to x} (f/g)(t) = \lambda/\mu$.

(e) This follows from Prop.1-10.1(b) and the obvious fact that $\lim_{t\to x} 0 = 0$.

(f)&(g) That $\lim_{t\to x} |f(t)| = |\lambda|$ is a simple consequence of the inequality $||f(t)| - |\lambda|| \le |f(t) - \lambda|$. Part (f) follows from this, the equalities

$$\min\{f,g\} = (\tfrac{1}{2})[(f+g) - |f-g|], \qquad \max\{f,g\} = (\tfrac{1}{2})[(f+g) + |f-g|],$$

[see 1-4.P12], and parts (a) and (c).

(h) Let $g(t) = K$ for every $t \in B$ and f_B be the restriction of f to B. Then $\lim_{t\to x} g(t) = K$ and $\min\{f_B, g\} = f_B$, so that $\lim_{t\to x} [\min\{f_B, g\}(t)] = \lambda$. Therefore by part (f), we have $\lambda = \min\{\lambda, K\}$, so that $\lambda \le K$. ☺

A familiar application of part (a) of the above Proposition would be that $\lim_{t\to 0} (t + \sqrt{t}) = 0 + 0 = 0$, on the grounds that $\lim_{t\to 0} t = 0$ and $\lim_{t\to 0} \sqrt{t} = 0$. However, it should be noted that the latter limit concerns the function $\sqrt{t}$, which has domain $\{t \in \mathbb{R} : t \ge 0\}$, or some subset thereof that has 0 as a limit point; in order to apply Theorem 7-1.9(a), it is necessary to restrict the function t also to the same domain, although we need not specify that domain in view of Remark 7-1.5. A possible pitfall is indicated in 7-1.P3.

Problem Set 7-1

7-1.P1. For the set $(5,6] \cup \{7\}$, find (a) the set of limit points and (b) the set of isolated points.

7-1.P2. Show that there cannot exist two distinct numbers λ_1 and λ_2, both satisfying the condition of Def.7-1.4.

7-1.P3. Which of the following statements is true?

$$\text{(A) } \lim_{t\to 0} \sqrt{t} = 0 \quad \text{(B) } \lim_{t\to 0} \sqrt{(-t)} = 0 \quad \text{(C) } \lim_{t\to 0} [\sqrt{t} + \sqrt{(-t)}] = 0.$$

7-1.P4. Show that the only limit point of $\{\frac{1}{k} : k \in \mathbb{N}\}$ is 0.

7-1.P5. Let f be a function with domain $A \subseteq \mathbb{R}$ and values in $\mathbb{R}$ or $\mathbb{C}$. Suppose x is a limit point of A such that, for every $\delta > 0$, there exists h satisfying $0 < h < \delta$ and $x + h \in A$ and there also exists h' satisfying $0 > h' > -\delta$ and $x + h' \in A$. Show that $\lim_{t\to x} f(t)$ exists if, and only if, the right limit $f(x+)$ and left limit $f(x-)$ both exist and are equal to each other, in which case both are equal to $\lim_{t\to x} f(t)$.

7-2 Definition of Derivative and Elementary Properties

The functions considered in this Section will be real valued with domain a subset of $\mathbb{R}$, unless specified otherwise. For any function $f: A \to \mathbb{R}$ and $x \in A$, there is an associated function which maps every nonzero h for which $x + h \in A$ into the quotient

$$\frac{f(x+h) - f(x)}{h},$$

often called the *difference quotient*. If x is a limit point of A, then 0 is a limit point of the domain of the difference quotient without being an element of it.

7-2.1. Definition. *Let $f: A \to \mathbb{R}$ be a function, where $A \subseteq \mathbb{R}$ and let $x \in A$ be a limit point. By* **derivative of** *f at x we mean*

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

if the limit exists. It is denoted by $f'(x)$. If the derivative $f'(x)$ exists, we say that f is **differentiable at** *the point x. If f is differentiable at each point of a subset B of its domain, we say that f is* **differentiable on** *B* [regardless of whether it is also differentiable at points outside B].

The process of obtaining a derivative is of course called *differentiation*. The meanings of the terms *second derivative*, *higher derivative* and of the symbols f'', $f^{(k)}$ will be as understood in Calculus.

7-2.2. Remarks. The derivative $f'(x)$ is the number λ (if any) having the property that for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\left| \frac{f(x+h) - f(x)}{h} - \lambda \right| < \varepsilon \text{ whenever } 0 < |h| < \delta \text{ and } x + h \in A.$$

It is sometimes more convenient to express the above Definition without explicit reference to h by introducing $t = x + h$:

$$f'(x) = \lim_{t \to x} \frac{f(t) - f(x)}{t - x}.$$

If one has an expression $f(x)$ describing the function, such as $7x^3 - 4$, but not a symbol for it, then it is convenient to use the notation $\frac{d}{dx} f(x)$ instead of $f'(x)$.

The derivative (when it exists) is *unique* in the sense that there cannot be two different numbers λ having the property just described; this is because any limit that exists has to be unique, as remarked immediately after Def.7-1.4 .

If we replace the limit in the above Definition by the right [resp. left] limit, we get the **right derivative** [resp. **left derivative**]. If the domain of f is an interval $[a,b]$ and $x = a$, then $x + h$ can be in the domain $[a,b]$ only when $h > 0$. Therefore the limit is necessarily the right limit and therefore the derivative $f'(a)$ is necessarily the right derivative. Similarly, the derivative $f'(b)$ is necessarily the left derivative.

Example. The simplest instance of the non-existence of a derivative at one point of the domain is given by the function $f(x) = |x|$. Since

$$\frac{f(0+h) - f(0)}{h} = \frac{|h|}{h} = 1 \text{ or } -1 \text{ according as } h > 0 \text{ or } h < 0,$$

the right derivative at 0 is 1 while the left derivative is -1. In particular, the derivative does not exist at 0. In the next Section, we shall see an instance when neither a left nor a right derivative exists.

As with limits, the derivative at x of the restriction of f to a subset having x as a limit point is the same as the derivative at x of the original function f. When a restriction is intended, it is to be understood from the context.

7-2.3. Proposition. *If a function f is differentiable at a point x of its domain, then it is continuous at x.*

Proof. From the equality

$$f(x+h) - f(x) = h \cdot \frac{f(x+h) - f(x)}{h},$$

along with the fact that $\lim\limits_{h\to 0} h = 0$, and from Theorem 7-1.9(b), we have

$$\lim_{h\to 0} [f(x+h) - f(x)] = 0 \cdot f'(x) = 0.$$

By Theorem 7-1.9(a), it now follows that $\lim\limits_{h\to 0} f(x+h) = \lim\limits_{h\to 0} f(x) = f(x)$. Therefore f is continuous at x by Prop.7-1.6. ☺

The converse of the above result is not true. The function $f(x) = |x|$ is continuous at $x = 0$ but not differentiable there. Continuity is a consequence of the equality $|f(0+h) - f(0)| = |h - 0| = |h|$. Indeed, it follows from the equality that, for any $\varepsilon > 0$, the number $\delta = \varepsilon$ has the property that $|h| < \delta \Rightarrow |f(0+h) - f(0)| < \varepsilon$. That f is not differentiable at 0 has been argued above.

7-2.4. Theorem. *Let x be a limit point of A. Suppose f and g are real or complex valued functions with domain A that are differentiable at x. Then*

(a) $f + g$ *is differentiable at x and* $(f+g)'(x) = f'(x) + g'(x)$;

(b) *fg is differentiable at x and* $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$;

(c) *for any real number* α, *the function* αf *is differentiable at x and* $(\alpha f)'(x) = \alpha f'(x)$;

(d) *if* $g(x) \neq 0$, *then* $B = \{t \in A : g(t) \neq 0\}$ *has x as a limit point*; *and the restriction of f/g to any subset of B having x as a limit point is differentiable at x with* $(f/g)'(x) = [f'(x)g(x) - f(x)g'(x)]/g(x)^2$.

Proof. (a) Whenever $x + h \in A$, we have

$$[(f+g)(x+h) - (f+g)(x)]/h = \frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h}.$$

It follows by Theorem 7-1.9(a) that $(f + g)'(x)$ exists and equals $f'(x) + g'(x)$.

(b) Whenever $x + h \in A$, we have

$$\frac{(fg)(x+h) - (fg)(x)}{h} = \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h}$$

$$= g(x+h) \cdot \frac{f(x+h) - f(x)}{h} + f(x) \cdot \frac{g(x+h) - g(x)}{h}.$$

Since g is differentiable at x, it is continuous there by Prop.7-2.3; it follows that $\lim\limits_{h \to 0} g(x+h) = g(x)$. This, together with the equality displayed above and Theorem 7-1.9(a)–(c), implies that fg is differentiable at x and that its derivative is given by $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$.

(c) Whenever $x + h \in A$, we have

$$\frac{\alpha f(x+h) - \alpha f(x)}{h} = \alpha \frac{f(x+h) - f(x)}{h}.$$

Hence by Theorem 7-1.9(c), the derivative $(\alpha f)'(x)$ exists and equals $\alpha f'(x)$.

(d) By Lemma 7-1.8, x is a limit point of B. Let B_1 denote any subset of B having x as a limit point and let ϕ denote the restriction of f/g to B_1. Then

$$\frac{\phi(x+h) - \phi(x)}{h} = \frac{1}{g(x)g(x+h)} \left[g(x) \frac{f(x+h) - f(x)}{h} - f(x) \frac{g(x+h) - g(x)}{h} \right],$$

whenever $x + h \in B_1$. By Prop.7-2.3, $\lim\limits_{h \to 0} g(x+h) = g(x)$. It now follows by Theorem 7-1.9(a)–(d), that ϕ is differentiable at x and that its derivative there is $[f'(x)g(x) - f(x)g'(x)]/g(x)^2$. ☺

It is straightforward to show that, if a function f is constant on a domain having a limit point x, then $f'(x) = 0$. Similarly, if $f(x) = x$ whenever $x \in B$ and x

is a limit point of B, then $f'(x) = 1$. This is most conveniently expressed without explicit reference to any domain as simply $\frac{d}{dx}x = 1$. From what we show later, it will follow that $\frac{d}{dx}\sqrt{x} = \frac{1}{2}\sqrt{x}$ and $\frac{d}{dx}\sqrt{(-x)} = -\frac{1}{2}\sqrt{(-x)}$. However, one cannot apply Theorem 7-2.4(a) and claim $\frac{d}{dx}[\sqrt{x} + \sqrt{(-x)}] = \frac{1}{2}\sqrt{x} - \frac{1}{2}\sqrt{(-x)}$, because there is a domain mismatch.

From the equality $\frac{d}{dx}x = 1$ and from Theorem 7-2.4(b), it follows by an induction argument that $\frac{d}{dx}x^n = nx^{n-1}$ for every $n \in \mathbb{N}$. For negative integers n, the formula (with $x \neq 0$) follows from part (d). The derivative of any polynomial function can now be computed as usual on the basis of parts (a) and (c). Details are omitted.

In Def.6-2.1 of a polynomial function, we did not speak of the degree of a polynomial or its coefficients. As mentioned then, the reason was that we had not ruled out the possibility that two different finite sequences could correspond to the same function. We do so now.

Consider any polynomial function $f(x) = a_0 + a_1x + \cdots + a_nx^n$ on an interval $[a, b]$. If there also exists a finite sequence $b_0, b_1, \ldots, b_n$ such that $f(x) = b_0 + b_1x + \cdots + b_nx^n$ for each $x \in [a, b]$, we shall show that $a_j = b_j$ for $0 \leq j \leq n$. In case $n = 0$, then $f(x) = a_0$ as well as $f(x) = b_0$, so that $a_0 = b_0$. Assume the result for some $n \geq 0$ (induction hypothesis) and suppose that $a_0, a_1, \ldots, a_{n+1}$ and $b_0, b_1, \ldots, b_{n+1}$ are finite sequences such that

$$a_0 + a_1x + \cdots + a_{n+1}x^{n+1} = f(x) = b_0 + b_1x + \cdots + b_{n+1}x^{n+1} \quad \text{for every } x \in [a, b].$$

Upon differentiating, we get

$$\sum_{j=1}^{n+1} ja_jx^{j-1} = f'(x) = \sum_{j=1}^{n+1} jb_jx^{j-1} \quad \text{for every } x \in [a, b].$$

Therefore the sequences $\alpha_j = (j+1)a_{j+1}$ and $\beta_j = (j+1)b_{j+1}$, $0 \leq j \leq n$, generate the same polynomial. By the induction hypothesis, we have $(j+1)a_{j+1} = (j+1)b_{j+1}$ for $0 \leq j \leq n$. Thus $a_j = b_j$ for $1 \leq j \leq n+1$. This implies that

$$a_1x + \cdots + a_{n+1}x^{n+1} = b_1x + \cdots + b_{n+1}x^{n+1} \quad \text{for every } x \in [a, b]$$

and hence $a_0 = b_0$ as well.

In case $a_0 + a_1x + \cdots + a_nx^n = f(x) = b_0 + b_1x + \cdots + b_mx^m$, where $m > n$, we take $a_j = 0$ for $n < j \leq m$ and apply what we have just proved, thereby obtaining $a_j = b_j$ for $0 \leq j \leq m$.

This shows that the numbers a_k are unique and we can speak of a_k as **the coefficient of the k^{th} degree term** and the largest k with $a_k \neq 0$ (if any) as **the degree of** the polynomial function. The polynomial $f(x) = 0$ is the only one with no degree.

In the Proposition below, the hypothesis of continuity of the inverse can be dropped if the domain is assumed to be an interval and the function assumed continuous everywhere. This refinement will be carried out in the next Section. For now, see 7-2.P1.

7-2.5. Theorem. *Let $f: A \to \mathbb{R}$ be an injective function continuous at a point $u \in A$. If u is a limit point of A, then $f(u)$ is a limit point of the range $f(A)$.*

If f is differentiable at u with $f'(u) \neq 0$ and f^{-1} is continuous at $f(u)$, then f^{-1} is also differentiable at $f(u)$ with derivative $(f^{-1})'(f(u)) = 1/f'(u)$, i.e.

$$(f^{-1})'(x) = 1/f'(f^{-1}(x)), \ \text{where} \ x = f(u).$$

Proof. In order to show that $f(u)$ is a limit point of the range $f(A)$, consider any $\delta > 0$. We must show that there exists k with $0 < |k| < \delta$ and $f(u) + k \in f(A)$. By continuity of f at u, there exists $\delta_1 > 0$ such that

$$|h| < \delta_1, \ u + h \in A \ \Rightarrow \ |f(u+h) - f(u)| < \delta.$$

Since u is a limit point of A, there exists h such that

$$0 < |h| < \delta_1 \ \text{and} \ u + h \in A.$$

Hence $|f(u+h) - f(u)| < \delta$. But $h \neq 0$ and therefore $f(u+h) - f(u) \neq 0$. Let $k = f(u+h) - f(u)$. Then $0 < |k| < \delta$ and $f(u) + k = f(u+h) \in f(A)$. Thus $f(u)$ is a limit point of $f(A)$.

To prove the second part, we need to show that, for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < |k| < \delta, f(u) + k \in f(A) \ \Rightarrow \ \left| \frac{f^{-1}(f(u)+k) - f^{-1}(f(u))}{k} - \frac{1}{f'(u)} \right| < \varepsilon \qquad \text{(A)}$$

So, consider any $k \neq 0$ such that $f(u) + k \in f(A)$ and set

$$h = f^{-1}(f(u) + k)) - u \qquad (1)$$

$$= f^{-1}(f(u) + k) - f^{-1}(f(u)). \qquad (2)$$

From (1), we have $f^{-1}(f(u) + k) = u + h$. Applying f to both sides, we get $f(u) + k = f(u+h)$ and hence

$$k = f(u+h) - f(u). \qquad (3)$$

By (2) and (3), we have

$$\frac{f^{-1}(f(u)+k)-f^{-1}(f(u))}{k} = \frac{h}{f(u+h)-f(u)}. \tag{4}$$

Now consider any $\varepsilon > 0$. By Def.7-2.1 and Theorem 7-1.9(d), there exists $\delta_1 > 0$ such that

$$0 < |h| < \delta_1 \quad \Rightarrow \quad \left|\frac{h}{f(u+h)-f(u)} - \frac{1}{f'(u)}\right| < \varepsilon. \tag{5}$$

By continuity of f^{-1} at $f(u)$, there exists $\delta > 0$ such that

$$|k| < \delta \Rightarrow |f^{-1}(f(u)+k)-f^{-1}(f(u))| < \delta_1.$$

Since $k \neq 0$, the injectivity of f ensures that

$$f^{-1}(f(u)+k)-f^{-1}(f(u)) \neq 0$$

and therefore the preceding statement can be strengthened as

$$\begin{aligned} 0 < |k| < \delta &\Rightarrow 0 < |f^{-1}(f(u)+k)-f^{-1}(f(u))| < \delta_1 \\ &\Rightarrow 0 < |h| < \delta_1 \quad \text{by (2).} \end{aligned}$$

In conjunction with (5) and then with (4), this implies (A). ☺

Our proof of the next result considers two cases. An alternative proof that does not require considering two cases may be found in [1] or [8]. In the proof of Case 1, the reader may wonder why we have not substituted $k = f(x+h)-f(x)$ immediately after statement (2), and then just used the fact that $k \to 0$. Our reason for not trying to simplify the argument in this manner is to avoid the kind of flaw illustrated in 7-2.P2.

7-2.6. Theorem. (Chain Rule) *Suppose $f: A \to \mathbb{R}$ is differentiable at a point $x \in A$, and that $g: f(A) \to \mathbb{R}$ is differentiable at the point $f(x)$. Then the composition $g \circ f: A \to \mathbb{R}$ is differentiable at x and $(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$.*

Proof. To begin with, we note that when $h \neq 0 \neq f(x+h) - f(x)$,

$$\frac{(g \circ f)(x+h) - (g \circ f)(x)}{h} = \frac{g(f(x+h)) - g(f(x))}{f(x+h) - f(x)} \cdot \frac{f(x+h) - f(x)}{h}. \tag{1}$$

Case 1: $f'(x) \neq 0$. By Prop.7-1.7, there exists $\delta > 0$ such that

$$0 < |h| < \delta \Rightarrow f(x+h) - f(x) \neq 0. \tag{2}$$

Since $g'(f(x))$ exists, for any $\varepsilon > 0$, there exists $\delta_1 > 0$ such that

$$0 < |k| < \delta_1 \Rightarrow \left|\frac{g(f(x)+k) - g(f(x))}{k} - g'(f(x))\right| < \varepsilon. \tag{3}$$

Since f is continuous at x, there exists $\delta_2 > 0$ such that

$$0 < |h| < \delta_2 \Rightarrow |f(x+h) - f(x)| < \delta_1. \tag{4}$$

Now let $k = f(x+h) - f(x)$. By (2) and (4), $0 < |h| < \min\{\delta, \delta_2\} \Rightarrow 0 < |k| < \delta_1$ and hence by (3),

$$0 < |h| < \min\{\delta, \delta_2\} \Rightarrow \left|\frac{g(f(x)+k) - g(f(x))}{k} - g'(f(x))\right| < \varepsilon$$
$$\Rightarrow \left|\frac{g(f(x+h)) - g(f(x))}{f(x+h) - f(x)} - g'(f(x))\right| < \varepsilon.$$

Therefore

$$\lim_{h\to 0} \frac{g(f(x+h)) - g(f(x))}{f(x+h) - f(x)} = g'(f(x)).$$

This together with (1) and Theorem 7-1.9(b) leads to $(g\circ f)'(x) = g'(f(x)) \cdot f'(x)$.

Case 2: $f'(x) = 0$. We need only show that $(g\circ f)'(x) = 0$. By Prop.7-1.7, there exist $M > 0$ and $\delta_1 > 0$ such that

$$0 < |k| < \delta_1 \Rightarrow \left|\frac{g(f(x)+k) - g(f(x))}{k}\right| < M. \tag{5}$$

Since f is continuous at x, there exists $\delta_2 > 0$ such that

$$|h| < \delta_2 \Rightarrow |f(x+h) - f(x)| < \delta_1. \tag{6}$$

Consider any $\varepsilon > 0$. There exists $\delta_3 > 0$ such that

$$0 < |h| < \delta_3 \Rightarrow \left|\frac{f(x+h) - f(x)}{h}\right| < \frac{\varepsilon}{M}. \tag{7}$$

Now let $k = f(x+h) - f(x)$ and $\delta = \min\{\delta_1, \delta_2, \delta_3\}$. Consider any h for which $0 < |h| < \delta$. Then by (6), $|k| < \delta_1$. In the event that $k = f(x+h) - f(x) \neq 0$, we further have $0 < |k| < \delta_1$ and hence by (1), (5) and (7), we have

$$\left|\frac{(g\circ f)(x+h) - (g\circ f)(x)}{h}\right| < \varepsilon.$$

If instead $k = f(x+h) - f(x) = 0$, the same inequality still holds because the left side is equal to 0. This shows that

$$\lim_{h\to 0} \frac{(g\circ f)(x+h) - (g\circ f)(x)}{h} = 0,$$

so that $(g\circ f)'(x) = 0$. ☺

It is immediate from Def.7-2.1 that a function f having derivative 0 at each point of an interval $[a,b]$ has the property that, for every $\varepsilon > 0$ and every $x \in [a,b]$, there exists $\delta > 0$ such that

$$|f(y) - f(x)| < \frac{\varepsilon}{(b-a)}|y-x| \quad \text{whenever} \quad |y-x| < \delta.$$

It therefore follows from the Proposition proved in the appendix to Chapter 1 that, for every $\varepsilon > 0$,

$$|f(\alpha) - f(\beta)| < \frac{\varepsilon}{(b-a)} |\alpha - \beta| \le \varepsilon \text{ for all } \alpha, \beta \in [a, b].$$

This has the consequence [see 1-4.P9] that $f(\alpha) = f(\beta)$ for all $\alpha, \beta \in [a, b]$, i.e. f is a constant. This result that a function with derivative 0 at every point of $[a, b]$ must be constant will be obtained later from the Mean Value Theorem, which also has other important uses.

Problem Set 7-2

7-2.P1. (a) Show that the function $f: A \to \mathbb{R}$, where $A = [0, \frac{1}{2}] \cup \mathbb{N}$ and $f(x) = x$ when $x \in [0, \frac{1}{2}]$ and $f(x) = -1/x$ when $x \in \mathbb{N}$, is injective and satisfies $f'(0) = 1$, but its inverse is not continuous at $f(0)$.
(b) Show that the function $f: [-1, 1] \to \mathbb{R}$, such that $f(x) = x$ when $x \in [-1, 0)$ and $f(x) = 1 - x$ when $x \in [0, 1]$ is injective and satisfies $f'(1) = -1$, but its inverse is not continuous at $f(1)$. [Here the domain of f is an interval; but f is not continuous at the point 0.]

7-2.P2. Give an example to show that, if $\lim_{h \to 0} f(h) = 0$ and $\lim_{k \to 0} g(k) = \alpha$, it does *not* follow that $\lim_{h \to 0} g(f(h)) = \alpha$. State an additional condition under which it does follow that $\lim_{h \to 0} g(f(h)) = \alpha$.

7-2.P3. For the function $f(x) = |x|^3$, show that $f'(0)$ and $f''(0)$ exist but $f'''(0)$ does not exist.

7-2.P4. Let f be differentiable on $[a, b]$. Show that f' is the pointwise limit of a sequence of continuous functions on $[a, b]$.

7-2.P5. Suppose for some x that $f'(x)$ and $g'(x)$ exist, $g'(x) \ne 0$ and $f(x) = g(x) = 0$. If also $g(t) \ne 0$ whenever $t \ne x$, then show that $\lim_{t \to x} \frac{f(t)}{g(t)} = \frac{f'(x)}{g'(x)}$.

7-2.P6. Let f be differentiable at c. Prove that

$$\lim_{h \to 0} \frac{f(c+h) - f(c-h)}{2h} = f'(c).$$

If this limit exists, does it follow that f is differentiable at c? Justify.

7-3 Powers, Trigonometric Functions and Their Inverses

The functions whose derivatives were obtained in Section 7-2 were polynomials and their quotients, which are called *rational functions*. We have yet to establish what the derivative of x^r is when r is a rational number. [The question of irrational r arises only after we define what we mean by an irrational power of a number; note that this definition is yet to be stated.] We shall now establish the usual formulas for the derivative of x^r ($r \in \mathbb{Q}$) and also of trigonometric functions, sine, cosine, tangent together with their inverses. The other three trigonometric functions and their inverses can then be easily handled and details are omitted.

7-3.1. Definition. *A function $f\colon I \to \mathbb{R}$, where I is an interval, is said to be* **strictly increasing** *if*

$$x_1 < x_2 \;\Rightarrow\; f(x_1) < f(x_2)$$

and is said to be **strictly decreasing** *if*

$$x_1 < x_2 \;\Rightarrow\; f(x_1) > f(x_2).$$

A function is **strictly monotone** *if it is either strictly increasing or strictly decreasing.*

A function which is strictly increasing [resp. strictly decreasing] is also increasing [resp. decreasing] in the sense of Def.6-2.5. The converse is of course false: A constant function is simultaneously increasing and decreasing but neither strictly increasing nor strictly decreasing. The step function f on the domain $[1,4]$ such that $f(1) = 5, f(3) = 7, f(x) = 6$ when $1 < x < 3$ and $f(x) = 20$ when $3 < x \le 4$ is increasing but not strictly increasing. However if $f(3)$ is changed to 21, then it is no longer increasing, nor does it become decreasing.

7-3.2. Theorem. *Let I be an interval and $f\colon I \to \mathbb{R}$ be a continuous strictly monotone function. If f is differentiable at $u = f^{-1}(x) \in I$ with $f'(u) \ne 0$, then f^{-1} is also differentiable at $x = f(u)$ and the derivative is $(f^{-1})'(x) = 1/f'(f^{-1}(x))$.*

Proof. By Theorem 2-2.1, f^{-1} is continuous on its domain. The rest follows by Theorem 7-2.5. ☺

Remark. If I is an interval, then any injective continuous function $f\colon I \to \mathbb{R}$ must be strictly monotone, as can be argued by using the Intermediate Value Theorem. By Theorem 7-3.2, it will follow that, if $f'(u) \ne 0$ at some $u \in I$, then f^{-1} is differentiable at $x = f(u)$ with derivative $(f^{-1})'(x) = 1/f'(f^{-1}(x))$.

Although we have so far defined a q^{th} root $a^{1/q}$ only for the case when $a \geq 0$, it is easy to see that when q is odd, a negative number a has a unique q^{th} root given by $a^{1/q} = -(-a)^{1/q}$. The resulting function $\phi(x) = x^{1/q}$ has domain $\mathbb{R}$ when $q > 0$ and is the inverse of the strictly monotone function $f(x) = x^q$. If p is a positive integer, $x^{p/q}$ describes a function with domain $\mathbb{R}$, while if p is a negative integer, $x^{p/q}$ describes a function with domain $\{x \in \mathbb{R} : x \neq 0\}$. Part (d) of the next result refers to the latter function. The validity of parts (c) and (d) will later be extended to include $x = 0$ when $r > 1$ [see Example 7-5.8(a)].

7-3.3. Proposition. *Let r be a rational number.*

(a) *If r is a positive integer, then* $\frac{d}{dx}x^r = rx^{r-1}$ *for all* $x \in \mathbb{R}$.

(b) *If r is a negative integer, then* $\frac{d}{dx}x^r = rx^{r-1}$ *for all* $x \neq 0$.

(c) *If r is a nonzero rational number, then* $\frac{d}{dx}x^r = rx^{r-1}$ *for all* $x > 0$.

(d) *If* $r = \frac{p}{q}$, *where p and q are nonzero integers and* $q > 0$ *is odd, then* $\frac{d}{dx}x^r = rx^{r-1}$ *for all* $x \neq 0$.

(e) *If* $r = 0$, *then* $\frac{d}{dx}x^r = 0$ *for all* $x \in \mathbb{R}$ (although x^{r-1} is now undefined for $x = 0$).

Proof. In function notation, the statement to be proved is roughly that, if ϕ is the function for which $\phi(x) = x^r$, then $\phi'(x) = rx^{r-1}$ for certain x.

(a) If r is a positive integer, then the equality to be proved is a consequence [as noted before] of the formula for differentiating a product and the easily proven fact that $\frac{d}{dx}x = 1$.

(b) If r is a negative integer, say $-s$, then the rule for differentiating a quotient [Theorem 7-2.4(d)] leads to

$$\frac{d}{dx}x^r = \frac{d}{dx}\left(1/x^s\right) = \left[0\cdot(x^s) - 1\cdot(sx^{s-1})\right]/x^{2s} = -sx^{-s-1} = rx^{r-1} \quad (x \neq 0).$$

(c) Now, suppose $r = \frac{1}{q}$, where q is a positive integer. We seek the derivative of $\phi = f^{-1}$, where the function f is given on the domain $[0, \infty)$ by $f(u) = u^q$. The latter is strictly monotone and has derivative $f'(u) = qu^{q-1}$, which is nonzero when $u \neq 0$. Therefore, when $x = f(u)$ with $u \neq 0$, i.e. $x > 0$, we have

$$\frac{d}{dx}x^r = (f^{-1})'(x) = 1/f'(f^{-1}(x)) \quad \text{by Theorem 7-3.2}$$

$$= 1/q(f^{-1}(x))^{q-1} = \frac{1}{q}(f^{-1}(x))^{1-q} = rx^{r(1-q)} = rx^{r-rq} = rx^{r-1}.$$

Finally, suppose $r = \frac{p}{q}$, where p and q are both integers and q is positive. Then $\phi(x) = x^r = (g \circ G)(x)$, where the functions G and g are given by $G(x) = x^{\frac{1}{q}}$ and $g(x) = x^p$. By what has already been shown, $g'(x) = px^{p-1}$ and $G'(x) = \frac{1}{q}x^{(\frac{1}{q}-1)}$ when $x > 0$. Therefore by the Chain Rule,

$$\phi'(x) = (g \circ G)'(x) = g'(G(x)) \cdot G'(x) = p(G(x))^{p-1} \cdot G'(x) = p(x^{\frac{1}{q}})^{p-1} \cdot \frac{1}{q}x^{(\frac{1}{q}-1)}$$
$$= \frac{p}{q}x^{(\frac{p}{q}-1)} = rx^{r-1}, \text{ provided } x > 0.$$

(d) This is argued just like part (c), but keeping in mind that $x^{\frac{1}{q}}$ now describes the inverse of the function f given by $f(u) = u^q$ on the domain $\mathbb{R}$.

(e) The case $r = 0$ is trivial because $x^0 = 1$ on $\mathbb{R}$, including 0. ☺

We go on to consider the trigonometric functions introduced in Section 5-3. The following identities, which were recorded there, will be used shortly:

$$\cos u - \cos v = -2\sin[\tfrac{1}{2}(u+v)]\sin[\tfrac{1}{2}(u-v)]$$
$$\sin u - \sin v = 2\cos[\tfrac{1}{2}(u+v)]\sin[\tfrac{1}{2}(u-v)].$$

7-3.4. Proposition. *The function* sin *is strictly increasing on* $[-\frac{\pi}{2}, \frac{\pi}{2}]$ *and* cos *is strictly decreasing on* $[0, \pi]$.

Proof. Suppose $-\frac{\pi}{2} \le v < u \le \frac{\pi}{2}$. Then

$$-\tfrac{\pi}{2} < \tfrac{1}{2}(u+v) < \tfrac{\pi}{2} \text{ and } 0 < \tfrac{1}{2}(u-v) < \tfrac{\pi}{2}.$$

It follows by Theorem 5-3.14(a) that $\cos[\frac{1}{2}(u+v)] > 0$ and $\sin[\frac{1}{2}(u-v)] > 0$. Therefore

$$\sin u - \sin v = 2\cos[\tfrac{1}{2}(u+v)]\sin[\tfrac{1}{2}(u-v)] > 0.$$

Now suppose $0 \le v < u \le \pi$. Then

$$0 < \tfrac{1}{2}(u+v) < \pi \text{ and } 0 < \tfrac{1}{2}(u-v) < \tfrac{\pi}{2}.$$

It follows by Theorem 5-3.14(a) that $\sin[\frac{1}{2}(u+v)] > 0$ and $\sin[\frac{1}{2}(u-v)] > 0$. Therefore

$$\cos u - \cos v = -2\sin[\tfrac{1}{2}(u+v)]\sin[\tfrac{1}{2}(u-v)] < 0. ☺$$

The values of $\sin(\pm\pi/2)$, $\cos 0$ and $\cos 2\pi$ are known from Theorem 5-3.7 and Theorem 5-3.14; they will be used in the next proof.

7-3.5. Proposition. *The restriction of the function* sin *to the interval* $[-\frac{\pi}{2}, \frac{\pi}{2}]$ *has range* $[-1, 1]$ *and it is a bijection, so that it has an inverse; the inverse is continuous and strictly increasing. Similarly, the restriction of the function* cos *to* *the interval* $[0, \pi]$ *has range* $[-1, 1]$ *and it is a bijection, so that it has an inverse; the inverse is continuous and strictly decreasing.*

Proof. Since $\sin(-\frac{\pi}{2}) = -1$, $\sin(\frac{\pi}{2}) = 1$, and sin is strictly increasing [see Prop.7-3.4] on the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$, the range of the restriction must be a subset of $[-1, 1]$. By the Intermediate Value Theorem, the range must be the whole of $[-1, 1]$. Any strictly increasing function is a bijection from its domain to its range, the inverse being also strictly increasing. Since sin is strictly increasing, the continuity of its inverse follows from Theorem 2-2.1.

The argument for cos is similar. ☺

7-3.6. Definition. *The inverse of the restriction of* sin *to* $[-\frac{\pi}{2}, \frac{\pi}{2}]$ *is called the* **inverse sine** *function and is denoted by either* **arcsin** *or* $\mathbf{sin^{-1}}$. *The inverse of the restriction of* cos *to* $[0, \pi]$ *is called the* **inverse cosine** *function and is denoted by either* **arccos** *or* $\mathbf{cos^{-1}}$.

Remark. The statement that $y = \sin^{-1}x$ is equivalent to the conjunction

$$x = \sin y \quad \underline{\text{and}} \quad -\tfrac{\pi}{2} \le y \le \tfrac{\pi}{2}.$$

Similarly, the statement that $y = \cos^{-1}x$ is equivalent to the conjunction

$$x = \cos y \quad \underline{\text{and}} \quad 0 \le y \le \pi.$$

7-3.7. Proposition. $\cos(\sin^{-1}x) = \sqrt{(1-x^2)} = \sin(\cos^{-1}x)$ *for all* $x \in [-1, 1]$.

Proof. Since $\sin^{-1}x$ always lies between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$, and [by Theorem 5-3.14(a)] cos remains nonnegative on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, therefore $\cos(\sin^{-1}x) \ge 0$ for any $x \in [-1, 1]$. Similarly, $\sin(\cos^{-1}x) \ge 0$ for any $x \in [-1, 1]$. Now, for $x \in [-1, 1]$, we have

$$1 = (\cos(\sin^{-1}x))^2 + (\sin(\sin^{-1}x))^2 = (\cos(\sin^{-1}x))^2 + x^2, \text{ so that}$$

$$(\cos(\sin^{-1}x))^2 = 1 - x^2.$$

Since $\cos(\sin^{-1}x) \ge 0$, it follows that $\cos(\sin^{-1}x) = \sqrt{(1-x^2)}$. The argument in the case of $\sin(\cos^{-1}x)$ is similar. ☺

7-3.8. Proposition. $\lim\limits_{h\to 0} \dfrac{\sin h}{h} = 1.$

Proof. From the representation in Theorem 5-3.9 of sin as a series and from 1-11.P2(a), we have

$$\left|\frac{\sin h}{h} - 1\right| = \left|\sum_{k=1}^{\infty}(-1)^k \frac{h^{2k}}{(2k+1)!}\right| \le \sum_{k=1}^{\infty} \frac{h^{2k}}{2^{2k}} = \sum_{k=1}^{\infty}\left(\frac{h}{2}\right)^{2k}$$

$$= \left(\frac{h^2}{4}\right)\frac{1}{1-\frac{h^2}{4}} \le h^2 \text{ when } 0 < |h| \le \sqrt{3}.$$

For any $\varepsilon > 0$ therefore, $\delta = \min\{\sqrt{\varepsilon}, \sqrt{3}\}$ has the property that

$$0 < h < \delta \Rightarrow \left|\frac{\sin h}{h} - 1\right| \le h^2 < \varepsilon. \;☺$$

7-3.9. Theorem. *The functions* sin *and* cos (*with domain* $\mathbb{R}$) *are differentiable and* $\frac{d}{dx}\sin x = \cos x$ *while* $\frac{d}{dx}\cos x = -\sin x$.

Proof. We first note that

$$\lim_{h\to 0} \frac{2}{h}\sin\frac{h}{2} = 1.$$

This follows from the preceding Proposition. By Theorem 5-3.11, sin and cos are continuous on $\mathbb{R}$. Therefore

$$\lim_{h\to 0} \sin\left(x+\frac{h}{2}\right) = \sin x \text{ and } \lim_{h\to 0} \cos\left(x+\frac{h}{2}\right) = \cos x.$$

Using the identities mentioned just before Prop.7-3.4, we have

$$\frac{\sin(x+h)-\sin x}{h} = \frac{2}{h}\cos\left(x+\frac{h}{2}\right)\sin\frac{h}{2}.$$

Therefore Theorem 7-3.8 and the second of the limits noted in the preceding paragraph lead to the formula for the derivative of $\sin x$. Similarly, the equality

$$\frac{\cos(x+h)-\cos x}{h} = -\frac{2}{h}\sin\left(x+\frac{h}{2}\right)\sin\frac{h}{2}.$$

leads to the formula for the derivative of $\cos x$. ☺

Examples. (a) The function f defined to be $x\sin\frac{1}{x}$ when $x \ne 0$ and to be 0 when $x = 0$ does not have a right or left derivative at 0. This is because

$$\frac{f(0+h) - f(0)}{h} = \sin\frac{1}{h},$$

which has neither a right limit nor a left limit at 0.

(b) Now consider the function f defined to be $x^2\sin\frac{1}{x}$ when $x \ne 0$ and 0 when $x = 0$. Its derivative at 0 exists and equals 0 because

$$\left|\frac{f(0+h) - f(0)}{h} - 0\right| = \left|h\sin\frac{1}{h}\right| \le |h|.$$

For $x \ne 0$, the derivative is $f'(x) = 2x\sin\frac{1}{x} - \cos\frac{1}{x}$, which shows that

$$|f'(0+h) - f'(0)| = \left|2h\sin\frac{1}{h} - \cos\frac{1}{h}\right| \ge \left|\cos\frac{1}{h}\right| - 2|h|.$$

This means that f' is *not* continuous at 0.

7-3.10. Theorem. *The functions* $\sin^{-1}$ *and* $\cos^{-1}$ *are differentiable at every* $x \neq \pm 1$, *and*

$$\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}} = -\frac{d}{dx}\cos^{-1}x.$$

Proof. By Prop.7-3.5, both the functions are continuous and hence by Theorem 7-3.2, both are also differentiable at every $x \neq \pm 1$, with

$$\frac{d}{dx}\sin^{-1}x = \frac{1}{\cos(\sin^{-1}x)}$$

$$= \frac{1}{\sqrt{1-x^2}} \qquad \text{by Prop.7-3.7,}$$

and similarly for $\frac{d}{dx}\cos^{-1}x$. ☺

As usual, we take the **tangent function** to be the one defined by

$$\tan x = \sin x/\cos x \text{ on the domain } \{x \in \mathbb{C} : \cos x \neq 0\}.$$

However, we shall be concerned with real x only. On the basis of the properties we have already proved for sin and cos, it is easy to derive the well known elementary properties of tan; in particular, its domain includes the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$.

7-3.11. Theorem. *The restriction of* tan *to the interval* $(-\frac{\pi}{2}, \frac{\pi}{2})$ *is strictly increasing and differentiable with* $\frac{d}{dx}\tan x = 1/\cos^2 x$. *The inverse*, *denoted by* $\tan^{-1}$ *or by* arctan, *is differentiable with derivative*

$$\frac{d}{dx}\tan^{-1}x = 1/(1+x^2).$$

Proof. From the formula for $\sin(u-v)$, we have

$$\tan u - \tan v = \sin(u-v)/(\cos u \cos v).$$

Suppose $-\frac{\pi}{2} < v < u < \frac{\pi}{2}$. Then $0 < u - v < \pi$. It follows from by Theorem 5-3.14(a) that $\cos u > 0$, $\cos v > 0$ and $\sin(u-v) > 0$. Hence $\tan u - \tan v > 0$. Thus tan is strictly increasing on $(-\frac{\pi}{2}, \frac{\pi}{2})$. Since sin and cos have been shown to be differentiable [see Theorem 7-3.9], it follows from Theorem 7-2.4(d) that their quotient, tan must also be differentiable wherever it is defined; moreover, the derivative is easily computed to be $1/\cos^2 x$. By Theorem 7-3.2, and the fact that

$$\frac{d}{dx}\tan x = 1/\cos^2 x,$$

the derivative of $\tan^{-1}$ is

$$\frac{d}{dx}\tan^{-1}x = 1/[1/\cos^2(\tan^{-1}x)] = \cos^2(\tan^{-1}x).$$

Now, $\cos^2(\tan^{-1}x) + \sin^2(\tan^{-1}x) = 1$. Dividing by $\cos^2(\tan^{-1}x)$, we get

$$1+\tan^2(\tan^{-1}x) = 1/\cos^2(\tan^{-1}x), \quad \text{i.e. } 1+x^2 = 1/\cos^2(\tan^{-1}x).$$

Together with the equality displayed above, this implies what we want. ☺

Problem Set 7-3

7-3.P1. Prove that $\lim\limits_{h\to 0}\dfrac{1-\cos h}{h} = 0$ by using the series for $\cos h$, as was done for $\sin h$ in Prop.7-3.8.

7-3.P2. Show that $\tan^{-1}\sqrt{3} = \frac{\pi}{3}$.

7-3.P3. If $\tan^{-1}x = \alpha$, $\tan^{-1}y = \beta$ and $xy < 1$, show that $-\frac{\pi}{2} < \alpha+\beta < \frac{\pi}{2}$.

7-3.P4. If $xy < 1$, show that $\tan^{-1}\big((x+y)/(1-xy)\big) = \tan^{-1}x + \tan^{-1}y$.

7-3.P5. If $x = \sqrt{3}$ and $y = 1$, show that

$$\tan^{-1}\big((x+y)/(1-xy)\big) \neq \tan^{-1}x + \tan^{-1}y.$$

7-4 Exponential Function and its Inverse

In this Section, we shall begin by considering the exponential function only on $\mathbb{R}$. As usual, we denote the restriction to $\mathbb{R}$ also by exp.

7-4.1. Proposition. *The function* exp *has range* $(0,\infty)$ *and it is strictly increasing*, *so that it is a bijection and has an inverse*; *the inverse is continuous and strictly increasing*.

Proof. We begin by noting that

$$\exp x = 1 + x + \sum_{k=2}^{\infty}(x^k/k!) > x \quad \text{whenever } x > 0. \tag{1}$$

Since $\exp 0 = 1$, the range contains the number 1. Let $\alpha > 1$. The range of exp contains the number $\exp\alpha$, which is greater than α by (1). Thus the range of exp contains the number $1 < \alpha$, and also contains the number $\exp\alpha > \alpha$. Also, exp is continuous by Theorem 5-3.11. It follows by the Intermediate Value Theorem that the range of exp contains α. This shows that every number greater than 1 is in the range of exp. Now, consider any positive number $\alpha < 1$. Then $1/\alpha > 1$ and is therefore in the range of exp. Let $\exp u = 1/\alpha$. Then $\exp(-u) = \alpha$, so that any positive number $\alpha < 1$ is also in the range of exp. Thus the range of exp contains $(0,\infty)$.

By (1), we have $\exp x > 0$ whenever $x > 0$. It follows that, when $x < 0$, we have $\exp x = 1/\exp(-x) > 0$; finally $\exp 0 = 1$. Thus $\exp x > 0$ for all $x \in \mathbb{R}$. This shows that the range of exp is precisely $(0, \infty)$.

When $v < u$, we have $\exp(u-v) > 1$ by (1) and hence

$$\exp u = (\exp v)\cdot[\exp(u-v)] > \exp v.$$

Therefore exp is strictly increasing and is thus a bijection. Any strictly increasing function is a bijection from its domain to its range, the inverse being also strictly increasing. The continuity of its inverse follows from Theorem 2-2.1. ☺

7-4.2. Definition. *The inverse of* $\exp:\mathbb{R}\to(0,\infty)$ *is called the* **natural logarithm** *function and is denoted by* **ln**.

7-4.3. Proposition. $\lim_{h\to 0}\dfrac{\exp h - 1}{h} = 1.$

Proof. Since $\exp h = \sum_{k=0}^{\infty}\dfrac{h^k}{k!} = 1 + h + h\left(\sum_{k=1}^{\infty}\dfrac{h^k}{(k+1)!}\right)$, therefore

$$\frac{\exp h - 1}{h} - 1 = \sum_{k=1}^{\infty}\frac{h^k}{(k+1)!}.$$

Now, $(k+1)! \geq 2^k$ when $k \geq 1$. Therefore

$$\left|\frac{\exp h - 1}{h} - 1\right| \leq \left|\sum_{k=1}^{\infty}\left(\frac{h}{2}\right)^k\right| \leq \frac{|h|}{2-|h|} < |h| \text{ when } |h| < 1.$$

Hence, for any $\varepsilon > 0$, the positive number $\delta = \min\{1, \varepsilon\}$ has the property that

$$0 < |h| < \delta \Rightarrow \left|\frac{\exp h - 1}{h} - 1\right| < \varepsilon. \text{ ☺}$$

7-4.4. Theorem. *The functions* exp *and* ln *are differentiable and*

$$\frac{d}{dx}\exp x = \exp x \quad \textit{while} \quad \frac{d}{dx}\ln x = 1/x \;\; (x > 0).$$

Proof. $\dfrac{\exp(x+h) - \exp x}{h} = \dfrac{(\exp x)(\exp h) - \exp x}{h} = (\exp x)\dfrac{\exp h - 1}{h}.$

Hence by Prop.7-4.3, $\frac{d}{dx}\exp x = \exp x$.

Since ln, the inverse of exp, is continuous and strictly monotone by Prop.7-4.1, it follows from Theorem 7-3.2 that it is differentiable and that

$$\frac{d}{dx}\ln x = 1/(\exp(\ln x)) = 1/x. \text{ ☺}$$

7-4.5. Definition. *Let f be a real valued function whose domain contains an interval (a,∞) for some $a\in\mathbb{R}$. Then* $\lim\limits_{x\to\infty} f(x)=\infty$ *means*

$$\forall\ A>0\ \exists\ M \text{ such that } x>M\Rightarrow f(x)>A.$$

7-4.6. Proposition. *Let n be any integer and $A>0$. Then there exists some $M>0$ such that* $x>M\Rightarrow \frac{\exp x}{x^n}>A$. *In other words, for any integer n,*

$$\lim_{x\to\infty}\frac{\exp x}{x^n}=\infty.$$

Proof. When $x>0$, all the terms in the series that defines $\exp x$ are positive. Also, one of the terms is x. Therefore $\exp x>x$ when $x>0$. When n is a negative integer, we have $x^{-n}>1$ for $x>1$ and hence $\frac{\exp x}{x^n}>\exp x>x$. Therefore, when $M=\max\{A,1\}$ we have $x>M\Rightarrow \frac{\exp x}{x^n}>A$.

Now suppose the integer n is nonnegative. Then $\frac{x^{n+1}}{(n+1)!}$ is one of the terms in the series that defines $\exp x$. It follows that

$$\frac{\exp x}{x^n}>\frac{x^{n+1}}{(n+1)!}>\frac{x}{(n+1)!} \text{ whenever } x>1.$$

With $M=\max\{1,(n+1)!A\}$, we have $x>M\Rightarrow \frac{\exp x}{x^n}>A$. ☺

The next result is essential for 7-4.P3, which will be used later on in connection with Taylor series.

7-4.7. Corollary. *Let n be any integer. Then* $\lim\limits_{h\to 0}\dfrac{\exp\left(-\dfrac{1}{h^2}\right)}{h^n}=0.$

Proof. Consider any $\varepsilon>0$. By the preceding Proposition, there exists $M>0$ such that $|1/h|>M\Rightarrow \frac{\exp(1/h^2)}{|1/h^n|}>1/\varepsilon$. Let $\delta=1/M$. Then

$$0<|h|<\delta \quad\Rightarrow\quad \frac{|1/h^n|}{\exp(1/h^2)}<\varepsilon \quad\Rightarrow\quad \left|\frac{\exp(-1/h^2)}{h^n}\right|<\varepsilon. \text{☺}$$

7-4.8. Remark. It is trivial that $\exp(1\cdot u)=(\exp u)^1$ for every $u\in\mathbb{R}$. The property $\exp(u+v)=(\exp u)(\exp v)$ implies that

$$\exp((p+1)u)=(\exp(pu))\cdot(\exp u),$$

which completes the induction argument that, for any $p\in\mathbb{N}$, we have

$$\exp(pu)=(\exp u)^p \text{ for every } u\in\mathbb{R}. \tag{1}$$

Now let p represent any positive rational number $\frac{m}{n}$, where $m \in \mathbb{N}$, $n \in \mathbb{N}$. Then by (1), we have

$$[\exp(pu)]^n = [\exp(\tfrac{m}{n}u)]^n = \exp(n \cdot \tfrac{m}{n}u) = \exp(mu) = (\exp u)^m,$$

and hence

$$\exp(pu) = (\exp u)^{m/n} = (\exp u)^p \quad \text{for every } u \in \mathbb{R}.$$

Thus (1) is valid for any positive rational number p. It follows for negative rational p by using the facts that $\exp(-pu) = 1/\exp(pu)$ and $(\exp u)^{-p} = 1/(\exp u)^p$, and is obvious for $p = 0$. Thus it holds for all rational p.

If we take $u = \ln x$ in (1), we get

$$\exp(p \ln x) = [\exp(\ln x)]^p = x^p. \tag{2}$$

The left side of this makes sense even for irrational p, but the right side is then an irrational power and is yet to be defined. We do so now essentially by a "decree" that (2) be required to hold even when p is irrational!

7-4.9. Definition. *If x is any positive real number and α any irrational real number, then* $\mathbf{x}^{\boldsymbol{\alpha}}$ *is defined to be* $\exp(\alpha \ln x)$. *Also,* $\mathbf{0}^{\boldsymbol{\alpha}}$ *is defined to be* 0 *when* α *is a positive real number.*

The equality $x^\alpha = \exp(\alpha \ln x)$ is now valid for *all* real α as long as x is positive. It is left to the reader in 7-4.P6 to prove the "laws of indices" when some or all indices involved are irrational and $x > 0$. This was done for rational indices in 2-4.P7.

The properties $\ln(xy) = \ln x + \ln y$ and $\ln(x^\alpha) = \alpha \ln x$, where $x,y > 0$ and α is any real number, are easy to derive.

In the discussion of the derivative of x^r in Prop.7-3.3, we had to restrict ourselves to rational values of r because x^r had not been defined for irrational r at that stage. It is time now to establish that the same formula holds for irrational r as well. The case $x = 0$ will be handled in Example 7-5.8(a).

7-4.10. Theorem. *If r is any real number, then* $\frac{d}{dx}x^r = rx^{r-1}$ *for* $x > 0$.

Proof. When r is rational, the equality holds by Prop.7-3.3. When r is irrational (and $x > 0$), $x^r = \exp(r \ln x)$, and by the Chain Rule and Theorem 7-4.4, we have

$$\begin{aligned}\frac{d}{dx}x^r &= \frac{d}{dx}\exp(r\ln x) = \exp(r\ln x)\frac{d}{dx}(r\ln x) = \exp(r\ln x)\,rx^{-1}\\ &= r\exp(r\ln x)\exp((-1)\cdot\ln x) = r\exp(r\ln x)\exp(-\ln x)\\ &= r\exp((r-1)\ln x) = rx^{r-1}. ☺\end{aligned}$$

The argument based on the equality $x^r = \exp(r\ln x)$ does not apply to all the cases covered earlier by Prop.7-3.3 because the latter included such cases as $r = \frac{1}{3}$ when x could be negative.

7-4.11. Definition. *The number* $\exp 1$ *is denoted by* $\boldsymbol{e}$.

Since $e = \exp 1$, we have $\ln e = 1$. Also, the function exp has only positive numbers in its range; so, the number e is positive. Therefore we may take $x = e$ in the equality $x^\alpha = \exp(\alpha \ln x)$, which is valid for all real α and positive x. Upon doing so, we obtain

$$e^\alpha = \exp(\alpha \ln e) = \exp\alpha \qquad \text{for every } \alpha \in \mathbb{R}.$$

This permits us to interchange e^x with $\exp x$ according to convenience. The number e is irrational [see 7-4.P5].

7-4.12. Proposition. $2.718 < e < 2.7186$.

Proof. From the series $e = \exp 1 = \sum_{k=0}^{\infty} \frac{1}{k!}$, we have

$$e > \sum_{k=0}^{6} \frac{1}{k!}. \tag{1}$$

Also,
$$0 < e - \sum_{k=0}^{6} \frac{1}{k!} = \sum_{k=7}^{\infty} \frac{1}{k!}. \tag{2}$$

Now, $7! = 5040 > 4096 = 2^{12} = 2^{7+5}$; an easy induction argument yields $k! > 2^{k+5}$ for $k \geq 7$. Therefore

$$\sum_{k=7}^{\infty} \frac{1}{k!} \leq \sum_{k=7}^{\infty} \frac{1}{2^{k+5}} = \frac{1}{2^{12}} \frac{1}{1-\frac{1}{2}} = \frac{1}{2^{11}} = \frac{1}{2048} < \frac{1}{2000} = 0.0005.$$

It now follows from (1) and (2) that

$$\sum_{k=0}^{6} \frac{1}{k!} < e < \sum_{k=0}^{6} \frac{1}{k!} + 0.0005. \tag{3}$$

It is easy to compute that $\sum_{k=0}^{6} \frac{1}{k!} = \frac{1957}{720}$ and that

$$2.718 < \frac{1957}{720} < 2.7181. \tag{4}$$

[For a full justification of such computations on the basis of the definition of the real number system, see the concluding paragraphs of Sections 1-11 and 4-1.] The required inequality now follows from (3) and (4). ☺

If we were to replace the 6 in $\sum_{k=0}^{6} \frac{1}{k!}$ by 5, the "bounds" obtained in place of 2.718 and 2.7186 would have differed by more than 10^{-3} [see 7-4.P4]. There is a systematic way to arrive at 6, given that we want the bounds to differ by less than 10^{-3}, but we shall not go into it.

7-4.13. Proposition. *Let* α *be any real number. Then for any* $A > 0$, *then there exists some* $M > 0$ *such that* $x > M \Rightarrow \frac{\exp x}{x^\alpha} > A$. *In other words,*

$$\lim_{x\to\infty} \frac{\exp x}{x^\alpha} = \infty \qquad \textit{for every } \alpha \in \mathbb{R}.$$

Proof. Let n be an integer greater than α. Then $x > 1 \Rightarrow 0 < x^\alpha < x^n$ (by one of the laws of indices) $\Rightarrow \frac{\exp x}{x^\alpha} > \frac{\exp x}{x^n}$. By Prop.7-4.6, there exists $M_1 > 0$ such that $\frac{\exp x}{x^\alpha} > A$ for $x > M_1$.

Let $M = \max\{1, M_1\}$. Then $x > M \Rightarrow \frac{\exp x}{x^\alpha} > A$. ☺

The sequence $\{a_n\}$ given by $a_n = (1 + \frac{1}{n})^n$ was shown in Example 3-3.4 to be a bounded monotone sequence and its convergence established. We now show that the limit is the number e: Since $f(x) = \ln x$ satisfies $f'(1) = 1$, we have

$$\lim_{h\to 0} \frac{\ln(1+h)}{h} = 1 \text{ and hence } \frac{\ln(1+\frac{1}{n})}{\frac{1}{n}} \to 1.$$

But $\ln a_n = n\ln(1+\frac{1}{n}) = \frac{\ln(1+\frac{1}{n})}{\frac{1}{n}}$. Therefore $\ln a_n \to 1$. Since exp is a continuous function, it follows that $\exp(\ln a_n) \to \exp 1$, i.e. $a_n \to e$.

It is worth noting that $a_{1000} = 2.716923\ldots$ and $a_{4500} = 2.717979\ldots$.

7-4.14. Proposition. *If* $\alpha > 0$, *then* $\lim_{x\to 0} x^\alpha = 0$. *The function* $f: [0,\infty) \to \mathbb{R}$ *defined by* $f(x) = x^\alpha$ *is therefore continuous at* 0.

Proof. For any $\varepsilon > 0$, let $\delta = \min\{1, \exp(-1/\varepsilon\alpha)\} > 0$. We shall show that $x^\alpha < \varepsilon$ whenever $0 < x < \delta$. The fact that $\exp u > u$ for $u > 0$ will be needed [see (1) of Prop.7-4.1]. For any x such that $0 < x < \delta$, we have

$$0 < x < \exp(-1/\varepsilon\alpha),\ \ln x < -1/\varepsilon\alpha,\ -\ln x > 1/\varepsilon\alpha,\ -\alpha\ln x > 1/\varepsilon > 0.$$

Since $\delta < 1$, we also have $u = -\alpha\ln x > 0$. Using the fact about exp mentioned above, we get $\exp(-\alpha\ln x) > -\alpha\ln x > 1/\varepsilon > 0$. Hence $0 < \exp(\alpha\ln x) < \varepsilon$, i.e. $0 < x^\alpha < \varepsilon$. ☺

For the rest of this Section, we consider exp on all of $\mathbb{C}$. The material about to be presented now was used in 4-4.P4 but will not be used in the sequel.

First of all, we note that the statement (1) of Remark 7-4.8 above is actually valid for all $u \in \mathbb{C}$: $\exp(pz) = (\exp z)^p$ for all $z \in \mathbb{C}$ and all $p \in \mathbb{N}$. In view of the identity $\exp(iz) = \cos z + i \sin z$ [see Remark preceding Theorem 5-3.7], this implies

$$\begin{aligned}(\cos z + i \sin z)^p &= (\exp(iz))^p = \exp(ipz)\\ &= \cos(pz) + i \sin(pz) \text{ for all } z \in \mathbb{C} \text{ and all } p \in \mathbb{N}.\end{aligned}$$

This identity is called **de Moivre's Theorem** and is particularly useful when z is real. We illustrate one of its uses. It will be crucial that, for real x, the complex conjugate of $\exp(ix) = \cos x + i \sin x$ is $\cos x - i \sin x = \exp(-ix)$. We shall also use the fact that $\exp(ib) = 1$ is equivalent to $\sin\frac{b}{2} = 0$; this is because $\exp(ib) = \cos b + i \sin b = (1 - 2\sin^2\frac{b}{2}) + i(2\sin\frac{b}{2}\cos\frac{b}{2})$.

Example. To find the finite sum $C = \sum_{k=1}^{n} \cos(a + (k-1)b)$ with a and b real, we introduce the related finite sum $S = \sum_{k=1}^{n} \sin(a + (k-1)b)$, and use the fact that

$$C + iS = \sum_{k=1}^{n} \exp[i(a + (k-1)b)],$$

which is a geometric progression with "common ratio" $\exp(ib)$. When $\exp(ib) \neq 1$, or equivalently $\sin\frac{b}{2} \neq 0$, we may use the formula for such finite sums and get

$$\begin{aligned}C + iS &= e^{ia}\,\frac{1-(e^{ib})^n}{1-e^{ib}} = e^{ia}\,\frac{(1-e^{inb})(1-e^{-ib})}{(1-e^{ib})(1-e^{-ib})} = e^{ia}\,\frac{1-e^{-ib}-e^{inb}+e^{i(n-1)b}}{1-e^{ib}-e^{-ib}+1}\\ &= \frac{e^{ia}-e^{-i(b-a)}-e^{i(nb+a)}+e^{i((n-1)b+a)}}{1-e^{-ib}-e^{ib}+1}.\end{aligned}$$

The denominator here is $2(1-\cos b) = 4\sin^2\frac{b}{2}$ (assumed to be nonzero) and the real part of the numerator is

$$\cos a - \cos(b-a) - \cos(nb+a) + \cos((n-1)b+a).$$

Upon using the formula $\cos C + \cos D = 2\cos\frac{1}{2}(C+D)\cos\frac{1}{2}(C-D)$, this numerator becomes

$$\begin{aligned}&2\cos((n-1)\tfrac{b}{2}+a)\cos(n-1)\tfrac{b}{2} - \cos(n+1)\tfrac{b}{2}\cos((n-1)\tfrac{b}{2}+a)\\ &= 2\cos((n-1)\tfrac{b}{2}+a)\Big(\cos(n-1)\tfrac{b}{2} - \cos(n+1)\tfrac{b}{2}\Big).\end{aligned}$$

Upon using the formula $\cos C - \cos D = -2\sin\frac{1}{2}(C+D)\sin\frac{1}{2}(C-D)$, this further becomes

$$4\cos((n-1)\tfrac{b}{2}+a)\sin n\tfrac{b}{2}\sin\tfrac{b}{2}.$$

Taking into account that the denominator is $4\sin^2\frac{b}{2}$, the real part of $C+iS$ is seen to be

$$C = \frac{\cos\left((n-1)\frac{b}{2}+a\right)\sin n\frac{b}{2}}{\sin\frac{b}{2}} \quad \text{when} \quad \sin\frac{b}{2}\neq 0,$$

which is the required sum. [This sum can also be obtained without using de Moivre's Theorem.] The expression we have obtained for C involves the same number of algebraic and trigonometric operations regardless of the value of n, while the number of operations in the original expression using Σ increases with n. This is the sense in which we have "found" the sum.

When $a = 0$, this becomes

$$\sin\frac{b}{2}\sum_{k=1}^{n-1}(1+\cos kb) = \cos\left((n-1)\frac{b}{2}\right)\sin n\frac{b}{2}$$

$$= \tfrac{1}{2}\left(\sin\left(n-\tfrac{1}{2}\right)b+\sin\tfrac{b}{2}\right), \quad \text{so that}$$

$$\sin\frac{b}{2}\sum_{k=1}^{n-1}\cos kb = \tfrac{1}{2}\left(\sin\left(n-\tfrac{1}{2}\right)b+\sin\tfrac{b}{2}\right)-\sin\tfrac{b}{2} = \tfrac{1}{2}\left(\sin\left(n-\tfrac{1}{2}\right)b-\sin\tfrac{b}{2}\right)$$

and hence $\displaystyle\sum_{k=1}^{n-1}\cos kb = \tfrac{1}{2}\left(\sin\left(n-\tfrac{1}{2}\right)b-\sin\tfrac{b}{2}\right)\Big/\sin\frac{b}{2}$ when $\sin\frac{b}{2}\neq 0$.

Problem Set 7-4

7-4.P1. Show that the function $f\colon\{x\in\mathbb{R} : x\neq 0\}\to\mathbb{R}$, where $f(x) = \ln|x|$, is differentiable with $f'(x) = 1/x$.

7-4.P2. For $x < 0$ and $p\in\mathbb{Z}$, show that $x^p = (-1)^p\exp(p\ln|x|)$. Does this hold for other rational values of p?

7-4.P3. If P is any polynomial function, show that $\frac{d}{dx}\left[P\left(\frac{1}{x}\right)\exp\left(-\frac{1}{x^2}\right)\right] = Q\left(\frac{1}{x}\right)\exp\left(-\frac{1}{x^2}\right)$ for some polynomial function Q. Show that the function f defined for $x\neq 0$ by $f(x) = P\left(\frac{1}{x}\right)\exp\left(-\frac{1}{x^2}\right)$ satisfies $\lim_{h\to 0} f(h) = 0$ and deduce that it also satisfies $f'(0) = 0$ if we take $f(0) = 0$.

7-4.P4. If you repeat the argument of Prop.7-4.12 with $\sum_{k=0}^{5}\frac{1}{k!}$ in (1), what inequalities do you get up to three decimal places for e?

7-4.P5. Prove that e is irrational.

7-4.P6. Prove that when x, y are positive real numbers, and α,β are any real numbers:

(a) $(x^\alpha)^\beta = x^{\alpha\beta}$ (b) $x^\alpha x^\beta = x^{\alpha+\beta}$ (c) $x^\alpha y^\alpha = (xy)^\alpha$ (d) $x > y > 0,\ \alpha > 0 \Rightarrow x^\alpha > y^\alpha$

(e) $x > 1, \alpha > \beta \Rightarrow x^\alpha > x^\beta$ (f) $0 < x < 1, \alpha > \beta \Rightarrow x^\alpha < x^\beta$.

7-4.P7. Prove the convergence of the series $\sum_{k=1}^{\infty}(1/k^p)$ when p is irrational and greater than 1.

7-4.P8. Prove the divergence of the series $\sum_{k=2}^{\infty}(1/\ln k)^\alpha$ for any real α.

7-4.P9. Find the finite sum $S = \sum_{k=1}^{n} \sin(a + (k-1)b)$ with a and b real. Hence show that,

$$\sum_{k=1}^{n-1} \sin kb = -\tfrac{1}{2}\left(\cos\left(n-\tfrac{1}{2}\right)b - \cos\tfrac{b}{2}\right)\Big/\sin\tfrac{b}{2} \quad \text{when} \quad \sin\tfrac{b}{2} \neq 0.$$

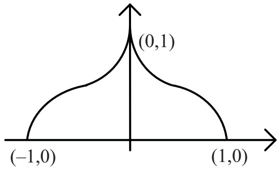
7-4.P10. From Def.7-4.9, show that $x > 0, b > 1 \Rightarrow 0 < b^{-x} < 1$.

7-4.P11. Show that the equation $e^x \tan^{-1}x = x^2 + \frac{1}{2}$ has a solution between 0 and 1. [In other words, $\exists\, x \in (0,1) \ni e^x \tan^{-1}x = x^2 + \frac{1}{2}$.]

7-4.P12. Let $f:\mathbb{R}\to\mathbb{R}$ have the property that $f' = f$ everywhere. Show that $f(x) = f(0)e^x$.

7-4.P13. By means of a single formula applicable on the entire domain, describe a function $f:[-1,1]\to\mathbb{R}$ such that

(a) $f(x) \geq 0 \quad \forall\, x \in [-1,1]$
(b) $f(-x) = f(x) \quad \forall\, x \in [-1,1]$
(c) f is differentiable on $(0,1)$ and
(d) $f'(0+) = -\infty = f'(1-)$.



7-5 Lagrange Mean Value Theorem

An informal statement of the first result below would be this: If the derivative of a function is positive at some point, then the value of the function there is greater [resp. less] than at any sufficiently close by point to the left [resp. right]; correspondingly when the derivative is negative. In order to prove it however, we need a technically precise statement.

7-5.1. Lemma. *Suppose the function $f:A\to\mathbb{R}$ is differentiable at a point $c \in A$. If $f'(c) > 0$, then there exists some $\delta > 0$ such that*

(A) $0 < h < \delta,\ c - h \in A \Rightarrow f(c-h) < f(c)$

(B) $0 < h < \delta,\ c + h \in A \Rightarrow f(c+h) > f(c)$.

If $f'(c) < 0$, then there exists some $\delta > 0$ such that

(A') $0 < h < \delta,\ c-h \in A \Rightarrow f(c-h) > f(c)$

(B') $0 < h < \delta,\ c+h \in A \Rightarrow f(c+h) < f(c)$.

Proof. Suppose $f'(c) > 0$. By the definition of derivative as a limit, taking $\varepsilon = f'(c)$, there exists $\delta > 0$ such that

$$0 < |h| < \delta,\ c+h \in A \Rightarrow \left|\frac{f(c+h)-f(c)}{h} - f'(c)\right| < f'(c)$$

$$\Rightarrow \frac{f(c+h)-f(c)}{h} - f'(c) > -f'(c)$$

$$\Rightarrow \frac{f(c+h)-f(c)}{h} > 0$$

$$\Rightarrow f(c+h) - f(c) \gtrless 0, \text{ according as } h \gtrless 0.$$

This shows that (A) and (B) hold for some $\delta > 0$ when $f'(c) > 0$. The case when $f'(c) < 0$ follows by applying the previous case to the function $-f$. ☺

7-5.2. Proposition. *Let* $f: A \to \mathbb{R}$ *be differentiable at* $c \in A$ *and*

either $f(x) \le f(c)$ *whenever* $x \in A$, i.e. *f has a maximum value at c*
or $f(x) \ge f(c)$ *whenever* $x \in A$, i.e. *f has a minimum value at c*.

If there exists $\delta > 0$ *such that* $[c-\delta, c+\delta] \subseteq A$, *then* $f'(c) = 0$.

Proof. We need consider only the case when f has a maximum at c. The other case follows by applying this one to $-f$. So, suppose $f(x) \le f(c)$ whenever $x \in A$. This means

$$f(c+h) - f(c) \le 0 \text{ whenever } c+h \in A. \tag{1}$$

Suppose, if possible, that $f'(c) > 0$. By Lemma 7-5.1, there exists some $\delta' > 0$ such that

$$f(c+h) - f(c) > 0 \text{ whenever } 0 < h < \delta' \text{ and } c+h \in A. \tag{2}$$

Let $0 < h < \min\{\delta, \delta'\}$, so that $c+h \in A$ and $0 < h < \delta'$. Then, on the one hand, we have $f(c+h)-f(c) \le 0$ by (1), and on the other hand, we also have $f(c+h)-f(c) > 0$ by (2). This contradiction shows that it is not possible that $f'(c) > 0$. Thus $f'(c) \le 0$. A similar argument shows that $f'(c) \ge 0$. It follows that $f'(c) = 0$. ☺

7-5.3. Examples. (a) Suppose we wish to find the minimum value of the function f given by $f(x) = xe^x$, $-1 \le x \le 0$. By the Extreme Value Theorem, there does exist at least one point in the interval $[-1, 0]$ where f takes a minimum value. If the point is in the open interval $(-1, 0)$, the derivative at that point must be 0 by

Prop.7-5.2. However, the derivative is found to be $(1+x)e^x$, which is positive when $x > -1$. Therefore the minimum value is taken at an endpoint. Comparing the values $f(-1)$ and $f(0)$, which are $-1/e$ and 0 respectively, we find that the minimum value is $f(-1) = -1/e$.

(b) We have shown that $1 < \frac{\pi}{2}$. Let $1 < a < \frac{\pi}{2}$ and f be the function defined on $[1,a]$ by $f(x) = x^{-1}\tan x$. Its derivative is

$$f'(x) = \frac{x - \sin x \cos x}{x^2 \cos^2 x},$$

which is positive when $x \in [1,a]$. By Prop.7-5.2, f cannot take a maximum or minimum value in the open interval $(1,a)$. So, the minimum value is the lesser of the two values $f(1) = \tan 1$ and $f(a) = a^{-1}\tan a$. However, deciding which of these is the lesser is not an attractive prospect! Since $f'(1)$ and $f'(a)$ are both positive, Lemma 7-5.1 guarantees that $f(1)$ is not the maximum value and $f(a)$ is not the minimum value. Hence $f(1)$ is the minimum value and $f(a)$ the maximum. [Alternatively, one may use Prop.7-5.9 below].

7-5.4. Rolle's Theorem. *Let the function* $f:[a,b]\to\mathbb{R}$ *be continuous on* $[a,b]$ *and differentiable on* (a,b), *while* $f(a) = f(b)$. *Then there exists some number* $c \in (a,b)$ *such that* $f'(c) = 0$.

Proof. Since f is continuous on the closed and bounded interval $[a,b]$, by the Extreme Value Theorem [see Theorem 2-3.3], it must have a maximum value as well as a minimum value at some point of the interval. If both these values are taken at an endpoint, then the function must be constant in view of the hypothesis that $f(a) = f(b)$. In this event, its derivative is 0 at *every* point of (a,b). In the contrary case, either the maximum or the minimum value is taken at a point $c \in (a,b)$, and therefore $f'(c) = 0$ by Prop.7-5.2. ☺

The above Theorem is an important theoretical tool, but it can also provide useful information in concrete situations. For instance, the function $f(x) = x^3 - 2x^2 + \cos x$ satisfies $f(0) = 1 > 0$ and $f(1) = -1 + \cos 1 < 0$. By the Intermediate Value Theorem, there must exist $\alpha \in (0,1)$ such that $f(\alpha) = 0$. The traditional way to say this is that the equation $x^3 - 2x^2 + \cos x = 0$ has a solution lying between 0 and 1. The question arises as to how many such solutions it has. If there were two or more, then by Rolle's Theorem, there would have to exist $c \in (0,1)$ such that $f'(c) = 0$. However,

$$f'(x) = 3x^2 - 4x - \sin x = 3x(x - \tfrac{4}{3}) - \sin x < 0 \text{ when } 0 < x < 1,$$

because for these values of x, we have $x > 0$, $x-\frac{4}{3} < 0$ and $\sin x > 0$. The contradiction shows that there cannot exist two (or more) solutions.

In the proof the next result, the function ϕ is of the form $\phi(x) = f(x)-\lambda(x-a)$, where the constant λ has been obtained by solving the simple linear equation $\phi(a) = \phi(b)$, i.e. $f(a) = f(b)-\lambda(b-a)$.

7-5.5. Lagrange Mean Value Theorem. *Let the function $f:[a,b]\to\mathbb{R}$ be continuous on $[a,b]$ and differentiable on (a,b). Then there exists some number $c \in (a,b)$ such that*

$$f'(c) = \frac{f(b)-f(a)}{b-a}.$$

Proof. Define the function $\phi:[a,b]\to\mathbb{R}$ as

$$\phi(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x-a).$$

Then ϕ is continuous on $[a,b]$ and differentiable on (a,b). Besides, $\phi(a) = f(a) = \phi(b)$, so that ϕ satisfies the conditions of Rolle's Theorem 7-5.4. Hence there exists $c \in (a,b)$ such that $\phi'(c) = 0$. However,

$$\phi'(c) = f'(c) - \frac{f(b)-f(a)}{b-a}. \ ☺$$

It is now a simple matter to prove the next result, which is at the heart of many scientific applications of the idea of derivative.

7-5.6. Theorem. *Suppose the functions $f:[a,b]\to\mathbb{R}$ and $g:[a,b]\to\mathbb{R}$ are continuous on $[a,b]$ and $f'(x) = g'(x)$ for each $x \in (a,b)$. Then $f-g$ is constant on $[a,b]$, i.e.*

$$f(u)-g(u) = f(v)-g(v) \quad \textit{for all } u, v \in [a,b].$$

Proof. Consider any $u, v \in [a,b]$. We shall consider only the case $u < v$, because the case when $u > v$ is similar. Then $[u,v] \subseteq [a,b]$. Also, and $\phi = f-g$ is continuous on the closed interval $[u,v]$ and differentiable on the open interval (u,v). By the Lagrange Mean Value Theorem, there exists some $c \in (u,v)$ such that $\phi'(c) = \frac{\phi(v)-\phi(u)}{v-u}$. Since $\phi'(c) = 0$, we have $\phi(u) = \phi(v)$. ☺

The hypothesis that the domain be an interval is essential to the above Theorem. As mentioned in the opening paragraphs of this book, the function f defined to be $1+\ln|x|$ for $x > 0$ and $\ln|x|$ for $x < 0$ has derivative $\frac{1}{x}$ everywhere, which is the same derivative as that of the function g defined to be $\ln|x|$ for $x > 0$ as well as for $x < 0$. Yet the difference between f and g is not a constant.

It was mentioned at the end of Section 7-2 that the result of Theorem 7-5.6 can also be obtained from the more elementary Proposition of the appendix to Chapter 1. The ideas in the appendix can be used for obtaining the Lagrange Mean Value Theorem and a couple of other results to be presented in this Chapter. (For details, see [10].)

7-5.7. Proposition. *Let the function* $f:[a,b]\to\mathbb{R}$ *be continuous on* $[a,b]$ *and differentiable on* (a,b). *If* $\lim_{x\to a} f'(x)$ *exists, then* $f'(a)$ *exists and is equal to* $\lim_{x\to a} f'(x)$. *If* $\lim_{x\to b} f'(x)$ *exists, then* $f'(b)$ *exists and is equal to* $\lim_{x\to b} f'(x)$.

Proof. Let $L = \lim_{x\to a} f'(x)$. Then for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < x-a < \delta,\ x \in (a,b) \ \Rightarrow\ |f'(x)-L| < \varepsilon. \tag{1}$$

Consider any h such that $0 < h < \delta$ and $a+h \in (a,b)$. By the Lagrange Mean Value Theorem, we have

$$\frac{f(a+h)-f(a)}{h} = f'(c) \text{ for some } c \in (a, a+h). \tag{2}$$

Now, $0 < c-a < h < \delta$ and therefore $|f'(c)-L| < \varepsilon$ by (1). Combining this with (2), we get

$$\left|\frac{f(a+h)-f(a)}{h} - L\right| < \varepsilon.$$

This means that $f'(a)$ exists and is equal to $\lim_{x\to a} f'(x)$. The proof of the second part is similar. ☺

7-5.8. Examples. (a) If $f(x) = x^r$ for $x \geq 0$, where $r > 1$, then $f'(x) = rx^{r-1}$ for $x > 0$ by Prop.7-4.10. By Prop.7-4.14, f is continuous at 0 and f' has limit 0 at 0. By Prop.7-5.7, it now follows that $f'(0) = 0$; in other words, the formula $\frac{d}{dx}x^r = rx^{r-1}$ is valid for $x = 0$ as well.

If $1 < r = p/q$, where p and q are positive integers with q odd, then by Prop.7-3.3(d), the function $f(x) = x^r$ on $\mathbb{R}$ has derivative $f'(x) = rx^{r-1}$ for all $x \neq 0$. Since $|x^r| = |x|^r$, it follows by Prop.7-4.14 that f is continuous at 0 and f' has limit 0 at 0. Again, Prop.7-5.7 leads to the same conclusion as in the preceding paragraph.

(b) Consider the function

$$f(x) = \frac{x}{1+\sqrt{1-x^2}} - \sin^{-1} x\,.$$

It is a sum of two functions, both continuous on $[-1,1]$ but differentiable only on $(-1,1)$. Therefore it is possible to compute its derivative on $(-1,1)$ by the usual formulas, and one gets

$$\frac{1}{\sqrt{1-x^2}\,(1+\sqrt{1-x^2})}-\frac{1}{\sqrt{1-x^2}},$$

which simplifies to

$$\frac{1-(1+\sqrt{1-x^2})}{\sqrt{1-x^2}\left(1+\sqrt{1-x^2}\right)}=-\frac{1}{1+\sqrt{1-x^2}}.$$

This function has limit -1 at ±1. Since the computation of the derivative is valid only for $-1 < x < 1$, it does not reveal what the derivative is, if at all it exists, when $x = -1$ and when $x = 1$. However, Prop.7-5.7 shows that the derivative is indeed -1 at these points.

7-5.9. Proposition. *Let the function* $f:[a,b]\to\mathbb{R}$ *be continuous on* $[a,b]$ *and suppose its derivative* $f'(x)$ *exists for every* $x\in(a,b)$. *If* $f'(x) > 0$ *for all* $x\in(a,b)$, *then* f *is strictly increasing on any closed subinterval* $[x_1,x_2]$ *of* $[a,b]$, *and in particular*, *on* $[a,b]$. *And if* $f'(x) < 0$ *for every* (a,b), *then* f *is strictly decreasing on any closed subinterval* $[x_1,x_2]$ *of* $[a,b]$, *and in particular*, *on* $[a,b]$.

Proof. Using the Lagrange Mean Value Theorem on $[x_1,x_2]$, we get $f(x_2)-f(x_1) = (x_2-x_1)f'(c)$ for some $c\in(x_1,x_2)$. However, $f'(c) > 0$ by the hypothesis of the first part, and also $x_2-x_1 > 0$. It follows that $f(x_2)-f(x_1) > 0$. The second part has a similar proof. ☺

7-5.10. Examples. (a) Consider the function $f(x) = x^{-1}\tan x$ of Example 7-5.3(b) above. Its domain is $[1,a]$, where $1 < a < \frac{\pi}{2}$, and as noted before, the derivative is positive on $(1,a)$. By Prop.7-5.9, $f(a) > f(1)$.

(b) We shall show that $0 < a < x \le 1/e \Rightarrow a\ln a > x\ln x$. First of all, $a < u < x \Rightarrow 0 < u < 1/e \Rightarrow \ln u < \ln(1/e) = -1 \Rightarrow 1+\ln u < 0$. Therefore the function $f:[a,x]\to\mathbb{R}$ defined by $f(u) = u\ln u$ has the property that $f'(u) = 1+\ln u < 0$ when $u \in(a,x)$. By Prop.7-5.9, we have $f(a) > f(x)$, as claimed.

(c) Let $f(x) = \tan x - x$, $0 \le x < \frac{\pi}{2}$. Then $f'(u) = \sec^2 u - 1 > 0$ for $0 < u < \frac{\pi}{2}$. By Prop.7-5.9, applied on an interval $[0,x]$ with $0 < x < \frac{\pi}{2}$, we get $f(x) > f(0) = 0$, i.e. $\tan x > x$ for all $x\in(0,\frac{\pi}{2})$.

(d) The area of a minor (or semicircular) segment of a circle, with the length of its arc given to be c, can be shown to be $\frac{1}{2}r^2(x-\sin x)$, where r is the radius of the circle and x is the central angle, so that $r = c/x$ and $0 < x \le \pi$. This is because the associated sector has area $\frac{1}{2}r^2x$ and the triangle which must be removed from it in order to get the segment has area $(r\sin\frac{1}{2}x)(r\cos\frac{1}{2}x)$. It should be borne in mind that the radius r and the angle x are variables here and the length c is constant. As a function of x, twice the area is $f(x) = c^2(x - \sin x)/x^2$. A visual estimate suggests that $\lim_{x\to 0} f(x) = 0$. This can be confirmed by using the inequality $\left|\frac{\sin x}{x} - 1\right| \le x^2$ for $0 < |x| \le \sqrt{3}$ that was derived in the proof of Prop.7-3.8.Therefore f becomes continuous on $[0, \pi]$ if we set $f(0) = 0$. Suppose we wish to find its maximum value. The derivative is found to be

$$f'(x) = \frac{c^2}{x^3}(2\sin x - x\cos x - x) = \frac{c^2}{x^3}4\cos^2\frac{x}{2}\left(\tan\frac{x}{2} - \frac{x}{2}\right),$$

which is positive when $0 < x < \pi$, as seen in the preceding Example. By Prop.7-5.9, $f(x) > f(u)$ whenever $0 < u < x \le \pi$. Thus the maximum value of f occurs when $x = \pi$ and the segment will have greatest area when it is a semicircle.

(e) We shall show that $x < \frac{\pi}{2}\sin x$ for $0 < x < \frac{\pi}{2}$. Consider the function defined on $[0, \frac{\pi}{2}]$ by $f(x) = x - \frac{\pi}{2}\sin x$. It satisfies $f(0) = 0$ and its derivative is $f'(x) = 1 - \frac{\pi}{2}\cos x$. Now $f'(0) = 1 - \frac{\pi}{2} < 0$. If we could show that $f'(x) < 0$ for all $x \in (0, \frac{\pi}{2})$, we would be able to use Prop.7-5.9 and conclude immediately as in Example (b) that $0 = f(0) > f(x)$ whenever $0 < x < \frac{\pi}{2}$. However, it cannot be true that $f'(x) < 0$ for all $x \in (0, \frac{\pi}{2})$, because then it would follow that $f(0) > f\left(\frac{\pi}{2}\right)$, which is false in view of the fact that both are 0. The situation at hand calls for a more sophisticated argument, which we now present:

To begin with, we note that the conclusion of Prop.7-5.9 can be rephrased as saying that, if $f(a) \ge f(b)$, then $f'(c) \le 0$ for some $c \in (a, b)$; and if $f(a) \le f(b)$, then $f'(c) \ge 0$ for some $c \in (a, b)$.

Now, returning to the function $f(x) = x - \frac{\pi}{2}\sin x$, suppose if possible, that $f(c) \ge 0$ for some $c \in (0, \frac{\pi}{2})$. Applying Prop.7-5.9 to f on the intervals $[0, c]$ and $[c, \frac{\pi}{2}]$, we find that

$$f'(c_1) \ge 0 \text{ for some } c_1 \in (0, c)$$

and also that

$$f'(c_2) \le 0 \text{ for some } c_2 \in (c, \tfrac{\pi}{2}).$$

Therefore $c_1 < c_2$ and $f'(c_1) \ge f'(c_2)$. Applying the same Proposition once again, but this time to f' on $[c_1, c_2]$, we find that

$$f''(c_3) \le 0 \text{ for some } c_3 \in (c_1, c_2).$$

However, $f''(x) = \frac{\pi}{2}\sin x > 0$ on $(0, \frac{\pi}{2})$, a contradiction.

The argument given in Example 7-5.10(e) works for any function $f: [a, b] \to \mathbb{R}$ with $f''(x) > 0$ whenever $x \in (a, b)$. The inequality can also be proved by means of the result which associates positivity of f'' with what is called a local minimum. We shall now establish such a result.

7-5.11. Definition. *A function* $f: A \to \mathbb{R}$ *is said to have a* **local strict minimum** *at* $c \in A$ *if there exists some* $\delta > 0$ *such that* $(c-\delta, c+\delta) \subseteq A$ *and*

$$0 < |h| < \delta \;\Rightarrow\; f(c+h) > f(c).$$

It is said to have a **local strict maximum** *at* $c \in A$ *if there exists some* $\delta > 0$ *such that* $(c-\delta, c+\delta) \subseteq A$ *and*

$$0 < |h| < \delta \;\Rightarrow\; f(c+h) < f(c).$$

Remarks. (i) If the final inequality in the condition above is replaced by $f(c+h) \geq f(c)$ [resp. $f(c+h) \leq f(c)$], we speak of a *local minimum* [resp. *local maximum*], without the word "strict".

(ii) When the domain is an interval, a local strict maximum or minimum, as defined above, cannot occur at an endpoint of that interval. However, some authors may define it differently, so as to include the possibility of a local strict maximum or minimum at an endpoint.

(iii) When a function has either a maximum [resp. minimum] value or a local maximum [resp. minimum] at some point of its domain, it cannot also have a local strict minimum [resp. maximum] at that point; it may however have a nonstrict local minimum [resp. maximum], as for example a constant function does.

7-5.12. Proposition. *Suppose* c *is a point of the set* $A \subseteq \mathbb{R}$ *and there exists* $\delta > 0$ *such that* $(c-\delta, c+\delta) \subseteq A$. *If a function* $f: A \to \mathbb{R}$ *satisfies* $f'(c) = 0$ *and* $f''(c) > 0$ [*resp.* $f''(c) < 0$], *and if* f' *exists on* $(c-\delta, c+\delta)$, *then* f *has a local strict minimum* [*resp. maximum*] *at* c.

Proof. Suppose $f''(c) > 0$. By Lemma 7-5.1 applied to f', there exists $\delta' > 0$ such that

$$0 < h < \delta',\ c-h \in A \;\Rightarrow\; f'(c-h) < f'(c)$$

and

$$0 < h < \delta',\ c+h \in A \;\Rightarrow\; f'(c+h) > f'(c).$$

It follows that for $\delta_1 = \frac{1}{2}\min\{\delta, \delta'\}$, we have

$$c-\delta_1 < x < c \Rightarrow f'(x) < f'(c) = 0 \quad \text{and} \quad c < x < c+\delta_1 \Rightarrow f'(x) > f'(c) = 0.$$

Besides, $[c-\delta_1, c+\delta_1] \subseteq A$, so that f and f' are defined on $[c-\delta_1, c+\delta_1]$. Therefore we may apply Prop.7-5.9 to f on any interval $[c-h, c]$ as well as on any interval $[c, c+h]$, as long as $0 < h < \delta_1$. Upon doing so, we find that

$$0 < h < \delta_1 \;\Rightarrow\; f(c-h) > f(c)$$

and

$$0 < h < \delta_1 \;\Rightarrow\; f(c) < f(c+h).$$

Therefore

$$[c-\delta_1, c+\delta_1] \subseteq A \text{ and } 0 < |h| < \delta_1 \Rightarrow f(c+h) > f(c).$$

The existence of such a positive number δ_1 means that f has a local strict minimum at c.

The case when $f''(c) < 0$ can either be argued analogously or obtained by applying the preceding case to $-f$. ☺

We return to Example 7-5.10(e) and show how to obtain the same inequality by using Prop.7-5.12 instead of Prop.7-5.9. By the Extreme Value Theorem, the function $f(x) = x - \frac{\pi}{2}\sin x$ must have a maximum value at some point c in the domain $[0, \frac{\pi}{2}]$. If $c \in (0, \frac{\pi}{2})$, then $f'(c) = 0$ by Prop.7-5.2. Now, $f''(x) = \frac{\pi}{2}\sin x > 0$ $\forall\ x \in (0, \frac{\pi}{2})$, so that $f''(c) > 0$. By Prop.7-5.12, it follows that f has a local strict minimum at c. However this contradicts the fact that f has a maximum value at c. Therefore the maximum value cannot be taken at any point of $(0, \frac{\pi}{2})$, so that $x \in (0, \frac{\pi}{2}) \Rightarrow f(x) < f(0) = 0$.

Problem Set 7-5

7-5.P1. Show that the equation $2\sin x - x = \sqrt{3} - \frac{4}{3}$ has precisely one solution between 0 and $\frac{\pi}{2}$.

7-5.P2. For the function $f(x) = \ln(1+\sqrt{1-x}) - \sqrt{1-x}$ defined for $x \le 1$, show that $f'(1) = \frac{1}{2}$, although the two terms $\ln(1+\sqrt{1-x})$ and $\sqrt{1-x}$ both represent functions that are not differentiable at 1.

7-5.P3. (a) Show that, if $0 < a < b$, then

$$\frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}.$$

(b) Show that, if $0 < a < b \le 1$, then

$$b\tan^{-1} b - a\tan^{-1} a > (b-a)\left(\tan^{-1} a + \frac{a}{1+a^2}\right).$$

What modification of the second factor on the right side is required if $1 \le a < b$?

7-5.P4. (a) Let the continuous function $f: [a, b] \to \mathbb{R}$ satisfy $f(a) = f(b) = 0$ and $f''(x) > 0$ whenever $x \in (a, b)$. Show that $f(x) < 0$ for all $x \in (a, b)$.

(b) Let the continuous function $f: [a, b] \to \mathbb{R}$ satisfy $f''(x) > 0$ whenever $x \in (a, b)$. Show that $f(x) < \max\ \{f(a), f(b)\}$ for all $x \in (a, b)$.

7-5.P5. (a) Show that $\tan^{-1} x - \frac{\pi}{4}x > 0$ for $0 < x < 1$.

(b) Show that $xe^x < \frac{1}{2}(e + \frac{1}{e})x + \frac{1}{2}(e - \frac{1}{e})$ when $-1 < x < 1$.

7-5.P6. Use the Lagrange Mean Value Theorem and appropriate properties of trigonometric functions to show that, for any $x > 0$, there exists $c \in (0,x)$ such that $x - \sin x = 2x\sin^2 \frac{c}{2}$. Hence show that the right limit of $(x - \sin x)/x^2$ at 0 is 0.

7-5.P7. Show that $2\sin x - x\cos x - x > 0$ for $0 < x < \pi$, without expressing it in terms of $\frac{x}{2}$, and without depending on Example (c) of 7-5.10. Then formulate a general result concerning a continuous function $f:[a,b]\to\mathbb{R}$, from which the preceding inequality can be derived instantly. It may be assumed that f'' exists on $[a,b]$.

7-5.P8. (a) For the function $f:[a,b]\to\mathbb{R}$ defined by $f(x) = x^3$, show that, when $a = -b$, there are two numbers $c \in [a,b]$ such that $f'(c) = \dfrac{f(b)-f(a)}{b-a}$.

(b) Give an example of a function $f:[a,b]\to\mathbb{R}$ for which there are infinitely many numbers $c \in [a,b]$ such that $f'(c) = \dfrac{f(b)-f(a)}{b-a}$.

7-5.P9. Let $f:[-1,1]\to\mathbb{R}$ be defined by $f(x) = 2x + x^2 + \sin^{-1}x$. Show that f is a strictly increasing function with range $[a,b]$, where $a = f(-1)$ and $b = f(1)$, and that the inverse $f^{-1}:[a,b]\to[-1,1]$ is differentiable with derivative 0 at a as well as at b. [The reader will surely agree that it is futile to try computing f^{-1} explicitly.]

7-5.P10. Prove the following theorem of Darboux to the effect that a derivative has the intermediate value property (regardless of whether it is continuous): If $f:[a,b]\to\mathbb{R}$ is differentiable at each point of $[a,b]$ and $f'(a) < 0 < f'(b)$, then there exists some $c \in (a,b)$ such that $f'(c) = 0$.

7-5.P11. Prove the following corollary to the Lagrange Mean Value Theorem: Let the functions $f:[a,b]\to\mathbb{R}$ and $g:[a,b]\to\mathbb{R}$ both be continuous on $[a,b]$ and differentiable on (a,b). Suppose that $g'(c) \neq 0$ for all $c \in (a,b)$. Then $g(b) \neq g(a)$ and there exists $c \in (a,b)$ such that $\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$.

7-5.P12. (a) If $p > \frac{1}{e}$, show that $\ln x < x^p$ for every $x > 0$.

(b) If $p = \frac{1}{e}$, show that $\ln x \le x^p$ whenever $x > 0$ and that $\ln x = x^p \Leftrightarrow x = e^e$.

The following table of values is relevant here:—

x	12	13	14	15	16	17
$\ln x$	2.48491	2.56494	2.63906	2.70805	2.77259	2.83321
$x^{1/e}$	2.49464	2.56919	2.64020	2.70807	2.77313	2.83568

(c) If $0 < p < \frac{1}{e}$, show that there exists $x > 0$ such that $\ln x = x^p$.

***7-5.P13.** Prove the following sharpening of 2-4.P11: If $a > 1$, then $a^x - a^y < a^{x+1/e}(x-y)$ for any real numbers x, y with $x > y$.

7-5.P14. Suppose the continuous function $f:[0,1]\to\mathbb{R}$ is differentiable at 0 with $f'(0) > 0$ and let $f(0) = f(1) = 0$. Show that the equation $f(x) = x^2$ has a solution in (0,1).

7-5.P15. Show that $x^2e^{-x} \le 4/e^2$ for $x \ge 0$. [This has relevance to 5-1.P11.] Also, show that the equation $x^2e^{-x} = a$ has three real solutions when $0 < a < 4/e^2$, two of them being positive.

7-5.P16. If u, v, w are positive numbers and $\max\{u, v, w\} = u$, show that there exists a unique positive x such that $f(x) = x^3 - (u^2 + v^2 + w^2)x - 2uvw = 0$ and that this number x satisfies $u < x \le 2u$. When is $x = 2u$?

7-5.P17. Let p and q be positive numbers such that $1/p + 1/q = 1$. Show that

$$ab \le {a^p}/{p} + {b^q}/{q} \quad \text{for any nonnegative } a \text{ and } b.$$

7-5.P18. If $c_0, c_1, \ldots, c_n$ are real numbers such that

$$c_0 + \frac{c_1}{2} + \cdots + \frac{c_{n-1}}{n} + \frac{c_n}{n+1} = 0,$$

show that the equation

$$c_0 + c_1x + c_2x^2 + \cdots + c_nx^n = 0$$

has at least one real root between 0 and 1.

7-5.P19. Let $x > 0$. Show that $\ln(x+1) - \ln(x) = \frac{1}{t}$ for some $t \in (x, x+1)$.

7-5.P20. If f is defined on $[a, b]$ and $|f(x) - f(y)| \le A|x - y|^2$ for all $x, y \in [a, b]$, where A is a constant, show that f is constant on $[a, b]$.

7-5.P21. Suppose

(i) f is continuous for $x \ge 0$;

(ii) f' exists for $x > 0$;

(iii) $f(0) = 0$ and

(iv) f' is monotonically increasing for $x > 0$.

If $g(x) = f(x)/x$ for $x > 0$, show that g is monotonically increasing for $x > 0$.

7-5.P22. Suppose f' is continuous on $[a, b]$ and $\varepsilon > 0$. Prove that there exists $\delta > 0$ such that

$$\left|\frac{f(t) - f(x)}{t - x} - f'(x)\right| < \varepsilon \quad \text{whenever} \quad 0 < |t - x| < \delta \quad \text{and} \quad t, x \in [a, b].$$

7-5.P23. Without employing Rolle's Theorem, prove that if $P(x)$ is a polynomial, then between any two roots of $P(x) = 0$ lies a root of $P'(x) = 0$.

7-5.P24. If $P(x)$ is a polynomial, prove that there is a root of $P'(x)+kP(x) = 0$ between any two consecutive roots of $P(x) = 0$.

7-5.P25. A function f is said to satisfy a **Lipschitz condition** on an interval I, if there exists $M > 0$ such that $|f(x_2)-f(x_1)| \leq M|x_2-x_1|$ for all $x_1, x_2 \in I$. Let f be a function on an interval I such that f' is defined and bounded on I. Show that f satisfies a Lipschitz condition on I. Prove conversely that, if f satisfies a Lipschitz condition and f' is defined on I then f' is bounded on I.

7-5.P26. If f, g, h are continuous on $[a,b]$ and differentiable on (a,b), then show that there is a point $x_0 \in (a,b)$ such that

$$\begin{vmatrix} f'(x_0) & g'(x_0) & h'(x_0) \\ f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \end{vmatrix} = 0.$$

7-5.P27. Let f be an infinitely differentiable function from $\mathbb{R}$ to $\mathbb{R}$ and suppose that, for some positive integer n,

$$f(1) = f(0) = f'(0) = \cdots = f^{(n)}(0) = 0.$$

Prove that $f^{(n+1)}(x) = 0$ for some $x \in (0,1)$.

7-5.P28. Suppose $f:[0,1]\to\mathbb{R}$ is continuous with $f(0) = 0$ and let f be differentiable on $(0,1)$ with $0 \leq f'(x) \leq Mf(x)$ for some $M > 0$. Prove that f is identically zero.

7-5.P29. Suppose $f:\mathbb{R}\to\mathbb{R}$ is differentiable, $f'(x) > f(x)$ everywhere and $f(x_0) = 0$. Prove that $f(x) > 0$ for $x > x_0$.

7-6 Taylor's Theorem

The most convenient numbers to work with in practice are rational numbers. It is for this reason that one seeks rational approximations to numbers like √2 and e. Among functions, the most convenient ones to work with in practice are polynomials, because the representation as $f(x) = a_0 + \cdots + a_n x^n$ provides an explicit procedure for computing the value $f(x)$ in finitely many steps when x is known. This is in contrast to such functions as $\tan^{-1}x$ or $\ln(1+x)$. It is therefore natural to seek polynomial approximations to functions.

Recall that a rational approximation to a number x is not one single rational number r unless the level of accuracy and the form of the number (e.g. number of decimal digits) are specified. When the level of accuracy is specified, say as 10^{-3}, a rational approximation would be any rational number r for which $|x-r| < 10^{-3}$. Since the possibility of sharper and sharper levels of accuracy being specified has to be kept open, what we actually mean by "rational approximation" is a *sequence* r_n of rational numbers converging to x. Similarly, by a "polynomial approximation" to a function, we mean a *sequence* of polynomials converging to that function, where the convergence may be pointwise or uniform. In this Section, we shall be concerned primarily with uniform convergence.

By Prop.5-1.3, a uniform limit of a sequence of polynomials has to be continuous. So, it would make no sense to seek a polynomial approximation unless the function is continuous in the first place. The Stone-Weierstrass Approximation Theorem [see 6-3.6] guarantees that, on a closed and bounded interval, any continuous function does indeed have a uniform polynomial approximation. So, looking for the approximation is no wild goose chase. How we go about obtaining it depends on what we know about the function [see Example 7-6.2(a) below].

We can take a cue from the following property of all polynomial functions: Let f be a polynomial function given by $f(x) = a_0 + a_1x + \cdots + a_nx^n$, where $a_0, \ldots, a_n$ are real numbers. Consider it on any interval containing 0. Then it is easy to check that $f(0) = a_0$, $f'(0) = a_1$, $f''(0) = 2a_2$, and so forth. Using the symbol $f^{(k)}$ to denote the k^{th} derivative as usual, we have $f^{(k)}(0) = k!a_k$. This can be rewritten as $a_k = f^{(k)}(0)/k!$, so that we have

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!}x^n .$$

Since the summation on the right side here is a polynomial, the equality can only be expected to hold when f is itself as polynomial. Nevertheless, one can ask what relation the polynomial on the right side has to the function f when $f^{(k)}$ exists for all $k \in \mathbb{N}$. More precisely, when (if not always) does the sequence of polynomials represented by the summation, i.e. the sequence of partial sums of the series $\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}x^k$ converge to $f(x)$? Of course, this is the same as asking when the sum of the series is $f(x)$. A satisfactory answer is available in some special instances.

Before proceeding further, it would be useful to note a simple variation that removes the special position that 0 seems to have in the considerations in the preceding paragraph. To this end, consider any real a and introduce $u = x-a$. Then $f(x) = f(a+u) = g(u)$, say, and $g^{(k)}(u) = f^{(k)}(x)$ for all k. Since g is also polynomial with as many "terms" as f, the arguments of the preceding paragraph lead to

$$g(u) = \sum_{k=0}^{n} \frac{g^{(k)}(0)}{k!} u^k, \quad \text{or} \quad f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

The latter equality says that, for any $a \in \mathbb{R}$, the function which maps $x \in \mathbb{R}$ into the summation on the right is the same as f. Since there is no restriction intended either on x or on a, the latter equality could equally well have been written in the form

$$f(b) = \sum_{k=0}^{n} \frac{f^{(k)}(x)}{k!} (b-x)^k \quad \text{for all } b, x \in \mathbb{R}.$$

Stated in this form, the equality says that, for each $b \in \mathbb{R}$, the function which maps $x \in \mathbb{R}$ into the summation on the right is the constant function $f(b)$. Such a switch in notation will prove useful below.

7-6.1. Taylor's Theorem. *Suppose the function* $f: [a, b] \to \mathbb{R}$ *satisfies the following conditions*:

(i) *the derivatives* $f', f'', \dots, f^{(n-1)}$ *exist on* $[a, b]$;

(ii) $f^{(n-1)}$ *is continuous on* $[a, b]$ *and its derivative* $f^{(n)}$ *exists on* (a, b).

Then (I) *there exists some* $c \in (a, b)$ *such that*

$$f(b) = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!} (b-a)^k + \frac{f^{(n)}(c)}{n!} (b-a)^n,$$

and (II) *there also exists some* $c_1 \in (a, b)$ *such that*

$$f(b) = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!} (b-a)^k + \frac{f^{(n)}(c_1)}{(n-1)!} (b-c_1)^{n-1}(b-a).$$

Proof. First we shall show by induction that, for any $g: (a, b) \to \mathbb{R}$ such that $g^{(n)}$ exists at each point of (a, b), the function $G_n: (a, b) \to \mathbb{R}$ defined by

$$G_n(x) = \sum_{k=0}^{n-1} \frac{g^{(k)}(x)}{k!} (b-x)^k$$

has derivative

$$G_n'(x) = \frac{g^{(n)}(x)}{(n-1)!} (b-x)^{n-1} \text{ on } (a, b). \tag{1}$$

For $n = 1$, this states that, if g' exists on (a, b), then the derivative of $G_1(x) = g(x)$ is $G_1'(x) = \frac{g'(x)}{0!}(b-x)^0$, which is true. Assuming that (1) holds for some $n \in \mathbb{N}$

(induction hypothesis), consider any $g:(a,b)\to\mathbb{R}$ such that $g^{(n+1)}$ exists at each point of (a,b) and let

$$G_{n+1}(x)=\sum_{k=0}^{n}\frac{g^{(k)}(x)}{k!}(b-x)^k=G_n(x)+\frac{g^{(n)}(x)}{n!}(b-x)^n.$$

Then
$$G_{n+1}'(x)=G_n'(x)-\frac{g^{(n)}(x)}{(n-1)!}(b-x)^{n-1}+\frac{g^{(n+1)}(x)}{n!}(b-x)^n$$
$$=\frac{g^{(n+1)}(x)}{n!}(b-x)^n \text{ by the induction hypothesis.}$$

This completes the proof of (1) by induction.

To prove (I), set

$$F(x)=\sum_{k=0}^{n-1}\frac{f^{(k)}(x)}{k!}(b-x)^k+A\frac{(b-x)^n}{n!},$$

where A is a constant such that $F(b)=F(a)$. Since $F(b)=f(b)$, this is equivalent to saying that

$$f(b)=\sum_{k=0}^{n-1}\frac{f^{(k)}(a)}{k!}(b-a)^k+A\frac{(b-a)^n}{n!}. \tag{2}$$

Since $b-a\neq 0$, such a number A exists. By what we have proved above and by hypothesis (i), F' exists and

$$F'(x)=\frac{f^{(n)}(x)}{(n-1)!}(b-x)^{n-1}-A\frac{(b-x)^{n-1}}{(n-1)!}$$
$$=(f^{(n)}(x)-A)\frac{(b-x)^{n-1}}{(n-1)!}.$$

By hypothesis (ii) and the fact that $F(b)=F(a)$, Rolle's Theorem is applicable. Therefore there exists some $c\in(a,b)$ such that $F'(c)=0$, i.e.

$$(f^{(n)}(c)-A)\frac{(b-c)^{n-1}}{(n-1)!}=0.$$

Since $c\in(a,b)$, we have $b-c\neq 0$ and hence $f^{(n)}(c)-A=0$, i.e. $A=f^{(n)}(c)$. Upon substituting this in (2), we get the required equality in (I).

The proof of (II) is similar to the one above; F is taken as

$$F(x)=\sum_{k=0}^{n-1}\frac{f^{(k)}(x)}{k!}(b-x)^k+A(b-x).$$

Further details are left to the reader. ☺

The two expressions for $f(b) - \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!}(b-a)^k$ are known as the **Lagrange** and **Cauchy** forms of the **remainder** respectively. The series $\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(b-a)^k$ is called the **Taylor series of** f at a whether it converges or not. Of course, f cannot have a Taylor series at a unless the derivatives $f^{(n)}(a)$ all exist for $n = 0, 1, 2, \ldots$. The finite sum $\sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!}(b-a)$ is called the **Taylor expansion** to n terms.

There is an obvious version in which the roles of a and b are interchanged. We shall not state it explicitly, but shall use it freely as and when needed.

When a function f satisfies the hypotheses of Taylor's Theorem on some interval $[a, b]$, it also satisfies them on every subinterval $[a, x]$, where $x \in (a, b]$. Therefore, for every $x \in (a, b]$, there exists $c \in (a, x)$ such that

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n)}(c)}{n!}(x-a)^n,$$

and there also exists $c_1 \in (a, x)$ such that

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n)}(c_1)}{(n-1)!}(x-c_1)^{n-1}(x-a).$$

In this context, when one may wish to consider various values of x, the Taylor expansion to n terms represents a function of x, which is a polynomial function. This polynomial function is called the **Taylor polynomial** at a to n terms.

When $a = 0$, the equalities displayed above become

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(0)}{k!} x^k + \frac{f^{(n)}(c)}{n!} x^n, \quad \text{where } 0 < c < x$$

and

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(0)}{k!} x^k + \frac{f^{(n)}(c_1)}{(n-1)!}(x-c_1)^{n-1} x, \quad \text{where } 0 < c_1 < x.$$

This assumes $x > 0$. In case $x < 0$, one can consider $f(-x)$ and obtain the same equalities with $x < c < 0$ and $x < c_1 < 0$. These versions of the Taylor series [resp. expansion] are known as **Maclaurin series** [resp. **expansion**].

If we can show that the remainder in either version, as a function on $[a, b]$, tends to 0 uniformly as $n \to \infty$, the polynomials in the summation will have been shown to converge to f uniformly. Unfortunately, this is rarely feasible in practice and we shall consider other methods of obtaining polynomial approximations later. However, the general result above can be put to other uses, as we shall illustrate later on in the next Chapter.

7-6.2. Examples. (a) Each of the functions sin, cos and exp is the sum of a power series converging on all of $\mathbb{C}$ [see Definition 5-3.2 and Theorem 5-3.9]. It follows by Theorem 5-2.7(b) that they converge uniformly on any interval $[a,b]$. Since each partial sum of a power series is a polynomial, the sequence of partial sums of the defining series already provides a sequence of polynomials converging to the function uniformly. Thus we have approximating polynomials for sin, cos and exp independently of Taylor's Theorem.

It is left to the reader to verify for each of these functions that the n^{th} partial sum agrees with $\sum_{k=0}^{n-1} \frac{f^{(k)}(0)}{k!} x^k$. We also note that, if the functions are defined in some other manner and the formulas for differentiating them derived therefrom, then one can indeed *prove* that the remainder tends to 0 and *hence derive the fact* that the power series of our Def.5-3.2 and Theorem 5-3.9 converge to the respective functions.

(b) Recall from Example 3-2.11 that when $|a| < 1$, we have

$$\lim_{n\to\infty} \binom{p}{n} a^n = 0 \qquad \text{whenever } p \in \mathbb{R}, \tag{1}$$

where $\binom{p}{n}$ is defined inductively by setting

$$\binom{p}{0} = 1 \qquad \text{and} \qquad \binom{p}{n+1} = \binom{p}{n}\frac{p-n}{n+1}.$$

We shall use this to show that the remainder of the Maclaurin series for the function $f(x) = (1+x)^p$ has limit 0 if $-1 < x < 1$. It is a straightforward computation that $\frac{f^{(k)}(x)}{k!} = \binom{p}{k}(1+x)^{p-k}$. So, the Maclaurin expansion to n terms is

$$\sum_{k=0}^{n-1} \binom{p}{k} x^k$$

and the remainder in Cauchy form (with c_1 denoted by c) is

$$p\binom{p-1}{n-1}(1+c)^{p-n}(x-c)^{n-1}x,$$

which can be put into the form

$$xp(1+c)^{p-1}\binom{p-1}{n-1}\left(\frac{x-c}{1+c}\right)^{n-1}. \tag{2}$$

We now argue that

$$\left|\frac{x-c}{1+c}\right| < |x| \quad \text{for } 0 < c < x \quad \text{as well as} \quad \text{for } -1 < x < c < 0.$$

When $0 < c < x$, we have $\left|\frac{x-c}{1+c}\right| = \frac{x-c}{1+c} < x - c < x = |x|$. Suppose $-1 < x < c < 0$. Since $-1 < x$ and $c < 0$, we have $-c > cx$, so that $c < -cx$ and hence $c - x < -cx - x = -x(1+c)$. Since $1 + c > 0$, it follows that $\frac{c-x}{1+c} < -x$. But $\left|\frac{x-c}{1+c}\right| = \frac{c-x}{1+c}$ and $|x| = -x$.

In view of (2) and the inequality just proved, the absolute value of the remainder does not exceed

$$|xp|\,(1+c)^{p-1}\left|\binom{p-1}{n-1}\right||x|^{n-1}.$$

It follows that, if $0 < a < 1$, then for all $x \in [-a, a]$ the absolute value of the remainder does not exceed

$$|ap|\,(1+c)^{p-1}\left|\binom{p-1}{n-1}\right||a|^{n-1}.$$

Since $0 < 1-|a| < 1+c < 1+|a|$, it further follows that the remainder does not exceed

$$ap\,(1-|a|)^{p-1}\left|\binom{p-1}{n-1}\right||a|^{n-1} \quad \text{or} \quad ap\,(1+|a|)^{p-1}\left|\binom{p-1}{n-1}\right||a|^{n-1}$$

according as $p-1 < 0$ or $p-1 \geq 0$. Together with (1), this shows that the remainder tends to 0 uniformly on $[-a, a]$ as $n \to \infty$. Therefore

$$(1+x)^p = \sum_{k=0}^{\infty}\binom{p}{n}x^k \qquad \text{for every } x \in (-1, 1),$$

the convergence of the Maclaurin series being uniform on any interval $[-a, a]$, where $0 < a < 1$. Thus the partial sums of this series provide a sequence of polynomials converging uniformly to $(1+x)^p$ on $[-a, a]$ whenever $0 < a < 1$. The Maclaurin series is known as th **binomial series**.

(c) The function f given by $f(0) = 0$ and $f(x) = \exp(-1/x^2)$ for $x \neq 0$ provides a sharp contrast to Example (b). Every term in its Maclaurin expansion is 0; consequently the remainder, far from having limit 0, is actually equal to the function for every n.

To see why, recall from 7-4.P3 that the derivative of any function g with $g(0) = 0$ and $g(x) = P(1/x)f(x)$ for $x \neq 0$, where P is a polynomial function, is again of the same form. It follows by induction that any such function g satisfies $g^{(k)}(0) = 0$ for every $k \in \mathbb{N}$. Since f itself is of this form with $P(t) = 1$ everywhere, it follows that $f^{(k)}(0) = 0$ for all $k \in \mathbb{N}$.

This example illustrates that, although a polynomial approximation must exist according to the famous theorem of Weierstrass, it cannot always be obtained from the Maclaurin expansion.

We note in passing that the function just discussed plays a crucial role elsewhere in Mathematics, precisely because of its unusual property that $f(x) > 0$ when $x \neq 0$, while at the same time, $f^{(k)}$ exists everywhere and $f^{(k)}(0) = 0$ for every $k \in \mathbb{N}$.

Problem Set 7-6

7-6.P1. (a) By computing derivatives, find the Taylor polynomial of f at 3 when (i) $f(x) = x$ to 2 terms and (ii) $f(x) = x^2 + 4x + 1$ to 3 terms. Then use elementary algebraic computations to "simplify" the Taylor polynomials to obtain $f(x)$.
(b) By using only elementary algebraic computations (no differentiation), obtain the Taylor polynomial of $f(x) = x^2$ at 5 to 3 terms.

7-6.P2. The results here will be used in proving the irrationality of π later.

(a) If $f: \mathbb{R} \to \mathbb{R}$ has the property that for all $k \geq 0$, $f^{(k)}$ exists everywhere and $f^{(k)}(0)$ is an integer, show that the function $g: \mathbb{R} \to \mathbb{R}$ defined by $g(x) = f(1-x)$ has the property that for all $k \geq 0$, $g^{(k)}$ exists everywhere and $g^{(k)}(1)$ is an integer.

(b) If $f(x) = \frac{x^n(1-x)^n}{n!}$ on $\mathbb{R}$, show that $f^{(k)}(0)$ and $f^{(k)}(1)$ are integers for every $k \geq 0$.

7-6.P3. (a) Show that the number c in the Lagrange remainder need not be unique.

(b) If $f^{(n+1)}$ is continuous on $[a,b]$ and $f^{(n+1)}(a) \neq 0$, show that, for some $\delta > 0$, the number c in the Lagrange remainder $\frac{f^{(n)}(c)}{n!}(x-a)^n$ after n terms is unique whenever $a < x \leq a + \delta$.

7-6.P4. (a) Let $f: [a,b] \to \mathbb{R}$ satisfy the condition that $f(a) = 0$, $f'(a) = 0, \ldots,$ $f^{(n-1)}(a) = 0$ and $f^{(n)}(a) > 0$. Show that there exists $\delta > 0$ such that $a < x < a + \delta \Rightarrow$ $f(x) > 0$.

(b) **Young's form of Taylor's Theorem.** Suppose the function $f: [a,b] \to \mathbb{R}$ satisfies the following conditions:

(i) the derivatives $f', f'', \ldots, f^{(n-1)}$ exist on $[a,b]$;
(ii) $f^{(n)}$ exists at a.

For any $\varepsilon > 0$, show that there exists $\delta > 0$ such that

$$\left| f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k \right| < \varepsilon \frac{(x-a)^n}{n!} \text{ whenever } a < x < a + \delta.$$

7-6.P5. Let $f^{(n+1)}$ be continuous on $[a, b]$ and $f^{(n+1)}(a) \neq 0$. Show that $\exists\ \delta > 0$ such that, whenever $a < x < a + \delta$, the number 'c' in the Lagrange remainder $\dfrac{f^{(n)}(c)}{n!}(x-a)^n$ after n terms is a function of x, and that $\lim\limits_{x \to a} \dfrac{c-a}{x-a} = \dfrac{1}{n+1}$.

7-6.P6. For a polynomial function $f(x) = a_0 + a_1 x + \cdots + a_n x^n$, prove by induction on k that $a_k = f^{(k)}(0)/k!$, where a_k means 0 when $k > n$.

7-6.P7. Let f be n times differentiable on an open interval I which contains the point ξ and suppose that $f^{(n)}$ is continuous at ξ. Assume that

$$f'(\xi) = f''(\xi) = \cdots = f^{(n-1)}(\xi) = 0 \text{ but } f^{(n)}(\xi) \neq 0.$$

Discuss the behaviour of f close to ξ under the following conditions:

(i) $f^{(n)}(\xi) > 0$ and n is even,

(ii) $f^{(n)}(\xi) > 0$ and n is odd,

(iii) $f^{(n)}(\xi) < 0$ and n is even,

(iv) $f^{(n)}(\xi) < 0$ and n is odd.

Chapter 8

*More About Differentiation

8-1 Applications of Taylor's Theorem

Taylor's Theorem has applications beyond the search for polynomial approximations. We begin with a simple illustration. Let f' be continuous on $[0, 1]$ and f'' exist on $(0, 1)$. If $f'(0) = f'(1) = f(0) = 0$ and $f(1) = 1$, it can be shown that $|f''(c)| \geq 4$ for some $c \in (0, 1)$.

The hypotheses of the Theorem hold with $n = 2$ on the interval $[0, \frac{1}{2}]$ as well as on $[\frac{1}{2}, 1]$. Accordingly, we have

$$f(\tfrac{1}{2}) = f''(c_1)\,(\tfrac{1}{2})^2(\tfrac{1}{2!}), \text{ where } 0 < c_1 < \tfrac{1}{2}$$

and

$$f(\tfrac{1}{2}) = 1 + f''(c_2)(\tfrac{1}{2})^2(\tfrac{1}{2!}), \text{ where } \tfrac{1}{2} < c_2 < 1.$$

It follows that $f''(c_1) - f''(c_2) = 8$ and hence either $|f''(c_1)| \geq 4$ or $|f''(c_2)| \geq 4$.

For readers inclined towards Mechanics, we note that the above example can be rephrased thus: If a particle moving on a straight line covers unit distance in unit time, starting and ending at rest, then its absolute acceleration must exceed or equal 4 at some instant.

Upon substituting $h = x - a$, the Taylor polynomial up to n terms for a function f can be expressed as $\sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!} h^k$. Using θ to represent the number $(c - a)/(b - a)$, which lies between 0 and 1, the Lagrange form of the remainder can be expressed as $\frac{f^{(n)}(a + \theta h)}{n!} h^n$. Using this notation, the final conclusion of Taylor's Theorem with Lagrange's form of the remainder can be written as

$$f(a+h) = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!} h^k + \frac{f^{(n)}(a+\theta h)}{n!} h^n, \quad \text{where } 0 < \theta < 1.$$

Similarly, when the remainder is required in the Cauchy form, one can write

$$f(a+h) = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!} h^k + \frac{f^{(n)}(a+\theta h)}{(n-1)!} (1-\theta)^{n-1} h^n, \quad \text{where } 0 < \theta < 1.$$

The result of Prop.7-5.12 is known in Calculus as the "second derivative test" for local extrema. The test is inapplicable when the second derivative $f''(c)$ is 0 but by using higher derivatives one can sometimes obtain a definite conclusion one way or the other. This is made possible by the following extension of the result.

8-1.1. Proposition. *Suppose c is a point of the set $A \subseteq \mathbb{R}$ and there exists $\delta > 0$ such that $(c-\delta, c+\delta) \subseteq A$. Let $f: A \to \mathbb{R}$ be a function such that the derivatives $f', f'', \dots, f^{(n)}$ exist and are continuous on $(c-\delta, c+\delta)$ and $f(c) = 0, f'(c) = 0, \dots, f^{(n-1)}(c) = 0$ but $f^{(n)}(c) > 0$ [resp. $f^{(n)}(c) < 0$]. Then*

(a) *if n is even, f has a local strict minimum [resp. maximum] at c;*

(b) *if n isodd, f has neither a local minimum nor a local maximum at c.*

Proof. For any positive $\delta_1 < \delta$, the conditions of Taylor's Theorem hold on $[c-\delta_1, c+\delta_1]$. Therefore, for $0 < h < \delta_1$, we have

$$f(c+h) = f(c) + \frac{h^n}{n!} f^{(n)}(c+\theta h), \quad \text{where } 0 < \theta < 1. \tag{1}$$

Suppose first that $f^{(n)}(c) > 0$. By continuity of $f^{(n)}$ at c, there exists some $\delta_2 > 0$ such that $f^{(n)}(c+h_1) > 0$ for $0 < |h_1| < \delta_2$. Therefore $\frac{h^n}{n!} f^{(n)}(c+\theta h) > 0$ when $0 < h < \delta_2$. If n is even, then the same inequality holds when $-\delta_2 < h < 0$ and it follows from (1) in this case that $f(c+h) > f(c)$ when $0 < |h| < \delta_2$, so that f has a local strict minimum at c. But if n is odd, then the reverse inequality holds when $-\delta_2 < h < 0$, so that f has neither a local minimum nor a local maximum at c. This proves (a) and (b) when $f^{(n)}(c) > 0$. A similar argument proves (a) and (b) when $f^{(n)}(c) < 0$. ☺

Remark. The above argument uses the continuity of $f^{(n)}$ only at c. The special case with $n = 2$ was proved in Prop.7-5.12 assuming only the existence of f'' at c but not its continuity. It is possible to drop the assumption of continuity of $f^{(n)}$ at c [see 8-1.P3].

Examples. (a) The function $f: \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^4$ satisfies $f^{(k)}(0) = 0$ for $k = 1, 2, 3$, but $f^{(k)}(0) = 24 > 0$ when $k = 4$, an even integer. By Prop.8-1.1, f must have a local strict minimum at 0, which is easy to verify independently. The function $f: \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^3$ satisfies $f^{(k)}(0) = 0$ for $k = 1, 2$, but $f^{(k)}(0) = 6 > 0$ when $k = 3$, an odd integer. By Prop.8-1.1, f has neither a local minimum nor a local maximum at 0, which is also easy to verify independently.

(b) Let $f: \mathbb{R} \to \mathbb{R}$ given by $f(0) = 0$ and $f(x) = \exp(-1/x^2)$ for $x \neq 0$. As noted in Example (c) of 7-6.2, this function satisfies $f^{(k)}(0) = 0$ for all $k \in \mathbb{N}$. Therefore Prop.8-1.1 is not applicable. Nevertheless one can verify directly that f has a strict local minimum at 0.

Now we apply Taylor's Theorem to discuss the convergence of Newton's Method for an approximate solution of an equation $f(x) = 0$. It may be recalled that this method consists in generating the sequence $\{x_n\}$, where x_1 may be any number (in principle) and

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

It is well known that the sequence does not always converge to a solution of the equation, unless x_1 is sufficiently close to the solution. A "convergence theorem" in this context would be any result that gives a condition from which it follows that the sequence converges to a solution, i.e. a sufficient condition. Here is a convergence theorem for Newton's Method.

8-1.2. Theorem. *Suppose I is an interval and $f{:}I \to \mathbb{R}$ a function such that f'' exists on I and there exist positive numbers m and M for which*

$$|f'(x)| > m \text{ and } |f''(x)| < M \quad \text{for every } x \in I.$$

Let $f(r) = 0$, where $r \in I$ but is not an endpoint. Then there exists $\delta > 0$ such that $(r-\delta, r+\delta) \subseteq I$ and for any $\alpha \in (r-\delta, r+\delta)$, the sequence $\{x_n\}$ defined inductively by

$$x_1 = \alpha, \qquad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

lies in $(r-\delta, r+\delta)$ and satisfies

$$|x_{n+1} - r| \le \frac{M}{2m}|x_n - r|^2 \qquad \text{and} \qquad \lim_{n\to\infty} x_n = r.$$

Proof. For any $x \in I$ such that $x \neq r$, the second derivative f'' exists on $[x,r]$ or $[r,x]$ according as $x < r$ or $x > r$. By Taylor's Theorem,

$$0 = f(r) = f(x) + (r-x)f'(x) + \tfrac{1}{2}(r-x)^2 f''(c), \quad \text{where } x \lesseqgtr c \lesseqgtr r.$$

Since $|f'(x)| > m > 0$ for each $x \in I$, it follows upon dividing by $f'(x)$ that

$$0 = \frac{f(x)}{f'(x)} + (r-x) + \tfrac{1}{2}(r-x)^2 \frac{f''(c)}{f'(x)},$$

so that
$$x - \frac{f(x)}{f'(x)} - r = \tfrac{1}{2}(r-x)^2 \frac{f''(c)}{f'(x)}.$$

Hence
$$\left| x - \frac{f(x)}{f'(x)} - r \right| \le \frac{M}{2m}|x-r|^2.$$

Denote the positive number $\dfrac{M}{2m}$ by K. Then

$$\left|x-\frac{f(x)}{f'(x)}-r\right|\leq K|x-r|^2. \tag{1}$$

Let δ be any number such that $0<\delta<1/K$ as well as $(r-\delta,r+\delta)\subseteq I$, and consider any $x\in(r-\delta,r+\delta)$. If $x=r$, then $f(x)=0$ and hence

$$x-\frac{f(x)}{f'(x)}=x=r,$$

so that $x-\dfrac{f(x)}{f'(x)}\in(r-\delta,r+\delta)$.

If $x\neq r$, then by (1),

$$\left|x-\frac{f(x)}{f'(x)}-r\right|\leq K\delta^2=(K\delta)\delta<\delta \text{ (recalling that } K\delta<1),$$

so that $x-\dfrac{f(x)}{f'(x)}\in(r-\delta,r+\delta)$ in this case too. This shows that

$$x\in(r-\delta,r+\delta) \;\Rightarrow\; x-\frac{f(x)}{f'(x)}\in(r-\delta,r+\delta).$$

It follows by an induction argument that, for any $\alpha\in(r-\delta,r+\delta)$, the sequence $\{x_n\}$ defined inductively by

$$x_1=\alpha, \qquad x_{n+1}=x_n-\frac{f(x_n)}{f'(x_n)}$$

lies in the interval $(r-\delta,r+\delta)$. Furthermore, by (1), it satisfies

$$|x_{n+1}-r|\leq K|x_n-r|^2=\frac{M}{2m}|x_n-r|^2.$$

Since this inequality implies $|x_{n+1}-r|\leq K\delta|x_n-r|$, it follows by an induction argument that $|x_{n+1}-r|\leq (K\delta)^n|x_1-r|=(K\delta)^n|\alpha-r|$. Since $0<K\delta<1$, this further implies that $\lim\limits_{n\to\infty}x_n=r$. ☺

The conclusion of the above result can be interpreted as saying that Newton's Method converges to the solution if the starting term x_1 of the sequence is in the intersection of I with the interval $|x-r|<2m/M$.

8-1.3. Examples. (a) Suppose we wish to approximate $2^{1/3}$, in other words, find an approximate solution of $f(x)=x^3-2=0$. Since the solution obviously lies in (1,2), we may take this interval as I. Then $|f'(x)|>3$ and $|f''(x)|<12$ for all $x\in I$. So, we may take m to be 3 and M to be 12. Then $2m/M=\frac{1}{2}$. According to Theorem 8-1.2, we may take x_1 to be any number in the intersection of I with the interval $J=(2^{1/3}-\frac{1}{2},2^{1/3}+\frac{1}{2})$. This interval has length 1. Since $\left(\frac{3}{2}\right)^3>2$, we know that $2^{1/3}$ actually lies in $(1, \frac{3}{2})$, which has length $\frac{1}{2}$. Therefore this interval is contained in J. So, the sequence given by

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 - 2}{3x_n^2} = \tfrac{2}{3}\left(x_n + \frac{1}{x_n^2}\right),\ x_1 \in (1, \tfrac{3}{2})$$

converges to $2^{1/3}$. However, the reader can check by the methods used for Example 3-5.3(e) that x_1 may, in fact, be chosen to be any positive number. It is also instructive to check with a calculator that the sequence of that Example converges slowly as compared to the one generated here by Newton's Method.

(b) This example is artificial in the sense that no one would want to use Newton's Method to solve the equation $f(x) = \frac{x}{1+x^2} = 0$, since its obvious solution is $x = 0$. However, it provides an instance when the sequence generated by Newton's Method fails to converge at all. Here

$$f'(x) = \frac{1-x^2}{(1+x^2)^2} \quad \text{and} \quad f''(x) = -\frac{2x(3-x^2)}{(1+x^2)^3}$$

and the Newton sequence is given by

$$x_{n+1} = x_n - \frac{x_n(1+x_n^2)}{1-x_n^2}.$$

It is straightforward to verify that if $x_1 = \pm 1/\sqrt{3}$, then $x_{n+1} = -x_n$; thus the sequence "oscillates". If $|x| < 1/\sqrt{3}$, then $|f'(x)| > m$ and $|f''(x)| < M$, where $m = 3/8$ and $M = \frac{1}{4(3-2\sqrt{2})}$. So, the sequence converges to the solution 0 if $|x_1| < 2m/M = 3(3-2\sqrt{2})$. Note that $3(3-2\sqrt{2}) < 1/\sqrt{3}$.

For readers who may be aware of the condition $|f(x)f''(x)/f'(x)^2| < 1$ for convergence, we note that this holds only on the *smaller* interval given by $|x| < \sqrt{[\frac{1}{3}(4-\sqrt{13})]}$.

Problem Set 8-1

8-1.P1. Give an example of a function having derivatives of all orders in an interval, having a local strict maximum at a point in the interval, and yet not satisfying the hypotheses of Prop.8-1.1.

8-1.P2. Let $f^{(n+1)}$ be continuous on $[a, b]$ and $f^{(n+1)}(a) \neq 0$. Show that there exists $\delta > 0$ such that, whenever $0 < h < \delta$, the number 'θ' in the Lagrange remainder $\frac{f^{(n)}(a+\theta h)}{n!} h^n$ after n terms is a function of h, and that $\lim_{h\to 0} \theta = \frac{1}{n+1}$.

8-1.P3. Prove Prop.8-1.1 without assuming continuity of $f^{(n)}$ at c but only that $f^{(n)}(c) \neq 0$. [This means that all derivatives of lower order must exist on a subset having c as a limit point!]

8-1.P4. Let f'' exist on (a, b) and $f'(a) = f'(b) = 0$. show that there exists a point $c \in (a, b)$ such that

$$f''(c) \geq \frac{4}{(b-a)^2}[f(b) - f(a)].$$

8-1.P5. Let $|f(x)| \leq A$ and $|f''(x)| \leq B$ for all $x \geq 0$, where A and B are positive constants. Show that $|f'(x)| \leq 2\sqrt{(AB)}$ for all $x \geq 0$.

8-1.P6. Suppose f is a three times differentiable function on $[-1, 1]$ such that $f(-1) = 0$, $f(1) = 1$ and $f'(0) = 0$. Prove that $f'''(x) \geq 3$ for some $x \in (-1, 1)$.

8-2 Cauchy Mean Value Theorem

8-2.1. Cauchy Mean Value Theorem. *Let the functions* $f: [a, b] \to \mathbb{R}$ *and* $g: [a, b] \to \mathbb{R}$ *both be continuous on* $[a, b]$ *and differentiable on* (a, b). *Suppose that* $g(b) \neq g(a)$ *and that there is no* $x \in (a, b)$ *where* $f'(x)$ *and* $g'(x)$ *are both* 0. *Then there exists* $c \in (a, b)$ *such that*

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Proof. Define the function $\phi: [a, b] \to \mathbb{R}$ as

$$\phi(x) = f(x) - \frac{f(b) - f(a)}{g(b) - g(a)}(g(x) - g(a)).$$

Then ϕ is continuous on $[a, b]$ and differentiable on (a, b). Besides, $\phi(a) = f(a) = \phi(b)$, so that ϕ satisfies the conditions of Rolle's Theorem. Hence there exists $c \in (a, b)$ such that $\phi'(c) = 0$. However, $\phi'(c) = f'(c) - g'(c)\frac{f(b)-f(a)}{g(b)-g(a)}$. Therefore $f'(c) = g'(c)\frac{f(b)-f(a)}{g(b)-g(a)}$. If $g'(c)$ were to be 0, then $f'(c)$ would also be 0, which is not possible according to the hypothesis. Therefore $g'(c) \neq 0$ and we get the required result by dividing both sides of the preceding equality by $g'(c)$. ☺

The limit as $x \to 0$ of a quotient such as $\frac{x\cos x - \ln(1+x)}{x^2}$ is not so simple to find because the numerator and denominator both have limit 0. Such a situation is called an *indeterminate form* $\frac{0}{0}$ and can often be evaluated by using one of several related theorems, all known by the same name:

8-2.2. Theorem. l'Hôpital's Rule for an indeterminate form $\frac{0}{0}$. *Suppose the functions f and g are differentiable on the interval* (a, b), *and* $g' \neq 0$ *everywhere. If*

$$\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0 \tag{A}$$

and

$$\lim_{x\to a} \frac{f'(x)}{g'(x)} = L, \tag{B}$$

then $g \neq 0$ *on* (a, b) *and*

$$\lim_{x\to a} \frac{f(x)}{g(x)} = L.$$

[Existence of the last mentioned limit is part of the assertion; it is *not assumed* to exist!]

Proof. Extend both the functions to the domain $[a, b)$ by setting $f(a) = g(a) = 0$. In view of (A), these extended functions are continuous on $[a,x]$ for any $x \in (a, b)$ and differentiable on (a,x) with $g' \neq 0$ everywhere. It follows by Rolle's Theorem that g cannot be 0 at two points of $[a,x]$. Thus there can exist at most one point of $[a,x]$ where g takes the value 0. Since this is true for any $x \in (a, b)$, it follows that $g \neq 0$ on (a, b).

In particular, the Cauchy Mean Value Theorem 8-2.1 [or 7-5.P11] is applicable to f and g on the interval $[a,x]$ for any $x \in (a, b)$. Consequently,

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)} \text{ for some } c \in (a,x). \tag{1}$$

First suppose that $L \in \mathbb{R}$ and consider any $\varepsilon > 0$. By (B), there exists some $\delta > 0$ such that

$$\left| \frac{f'(c)}{g'(c)} - L \right| < \varepsilon \quad \text{whenever} \quad c \in (a, a+\delta). \tag{2}$$

We shall show that

$$\left| \frac{f(x)}{g(x)} - L \right| < \varepsilon \quad \text{whenever} \quad x \in (a, a+\delta).$$

If $x \in (a, a+\delta)$, we have $(a,x) \subset (a, a+\delta)$ and hence any number c that satisfies (1) also satisfies $c \in (a, a+\delta)$, as stipulated in (2). Thus the required inequality follows immediately from (1) and (2).

Before going on to the case when $L = \pm\infty$, we observe the following consequence of (1): If $\frac{f'}{g'} > 0$ [or < 0] on some subinterval with one endpoint a, i.e. $(a, a+\delta')$ with some positive δ', then the same inequality holds for the quotient $\frac{f}{g}$ as well.

Now suppose that $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$ and $\lim_{x\to a} \frac{f'(x)}{g'(x)} = \infty$ [resp. $-\infty$]. Then $f' \neq 0$ on some interval (a, c) and the roles of f and g can be interchanged on (a, c).

We have $\lim_{x\to a}\frac{g'(x)}{f'(x)} = 0$ as well as $\frac{g'(x)}{f'(x)} > 0$ [resp. < 0] on some subinterval with one endpoint a. It follows from what has been proved for the case when $L \in \mathbb{R}$ and by the observation in the preceding paragraph that $\lim_{x\to a}\frac{g(x)}{f(x)} = 0$ and that $\frac{g(x)}{f(x)} > 0$ [resp. < 0] on some subinterval with one endpoint a. Consequently, we have $\lim_{x\to a}\frac{f(x)}{g(x)} = \infty$ [resp. $-\infty$]. ☺

Remarks. In the above proof, it may seem easier to avoid explicit mention of inequalities and proceed directly from (2) by saying "$\lim_{x\to a} c = a$". This would, in some sense, be correct though technically faulty, because a limit applies to functions, while c is not a function of anything (it is not even unique). The only sense in which it would be correct is that one can recast the idea in terms of inequalities, as we have done.

The limits in the above Theorem are right limits; there is an obvious version when the limits are left limits (at b) and hence also when they are two sided limits. These versions will be taken for granted without explicit mention.

8-2.3. Examples. (a) Consider the limit mentioned earlier:

$$\lim_{x\to 0}\frac{x\cos x - \ln(1+x)}{x^2}.$$

If we let $f(x) = x\cos x - \ln(1+x)$ and $g(x) = x^2$ on $(-1,\infty)$, then $\lim_{x\to 0} f(x) = 0 = \lim_{x\to 0} g(x)$. Since $g'(x) = 2x \neq 0$ on $(-1,0)\cup(0,\infty)$, l'Hôpital's Rule for $\frac{0}{0}$ will be applicable if $\lim_{x\to 0}\frac{f'(x)}{g'(x)}$ exists. Upon computing

$$f'(x) = \cos x - x\sin x - \frac{1}{1+x} \text{ and } g'(x) = 2x,$$

we find once again that $\lim_{x\to 0} f'(x) = 0 = \lim_{x\to 0} g'(x)$. Therefore l'Hôpital's Rule for $\frac{0}{0}$ will be applicable if $\lim_{x\to 0}\frac{f''(x)}{g''(x)}$ exists and g'' does not vanish. Now

$$f''(x) = -2\sin x - x\cos x + \frac{1}{(1+x)^2} \quad \text{and} \quad g''(x) = 2.$$

Therefore $\lim_{x\to 0}\frac{f''(x)}{g''(x)}$ exists and equals $\frac{1}{2}$. Hence the required limit also exists and equals $\frac{1}{2}$.

(b) The limit $\lim_{x\to 0}\left(\frac{\sin x}{x^2} - \frac{1}{x}\right)$ is an indeterminate form of type "$\infty - \infty$".

Since $\frac{\sin x}{x^2} - \frac{1}{x} = \frac{\sin x - x}{x^2}$, it is converted to a $\frac{0}{0}$ type. It can now be evaluated by applying l'Hôpital's Rule for the form $\frac{0}{0}$ twice in a row:

$$\lim_{x\to 0} \frac{\sin x - x}{x^2} = \lim_{x\to 0} \frac{\cos x - 1}{2x} \quad \text{if this limit exists}$$

$$= \lim_{x\to 0} \frac{-\sin x}{2} \quad \text{if this limit exists.}$$

The last mentioned limit indeed exists and is 0. So, $\lim_{x\to 0} \left(\frac{\sin x}{x^2} - \frac{1}{x} \right) = 0.$

(c) The indeterminate form $\lim_{x\to 0} \frac{\sin x - x\cos x}{x^2 \sin x}$ is best evaluated as

$$\lim_{x\to 0} \frac{\sin x - x\cos x}{x^2 \sin x} = \lim_{x\to 0} \frac{\cos x - \cos x + x\sin x}{2x\sin x + x^2 \cos x} \quad \text{if this limit exists}$$

$$= \lim_{x\to 0} \frac{\sin x}{2\sin x + x\cos x}$$

$$= \lim_{x\to 0} \frac{1}{2 + \left(\frac{x}{\sin x}\right)\cos x},$$

which indeed exists, and equals $\frac{1}{3}$. Therefore

$$\lim_{x\to 0} \frac{\sin x - x\cos x}{x^2 \sin x} = \frac{1}{3}.$$

(d) It can happen that (A) of l'Hôpital's Rule holds but (B) does not, which renders the Rule inapplicable. Suppose we wish to find

$$\lim_{x\to 0} \frac{x^2 \sin \frac{1}{x}}{\sin x}.$$

Here (A) holds and we could proceed as

$$\lim_{x\to 0} \frac{x^2 \sin \frac{1}{x}}{\sin x} = \lim_{x\to 0} \frac{2x\sin \frac{1}{x} - \cos \frac{1}{x}}{\cos x} \quad \text{if this limit exists.}$$

However, this limit does not exist, because the denominator has limit 1 and the first term in the numerator has limit 0, but the second term in the numerator does not. Thus (B) breaks down, thereby making l'Hôpital's Rule inapplicable, and therefore the limit has to be found in some other manner. In fact, since

$$\frac{x^2 \sin \frac{1}{x}}{\sin x} = \left(\frac{x}{\sin x}\right) x\sin \frac{1}{x}, \quad \lim_{x\to 0} \frac{x}{\sin x} = 1 \quad \text{and} \quad \left|x\sin \tfrac{1}{x}\right| \le x,$$

we have

$$\lim_{x\to 0}\frac{x^2\sin\frac{1}{x}}{\sin x} = 0.$$

(e) An instance when both (A) and (B) hold and yet the Rule is inapplicable is $\lim_{x\to 0}\frac{\sqrt{\sin x}}{\sqrt{x}}$. The limit is immediately seen to be 1. However, upon taking $f(x) = \sqrt{\sin x}$ and $g(x) = \sqrt{x}$, we find that (A) holds and that $\frac{f'(x)}{g'(x)} = \frac{\sqrt{x}}{\sqrt{\sin x}}\cos x$, so that verifying (B) is the same as finding the answer to the original question!

We go on to consider indeterminate forms $\frac{\infty}{\infty}$, such as $\lim_{x\to 0} x\ln x = \lim_{x\to 0}\frac{\ln x}{1/x}$. Rewriting the fraction as $\frac{x}{1/\ln x}$, in which the numerator and denominator both have limit 0, converts it to the form $\frac{0}{0}$. It may seem at first sight that this makes it possible to find the limit by proceeding as before, but an attempt to do so will convince the reader otherwise. Therefore it is useful to have another version of l'Hôpital's Rule for $\frac{\infty}{\infty}$, which makes it possible to proceed without converting to $\frac{0}{0}$.

In the $\frac{\infty}{\infty}$ version, one need only assume that the denominator approaches $\pm\infty$. What may seem surprising is that the result can be established without recourse to the Cauchy Mean Value Theorem [or to 7-5.P11].

8-2.4. Theorem. l'Hôpital's Rule for an indeterminate form $\frac{\infty}{\infty}$. *Suppose the functions f and g are differentiable on the interval* (a,b), *and* $g' \neq 0$ *everywhere. If*

$$\lim_{x\to a} g(x) = \pm\infty \tag{A}$$

and

$$\lim_{x\to a}\frac{f'(x)}{g'(x)} = L, \tag{B}$$

then $g \neq 0$ *on some interval* (a,u) *and*

$$\lim_{x\to a}\frac{f(x)}{g(x)} = L.$$

[Existence of the limit here is part of the assertion; it is *not assumed* to exist!]

Proof. It is obvious from (A) that $g \neq 0$ on some interval (a,u).

Since g' is nonzero everywhere, it follows by Darboux's Theorem [see 7-5.P10], that g' cannot take both positive and negative values. So let us suppose that g' is negative everywhere. The contrary case can then be derived by considering $-g$ in place of g. It follows from Prop.7-5.9 that

$$g(x) \geq g\left(\tfrac{a+b}{2}\right) \text{ when } a < x < \tfrac{a+b}{2},$$

which rules out the possibility that the limit in (A) is $-\infty$. Therefore the limit in (A) is ∞.

First suppose that $L \in \mathbb{R}$ and consider any $\varepsilon > 0$. By (A) and (B), there exists some $\delta_1 > 0$ such that

$$g(x) > 0 \text{ and } L - \tfrac{\varepsilon}{2} < \frac{f'(x)}{g'(x)} < L + \tfrac{\varepsilon}{2} \qquad \forall\, x \in (a, a+\delta_1). \tag{1}$$

Let $y \in (a, a+\delta_1)$. Then from (1) and the fact that $g' < 0$, we have

$$f'(x) - (L - \tfrac{\varepsilon}{2})g'(x) < 0 < f'(x) - (L + \tfrac{\varepsilon}{2})g'(x) \quad \forall\, x \in (a, y).$$

It follows that $f - (L - \frac{\varepsilon}{2})g$ is strictly decreasing on $(a,y]$ and that $f - (L + \frac{\varepsilon}{2})g$ is strictly increasing on $(a,y]$. Therefore

$$f(x) - (L - \tfrac{\varepsilon}{2})g(x) > f(y) - (L - \tfrac{\varepsilon}{2})g(y) \qquad \forall\, x \in (a, y) \tag{2}$$

and

$$f(x) - (L + \tfrac{\varepsilon}{2})g(x) < f(y) - (L + \tfrac{\varepsilon}{2})g(y) \qquad \forall\, x \in (a, y). \tag{3}$$

Since $(a,y) \subset (a, a+\delta_1)$, the inequalities in (1) are valid for $x \in (a, y)$. It follows from (2) and the first inequality in (1) that

$$\frac{f(x)}{g(x)} - (L - \tfrac{\varepsilon}{2}) > \frac{f(y)}{g(x)} - (L - \tfrac{\varepsilon}{2})\frac{g(y)}{g(x)} \qquad \forall\, x \in (a, y).$$

By (A), there exists $\delta_2 > 0$ such that the right side of this inequality is greater than $-\frac{\varepsilon}{2}$ whenever $x \in (a, a+\delta_2)$. If $\delta_3 = \min\{\delta_1, \delta_2\}$, we have

$$\frac{f(x)}{g(x)} - (L - \tfrac{\varepsilon}{2}) > -\tfrac{\varepsilon}{2}, \text{ i.e. } \frac{f(x)}{g(x)} > L - \varepsilon \qquad \forall\, x \in (a, a+\delta_3). \tag{4}$$

Similarly, upon using (3) and the first inequality in (1), it follows that there exists some $\delta_4 > 0$ such that

$$\frac{f(x)}{g(x)} - (L + \tfrac{\varepsilon}{2}) < \tfrac{\varepsilon}{2}, \text{ i.e. } \frac{f(x)}{g(x)} < L + \varepsilon \qquad \forall\, x \in (a, a+\delta_4). \tag{5}$$

It is a consequence of (4) and (5) that $\lim\limits_{x \to a} \frac{f(x)}{g(x)}$ exists and is L.

Now suppose that $L = \infty$. Then $\frac{f'(x)}{g'(x)} > 1$ on some interval (a,c). Suppose also that g' is negative on this interval (otherwise we may change the signs of both f and g). It follows that $f'(x) - g'(x) < 0$ on (a,c) and hence $f(x) - g(x) > f\left(\frac{a+c}{2}\right) - g\left(\frac{a+c}{2}\right)$ when $a < x < \frac{a+c}{2}$. Also, as was shown above, $\lim\limits_{x \to a} g(x) = \infty$.

Together with the preceding inequality, this implies that $\lim_{x\to a} f(x) = \infty$ as well. One consequence of this is that $\frac{g(x)}{f(x)} > 0$ on some subinterval with one endpoint a; another consequence is that it is possible to interchange the roles of f and g. Since $L = \infty$, we have $\lim_{x\to a}\frac{g'(x)}{f'(x)} = 0$ as well as $\lim_{x\to a} f(x) = \infty$. It follows from what has been proved for the case when $L \in \mathbb{R}$ that $\lim_{x\to a}\frac{g(x)}{f(x)} = 0$. However, $\frac{g(x)}{f(x)} > 0$ on some subinterval with one endpoint a, and hence $\lim_{x\to a}\frac{f(x)}{g(x)} = \infty$. This establishes the result for the case $L = \infty$; the case when $L = -\infty$ follows by considering $-f$ in place of f. ☺

Remark. Here again, the limit concerned is a right limit and there are similar results for left limits and for two sided limits.

8-2.5. Example. Consider $\lim_{x\to 0} x\ln x$. If we take $f(x) = \ln x$ and $g(x) = 1/x$, then $f(x)$ approaches $-\infty$, whereas $g(x)$ approaches ∞ as $x\to 0$.

Since $f'(x) = 1/x$ and $g'(x) = -1/x^2$, by l'Hôpital's Rule for the form $\frac{\infty}{\infty}$, we have

$$\lim_{x\to 0} x\ln x = \lim_{x\to 0}\frac{f'(x)}{g'(x)} \qquad \text{if this limit exists}$$

$$= \lim_{x\to 0}\frac{1/x}{-1/x^2} = 0.$$

Problem Set 8-2

8-2.P1. (a) For the functions $f: [-1, 1]\to\mathbb{R}$ and $g: [-1, 1]\to\mathbb{R}$ defined by

$$f(x) = x^2 \text{ and } g(x) = x^{\frac{1}{3}},$$

show that there is no $c \in (-1, 1)$ for which $\frac{f'(c)}{g'(c)} = \frac{f(1)-f(-1)}{g(1)-g(-1)}$. Explain why this does not contradict the Cauchy Mean Value Theorem 8-2.1.

(b) Suppose f and g are continuous real valued functions on $[a, b]$ such that g is injective and $\lim_{x\to c}\frac{f(x)-f(c)}{g(x)-g(c)}$ exists for every $c \in (a, b)$. Show that there exists $c \in (a, b)$ such that $\frac{f(b)-f(a)}{g(b)-g(a)} = \lim_{x\to c}\frac{f(x)-f(c)}{g(x)-g(c)}$.

(c) In each of the following cases, determine whether the hypotheses of the result in part (b) are satisfied and also whether those of the Cauchy Mean Value Theorem are satisfied on the interval $[-1, 1]$:

$$\text{(i) } f(x) = x \text{ and } g(x) = x^3 \qquad \text{(ii) } f(x) = x^2 \text{ and } g(x) = x^{\frac{1}{3}}.$$

8-2.P2. Suppose that the function g is differentiable on (a, b), $\lim\limits_{x\to a} g(x) = 0$ and that $g' \neq 0$ everywhere. Show that $g \neq 0$ everywhere.

8-2.P3. (a) Find $\lim\limits_{x\to 0} \dfrac{e^x - e^{-x} - 2\cos x}{x \sin x}$.

(b) Find $\lim\limits_{x\to 0} \dfrac{\sqrt{2\cos x - 2 + x^2}}{x^2}$.

(c) Show that $\lim\limits_{x\to 0} \dfrac{\sqrt{1-\cos x}}{\sin x}$ does not exist.

8-2.P4. Show that the function $f(x) = \frac{\sin x}{x}$ with $f(0) = 1$ has a second derivative for nonzero x and find its Maclaurin polynomial of two terms with Lagrange's form of the remainder. [Terms that are 0 need not be written of course.]

8-2.P5. If f'' exists and is continuous in an interval centred at x, show that

$$\lim_{h\to 0} \frac{f(x+2h) - 2f(x+h) + f(x)}{h^2} = f''(x).$$

8-3 Differentiability of Uniform Limits

Since a uniform limit of a sequence of continuous functions is continuous [see Theorem 5-1.3], one may ask whether a uniform limit of differentiable functions must be differentiable. The answer is No, as we now illustrate:

8-3.1. Example. Let $f_n(x) = x/(1 + n^4x^2)$, $n \in \mathbb{N}$ and $x \in \mathbb{R}$. Then f_n converges uniformly to $f(x) = 0$ on $\mathbb{R}$, because

$$|f_n(x)| = \frac{|x|}{(1-n^2|x|)^2 + 2n^2|x|} \leq \frac{1}{2n^2}.$$

On the other hand,

$$f_n'(x) = \frac{1-n^4x^2}{(1+n^4x^2)^2},$$

which shows that $\lim\limits_{n\to\infty} f_n'(x) = 0$ for $x \neq 0$ whereas $\lim\limits_{n\to\infty} f_n'(0) = 1 \neq f'(0)$.

The functions f_n' are continuous and the fact that their limit is not continuous at 0 shows [in view of Theorem 5-1.3] that they converge only pointwise and not uniformly on any interval containing 0. It is worth noting that they do converge uniformly to 0 on any interval $[a, \infty)$ with $a > 0$. This is true because

$$|f_n'(x)| = \left|\frac{1-n^4x^2}{(1+n^4x^2)^2}\right| \le \frac{1+n^4x^2}{(1+n^4x^2)^2} \le \frac{1}{1+n^4a^2} \text{ when } x \ge a.$$

8-3.2. Theorem. *Let $\{f_n\}$ be a sequence of real valued differentiable functions on an interval I and suppose there exists some $x_0 \in I$ such that the real sequence $\{f_n(x_0)\}$ converges. If the sequence of derivatives $\{f_n'\}$ converges uniformly to a limit function g, then there exists a differentiable function f on I such that*

(a) *the sequence $\{f_n\}$ converges uniformly to f on every bounded subinterval of I and*

(b) $f'(x) = g(x)$ *whenever $x \in I$.*

Proof. For each $n \in \mathbb{N}$ and each $c \in I$, let g_n be the function such that

$$g_n(x) = \begin{cases} \dfrac{f_n(x) - f_n(c)}{x - c} & \text{if } x \ne c \\ f_n'(c) & \text{if } x = c. \end{cases}$$

Then g_n is continuous at each point of I (including c). Also,

$$\begin{aligned}(x-c)\big(g_n(x) - g_m(x)\big) &= \big(f_n(x) - f_m(x) - f_n(c) + f_m(c)\big) \\ &= \big((f_n - f_m)(x) - (f_n - f_m)(c)\big) \\ &= (x-c)(f_n' - f_m')(\xi), \quad \text{where } \xi \text{ lies between } c \text{ and } x.\end{aligned}$$

Since $\{f_n'\}$ converges uniformly on I, it follows from the above equality that $\{g_n\}$ is uniformly Cauchy and hence uniformly convergent on $\{x \in I : x \ne c\}$. But $\{g_n(c)\} = \{f_n'(c)\}$ and is also convergent and it follows that $\{g_n\}$ is uniformly convergent on the whole of I [see 5-1.P2]. Since each g_n is continuous, the uniform limit $G = \lim_{n\to\infty} g_n$ is continuous on I.

Now, we use the uniform convergence of $\{g_n\}$ to show that $\{f_n\}$ converges uniformly on I. To this end, consider the sequence $\{g_n\}$ corresponding to $c = x_0$. By the definition of g_n,

$$f_n(x) - f_m(x) = f_n(x_0) - f_m(x_0) + (x - x_0)\big(g_n(x) - g_m(x)\big).$$

Since $\{f_n(x_0)\}$ is Cauchy and $\{g_n\}$ is uniformly Cauchy on I, this equality implies that $\{f_n\}$ is uniformly Cauchy on every bounded subinterval of I. Therefore there is a function f on I to which $\{f_n\}$ converges uniformly on every bounded subinterval of I. It remains to show that $f'(x)$ exists and equals $g(x)$ for every $x \in I$.

Again consider the sequence g_n for some $c \in I$. Since G has been shown to be continuous,

$$\lim_{x\to c} G(x) = G(c). \tag{1}$$

But, when $x \neq c$,

$$G(x) = \lim_{n\to\infty} g_n(x) = \lim_{n\to\infty} \frac{f_n(x) - f_n(c)}{x - c} = \frac{f(x) - f(c)}{x - c}.$$

Therefore from (1), we have $\displaystyle\lim_{x\to c} \frac{f(x) - f(c)}{x - c} = G(c)$,

i.e. $$f'(c) = \lim_{n\to\infty} g_n(c) = \lim_{n\to\infty} f_n'(c) = g(c).$$

Since this holds for every $c \in I$, we have $f'(x) = g(x)$ for all $x \in I$. ☺

8-3.3. Examples. It can happen that (a) and (b) of the above Theorem hold even when $\{f_n'\}$ converges only pointwise. Let $f_n(x) = \frac{1}{n} e^{-n^2x^2}$. Then $0 \leq f_n(x) \leq \frac{1}{n}$ for every $x \in \mathbb{R}$, and therefore $\{f_n\}$ converges uniformly to 0 on $\mathbb{R}$. Also $f_n'(x) = -2nxe^{-n^2x^2}$, which converges pointwise to 0 for each $x \in \mathbb{R}$, in view of the property of the exponential function that $\lim_{u\to\infty} ue^{-u^2} = 0$. However this convergence is not uniform, because $|f_n'(\frac{1}{n}) - 0| = \frac{2}{e}$ and therefore the inequality $|f_n(x) - 0| < \varepsilon$ cannot hold for all $x \in \mathbb{R}$ if $\varepsilon = \frac{2}{e}$.

Now consider $f_n(x) = -\sum_{k=1}^{n} (-x)^k/k$. We have $f_n(0) = 0$ and $f_n'(x) = \sum_{k=1}^{n} (-x)^{k-1} = \sum_{k=0}^{n-1} (-x)^k$, which is known to converge to $\frac{1}{1+x}$ uniformly on $[0, a]$ whenever $0 < a < 1$. By Theorem 8-3.2, f_n converges uniformly to f on $[0, a]$, where $f'(x) = \frac{1}{1+x}$. Since $f(0) = 0$, it follows that $f(x) = \ln(1+x)$. As this holds on $[0, a]$ whenever $0 < a < 1$,we have

$$\ln(1+x) = \lim_{n\to\infty} f_n(x) = -\sum_{k=1}^{\infty} (-x)^k/k \qquad \text{for} \quad 0 \leq x < 1. \tag{1}$$

Leibnitz' Theorem 4-1.7 guarantees the convergence of this series to a limit $F(x)$ such that $|f_n(x) - F(x)| \leq 1/(n+1)$ for all $x \in [0, 1]$. Thus f_n converges uniformly to F on this domain and hence F must be continuous [see Theorem 5-1.3]. But $F(x) = f(x) = \ln(1+x)$ for $0 \leq x < 1$ as shown above, and it now follows by continuity that this holds even when $x = 1$. Therefore (1) holds for $x = 1$ as well, thereby yielding the equality

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - + \cdots.$$

A similar argument starting with $f_n(x) = \sum_{k=1}^{n}(-1)^{k-1}x^{2k-1}/(2k-1)$ and $f(x) = \tan^{-1}x$ yields

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - + \cdots.$$

8-3.4. Theorem. *Let R be the radius of convergence of the power series* $\sum_{k=0}^{\infty} a_k z^k$. *Then the "differentiated" power series* $\sum_{k=0}^{\infty} k a_k z^{k-1}$ *also has radius of convergence* R. *If* $R > 0$ *and each* a_k *is real, then the functions f and g defined by their sums,*

$$f(x) = \sum_{k=0}^{\infty} a_k x^k \quad \textit{and} \quad g(x) = \sum_{k=0}^{\infty} k a_k x^{k-1} \qquad \textit{for every } x \in (-R, R)$$

have the relation $f'(x) = g(x)$ *everywhere on* $(-R, R)$.

Proof. By Theorem 5-2.7, $R = 1/\limsup_{k\to\infty} |a_k|^{1/k}$, while the radius of convergence of the differentiated series is $1/\limsup_{k\to\infty} |(k+1)a_{k+1}|^{1/k}$. From the inequality

$$k^{1/k} \leq (k+1)^{1/k} \leq (k+1)^{2/(k+1)}$$

and Prop.3-1.6, it follows that $\limsup_{k\to\infty} |(k+1)a_{k+1}|^{1/k} = \limsup_{k\to\infty} |a_k|^{1/k}$. Therefore the power series $\sum_{k=0}^{\infty} k a_k z^{k-1}$ has radius of convergence R.

Now suppose $R > 0$ and each a_k is real. By Theorem 5-2.7, the sequence of partial sums of $\sum_{k=0}^{\infty} k a_k x^{k-1}$ converges to g uniformly on $[-\rho, \rho]$ whenever $0 < \rho < R$. But this sequence is the sequence of derivatives of the partial sums of $\sum_{k=0}^{\infty} a_k x^k$ and the latter converge for $x = 0 \in [-\rho, \rho]$. By part (b) of Theorem 8-3.2, $f'(x) = g(x)$ whenever $x \in [-\rho, \rho]$. Since this is true for any positive $\rho < R$, it follows that $f'(x) = g(x)$ for every $x \in (-R, R)$. ☺

Problem Set 8-3

8-3.P1. Let $f_n(x) = \frac{1}{2n^2}\ln(1 + n^4x^2)$, where $x \in [0,1]$. Determine whether the sequences $\{f_n\}$ and $\{f_n'\}$ converge pointwise or uniformly, and if they do, whether the latter converges to the derivative of $\lim_{n\to\infty} f_n$.

8-3.P2. Let R be the radius of convergence of the power series $\sum_{k=0}^{\infty} a_k z^k$. Then show that the "integrated" power series $\sum_{k=0}^{\infty} a_k z^{k+1}/(k+1)$ also has radius of convergence R. If $R > 0$ and each a_k is real, then show that the functions f and g defined by their sums,

$$f(x) = \sum_{k=0}^{\infty} a_k x^k \text{ and } g(x) = \sum_{k=0}^{\infty} a_k x^{k+1}/(k+1) \qquad \text{for } x \in (-R, R)$$

have the relation $f(x) = g'(x)$ everywhere on $(-R, R)$.

8-3.P3. Show that the series $\sum_{k=1}^{\infty} \frac{1}{k^3 + k^4 x^2}$ can be "differentiated term by term", i.e. the series obtained by differentiating each term converges to a sum which is the derivative of the sum of the given series.

8-3.P4. Starting with $f_n(x) = \sum_{k=1}^{n} (-1)^{k-1} x^{3k-2}/(3k-2)$ and

$$f(x) = \frac{1}{6} \ln \frac{x^2 + 2x + 1}{x^2 - x + 1} + \frac{1}{\sqrt{3}} \tan^{-1} \frac{2}{\sqrt{3}} (x - \tfrac{1}{2}),$$

use Theorem 8-3.2 and Leibnitz' Theorem to show that

$$\frac{1}{3}(\ln 2 + \frac{\pi}{\sqrt{3}}) = 1 - \frac{1}{4} + \frac{1}{7} - \frac{1}{10} + \frac{1}{13} - + \cdots .$$

8-3.P5. Starting with $f_n(x) = \sum_{k=1}^{n} (-1)^{k-1} x^{4k-3}/(4k-3)$ and

$$f(x) = \begin{cases} \frac{1}{2\sqrt{2}} \tan^{-1}(x - \frac{1}{x}) + \frac{1}{4\sqrt{2}} \ln \frac{x^2 + \sqrt{2}x + 1}{x^2 - \sqrt{2}x + 1} + \frac{\pi}{4\sqrt{2}} & \text{if } x \neq 0 \\ 0 \text{ if } x = 0 \end{cases},$$

use Theorem 8-3.2 and Leibnitz' Theorem to show that

$$\frac{1}{2\sqrt{2}} \left(\ln(1 + \sqrt{2}) + \frac{\pi}{2} \right) = 1 - \frac{1}{5} + \frac{1}{9} - \frac{1}{13} + \frac{1}{15} - + \cdots .$$

8-4 Continuous Nowhere Differentiable Function

It is easy to describe a continuous function $h: \mathbb{R} \to \mathbb{R}$ which is not differentiable at certain special points. For example,

$$h(x) = \min \{x - [x], 1 - (x - [x])\},$$

or what is the same thing,

$$h(x) = \begin{cases} x - [x] & \text{if } [x] \leq x < [x] + \frac{1}{2} \\ 1 - (x - [x]) & \text{if } [x] + \frac{1}{2} \leq x < [x] + 1, \end{cases}$$

is continuous on $\mathbb{R}$ [see 2-1.P5 for a graph] but not differentiable at n or $n + \frac{1}{2}$, where n is an integer. However, it is quite difficult to describe a function continuous on $\mathbb{R}$ but differentiable nowhere.

A relatively easy construction of such a function, due to B.L.van der Waerden, makes use of the decimal representation of numbers, in particular, the property that a real number x satisfies $[x] \le x < [x] + \frac{1}{2}$ if, and only if, its "first decimal digit" is less than 5.

Readers who are willing to accept such properties of decimal representations can content themselves with the construction in Theorem 8-4.1 below, whereas those who prefer to question how the properties follow from the axioms for $\mathbb{R}$ will find the answer in the material presented subsequently.

It will be convenient to call an interval of the form $[k, k+1)$, where k is an integer, as an integer interval. Also, $[k, k+\frac{1}{2})$ will be called the left half of the integer interval $[k, k+1)$ and $[k+\frac{1}{2}, k+1)$ its right half. Note that any real x belongs either to the left half of $[[x], [x]+1)$ or to its right half, but not to both.

8-4.1. Theorem. *There exists a function $f:\mathbb{R}\to\mathbb{R}$ that is continuous everywhere but differentiable nowhere.*

Proof. Let $h(x) = \min\{x-[x], 1-(x-[x])\}$. Then h is continuous on $\mathbb{R}$ and $0 \le h(x) \le \frac{1}{2}$ for every $x \in \mathbb{R}$ [see 2-1.P5]. Besides, it satisfies $h(x+p) = h(x)$ for every $p \in \mathbb{N}$ and $x \in \mathbb{R}$. Define $f:\mathbb{R}\to\mathbb{R}$ as

$$f(x) = \sum_{n=0}^{\infty} \frac{h(10^n x)}{10^n}.$$

Since $0 \le h(10^n)/10^n \le \frac{1}{2}(1/10^n)$ and the geometric series $\sum_{n=0}^{\infty} 1/10^n$ converges, it follows by the Weierstrass M-test [Theorem 5-2.3] that the defining series for f converges uniformly, and therefore the sum f is continuous by Theorem 5-1.3. We shall show that it is nowhere differentiable.

As h satisfies $h(x+p) = h(x)$, $p \in \mathbb{N}$, so does f. We therefore restrict attention to $[0, 1)$. Accordingly, consider any $a \in [0, 1)$ with decimal representation

$$a = .a_1 a_2 \cdots = \sum_{k=1}^{\infty} a_k(10^{-k}).$$

For each $n \in \mathbb{N}$, let

$$x_n = .a_1 a_2 \cdots a_{n-1} b_n a_{n+1} \cdots,$$

where $b_n = a_n + 1$ if $a_n \ne 4$ or 9 while $b_n = a_n - 1$ if $a_n = 4$ or 9. Then $x_n - a = \pm 10^{-n}$ and so, $x_n \to a$ as $n \to \infty$. To complete the proof, it will be sufficient to show that

$$\lim_{n\to\infty} \frac{f(x_n) - f(a)}{x_n - a} \text{ does not exist.}$$

If $0 \le m < n-1$, then $10^m x_n$ and $10^m a$ have the same integer part $a_1 a_2 \cdots a_m$ and the same first decimal digit a_{m+1}. Therefore both belong to the same half of the integer interval $[a_1 a_2 \cdots a_m, a_1 a_2 \cdots a_m + 1)$. Consequently,

$$h(10^m x_n) - h(10^m a) = \pm(10^m x_n - 10^m a) = \pm 10^m (10^{-n}) = \pm 10^{m-n}.$$

If $m = n-1$, then the integer parts of $10^m x_n$ and $10^m a$ are still the same but the first decimal digits are b_n and a_n respectively; however b_n has been so chosen that either b_n and a_n are both less than 5 or both not less than 5. Therefore it is still true that $10^m x_n$ and $10^m a$ belong to the same half of the same integer interval and hence that

$$h(10^m x_n) - h(10^m a) = \pm 10^{m-n}.$$

Finally, for $m \geq n$, $10^m x_n - 10^m a$ is an integer and therefore $h(10^m x_n) - h(10^m a) = 0$.

Using the above facts about $h(10^m x_n) - h(10^m a)$ when $m \leq n-1$ and $m \geq n$, we find that

$$\begin{aligned}\frac{f(x_n)-f(a)}{x_n-a} &= \sum_{m=0}^{\infty} \frac{h(10^m x_n) - h(10^m a)}{10^m (x_n - a)} \\ &= \sum_{m=0}^{n-1} \frac{\pm 10^{m-n}}{10^m (10^{-n})} = \sum_{m=0}^{n-1} \pm 1.\end{aligned}$$

Since there are n terms in this summation, each equal to 1 or -1, the sum is an even integer when n is even and an odd integer when n is odd. Consequently, it cannot converge as $n \to \infty$. ☺

The rest of the material in this Section is optional reading, because it is devoted exclusively to justifying some of the facts about decimal representations that were taken for granted in the above proof.

The existence of a "decimal" type representation of any $x \in [0,1)$ with an arbitrary integer base $b > 1$ was proved in Theorem 4-1.9. For a general $x \in \mathbb{R}$, we have $x - [x] \in [0,1)$, and hence $x = [x] + \sum_{k=1}^{\infty} a_k b^{-k}$, where the sequence $\{a_k\}$ satisfies (B) and (C) of that Theorem. When $x \geq 0$, this will be called a *decimal representation of x with base b* (even though b may not be 10). The series $\sum_{k=1}^{\infty} a_k b^{-k}$ will be called the *decimal part* of x. The terminology and notation introduced in this paragraph will be used throughout the rest of this Section.

We begin by proving a result which justifies why $10^3(1.73268)$ has integer part 1732 and has decimal representation 1732.68.

8-4.2. Proposition. *For $n \geq 1$,*

$$[b^{n-1}x] = b^{n-1}[x] + \sum_{k=1}^{n-1} a_k b^{n-k-1}$$

with the understanding that when $n = 1$, the sum is 0. Also,

$$b^{n-1}x = [b^{n-1}x] + \sum_{k=n}^{\infty} a_k b^{n-k-1}$$

is a decimal representation of $b^{n-1}x$.

Proof. This is obvious when $n = 1$ and we need consider only $n > 1$. Since $x = [x] + \sum_{k=1}^{\infty} a_k b^{-k}$, we have

$$\begin{aligned} b^{n-1}x &= b^{n-1}[x] + \sum_{k=1}^{\infty} a_k b^{n-k-1} \\ &= b^{n-1}[x] + \sum_{k=1}^{n-1} a_k b^{n-k-1} + \sum_{k=n}^{\infty} a_k b^{n-k-1}. \end{aligned} \tag{1}$$

By (B) and (C) of Theorem 4-1.9, the third term satisfies the double inequality

$$0 \le \sum_{k=n}^{\infty} a_k b^{n-k-1} < \sum_{k=n}^{\infty} (b-1) b^{n-k-1} = 1,$$

where we have taken the sum of the geometric series. Together with (1) and the fact that the first two terms in it represent integers, this double inequality implies that

$$[b^{n-1}x] = b^{n-1}[x] + \sum_{k=1}^{n-1} a_k b^{n-k-1}$$

and that

$$b^{n-1}x = [b^{n-1}x] + \sum_{k=n}^{\infty} a_k b^{n-k-1}.$$

Since the sequence $\{a_n, a_{n+1}, \dots\}$ satisfies (B) and (C) of Theorem 4-1.9, the last mentioned equality gives a decimal representation of $b^{n-1}x$. ☺

8-4.3. Corollary. *In any decimal representation of* x,

$$a_n = [b(b^{n-1}x - [b^{n-1}x])] \quad \textit{for all } n \ge 1.$$

In particular, the decimal representation is unique.

Proof. By Prop.8-4.2, $b^{n-1}x - [b^{n-1}x] = \sum_{k=n}^{\infty} a_k b^{n-k-1}$. Therefore

$$b(b^{n-1}x - [b^{n-1}x]) = a_n + \sum_{k=n+1}^{\infty} a_k b^{n-k}.$$

Arguing as in the preceding proof, we can show that $0 \le \sum_{k=n+1}^{\infty} a_k b^{n-k} < 1$. As before, it follows that $a_n = [b(b^{n-1}x - [b^{n-1}x])]$. ☺

The integer a_n in the decimal representation $x = [x] + \sum_{k=1}^{\infty} a_k b^{-k}$ will be called the ***n*^th^ decimal digit of** x. According to Prop.8-4.2, the k^{th} decimal digit of $b^{n-1}x$ is the $(n+k-1)^{\text{th}}$ decimal digit of x.

We now come to the question of how to tell from the first decimal digit of x whether $[x] \le x < [x] + \frac{1}{2}$ or not, in other words, whether it belongs to the left half of the integer interval $[[x], [x]+1)$ or to its right half. Note that in base 5, an odd integer, the number .23 is not in the left half of an integer interval despite its first decimal digit being less than half the base.

8-4.4. Proposition. *Suppose the base b is an even integer. Then x belongs to the left half of* $[[x], [x]+1)$ *if, and only if, its first decimal digit is less than* $\frac{b}{2}$. *More* *generally*, $b^{n-1}x$ *belongs to the left half of* $[[b^{n-1}x], [b^{n-1}x]+1)$ *if, and only if, the* n^{th} *decimal digit of* x *is less than* $\frac{b}{2}$.

Proof. By Cor.8-4.3, the first decimal digit of x is

$$a_1 = [b(x-[x])].$$

If $a_1 < \frac{b}{2}$, then $[b(x-[x])] < \frac{b}{2}$ and, since $\frac{b}{2}$ is an integer, it further follows that $b(x-[x]) < \frac{b}{2}$, i.e. $(x-[x]) < \frac{1}{2}$. Thus x belongs to the left half when $a_1 < \frac{b}{2}$.

Conversely, suppose x belongs to the left half, i.e. $x - [x] < \frac{1}{2}$. Then $b(x-[x]) < \frac{b}{2}$ and therefore $[b(x-[x])] < \frac{b}{2}$, which means that $a_1 < \frac{b}{2}$.

By Prop.8-4.2, the first decimal digit of $b^{n-1}x$ is the n^{th} decimal digit of x. Therefore the second part follows upon applying to $b^{n-1}x$ what has just been proved for x. ☺

8-4.5. Corollary. *Suppose the base* b *is an even integer. Let* $[x] = [y]$ *and the decimal representations of* x *and* y *have the same* k^{th} *decimal digit for* $1 \le k \le n-1$. *Then for any integer* m *such that* $1 \le m \le n-1$, *the numbers* $b^{m-1}x$ *and* $b^{m-1}y$ *belong to the same half of the same integer interval.*

Proof. Since $m \le n-1$, it follows from Prop.8-4.2 that $b^{m-1}x$ and $b^{m-1}y$ not only have the same integer part $b^{m-1}[x] + \sum_{k=1}^{m-1} a_k b^{-k}$ but also the same first decimal digit a_m. By Prop.8-4.4, one of them is in the left half if, and only if, the other one is. ☺

Finally, we note certain properties of the function h occurring in the proof of Theorem 8-4.1 that have been implicitly used in the argument.

1. If x and y belong to the left half of the same integer interval $[k, k+1)$, then it is immediate from the definition of h that $h(x) = x-k$ and $h(y) = y-k$, and hence $h(x)-h(y) = x-y$.
2. If x and y belong to the right half of the same integer interval $[k, k+1)$, then it is immediate from the definition of h that $h(x) = 1-(x-k)$ and $h(y) = 1-(y-k)$, and hence $h(x)-h(y) = y-x$.
3. If x and y belong to the same half of the same integer interval $[k, k+1)$, then $h(x)-h(y) = \pm(x-y)$.
4. Suppose the base b is even and that the numbers x and y have decimal representations that differ only in the n^{th} decimal place. Then, for $0 \le m < n-1$, the numbers $b^m x$ and $b^m y$ belong to the same half of the same integer interval by Cor.8-4.5, so that $h(b^m x)-h(b^m y) = \pm(b^m x - b^m y)$. For $m \ge n$, the numbers $b^m x$ and $b^m y$ differ by an integer and $h(b^m x)-h(b^m y) = 0$. When $m =$

$n-1$, the numbers $b^m x$ and $b^m y$ have the same integer part but different first decimal digits. In case both first decimal digits are less than $\frac{b}{2}$ or both not less than $\frac{b}{2}$, the numbers $b^m x$ and $b^m y$ belong to the same half of the same integer interval and $h(b^m x) - h(b^m y) = \pm(b^m x - b^m y)$ in this case too.

Problem Set 8-4

8-4.P1. Can Theorem 8.4.1 be proved by using a base other than 10?

Appendix 1 (Irrationality of π)

Theorem. *The number* π *is irrational.*

Proof. It is enough to show that π^2 is irrational. Suppose, if possible, that $\pi^2 = p/q$, where p and q are positive integers. Then $q^n(\pi^2)^{n-k} = p^{n-k}q^k$ and is therefore an integer whenever k and n are integers such that $0 \le k \le n$.

Since the series $\sum_{k=0}^{\infty}(p^k/k!)$ converges according to Prop.5-3.1, we have $\lim_{n\to\infty}(p^n/n!) = 0$ by Prop.4-1.3. So, there exists a positive integer n such that

$$0 < \pi p^n/n! < 1. \tag{1}$$

Let

$$f(x) = \frac{x^n(1-x)^n}{n!},$$

a polynomial of degree $2n$. Then $f^{(2n+1)}(x) = 0 = f^{(2n+2)}(x)$, and as seen in 7-6.P2, $f^{(j)}(0)$ and $f^{(j)}(1)$ are both integers for all $j \in \mathbb{N}$. Besides,

$$0 < f(x) < \frac{1}{n!} \text{ when } 0 < x < 1. \tag{2}$$

Define

$$\begin{aligned}\phi(x) &= q^n[(\pi^2)^n f(x) - (\pi^2)^{n-1} f''(x) + \cdots + (-1)^n f^{(2n)}(x)]\\ &= q^n[\sum_{k=0}^{n}(-1)^k(\pi^2)^{n-k} f^{(2k)}(x)].\end{aligned}$$

Since $q^n(\pi^2)^{n-k}$, $f^{(2k)}(0)$ and $f^{(2k)}(1)$ are always integers, it follows that $\phi(0)$ and $\phi(1)$ are both integers. [We shall now take advantage of the fact that the second derivative of each term is the next term multiplied by $-\pi^2$, the last term having derivative 0. The reader may wish to rewrite the computation below in terms of $+\cdots+$ instead of the sigma notation.]

$$\begin{aligned}\phi''(x) &= q^n[\sum_{k=0}^{n}(-1)^k(\pi^2)^{n-k} f^{(2k+2)}(x)]\\ &= q^n[\sum_{k=0}^{n-1}(-1)^k(\pi^2)^{n-k} f^{(2k+2)}(x)] \quad \text{because } f^{(2n+2)}(x) = 0\end{aligned}$$

$$= q^n[\sum_{k=1}^{n}(-1)^{k-1}(\pi^2)^{n-k+1} f^{(2k)}(x)]$$
$$= -\pi^2 q^n[\sum_{k=1}^{n}(-1)^{k}(\pi^2)^{n-k} f^{(2k)}(x)]$$
$$= -\pi^2 q^n[\sum_{k=0}^{n}(-1)^{k}(\pi^2)^{n-k} f^{(2k)}(x) - (\pi^2)^n f(x)]$$
$$= -\pi^2 \phi(x) + \pi^2(\pi^2)^n q^n f(x) \text{ by the definition of } \phi.$$

Therefore

$$\phi''(x) + \pi^2\phi(x) = \pi^2(p/q)^n q^n f(x) = \pi^2 p^n f(x). \tag{3}$$

Now define

$$\psi(x) = \phi'(x)\sin\pi x - \pi\,\phi(x)\cos\pi x.$$

Elementary computation, using (3), shows that, on the one hand

$$\psi'(x) = \pi^2 p^n f(x)\sin\pi x$$

and on the other hand

$$\psi(1) - \psi(0) = \pi\,(\phi(1) + \phi(0)).$$

Together with the Lagrange Mean Value Theorem, this implies that

$$\phi(1) + \phi(0) = \pi\, p^n f(c)\sin\pi c \quad \text{for some } c \in (0,1).$$

But

$$0 < \pi\, p^n f(c)\sin\pi c \le \pi\, p^n f(c)$$
$$< \pi\, p^n/n! \quad \text{by (2)}$$
$$< 1 \quad \text{by (1)},$$

although $\phi(1) + \phi(0)$ is an integer. This is a contradiction. ☺

Appendix 2 (Pre-Calculus Functions and ℝ)

The need for having a number system that goes beyond rational numbers generally becomes evident right at the pre-Calculus stage, if not earlier. Our definitions of sine, cosine and exponential functions make sense only in the context of ℝ, because the convergence of the series defining them depends upon the Completeness Axiom of the real number system. It should not be concluded however that these functions would not have been possible in any smaller number system than ℝ.

To clarify the matter, it is necessary to keep in mind that a *number system* is understood in the present context to be an ordered field as in Def.1-4.1. It is easy to see that a subset of ℝ that fulfils the conditions of Def.1-3.5 automatically fulfils the conditions (O1)—(O4) of Section 1-4 as well, and is therefore an ordered field; in particular, it provides a number system. It will make matters easier to restrict attention to such number systems, which are called **subfields** of

$\mathbb{R}$. What we have said in the above paragraph is that the functions mentioned are possible in some subfield of $\mathbb{R}$ that is actually smaller than $\mathbb{R}$, but of course bigger than $\mathbb{Q}$. There are many subfields in-between $\mathbb{Q}$ and $\mathbb{R}$, and the reader who is aware of at least one instance of such a subfield would be better off skipping the example in the next paragraph.

For an example of a subfield of $\mathbb{R}$ that includes $\mathbb{Q}$ but is not the whole of $\mathbb{R}$, consider the subset of $\mathbb{R}$ defined as $F = \{a+b\sqrt{2} : a \in \mathbb{Q},\ b \in \mathbb{Q}\}$. Simple computations show that all field axioms listed in Section 1-3 are satisfied, though (M5) may seem a little doubtful. To see that (M5) indeed holds, consider any $a+b\sqrt{2} \neq 0$. Then $a^2-2b^2 \neq 0$ (because otherwise $\sqrt{2}$ would be equal to the rational number $|a/b|$), and

$$(a+b\sqrt{2})\left(\frac{a}{a^2-2b^2}-\frac{b\sqrt{2}}{a^2-2b^2}\right)=1.$$

Since the numbers $\frac{a}{a^2-2b^2}$ and $\frac{b}{a^2-2b^2}$ are both rational, the above equality shows a multiplicative inverse for $a+b\sqrt{2}$ that belongs to F.

The reason why even a simple function like $f(x) = \sqrt{x}$ is not "possible" in $\mathbb{Q}$ is that it maps some (positive) numbers of the rational field to numbers outside the field. Similarly, the function exp maps the rational number 1 to e, which is outside the rational field. Another way to express this is that the subfield of rational numbers fails to be "adequate for" the functions concerned in the following sense:

Definition. *Given an* $\mathbb{R}$*-valued function f with domain contained in* $\mathbb{R}$*, a subfield K of* $\mathbb{R}$ *is said to be* **adequate for** *f if*

$$x \in K \cap (\text{domain of } f) \Rightarrow f(x) \in K.$$

Question. Is there a set of commonly used functions like the ones mentioned above, which can be understood at the pre-Calculus level, such that the only subfield of $\mathbb{R}$ which is adequate for all functions belonging to the set is the entire field $\mathbb{R}$?

Main Result. Given a countable family Φ of $\mathbb{R}$-valued functions with domains contained in $\mathbb{R}$, there exists a countable subfield F of $\mathbb{R}$ which is adequate for all the functions belonging to Φ.

Proof. Let E be a countable subfield of $\mathbb{R}$. The set $S = \{f(x) : x \in E \cap (\text{domain of } f)$ for some $f \in \Phi\}$ is countable. Therefore there exists a countable subfield E_1 of $\mathbb{R}$ containing E as well as S. It has the property that

$$x \in E \cap (\text{domain of } f) \Rightarrow f(x) \in E_1 \quad \text{for all } f \in \Phi.$$

E_1 will be referred to as the *preliminary extension* of E. Now E_1 is itself a countable subfield of $\mathbb{R}$. Therefore it has a preliminary extension E_2 which is again countable. In this manner, we get a sequence $E_1 \subseteq E_2 \subseteq E_3 \subseteq \dots$ of countable subfields of $\mathbb{R}$ such that

$$x \in E_n \cap (\text{domain of } f) \Rightarrow f(x) \in E_{n+1} \quad \text{for all } f \in \Phi.$$

The union F of all the subfields E_n is then a countable subfield of $\mathbb{R}$ with the property that

$$x \in F \cap (\text{domain of } f) \Rightarrow f(x) \in F \quad \text{for all } f \in \Phi.$$

Thus F is adequate for all the functions in Φ. ☺

Conclusion: If we take E to be the rational number field, the countable subfield F as obtained above seems to fulfil all the needs of pre-Calculus Mathematics. The need to work with a field satisfying the Completeness Axiom becomes compelling only at the stage of Integral Calculus, when one wants every function whose derivative vanishes throughout an interval to be constant on that interval. Admittedly, our description of the subfield F requires prior knowledge of $\mathbb{R}$.

Chapter 9
The Riemann Integral

9-1 Introduction

Anyone who has tried to evaluate $\int e^{x^2} dx$ is aware that no simple expression can be found that represents a function $f(x)$ with derivative e^{x^2}. One can legitimately ask whether such a function exists at all. The answer, as we shall see later in this Chapter, is that such a function does indeed exist. This answer is only one of the applications of the powerful concept of integration discussed here.

The term "integration" will not mean finding a function whose derivative is known. This process will be referred to as "antidifferentiation". Thus a function f is an **antiderivative** of f' and, when the domain of f is an interval, any other antiderivative of f' must be of the form $f + C$, where C is a constant function [see Theorem 7-5.6]. Under certain broad conditions, the integral of a function can be evaluated via its antiderivative.

Recall from Def.6-2.3 that a partition P of an interval $[a,b]$ is any finite sequence of points $x_0, x_1, \ldots, x_n$ such that $a = x_0 < x_1 < \cdots < x_n = b$.

9-1.1. Definition. *Let the finite sequence P of points* $x_0, x_1, \ldots, x_n$ *be a partition of* $[a,b]$. *The subinterval* $[x_{j-1}, x_j]$ *is called the* $\mathbf{j^{th}}$ **subinterval** *of the partition. The length of the longest subinterval*,

$$\max\{x_j - x_{j-1} : 1 \le j \le n\},$$

is called the **norm** *of the partition P, and is denoted by* $|P|$.

Examples. (a) The sequence $\{x_0, x_1, x_2, x_3\}$, where $x_0 = 0$, $x_1 = 1$, $x_2 = 3$, $x_3 = 8$ is a partition of $[0,8]$ with three subintervals $[0,1], [1,3], [3,8]$. Their lengths are $1, 2, 5$. Therefore the norm of the partition is 5.

(b) The sequence $0, 1, 1\frac{1}{2}, 3, 8$ is a partition P of $[0,8]$. The third subinterval is $[1\frac{1}{2}, 3]$ and the norm is $|P| = 5$.

(c) If $0 < \varepsilon < 1$, the sequence $0, 1-\varepsilon, 1+\varepsilon, 5$ is a partition of $[0,5]$. The norm is $4 - \varepsilon$.

(d) For the interval $[0,5]$, the sequence

$$0,\ 1-\tfrac{1}{6},\ 1+\tfrac{1}{6},\ 2-\tfrac{1}{6},\ 2+\tfrac{1}{6},\ 3-\tfrac{1}{6},\ 3+\tfrac{1}{6},\ 5$$

is a partition. The subintervals containing 1, 2, 3 are

$$\left[1-\tfrac{1}{6},\ 1+\tfrac{1}{6}\right],\ \left[2-\tfrac{1}{6},\ 2+\tfrac{1}{6}\right] \text{ and } \left[3-\tfrac{1}{6},\ 3+\tfrac{1}{6}\right].$$

Their total length is 1.

(e) For the interval $[0,5]$ and a positive number ε, one possible partition such that the subintervals containing $1,2,3$ have total length less than or equal to ε is

$$0,\ \ 1-\tfrac{\varepsilon}{6},\ 1+\tfrac{\varepsilon}{6},\ 2-\tfrac{\varepsilon}{6},\ 2+\tfrac{\varepsilon}{6},\ 3-\tfrac{\varepsilon}{6},\ 3+\tfrac{\varepsilon}{6},\ 5,$$

but only if these numbers are in increasing order, that is to say, $\varepsilon < 3$. If it is not known whether $\varepsilon < 3$, then one takes a similar sequence but with ε replaced by $\min\{3,\varepsilon\}$.

For any partition $P : a = x_0 < x_1 < \cdots < x_n = b$ of $[a,b]$, the total length of all its subintervals is $b-a$. In other words, $\sum_{j=1}^{n}(x_j - x_{j-1}) = b-a$.

9-1.2. Proposition. [Needed in Prop.9-3.7] *Let $a = c_0 < c_1 < \cdots < c_n = b$ be a partition of $[a,b]$ and let $\varepsilon > 0$. Then there exists a partition of $[a,b]$ such that the total length of the subintervals containing $c_0, c_1, \ldots, c_n$ is less than or equal to ε and the remaining subintervals are n in number.*

Proof. Let $\eta > 0$ and consider the numbers

$$c_0,\ c_0+\eta,\ c_1-\eta,\ c_1+\eta,\ c_2-\eta,\ c_2+\eta,\ \ldots,\ c_n-\eta,\ c_n. \tag{1}$$

For them to be in increasing order, and thus form a partition of $[a,b]$, it is sufficient that

$$c_j+\eta < c_{j+1}-\eta \quad \text{for} \quad 0 \le j \le n-1,$$

which we ensure by requiring that

$$\eta < \tfrac{1}{2}\cdot\min\{c_{j+1}-c_j : 0 \le j \le n-1\}. \tag{2}$$

In the partition of $[a,b]$ formed by the numbers in (1), the subintervals $[c_0, c+\eta]$ and $[c_n-\eta, c_n]$ have length η, and the subintervals $[c_j-\eta, c_j+\eta]$ with $0 < j < n$ have length 2η each. Therefore the total length of the subintervals containing $c_0, c_1, \ldots, c_n$ is $2n\eta$. So, we also require that

$$\eta < \frac{\varepsilon}{2n}. \tag{3}$$

The remaining subintervals (those not containing any c_j) are $[c_{j-1}+\eta, c_j-\eta]$, where $1 \le j \le n$. These are n in number.

Thus by choosing the positive number η in accordance with (2) as well as (3), we find that the numbers in (1) provide the required partition. ☺

9-1.3. Definition. *Let $f: [a,b] \to \mathbb{R}$ be bounded and*

$$P: a = x_0 < x_1 < \cdots < x_n = b$$

be a partition of $[a,b]$. For $j = 1, \ldots, n$, let

$$m_j = \inf\{f(x) : x_{j-1} \le x \le x_j\} \quad \textit{and} \quad M_j = \sup\{f(x) : x_{j-1} \le x \le x_j\}.$$

Then the **lower sum of** *f over the partition P is*

$$L(f,P) = \sum_{j=1}^{n} m_j(x_j - x_{j-1})$$

and the **upper sum** *is*

$$U(f,P) = \sum_{j=1}^{n} M_j(x_j - x_{j-1}).$$

9-1.4. Remark. Since $m_j \le M_j$ for all j, we have $L(f,P) \le U(f,P)$ for any bounded function f and any partition P.

9-1.5. Examples. (a) Let $f: [1,5] \to \mathbb{R}$ be given by $f(x) = x^2$ and P be the partition consisting of the numbers $x_0 = 1$, $x_1 = 2$, $x_2 = 4$, $x_3 = 5$. Then

$$m_1 = \inf\{x^2 : 1 \le x \le 2\} = 1, \quad m_2 = \inf\{x^2 : 2 \le x \le 4\} = 4$$
$$\text{and} \quad m_3 = \inf\{x^2 : 4 \le x \le 5\} = 16.$$

Therefore
$$m_1(x_1 - x_0) = 1 \cdot (2-1) = 1,$$
$$m_2(x_2 - x_1) = 4 \cdot (4-2) = 8,$$
and
$$m_3(x_3 - x_2) = 16 \cdot (5-4) = 16,$$
so that
$$L(f,P) = 1 + 8 + 16 = 25.$$
Similarly,
$$M_1 = \sup\{x^2 : 1 \le x \le 2\} = 4, \quad M_2 = \sup\{x^2 : 2 \le x \le 4\} = 16$$
$$\text{and} \quad M_3 = \sup\{x^2 : 4 \le x \le 5\} = 25.$$
Therefore
$$M_1(x_1 - x_0) = 4 \cdot (2-1) = 4,$$
$$M_2(x_2 - x_1) = 16 \cdot (4-2) = 32,$$
and
$$M_3(x_3 - x_2) = 25 \cdot (5-4) = 25,$$
so that
$$U(f,P) = 4 + 32 + 25 = 61.$$

(b) Suppose f is as above but P contains one more point 3. Then there are 4 subintervals instead of 3; also the first and last subintervals are the same as above but there are new subintervals, namely $[2,3]$ and $[3,4]$. Their union is $[2,4]$, which was a subinterval of the partition above. Naturally, $m_1 = 1$, $m_2 = 4$ and $m_4 = 16$. However, $m_3 = 9$. Consequently

$$L(f,P) = 1 \cdot (2-1) + 4 \cdot (3-2) + 9 \cdot (4-3) + 16 \cdot (5-4)$$
$$= 1 + 4 + 9 + 16 = 30.$$

This is a larger value than what we obtained above for the partition having one point less than the present one. That value was $25 = 1 + 8 + 16$ while in the present situation, the 8 has been replaced by $4 + 9$. It is easy to see why a single term has been replaced by two terms, but there is a reason why the sum of the two terms is greater than the single term they replace. The reason will soon be discussed formally, but the reader is invited to think about it now.

It can also be verified that $U(f,P) = 4+9+16+25 = 54$. Thus the single term 32 in the sum for $U(f,P)$ above has been replaced by the two terms 9 + 16, whose sum is *smaller* than the single term they replace. The value of $U(f,P)$ now is therefore less than what it was above. Here again, there is a reason behind the decrease.

(c) Let $f: [a,b] \to \mathbb{R}$ be such that $f(x) = 0$ if $x \in \mathbb{Q}$ and $f(x) = 1$ otherwise. Since every interval of positive length contains a rational number as well as an irrational number, we have $m_j = 0$ and $M_j = 1$ for every j, whatever the partition P may be. It follows that $L(f,P) = 0$ and $U(f,P) = \sum_{j=1}^{n}(x_j - x_{j-1}) = b-a$ for every partition P.

In case the above function f seems somehow "artificial", it can be dressed up to look more natural by describing it without separating the cases $x \in \mathbb{Q}$ and $x \notin \mathbb{Q}$. Consider the sequence $\{f_m\}$ of functions defined on $\mathbb{R}$ by

$$f_m(x) = \lim_{n\to\infty}(\cos(m!\pi x))^{2n}.$$

For irrational x, the product $m!x$ is never an integer and hence $|\cos(m!\pi x)| < 1$, so that $f_m(x) = 0$. In contrast, for rational x, say p/q with $q \in \mathbb{N}$, $f_m(x) = 1$ for $m \geq q$. It follows that the sequence $\{f_m\}$ has pointwise limit f. Thus

$$f(x) = \lim_{m\to\infty}[\lim_{n\to\infty}(\cos(m!\pi x))^{2n}] \quad \text{for all real } x.$$

(d) Let $f: [0,1] \to \mathbb{R}$ be the function given by $f(0) = 2$ and $f(x) = 3$ for $0 < x \leq 1$. Consider the partition P: 0, ε, $1-\varepsilon$, 1, where $0 < \varepsilon < \frac{1}{2}$. We have

$$\begin{aligned}
m_1 &= \inf\{f(x) : 0 \leq x \leq \varepsilon\} = 2,\\
m_2 &= \inf\{f(x) : \varepsilon \leq x \leq 1-\varepsilon\} = 3,\\
m_3 &= \inf\{f(x) : 1-\varepsilon \leq x \leq 1\} = 3.
\end{aligned}$$

Hence $L(f,P) = 2\varepsilon + 3(1-\varepsilon) = 3-\varepsilon$. Also, $U(f,P') = 3$ for any partition P'. Thus $L(f,P) \leq U(f,P')$. Note the additional feature that

$$0 \leq U(f,P') - L(f,P) = \varepsilon.$$

(e) Let $f: [0,1] \to \mathbb{R}$ be the function given by $f(0) = 2$, $f(1) = 8$ and $f(x) = 3$ for $0 < x < 1$. Consider any partition P: $0 = x_0 < x_1 < \cdots < x_n = 1$. If $n = 1$, then $L(f,P) = 2$ and $U(f,P) = 8$. But if $n > 1$, we have

$$\begin{aligned}
m_1 &= \inf\{f(x) : x_0 \leq x \leq x_1\} = 2, & M_1 &= \sup\{f(x) : x_0 \leq x \leq x_1\} = 3\\
m_n &= \inf\{f(x) : x_{n-1} \leq x \leq x_n\} = 3, & M_n &= \sup\{f(x) : x_{n-1} \leq x \leq x_n\} = 8
\end{aligned}$$

and

$$m_j = \inf\{f(x) : x_{j-1} \leq x \leq x_j\} = \sup\{f(x) : x_{j-1} \leq x \leq x_j\} = M_j = 3 \text{ for } 1 < j < n.$$

It follows that

$$L(f,P) = 2(x_1 - x_0) + 3(x_n - x_1) = 3 - x_1 < 3$$

and $$U(f,P) = 3(x_{n-1} - x_0) + 8(x_n - x_{n-1}) = 8 - 5x_{n-1} > 3.$$

Since each of these inequalities holds true for an arbitrary partition P (even if $n = 1$), it follows that $L(f,P) \leq 3 \leq U(f,P')$ for any partitions P and P'.

For any $\varepsilon > 0$, there exists a partition P such that

$$0 < x_1 < \varepsilon \quad \text{and} \quad 0 < 1 - x_{n-1} < \varepsilon.$$

For such a partition P, we have

$$0 < 3 - L(f,P) < \varepsilon \quad \text{and} \quad 0 < U(f,P) - 3 < 5\varepsilon,$$

or equivalently,

$$3 - \varepsilon < L(f,P) < 3 < U(f,P) < 3 + 5\varepsilon.$$

Problem Set 9-1

9-1.P1. (a) What is the norm of the partition of $[1,6]$ given by $1,2,4,6$?
(b) Find all values of x such that the sequence $1,2,x,5,6$ is a partition of $[1,6]$ having norm 2.

9-1.P2. For the interval $[0,5]$ and a number $\varepsilon > 0$, give one example of a partition such that the total length of the intervals containing $0, 1, 4\frac{1}{2}, 5$ is less than ε.

9-1.P3. (a) For the function $f: [-1,1] \to \mathbb{R}$ defined by $f(x) = x^3$ and partition P consisting of $-1, 0, \frac{1}{2}, 1$, find $L(f,P)$ and $U(f,P)$.
(b) P is a partition of $[0,5]$ such that 1 and 3 are consecutive points of it and $L(f,P) = 43$, where $f(x) = x^2$ for all $x \in [0,5]$. For the partition P' obtained from P by including the point 2, find $L(f,P')$.

9-1.P4. (a) Let $f: [0,1] \to \mathbb{R}$ be the function given by $f(0) = 8$ and $f(x) = 3$ for $0 < x \leq 1$. Consider the partition P: $0, \varepsilon, 1-\varepsilon, 1$, where $0 < \varepsilon < \frac{1}{2}$. Show that $U(f,P) = 3 + 5\varepsilon$ and that $L(f,P') = 3$ for any partition P'.
(b) Let $f: [0,1] \to \mathbb{R}$ be the function given by $f(1) = 7$ and $f(x) = 3$ for $0 \leq x < 1$. Consider the partition P: $0, \varepsilon, 1-\varepsilon, 1$, where $0 < \varepsilon < \frac{1}{2}$. Show that $U(f,P) = 3 + 4\varepsilon$ and that $L(f,P') = 3$ for any partition P'.
(c) Let $f: [0,1] \to \mathbb{R}$ be the function given by $f(1) = 1$ and $f(x) = 3$ for $0 \leq x < 1$. Consider the partition P: $0, \varepsilon, 1-\varepsilon, 1$, where $0 < \varepsilon < \frac{1}{2}$. Show that $L(f,P) = 3 - 2\varepsilon$ and that $U(f,P') = 3$ for any partition P'.

9-1.P5. (a) Let $f: [0,2] \to \mathbb{R}$ be a function such that $f(x) = 3$ if $0 \leq x < 1$, $f(1) = 9$ and $f(x) = 4$ if $1 < x \leq 2$. For any positive $\varepsilon < 1$, the sequence

$$0,\ 1-\varepsilon, 1+\varepsilon, 2$$

is a partition P of $[0,2]$. Find $L(f,P)$ and $U(f,P)$.

(b) Let $f:[3,6]\to\mathbb{R}$ be a function such that $f(x) = 2$ if $3 \le x < 4$, $f(4) = 1$ and $f(x) = 4$ if $4 < x \le 6$. For any positive $\varepsilon < 1$, the sequence

$$3, 4-\varepsilon, 4+\varepsilon, 6$$

is a partition P of $[3,6]$. Find $L(f,P)$ and $U(f,P)$.

(b) $L(f,P) = 10-4\varepsilon$ and $U(f,P) = 10+2\varepsilon$.

9-1.P6. Let $f:[0,4]\to\mathbb{R}$ be a function such that $f(x) = 2$ if $0 \le x < 1$, $f(x) = 3$ if $1 < x < 3$, $f(x) = 5$ if $3 < x \le 4$, $f(1) = 8$ and $f(3) = 0$. For the partition

$$0, 1-\varepsilon, 1+\varepsilon, 3-\varepsilon, 3+\varepsilon, 4, \quad \text{where } 0<\varepsilon<1,$$

find $U(f,P)-L(f,P)$. Also, find a partition P' such that $U(f,P') - L(f,P') = 11\varepsilon$.

9-1.P7. Given the partition P: $x_j = \frac{j}{n}$, $0 \le j \le 4n$, of $[0,4]$, find $L(f,P)$ and $U(f,P)$, where $f(x) = x$ for every $x \in [0,4]$.

9-1.P8. (a) Let $f:[0,4]\to\mathbb{R}$ be defined as $f(x) = x$ if $x \ne 2$ and $f(2) = 6$. Find $L(f,P)$ and $U(f,P)$, with the partition

$$P: 0, 1, 2-\varepsilon, 2+\varepsilon, 3, 4, \quad \text{where } 0<\varepsilon<1.$$

(b) Let $f:[0,4]\to\mathbb{R}$ be defined as

$$f(x) = x \text{ if } x \ne 2 \text{ and } f(2) = A, \text{ where } A \text{ is a number greater than 3.}$$

Find $L(f,P)$ and $U(f,P)$ for the same partition as above.

9-1.P9. Let $f:[a,b]\to\mathbb{R}$ and $g:[a,b]\to\mathbb{R}$ be bounded functions and k be a real number such that $f(x) \le g(x)+k$ for $x \in [a,b]$. Show that, for any partition P of the interval, $U(f,P) \le U(g,P)+k(b-a)$. What is the corresponding inequality concerning $L(f,P)$ and $L(g,P)$?

9-1.P10. If $f:[a,b]\to\mathbb{R}$ is any bounded function, show that $-U(|f|,P) \le L(f,P) \le U(f,P) \le U(|f|,P)$ for every partition P.

9-1.P11. Let $f:[0,1]\to\mathbb{R}$ be the function defined by $f(x) = \frac{1}{1+x}$. For the partition P of $[0,1]$ consisting of the points $x_j = \frac{j}{n}$, $0 \le j \le n$, show that $U(f,P) = \sum_{j=0}^{n-1} \frac{1}{n+j}$. Find a similar expression for the lower sum.

9-1.P12. [Needed in 9-2.P7.] Let $f:[a,b]\to\mathbb{R}$ and $g:[a,b]\to\mathbb{R}$ be bounded nonnegative functions such that $L(g,P) = 0$ for some partition P of $[a,b]$. Show that $L(fg,P) = 0$.

9-2 Definition of the Riemann Integral

In Example (b) of 9-1.5, we saw that adding one more point to a partition increased the lower sum and decreased the upper sum. We shall now show that this must always happen but with the proviso that the actual quantum of increase or decrease can be zero.

9-2.1. Definition. *Let P be a partition of* $[a,b]$. *A* **refinement** *of P is a partition P' of* $[a,b]$ *such that every point of P is also a point of P'.*

It is easy to see that $|P'| \le |P|$ when P' is a refinement of P. The second part of the next result, involving the norm, will not be needed until Section 10-1.

9-2.2. Proposition. *Let P be a partition of* $[a,b]$ *and* $f:[a,b]\to\mathbb{R}$ *be a bounded function. For any refinement P' of P,*

$$L(f,P) \le L(f,P') \le U(f,P') \le U(f,P).$$

If P' contains k points more than P, then

$$L(f,P') - L(f,P) \le 2k\cdot M|P| \quad \textit{and} \quad U(f,P) - U(f,P') \le 2k\cdot M|P|,$$

where $M = \sup\{|f(x)| : a \le x \le b\}$.

Proof. Since $L(f,P') \le U(f,P')$ for any partition P', we need establish only the other four inequalities. In case $P' = P$, there is nothing to prove. We shall show that, if P' contains one more point than P, then the other inequalities hold. It will then follow by an induction argument [which need not be written out] that they also hold when P' has several more points than P.

Let P consist of the points $a = x_0 < x_1 < \cdots < x_n = b$ and P' consist of all these points and one more point ξ. This point ξ must lie in one of the subintervals $[x_{j-1},x_j]$, so that $[x_{j-1},x_j] = [x_{j-1},\xi]\cup[\xi,x_j]$. Therefore $[x_{j-1},x_j]$ is not a subinterval of P', but instead $[x_{j-1},\xi]$ and $[\xi,x_j]$ are. All other subintervals are the same for P and P'. Let

$$m_j = \inf\{f(x) : x_{j-1} \le x \le x_j\},\ \mu_1 = \inf\{f(x) : x_{j-1} \le x \le \xi\}$$
$$\text{and } \mu_2 = \inf\{f(x) : \xi \le x \le x_j\}.$$

Then

$$L(f,P') - L(f,P) = \mu_1(\xi - x_{j-1}) + \mu_2(x_j - \xi) - m_j(x_j - x_{j-1}). \tag{1}$$

Since $\{f(x) : x_{j-1} \le x \le \xi\}$ and $\{f(x) : \xi \le x \le x_j\}$ are both subsets of $\{f(x) : x_{j-1} \le x \le x_j\}$, we also have

$$m_j \le \mu_1 \quad \text{and} \quad m_j \le \mu_2. \tag{2}$$

It follows from (1) and (2) that

$$L(f,P') - L(f,P) \geq m_j(\xi - x_{j-1}) + m_j(x_j - \xi) - m_j(x_j - x_{j-1})$$
$$\geq m_j[(\xi - x_{j-1}) + (x_j - \xi) - (x_j - x_{j-1})] = m_j \cdot 0 = 0$$

and that

$$L(f,P') - L(f,P) = (\mu_2 - m_j)(x_j - \xi) + (\mu_1 - m_j)(\xi - x_{j-1})$$
$$\leq 2M((x_j - \xi) + (\xi - x_{j-1}))$$
$$\leq 2M(x_j - x_{j-1}) \leq 2M|P|.$$

The proofs concerning upper sums are similar. ☺

9-2.3. Proposition. *Suppose $f: [a,b] \to \mathbb{R}$ is a bounded function and let P and P' be partitions of $[a,b]$. Then $L(f,P) \leq U(f,P')$. In other words, any lower sum of f is less than or equal to any upper sum of f, even if the partitions involved are different.*

Proof. Let P'' be the partition consisting of all the points of P and P' together. Then P'' is a refinement of both P and P'. It follows from Prop.9-2.2 that $L(f,P) \leq L(f,P'') \leq U(f,P'') \leq U(f,P')$. ☺

9-2.4. Remark. For a bounded function $f: [a,b] \to \mathbb{R}$, it follows from the above Proposition that the set of lower sums generated by all possible partitions of $[a,b]$ is bounded above by any one upper sum, and therefore has a supremum; similarly, the set of all possible upper sums is bounded below by any one lower sum, and therefore has an infimum.

9-2.5. Definition. *For a bounded function $f: [a,b] \to \mathbb{R}$, the supremum of the set of all lower sums is called the* **lower Riemann integral of** *f over $[a,b]$ and is denoted by $\underline{\int_a^b} f$. The infimum of all upper sums is called the* **upper Riemann integral of** *f over $[a,b]$ and is denoted by $\overline{\int_a^b} f$.*

9-2.6. Proposition. *If $f: [a,b] \to \mathbb{R}$ is any bounded function,*

$$\underline{\int_a^b} f \leq \overline{\int_a^b} f.$$

Proof. Let P be any partition of $[a,b]$. Then by Prop.9-2.3,

$$L(f,P) \leq U(f,P') \text{ for any partition } P' \text{ of } [a,b].$$

Since this inequality holds for any partition P' of $[a,b]$, we have

$$L(f,P) \leq \overline{\int_a^b} f \text{ for any partition } P \text{ of } [a,b].$$

Since this, in turn, holds for any partition P of $[a,b]$, we have $\underline{\int_a^b} f \leq \overline{\int_a^b} f$. ☺

9-2.7. Examples. (a) Consider the function $f:[a,b]\to\mathbb{R}$ such that $f(x) = 0$ if $x \in \mathbb{Q}$ and $f(x) = 1$ otherwise. It was noted in Example 9-1.5(c) that $L(f,P) = 0$ and $U(f,P) = b-a$ for every partition P. It follows that $\underline{\int_a^b} f = 0$ and $\overline{\int_a^b} f = b-a$.

(b) Let $f:[0,1]\to\mathbb{R}$ be the function given by $f(0) = 2$ and $f(x) = 3$ for $0 < x \le 1$. Consider the partition P: $0, \varepsilon, 1-\varepsilon, 1$, where $0 < \varepsilon < \frac{1}{2}$. It was shown in Example 9-1.5(d) that $L(f,P) = 3-\varepsilon$ and also that $U(f,P') = 3$ for any partition P'. Thus $L(f,P) \le U(f,P')$. In view of the fact that ε here can be *any* positive number less than $\frac{1}{2}$, the supremum of the set of all lower sums is greater than or equal to $\sup\{3-\varepsilon : \varepsilon > 0\}$, which is 3. Hence $\underline{\int_0^1} f \ge 3$. Also, $\overline{\int_0^1} f = 3$. It now follows from Prop.9-2.6 that $\underline{\int_0^1} f = \overline{\int_0^1} f = 3$.

Note that $0 \le U(f,P') - L(f,P) = \varepsilon$.

(c) Let $f:[0,1]\to\mathbb{R}$ be the function given by $f(0) = 2, f(1) = 8$ and $f(x) = 3$ for $0 < x < 1$. As seen in Example 9-1.5(e), there exists a partition P such that

$$3-\varepsilon < L(f,P) < 3 < U(f,P) < 3+5\varepsilon.$$

In view of the fact that ε here can be *any* positive number, the supremum of the set of all lower sums is greater than or equal to $\sup\{3-\varepsilon : \varepsilon > 0\}$, which is 3. Hence $\underline{\int_0^1} f \ge 3$. Similarly, the infimum of the set of all upper sums is less than or equal to $\inf\{3+5\varepsilon : \varepsilon > 0\}$, which is 3. Hence $\overline{\int_0^1} f \le 3$. It now follows from Prop.9-2.6 that $\underline{\int_0^1} f = \overline{\int_0^1} f = 3$.

Note that $0 \le U(f,P) - L(f,P) < 6\varepsilon$.

(d) Let $f:[0,2]\to\mathbb{R}$ be a function such that

$$f(x) = 4 \text{ if } 0 \le x < 1, \quad f(1) = 8 \quad \text{and} \quad f(x) = 6 \text{ if } 1 < x \le 2.$$

For any positive $\varepsilon < 1$, the sequence

$$0, 1-\varepsilon, 1+\varepsilon, 2$$

is a partition P of $[0,2]$. Moreover, $m_1 = 4$, $m_2 = 4$, $m_3 = 6$, while $M_1 = 4$, $M_2 = 8$, $M_3 = 6$. Therefore

$$L(f,P) = 4(1-\varepsilon) + 4(2\varepsilon) + 6(1-\varepsilon) = 10-2\varepsilon$$

and

$$U(f,P) = 4(1-\varepsilon) + 8(2\varepsilon) + 6(1-\varepsilon) = 10+6\varepsilon.$$

In view of the fact that ε here can be *any* positive number less than 1, the supremum of the set of all lower sums is greater than or equal to $\sup\{10-2\varepsilon : \varepsilon > 0\}$, which is 10. Hence $\underline{\int_0^2} f \ge 10$, and similarly, $\overline{\int_0^2} f \le 10$. It now follows from Prop.9-2.6 that $\underline{\int_0^2} f = \overline{\int_0^2} f = 10$.

We note that $0 \le U(f,P) - L(f,P) = 8\varepsilon$.

9-2.8. Definition. *A bounded function* $f:[a,b]\to\mathbb{R}$ *is called* **Riemann integrable** *on* $[a,b]$ *if* $\underline{\int_a^b} f = \overline{\int_a^b} f$. *We shall abbreviate this as "R-integrable" or simply "integrable" whenever convenient. The common value of the upper and lower integrals is denoted by* $\int_a^b f$ *and is called the* **Riemann integral of** *the function f* **from *a* to *b***.

It is customary to denote the Riemann integral of the restriction of f to a subinterval $[\alpha,\beta]$ of $[a,b]$ by the symbol $\int_\alpha^\beta f$ without introducing a symbol for the restriction. Often one works with a function through an expression for $f(x)$, such as x^3 without introducing a symbol for the function. It is then more convenient to write the integral as $\int_a^b x^3\,dx$, or more generally, $\int_a^b f(x)\,dx$.

Example (a) above describes a function that is not integrable, while (b) to (d) describe integrable functions. We shall later establish that all step functions are integrable.

9-2.9. Proposition. *A bounded function* $f:[a,b]\to\mathbb{R}$ *is integrable if, and only if, for every* $\varepsilon>0$, *there exists a partition P of* $[a,b]$ *such that*

$$U(f,P)-L(f,P)<\varepsilon. \tag{A}$$

Such a partition satisfies

$$\int_a^b f-\varepsilon<L(f,P)\le U(f,P)<\int_a^b f+\varepsilon. \tag{B}$$

Proof. First suppose f is integrable and let ε be any positive number. By definition of lower and upper Riemann integrals, there exists some partitions P_1 and P_2 such that

$$\underline{\int_a^b} f-L(f,P_1)<\tfrac{\varepsilon}{2} \quad\text{and}\quad U(f,P_2)-\overline{\int_a^b} f<\tfrac{\varepsilon}{2}.$$

Let P be the partition consisting of all the points of P_1 and P_2 together. Then P is a refinement of both P_1 and P_2. It follows from Prop.9-2.2 that

$$\underline{\int_a^b} f-L(f,P)<\tfrac{\varepsilon}{2} \quad\text{and}\quad U(f,P)-\overline{\int_a^b} f<\tfrac{\varepsilon}{2}.$$

When f is integrable [which means $\underline{\int_a^b} f=\overline{\int_a^b} f$], it follows from these inequalities that (A) holds.

Conversely, suppose that for every $\varepsilon>0$, a partition P satisfying (A) exists. By Prop.9-2.6 and the definition of lower and upper Riemann integrals,

$$L(f,P)\le\underline{\int_a^b} f\le\overline{\int_a^b} f\le U(f,P).$$

Hence $0 \le \overline{\int_a^b} f - \underline{\int_a^b} f < U(f,P) - L(f,P) < \varepsilon$. Since this is true for every $\varepsilon > 0$, $\underline{\int_a^b} f = \overline{\int_a^b} f$, i.e. f is integrable.

The inequalities $L(f,P) \le \underline{\int_a^b} f = \int_a^b f = \overline{\int_a^b} f \le U(f,P)$ lead to

$$\int_a^b f - L(f,P) \le U(f,P) - L(f,P) \quad \text{and} \quad U(f,P) - \int_a^b f \le U(f,P) - L(f,P),$$

which implies that (B) holds whenever (A) holds. ☺

Remark. In the trivial case when $b = a$, so that the interval $[a,b]$ consists of a single point and every function is bounded, we define $\underline{\int_a^b} f = 0 = \overline{\int_a^b} f$, with the consequence that $\int_a^b f = 0$. Thus $\int_c^c f = 0$ for any c in the domain of f.

9-2.10. Proposition. *If $a \le c \le b$ and $f:[a,b] \to \mathbb{R}$ is a function such that its restrictions to $[a,c]$ and to $[c,b]$ are both integrable, then f is also integrable and*

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

[The converse of this result will be dealt with in Prop.9-4.8.]

Proof. The cases when $c = a$ or $c = b$ are trivial in view of the Remark above. So we assume $a < c < b$.

Denote the restrictions to $[a,c]$ and $[c,b]$ by f_1 and f_2 respectively. Then $\int_a^c f$ and $\int_c^b f$ mean the same thing as $\int_a^c f_1$ and $\int_c^b f_2$ respectively.

By Prop.9-2.9, for any $\varepsilon > 0$, there exist partitions P_1 and P_2 of $[a,c]$ and $[c,b]$ respectively such that

$$0 \le U(f_1,P_1) - L(f_1,P_1) < \tfrac{\varepsilon}{2} \quad \text{and} \quad 0 \le U(f_2,P_2) - L(f_2,P_2) < \tfrac{\varepsilon}{2}.$$

Now the points of P_1 and P_2 taken together form a partition P of $[a,b]$ such that

$$L(f,P) = L(f_1,P_1) + L(f_2,P_2) \quad \text{and} \quad U(f,P) = U(f_1,P_1) + U(f_2,P_2).$$

Together with the foregoing inequalities, this shows that

$$0 \le U(f,P) - L(f,P) < \varepsilon.$$

Since such a partition has been shown to exist for any $\varepsilon > 0$, it follows by Prop.9-2.9 that f is integrable.

It also follows from the above inequalities that

$$L(f_1,P_1) > U(f_1,P_1) - \tfrac{\varepsilon}{2} \ge \overline{\int_a^c} f - \tfrac{\varepsilon}{2} = \int_a^c f - \tfrac{\varepsilon}{2},$$

and

$$L(f_2,P_2) > U(f_2,P_2) - \tfrac{\varepsilon}{2} \ge \overline{\int_c^b} f - \tfrac{\varepsilon}{2} = \int_c^b f - \tfrac{\varepsilon}{2}.$$

Hence $L(f,P) \geq \int_a^c f + \int_c^b f - \varepsilon$, so that $\underline{\int_a^b} f \geq \int_a^c f + \int_c^b f - \varepsilon$. Since f has already been shown to be integrable, this inequality states that

$$\int_a^b f \geq \int_a^c f + \int_c^b f - \varepsilon.$$

This holds for every $\varepsilon > 0$, and therefore $\int_a^b f \geq \int_a^c f + \int_c^b f$. A similar argument shows that $\int_a^b f \leq \int_a^c f + \int_c^b f$. ☺

Recall the following fact from the visual interpretation of the integral as area in elementary Calculus: The integral of a step function is the sum of areas of finitely many rectangles; the values taken only at the endpoints of subintervals of constancy play no role. A formal statement and proof of this fact based on Def.9-2.8 requires some effort, which we present now:

9-2.11. Theorem. *A step function* $f:[a,b]\to\mathbb{R}$ *is integrable. Moreover, if the numbers* $x_0, x_1, \ldots, x_n$ *form a partition of* $[a,b]$ *such that* $f(x) = T_j$ *on the open interval* (x_{j-1}, x_j), $1 \leq j \leq n$, *then*

$$\int_a^b f = \sum_{j=1}^{n} T_j(x_j - x_{j-1}). \tag{A}$$

Proof. We proceed by induction on n.

Suppose $n = 1$. Then the function takes a constant value T_1 on (a,b). Suppose that

$$M > \max\{|f(a)|, |f(b)|, |T_1|\},$$

i.e. $M > \sup|f(x)|$. Let ε be any positive number and $\eta = \min\{\frac{b-a}{2}, \varepsilon/4M\}$. Consider the partition P of $[a,b]$ consisting of the points $a, a+\eta, b-\eta, b$. We have

$$\inf\{f(x) : a \leq x \leq a+\eta\} \geq -M, \quad \inf\{f(x) : b-\eta \leq x \leq b\} \geq -M$$
$$\text{and} \quad \inf\{f(x) : a+\eta \leq x \leq b-\eta\} = T_1.$$

Consequently,
$$L(f,P) > -M\eta + T_1((b-\eta)-(a+\eta)) - M\eta$$
$$\geq -4M\eta + T_1(b-a) > -\varepsilon + T_1(b-a).$$

Since ε is any positive number, it follows from the definition of the lower Riemann integral that $\underline{\int_a^b} f \geq T_1(b-a)$. By a similar argument, it follows that $\overline{\int_a^b} f \leq T_1(b-a)$. By Prop.9-2.6, we conclude that

$$\underline{\int_a^b} f = \overline{\int_a^b} f = T_1(b-a) = \sum_{j=1}^{n} T_j(x_j - x_{j-1}) \text{ when } n = 1.$$

Now assume (induction hypothesis) for some $k \in \mathbb{N}$ that the result holds for any closed bounded interval and any step function on it that is constant on subintervals of a partition having k subintervals.

Let $f: [a, b] \to \mathbb{R}$ be a step function constant on the subintervals of the partition

$$a = x_0 < x_1 < \cdots < x_k < x_{k+1} = b$$

having $k+1$ subintervals, taking values T_j on (x_{j-1}, x_j), $1 \le j \le k+1$. By the induction hypothesis, the restriction of f to $[a, x_k]$ is integrable and its Riemann integral is

$$\int_a^{x_k} f = \sum_{j=1}^{k} T_j(x_j - x_{j-1}). \tag{1}$$

According to what has been proved for the case $k = 1$, the restriction of f to $[x_k, x_{k+1}]$ is also integrable and

$$\int_{x_k}^{b} f = \int_{x_k}^{x_{k+1}} f = T_{k+1}(x_{k+1} - x_k). \tag{2}$$

By Prop.9-2.10, f is integrable and $\int_a^b f = \int_a^{x_k} f + \int_{x_k}^{b} f$. Therefore in view of (1) and (2), we have $\int_a^b f = \sum_{j=1}^{k+1} T_j(x_j - x_{j-1})$, as required. ☺

Problem Set 9-2

9-2.P1. (a) Let $f: [0, 1] \to \mathbb{R}$ be the function given by $f(0) = 8$ and $f(x) = 3$ for $0 < x \le 1$. Consider the partition P: 0, ε, 1 − ε, 1, where $0 < \varepsilon < \frac{1}{2}$ and show that $\underline{\int_0^1} f = \overline{\int_0^1} f = 3$.

(b) Let $f: [0, 1] \to \mathbb{R}$ be the function given by $f(1) = 1$ and $f(x) = 3$ for $0 \le x < 1$. Consider the partition P: 0, ε, 1 − ε, 1, where $0 < \varepsilon < \frac{1}{2}$ and show that $\underline{\int_0^1} f = \overline{\int_0^1} f = 3$.

9-2.P2. (a) Let $f: [0, 2] \to \mathbb{R}$ be a function such that $f(x) = 3$ if $0 \le x < 1$, $f(1) = 9$ and $f(x) = 4$ if $1 < x \le 2$. Find $\underline{\int_0^2} f$ and $\overline{\int_0^2} f$.

(b) Let $f: [3, 6] \to \mathbb{R}$ be a function such that $f(x) = 2$ if $3 \le x < 4$, $f(4) = 1$ and $f(x) = 4$ if $4 < x \le 6$. Find $\underline{\int_3^6} f$ and $\overline{\int_3^6} f$.

9-2.P3. Find $\underline{\int_0^4} f$ and $\overline{\int_0^4} f$, where $f(x) = x$ for all $x \in [0, 4]$.

9-2.P4. Let $f: [a, b] \to \mathbb{R}$ be the function such that $f(x) = 0$ if $x \in \mathbb{Q}$ and $f(x) = 1$ otherwise. Find $\underline{\int_a^b} f$ and $\overline{\int_a^b} f$.

9-2.P5. Let $f:[a,b]\to\mathbb{R}$ be any bounded function and $a < c < b$. Show that $\underline{\int_a^b} f = \underline{\int_a^c} f + \underline{\int_c^b} f$. Determine whether the corresponding equality holds for upper Riemann integrals.

9-2.P6. For a bounded function $f:[a,b]\to\mathbb{R}$, prove that $|\overline{\int_a^b} f| \le \overline{\int_a^b} |f|$.

9-2.P7. Let $f:[a,b]\to\mathbb{R}$ and $g:[a,b]\to\mathbb{R}$ be bounded nonnegative functions such that $\int_a^b g = 0$. Show that $\int_a^b fg = 0$.

9-2.P8. Suppose f is a Riemann integrable function on $[a,b]$. Prove that

$$\lim_{m\to\infty} \int_a^b f(x)\sin mx\,dx = 0 \text{ and } \lim_{m\to\infty} \int_a^b f(x)\cos mx\,dx = 0.$$

9-2.P9. Let $I_1, I_2, \ldots, I_n$ be n subintervals of $[0,1]$. If each point of $[0,1]$ belongs to at least q of these intervals, show that one of the intervals has length greater than or equal to q/n.

9-2.P10. Suppose $f:[0,1]\to\mathbb{R}$ is Riemann integrable on $[b,1]$ for every $b > 0$. If f is bounded, show that it is Riemann integrable.

9-2.P11. If $p > 0$ and f is a bounded nonnegative function on $[a,b]$ with $\underline{\int_a^b} f^p = 0$, show that $\underline{\int_a^b} f = 0$.

9-3 Riemann Integrable Functions

At the end of the preceding Section, it was shown that every step function is integrable. We now do the spadework for deducing integrability of a larger class of functions that includes continuous as well as many other functions.

9-3.1. Theorem. *Let $f:[a,b]\to\mathbb{R}$ be a uniform limit of a sequence of integrable functions. Then f is integrable.* Briefly put: *A uniform limit of integrable functions is integrable.*

Moreover, $\int_a^b f = \lim_{n\to\infty} \int_a^b f_n$, *where* $\{f_n\}$ *is any sequence of integrable functions on* $[a,b]$ *converging uniformly to* f.

Proof. Let $\{f_n\}$ be a sequence of integrable functions on $[a,b]$ converging uniformly to f. Since each f_n is bounded [note that integrability is defined only for bounded functions], the uniform limit f is bounded by Prop.5-1.7.

Now consider any $\varepsilon > 0$ and let $\eta = \frac{1}{3}\left(\frac{\varepsilon}{b-a}\right)$. By definition of uniform convergence [see Def.5-1.1], there exists $n_0 \in \mathbb{N}$ such that

$$|f_n(x) - f(x)| < \eta \text{ whenever } n \ge n_0 \text{ and } x \in [a,b]. \tag{1}$$

Now consider any $n \geq n_0$. Since f_n is integrable, by Prop.9-2.9 there exists a partition P such that

$$U(f_n, P) - L(f_n, P) < \eta(b-a). \tag{2}$$

By (1), we have $f_n(x) - \eta \leq f(x) \leq f_n(x) + \eta$ for all $x \in [a, b]$. It follows that, for any subinterval J of the partition P of $[a, b]$,

$$\inf\{f_n(x) : x \in J\} - \eta \leq \inf\{f(x) : x \in J\}$$

and
$$\sup\{f(x) : x \in J\} \leq \sup\{f_n(x) : x \in J\} + \eta.$$

Consequently,
$$L(f, P) \geq L(f_n, P) - \eta(b-a) \tag{3}$$

and
$$U(f, P) \leq U(f_n, P) + \eta(b-a). \tag{4}$$

Together with (2), these two inequalities imply that

$$U(f, P) - L(f, P) < 3\eta(b-a) = \varepsilon.$$

The existence of such a partition P shows by Prop.9-2.9 that the function f is integrable.

To establish the second part, we note that

$$\int_a^b f = \overline{\int_a^b} f \leq U(f, P) \leq U(f_n, P) + \eta(b-a) \qquad \text{by (4)}$$

and
$$-\int_a^b f_n \leq -L(f_n, P).$$

Hence
$$\int_a^b f - \int_a^b f_n \leq U(f_n, P) - L(f_n, P) + \eta(b-a) < 2\eta(b-a) < \varepsilon$$

by (2). By a similar argument, using (3) in place of (4), we get

$$\int_a^b f - \int_a^b f_n > -\varepsilon.$$

From the preceding two inequalities, we conclude that $\left| \int_a^b f - \int_a^b f_n \right| < \varepsilon$ for all $n \geq n_0$. The existence of such an n_0 for any $\varepsilon > 0$ shows that

$$\int_a^b f = \lim_{n\to\infty} \int_a^b f_n. \ ☺$$

Example. If a sequence $\{f_n\}$ of integrable functions converges to a limit f pointwise but not uniformly, then f need not be integrable. Consider the sequence defined as follows. To begin with, take any sequence $\{r_n\}$ whose range consists of all the rational numbers in $[0, 1]$. Such a sequence exists because $\mathbb{Q}$ is denumerable by Theorem 3-3.8 and hence its infinite subset $\mathbb{Q} \cap [0, 1]$ is also denumerable. Let $f_n(x) = 0$ if $x = r_k$ for some $k \leq n$ and $f_n(x) = 1$ otherwise. Then each f_n is a step function and thus integrable. The pointwise limit of the sequence is the non-integrable function in Example 9-2.7(a). The reader may wish to note that the sequence $\{f_n\}$ converges "monotonically" in the sense that, for each $x \in [0, 1]$, we have $f_n(x) \geq f_{n+1}(x)$.

We are about to use the above Theorem to establish the integrability of continuous functions and of monotone functions using Prop.6-2.7. However, it is customary to prove their integrability without using the latter and we present such an argument in 9-3.5&6 below. Before proceeding, we note that a monotone function f with domain a closed bounded interval $[a, b]$ is bounded, because all its values lie between $f(a)$ and $f(b)$.

9-3.2. Corollary. *A function with domain* $[a, b]$ *and having a one sided limit at each point is Riemann integrable.*

***Proof.** By Prop.6-2.7, such a function is a uniform limit of step functions and step functions are Riemann integrable according to Prop.9-2.11. Hence such a function must be integrable by Theorem 9-3.1. ☺

It is true that a function on a closed bounded interval and having either a left limit at each point or a right limit at each point is Riemann integrable on that interval. However we shall not prove this here.

9-3.3. Theorem. *Continuous functions and monotone functions whose domain is a closed bounded interval are integrable on that domain.*

***Proof.** Suppose $f: [a, b] \to \mathbb{R}$ is continuous. Then its right limit at any point $\alpha \in [a, b)$ is $f(\alpha)$ and left limit at any $\alpha \in (a, b]$ is $f(\alpha)$. Thus it has a one sided limit at each point and, by Cor.9-3.2, it must be integrable.

Now suppose $f: [a, b] \to \mathbb{R}$ is monotone. To be specific, suppose it is increasing. For any $\alpha \in [a, b)$, let $A = \inf\{f(x) : \alpha < x < b\}$; we shall show that $f(\alpha+) = A$. For this purpose, consider any $\varepsilon > 0$. By the definition of inf, there exists some $x_0 > \alpha$ such that $A > f(x_0) - \varepsilon$. Let $\delta = x_0 - \alpha > 0$. Then

$$\alpha < x < \alpha + \delta \Rightarrow \alpha < x < x_0 \Rightarrow x \in [a, b) \quad \text{and also}$$
$$\Rightarrow A \leq f(x) \leq f(x_0) < A + \varepsilon \Rightarrow 0 \leq f(x) - A < \varepsilon \Rightarrow |f(x) - A| < \varepsilon.$$

Therefore $f(\alpha+) = A$. By a similar argument, $f(\alpha-) = \sup\{f(x) : a < x < \alpha\}$ for any $\alpha \in (a, b]$. Thus f has a one sided limit at each point if it is increasing. The same can be shown to be true of a decreasing function f. Thus any monotone function is integrable by Cor.9-3.2. ☺

9-3.4. Examples. (a) Let $f: [0, 2] \to \mathbb{R}$ be defined as $x - [x]$ for $x < 1$ and $2 - x$ for $1 \leq x$, where [ ] denotes the integer part function. Then f is neither continuous nor monotone. However the restrictions to $[0,1]$ and $[1,2]$ are both monotone and are therefore integrable by Theorem 9-3.3. It follows by Prop.9-2.10 that f is integrable. Alternatively, one can note that f has a one sided limit at each point and Cor.9-3.2 shows that it is integrable.

*(b) We now give two examples of functions neither of which satisfies the hypotheses of Theorem 9-3.3 on any subinterval. Their integrability can however be proved by using Cor.9-3.2. The first example is the restriction to $[a,b]$ of the function of 6-2.P4 with real c_j; as discussed there, the function has a one sided limit at each point and Cor.9-3.2 shows that it is integrable. As a second example, let f be the restriction to $[a,b]$ of the function of Example 2-1.4(d). Recall the special feature of this function that it is continuous at every irrational point and discontinuous at every rational point. Consequently it has a one sided limit at every irrational point and we shall now argue that it has a one sided limit at every rational point as well. More precisely, $f(c+) = f(c-) = 0$ for every rational c. Let $c = j/k$, where k is the least such positive integer, and let $\varepsilon > 0$. There exists some $n_0 \in \mathbb{N}$ such that $n_0 > \max\{k, \frac{1}{\varepsilon}\}$. Consider the interval $(c-1/2n_0^2, c+1/2n_0^2)$. For any p and q in this interval, $|p-q| < 1/n_0^2$. It follows that this interval can contain at most one rational number of the form $r = m/n$ with $n \le n_0$, because when $m_1 n_2 - m_2 n_1 \ne 0$, we have

$$n_1 \le n_0,\ n_2 \le n_0 \Rightarrow |m_1/n_1 - m_2/n_2| = |m_1 n_2 - m_2 n_1|/n_1 n_2 \ge 1/n_1 n_2 \ge 1/n_0^2.$$

For every rational number $r = m/n$ in the interval $(c-1/2n_0^2, c+1/2n_0^2)$ other than c, we therefore have $n > n_0$. Hence

$$0 < |x-c| < 1/2n_0^2 \Rightarrow \text{ either } f(x) = 0 \text{ or } f(x) = 1/n \text{ with } n > n_0 \Rightarrow |f(x)| < \varepsilon.$$

This shows that $f(c+) = f(c-) = 0$.

Since f has a one sided limit at each point of its domain, rational as well as irrational, it is integrable by Cor.9-3.2.

Here are the proofs of the integrability of continuous functions and of monotone functions without using Prop.6-2.7:

9-3.5. Proof that a continuous function is integrable. Consider a continuous function $f:[a,b]\to\mathbb{R}$ and any $\varepsilon > 0$. By the Boundedness Theorem 2-3.2, f is bounded. Moreover, by the Uniform Continuity Theorem 2-5.3, f is uniformly continuous. Hence there exists some $\delta > 0$ such that $|s-t| < \delta \Rightarrow |f(s)-f(t)| < \frac{\varepsilon}{b-a}$. Now, let $n > \frac{b-a}{\delta}$ and P be the partition consisting of $x_j = a + j\frac{b-a}{n}$, $0 \le j \le n$. Then for any j, we have

$$s,t \in [x_{j-1},x_j] \Rightarrow |s-t| \le |x_j - x_{j-1}| = \frac{b-a}{n} < \delta \Rightarrow |f(s)-f(t)| < \frac{\varepsilon}{b-a}.$$

Let $m_j = \inf\{f(x) : x_{j-1} \le x \le x_j\}$ and $M_j = \sup\{f(x) : x_{j-1} \le x \le x_j\}$. By Theorem 2-3.3, there exist s_j and t_j in $[x_{j-1},x_j]$ such that $f(s_j) = M_j$ and $f(t_j) = m_j$. We now have $0 \le M_j - m_j = f(s_j) - f(t_j) \le \frac{\varepsilon}{b-a}$. Therefore

$$U(f,P) - L(f,P) = \sum_{j=1}^{n}(M_j - m_j)(x_j - x_{j-1}) \le \frac{\varepsilon}{b-a}\sum_{j=1}^{n}(x_j - x_{j-1}) = \frac{\varepsilon}{b-a}(b-a) = \varepsilon.$$

By Prop.9-2.9, it follows that f is integrable. ☺

9-3.6. Proof that a monotone function is integrable. Let $f:[a,b]\to\mathbb{R}$ be increasing and $\varepsilon > 0$. Consider an integer

$$n > \frac{1}{\varepsilon}(b-a)(f(b)-f(a))$$

and let P be the partition consisting of the points $x_j = a + j\frac{b-a}{n}$, $0 \le j \le n$. Since f is increasing, for each j, we have

$$M_j = \sup\{f(x) : x_{j-1} \le x \le x_j\} = f(x_j)$$

and

$$m_j = \inf\{f(x) : x_{j-1} \le x \le x_j\} = f(x_{j-1}).$$

Therefore

$$\begin{aligned}U(f,P) - L(f,P) &= \sum_{j=1}^{n}(M_j - m_j)(x_j - x_{j-1}) = \sum_{j=1}^{n}(f(x_j) - f(x_{j-1}))(x_j - x_{j-1})\\ &= \frac{b-a}{n}\sum_{j=1}^{n}(f(x_j) - f(x_{j-1})) = \frac{1}{n}(b-a)(f(b)-f(a))\\ &< \varepsilon \quad \text{by choice of } n.\end{aligned}$$

By Prop.9-2.9, it follows that f is integrable. The argument for decreasing f is similar. ☺

The preceding theorems by no means describe all possible integrable functions. The bounded function $f:[0,1]\to\mathbb{R}$ defined by

$$f(0) = 0 \qquad \text{and} \qquad f(x) = 2x\sin\frac{1}{x} - \cos\frac{1}{x} \quad \text{for } x \ne 0$$

is not monotone, not continuous at 0 and the right hand limit $f(0+)$ does not exist. Yet it can easily be shown to be integrable by using the following result.

9-3.7. Proposition. *If a bounded function f having domain $[a,b]$ is discontinuous at only a finite number of points, then it is integrable.*

Proof. Let ε be any positive number. We shall show the existence of a partition P such that $U(f,P) - L(f,P) < \varepsilon$.

Consider the set consisting of all points of discontinuity, together with the endpoints a and b. Suppose the points of this set, arranged in increasing order, are $a = c_0, \dots, c_n = b$. Let M be any upper bound for $|f(x)|$, $x \in [a,b]$. By Prop.9-1.2, there exists a partition of $[a,b]$ such that the total length of the subintervals containing $c_0, \dots, c_n$ is less than or equal to $\frac{\varepsilon}{4M}$ and the remaining subintervals are n in number. Now f is continuous on each of the latter. Therefore each of these subintervals has a partition P_j such that $U(f,P_j) - L(f,P_j) < \frac{\varepsilon}{2n}$. For the partition P of $[a,b]$ consisting of all the points of all the P_j as well as a and b, it follows that

$$U(f,P) - L(f,P) < n \cdot \frac{\varepsilon}{2n} + \frac{\varepsilon}{4M}(2M) = \varepsilon.$$

Thus f is integrable by Prop.9-2.9. ☺

Problem Set 9-3

9-3.P1. Without using either Cor.9-3.2 or Prop.9-3.7, show that the following function $f: [0,1] \to \mathbb{R}$ is integrable: $f(0) = 5$ and $f(x) = x$ if $0 < x \leq 1$.

9-3.P2. Suppose a bounded function $f:[a,b] \to \mathbb{R}$ is integrable on every subinterval $[\alpha, b]$ with $a < \alpha < b$ and that $\lim\limits_{\alpha \to a} \int_\alpha^b f = A$. Show that f is integrable on $[a,b]$ with $\int_a^b f = A$.

9-3.P3. Define a function $f:[0,1] \to \mathbb{R}$ in terms of the integer part function [ ] by $f(0) = -3$ and $f(x) = 1 + (-1)^{[1/x]}$ when $0 < x \leq 1$. Show that $\int_0^1 f$ exists and equals 2 $\sum\limits_{k=1}^{\infty}\left(\frac{1}{2k} - \frac{1}{2k+1}\right)$.

9-3.P4. Prove that the function f defined on [0,1] by

$$f(x) = \begin{cases} \frac{1}{2^n} & \frac{1}{2^{n+1}} < x \leq \frac{1}{2^n}, \quad n = 0,1,\dots \\ 0 & x = 0 \end{cases}$$

is integrable on [0,1], although it has infinitely many discontinuities.

9-3.P5. Let $f(x) = \begin{cases} \frac{\sin x}{x} & x \neq 0 \\ 1 & x = 0 \end{cases}$. Show that $\int_0^1 f(x)\,dx$ exists (i.e. f is integrable over [0,1]).

9-4 Properties of the Riemann Integral

We turn our attention from the question "What properties of a function ensure that it is integrable?" to that of "If we already have some integrable functions, how can we invent new ones from them?" The obvious attempt would be to add or multiply two of them.

9-4.1. Proposition. *Let f and g be integrable functions on $[a,b]$. Then the function $f+g$ is integrable and* $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.

Proof. First consider any partition $P: a = x_0 < x_1 < \cdots < x_n = b$ of $[a,b]$ and let

$$m_j(f) = \inf\{f(x) : x_{j-1} \le x \le x_j\},$$

with corresponding meanings for $m_j(g)$ and $m_j(f+g)$. Now

$$f(x)+g(x) \ge m_j(f)+m_j(g) \text{ whenever } x_{j-1} \le x \le x_j,\ 1 \le j \le n.$$

Therefore $m_j(f+g) \ge m_j(f)+m_j(g)$, $1 \le j \le n$. This implies that

$$L(f+g,P) \ge L(f,P)+L(g,P) \text{ for any partition } P. \tag{1}$$

Similarly, $$U(f+g,P) \le U(f,P)+U(g,P) \text{ for any partition } P. \tag{2}$$

Now consider any $\varepsilon > 0$. Since f and g are integrable, by Prop.9-2.9, there exist partitions P_f and P_g such that

$$\int_a^b f - \frac{\varepsilon}{2} < L(f,P_f) \le U(f,P_f) < \int_a^b f + \frac{\varepsilon}{2}$$

and $$\int_a^b g - \frac{\varepsilon}{2} < L(g,P_g) \le U(g,P_g) < \int_a^b g + \frac{\varepsilon}{2}.$$

The partition P obtained by taking all the points of P_f and P_g together is a refinement of both and therefore [see Prop.9-2.2] both the inequalities displayed above hold if P_f and P_g are both replaced by P. By adding the two inequalties obtained by making the replacement, we get

$$\int_a^b f + \int_a^b g - \varepsilon < L(f,P)+L(g,P) \le U(f,P)+U(g,P) < \int_a^b f + \int_a^b g + \varepsilon.$$

Combining this with (1) and (2), we get

$$\int_a^b f + \int_a^b g - \varepsilon < L(f+g,P) \text{ and } U(f+g,P) < \int_a^b f + \int_a^b g + \varepsilon.$$

Since such a partition P has been shown to exist for any positive ε, it follows that $\int_a^b f + \int_a^b g \le \underline{\int_a^b}(f+g)$ and $\overline{\int_a^b}(f+g) \le \int_a^b f + \int_a^b g$. Since the lower Riemann integral never exceeds the upper Riemann integral [by Prop.9-2.6], it follows from the preceding inequalities that

$$\underline{\int_a^b}(f+g) = \overline{\int_a^b}(f+g) = \int_a^b f + \int_a^b g. \ ☺$$

Example. Let $\phi:[0,1]\to\mathbb{R}$ be the function for which $\phi(0) = 1$ and $\phi(x) = x$ when $x \in (0,1]$. Then ϕ is neither continuous nor monotone and its integrability does not follow immediately from Theorem 9-3.3. However, it is the sum of the functions f and g given by

$$f(0) = 1 \text{ while } f(x) = 0 \text{ when } x \in (0,1],$$

and
$$g(x) = x \text{ whenever } x \in [0,1].$$

Since f is monotone [in fact a step function] and g is continuous, both are integrable by Theorem 9-3.3 and their sum ϕ is integrable by Prop.9-4.1. Besides, its integral is $\int_0^1 f + \int_0^1 g = \int_0^1 g$.

9-4.2. Lemma. *Let $f: X \to \mathbb{R}$ be a bounded function on some domain X* [not necessarily a subset of $\mathbb{R}$] *and*

$$M = \sup\{f(s) : s \in X\},\ m = \inf\{f(s) : s \in X\}.$$

Then
$$M - m = \sup\{|f(s) - f(t)| : s, t \in X\}. \tag{A}$$

Proof. Since
$$f(s) \le M \text{ and } f(t) \ge m \ \forall\ s, t \in X,$$

we have
$$f(s) - f(t) \le M - m \ \forall\ s, t \in X.$$

Similarly,
$$f(t) - f(s) \le M - m \ \forall\ s, t \in X.$$

It follows that $|f(s) - f(t)| \le M - m$ for all $s, t \in X$. Therefore

$$\sup\{|f(s) - f(t)| : s, t \in X\} \le M - m. \tag{1}$$

To prove the reverse inequality, consider any $\varepsilon > 0$. There exist $s, t \in X$ such that $f(s) > M - \frac{\varepsilon}{2}$ and $f(t) < m + \frac{\varepsilon}{2}$. This implies $f(s) - f(t) > M - m - \varepsilon$. It follows from this inequality that $\sup\{|f(s) - f(t)| : s, t \in X\} \ge M - m - \varepsilon$. Since this is true for every positive ε, it further follows that $\sup\{|f(s) - f(t)| : s, t \in X\} \ge M - m$. Hence by (1), the equality (A) holds. ☺

9-4.3. Proposition. *Let f and g be integrable functions on $[a, b]$. Then the function fg is integrable.*

Proof. First consider any partition $P: a = x_0 < x_1 < \cdots < x_n = b$ of $[a, b]$ and let

$$m_j(f) = \inf\{f(x) : x_{j-1} \le x \le x_j\}, \quad M_j(f) = \sup\{f(x) : x_{j-1} \le x \le x_j\},$$

with corresponding meanings for $m_j(g)$, $M_j(g)$, $m_j(fg)$ and $M_j(fg)$. Denote by K the greater of $\sup\{|f(x)| : a \le x \le b\}$ and $\sup\{|g(x)| : a \le x \le b\}$. For any s and t belonging to $[x_{j-1}, x_j]$, we have

$$\begin{aligned}|f(s)g(s) - f(t)g(t)| &= |f(s)g(s) - f(t)g(s) + f(t)g(s) - f(t)g(t)| \\ &\le |g(s)||f(s) - f(t)| + |f(t)||g(s) - g(t)|\end{aligned}$$

$$\le K|f(s)-f(t)|+K|g(s)-g(t)|$$
$$\le K(M_j(f)-m_j(f))+K(M_j(g)-m_j(g)) \text{ by Lemma 9-4.2.}$$

This implies that

$$\sup\{|f(s)g(s)-f(t)g(t)| : s,t\in[x_{j-1},x_j]\} \le K(M_j(f)-m_j(f))+K(M_j(g)-m_j(g)).$$

Applying Lemma 9-4.2 once again, we conclude that

$$M_j(fg)-m_j(fg) \le K(M_j(f)-m_j(f))+K(M_j(g)-m_j(g)).$$

Upon multiplying by $x_j - x_{j-1}$ and taking the sum over $1 \le j \le n$, we obtain

$$U(fg,P)-L(fg,P) \le K[(U(f,P)-L(f,P))+(U(g,P)-L(g,P))]. \tag{1}$$

This has been shown to hold for any partition P.

Now consider any $\varepsilon > 0$. Since f and g are integrable, by Prop.9-2.9, there exist partitions P_f and P_g such that

$$U(f,P_f) - L(f,P_f) < \frac{\varepsilon}{2K} \qquad \text{and} \qquad U(g,P_g) - L(g,P_g) < \frac{\varepsilon}{2K}.$$

The partition P obtained by taking all the points of P_f and P_g together is a refinement of both and therefore [see Prop.9-2.2] both the inequalities displayed above hold if P_f and P_g are both replaced by P. The two inequalties obtained by making the replacement, when combined with (1), lead to

$$U(fg,P) - L(fg,P) < K\left(\frac{\varepsilon}{2K} + \frac{\varepsilon}{2K}\right) = \varepsilon.$$

By Prop.9-2.9, fg is integrable. ☺

9-4.4. Prop. *If f is an integrable function on $[a,b]$ and c is any constant, then the product cf is integrable and $\int_a^b cf = c\int_a^b f$.*

Proof. First consider any partition P: $a = x_0 < x_1 < \cdots < x_n = b$ of $[a,b]$ and let

$$m_j(f) = \inf\{f(x) : x_{j-1} \le x \le x_j\}, \quad M_j(f) = \sup\{f(x) : x_{j-1} \le x \le x_j\},$$

with corresponding meanings for $m_j(cf)$, $M_j(cf)$. Let $c > 0$. Then

$$m_j(cf) = cm_j(f) \quad \text{and} \quad M_j(cf) = cM_j(f). \tag{1}$$

Therefore $L(cf,P) = cL(f,P)$ and $U(cf,P) = cU(f,P)$ for any partition P. Hence, whenever f is any bounded function,

$$\underline{\int_a^b} cf = c\underline{\int_a^b} f \quad \text{and} \quad \overline{\int_a^b} cf = c\overline{\int_a^b} f. \tag{2}$$

This shows that, when f is integrable and $c > 0$, the function cf is also integrable and $\int_a^b cf = c\int_a^b f$. When $c < 0$, instead of (1) and (2), we have

$$m_j(cf) = cM_j(f) \quad \text{and} \quad M_j(cf) = cm_j(f)$$

and $$\underline{\int_a^b} cf = c\overline{\int_a^b} f \quad \text{and} \quad \overline{\int_a^b} cf = c\underline{\int_a^b} f.$$

This shows that, when f is integrable and $c < 0$, the function cf is also integrable and $\int_a^b cf = c\int_a^b f$. The case $c = 0$ is straightforward. ☺

For integrable f, it is a consequence of the preceding result that $-f$ is integrable. The analogue of this for multiplication would be about $1/f$ in place of $-f$; one would therefore have to assume f to be nonzero everywhere and $1/f$ to be bounded [see 9-4.P3&P4].

9-4.5.Prop. *If f and g are integrable functions on $[a,b]$ and $f(x) \le g(x)$ for every $x \in [a,b]$, then $\int_a^b f \le \int_a^b g$.*

Proof. Consider any partition $P: a = x_0 < x_1 < \cdots < x_n = b$ of $[a,b]$ and let

$$M_j(f) = \sup\{f(x) : x_{j-1} \le x \le x_j\}, \quad M_j(g) = \sup\{g(x) : x_{j-1} \le x \le x_j\}.$$

Since $f(x) \le g(x)$ whenever $x \in [a,b]$, we have $M_j(f) \le M_j(g)$. Upon multiplying by $x_j - x_{j-1}$ and taking the sum over $1 \le j \le n$, we find that $U(f,P) \le U(g,P)$. This is true for any partition P. Therefore, by taking the supremum of both sides, we get $\overline{\int_a^b} f \le \overline{\int_a^b} g$. When f and g are both integrable, this means $\int_a^b f \le \int_a^b g$. ☺

9-4.6. Prop. *If $f: [a,b] \to \mathbb{R}$ is integrable, then so is $|f|$, and*

$$\left|\int_a^b f\right| \le \int_a^b |f|.$$

Proof. First consider any partition $P: a = x_0 < x_1 < \cdots < x_n = b$ of $[a,b]$ and let

$$m_j(f) = \inf\{f(x) : x_{j-1} \le x \le x_j\}, \quad M_j(f) = \sup\{f(x) : x_{j-1} \le x \le x_j\},$$

with corresponding meanings for $m_j(|f|)$, $M_j(|f|)$. For any s and t belonging to $[x_{j-1}, x_j]$, by Lemma 9-4.2, we have

$$\big|\,|f(s)| - |f(t)|\,\big| \le |f(s) - f(t)| \le M_j(f) - m_j(f),$$

so that

$$\sup\{\big|\,|f(s)| - |f(t)|\,\big| : s,t \in [x_{j-1}, x_j]\} \le M_j(f) - m_j(f).$$

Applying Lemma 9-4.2 once again, we conclude that

$$M_j(|f|) - m_j(|f|) \le M_j(f) - m_j(f).$$

Upon multiplying by $x_j - x_{j-1}$ and taking the sum over $1 \le j \le n$, we obtain

$$U(|f|,P) - L(|f|,P) \le U(f,P) - L(f,P)$$

for any partition P and any bounded function f. By Prop.9-2.9, it now follows that, if f is integrable, then so is $|f|$.

To prove the inequality, note that $-|f(x)| \le f(x) \le |f(x)|$ for each $x \in [a,b]$. Since f and $|f|$ are both integrable, it follows in view of Prop.9-4.5 that

$$-\int_a^b |f| \le \int_a^b f \le \int_a^b |f|. ☺$$

The inequality in the above Proposition could also have been proved by using 9-2.P6 instead of Prop.9-4.5.

9-4.7. Corollary. *If f and g are integrable functions on* $[a,b]$, *then the same is true of* $\max\{f,g\}$ *and* $\min\{f,g\}$.

Proof. This follows immediately from the above results and the fact that [see 1-4.P12]

$$\max\{f,g\} = \tfrac{1}{2}[f+g+|f-g|] \text{ and } \min\{f,g\} = \tfrac{1}{2}[f+g-|f-g|]. ☺$$

We now come to a converse of Prop.9-2.10.

9-4.8. Proposition. *If* $f:[a,b]\to\mathbb{R}$ *is an integrable function and* $a \le c \le b$, *then its restrictions to* $[a,c]$ *and to* $[c,b]$ *are both integrable, and*

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Proof. The cases when $c = a$ or $c = b$ are trivial; so we assume $a < c < b$.

Denote the restrictions to $[a,c]$ and $[c,b]$ by f_1 and f_2 respectively. Then $\int_a^c f$ and $\int_c^b f$ mean the same thing as $\int_a^c f_1$ and $\int_c^b f_2$ respectively.

Since f is integrable, by Prop.9-2.9, for any $\varepsilon > 0$, there exists a partition P_0 such that $U(f,P_0) - L(f,P_0) < \varepsilon$. In case c is not already one of the points of P_0, then its refinement obtained by including c, call it P, will also satisfy the preceding inequality:

$$U(f,P) - L(f,P) < \varepsilon. \tag{1}$$

The points of P that belong to $[a,c]$ form a partition P_1 of $[a,c]$, and those that belong to $[c,b]$ form a partition P_2 of $[c,b]$. Besides,

$$L(f,P) = L(f_1,P_1) + L(f_2,P_2) \quad \text{and} \quad U(f,P) = U(f_1,P_1) + U(f_2,P_2)$$

and hence

$$U(f,P) - L(f,P) = U(f_1,P_1) - L(f_1,P_1) + U(f_2,P_2) - L(f_2,P_2).$$

Since $U(f_1,P_1) - L(f_1,P_1) \ge 0$ and $U(f_2,P_2) - L(f_2,P_2) \ge 0$, it follows from (1) that

$$U(f_1,P_1) - L(f_1,P_1) < \varepsilon \text{ and } U(f_2,P_2) - L(f_2,P_2) < \varepsilon.$$

The existence of such partitions of $[a,c]$ and $[c,b]$ for every positive $\varepsilon > 0$ shows that f_1 and f_2 are integrable [once again by Prop.9-2.9]. The equality now follows by Prop.9-2.10. ☺

It is possible to derive Props.9-4.3 and 9-4.4 as special cases of a more general result, which we give below but shall use only for some problems. We note in passing that the proof uses the Generalised Associative and Commutative Properties of Props.1-11.2 and 1-11.3.

***9-4.9. Proposition.** *Let $f:[a,b]\to\mathbb{R}$ be an integrable function and h be a uniformly continuous function on the range of f. Then the composition $h\circ f:[a,b]\to\mathbb{R}$ is integrable.* [No assumption that the domain of h is an interval.]

Proof. Since f is integrable, it is bounded. Therefore h is a uniformly continuous function whose domain is a bounded subset of $\mathbb{R}$, and hence h is also bounded [see 2-5.P4]. It follows that $h\circ f$ has some bound K, say. Then

$$|(h\circ f)(x)| \le K \text{ whenever } x \in [a,b]. \tag{1}$$

For any $\varepsilon > 0$, a partition P such that $U(h\circ f,P) - L(h\circ f,P) < \varepsilon$ will be shown to exist.

Let
$$\varepsilon' = \frac{\varepsilon}{b-a+2K}. \tag{2}$$

By uniform continuity of h, there exists $\delta > 0$ such that, whenever u and v are in the domain of h,

$$|u-v| < \delta \Rightarrow |h(u)-h(v)| < \varepsilon'. \tag{3}$$

Replacing δ by $\min\{\delta,\varepsilon'\}$, we may assume that

$$\delta \le \varepsilon'. \tag{4}$$

Since f is integrable, there exists a partition $P: a = x_0 < x_1 < \cdots < x_n = b$ such that [the reason for the square will become clear later in the proof]

$$U(f,P) - L(f,P) < \delta^2. \tag{5}$$

Let $\quad m_j(f) = \inf\{f(x) : x_{j-1} \le x \le x_j\}, \quad M_j(f) = \sup\{f(x) : x_{j-1} \le x \le x_j\},$

with corresponding meanings for $m_j(h\circ f)$, $M_j(h\circ f)$. Consider the following two subsets of $\{1,\ldots,n\}$:

$A = \{j \in \{1,\ldots,n\} : M_j(f) - m_j(f) < \delta\}, \quad B = \{j \in \{1,\ldots,n\} : M_j(f) - m_j(f) \ge \delta\}.$

These are disjoint subsets with union $\{1,\ldots,n\}$. Therefore

$$U(h\circ f,P) - L(h\circ f,P) = \sum_{j\in A}(M_j(h\circ f) - m_j(h\circ f))(x_j - x_{j-1})$$

$$+\sum_{j\in B}(M_j(h\circ f)-m_j(h\circ f))(x_j-x_{j-1}). \tag{6}$$

Now, if $j\in A$ and $s,t\in[x_{j-1},x_j]$, then by Lemma 9-4.2,

$$|f(s)-f(t)|\leq M_j(f)-m_j(f)<\delta,$$

so that by (3),

$$|(h\circ f)(s)-(h\circ f)(t)|<\varepsilon'$$

and hence by Lemma 9-4.2 again,

$$M_j(h\circ f)-m_j(h\circ f)\leq\varepsilon'.$$

It follows that

$$\sum_{j\in A}(M_j(h\circ f)-m_j(h\circ f))(x_j-x_{j-1})\leq\varepsilon'\sum_{j=1}^{n}(x_j-x_{j-1})=\varepsilon'(b-a). \tag{7}$$

If $j\in B$, then $M_j(h\circ f)-m_j(h\circ f)\leq 2K$ by (1), and $M_j(f)-m_j(f)\geq\delta$, whence

$$\sum_{j\in B}(x_j-x_{j-1})\leq\frac{1}{\delta}\Big(\sum_{j\in B}(M_j(f)-m_j(f))(x_j-x_{j-1})\Big)\leq\frac{1}{\delta}\big(U(f,P)-L(f,P)\big)$$
$$<\delta\leq\varepsilon' \text{ by (5) and (4).}$$

Therefore
$$\sum_{j\in B}(M_j(h\circ f)-m_j(h\circ f))(x_j-x_{j-1})<2K\varepsilon'.$$

Adding this to (7) and then using (6), we get

$$U(h\circ f,P)-L(h\circ f,P)<\varepsilon'(b-a+2K)=\varepsilon \quad \text{by (2).} ☺$$

The reader is invited to show [see 9-4.P9] that the hypothesis about h in the above Theorem cannot be replaced by the requirement that it be a monotone or a step function.

Problem Set 9-4

9-4.P1. For bounded functions $f:[a,b]\to\mathbb{R}$ and $g:[a,b]\to\mathbb{R}$, show that

$$\overline{\int_a^b}(f+g)\leq\overline{\int_a^b}f+\overline{\int_a^b}g.$$

9-4.P2. Let $f:[a,b]\to\mathbb{R}$ be a bounded function and $a<c<b$. Show that

$$\overline{\int_a^b}f=\overline{\int_a^c}f+\overline{\int_c^b}f \text{ and } \underline{\int_a^b}f=\underline{\int_a^c}f+\underline{\int_c^b}f.$$

9-4.P3. Is it true for every bounded function $f:[a,b]\to\mathbb{R}$ that $|\underline{\int_a^b}f|\leq\underline{\int_a^b}|f|$?

9-4.P4. Let $f:[a,b]\to\mathbb{R}$ be a bounded function, $m=\inf\{f(x):x\in[a,b]\}$ and P: $a=x_0<x_1<\cdots<x_n=b$ a partition of $[a,b]$.

(a) Suppose $m>0$. Show that $g=1/f$ is bounded. If

$$m_j(f)=\inf\{f(x):x_{j-1}\leq x\leq x_j\},\quad M_j(f)=\sup\{f(x):x_{j-1}\leq x\leq x_j\},$$

with corresponding meanings for $m_j(g)$, $M_j(g)$, show that $m_j(f) = 1/M_j(g)$ and $M_j(f) = 1/m_j(g)$. Hence show that $U(g,P) - L(g,P) \leq [U(f,P) - L(f,P)]/m^2$.

(b) If $m > 0$ and f is integrable, show that $g = 1/f$ is integrable. Must $1/f^2$ be integrable when $\mu = \inf\{|f(x)| : x \in [a,b]\} > 0$?

(c) If $\inf\{|f(x)| : x \in [a,b]\} > 0$ and f is integrable, show that $1/f$ is integrable by using the fact that

$$1/x = (\max\{x,0\} + \min\{x,0\})/x^2 \text{ whenever } x \neq 0.$$

9-4.P5. (Cauchy-Schwarz Inequality) Let f and g be integrable functions on $[a,b]$, so that f^2, g^2 and fg are also integrable. Prove that

(a) $\left(\int_a^b fg\right)^2 \leq \left(\int_a^b f^2\right)\left(\int_a^b g^2\right)$.

(b) If $\int_a^b (\lambda f - g)^2 = 0$ for some λ, then $\left(\int_a^b fg\right)^2 = \left(\int_a^b f^2\right)\left(\int_a^b g^2\right)$.

(c) If $\left(\int_a^b fg\right)^2 = \left(\int_a^b f^2\right)\left(\int_a^b g^2\right)$ and $\int_a^b f^2 \neq 0$, then there is a unique λ, namely $\left(\int_a^b fg\right)/\left(\int_a^b f^2\right)$, for which $\int_a^b (\lambda f - g)^2 = 0$.

(d) If $\int_a^b f^2 = \int_a^b g^2 = 0$, then every λ satisfies $\int_a^b (\lambda f - g)^2 = 0$.

9-4.P6. If f is a continuous nonnegative valued function on $[a,b]$ with $\int_a^b f = 0$, show that $f(x) = 0$ everywhere on $[a,b]$.

***9-4.P7.** (a) If f is a continuous function on $[0,1]$ such that $\int_0^1 f(x)x^n\,dx = 0$ for every integer $n \geq 0$, show that $f(x) = 0$ everywhere on $[0,1]$.

(b) If f is a continuous function on $[0,1]$ such that $\int_0^1 f(x)x^{3n}\,dx = 0$ for every integer $n \geq 0$, show that $f(x) = 0$ everywhere on $[0,1]$.

***9-4.P8.** [**Hölder Inequality**; needed in 10-2.P9 and 10-3.P9] Let f and g be integrable functions on $[a,b]$ and suppose p and q are positive numbers such that $1/p + 1/q = 1$. Show that $|fg|$, $|f|^p$ and $|g|^q$ are integrable and that

$$\int_a^b |fg| \leq \left(\int_a^b |f|^p\right)^{1/p}\left(\int_a^b |g|^q\right)^{1/q}.$$

***9-4.P9.** Show that the hypothesis of uniform continuity in Prop.9-4.9 cannot be replaced by the requirement that h be a monotone or a step function.

***9-4.P10.** (i) Supose f is a function defned on $[a,b]$ and f^2 is integrable on $[a,b]$. Show that $|f|$ must be integrable on $[a,b]$, and give an example to show that f need not be integrable.

(ii) Supose f is a function defned on $[a,b]$ and f^3 is integrable on $[a,b]$. Show that f must be integrable on $[a,b]$.

9-4.P11. (a) If f is integrable on $[a,b]$ and and $0 \le m \le f(x) \le M$ for all $x \in [a,b]$, show that

$$m \le \left[\frac{1}{b-a}\int_a^b f(x)^2\,dx\right]^{\frac{1}{2}} \le M.$$

(b) If f is also continuous and nonnegative on $[a,b]$ everywhere, show that there exists some $c \in [a,b]$ such that $f(c) = \left[\frac{1}{b-a}\int_a^b f(x)^2\,dx\right]^{\frac{1}{2}}$.

9-5 Fundamental Theorem of Calculus

The reader is undoubtedly familiar with the use of antidifferentiation for computing a Riemann integral. The close connection between the antiderivative and the Riemann integral—often called "indefinite" and "definite" integral—is valid only for a restricted class of functions. However, the class is large enough to overshadow the fact that there are functions outside it. For purposes of introductory Calculus, the distinction may be unimportant, but it would be inappropriate to ignore it in the study of Mathematical Analysis.

The reader is cautioned that, although the names "First" and "Second" Fundamental Theorems are commonly used, they do not always refer to the same two Theorems in different presentations of the subject.

9-5.1. First Fundamental Theorem of Calculus. *Let $f:[a,b]\to\mathbb{R}$ be integrable and $F:[a,b]\to\mathbb{R}$ be the function such that*

$$F(x) = \int_a^x f \qquad \textit{for every } x \in [a,b].$$

Then F is continuous on $[a,b]$. Moreover, if f is continuous at $c \in [a,b]$, then $F'(c) = f(c)$.

Proof. Since f is integrable, it is bounded, so that there exists $M > 0$ for which $|f(x)| \le M$ for all $x \in [a,b]$.

Consider any $c \in [a,b]$ and any $\varepsilon > 0$. We shall prove the existence of a number $\delta > 0$ such that

$$|F(x) - F(c)| < \varepsilon \quad \text{whenever} \quad x \in [a,b] \text{ and } |x - c| < \delta. \tag{A}$$

For this purpose, first suppose that $a \le c < b$ and consider any $x \in [a,b]$ such that $c < x < b$. Then we have

$$\begin{aligned} |F(x) - F(c)| &= \left|\int_a^x f - \int_a^c f\right| = \left|\int_c^x f\right| \quad \text{by Prop.9-4.8} \\ &\le \int_c^x |f| \quad \text{by Prop.9-4.6} \\ &\le \int_c^x M \quad \text{by Prop.9-4.5} \end{aligned}$$

$$\leq M(x-c) \text{ by Prop.9-4.4.}$$

This shows that the positive number $\delta = \varepsilon/M$ has the property that

$$|F(x)-F(c)| < \varepsilon \quad \text{whenever} \quad x \in [a,b] \text{ and } c < x < c+\delta. \tag{1}$$

By an analogous argument, when $a < c \leq b$, the positive number $\delta = \varepsilon/M$ has the property that

$$|F(x)-F(c)| < \varepsilon \quad \text{whenever} \quad x \in [a,b] \text{ and } c-\delta < x < c. \tag{2}$$

Now (A) follows immediately from (1) and (2). This completes the proof of the continuity of F at every point $c \in [a,b]$.

Now let c denote a point in $[a,b]$ where f is continuous. First suppose that $a \leq c < b$. For any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(t)-f(c)| < \varepsilon \quad \text{whenever} \quad t \in [a,b] \text{ and } c < t < c+\delta.$$

By Prop.9-4.6, it follows that

$$|\int_c^x (f-f(c))| \leq \varepsilon(x-c) \quad \text{whenever} \quad x \in [a,b] \text{ and } c < x < c+\delta.$$

But $\int_c^x (f-f(c)) = (\int_c^x f)-f(c)(x-c)$. Therefore for any x of the kind above,

$$|(\int_c^x f)-f(c)(x-c)| \leq \varepsilon(x-c)$$

and hence

$$\left|\frac{1}{x-c}\int_c^x f-f(c)\right| \leq \varepsilon.$$

As noted in the earlier part of this Proof, $\int_c^x f = F(x)-F(c)$. Therefore

$$\left|\frac{F(x)-F(c)}{x-c}-f(c)\right| \leq \varepsilon \quad \text{whenever } x \in [a,b] \text{ and } c < x < c+\delta.$$

The existence of such a positive δ for any positive ε means that the right derivative of F at c is $f(c)$ in case $a \leq c < b$. An analogous argument shows that the left derivative is $f(c)$ in case $a < c \leq b$. ☺

The second Example below answers a question raised in the opening paragraph of this Chapter.

9-5.2. Examples. *(a) Let f be the restriction to $[a,b]$ of the function of Example 2-1.4(d), wherein it was shown that this function is continuous at every irrational point. It was argued in Example 9-3.4(b) that it is integrable. By definition, it is 0 at every irrational point. Since every subinterval of any partition contains irrational numbers, all lower sums are 0 and hence $F(x) = \int_a^x f = 0$ whenever $x \in [a,b]$. It follows that $F'(x) = 0$ everywhere. At irrational points, f is continuous and F' is seen to agree with f, which is in accordance with the above Theorem.

(b) The function $f:[0,A]\to\mathbb{R}$ defined by $f(t) = e^{t^2}$ is continuous and hence integrable. By Theorem 9-5.1, the function $F:[0,A]\to\mathbb{R}$ defined by $F(x) = \int_0^x f$ satisfies $F'(x) = e^{x^2}$ whenever $x \in [0,A]$. Since A can be any positive number, we actually have a function on $[0,\infty)$ with e^{x^2} as its derivative. To extend this function to all of $\mathbb{R}$, one need only take $F(x) = -\int_0^{-x} f = -F(-x)$ for $x < 0$. The Chain Rule shows that this implies

$$F'(x) = -(-f(-x)) = e^{x^2} \text{ for } x < 0.$$

The left and right derivatives of F at 0 agree, because

$$\frac{F(0+h)-F(0)}{h} = \frac{-F(-h)-0}{h} = \frac{F(-h)}{-h} = \frac{F(-h)-F(0)}{-h} \text{ for } h<0.$$

Thus there does exist a function $F:\mathbb{R}\to\mathbb{R}$ with derivative e^{x^2} everywhere, whether we can compute it or not.

(c) The function $f:[0,A]\to\mathbb{R}$, where $A > 0$, defined by $f(t) = \sin\frac{1}{t}$ for $t \neq 0$ and $f(0) = 7$ [or any other real number] is continuous except at 0. By Prop.9-3.7, it is integrable. Let $F(x) = \int_0^x f$. Then $F'(x) = \sin\frac{1}{x}$ when x is not 0.

9-5.3. Second Fundamental Theorem of Calculus. *Let $f:[a,b]\to\mathbb{R}$ be integrable and $F:[a,b]\to\mathbb{R}$ be a continuous function such that*

$$F'(x) = f(x) \qquad \textit{for every } x \in (a,b).$$

Then $$\int_a^x f = F(x) - F(a) \qquad \textit{for every } x \in [a,b].$$

Proof. Let $\varepsilon > 0$ and $x \in [a,b]$. There exists a partition P: $a = x_0 < x_1 < \cdots < x_n = x$ such that

$$\int_a^x f - \varepsilon < L(f,P) \le U(f,P) \le \int_a^x f + \varepsilon. \tag{1}$$

As usual, let

$$m_j = \inf\{f(t) : x_{j-1} \le t \le x_j\} \quad \text{and} \quad M_j = \sup\{f(t) : x_{j-1} \le t \le x_j\}.$$

Now, by the Lagrange Mean Value Theorem,

$$\begin{aligned} F(x) - F(a) &= \sum_{j=1}^n (F(x_j) - F(x_{j-1})) \\ &= \sum_{j=1}^n f(u_j)(x_j - x_{j-1}), \text{ where } x_{j-1} < u_j < x_j \text{ for } 1 \le j \le n. \end{aligned} \tag{2}$$

Since $m_j \le f(u_j) \le M_j$, we have $L(f,P) \le \sum_{j=1}^n f(u_j)(x_j - x_{j-1}) \le U(f,P)$. Together with (1), this implies that

$$-\varepsilon < \sum_{j=1}^n f(u_j)(x_j - x_{j-1}) - \int_a^x f < \varepsilon,$$

and hence by (2),

$$-\varepsilon < F(x) - F(a) - \int_a^x f < \varepsilon.$$

Since this inequality holds for every ε > 0, we have $\int_a^x f = F(x) - F(a)$. ☺

To apply the above Theorem, one needs prior knowledge of the "antiderivative" F. Therefore it cannot be applied to the instances discussed in 9-5.2. It provides a basis for the familiar computations of Calculus, in which an antiderivative is used in order to find the "definite integral", as e.g. in

$$\int_a^b x\,dx = \tfrac{1}{2}(b^2 - a^2) \quad \text{because we know } \frac{d}{dx}\left(\tfrac{1}{2}x^2\right) = x.$$

9-5.4. Examples. (a) Let $f: [0, 1] \to \mathbb{R}$ be defined as follows:

$$f(x) = \begin{cases} n^2 x & \text{if } 0 \le x \le \dfrac{1}{2n} \\ -n^2 x + n & \text{if } \dfrac{1}{2n} < x \le \dfrac{1}{n} \\ 0 & \text{if } \dfrac{1}{n} < x \le 1. \end{cases}$$

It is easy to check that f is continuous. The integral over each of the three intervals

$$[0, \frac{1}{2n}],\ [\frac{1}{2n}, \frac{1}{n}] \text{ and } [\frac{1}{n}, 1]$$

can be computed in the familiar manner by using the Second Fundamental Theorem, leading to $\int_0^1 f = \frac{1}{4}$.

(b) In connection with Theorem 9-3.1, we have already given an example in which a sequence $\{f_n\}$ of integrable functions has limit f that is not integrable. It may be recalled that the functions f_n were not continuous. We now give an illustration in which not only is each f_n continuous but also the limit is continuous, and yet

$$\int_a^b f \neq \lim_{n\to\infty} \int_a^b f_n .$$

For each $n \in \mathbb{N}$, let f_n be the function in Example 9-5.4(a) above. As already seen, $\int_0^1 f_n = \frac{1}{4}$ for every n. Since $f_n(0) = 0$ for every n, we have $\lim_{n\to\infty} f_n(0) = 0$. For $0 < x \le 1$, there exists some natural number $N > 1/x$ and then $n \ge N \Rightarrow 1/n < x$, so that $f_n(x) = 0$. This shows that have $\lim_{n\to\infty} f_n(x) = 0$ also for $0 < x \le 1$. Hence $\lim_{n\to\infty} f_n$ is the zero function, which is continuous and has integral 0, not $\frac{1}{4}$.

It is convenient to define $\int_b^a f$ to mean $-\int_a^b f$ when $a < b$. Under the conditions of the Second Fundamental Theorem, we also have $\int_x^b f = F(b) - F(x)$, which can be written as $\int_b^x f = F(x) - F(b)$. This removes the need to distinguish the cases $\phi(\alpha) \le \phi(\beta)$ and $\phi(\alpha) \ge \phi(\beta)$ in the following:

9-5.5. Theorem. Substitution Rule. *Suppose*

$$F:[a,b]\to\mathbb{R} \quad \textit{and} \quad \phi:[\alpha,\beta]\to[a,b]$$

are differentiable functions such that

$$F':[a,b]\to\mathbb{R} \quad \textit{and} \quad (F'\circ\phi)\,\phi':[\alpha,\beta]\to\mathbb{R}$$

are both integrable. Then

$$\int_{\phi(\alpha)}^{\phi(\beta)} F' = \int_\alpha^\beta (F'\circ\phi)\,\phi'.$$

Proof. By the Chain Rule, we have

$$(F\circ\phi)' = (F'\circ\phi)\,\phi'.$$

Since this is integrable, by the Second Fundamental Theorem, we have

$$\int_\alpha^\beta (F'\circ\phi)\,\phi' = (F\circ\phi)(\beta) - (F\circ\phi)(\alpha) = F(\phi(\beta)) - F(\phi(\alpha)). \tag{1}$$

Similarly, since F' is also integrable, we have

$$\int_{\phi(\alpha)}^{\phi(\beta)} F' = F(\phi(\beta)) - F(\phi(\alpha)). \tag{2}$$

The required equality follows immediately from (1) and (2). ☺

9-5.6. Example. It is familiar from Calculus how to evaluate

$$\int_0^1 \frac{1}{1+\sqrt{t}}\,dt$$

by substituting $x = \sqrt{t}$. The purpose here is to clarify the role of the above Theorem in legitimising the computation. The function ϕ is given by $\phi(t) = \sqrt{t}$, for which $\phi'(t) = 1/2\sqrt{t}$ when $t \neq 0$. Upon rewriting

$$\frac{1}{1+\sqrt{t}} \quad \text{as the product} \quad \frac{2\sqrt{t}}{1+\sqrt{t}}\cdot\frac{1}{2\sqrt{t}},$$

we see that, for $t \neq 0$, it equals $f(\phi(t))\,\phi'(t)$, where $f(x) = 2x/(1+x)$. If we can find F such that $F'(x) = f(x)$, then the Theorem becomes applicable on any interval $[\varepsilon,1]$ with $\varepsilon > 0$. Of course, such an F is obtained by writing $f(x) = 2 - 2/(1+x)$, which leads to $F(x) = 2x - 2\ln(1+x)$. The integral over $[\varepsilon,1]$ is therefore $2[(1-\ln 2) - \sqrt{\varepsilon} + \ln(1+\sqrt{\varepsilon})]$. Taking the limit as $\varepsilon\to 0$ and using the continuity assured by the First Fundamental Theorem, we get the value. The process of taking the limit is implicit in the usual Calculus procedure which avoids explicit mention of it.

9-5.7. Theorem. Integration by Parts. *Let F and G be differentiable functions on $[a,b]$ such that F' and G' are both integrable. Then FG' and GF' are both integrable, and*

$$\int_a^b FG' = F(b)G(b) - F(a)G(a) - \int_a^b GF'.$$

Proof. Being differentiable, F and G are continuous and hence integrable. It follows by Prop.9-4.3 that FG' and GF' are both integrable, and further by Prop.9-4.1 that their sum $FG' + GF'$ is also integrable, with

$$\int_a^b (FG' + GF') = \int_a^b FG' + \int_a^b GF'.$$

Since $FG' + GF' = (FG)'$, the result now follows by the Second Fundamental Theorem of Calculus. ☺

Problem Set 9-5

9-5.P1. Let $f:(a,b)\to\mathbb{R}$ be continuous and bounded. Show that there exists a function $F:[a,b]\to\mathbb{R}$ such that $F'(x) = f(x)$ for each $x \in (a,b)$. [In summary: every continuous function is a derivative.]

9-5.P2. Let $f:[a,b]\to\mathbb{R}$ be bounded [but not necessarily integrable] and suppose $F:[a,b]\to\mathbb{R}$ is defined as $F(x) = \overline{\int_a^x} f$. Show that F is continuous on $[a,b]$. If f is continuous at $c \in [a,b]$, then show further that $F'(c) = f(c)$. Prove the same for the lower Riemann integral over $[a,x]$. Use this to show that a bounded function on $[a,b]$ which is continuous on (a,b) is integrable. [This is an alternative to 9-3.5 that avoids uniform continuity.]

9-5.P3. Suppose F and G are continuous functions on $[a,b]$ that are differentiable on (a,b) and such that G' and F' are both integrable when defined at a and b in some manner. Then

$$\int_a^b FG' = F(b)G(b) - F(a)G(a) - \int_a^b GF'.$$

9-5.P4. Find the Maclaurin polynomial of degree 2 for each of the functions (i) $f(x) = \int_0^x \sin(\cos t)\,dt$ (ii) $f(x) = \int_0^x \exp(t^2)\,dt$.

9-5.P5. If $f:[a,b]\to\mathbb{R}$ is continuous and there exists $K > 0$ such that $|f(x)| \le K\int_a^x |f|$ for all $x \in [a,b]$, then show that $f(x) = 0$ for all $x \in [a,b]$.

9-5.P6. Prove the following elementary version of **Gronwall's Lemma**: If $f:[a,b]\to\mathbb{R}$ is continuous and nonnegative valued and if there exist positive numbers c and K such that $f(x) \le c + K\int_a^x f$ whenever $x \in [a,b]$, then $f(x) \le c\exp(K(x-a))$ whenever $x \in [a,b]$.

9-5.P7. Let the function G on the domain $\{(x,t) \in \mathbb{R}^2 : 0 \le x \le 1, 0 \le t \le 1\}$ be the so called "Green's function":

$$G(x,t) = x(t-1) \text{ when } x \le t \quad \text{and} \quad G(x,t) = t(x-1) \text{ when } x > t.$$

If $f:[0,1]\to\mathbb{R}$ is continuous and $g:[0,1]\to\mathbb{R}$ is defined as $g(x) = \int_0^1 f(t)G(x,t)\,dt$, show that (i) $g(0) = g(1) = 0$ and that (ii) $g''(x) = f(x)$ everywhere on $[0,1]$.

9-5.P8. Show that (a) $\int_k^{k+1} \frac{1}{x}\,dx \le \frac{1}{k}$;

(b) $\ln n \le \sum_{k=1}^{n-1} \frac{1}{k}$;

(c) $\sum_{k=1}^{n} \frac{1}{k} - \ln n \ge \frac{1}{n}$;

(d) the sequence $s_n = \sum_{k=1}^{n} \frac{1}{k} - \ln n$ converges to a nonnegative limit. [The limit is called **Euler's constant**.]

9-5.P9. For $n \in \mathbb{N}$, define $f_n:[0,1]\to\mathbb{R}$ by $f_n(x) = \min\{x^{-1/2}, n^{1/2}\}$. In other words, $f_n(x) = n^{1/2}$ if $0 \le x \le 1/n$ and $f_n(x) = x^{-1/2}$ if $1/n \le x \le 1$, so that each f_n is continuous, and $f_n \ge f_m$ for $n \ge m$.

(a) Show that $\int_0^1 |f_n - f_m| = m^{-1/2} - n^{1/2}$ when $n > m$.

(b) Denote the integer part function by [ ]. For any continuous $f:[0,1]\to\mathbb{R}$ and $K > 0$ such that $|f(x)| \le K$ everywhere, show that $\varepsilon = 1/([K]+2)^2$ has the following property: There exists $N \in \mathbb{N}$ for which $n \ge N \Rightarrow \int_0^1 |f_n - f| > \varepsilon$.

9-5.P10. Suppose $f:[0,1]\to\mathbb{R}$ is continuous. Show that $\int_0^1 f(x)x^2\,dx = \frac{1}{3}f(\xi)$ for some $\xi \in [0,1]$.

9-5.P11. Evaluate $\int_0^1 \left(2x\sin\frac{1}{x} - \cos\frac{1}{x}\right)dx$, given that the function involved has value 0 when $x = 0$.

9-5.P12. Suppose that f is a continuously differentiable real function on $[a,b]$, $f(a) = f(b) = 0$ and that $\int_a^b f^2 = 1$. Prove that $\int_a^b xf(x)f'(x)\,dx = -\frac{1}{2}$ and $(\int_a^b f'(x)^2\,dx)(\int_a^b x^2 f(x)^2\,dx) > \frac{1}{4}$.

9-5.P13. Let $f:[a,b]\to\mathbb{R}$ be a differentiable function. Must the equation $\int_a^b f' = f(b) - f(a)$ hold? Justify your answer.

9-5.P14. Let L be the continuous function defined on $(0,\infty)$ by $L(x) = \int_1^x (1/t)\,dt$. Using properties of the Riemann integral but *not* of the natural logarithm function ln, show that $L(xy) = L(x) + L(y)$ and $L(1/x) = -L(x)$, where $x > 0$ and $y > 0$.

9-5.P15. Let f be continuous and positive on $[1,\infty]$. Suppose $F(x) = \int_1^x f(t)\,dt \leq f(x)^2$ for $x \geq 1$. Prove that $f(x) \geq \frac{1}{2}(x-1)$ for $x \geq 1$.

Chapter 10
More About the Riemann Integral

10-1 Integral as a Limit

Given a partition P: $a = x_0 < x_1 < \cdots < x_n = b$ of $[a,b]$ and a bounded function $f:[a,b]\to\mathbb{R}$, the lower sum $L(f,P)$ is less than or equal to any sum of the form $\sum_{j=1}^{n} f(t_j)(x_j - x_{j-1})$, where $x_{j-1} \le t_j \le x_j$ for $1 \le j \le n$. There are several such sums possible, depending upon which numbers $t_j \in [x_{j-1}, x_j]$ are taken. It will be convenient to have a name for selections of such numbers t_j and their associated sums:

10-1.1. Definition. *For any partition P: $a = x_0 < x_1 < \cdots < x_n = b$ of $[a,b]$, a finite sequence $\{t_j\}_{j=1}^{n}$ such that $x_{j-1} \le t_j \le x_j$ for $1 \le j \le n$ is called* **a dual of** *the partition P.*

Given a partition P : $a = x_0 < x_1 < \cdots < x_n = b$, a dual $\bar{P}$: $\{t_j\}_{j=1}^{n}$ and a function $f:[a,b]\to\mathbb{R}$, the **Riemann sum** *of f over P and $\bar{P}$ is*

$$\boldsymbol{S(f,P,\bar{P})} = \sum_{j=1}^{n} f(t_j)(x_j - x_{j-1}).$$

The above Definition does not presume that f is bounded. If f happens to be bounded, then $L(f,P) \le S(f,P,\bar{P}) \le U(f,P)$ for any partition P and any dual $\bar{P}$ of P.

10-1.2. Examples. (a) Let $f:[0,1]\to\mathbb{R}$ be the function defined by $f(x) = \frac{1}{1+x}$. For the partition P of $[0,1]$ consisting of the points $x_j = \frac{j}{n}$, $0 \le j \le n$, a dual $\bar{P}$ is given by the points $t_j = \frac{j}{n}$, $1 \le j \le n$. Then $S(f,P,\bar{P}) = \sum_{j=1}^{n} \frac{1}{n+j}$. Since f is decreasing, this sum also happens to be $L(f,P)$.

(b) Let $f:[0,1]\to\mathbb{R}$ be the function defined by $f(x) = (\frac{1}{2} - x)^2$ and P and $\bar{P}$ be as above. Then

$$S(f,P,\bar{P}) = \sum_{j=0}^{n} \frac{1}{n}\left(\frac{1}{2} - \frac{j}{n}\right)^2.$$

Here f is neither increasing nor decreasing on $[0,1]$, and this Riemann sum is neither an upper sum nor a lower sum.

10-1.3. Proposition. *Let $f:[a,b]\to\mathbb{R}$ be bounded and P: $a = x_0 < x_1 < \cdots < x_n = b$ a partition. Then*

and
$$L(f,P) = \inf\{S(f,P,\bar{P}) : \bar{P} \text{ a dual of } P\}$$
$$U(f,P) = \sup\{S(f,P,\bar{P}) : \bar{P} \text{ a dual of } P\}.$$

Proof. We need prove only the first equality. Since $L(f,P) \leq S(f,P,\bar{P})$ for any dual $\bar{P}$ of P, it is sufficient to show that
$$L(f,P) \geq \inf\{S(f,P,\bar{P}) : \bar{P} \text{ a dual of } P\}. \tag{A}$$

To this end, consider any $\varepsilon > 0$. As usual, let $m_j = \inf\{f(x) : x_{j-1} \leq x \leq x_j\}$. Then, for each j, there exists $t_j \in [x_{j-1}, x_j]$ such that $0 \leq f(t_j) - m_j < \frac{\varepsilon}{b-a}$. Upon multiplying by $x_j - x_{j-1}$ and adding, we get $0 \leq S(f,P,\bar{P}_1) - L(f,P) < \varepsilon$, where the dual $\bar{P}_1$ is $\{t_j\}_{j=1}^n$. Thus $L(f,P) + \varepsilon > S(f,P,\bar{P}_1) \geq \inf\{S(f,P,\bar{P}) : \bar{P} \text{ a dual of } P\}$. Since this holds for every $\varepsilon > 0$, (A) follows. ☺

***10-1.4. Proposition.** *Let* $f : [a,b] \to \mathbb{R}$ *be integrable. Then for every* $\varepsilon > 0$, *there exists a partition* P_0 *such that any refinement* P *of* P_0 *satisfies*
$$|S(f,P,\bar{P}) - \int_a^b f| < \varepsilon$$
whenever $\bar{P}$ *is any dual of* P.

Proof. By Prop.9-2.9, there exists a partition P_0 such that $\int_a^b f - \varepsilon < L(f,P_0) \leq U(f,P_0) < \int_a^b f + \varepsilon$. It then follows from the first part of Prop.9-2.2 that any refinement P of P_0 also satisfies the same inequality:
$$\int_a^b f - \varepsilon < L(f,P) \leq U(f,P) < \int_a^b f + \varepsilon.$$
But $L(f,P) \leq S(f,P,\bar{P}) \leq U(f,P)$ for any dual $\bar{P}$ of P. Therefore
$$\int_a^b f - \varepsilon < S(f,P,\bar{P}) < \int_a^b f + \varepsilon. \text{ ☺}$$

***10-1.5. Proposition.** *Let* $f : [a,b] \to \mathbb{R}$ *be any real valued function and* A *be a real number. Suppose that, for every* $\varepsilon > 0$, *there exists a partition* P
$$|S(f,P,\bar{P}) - A| < \varepsilon$$
whenever $\bar{P}$ *is any dual of* P. *Then* f *is integrable and*
$$A = \int_a^b f.$$

Proof. Suppose, if possible, that f is unbounded above. Then there exists a sequence $\{s_k\}$ in $[a,b]$ such that
$$f(s_k) \to \infty \qquad \text{and} \qquad f(s_{k+1}) > f(s_k) \text{ for every } k. \tag{1}$$
In particular $\{s_k\}$ has distinct terms in the sense that $j \neq k \Rightarrow s_j \neq s_k$. Now, (1) is easily seen to be true of any subsequence as well. By the Bolzano-Weierstrass

Theorem 3-4.7, $\{s_k\}$ has a convergent subsequence with some limit, say u. Then $u \in [a,b]$. Since the subsequence satisfies (1), it too has distinct terms. As is true of any convergent sequence with distinct terms, the aforementioned subsequence has some subsequence with the property either that all terms are less than the limit or that all terms are greater than the limit. We need consider only the former case. Then $u > a$. Denoting the last mentioned subsequence by $\{u_k\}$, we have

$$f(u_k)\to\infty, \quad u_k\to u \in (a,b], \quad \text{and} \quad u_k < u \text{ for each } k. \tag{2}$$

Now let P: $a = x_0 < x_1 < \cdots < x_n = b$ be a partition such that

$$|S(f,P,\overline{P}) - A| < 1$$

whenever $\overline{P}$ is any dual of P. Since $u \in (a,b]$, there exists j such that $x_{j-1} < u \le x_j$. By (2), there must exist k_1 such that

$$x_{j-1} < u_{k_1} < u \le x_j$$

and also k_2 such that

$$x_{j-1} < u_{k_2} < u \le x_j \text{ as well as } f(u_{k_2}) > f(u_{k_1}) + \frac{2}{x_j - x_{j-1}}.$$

Let $\overline{P}$ be a dual for which the point belonging to the subinterval $[x_{j-1}, x_j]$ is u_{k_1} and $\overline{P}'$ be a dual for which the point belonging to the subinterval $[x_{j-1}, x_j]$ is u_{k_2}. The remaining points are the same for both $\overline{P}$ and $\overline{P}'$ but otherwise arbitrary. Then

$$S(f,P,\overline{P}') - S(f,P,\overline{P}) = \big(f(u_{k_2}) - f(u_{k_1})\big)(x_j - x_{j-1}) > 2,$$

while at the same time,

$$|S(f,P,\overline{P}') - S(f,P,\overline{P})| \le |S(f,P,\overline{P}') - A| + |S(f,P,\overline{P}) - A| < 2.$$

This contradiction shows that f must be bounded above. A similar argument shows that it must be bounded below.

We shall now show that f is integrable with $\int_a^b f = A$. Consider any $\varepsilon > 0$. Let P be a partition such that $|S(f,P,\overline{P}) - A| < \frac{\varepsilon}{3}$ whenever $\overline{P}$ is a dual of P. Then $A - \frac{\varepsilon}{3} < S(f,P,\overline{P}) < A + \frac{\varepsilon}{3}$ for every dual $\overline{P}$. Since f has been shown to be bounded, it follows from Prop.10-1.3 that $A - \frac{\varepsilon}{3} \le L(f,P) \le U(f,P) \le A + \frac{\varepsilon}{3}$ and hence that $U(f,P) - L(f,P) < \varepsilon$. Therefore f is integrable [Prop.9-2.9]. Moreover,

$$A - \varepsilon < \int_a^b f < A + \varepsilon.$$

for every $\varepsilon > 0$. Therefore $\int_a^b f = A$. ☺

The preceding two Propositions show that $\int_a^b f$ is the only number A with the property that: for any $\varepsilon > 0$, there exists a partition P_0 such that any refinement P of it satisfies

$$|S(f,P,\bar{P}) - A| < \varepsilon$$

whenever $\bar{P}$ is any dual of P. Formulated in this manner, $\int_a^b f$ can be thought of as the limit of $S(f,P,\bar{P})$ as P "advances" in some manner. The clarification of how P "advances" will take us rather far afield and will not be undertaken here. The interested reader is referred to the study of "convergence of a net" on p.79 of [6]. We turn to another description of the concept of integral as a limit, one that bears a stronger resemblance to the familiar kind formulated in Def.7-1.4.

***10-1.6. Definition.** *Let* $f:[a,b]\to\mathbb{R}$ *be any real valued function. A real number* λ *is called a* **limit of $S(f,P,\bar{P})$ as $|P|\to 0$** *if for every* $\varepsilon > 0$, *there exists some* $\delta > 0$ *such that*

$$|S(f,P,\bar{P}) - \lambda| < \varepsilon \textit{ whenever } |P| < \delta \textit{ and } \bar{P} \textit{ is any dual of } P.$$

Remark. As with already familiar instances of limits, it is easy to prove that such a number λ, if it exists, must be unique. We therefore speak of *the* limit and denote it by $\lim\limits_{|P|\to 0} S(f,P,\bar{P})$.

***10-1.7. Darboux's Theorem.** *A real valued function* $f:[a,b]\to\mathbb{R}$ *is integrable if, and only if,* $\lim\limits_{|P|\to 0} S(f,P,\bar{P})$ *exists, in which case, this limit is equal to* $\int_a^b f$.

Proof. First suppose $\lim\limits_{|P|\to 0} S(f,P,\bar{P})$ exists and equals λ. We shall show that $\int_a^b f$ exists and equals λ. For this purpose, consider any $\varepsilon > 0$. By Def.10-1.6, there exists $\delta > 0$ such that $|S(f,P,\bar{P}) - \lambda| < \varepsilon$ whenever $|P| < \delta$ and $\bar{P}$ is any dual of P. Suppose $n \in \mathbb{N}$ satisfies $n > \frac{b-a}{\delta}$. Then the partition P consisting of the points $x_j = a + j\frac{b-a}{n}$ satisfies $|P| < \delta$ and consequently, $|S(f,P,\bar{P}) - \lambda| < \varepsilon$ whenever $\bar{P}$ is any dual of P. The existence of such a partition for every $\varepsilon > 0$ implies by Prop.10-1.5 that f is integrable and that $\int_a^b f = \lambda$.

Conversely, suppose that $\int_a^b f$ exists. We shall prove that $\lim\limits_{|P|\to 0} S(f,P,\bar{P})$ exists and equals $\int_a^b f$. In order to show this, consider any $\varepsilon > 0$. By Prop.9-2.9, there exists a partition Q such that

$$\int_a^b f - \tfrac{\varepsilon}{2} < L(f,Q) \le U(f,Q) < \int_a^b f + \tfrac{\varepsilon}{2}.$$

Denote the number of points of Q by $N+2$ and sup $\{|f(x)| : a \le x \le b\}$ by M, and let $0 < \delta < \varepsilon/4NM$. It is sufficient to show that whenever $|P| < \delta$,

$$\int_a^b f - \varepsilon < S(f,P,\bar{P}) < \int_a^b f + \varepsilon \text{ for any dual } \bar{P} \text{ of } P. \tag{A}$$

Let P be any partition such that $|P| < \delta$. To prove (A), consider the partition P' consisting of all the points of P as well as Q. Then on the one hand, since P' is a refinement of Q, we have by the first part of Prop.9-2.2

$$\int_a^b f - \frac{\varepsilon}{2} < L(f,P') \le U(f,P') < \int_a^b f + \frac{\varepsilon}{2}.$$

On the other hand, since P' contains at most N more points than P, we also have by the second part of the same Proposition,

$$L(f,P') - L(f,P) \le 2N\cdot M|P| \quad \text{and} \quad U(f,P) - U(f,P') \le 2N\cdot M|P|.$$

Hence

$$\begin{aligned} L(f,P) \ge L(f,P') - 2N\cdot M|P| &\ge \int_a^b f - \frac{\varepsilon}{2} - 2N\cdot M|P| \\ &> \int_a^b f - \varepsilon \end{aligned}$$

and similarly,

$$U(f,P) < \int_a^b f + \varepsilon.$$

Since $L(f,P) \le S(f,P,\bar{P}) \le U(f,P)$ for every dual $\bar{P}$ of P, it follows from the preceding two inequalities that (A) holds. ☺

***10-1.8. Corollary.** *If $f: [a,b] \to \mathbb{R}$ is integrable, then the sequences*

$$\frac{1}{n}\sum_{j=1}^{n} f\left(a + \frac{j}{n}(b-a)\right) \quad \textit{and} \quad \frac{1}{n}\sum_{j=0}^{n-1} f\left(a + \frac{j}{n}(b-a)\right)$$

both converge to the limit $\int_a^b f$.

Proof. The terms of the sequences are Riemann sums with partition

$$P : a, a + \frac{1}{n}(b-a), a + \frac{2}{n}(b-a), \ldots, a + \frac{n}{n}(b-a)$$

and respective duals

$$a + \frac{1}{n}(b-a), a + \frac{2}{n}(b-a), \ldots, a + \frac{n}{n}(b-a)$$

and $$a,\ a + \frac{1}{n}(b-a), a + \frac{2}{n}(b-a), \ldots, a + \frac{n-1}{n}(b-a).$$

The norm of P is $\frac{1}{n}$, which converges to 0 as $n \to \infty$. By Darboux's Theorem 10-1.7, it follows that Riemann sums converge to $\int_a^b f$ as $n \to \infty$. ☺

***10-1.9. Examples.** (a) As seen in Example 10-1.2(a) above, $\sum_{j=1}^{n} \frac{1}{n+j}$ is a Riemann sum for $f(x) = \frac{1}{1+x}$ with the partition of [0,1] given by the points $\frac{j}{n}$, $0 \le j \le n$, using the dual consisting of $t_j = \frac{j}{n}$, $1 \le j \le n$. By Cor.10-1.8, $\lim_{n\to\infty} \sum_{j=1}^{n} \frac{1}{n+j} = \int_0^1 f = \ln 2$.

(b) $\lim_{n\to\infty} \sum_{j=1}^{n} \frac{1}{n}(\frac{1}{2}-\frac{j}{n})^2 = \int_0^1 f$, where $f(x) = (\frac{1}{2}-x)^2$. This follows from Example 10-1.2(b) and Cor.10-1.8. This integral equals $\frac{1}{12}$.

Problem Set 10-1

10-1.P1. (a) For the function f: $[0,1]\to\mathbb{R}$ defined by $f(x) = x$ if $0 < x < 1$ and $f(0) = f(1) = \frac{1}{2}$ and partition consisting of 0 and 1 only, show that no Riemann sum is equal to the lower sum or upper sum.

(b) If f: $[a,b]\to\mathbb{R}$ is continuous and P is any partition of $[a,b]$, show that both $L(f,P)$ and $U(f,P)$ are Riemann sums.

10-1.P2. (a) Represent the sum $\sum_{j=1}^{n} \frac{n}{n^2+j^2}$ as a Riemann sum for a suitable function on the domain [0,1].

(b) Find $\lim_{n\to\infty} \sum_{j=1}^{n} \frac{n}{n^2+j^2}$.

(c) Find $\lim_{n\to\infty} \sum_{j=1}^{n} \frac{j}{n^2+j^2}$.

10-1.P3. Let f and g be integrable functions on $[a,b]$. For any $\varepsilon > 0$, show that there exists $\delta > 0$ such that whenever P: $a = x_0 < x_1 < \cdots < x_n = b$ is a partition with $|P| < \delta$ and $\bar{P}_1$: $\{t_j\}_{j=1}^{n}$ and $\bar{P}_2$: $\{s_j\}_{j=1}^{n}$ are any two duals of P, we have

$$\left|\sum_{j=1}^{n} f(s_j)g(t_j)(x_j - x_{j-1}) - \int_a^b fg\right| < \varepsilon.$$

[This is used in elementary Calculus for deriving the formula for the area of a surface of revolution.]

*10-2 Improper Integrals

The integral we have discussed so far has been over a bounded interval and for a bounded function. Just as it is useful to extend the concept of a finite sum to that of an "infinite sum" of a series, it is useful to extend the concept of integral to an unbounded interval and to an unbounded function.

10-2.1. Definition. *If the function* f: $[a,\infty)\to\mathbb{R}$ *is integrable on* $[a,b]$ *for every* $b > a$ *and* $\lim_{b\to\infty} \int_a^b f$ *exists, either as a real number or as* $\pm\infty$, *we denote the limit by*

$\int_a^\infty f$ *and call it the* **improper integral of *f* over [*a*,∞)**. *In case* $\lim_{b\to\infty}\int_a^b f$ *is a real number*, *we speak of the improper integral* $\int_a^\infty f$ *as being* **convergent**; *otherwise* **divergent**.

Note that when the limit does not exist, and therefore the integral symbol does not represent a real number or ±∞, we nonetheless speak of the "improper integral $\int_a^\infty f$" being divergent.

10-2.2. Examples. (a) $\int_1^\infty \frac{1}{x}dx$ is an improper integral with $\int_1^\infty \frac{1}{x}dx = \infty$, because $\lim_{b\to\infty}\int_1^b \frac{1}{x}dx = \lim_{b\to\infty}(\ln b) = \infty$. The improper integral is divergent.

(b) $\int_1^\infty \frac{1}{x^2}dx$ is an improper integral with $\int_1^\infty \frac{1}{x^2}dx = 1$. This is because $\lim_{b\to\infty}\int_1^b \frac{1}{x^2}dx = \lim_{b\to\infty}(1-\frac{1}{b}) = 1$. The improper integral is convergent.

(c) Since $\lim_{b\to\infty}\int_0^b \cos x\,dx$ does not exist, $\int_0^\infty \cos x\,dx$ is a divergent improper integral, although the symbol "$\int_0^\infty \cos x\,dx$" here represents neither a real number nor ±∞.

10-2.3. Proposition. *If* $\int_1^\infty f$ *converges*, *then the sequence* $\{s_n\}$ *defined by* $s_n = \int_1^n f$ *converges and has limit* $\int_1^\infty f$. *When* $f(x) \ge 0$ *on* $[1,\infty)$, *the following converse is also true*: *If* $\{s_n\}$ *converges*, *then* $\int_1^\infty f$ *converges and* $\int_1^\infty f = \lim_{n\to\infty} s_n$.

Proof. The first statement is clear. For the second statement, we begin by noting that, in view of the hypothesis that f has only nonnegative values, we have

$$1 < b < c \Rightarrow \int_1^b f \le \int_1^c f. \tag{1}$$

Denote $\lim_{n\to\infty} s_n$ by s, and consider any $\varepsilon > 0$. There exists $N \in \mathbb{N}$ with the property that

$$n \ge N \Rightarrow s-\varepsilon < s_n < s+\varepsilon. \tag{2}$$

Let [ ] be the greatest integer function. Then

$$\begin{aligned} b > N \Rightarrow N < b < [b]+1 \Rightarrow s-\varepsilon < s_N &= \int_1^N f && \text{by (2)} \\ &\le \int_1^b f && \text{by (1)} \\ &\le \int_1^{[b]+1} f = s_{[b]+1} && \text{by (1)} \\ &< s+\varepsilon. && \text{by (2)} \end{aligned}$$

Thus N has the property that $b > N \Rightarrow s-\varepsilon < \int_1^b f < s+\varepsilon$. ☺

10-2.4. Theorem. Integral Test. *Let the function* $f\colon[1,\infty)\to\mathbb{R}$ *be nonnegative valued and decreasing. Then the series* $\sum_{k=1}^{\infty} f(k)$ *is convergent if, and only if, the improper integral* $\int_1^\infty f$ *is convergent, in which case, the partial sums* $s_n = \sum_{k=1}^{n} f(k)$ *satisfy*

$$\int_{n+1}^{\infty} f \le \sum_{k=1}^{\infty} f(k) - s_n \le \int_n^\infty f. \tag{A}$$

Proof. Since f is decreasing, it is integrable on $[1,b]$ for any $b > 1$. Also, it satisfies $f(k) \le f(x) \le f(k-1)$ for $x \in [k-1,k]$. Therefore

$$f(k) \le \int_{k-1}^{k} f \le f(k-1) \qquad \text{for } k \ge 2. \tag{1}$$

Taking the sum over $2 \le k \le n$, we get

$$s_n - s_1 \le \int_1^n f \le s_{n-1}.$$

It follows from this double inequality that the sequence $\{s_n\}$ is bounded above if, and only if, the sequence $\{\int_1^n f\}$ is bounded above. Since f is nonnegative valued, each of these sequences is increasing and therefore, being convergent is equivalent to being bounded above [see Theorem 3-3.3]. Thus one is convergent if, and only if, the other is. However, by Prop.10-2.3, convergence of the sequence $\{\int_1^n f\}$ is equivalent to the convergence of the improper integral $\int_1^\infty f$. Thus the sequence $\{s_n\}$ of partial sums of $\sum_{k=1}^{\infty} f(k)$ converges if, and only if, the improper integral $\int_1^\infty f$ converges.

It remains to show that (A) holds. Consider any integer $n \ge 1$. Then $n+1 \ge 2$, so that (1) holds for $k \ge n+1$. We may therefore sum the first inequality in (1) over $n+1 \le k \le m$, thereby obtaining

$$s_m - s_n = \sum_{k=n+1}^{m} f(k) \le \int_n^m f.$$

Similarly, we may sum the second inequality in (1) over $n+2 \le k \le m+1$, thereby obtaining

$$\int_{n+1}^{m+1} f \le \sum_{k=n+2}^{m+1} f(k-1) = \sum_{k=n+1}^{m} f(k) = s_m - s_n.$$

Upon taking the limit as $m\to\infty$ in the two inequalities just derived and making use of Prop.10-2.3, we arrive at (A). ☺

10-2.5. Example. Let $f(x) = 1/x^2$. Since $\int_1^\infty f = 1$, the series $\sum_{k=1}^{n} (1/k^2)$ is convergent. Moreover, since $\int_b^\infty f = 1/b$, we also know from (A) of Theorem 10-2.4 that

$$\frac{1}{n+1} \le \sum_{k=1}^{\infty} \frac{1}{k^2} - \sum_{k=1}^{n} \frac{1}{k^2} \le \frac{1}{n}.$$

With $n = 4$, this leads to

$$\frac{1}{5}+\left(1+\frac{1}{4}+\frac{1}{9}+\frac{1}{16}\right) \le \sum_{k=1}^{\infty}\frac{1}{k^2} \le \frac{1}{4}+\left(1+\frac{1}{4}+\frac{1}{9}+\frac{1}{16}\right),$$

which simplifies to:

$$\frac{1169}{720} \le \sum_{k=1}^{\infty}\frac{1}{k^2} \le \frac{1205}{720}.$$

So far, we have checked the convergence of an improper integral directly from the definition by first obtaining $\int_a^b f$ and then finding its limit. This is somewhat akin to checking the convergence of an infinite series by first obtaining a convenient expression for the partial sum and then finding its limit. The reader may recall that our most far reaching use of infinite series, which was to set up the exponential function as the sum $\sum_{k=0}^{\infty} z^k/k!$, involved doing quite the reverse. At no stage did we seek an expression for the partial sum and the convergence of the series was established by other means instead. Something similar can be done with improper integrals: After establishing convergence without evaluating $\int_a^b f$, an improper integral can be used for defining a function, and a useful one too.

There are many convergence theorems, but we shall consider only a basic one.

10-2.6. Proposition. *Suppose* $f:[1,\infty)\to\mathbb{R}$ *is integrable on* $[1,b]$ *for every* $b > 1$, *so that* $\int_1^{\infty} f$ *is an improper integral. If* $f(x) \ge 0$ *on* $[1,b]$, *and there exists a number* $M > 0$ *such that*

$$\int_1^b f \le M \qquad \text{for every } b > 1,$$

then $\int_1^{\infty} f$ *is convergent.*

Proof. The hypothesis implies that the sequence $\{s_n\}$ defined by $s_n = \int_1^n f$ is bounded above. Since f is nonnegative valued, the sequence is also increasing and hence convergent. It follows by Prop.10-2.3 that the improper integral in question is convergent. ☺

10-2.7. Examples. (a) Let x be a real number and consider the improper integral $\int_1^{\infty} t^{x-1}e^{-t}\,dt$. Evaluation of $\int_1^b t^{x-1}e^{-t}\,dt$ would be a foolhardy undertaking. However, it can be shown that the integral is convergent by arguing as follows.

By Prop.7-4.13, $\lim_{t\to\infty} t^{x-1}e^{-t/2} = 0$. Therefore there exists L such that $t > L \Rightarrow t^{x-1}e^{-t/2} < 1$. By continuity, $t^{x-1}e^{-t/2}$ must be bounded on $[1,L]$. If K is large enough to be an upper bound and is also greater than 1, then we have $t^{x-1}e^{-t/2} \le K$ for any $t \in [1,\infty)$. It follows that $t^{x-1}e^{-t} \le Ke^{-t/2}$ for any $t \in [1,\infty)$. Now,

$$\int_1^b Ke^{-t/2}\,dt = 2K(e^{-1/2} - e^{-b/2}) < 2Ke^{-1/2}.$$

Therefore the preceding inequality implies that

$$\int_1^b t^{x-1}e^{-t}\,dt \le \int_1^b Ke^{-t/2}\,dt < 2Ke^{-1/2} \text{ whenever } b > 1.$$

Since $t^{x-1}e^{-t/2} > 0$ when $t \ge 1$, Prop.10-2.6 is applicable with $M = 2Ke^{-1/2}$.

(b) Consider the improper integral $\int_1^\infty u^{-x-1}e^{-1/u}\,du$, where x is some real number. Since $0 < e^{-1/u} < 1$ and $u^{-x-1} > 0$ when $u > 0$, we have

$$0 < u^{-x-1}e^{-1/u} < u^{-x-1} \qquad \text{for all } u \in [1,\infty).$$

If $x > 0$, then $0 < b^{-x} < 1$ for any $b > 1$ [see 7-4.P10]. Therefore $\int_1^b u^{-x-1}\,du = (1-b^{-x})/x < 1/x$. Thus Prop.10-2.6 is applicable with $M = 1/x$. This shows that the improper integral considered here is convergent when $x > 0$.

(c) $\int_1^\infty \frac{1}{1+x^5}\,dx$ can be shown to be convergent. Since $0 < \frac{1}{1+x^5} \le \frac{1}{x^5}$, and $\int_1^\infty \frac{1}{x^5}\,dx$ can be computed directly to be $\frac{1}{4}$, it follows from Prop.10-2.6 that the improper integral in question is convergent.

10-2.8. Definition. *If $f:(a,b]\to\mathbb{R}$ is unbounded but integrable on $[c,b]$ for every $c > a$ and $\lim\limits_{c\to a}\int_c^b f$ exists, either as a real number or as $\pm\infty$, we denote the limit by $\int_a^b f$ and call it the* **improper integral of *f* over [*a*, *b*]**. *In case $\lim\limits_{c\to a}\int_c^b f$ is a real number, we speak of the improper integral $\int_a^b f$ as being* **convergent**; *otherwise* **divergent**. *The same terminology and notation are used when $f:[a,b)\to\mathbb{R}$ is unbounded but integrable on $[a,c]$ for every $c < b$.*

Note that when the limit does not exist, and therefore the integral symbol does not represent a real number or $\pm\infty$, we nonetheless speak of the "improper integral $\int_a^b f$" being divergent.

Example. When $x < 1$, the function $f(t) = t^{x-1}e^{-t}$ is unbounded on $(0,1]$, but is bounded on $[\varepsilon,1]$ when $0 < \varepsilon < 1$. So we take the improper integral $\int_0^1 f$ to mean $\lim\limits_{\varepsilon\to 0}\int_\varepsilon^1 f = \lim\limits_{\varepsilon\to 0}\int_\varepsilon^1 t^{x-1}e^{-t}\,dt$ provided that the limit exists. The substitution $t = 1/u$ converts the integral $\int_\varepsilon^1 t^{x-1}e^{-t}\,dt$ into the integral $\int_1^{1/\varepsilon} u^{-x-1}e^{-1/u}\,du$. Therefore

$$\int_0^1 t^{x-1}e^{-t}\,dt = \int_1^\infty u^{-x-1}e^{-1/u}\,du$$

in the sense that if one limit exists and is a real number, the other one also exists and is equal to the former. As we have already seen [Example 10-2.7(b)], the improper integral on the right side is a real number when $x > 0$. Therefore $\int_0^1 t^{x-1}e^{-t}\,dt$ is a real number when $x > 0$. We have also seen [Example 10-2.7(a)] that $\int_1^\infty t^{x-1}e^{-t}\,dt$ is a real number for all real x. The **gamma function Γ** is defined as

$$\Gamma(x) = \int_0^1 t^{x-1} e^{-t}\,dt + \int_1^\infty t^{x-1} e^{-t}\,dt = \int_0^\infty t^{x-1} e^{-t}\,dt \qquad \text{for } x > 0.$$

10-2.9. Definition. *Suppose the function* $f: E\times[a,\infty)\to\mathbb{R}$, *where* E *is any set and* $a \in \mathbb{R}$, *has the property that, for each* $t \in E$, *the function* $f_t : [a,\infty)\to\mathbb{R}$ *defined by* $f_t(x) = f(t,x)$ *has a convergent improper integral* $\int_a^\infty f_t$. *Then this improper integral is said to be* **uniformly convergent on** *a subset* F *of* E *if, for every* $\varepsilon > 0$, *there exists* $M > 0$ *such that*

$$b \geq M \;\Rightarrow\; \left|\int_a^b f - \int_a^\infty f_t\right| < \varepsilon \textit{ for each } t \in F.$$

10-2.10. Examples. (a) If $k > 0$, the improper integral $\int_0^\infty \exp(-tx)\,dx$ is uniformly convergent on $[k,\infty)$. Explicit computation shows that

$$\int_0^b \exp(-tx)\,dx = \frac{\exp(-tb) - 1}{-t}, \text{ which has limit } \frac{1}{t} \text{ as } b\to\infty.$$

Therefore $\int_0^\infty \exp(-tx)\,dx = 1/t$ and

$$\left|\int_0^b \exp(-tx)\,dx - \int_0^\infty \exp(-tx)\,dx\right| = \left|\frac{\exp(-tb) - 1}{-t} - \frac{1}{t}\right| = \frac{\exp(-tb)}{t}.$$

The derivative of this quotient is $-(tb + 1)t^{-2}\exp(-tb)$, which is always negative. Therefore the quotient remains less than or equal to $k^{-1}\exp(-kb)$ for $t \geq k$. Now, $k^{-1}\exp(-kb)$ tends to 0 as $b\to\infty$ and therefore for any $\varepsilon > 0$ there exists $M > 0$ such that $b \geq M \Rightarrow k^{-1}\exp(-kb) < \varepsilon \Rightarrow \left|\int_0^b \exp(-tx)\,dx - \int_0^\infty \exp(-tx)\,dx\right| < \varepsilon$. This proves uniform convergence on $[k,\infty)$. Since this has been shown for each $k > 0$, the integral converges for each $t \in (0,\infty)$.

(b) The improper integral in part (a) is not uniformly convergent on $(0,\infty)$. As already seen, $\left|\int_0^b \exp(-tx)\,dx - \int_0^\infty \exp(-tx)\,dx\right| = \frac{\exp(-tb)}{t}$. When $b > 1$, we find that $t = 1/b \in (0,\infty)$ and

$$\frac{\exp(-tb)}{t} = b/e > 1/e.$$

Thus the integral does not converge uniformly on $(0,\infty)$, although it converges for each $t \in (0,\infty)$.

Problem Set 10-2

10-2.P1. Evaluate $\int_0^\infty \frac{1}{1+x^2}\,dx$ and $\int_2^\infty \frac{1}{x(\ln x)^2}\,dx$.

10-2.P2. (a) Show that $\frac{51739}{43200} \leq \sum_{k=1}^\infty \frac{1}{k^3} \leq \frac{52225}{43200}$.
(b) Show that

$$\frac{4}{5}+\frac{\pi}{2}-\tan^{-1}4 \le \sum_{k=1}^{\infty}\frac{1}{1+k^2} \le \frac{4}{5}+\frac{\pi}{2}-\tan^{-1}3.$$

10-2.P3. Show that $\int_0^\infty \frac{\sin^2 x}{x^2}\,dx$ and $\int_2^\infty \frac{\exp(x/(1+x))}{x^2}\,dx$ are convergent.

10-2.P4. Which of the following improper integrals are convergent?

(a) $\int_1^\infty \frac{\ln x}{x}\,dx$ (b) $\int_2^\infty \frac{1}{x\ln x}\,dx$ (c) $\int_1^\infty \frac{\exp(-x-1/x)}{2x^2+\sin x}\,dx$

10-2.P5. Show that $\int_1^\infty \frac{1}{x}\left(\frac{1}{x}-\frac{2}{e^x-e^{-x}}\right)dx$ is convergent.

10-2.P6. Is $\int_0^\infty \ln\left(1+\frac{4}{e^x+e^{-x}}\right)dx$ convergent?

10-2.P7. Determine which (if any) of the improper integrals $\int_1^2 \frac{1}{\ln x}\,dx$ and $\int_1^\infty \frac{dx}{x^2(\ln(x+1)-\ln x)}$ is convergent.

10-2.P8. Show that the series $\sum_{k=1}^{\infty} x/(1+k^2x^2)$ does not converge uniformly on $(0,1]$.

10-2.P9. Prove the following properties of the gamma function:

(a) $\Gamma(x+1)=x\Gamma(x)$ for $x>0$;

(b) $\Gamma(n+1)=n!$ for $n\in\mathbb{N}$;

(c) $\Gamma(\frac{x}{p}+\frac{y}{q}) \le \Gamma(x)^{1/p}\Gamma(y)^{1/q}$, where p and q are any positive numbers such that $1/p+1/q=1$

(d) the function f defined by $f(x)=\ln\Gamma(x)$ is "convex" in the sense that $f(\lambda x+(1-\lambda)y) \le \lambda f(x)+(1-\lambda)f(y)$ whenever $x,y\in(0,\infty)$ and for all $\lambda\in(0,1)$.

10-2.P10. Let $f(x)=\exp(x^2/2)\int_x^\infty \exp(-t^2/2)\,dt$, $x>0$. Show that $0<f(x)<1/x$ and that f is strictly increasing for $x>0$.

*10-3 Mean Value Theorems and Applications

For an integrable function $f:[a,b]\to\mathbb{R}$, with infimum m and supremum M, we have $m\le f(x)\le M$ on $[a,b]$ and hence $m(b-a)\le \int_a^b f \le M(b-a)$. So there must exist $\mu\in[m,M]$ such that $\int_a^b f=\mu(b-a)$. This number μ is called the mean value of f over the interval, although it may not be the value of f at any point. In case f is continuous [resp. a derivative], the Intermediate Value Theorem [resp. 7-5.P10] yields a number $\xi\in[a,b]$ such that $f(\xi)=\mu$, and we have

$$\int_a^b f = f(\xi)(b-a).$$

However, in this case we also have some $F:[a,b]\to\mathbb{R}$ such that $f = F'$ and $F(x) = \int_a^x f$ by the First [resp. Second] Fundamental Theorem, and the above equality states that $F(b)-F(a) = F'(\xi)(b-a)$. Thus it could also have been obtained from the Mean Value Theorem (MVT) of Lagrange.

A slight generalisation involving two functions is just as easy to prove and is known as the **First MVT** (**for Integrals**). With the additional hypotheses that the product of the two functions involved as well as one of the functions are both continuous [or are derivatives] and that the latter function is never 0, the result can also be obtained from the Cauchy MVT or 7-5.P11 [see 10-3.P1].

10-3.1. Theorem. First MVT. *If f and g are integrable functions on $[a,b]$ and either $g \geq 0$ everywhere or $g \leq 0$ everywhere, then there exists μ such that*

$$\inf f \leq \mu \leq \sup f \qquad \textit{and} \qquad \int_a^b fg = \mu \int_a^b g. \tag{A}$$

If f is continuous or a derivative, there exists $\xi \in [a,b]$ such that

$$\int_a^b fg = f(\xi)\int_a^b g\,.$$

Proof. If $g \geq 0$ everywhere, then $(\inf f)g(x) \leq f(x)g(x) \leq (\sup f)g(x)$ everywhere. Therefore

$$(\inf f)\int_a^b g \leq \int_a^b fg \leq (\sup f)\int_a^b g\,.$$

If $\int_a^b g = 0$, it follows that $\int_a^b fg = 0$ and hence (A) holds with any μ that satisfies $\inf f \leq \mu \leq \sup f$. Otherwise, $\int_a^b g > 0$ and the preceding inequality implies that (A) holds with $\mu = (\int_a^b fg)/(\int_a^b g)$.

When f is continuous, the last part follows by the Intermediate Value Theorem; if f is a derivative, we use 7-5.P10 instead. ☺

The next result has a more general version that does not use any MVT about derivatives. For details, we refer the reader to [9]. We restrict ourselves to a case adequate for concrete applications to improper integrals.

10-3.2. Second MVT (Weierstrass). *Suppose $f:[a,b]\to\mathbb{R}$ is integrable and $g:[a,b]\to\mathbb{R}$ is differentiable with a derivative that is never 0. Suppose also that f and fg are derivatives* [*e.g. continuous*]. *Then there exists $\xi \in (a,b)$ such that*

$$\int_a^b fg = g(a)\int_a^\xi f + g(b)\int_\xi^b f.$$

Proof. Let $f = F'$ and $fg = \phi'$. Now f and fg are both integrable and it follows by the Second Fundamental Theorem for any $\xi \in (a,b)$ that

$$F(\xi) - F(a) = \int_a^\xi f,\ F(b) - F(\xi) = \int_\xi^b f \quad \text{and that} \quad \phi(b) - \phi(a) = \int_a^b fg. \tag{1}$$

Also,

$$(\phi - Fg)' = \phi' - (F'g + Fg') = fg - (fg + Fg') = -Fg'. \tag{2}$$

The Cauchy MVT or 7-5.P11 applied to the pair of functions $\phi - Fg$ and g yields a number $\xi \in (a,b)$ such that

$$\begin{aligned}\phi(b) - F(b)g(b) - \phi(a) + F(a)g(a) &= \big(g(b) - g(a)\big)[(\phi - Fg)'(\xi)]\frac{1}{g'(\xi)} \\ &= -\big(g(b) - g(a)\big)F(\xi) \quad \text{by (2)}.\end{aligned}$$

Hence
$$\begin{aligned}\phi(b) - \phi(a) &= F(b)g(b) - F(a)g(a) - g(b)F(\xi) + g(a)F(\xi) \\ &= g(a)\big(F(\xi) - F(a)\big) + g(b)\big(F(b) - F(\xi)\big).\end{aligned}$$

Upon substituting into this from (1), we get precisely the required equality. ☺

10-3.3. Examples. (a) Consider $\int_a^b \frac{\sin x}{x}\,dx$, where $0 < a < b$. With $f(x) = \sin x$ and $g(x) = \frac{1}{x}$, we have $\frac{\sin x}{x} = f(x)g(x)$. Since fg and g are both continuous, they are derivatives. Besides, $g' < 0$ on $[a,b]$. It follows by the Second MVT above that

$$\int_a^b \frac{\sin x}{x}\,dx = \frac{1}{a}\int_a^\xi \sin x\,dx + \frac{1}{b}\int_\xi^b \sin x\,dx, \quad \text{where } a < \xi < b.$$

However, the integrals on the right both have absolute value no greater than 2. Hence

$$\left|\int_a^b \frac{\sin x}{x}\,dx\right| \le 2\left(\frac{1}{a} + \frac{1}{b}\right) < \frac{4}{a}.$$

In particular, for any $\varepsilon > 0$, the number $A = \frac{4}{\varepsilon}$ has the property that

$$b > a > A \Rightarrow \left|\int_a^b \frac{\sin x}{x}\,dx\right| < \varepsilon.$$

It may be noted that this cannot be derived by using the First MVT, because the latter leads to: $\int_a^b \frac{\sin x}{x}\,dx = \sin\xi(\ln b - \ln a)$.

(b) When $0 < a < b$, we have

$$\begin{aligned}\int_a^b \sin x^p\,dx &= \int_a^b \frac{1}{px^{p-1}}(px^{p-1}\sin x^p)\,dx \\ &= \frac{1}{pa^{p-1}}\int_a^\xi px^{p-1}\sin x^p\,dx + \frac{1}{pb^{p-1}}\int_\xi^b px^{p-1}\sin x^p\,dx.\end{aligned}$$

Assume $p > 1$. Then $|\int_a^b \sin x^p\, dx| \le \frac{4}{pa^{p-1}}$, and therefore for any $\varepsilon > 0$, the number $A = \left(\frac{4}{p\varepsilon}\right)^{1/(p-1)}$ has the property that

$$b > a > A \Rightarrow |\int_a^b \sin x^p\, dx| < \varepsilon.$$

The significance of the kind of property that each of the preceding Examples ends with is that it is the analogue of the Cauchy criterion for the existence of $\lim_{x\to\infty} F(x)$, where x need not be an integer. More precisely:

10-3.4. Proposition. *For a function* $F:[a,\infty)\to\mathbb{R}$, $\lim_{x\to\infty} F(x)$ *exists if, and only if,*

$$\forall\ \varepsilon > 0, \exists\ A \ge a \text{ such that: } x \ge A \text{ and } y \ge A \Rightarrow |F(x) - F(y)| < \varepsilon. \tag{I}$$

Proof. The proof is left an an exercise for the reader [see 10-3.P2]. ☺

In view of this Proposition, the illustrations in 10-3.3 show that the improper integrals $\int_1^\infty \frac{\sin x}{x}\, dx$ and $\int_1^\infty \sin x^p\, dx$ ($p > 1$) are convergent. The role of the Second MVT in the proof of convergence can be formalised into a general result:

10-3.5. Theorem. Dirichlet's Test. *Suppose* $f:[a,\infty)\to\mathbb{R}$ *is integrable on every bounded closed subinterval of its domain and that* $g:[a,\infty)\to\mathbb{R}$ *is differentiable with a derivative that is never* 0. *Suppose also that* f *and* fg *are derivatives* [*e.g. continuous*], *that* $\lim_{x\to\infty} g(x) = 0$ *and that there exists* $M > 0$ *such that* $|\int_a^b f\,| \le M$ *for all* $b > a$. *Then* $\int_a^\infty fg$ *is convergent.*

Proof. For $a < b < c$, we have from the Second MVT

$$\begin{aligned} |\int_b^c fg| &= \left|g(b)\int_b^\xi f + g(c)\int_\xi^c f\right| \text{ for some } \xi \in (b,c) \\ &= \left|g(b)\left(\int_a^\xi f - \int_a^b f\right) + g(c)\left(\int_a^c f - \int_a^\xi f\right)\right| \\ &\le 4M\cdot\max\{|g(b)|, |g(c)|\}. \end{aligned}$$

Since $\lim_{x\to\infty} g(x) = 0$, for any $\varepsilon > 0$, there exists $A > a$ such that $x \ge A \Rightarrow |g(x)| < \varepsilon/4M$. Hence $b, c \ge A \Rightarrow |\int_b^c fg| < \varepsilon$. By Prop.10-3.4, $\int_a^\infty fg$ converges. ☺

10-3.6. Examples. (a) $\int_0^\infty \frac{\sin x}{\sqrt{x}}\, dx$ is convergent according to Dirichlet's Test 10-3.5, where $f(x) = \sin x$ and $g(x) = 1/\sqrt{x}$. [Note that $(\sin x)/\sqrt{x}$ is continuous when taken to be 0 for $x = 0$.]

(b) $\int_1^\infty \frac{\sin x^p}{x^q}\,dx$ is convergent if $p + q > 1$. This is obvious when $p = 0$. For $p \neq 0$, express $\frac{\sin x^p}{x^q}$ as $\frac{px^{p-1}\sin x^p}{px^{p+q-1}}$ and apply Dirichlet's Test 10-3.5 with $f(x) = px^{p-1}\sin x^p$ and $g(x) = \frac{1}{px^{p+q-1}}$.

(c) Since e^{-ax}/x (where $a > 0$) has limit 0 as $x\to\infty$, and has derivative $-e^{-ax}(ax+1)/x^2$, which is never 0 on $[1,\infty]$, it follows by Dirichlet's Test 10-3.5 as in (a) above that

$$\int_1^\infty \frac{e^{-ax}}{x}\sin x\,dx \text{ is convergent.}$$

Since e^{-ax}/x has limit 1 as $x\to 0$, it further follows that

$$\int_0^\infty \frac{e^{-ax}}{x}\sin x\,dx \text{ is convergent.}$$

(d) Consider the improper integral

$$\int_0^\infty \frac{x}{1+x^2}\sin x\,dx.$$

Here the function $\frac{x}{1+x^2}$ has a nonzero derivative on $[2,\infty)$ but not on $[0,\infty)$, because the derivative is $\frac{1-x^2}{(1+x^2)^2}$. Also, its limit as $x\to\infty$ is 0. Therefore it follows by Dirichlet's Test that

$$\int_2^\infty \frac{x}{1+x^2}\sin x\,dx$$

is convergent. However, since the integral over $[0,2]$ is a "proper" integral, the given improper integral is convergent.

(e) It is possible to prove the analogues of the Cauchy criterion for uniform convergence and the Weierstrass M-test. We leave it to the reader to formulate and prove them. However, we shall present an application of the M-test and the reader would do well to formulate it before reading on.

Consider the integral $\int_0^\infty \frac{t}{t^2+x^2}\,dx$, where $0 < a \le t \le b$. Since

$$\left|\frac{t}{t^2+x^2}\right| \le \frac{b}{a^2+x^2} \qquad \text{whenever } t\in[a,b] \text{ and } x\in(0,\infty),$$

and $\int_0^\infty \frac{b}{a^2+x^2}\,dx = b\pi/2a$, a positive real number, it follows by the Weierstrass M-test that $\int_0^\infty \frac{t}{t^2+x^2}\,dx$ converges uniformly (and absolutely in the sense of 10-3.P10 below) on $[a,b]$.

Problem Set 10-3

10-3.P1. Deduce the First MVT (10-3.1) from the Cauchy MVT or 7-5.P11, assuming that fg and g are both derivatives and that g is never 0.

10-3.P2. Prove Prop.10-3.4.

10-3.P3. Determine whether $\int_1^\infty \frac{\cos x}{(x^2+x)^{\frac{1}{2}}}\,dx$ converges.

10-3.P4. For which values of p is $\int_1^\infty \frac{\cos x}{x^p}\,dx$ convergent?

10-3.P5. For which values of p is $\int_1^\infty \frac{\cos(x+x^2)}{x^p}\,dx$ convergent?

10-3.P6. (a) Let $f:[a,b]\to\mathbb{R}$ be continuous on its domain and differentiable on $[a,b)$. If $0 \le a < b$, show that there exists $\xi \in (a,b)$ such that $f(b)-f(a) = bf'(\xi)-af'(a)$.

(b) Let the continuous functions $f:[a,b]\to\mathbb{R}$ and $g:[a,b]\to\mathbb{R}$ be differentiable on $[a,b)$, with $g' > 0$ everywhere and $g(a) \ge 0$. Show that there exists $\xi \in (a,b)$ such that

$$f(b)-f(a) = \frac{g(b)f'(\xi)}{g'(\xi)} - \frac{g(a)f'(a)}{g'(a)}.$$

(c) Prove the following version of the Second MVT, known as **Bonnet's** form:

Under the conditions of 10-3.2, if g is also strictly increasing with $g(a) \ge 0$, then $\int_a^b fg = g(b)\int_\xi^b f$ for some $\xi \in (a,b)$.

10-3.P7. Show that $\int_0^\infty \frac{\sin tx}{1+x^2}\,dx$ is uniformly convergent on $\mathbb{R}$.

10-3.P8. Show that the improper integral $\int_0^1 \frac{1}{x}e^{-ax}\sin\frac{1}{x}\,dx$ over $[0,1]$ is convergent when $a > 0$.

***10-3.P9.** (a) Show that the analogue of 9-4.P10(i) holds for improper integrals over bounded domains but not over unbounded domains.
(b) Show that the analogue of 9-4.P10(ii) holds for improper integrals over bounded domains provided that the function is nonnegative, but not over unbounded domains.

10-3.P10. If the function $f:[a,\infty)\to\mathbb{R}$ is integrable on $[a,b]$ for every $b > a$, then so is $|f|$. If $\int_a^\infty |f|$ converges, we say that $\int_a^\infty f$ **converges absolutely**. Show that, if $\int_a^\infty f$ converges absolutely, then it converges and give an example to show that the converse is false. An improper integral that converges but not absolutely is said to **converge conditionally**.

10-3.P11. Let $f: [0,\infty)\to\mathbb{R}$ be given by $f(x) = \frac{(-1)^{n+1}}{n}$, where $n-1 \le x \le n$, (i.e. $n-1$ is the integer part of x). Show that $\int_0^\infty f$ is convergent but not absolutely [for definition, see 10-3.P10].

10-3.P12. Let f be a continuous nonnegative valued function on $(0,\infty)$ such that $\int_0^\infty f < \infty$. Prove that $\frac{1}{n}\int_0^\infty xf(x)\,dx \to 0$ as $n\to\infty$.

10-3.P13. Let $f: [0,\infty)\to\mathbb{R}$ be uniformly continuous and $\int_0^\infty f$ be convergent. Show that $\lim\limits_{x\to\infty} f(x) = 0$.

*10-4 $\int_0^\infty \exp(-x^2)\,dx = \sqrt{\pi}/2$

Although there is no simple expression for $\int_0^b \exp(-x^2)\,dx$ in terms of b, it is still possible to find its limit as $b\to\infty$. Readers familiar with what is called the normal distribution in Statistics will be aware that the value of this improper integral plays a crucial role.

10-4.1. Theorem. $\int_0^\infty \exp(-x^2)\,dx = \frac{1}{2}\sqrt{\pi}$.

Proof. Let $f: \mathbb{R}\to\mathbb{R}$ have a continuous second derivative everywhere. By Taylor's Theorem,

$$\left|\frac{f(x+h)-f(x)}{h} - f'(x)\right| \le \frac{|h|}{2}\,|f''(c)|,$$

where $h \ne 0$ and c is a number between x and $x+h$. In particular, if $b > 0$ and $M_b = \sup\{|f''(t)| : 0 \le t \le b\sqrt{2}\}$ [which must be a real number because of the assumed continuity of f''], then

$$\left|\frac{f(x+h)-f(x)}{h} - f'(x)\right| \le \frac{|h|}{2}\,M_b,$$

as long as $x, x+h \in [0, b\sqrt{2}]$ and $h \ne 0$. Now any $\theta \in [0, \pi/4]$ must satisfy $1 \le \sec^2\theta \le 2$. It follows for $x, x+h \in [0, b]$ with $h \ne 0$ and every $\theta \in [0, \pi/4]$ that

$$\left|\frac{f((x+h)\sec\theta) - f(x\sec\theta)}{h} - f'(x\sec\theta)\sec\theta\right| \le \frac{|h\sec\theta|}{2}\,M_b\cdot\,\sec\theta.$$

For the function $g: [0,\infty)\to\mathbb{R}$ defined by $g(x) = \int_0^{\pi/4} f(x\sec\theta)\,d\theta$, the above inequality implies that

$$\left|\frac{g(x+h)-g(x)}{h} - \int_0^{\pi/4} f'(x\sec\theta)\sec\theta\,d\theta\right| \le \frac{|h|}{2} M_b.$$

[We have used $\int_0^{\pi/4}\sec^2\theta\,d\theta = 1$.] Since this holds for all x, $x+h \in [0,b]$ with $h \ne 0$, it follows that $g'(x)$ exists for $x \in [0,b]$ and that

$$g'(x) = \int_0^{\pi/4} f'(x\sec\theta)\sec\theta\,d\theta. \tag{1}$$

But this is true for all $b > 0$. Therefore it holds for all $x \in [0,\infty)$.

Choose $f(x) = \exp(-x^2)$. Then $f'(x) = -2x\exp(-x^2)$ and by (1),

$$g'(x) = \int_0^{\pi/4} -2x\sec^2\theta\,\exp(-x^2\sec^2\theta)\,d\theta, \tag{2}$$

where

$$g(x) = \int_0^{\pi/4}\exp(-x^2\sec^2\theta)\,d\theta.$$

On the other hand, by the First Fundamental Theorem, the function

$$\phi(x) = \left(\int_0^x \exp(-t^2)\,dt\right)^2$$

satisfies

$$\begin{aligned}\phi'(x) &= 2\exp(-x^2)\int_0^x \exp(-t^2)\,dt\\ &= \int_0^{\pi/4} 2x\sec^2\theta\,\exp(-x^2\sec^2\theta)\,d\theta \quad \text{upon substituting } t = x\tan\theta.\end{aligned}$$

Comparing with (2), we find that $g'(x) + \phi'(x) = 0$ on $[0,\infty)$. Thus $g+\phi$ is a constant. Now, an easy evaluation leads to $g(0) + \phi(0) = \pi/4$ and therefore

$$\lim_{x\to\infty}(g(x)+\phi(x)) = \pi/4.$$

On the other hand, since $0 \le x^2 \le x^2\sec^2\theta$, the definition of g shows that

$$g(x) \le (\pi/4)\exp(-x^2), \text{ so that } \lim_{x\to\infty} g(x) = 0.$$

Therefore $\lim_{x\to\infty}\phi(x) = \pi/4$, which is to say that

$$\left(\int_0^\infty \exp(-t^2)\,dt\right)^2 = \pi/4. \text{ ☺}$$

Problem Set 10-4

10-4.P1. Let $f:\mathbb{R}\to\mathbb{R}$ have a continuous first derivative on $\mathbb{R}$ and let $b > 0$. For any $\varepsilon > 0$, show that there exists $\delta > 0$ such that

$$x \in [0,b],\ x+h \in [0,b],\ 0 < |h| < \delta \ \Rightarrow\ \left|\frac{f(x+h)-f(x)}{h} - f'(x)\right| < \varepsilon.$$

10-4.P2. Let $f:\mathbb{R}\to\mathbb{R}$ have a continuous first derivative on $\mathbb{R}$. For $g(x) = \int_0^{\pi/4} f(x\sec\theta)\sec\theta\,d\theta$, show that $g'(x) = \int_0^{\pi/4} f'(x\sec\theta)\sec\theta\,d\theta$ for $x \ge 0$.

Chapter 11

*The Stieltjes Integral

11-1 Introduction

The reader has undoubtedly noticed some sort of similarity between the two "infinite summation" proceses of series and Riemann integrals. We make the similarity more precise in the final Theorem of this Chapter and of the book. To do so, we employ a generalisation of the Riemann integral, called the Riemann-Stieltjes integral, or simply "Stieltjes integral" for short. The concept is relevant elsewhere too; in Probability Theory, what is known as "the expected value of a random variable" is best defined either as a Stieltjes integral or something closely related.

11-1.1. Definition. *Let* $f:[a,b]\to\mathbb{R}$ *be bounded*, $\alpha:[a,b]\to\mathbb{R}$ *be increasing and*

$$P: a = x_0 < x_1 < \cdots < x_n = b$$

be a partition of $[a,b]$. *For* $j = 1, \ldots, n$, *let*

$$m_j = \inf\{f(x) : x_{j-1} \le x \le x_j\} \quad \textit{and} \quad M_j = \sup\{f(x) : x_{j-1} \le x \le x_j\}.$$

Then the **lower sum of** *f over the partition P* **with respect to** α *is*

$$L(f,P,\alpha) = \sum_{j=1}^{n} m_j(\alpha(x_j) - \alpha(x_{j-1}))$$

and the **upper sum** *is*

$$U(f,P,\alpha) = \sum_{j=1}^{n} M_j(\alpha(x_j) - \alpha(x_{j-1})).$$

11-1.2. Remark. Since $m_j \le M_j$ and $\alpha(x_{j-1}) \le \alpha(x_j)$ for each j, we have $L(f,P,\alpha) \le U(f,P,\alpha)$ for any bounded function f and any partition P.

In the case when $\alpha(x) = x$ for every $x \in [a,b]$, the lower and upper sums with respect to α are the same as what were called lower sum and upper sum in the preceding two Chapters and were denoted by $L(f,P)$ and $U(f,P)$ respectively.

11-1.3. Examples. (a) Let $f:[1,5]\to\mathbb{R}$ and $\alpha:[1,5]\to\mathbb{R}$ be given by $f(x) = x^2$ and $\alpha(x) = x^3$ respectively, and let P be the partition consisting of the numbers $x_0 = 1$, $x_1 = 2$, $x_2 = 4$, $x_3 = 5$. Then

$$m_1 = \inf\{x^2 : 1 \le x \le 2\} = 1, \quad m_2 = \inf\{x^2 : 2 \le x \le 4\} = 4$$

$$\text{and} \quad m_3 = \inf\{x^2 : 4 \le x \le 5\} = 16.$$

Therefore $$m_1(\alpha(x_1) - \alpha(x_0)) = 1\cdot(\alpha(2) - \alpha(1)) = 1(8-1) = 7,$$

$$m_2(\alpha(x_2) - \alpha(x_1)) = 4\cdot(\alpha(4) - \alpha(2)) = 4(64-8) = 224,$$

and $\qquad m_3(\alpha(x_3)-\alpha(x_2)) = 16\cdot(\alpha(5)-\alpha(4)) = 16(125-64) = 976,$

so that $\qquad L(f,P,\alpha) = 7+224+976 = 1207.$

Similarly,

$$M_1 = \sup\{x^2 : 1 \le x \le 2\} = 4, \quad M_2 = \sup\{x^2 : 2 \le x \le 4\} = 16$$
$$\text{and} \quad M_3 = \sup\{x^2 : 4 \le x \le 5\} = 25.$$

Therefore $\qquad M_1(\alpha(x_1)-\alpha(x_0)) = 4\cdot(8-1) = 28,$

$$M_2(\alpha(x_2)-\alpha(x_1)) = 16\cdot(64-8) = 896,$$

and $\qquad M_3(\alpha(x_3)-\alpha(x_2)) = 25\cdot(125-64) = 1525,$

so that $\qquad U(f,P,\alpha) = 28+896+1525 = 2449.$

(b) Suppose f is as above but P contains one more point 3. Then there are 4 subintervals instead of 3; also the first and last subintervals are the same as above but there are new subintervals, namely $[2,3]$ and $[3,4]$. Their union is $[2,4]$, which was a subinterval of the partition above. Naturally, $m_1 = 1$, $m_2 = 4$ and $m_4 = 16$. However, $m_3 = 9$. Consequently

$$\begin{aligned} L(f,P,\alpha) &= 1\cdot(8-1)+4\cdot(27-8)+9\cdot(64-27)+16\cdot(125-64) \\ &= 7+76+333+976 = 1392. \end{aligned}$$

This is a larger value than what we obtained above for the partition having one point less than the present one. That value was $1207 = 7+224+976$ while in the present situation, the 224 has been replaced by $76+333$. A single term has been replaced by two terms, and the reason why the sum of the two terms is greater than the single term they replace is similar to the one discussed in Chapter 9 when $\alpha(x) = x$.

The analogous thing happens for upper sums.

(c) Let $f\colon[0,1]\to\mathbb{R}$ be the function given by $f(0) = 2$ and $f(x) = 3$ for $0 < x \le 1$ and $\alpha\colon[0,1]\to\mathbb{R}$ be given by $\alpha(x) = x^2$. Consider the partition P: 0, ε, $1-\varepsilon$, 1, where $0 < \varepsilon < \frac{1}{2}$. We have

$$\begin{aligned} m_1 &= \inf\{f(x) : 0 \le x \le \varepsilon\} = 2, \\ m_2 &= \inf\{f(x) : \varepsilon \le x \le 1-\varepsilon\} = 3, \\ m_3 &= \inf\{f(x) : 1-\varepsilon \le x \le 1\} = 3. \end{aligned}$$

Hence $L(f,P,\alpha) = 2[\varepsilon^2-0^2]+3[(1-\varepsilon)^2-\varepsilon^2]+3[1^2-(1-\varepsilon)^2] = 3-\varepsilon^2$. Also, $U(f,P',\alpha) = 3$ for every partition P'. Thus $L(f,P,\alpha) \le U(f,P',\alpha)$. Note the additional feature that

$$0 \le U(f,P',\alpha)-L(f,P,\alpha) = \varepsilon^2.$$

(d) Let $f:[0,1]\to\mathbb{R}$ be as in (c) above but $\alpha:[0,1]\to\mathbb{R}$ be given by $\alpha(0) = -1$ and $\alpha(x) = x^2$ for $0 < x \leq 1$. Consider the same partition again, i.e.

$$P: 0,\ \varepsilon,\ 1-\varepsilon,\ 1, \text{ where } 0 < \varepsilon < \tfrac{1}{2}.$$

This time, $L(f,P,\alpha) = 2[\varepsilon^2 - (-1)] + 3[(1-\varepsilon)^2 - \varepsilon^2] + 3[1^2 - (1-\varepsilon)^2] = 5 - \varepsilon^2$. Also, $U(f,P',\alpha) = 6$ for every partition P'. Thus $L(f,P,\alpha) < U(f,P',\alpha)$. However,

$$U(f,P',\alpha) - L(f,P,\alpha) = 1 + \varepsilon^2 > 1.$$

Problem Set 11-1

11-1.P1. (a) For the functions $f:[-1,1]\to\mathbb{R}$ and $\alpha:[-1,1]\to\mathbb{R}$ defined by $f(x) = x^3$, $\alpha(x) = (x+1)^2$, find $L(f,P,\alpha)$ and $U(f,P,\alpha)$ when the partition P consists of $-1, 0, \frac{1}{2}, 1$.

(b) P is a partition of $[0,5]$ such that 1 and 3 are consecutive points of it and $L(f,P,\alpha) = 908$, where $f(x) = x^2$ and $\alpha(x) = x^3$ for $x \in [0,5]$. For the partition P' obtained from P by including the point 2, find $L(f,P',\alpha)$.

11-1.P2. (a) Let $f:[0,1]\to\mathbb{R}$ be the function given by $f(0) = 8$ and $f(x) = 3$ for $0 < x \leq 1$ and $\alpha:[0,1]\to\mathbb{R}$ be given by $\alpha(x) = x^2$. Consider the partition P: 0, ε, $1-\varepsilon$, 1, where $0 < \varepsilon < \frac{1}{2}$. Show that $U(f,P,\alpha) = 3 + 5\varepsilon^2$ and that $L(f,P',\alpha) = 3$ for any partition P'.

(b) Let $f:[0,1]\to\mathbb{R}$ be the function given by $f(1) = 7$ and $f(x) = 3$ for $0 \leq x < 1$ and $\alpha:[0,1]\to\mathbb{R}$ be given by $\alpha(x) = x^2$. Consider the partition P: 0, ε, $1-\varepsilon$, 1, where $0 < \varepsilon < \frac{1}{2}$. Show that $U(f,P,\alpha) = 3 + 4\varepsilon^2$ and that $L(f,P',\alpha) = 3$ for any partition P'.

(c) Let $f:[0,1]\to\mathbb{R}$ be the function given by $f(1) = 1$ and $f(x) = 3$ for $0 \leq x < 1$ and $\alpha:[0,1]\to\mathbb{R}$ be given by $\alpha(x) = x^2$. Consider the partition P: 0, ε, $1-\varepsilon$, 1, where $0 < \varepsilon < \frac{1}{2}$. Show that $L(f,P,\alpha) = 3 - 2\varepsilon^2$ and that $U(f,P',\alpha) = 3$ for any partition P'.

11-1.P3. Let $f:[0,2]\to\mathbb{R}$ be a function such that $f(x) = 3$ if $0 \leq x < 1$, $f(1) = 8$ and $f(x) = 4$ if $1 < x \leq 2$. Also, let $\alpha:[0,2]\to\mathbb{R}$ be given by $\alpha(x) = x^3$. For any positive $\varepsilon < 1$, the sequence

$$0,\ 1-\varepsilon, 1+\varepsilon, 2$$

is a partition P of $[0,2]$. Find $L(f,P,\alpha)$ and $U(f,P,\alpha)$.

11-1.P4. Let $f:[1,3]\to\mathbb{R}$ be a function such that $f(x) = x$ whenever $x \in [1,3]$ and $\alpha:[1,3]\to\mathbb{R}$ be given by $\alpha(x) = 10$ for $1 \leq x < 2$ and $\alpha(x) = 17$ for $2 \leq x \leq 3$. For the partition

$$P: 1, 2-\varepsilon, 2+\varepsilon, 3, \quad \text{where } 0 < \varepsilon < 1,$$

find $U(f,P,\alpha) - L(f,P,\alpha)$. Find a partition P' such that $U(f,P',\alpha) - L(f,P',\alpha) = 7\varepsilon$.

11-1.P5. Given the partition P: $x_j = \frac{j}{n}$, $0 \le j \le 4n$, of $[0,4]$, find $L(f,P,\alpha)$ and $U(f,P,\alpha)$, where $f(x) = x$ and $\alpha(x) = x^2$ for $x \in [0,4]$.

11-1.P6. Suppose $f: [0,2] \to \mathbb{R}$ is defined as $f(x) = x$ if $x \ne 1$ while $f(1) = 3$, and $\alpha: [0,2] \to \mathbb{R}$ as $\alpha(x) = -2$ for $0 \le x < 1$ while $\alpha(x) = 5$ for $1 \le x \le 2$. Let P: $0 = x_0 < x_1 < \cdots < x_n = 2$ be any partition of $[0,2]$. If $x_{k-1} < 1 \le x_k$ for some k, show that $L(f,P,\alpha) = 7x_{k-1}$, $U(f,P,\alpha) = 21$ and hence that $U(f,P,\alpha) - L(f,P,\alpha) > 14$.

11-1.P7. Let f and α be as in 11-1.P6, except that $\alpha(1) = 4$, not 5. If $x_{k-1} < 1 < x_k$ for some k, show that $L(f,P,\alpha) = 7x_{k-1}$, $U(f,P,\alpha) = 21$ and hence that $U(f,P,\alpha) - L(f,P,\alpha) > 14$. If on the other hand $x_k = 1$ for some k (in which case $k-1 \ge 0$), show that $L(f,P,\alpha) = 6x_{k-1} + 1$, $U(f,P,\alpha) = 21$ and hence that $U(f,P,\alpha) - L(f,P,\alpha) > 14$.

11-1.P8. If $f: [a,b] \to \mathbb{R}$ is any bounded function function, show that

$$-U(|f|,P,\alpha) \le L(f,P,\alpha) \le U(f,P,\alpha) \le U(|f|,P,\alpha)$$

for every partition P.

11-2 Definition of the Stieltjes Integral

The definition and many of the elementary properties of the Riemann-Stieltjes integral are very similar to those of the Riemann integral. However, not every step function or monotone function is integrable [see Theorem 11-2.10 below].

We shall often abbreviate the phrase "Riemann-Stieltjes" to simply "Stieltjes".

11-2.1. Proposition. *Let P be a partition of $[a,b]$, $f: [a,b] \to \mathbb{R}$ be bounded function and $\alpha: [a,b] \to \mathbb{R}$ an increasing function. For any refinement P' of P,*

$$L(f,P,\alpha) \le L(f,P',\alpha) \le U(f,P',\alpha) \le U(f,P,\alpha).$$

If P' contains k points more than P, then

$$L(f,P',\alpha) - L(f,P,\alpha) \le 2^k \cdot M \max \{\alpha(x_j) - \alpha(x_{j-1})) : 1 \le j \le n\}$$

and $$U(f,P,\alpha) - U(f,P',\alpha) \le 2^k \cdot M \max \{\alpha(x_j) - \alpha(x_{j-1})) : 1 \le j \le n\},$$

where $M = \sup \{|f(x)| : a \le x \le b\}$.

Proof. Similar to the proof of Prop.9-2.2. ☺

11-2.2. Proposition. *Suppose* $f:[a,b]\to\mathbb{R}$ *is a bounded function and* $\alpha:[a,b]\to\mathbb{R}$ *an increasing function, and let* P *and* P' *be partitions of* $[a,b]$. *Then* $L(f,P,\alpha)\le U(f,P',\alpha)$. *In other words, any lower sum of f is less than or equal to any upper sum of f, even if the partitions involved are different.*

Proof. Similar to the proof of Prop.9-2.3. ☺

11-2.3. Remark. For a bounded function $f:[a,b]\to\mathbb{R}$, it follows from the above Proposition that the set of lower sums generated by all possible partitions of $[a,b]$ is bounded above by any one upper sum; similarly, the set of all possible upper sums is bounded below by any one lower sum. In particular, the set of all possible lower sums of f is a nonempty set having an upper bound; therefore it has a supremum. Similarly, the set of all upper sums has an infimum.

11-2.4. Definition. *When* $f:[a,b]\to\mathbb{R}$ *is bounded and* $\alpha:[a,b]\to\mathbb{R}$ *is increasing, the supremum of the set of all lower sums* $L(f,P,\alpha)$ *is called the* **lower (Riemann-)Stieltjes integral of** f *over* $[a,b]$ *and is denoted by* $\underline{\int_a^b} f\,d\alpha$. *The infimum of the set of all upper sums* $U(f,P,\alpha)$ *is called the* **upper (Riemann-)Stieltjes integral of** f *over* $[a,b]$ *and is denoted by* $\overline{\int_a^b} f\,d\alpha$.

11-2.5. Proposition. *Suppose* $f:[a,b]\to\mathbb{R}$ *is bounded and* $\alpha:[a,b]\to\mathbb{R}$ *is increasing. Then*

$$\underline{\int_a^b} f\,d\alpha \le \overline{\int_a^b} f\,d\alpha .$$

Proof. Similar to the proof of Prop.9-2.6. ☺

11-2.6. Examples. (a) Let $f:[0,1]\to\mathbb{R}$ be the function given by $f(0)=2$ and $f(x)=3$ for $0<x\le 1$ and $\alpha:[0,1]\to\mathbb{R}$ be given by $\alpha(x)=x^2$. Consider the partition P: $0,\ \varepsilon,\ 1-\varepsilon,\ 1$, where $0<\varepsilon<\frac{1}{2}$. It was shown in Example (c) of 11-1.3 that $L(f,P,\alpha)=3-\varepsilon^2$ and also that $U(f,P',\alpha)=3$ for any partition P'. Thus $L(f,P,\alpha)\le U(f,P',\alpha)$. In view of the fact that ε here can be *any* positive number, the supremum of the set of all lower sums is greater than or equal to $\sup\{3-\varepsilon^2 : \varepsilon>0\}$, which is 3. Hence $\underline{\int_0^1} f\,d\alpha \ge 3$; moreover, $\overline{\int_0^1} f\,d\alpha = 3$. It now follows from Prop.11-2.5 that $\underline{\int_0^1} f\,d\alpha = \overline{\int_0^1} f\,d\alpha = 3$.

Note that $0\le U(f,P',\alpha)-L(f,P,\alpha)=\varepsilon^2$.

(b) Let $f:[0,2]\to\mathbb{R}$ be the function for which $f(x)=3$ if $0\le x<1$, $f(1)=8$ and $f(x)=4$ if $1<x\le 2$. Also, let $\alpha:[0,2]\to\mathbb{R}$ be given by $\alpha(x)=x^3$. For any positive $\varepsilon<1$, the sequence

$$0,\ 1-\varepsilon, 1+\varepsilon, 2$$

is a partition P of $[0,2]$. As seen in 11-1.P3, $L(f,P,\alpha) = 31-3\varepsilon-3\varepsilon^2-\varepsilon^3$ and $U(f,P,\alpha) = 31+27\varepsilon-3\varepsilon^2+9\varepsilon^3$. Therefore $\underline{\int_0^2} f\,d\alpha = 31 = \overline{\int_0^2} f\,d\alpha$.

Note that $0 \le U(f,P,\alpha)-L(f,P,\alpha) = 30\varepsilon+10\varepsilon^3$.

(c) Suppose $f:[0,2]\to\mathbb{R}$ is defined as $f(x) = x$ if $x \neq 1$ while $f(1) = 3$, and $\alpha:[0,2]\to\mathbb{R}$ as $\alpha(x) = -2$ for $0 \le x < 1$ while $\alpha(x) = 5$ for $1 \le x \le 2$. Let P: $0 = x_0 < x_1 < \cdots < x_n = 2$ be any partition of $[0,2]$. Since $1 < 2 = x_n$, there exists at least one integer j for which $1 \le x_j$. Moreover, any such integer must be different from 0 because $x_0 = 0 < 1$. Let k be the smallest integer of the kind. Then $k-1 \ge 0$ and $x_{k-1} < 1 \le x_k$ and it follows [see 11-1.P6] that $\underline{\int_0^2} f\,d\alpha = 7$ and $\overline{\int_0^2} f\,d\alpha = 21$.

(d) Let $f:[0,1]\to\mathbb{R}$ be the function for which $f(0) = 0$ and $f(x) = 1$ if $0 < x \le 1$ and let $\alpha:[0,1]\to\mathbb{R}$ be the same as f. For any partition P: $0 = x_0 < x_1 < \cdots < x_n = 1$ of $[0,1]$, we have $\alpha(x_1)-\alpha(0) = 1-0 = 1$ and $\alpha(x_j)-\alpha(x_{j-1}) = 0$ for $j > 1$. It follows that

$$L(f,P,\alpha) = 0\cdot(\alpha(x_1)-\alpha(0)) = 0 \text{ and } U(f,P,\alpha) = 1\cdot(\alpha(x_1)-\alpha(0)) = 1.$$

Therefore $\underline{\int_0^2} f\,d\alpha = 0$ and $\overline{\int_0^2} f\,d\alpha = 1$.

11-2.7. Definition. *A bounded function* $f:[a,b]\to\mathbb{R}$ *is called* **(Riemann-)Stieltjes integrable** *on* $[a,b]$ **with respect to** *an increasing function* $\alpha:[a,b]\to\mathbb{R}$ *if* $\underline{\int_a^b} f\,d\alpha = \overline{\int_a^b} f\,d\alpha$. *We shall abbreviate this as "RS-integrable" or simply "Stieltjes integrable" whenever convenient. The common value of the upper and lower integrals is denoted by* $\int_a^b f\,d\alpha$ *and is called the* **(Riemann-)Stieltjes integral of** *the function* f **with respect to** α (*or* $d\alpha$) **from** a **to** b.

It is customary to denote the Stieltjes integral of the restriction of f to a subinterval $[\alpha,\beta]$ of $[a,b]$ by the symbol $\int_\alpha^\beta f\,d\alpha$ without introducing a symbol for the restriction. Often one works with a function through an expression for $f(x)$, such as x^3 without introducing a symbol for the function. It is then more convenient to write the integral as $\int_a^b x^3\,d\alpha(x)$, or more generally, $\int_a^b f(x)\,d\alpha(x)$.

Examples (c) and (d) above describe functions that are not Stieltjes integrable, while (a) and (b) describe Stieltjes integrable functions.

11-2.8. Proposition. *A bounded function* $f:[a,b]\to\mathbb{R}$ *is Stieltjes integrable with respect to an increasing function* $\alpha:[a,b]\to\mathbb{R}$ *if, and only if, for every* $\varepsilon > 0$, *there exists a partition P of* $[a,b]$ *such that*

$$U(f,P,\alpha)-L(f,P,\alpha) < \varepsilon. \tag{A}$$

Such a partition satisfies

$$\int_a^b f\,d\alpha - \varepsilon < L(f,P,\alpha) \le U(f,P,\alpha) < \int_a^b f\,d\alpha + \varepsilon. \qquad \text{(B)}$$

Proof. Similar to that of Prop.9-2.9. ☺

Remark. In the trivial case when $b = a$, so that the interval $[a,b]$ consists of a single point and every function is bounded, we define $\underline{\int_a^b} f\,d\alpha = 0 = \overline{\int_a^b} f\,d\alpha$, so that $\int_a^b f\,d\alpha = 0$. Thus $\underline{\int_c^c} f\,d\alpha = 0$ for any c in the domain of f.

11-2.9. Proposition. *If $a \le c \le b$ and $f:[a,b]\to\mathbb{R}$ is a function such that its restrictions to $[a,c]$ and to $[c,b]$ are both Stieltjes integrable with respect to an increasing function $\alpha:[a,b]\to\mathbb{R}$, then so is f, and furthermore*

$$\int_a^b f\,d\alpha = \int_a^c f\,d\alpha + \int_c^b f\,d\alpha.$$

[The converse of this result will be dealt with in Prop.11-4.7.]

Proof. Similar to that of Prop.9-2.10. ☺

It is natural to call $f:[a,b]\to\mathbb{R}$ **right continuous** at a point $x \in [a,b)$ if $f(x+) = f(x)$. What we saw in Example 11-2.6 (d) above is that a step function need not be Stieltjes integrable with respect to an increasing function α when it shares a point of right discontinuity—meaning of this should be obvious!— with the latter; a similar example illustrates the same with regard to left discontinuity. We shall now show that when this does not happen, integrability indeed obtains:

11-2.10. Theorem. *Let $f:[a,b]\to\mathbb{R}$ be a step function for which the numbers $x_0, x_1, \ldots, x_n$ form a partition of $[a,b]$ such that $f(x) = T_j$ on the open interval (x_{j-1}, x_j), $1 \le j \le n$. Suppose that $\alpha:[a,b]\to\mathbb{R}$ is an increasing function such that for $1 \le j \le n$, either f or α is left continuous at x_j, and for $0 \le j \le n-1$, either f or α is right continuous at x_j. Then*

$$\int_a^b f\,d\alpha = \sum_{j=1}^{n} T_j\,(\alpha(x_j) - \alpha(x_{j-1})). \qquad \text{(A)}$$

Proof. We proceed by induction on n. Suppose $n = 1$. Then the function takes the constant value T_1 on (a,b). Moreover, $f(a+) = T_1 = f(b-)$, so that f is right continuous at a if, and only if, $T_1 = f(a)$ and is left continuous at b if, and only if, $T_1 = f(b)$. For a positive $\eta < \frac{b-a}{2}$, consider the partition P of $[a,b]$ consisting of the points a, $a+\eta$, $b-\eta$, b. On the three subintervals of this partition, the infima of f are respectively $\min\{f(a), T_1\}$, T_1 and $\min\{f(b), T_1\}$. Therefore

$$L(f,P,\alpha) = \min\{f(a),T_1\}\cdot(\alpha(a+\eta) - \alpha(a)) + T_1\cdot(\alpha(b-\eta) - \alpha(a+\eta)) \\ + \min\{f(b),T_1\}\cdot(\alpha(b) - \alpha(b-\eta)).$$

A straightforward computation shows that this is the same as

$$L(f,P,\alpha) = T_1(\alpha(b)-\alpha(a)) + (\min\{f(a),T_1\}-T_1)\cdot(\alpha(a+\eta)-\alpha(a)) + (\min\{f(b),T_1\}-T_1)\cdot(\alpha(b)-\alpha(b-\eta)). \qquad (1)$$

Now consider any $\varepsilon > 0$. We shall first show that there exist positive δ_1 and δ_2 such that

$$0 < \eta < \min\{\delta_1, \frac{b-a}{2}\} \;\Rightarrow\; |\min\{f(a),T_1\}-T_1|\cdot(\alpha(a+\eta)-\alpha(a)) < \frac{\varepsilon}{2}$$

and $$0 < \eta < \min\{\delta_2, \frac{b-a}{2}\} \;\Rightarrow\; |\min\{f(b),T_1\}-T_1|\cdot(\alpha(b)-\alpha(b-\eta)) < \frac{\varepsilon}{2}.$$

We need prove only the existence of such a positive δ_1. In case $\min\{f(a),T_1\}-T_1 = 0$, there is nothing to prove. So, suppose $|\min\{f(a),T_1\}-T_1| > 0$. This implies $T_1 \neq f(a)$ and hence f is not right continuous at a. According to the hypothesis, α must then be right continuous at a, so that there exists $\delta_1 > 0$ for which

$$0 < \eta < \delta_1,\, a+\eta < b \quad\Rightarrow\quad \alpha(a+\eta)-\alpha(a) < {}^{\varepsilon}\!/\!{}_{2}|\min\{f(a),T_1\}-T_1|.$$

This δ_1 clearly has the required property.

It now follows from (1) that whenever $0 < \eta < \min\{\delta_1, \delta_2, \frac{b-a}{2}\}$, the partition P satisfies

$$L(f,P,\alpha) > T_1(\alpha(b)-\alpha(a)) - \varepsilon.$$

Since such a partition P exists for every $\varepsilon > 0$, we have

$$\underline{\int_a^b} f\,d\alpha \geq T_1(\alpha(b)-\alpha(a)).$$

An analogous argument shows that

$$\overline{\int_a^b} f\,d\alpha \leq T_1(\alpha(b)-\alpha(a)).$$

By Prop.11-2.5, we conclude that

$$\underline{\int_a^b} f\,d\alpha = \overline{\int_a^b} f\,d\alpha = T_1(\alpha(b)-\alpha(a)) = \sum_{j=1}^{n} T_j\,(\alpha(x_j)-\alpha(x_{j-1})) \text{ when } n = 1.$$

The induction argument can now be completed along the lines of Prop.9-2.11. ☺

11-2.11. Proposition. [Needed in Theorem 11-4.12] *Let* $I:\mathbb{R}\to\mathbb{R}$ *be defined to be* 0 *when* $x \leq 0$ *and to be* 1 *when* $x > 0$. *Then the function* $\alpha:[a,b]\to\mathbb{R}$ *defined by* $\alpha(x) = I(x-s)$, *where* $s \in \mathbb{R}$, *is an increasing function. If* $s \in [a,b)$ *and* $f:[a,b]\to\mathbb{R}$ *is right continuous at* s, *then* $\int_a^b f\,d\alpha = f(s)$. *The same holds when* $I(0)$ *is taken to be* 1, $s \in (a,b]$ *and* f *is left continuous at* s.

Proof. It is immediate that α is increasing, because I is. Let $\varepsilon > 0$. By the right continuity of f at s, there exists a positive $\delta < b-s$ such that

$$s \le x < s+\delta \Rightarrow |f(x)-f(s)| < \tfrac{\varepsilon}{2}. \qquad (1)$$

First suppose $s > a$. Then the numbers $x_0 = a$, $x_1 = s$, $x_2 = s+\delta$, $x_3 = b$ form a partition P of $[a,b]$. As usual, denote the infimum and supremum of f on the j^{th} subinterval of P by m_j and M_j respectively. It follows from (1) that $|m_2 - f(s)| \le \frac{\varepsilon}{2}$ and $|M_2 - f(s)| \le \frac{\varepsilon}{2}$. But $L(f,P,\alpha) = m_2$ and $U(f,P,\alpha) = M_2$, and hence $|L(f,P,\alpha) - f(s)| < \varepsilon$ and $|U(f,P,\alpha) - f(s)| < \varepsilon$. Consequently, $f(s) - \varepsilon < \underline{\int_a^b} f\,d\alpha \le \overline{\int_a^b} f\,d\alpha < f(s) + \varepsilon$. Since this holds for every positive ε, it follows when s > a that

$$\underline{\int_a^b} f\,d\alpha = \overline{\int_a^b} f\,d\alpha = f(s).$$

In case $s = a$, we take the partition P to consist of $x_0 = a = s$, $x_1 = s+\delta$, $x_2 = b$ and proceed as above.

The argument for the remaining part is similar. ☺

Problem Set 11-2

11-2.P1. (a) Let $f:[0,2]\to\mathbb{R}$ be the function given by $f(x) = 4$ on $[0,1)$ and 5 on $[1,2]$, and let $\alpha:[0,2]\to\mathbb{R}$ be given by $\alpha(x) = 0$ on $[0,1]$ and 1 on $(1,2]$. Find $\underline{\int_0^2} f\,d\alpha$ and $\overline{\int_0^2} f\,d\alpha$.

(b) Let f and α be as above, except that $f(1) = 4$, not 5. Find $\underline{\int_0^2} f\,d\alpha$ and $\overline{\int_0^2} f\,d\alpha$.

11-2.P2. Find $\underline{\int_0^4} f\,d\alpha$ and $\overline{\int_0^4} f\,d\alpha$, where $f(x) = x$ and $\alpha(x) = x^2$ for $x \in [0,4]$.

11-2.P3. Let $f:[a,b]\to\mathbb{R}$ be the function such that $f(x) = 0$ if $x \in \mathbb{Q}$ and $f(x) = 1$ otherwise. Find $\underline{\int_a^b} f\,d\alpha$ and $\overline{\int_a^b} f\,d\alpha$, where $\alpha(x) = x^{10}$ for all $x \in [a,b]$.

11-2.P4. Let $f:[a,b]\to\mathbb{R}$ be any bounded function and $a < c < b$. Show that $\underline{\int_a^b} f\,d\alpha = \underline{\int_a^c} f\,d\alpha + \underline{\int_c^b} f\,d\alpha$, whenever $\alpha:[a,b]\to\mathbb{R}$ is an increasing function. Determine whether the corresponding equality holds for upper Riemann-Stieltjes integrals.

11-2.P5. For a bounded function $f:[a,b]\to\mathbb{R}$ and any increasing function $\alpha:[a,b]\to\mathbb{R}$, prove that $\left|\overline{\int_a^b} f\,d\alpha\right| \le \overline{\int_a^b} |f|\,d\alpha$.

11-2.P6. Let $I:\mathbb{R}\to\mathbb{R}$ be defined to be 0 when $x < 0$ and to be 1 when $x \ge 0$. Then the function $\alpha:[a,b]\to\mathbb{R}$ defined by $\alpha(x) = I(x-s)$, where $s \in \mathbb{R}$, is increasing. If $s \in (a,b)$ and the function $f:[a,b]\to\mathbb{R}$ is left continuous at s, then show that $\int_a^b f\,d\alpha = f(s)$.

11-2.P7. Let α be an increasing function on $[a, b]$ and be continuous at $c \in [a, b]$. If $f(c) = 1$ and $f(x) = 0$ for $x \neq c$, show without using Theorem 11-2.10 that f is Stieltjes integrable with respect to α and $\int_a^b f\,d\alpha = 0$.

11-2.P8. Let f be a bounded real valued function on $[-1, 1]$ and let α be defined to be 0 on $(-\infty,0]$ and 1 on $(0,\infty)$. Without using Prop.11-2.11, prove that f is Riemann-Stieltjes integrable with respect to α if, and only if, $f(0+) = f(0)$ and in that case $\int_{-1}^1 f\,d\alpha = f(0)$.

11-3 Stieltjes Integrable Functions

Towards the end of the preceding Section, it was shown that a step function is Riemann-Stieltjes integrable with respect to an increasing function α when it shares no point of right or left discontinuity with the latter. We now do the spadework for deducing integrability of a larger class of functions that includes continuous as well as many other functions.

11-3.1. Theorem. *Let $f: [a, b] \to \mathbb{R}$ be the uniform limit of a sequence of functions each of which is Stieltjes integrable with respect to* α. *Then f is Stieltjes integrable with respect to* α. Briefly put: *A uniform limit of Stieltjes integrable functions is Stieltjes integrable.*

Moreover, $\int_a^b f\,d\alpha = \lim_{n\to\infty} \int_a^b f_n\,d\alpha$, *where $\{f_n\}$ is any sequence of Stieltjes integrable functions on $[a, b]$ converging uniformly to f.*

Proof. The argument is completely analogous to that in Theorem 9-3.1, the only change being that $b - a$ is to be replaced by $\alpha(b) - \alpha(a)$ in appropriate places. The case when $\alpha(b) - \alpha(a) = 0$ is trivial, because the increasing function α must then be a constant, in which case *all* bounded functions f are Stieltjes integrable with integral 0. ☺

We are about to use the above Theorem to establish the integrability of continuous functions and of monotone functions (the latter under a certain additional condition) by using Prop.6-2.7. However, it is customary to prove their integrability directly and we comment on the proofs in Remark 11-3.6 below; it is possible to proceed straight to it from here without loss of continuity.

11-3.2. Proposition. *For any increasing function $\alpha: [a, b] \to \mathbb{R}$, the set of points of discontinuity is countable. The same holds if α is decreasing.*

Proof. Being an increasing function, α possesses one sided limits at each point [see proof of Theorem 9-3.3]. When $a < p < b$, we have

$$\alpha(a) \le \alpha(p-) \le \alpha(p+) \le \alpha(b)$$

and hence

$$\alpha(b) - \alpha(a) \ge \alpha(p+) - \alpha(p-).$$

It follows by an induction argument that, for distinct $p_j \in (a, b)$, $1 \le j \le n$, we have

$$\alpha(b) - \alpha(a) \ge \sum_{j=1}^{n} (\alpha(p_j+) - \alpha(p_j-)).$$

If α is discontinuous at each p_j, then each term on the right is positive and hence those among them that exceed $\frac{1}{k}$, where $k \in \mathbb{N}$, are at most $k(\alpha(b) - \alpha(a))$ in number. This establishes that, for each $k \in \mathbb{N}$, the number of points $p \in (a, b)$ where $\alpha(p+) - \alpha(p-) > \frac{1}{k}$ is finite. Now, every point p of discontinuity must satisfy $\alpha(p+) - \alpha(p-) > 0$ because α is increasing, and must further satisfy $\alpha(p+) - \alpha(p-) > \frac{1}{k}$ for some $k \in \mathbb{N}$ because of the archimedean property of $\mathbb{R}$. Therefore the set of points of discontinuity is the union of the finite sets

$$\{p \in (a, b) : \alpha(p+) - \alpha(p-) > \frac{1}{k}\},\ k \in \mathbb{N}.$$

These sets are countable in number and therefore the union is countable, as claimed. ☺

Before stating the forthcoming corollary to Theorem 11-3.1, we remind the reader that functions f and α with domain $[a, b]$ that have a one sided limit at each point are said to *share a point of left discontinuity* if there exists $x \in (a, b]$ such that $f(x-) \ne f(x)$ as well as $\alpha(x-) \ne \alpha(x)$. Similarly for right discontinuity.

11-3.3. Corollary. *A function with domain* $[a, b]$ *and having a one sided limit at each point is Stieltjes integrable with respect to any increasing function* α *with which it does not share a point of left or right discontinuity.*

Proof. By Prop.6-2.7, such a function f is a uniform limit of step functions. We shall indicate how the argument in that proof can be refined in order to show that the step functions can be so chosen that they too do not share a point of left or right discontinuity with α.

In the notation of that proof, it follows from the definition of g_U that, for $x \in U = (c - \delta_2, c + \delta_1)$ or $(b - \delta_2, b]$, we have $g_U(x-) \ne g_U(x) \Rightarrow f(x-) \ne f(x) \Rightarrow \alpha(x-) = \alpha(x)$, and correspondingly for right limits. Therefore, it is only at an endpoint of one of the intervals U, but not a or b, that the step function s can share a point of discontinuity with α. Thus it suffices to show that the endpoints $c - \delta_2$ and $c + \delta_1$ of U can be chosen not to be points of discontinuity of α. Since δ_1 and δ_2 can be replaced by any smaller positive numbers, there are uncountably many replacements available by Theorem 3-3.7. Correspondingly, there are uncountably many replacements available for the endpoints of U. Those among them that are points of discontinuity of α are countable by Prop.11-3.2. Therefore the endpoints $c-\delta_2$ and $c+\delta_1$ of U can be chosen not to be points of discontinuity of α.

Step functions that do not share a point of left or right discontinuity with α are Stieltjes integrable with respect to α according to Theorem 11-2.10. Hence by Theorem 11-3.1, so is f. ☺

11-3.4. Theorem. *If α is an increasing function on $[a,b]$, then any function on $[a,b]$ that is either continuous or is monotone but does not share a point of left or right discontinuity with α is Stieltjes integrable with respect to α.*

Proof. As shown in the proof of Theorem 9-3.3, such functions have a one sided limit at each point. Stieltjes integrability follows by Cor.11-3.3. ☺

11-3.5. Examples. (a) Let $f:[0,2]\to\mathbb{R}$ be defined to be x for $x<1$ and 0 for $1\le x$. Then f is neither continuous nor monotone. However, f has a one sided limit at each point with a single left discontinuity at 1, and by Cor.11-3.3, is integrable with respect to any increasing α having no left discontinuity at 1.

(b) We now give two examples when integrability cannot be proved by applying Theorem 11-3.4 and Prop.11-2.9. The first example is the restriction to $[a,b]$ of the function f of 6-2.P4; as discussed there, the function has a one sided limit at each point. Since the function called I and used in defining f [see 5-2.P5] is continuous everywhere except for a right discontinuity at 0, the function f is continuous everywhere except possibly for right discontinuities at x_j. It follows that $\int_a^b f\,d\alpha$ exists whenever α is right continuous at each x_j.

As a second example, let f be the restriction to $[a,b]$ of the function of Example 2-1.4(d). Recall the special feature of this function that it is continuous at every irrational point and discontinuous at every rational point. It was argued in Example 9-3.4(b) that it has a one sided limit at every point. It follows that it is Stieltjes integrable with respect to any increasing function that is continuous at each rational point

11-3.6. Remark. We comment here on the proofs of the integrability of continuous functions and of monotone functions without using Prop.6-2.7. For continuous functions, one need only repeat the argument of 9-3.5 with $b-a$ replaced by $\alpha(b)-\alpha(a)$, keeping in mind that, if this difference were to be 0, then all bounded functions would be integrable.

For monotone functions, one assumes that α is continuous and replaces the partition used in 9-3.6 by any partition consisting of points x_j for which

$$\alpha(x_j) = \alpha(a) + j\frac{\alpha(b) - \alpha(a)}{n}\text{, where } n > \frac{1}{\varepsilon}(\alpha(b) - \alpha(a))(f(b) - f(a)).$$

The existence of such a partition is ensured by the Intermediate Value Theorem.

11-3.7. Proposition. [Needed in Theorem 11-4.12] *Let* $\{\alpha_n\}$ *be a sequence of increasing functions and* α *an increasing function on* $[a, b]$. *Suppose that*

(i) $\lim_{n\to\infty} \alpha_n(a) = \alpha(a)$, $\lim_{n\to\infty} \alpha_n(b) = \alpha(b)$ *and*

(ii) $(\alpha - \alpha_n):[a, b]\to\mathbb{R}$ *is increasing for each* $n \in \mathbb{N}$.

Then $\alpha_n \to \alpha$ *uniformly on* $[a, b]$ *and any bounded function* $f:[a, b]\to\mathbb{R}$ *that is Stieltjes integrable with respect to each* α_n *is Stieltjes integrable with respect to* α; *moreover*,

$$\int_a^b f\,d\alpha = \lim_{n\to\infty} \int_a^b f\,d\alpha_n.$$

Proof. By (ii), we have

$$\alpha(a) - \alpha_n(a) \le \alpha(x) - \alpha_n(x) \le \alpha(b) - \alpha_n(b) \quad \text{for all } x \in [a, b] \text{ and all } n \in \mathbb{N}.$$

This implies the uniform convergence upon using (i).

For the rest, it suffices to show that $\overline{\int_a^b} f\,d\alpha = \lim_{n\to\infty} \overline{\int_a^b} f\,d\alpha_n$. A similar argument establishes the corresponding result for lower integrals.

Let P: $a = x_0 < x_1 < \cdots < x_m = b$ be any partition of $[a, b]$. As usual, denote $\sup\{f(x) : x_{j-1} \le x \le x_j\}$ by M_j; also, denote $\sup\{|f(x)| : a \le x \le b\}$ by M. Then

$$\begin{aligned} U(f, P, \alpha) - U(f, P, \alpha_n) &= \sum_{j=1}^{m} M_j\big((\alpha(x_j) - \alpha(x_{j-1})) - (\alpha_n(x_j) - \alpha_n(x_{j-1}))\big) \\ &= \sum_{j=1}^{m} M_j\big((\alpha(x_j) - \alpha_n(x_j)) - (\alpha(x_{j-1}) - \alpha_n(x_{j-1}))\big). \end{aligned}$$

By (ii), the factor multiplied to M_j in the preceding summation is nonnegative. It follows for any $n \in \mathbb{N}$ that

$$\begin{aligned} |U(f, P, \alpha) - U(f, P, \alpha_n)| &\le M\sum_{j=1}^{m} \big((\alpha(x_j) - \alpha_n(x_j) - (\alpha(x_{j-1}) - \alpha_n(x_{j-1}))\big) \\ &\le M\big((\alpha(b) - \alpha_n(b)) - (\alpha(a) - \alpha_n(a))\big). \end{aligned} \tag{1}$$

Note that (1) holds for *every* $n \in \mathbb{N}$ and *every* partition P.

Now consider any $\varepsilon > 0$. By (i), there exists $n_0 \in \mathbb{N}$ such that

$$n \ge n_0 \Rightarrow M\big((\alpha(b) - \alpha_n(b)) - (\alpha(a) - \alpha_n(a))\big) < \frac{\varepsilon}{3}. \tag{2}$$

But, for any $n \ge n_0$, there exists a partition P such that

$$|U(f, P, \alpha) - \overline{\int_a^b} f\,d\alpha| < \frac{\varepsilon}{3} \quad \text{and} \quad |U(f, P, \alpha_n) - \overline{\int_a^b} f\,d\alpha_n| < \frac{\varepsilon}{3}.$$

Hence by (1) and (2), $|\overline{\int_a^b} f\,d\alpha - \overline{\int_a^b} f\,d\alpha_n| < \varepsilon$. ☺

Problem Set 11-3

11-3.P1. Let $\alpha:[0,2]\to\mathbb{R}$ be an increasing function which is 0 on $[0,1)$ and is 1 on $(1,2]$. Then $0 \le \alpha(1) \le 1$. Also, let $f:[0,2]\to\mathbb{R}$ be any bounded function. Show that:

(a) If $\alpha(1) = 0$, then f is Stieltjes integrable with respect to α if, and only if, $f(1+) = f(1)$, in which case $\int_0^2 f\, d\alpha = f(1)$.

(b) If $\alpha(1) = 1$, then f is Stieltjes integrable with respect to α if, and only if, $f(1-) = f(1)$, in which case $\int_0^2 f\, d\alpha = f(1)$.

(c) If $0 < \alpha(1) < 1$, then f is Stieltjes integrable with respect to α if, and only if, $f(1-) = f(1) = f(1+)$, in which case $\int_0^2 f\, d\alpha = f(1)$.

11-3.P2. When k is any integer, show that $\int_k^{k+1} x\, d[x] = k+1$, where [ ] denotes the integer part function. That is, $\int_k^{k+1} f\, d\alpha = k+1$ when f and α are the functions defined by $f(x) = x$ and $\alpha(x) = [x]$.

11-3.P3. Suppose a bounded function $f:[a,b]\to\mathbb{R}$ is Stieltjes integrable with respect to α on every subinterval $[c,b]$ with $a < c < b$ and that $\lim_{c\to a} \int_c^b f\, d\alpha = A$. If α is continuous at a, show that f is integrable on $[a,b]$ with $\int_a^b f\, d\alpha = A$.

11-3.P4. Give an example of a sequence $\{\alpha_n\}$ of increasing functions converging to a limit α on $[0,1]$, and of a bounded function f on the same domain such that $\alpha - \alpha_n$ is not necessarily increasing and f is Stieltjes integrable with respect to each α_n but not with respect to α.

11-4 Properties of the Stieltjes Integral

We turn our attention to the question of constructing new Stieltjes integrable functions from known ones, and other matters.

11-4.1. Proposition. *Let the functions f and g be Stieltjes integrable with respect to α on $[a,b]$. Then the function $f+g$ is also Stieltjes integrable and* $\int_a^b (f+g)\, d\alpha = \int_a^b f\, d\alpha + \int_a^b g\, d\alpha$.

Proof. Analogous to Prop.9-4.1. ☺

11-4.2. Proposition. *Let the functions f and g be Stieltjes integrable with respect to α on $[a,b]$. Then the function fg is also Stieltjes integrable with respect to α on $[a,b]$.*

Proof. Analogous to Prop.9-4.3. ☺

11-4.3. Prop. *If f is Stieltjes integrable on* $[a,b]$ *with respect to* α, *and c is any constant, then the product cf is integrable and* $\int_a^b cf\, d\alpha = c\int_a^b f\, d\alpha$.

Proof. Analogous to Prop.9-4.4. ☺

For integrable f, it is a consequence of the preceding result that $-f$ is integrable. The analogue of this for multiplication would be about $1/f$ in place of $-f$; one would therefore have to assume f to be nonzero everywhere and $1/f$ to be bounded.

11-4.4. Prop. *If the functions f and g are Stieltjes integrable with respect to* α *on* $[a,b]$ *and* $f(x) \le g(x)$ *for every* $x \in [a,b]$, *then* $\int_a^b f\, d\alpha \le \int_a^b g\, d\alpha$.

Proof. Analogous to Prop.9-4.5. ☺

11-4.5. Prop. *If* $f: [a,b] \to \mathbb{R}$ *is Stieltjes integrable with respect to* α *on* $[a,b]$, *then so is* $|f|$, *and*

$$\left| \int_a^b f\, d\alpha \right| \le \int_a^b |f|\, d\alpha.$$

Proof. Analogous to Prop.9-4.6. ☺

11-4.6. Corollary. *If f and g are integrable functions on* $[a,b]$, *then so are* $\max\{f,g\}$ *and* $\min\{f,g\}$.

Proof. Analogous to Cor.9-4.7. ☺

We now come to a converse of Prop. 11-2.9.

11-4.7. Proposition. *If the function* $f: [a,b] \to \mathbb{R}$ *is Stieltjes integrable with respect to* α *on* $[a,b]$ *and* $a \le c \le b$, *then its restrictions to* $[a,c]$ *and to* $[c,b]$ *are both Stieltjes integrable with respect to* α, *and*

$$\int_a^b f\, d\alpha = \int_a^c f\, d\alpha + \int_c^b f\, d\alpha.$$

Proof. Analogous to Prop.9-4.8. ☺

It is possible to derive Props. 11-4.2&3 as special cases of a more general result, which is included below only for the more enthusiastic reader. It will be needed only in 11-4.P5.

11-4.8. Proposition. *Let the function* $f: [a,b] \to \mathbb{R}$ *be Stieltjes integrable with respect to* α *on* $[a,b]$ *and h be a uniformly continuous function on the range of f. Then the composition* $h \circ f: [a,b] \to \mathbb{R}$ *is Stieltjes integrable with respect to* α *on* $[a,b]$. [No assumption that the domain of h is an interval.]

Proof. Analogous to Prop.9-4.9. ☺

11-4.9. Remark. The following two equalities are easy to formulate and prove under appropriate hypotheses:

$$\int_a^b f\,d(\alpha_1+\alpha_2) = \int_a^b f\,d\alpha_1 + \int_a^b f\,d\alpha_2\ ;\ \ \int_a^b f\,d(c\alpha) = c\int_a^b f\,d\alpha.$$

Given a partition P of $[a,b]$ and a dual $\overline{P}$, it is clear what $S(f,P,\overline{P},\alpha)$ means [see Def.10-1.1]. For the next result, we shall need the following analogue of Prop.10-1.3:

$$L(f,P,\alpha) = \inf\{S(f,P,\overline{P},\alpha) : \overline{P} \text{ a dual of } P\}$$

and

$$U(f,P,\alpha) = \sup\{S(f,P,\overline{P},\alpha) : \overline{P} \text{ a dual of } P\}.$$

11-4.10. Theorem. *Suppose $f:[a,b]\to\mathbb{R}$ is a bounded function and $\alpha:[a,b]\to\mathbb{R}$ an increasing function having a Riemann integrable derivative α' on $[a,b]$. Then f is Stieltjes integrable with respect to α if, and only if, the product $f\alpha'$ is Riemann integrable, in which case $\int_a^b f\,d\alpha = \int_a^b (f\alpha')$.*

Proof. Let M be an upper bound for $|f|$. Since α' is Riemann integrable, for any $\varepsilon > 0$, there exists a partition P_0 such that

$$0 \le U(\alpha',P) - L(\alpha',P) < \frac{\varepsilon}{2M} \text{ for any refinement } P \text{ of } P_0. \tag{1}$$

Consider any dual $\overline{P}$ of such a refinement P, consisting of numbers $s_j \in [x_{j-1}, x_j]$. By the Lagrange Mean Value Theorem 7-5.5, the partition P: $a = x_0 < x_1 < \cdots < x_n = b$ has a dual $\overline{P}_1$ consisting of numbers $t_j \in [x_{j-1}, x_j]$ such that

$$\alpha(x_j) - \alpha(x_{j-1}) = \alpha'(t_j)(x_j - x_{j-1}), \qquad 1 \le j \le n. \tag{2}$$

By applying Lemma 9-4.2 to each interval $[x_{j-1}, x_j]$ and adding up, we get

$$\sum_{j=1}^n |\alpha'(s_j) - \alpha'(t_j)|(x_j - x_{j-1}) \le U(\alpha',P) - L(\alpha',P).$$

By (1), this implies that, for any refinement P of P_0 and any dual $\overline{P}$ consisting of numbers $s_j \in [x_{j-1}, x_j]$, there exists a dual $\overline{P}_1$ consisting of numbers $t_j \in [x_{j-1}, x_j]$ such that (2) holds and

$$\sum_{j=1}^n |\alpha'(s_j) - \alpha'(t_j)|(x_j - x_{j-1}) < \frac{\varepsilon}{2M}. \tag{3}$$

Since (2) holds, it follows that $S(f,P,\overline{P},\alpha) = \sum_{j=1}^n f(s_j)\alpha'(t_j)(x_j - x_{j-1})$ and hence

$$\begin{aligned}|S(f,P,\overline{P},\alpha) - S(f\alpha',P,\overline{P})| &= \left|\sum_{j=1}^n f(s_j)\alpha'(t_j)(x_j - x_{j-1}) - \sum_{j=1}^n f(s_j)\alpha'(s_j)(x_j - x_{j-1})\right| \\ &= \left|\sum_{j=1}^n f(s_j)(\alpha'(t_j) - \alpha'(s_j))\cdot(x_j - x_{j-1})\right|\end{aligned}$$

$$\leq M\sum_{j=1}^{n}|\alpha'(s_j)-\alpha'(t_j)|(x_j-x_{j-1})$$
$$< \tfrac{\varepsilon}{2} \qquad \text{by (3).}$$

Therefore $S(f,P,\bar{P},\alpha) < S(f\alpha',P,\bar{P})+\frac{\varepsilon}{2}$ and $S(f\alpha',P,\bar{P}) < S(f,P,\bar{P},\alpha)+\frac{\varepsilon}{2}$.

Thus for any $\varepsilon > 0$, there exists a partition P_0 such that for any refinement P of it and any dual $\bar{P}$ of P,

$$S(f,P,\bar{P},\alpha) < S(f\alpha',P,\bar{P})+\tfrac{\varepsilon}{2} \quad \text{and} \quad S(f\alpha',P,\bar{P}) < S(f,P,\bar{P},\alpha)+\tfrac{\varepsilon}{2}.$$

It follows from Prop.10-1.3 and its analogue for $S(f,P,\bar{P},\alpha)$ that

$$S(f,P,\bar{P},\alpha) < U(f\alpha',P)+\tfrac{\varepsilon}{2} \quad \text{and} \quad S(f\alpha',P,\bar{P}) < U(f,P,\alpha)+\tfrac{\varepsilon}{2}.$$

Since this holds for any dual $\bar{P}$ of P, it follows by again applying Prop.10-1.3 and its analogue that

$$U(f,P,\alpha) \leq U(f\alpha',P)+\tfrac{\varepsilon}{2} \quad \text{and} \quad U(f\alpha',P) \leq U(f,P,\alpha)+\tfrac{\varepsilon}{2},$$

so that
$$|U(f,P,\alpha)-U(f\alpha',P)| \leq \tfrac{\varepsilon}{2} < \varepsilon.$$

Thus for any $\varepsilon > 0$, there exists a partition P_0 such that any refinement P of it satisfies the inequality $|U(f,P,\alpha)-U(f\alpha',P)| < \varepsilon$. Consequently, $\overline{\int_a^b} f\,d\alpha = \overline{\int_a^b}(f\alpha')$.

A similar argument shows that $\underline{\int_a^b} f\,d\alpha = \underline{\int_a^b}(f\alpha')$. ☺

If in the Theorem below, we further assume that α and β have Riemann integrable derivartives, then the equality asserted therein can be recast in the light of Theorem 11-4.10. The resulting equality is the familiar formula of integration by parts for Riemann integrals.

11-4.11. Theorem. (Integration by Parts) *Suppose* α *and* β *are both increasing functions on* $[a,b]$ *and that each is Stieltjes integrable with respect to the other. Then*

$$\int_a^b \beta\,d\alpha = \beta(b)\alpha(b)-\beta(a)\alpha(a)-\int_a^b \alpha\,d\beta.$$

Proof. Since α and β are both increasing, for any partition P: $a = x_0 < x_1 < \cdots < x_n = b$, by Prop.1-11.4 we have

$$\begin{aligned} U(\beta,P,\alpha)+L(\alpha,P,\beta) &= \sum_{j=1}^{n}\beta(x_j)(\alpha(x_j)-\alpha(x_{j-1}))+\sum_{j=1}^{n}\alpha(x_{j-1})(\beta(x_j)-\beta(x_{j-1})) \\ &= \sum_{j=1}^{n}\big(\beta(x_j)\alpha(x_j)-\beta(x_j)\alpha(x_{j-1})+\alpha(x_{j-1})\beta(x_j)-\alpha(x_{j-1})\beta(x_{j-1})\big) \end{aligned}$$

$$= \sum_{j=1}^{n} \big(\beta(x_j)\alpha(x_j) - \alpha(x_{j-1})\beta(x_{j-1})\big)$$
$$= \beta(b)\alpha(b) - \beta(a)\alpha(a). \tag{1}$$

From the integrability of α and β with respect to each other, it follows that, for every ε > 0, there exists a partition P such that

$$|L(\alpha,P,\beta) - \int_a^b \alpha\, d\beta| < \frac{\varepsilon}{2} \quad \text{and} \quad |U(\beta,P,\alpha) - \int_a^b \beta\, d\alpha| < \frac{\varepsilon}{2},$$

which, together with (1), implies that

$$\Big| \int_a^b \alpha\, d\beta + \int_a^b \beta\, d\alpha - \beta(b)\alpha(b) + \beta(a)\alpha(a) \Big| < \varepsilon.$$

Since this holds for every ε > 0, the left side must be 0. ☺

As an illustration of the Theorem, consider $\int_0^2 x\, d[x]$. By Prop.11-2.9, this integral equals $\int_0^1 x\, d[x] + \int_1^2 x\, d[x]$, which equals $1+2 = 3$ by 11-3.P2. On the other hand, it also equals

$$2\cdot[2] - 0\cdot[0] - \int_0^2 [x]\, dx = 4 - 0 - 1 = 3.$$

Our final result shows that the Stieltjes integral unifies the concepts of an integral and a sum:

11-4.12. Theorem. *Let $I:\mathbb{R}\to\mathbb{R}$ be defined to be 0 when $x \le 0$ and to be 1 when $x > 0$. Suppose that Σc_j is a convergent series of nonnegative terms, $\{t_j\}$ a sequence of numbers in $[a,b)$ and*

$$\alpha(x) = \sum_{j=1}^{\infty} c_j I(x - t_j).$$

Then any bounded function $f:[a,b]\to\mathbb{R}$ that is right continuous at each t_j is Stieltjes integrable with respect to α and

$$\int_a^b f\, d\alpha = \sum_{j=1}^{\infty} c_j f(t_j).$$

The same holds when $I(0)$ is taken to be 1, the sequence $\{t_j\}$ lies in $(a,b]$ and the function f is left continuous at each t_j.

Proof. Let $\alpha_n(x) = \sum_{j=1}^{n} c_j I(x - t_j)$. Since each c_j is nonnegative and I is increasing, each α_n must be increasing. The convergence of Σc_j implies the uniform convergence of the series for α in view of the Weierstrass M-test [see Theorem 5-2.3]. By Prop.11-2.11 and Remark 11-4.9, f is Stieltjes integrable with respect to α_n with

$$\int_a^b f\, d\alpha_n = \sum_{j=1}^{n} c_j f(t_j).$$

Moreover, $\alpha(x)-\alpha_n(x) = \sum_{j=n+1}^{\infty} c_j I(x-t_j)$ and therefore $\alpha-\alpha_n$ is also increasing. So, the result follows from Prop.11-3.7. ☺

Problem Set 11-4

11-4.P1. (a) Show that $\int_0^1 x^2\,dx^3 = \frac{3}{5}$.
(b) Let $f(x) = x$ and $\alpha(x) = x^3$ for $x < 1$ but $x^3 + 5$ for $x \geq 1$. Find $\int_0^2 f\,d\alpha$.

11-4.P2. Let $\{\alpha_j\}$ be a sequence of increasing functions on $[a,b]$ such that $\sum_{j=1}^{\infty} \alpha_j = \alpha$. Show that any bounded function $f:[a,b]\to\mathbb{R}$ that is Stieltjes integrable with respect to each α_n is Stieltjes integrable with respect to α and that $\int_a^b f\,d\alpha = \sum_{n=1}^{\infty}\int_a^b f\,d\alpha_n$.

11-4.P3. Evaluate $\int_0^4 x\,d([x]-x)$, where [ ] denotes the integer part.

11-4.P4. Let f be a bounded real valued function on $[-1,1]$ and let α be defined on $[-1,1]$ by

$$\alpha(x) = \begin{cases} 0 & \text{if } -1 < x < 0 \\ \frac{1}{3} & \text{if } x = 0 \\ 1 & \text{if } 0 < x < 1 \end{cases}.$$

If f is Riemann-Stieltjes integrable with respect to α, show that f is continuous at 0 and that $\int_{-1}^1 f\,d\alpha = f(0)$.

11-4.P5. Using the Stieltjes versions of the Cauchy-Schwarz (or Hölder) Inequality (proved analogously) and Prop.11-4.8, show the following: Suppose f and g are Stieltjes integrable with respect to an increasing function α on [0,1]. If f and are nonnegative valued and $fg \geq 1$ everywhere, then $(\int_0^1 f\,d\alpha)(\int_0^1 g\,d\alpha) \geq [\alpha(1)-\alpha(0)]^2$.

Hints

Problem Set 1-3

1-3.P2. (a) Start out as follows: $x(y-z) = x(y+(-z))$ definition of subtraction.

1-3.P3. Use appropriate parts of Prop.1-3.4, Prop.1-3.3, Prop.1-3.6, and of course, Def.1-3.7.

1-3.P5. (a) Start with $2x = (1+1)x$ [by definition of 2].

1-3.P6. Start with $a/x = b/y \Rightarrow ax^{-1} = by^{-1}$.

1-3.P8. (b) Subtract one equation from the other.

Problem Set 1-4

1-4.P1. (a) Start by using Prop.1-4.2(a), Prop.1-4.2(f), Prop.1-3.4(c).
(b) Use Props.1-4.2(h) and 1-3.1(e) and others.

1-4.P2. Use Prop.1-4.8, (O3) and properties among (A1)–(A5).

1-4.P3. Apply (O3) twice at the outset.

1-4.P5. Consider the cases $a < 1$ and $a > 1$ separately, using Prop.1-4.2(g) and (O4).

1-4.P6. Let $a < 1$. Then at least one among b and c must be greater than 1 in order to have $abc = 1$. Suppose $b > 1$. Apply (O4) to arrive at $ab < a+b-1$. Then use $c = 1/ab$ in conjunction with 1-4.P5 to obtain another inequality and combine it with the inequality already arrived at.

1-4.P11. Separate the cases $x_1 \le x_2$ and $x_1 > x_2$. When $x_1 \le x_2$, further separate the cases $x_1 \le p$ and $x_1 > p$.

1-4.P16. $a(a-1) - b(b-1) = (a-b)(a+b-1)$.

Problem Set 1-5

1-5.P2. (b) $x \in S \Rightarrow x \ge a$ and $x \in T \Rightarrow x \ge b \ge a$.

1-5.P3. If $a < s \in S$, then $a < \frac{s+a}{2} < s$ and $\varepsilon = \frac{s-a}{2} > 0$.

1-5.P7. Consider $s \in S$ such that $s > a$.

1-5.P12. For the first part, use Prop.1-4.4, Prop.1-4.7(b) and Prop.1-4.8.

1-5.P16. Use Problem 1-4.P3.

Problem Set 1-6

In this Problem Set, the symbol F denotes a complete ordered field.

1-6.P4. Completeness of F is needed only for the "$\Rightarrow$" part.

1-6.P5. Prop.1-6.5 and Prop.1-6.6 can be used.

1-6.P6. Problem 1-5.P15(c) and Prop.1-6.5 are useful.

1-6.P11. Use Prop.1-6.5.

1-6.P14. For the first part, prove that (1) $x \leq \sup T \ \forall \ x \in S \cup T$ and (2) $\forall \ b < \sup T$, $\exists \ t \in S \cup T \ni b < t$ and then use Prop.1-6.5.

Problem Set 1-7

1-7.P3. Take any $y \in I$ such that $y > a$, and separate the cases $y - a < \varepsilon$ and $y - a \geq \varepsilon$. Prop.1-7.3 is relevant.

1-7.P4. It is enough to show that $x \in [0,1) \Rightarrow a \notin [2,3]$; if $x \in [0,1)$ and $a \in [2,3]$, what can be deduced about $|x - a|$?

Problem Set 1-8

1-8.P1. Let $T = \{k \in \mathbb{N} : \{j \in \mathbb{N} : 1 \leq j \leq k\} \subseteq S\}$. Use Remark 1-8.3 to show that T is inductive.

1-8.P2. It is enough to prove that $n \in \mathbb{N}$, $n > m \Rightarrow n \in S$. Show by induction that $\{j \in \mathbb{N} : m + j \in S\} = \mathbb{N}$ and use the first part of Remark 1-8.3.

1-8.P4. (a) Remark 1-8.2 and the second part of Remark 1-8.3 are relevant. (b) Induction starting with m, or induction on $m - n$.

1-8.P5. Let q be the integer part of the quotient a/b and $r = a - bq$. To prove uniqueness, use the fact that if $|bn| < b$, where n is an integer, then n must be 0.

1-8.P7. From the double inequality for $[\frac{k-1}{3}]$ in accordance with the definition of integer part, obtain one for $k - 3[\frac{k-1}{3}]$; note that the latter has to be an integer.

1-8.P8. For distinct integers k_1 and k_2, we must have $4(k_1 - k_2) \neq 2$.

1-8.P9. In proving surjectivity, the result of Problem 1-8.P8 may be used.

1-8.P12. (a) Since $N > n$, we have $N \geq n + 1$ by Remark 1-8.3. Use 1-4.P16.

Problem Set 1-9

1-9.P2. Theorem 1-9.4 and Prop.1-4.4 are relevant. Either use Prop.1-6.9 together with an appropriate modification of Prop.1-9.6 for the case when $x > 0$ or adapt the proof of Prop.1-6.9.

1-9.P3. By Prop.1-9.4, $\forall\ t \in C\ \exists\ s \in \mathbb{Q} \ni t - x < s < y$.

1-9.P7. If $p^2 = 5q^2$, then $p/q = (5q - 2p)/(p - 2q)$.

Problem Set 1-10

1-10.P1. Trivially, $z = \Re z + i\Im z$. Compute $(a \pm ib)^2$ to get $\Re z \pm i|\Im z|$. If $z = \pm i$, we have $a = b = 1/\sqrt{2}$.

***1-10.P4.** $x^3 + y^3 + z^3 - 3xyz = \frac{1}{2}(x + y + z)[(y - z)^2 + (z - x)^2 + (x - y)^2]$.

***1-10.P5.** $(1 - z)(1 + z + z^2 + \cdots + z^k) = 1 - z^{k+1}$ and $1 - z \neq 0$.

***1-10.P8.** Let the real number be a. Express z^3 in terms of a and see what the hypothesis that $a - 1 > 0$ tells us about z^3.

Problem Set 1-11

1-11.P1. In working with the "additional hypothesis", use induction on n and distinguish the cases $a_{n+1} < b_{n+1}$ and $a_{n+1} = b_{n+1}$.

1-11.P2. (c) Use part (b).

1-11.P3. Show that a sum of n terms, each of which is equal to x, is nx. In other

1-11.P4. To prove $x > y > 0 \ \Rightarrow x^n > y^n$, use either induction or the factorisation of $x^n - y^n$.

1-11.P5. Use Problem 1-11.P1 to show that one among $a_1, \ldots, a_n$ is > 1 and one is 1; we may assume $a_1 < 1$ and $a_2 > 1$. This implies $(1 - a_1)(1 - a_2) < 0$, i.e. $a_1 + a_2 > 1 + a_1 a_2$. To argue by induction, apply the induction hypothesis to the n-term sequence $(a_1 a_2), a_3, \ldots, a_{n+1}$.

1-11.P10. Induction argument, using $nx^2 \geq 0$.

1-11.P11. Induction arguments, using $\frac{1}{2}n(n-1)x^3 \geq 0$ and $\frac{1}{2}(n+2)(n+1)x^2 > 0$.

Problem Set 2-1

2-1.P1. $|x - 3| < \min\ \{1, \frac{\varepsilon}{7}\} \Rightarrow |x - 3| < 1 \Rightarrow 2 < x < 4 \Rightarrow 5 < x + 3 < 7$.

2-1.P2. $|x - 2| < \min\ \{1, \frac{\varepsilon}{19}\} \Rightarrow |x - 2| < 1 \Rightarrow 1 < x < 3 \Rightarrow x^2 + 2x + 4 < 19$.

2-1.P3. Imitate the argument for g at the end of the Example (a) of 2-1.4.

2-1.P4. Choose $\delta = \min\{a-[a], 1-(a-[a])\}$.

2-1.P5. $n-\frac{1}{2} \le x < n \Rightarrow f(x) = n-x = |x-n|$; $n \le x \le n+\frac{1}{2} \Rightarrow f(x) = x-n = |x-n|$.

2-1.P7. $f(c) = 0$ for some $c \in \mathbb{R} \Rightarrow f(x) = 0\ \forall\ x \in \mathbb{R}$; suppose $f(x) \ne 0\ \forall\ x \in \mathbb{R}$; fix $a \in \mathbb{R}$ and note that $f(b-a) \ne 0\ \forall\ b \in \mathbb{R}$; use continuity of f at b.

Problem Set 2-2

2-2.P1. $f(g(y)) = y$ and $g(f(x)) = x$. For $y_1 \in [0,1)$ and $y_2 \in [2,3]$, $g(y_1) = y_1 < 1 \le y_2 - 1 = g(y_2)$.

2-2.P2. To prove continuity of g, use Theorem 2-2.2. For $x = 1/(n+\frac{1}{2})$, note that $h(x) = \frac{1}{2}$.

2-2.P3. The function $g(x) = x^5 + x + 1$ is bijective. The given function f is the inverse of g and satisfies $g(f(x)) = x$.

2-2.P4. Consider $x = b$. For $\varepsilon > 0$, $\exists\ \delta_1, \delta_2 > 0 \ni |f(x)-f(b)| < \varepsilon$ whenever $|x-b| < \delta_1$, $x \in [a,b]$ and $|g(x)-g(b)| < \varepsilon$ whenever $|x-b| < \delta_2$, $x \in [b,c]$. It then follows that $|\phi(x)-\phi(b)| < \varepsilon$ whenever $|x-b| < \min\{\delta_1, \delta_2\}$, $x \in [a,c]$.

2-2.P5. Proceed as in 2-2.P4.

2-2.P6. Neither S nor T can contain an interval.

2-2.P7. Any $x \in \mathbb{R}$ satisfies $|x| \le n$ for some $n \in \mathbb{N}$ by the Archimedean property. Using the hypothesis, there is a unique y such that $f_n(x) = y\ \forall\ n \ge |x|$. Define $f(x) = f_n(x)$ whenever $n \ge |x|$.

Problem Set 2-3

2-3.P2. Consider its values at $1/n$ and $1/(n+\frac{1}{2})$.

2-3.P3. The corresponding result for continuous [bounded] functions are relevant.

2-3.P4. Apply the Extreme Value Theorem.

2-3.P5. $0 < 1/(1+[x]^2) \le 1\ \forall\ x \in \mathbb{R}$.

2-3.P6. $1/[x]^2$ is bounded above on $(-\infty, 0)$.

Problem Set 2-4

2-4.P1. Apply the Intermediate Value Theorem to the function $g: [\frac{1}{2}, 1] \to \mathbb{R}$, where $g(x) = f(x) - f(x-\frac{1}{2})$.

2-4.P2. Consider separately the cases when $a > 0$ and $a = 0$.

2-4.P3. Apply the Intermediate Value Theorem to $f(x) = x^3 + x^2 - 1$.

2-4.P4. Proceed as in 2-2.P3.

2-4.P5. The function $G(x) = xf(x)$ is a nonnegative increasing function and satisfies $G(u) = uf(u) > uf(1) > A$, where A is any nonnegative number and $u > \max\{1, \frac{A}{f(1)}\}$. Conclude that the range of G is $[0,\infty)$ and proceed as in 2-2.P3.

***2-4.P6.** Apply the Intermediate Value Theorem to $g(x) = f(x) - kx$, where $k \in (0, K]$ and $K = 2f(\frac{1}{2})$.

2-4.P8. $x^{1/n} \leq y^{1/n} \Rightarrow (x^{1/n})^n \leq (y^{1/n})^n \Rightarrow x \leq y$.

2-4.P9. Let $a_k = 1 + x$ if $1 \leq k \leq m$ and $a_k = 1$ if $m < k$. Apply AM-GM Inequality to the numbers $a_1, \dots, a_n$.

2-4.P10. Assume $1 - ra > 0$. Then $a^r = (1 + (a-1))^r \leq 1 + r(a-1)$. To see the relation between $1 + r(a-1)$ and $1 - ra$, multiply them.

Problem Set 2-5

2-5.P1. (a) Imitate the argument of Theorem 2-1.3, noting that Proposition 2-1.2 is not needed.
(b) Consider $f(x) = x$.For uniform continyuity of $1/f$, choose $\eta = \varepsilon/M^2$, where $|1/f| \leq M$.

2-5.P2. (a) $|x_1^k x_2^{4-k}| \leq M^4$ for $0 \leq k \leq 4$.

(b) $\frac{x}{1+x^2} - \frac{y}{1+y^2} = \frac{(1-xy)(x-y)}{1+x^2+y^2+x^2y^2}$.

2-5.P3. Take disjoint S_1 and S_2, and let f be two different constants on them.

2-5.P4. Consider the set $S = \{s \in \mathbb{R} : f \text{ is bounded on } A \cap (-\infty, s]\}$. Show that it is nonempty and not bounded above.

Problem Set 3-1

3-1.P1. (c) $1/[n^{\frac{1}{2}} + (n+1)^{\frac{1}{2}}] < 1/n^{\frac{1}{2}} < \varepsilon$, provided $n > 1/\varepsilon^2$.

3-1.P2. For $z = 1$, $\sum_{k=1}^{n} z^{k-1} = n$ and for $|z| > 1$, $|\sum_{k=1}^{n} z^{k-1}| > (|z^n| - 1)/|1 - z| \geq n(|z| - 1)/|1 - z| \; \forall \; n$.

3-1.P3. (c) $x_n = 2n$ for $n = 2, 4, 6, \dots$.
(d) For even n, $x_n = \frac{1}{5}$.

3-1.P6. $|x_4 - x| = \frac{1}{2}$, whereas $|x_3 - x| = 0$.

3-1.P7. To be proved: $8^{100} < 3^n$ for all $n \geq 4(8^{99})$. Use Bernoulli's Inequality.

Problem Set 3-2

3-2.P1. Take $z_n = (-1)^n$. For the second part, note that $|z_n - z| = |z_n^2 - z^2|/|z + z_n|$.

3-2.P3. Consider $x_n = 1 - \frac{1}{n}$ and $a = 1$. For the second part, prove $\varlimsup_{n\to\infty} x_n < a + \varepsilon$ for any $\varepsilon > 0$.

3-2.P4. $0 \le \min\{x - [x], 1 - (x - [x])\} \le \frac{1}{2}$.

3-2.P5. $|\Re z_n| \le \frac{2}{n}$ and $|\Im z_n| \le \frac{2}{n}$.

3-2.P6. Write x_n in terms of y_n.

3-2.P7. If $\frac{2}{3} < b < 1$, then $x_{n+1} < bx_n$ for sufficiently large n.

Problem Set 3-3

3-3.P1. (a) $x_{n+2} - x_{n+1} = \frac{1}{2}(x_{n+1} - x_n)(x_{n+1} + x_n)$.
(c) If $x_1 = 1$, then $x_2 = 3/4$. Use part (a).

3-3.P2. Example 3-3.4 is relevant.

3-3.P3. Example 3-3.4 is relevant.

3-3.P4. Prove $\sum_{k=1}^{n} x_n \le nx_{n+1}$.

3-3.P5. Note that $x_{n+1} - x_n = (x_{n+2} - x_{n+1})(x_{n+2} + x_{n+1})$.

3-3.P6. Prove $x_2 - x_1 < 0$ and use induction.

3-3.P7. (a) Check that $x_n > b\ \forall\ n$. For the second part, note that $x_1 = a + b = (a^2 - b^2)/(a - b)$ and use induction.
(b) The limit always exists and is $\max\{p, t\}$.

Problem Set 3-4

3-4.P2. Use the definition of limit.

3-4.P3. (b) Use part (a).

3-4.P4. Use Prop.3-4.4.

3-4.P5. Use Prop.3-4.6.

3-4.P6. (b) Use Prop.3-4.6.

3-4.P7. Imitate the proof of Prop.3-4.6.

3-4.P8. Apply Prop.3-4.2 to the real or to the imaginary part of the sequence.

Problem Set 3-5

3-5.P1. $\lim_{n\to\infty}(-\frac{1}{2})^{n-2} = 0$.

3-5.P2. Begin by using Theorem 3-5.2.

3-5.P4. First prove $|z_{n+1}-z_n| \le \alpha^{n-1}|z_2-z_1|\ \forall\ n \in \mathbb{N}$. Then use $z_m-z_n = \sum_{j=n}^{m-1}(z_{j+1}-z_j)$.

3-5.P5. Try $z_n = \sum_{j=1}^{n}\frac{1}{j}$.

3-5.P6. Show $|z_{k+1}-z_k| \le (3^{k-1}/(k-1)!)|z_2-z_1|$.

Problem Set 4-1

4-1.P1. $3+(-1)^k \ge 2$ for all k.

4-1.P2. Leibnitz' Alternating Series Test.

4-1.P4. The second term is greater than the first; the Leibnitz Test applies only after the first term is dropped. So, start with the series $2^2/2! - 2^4/4! + - \cdots$ and show that its sum is not less than 4/3, the second partial sum. Then revert to the given series.

4-1.P7. Argue first that $(1+\frac{1}{k})^{-k2} \le 2k^{-2}$.

Problem Set 4-2

4-2.P1. (i) Consider sufficiently large k (ii) $\frac{1}{2}+|z_k|^{1/9} > \frac{1}{2}\ \forall\ k$.

4-2.P2. (a) Sufficient to consider $z_k \ge 0$. The proof of Theorem 4-2.4 is relevant.

(b) The series formed by selecting every alternate term from the convergent series $\sum_{k=1}^{\infty}(-1)^k k^{-1}$ is divergent.

4-2.P3. (a) Consider large enough k to ensure that $|z_k| < \frac{1}{2}$ (or any other positive number less than 1).

Problem Set 4-3

4-3.P1. (i) Root Test (ii) Ratio Test (iii) Ratio Test (iv) $a_k > a_1 = 2$ (v) Compare with (iv).

4-3.P2. Use the Cauchy-Schwarz Inequality.

4-3.P3. Apply the Ratio Test.

4-3.P4. Use the Root Test.

4-3.P6. (a) Prove $(k-1)a_k - ka_{k+1} > (\lambda-1)a_k$ for $k \ge N$ and take the sum over $N \le k \le m$.

(b) Show that the sequence $(k-1)a_k$ is increasing from $k = N$ onwards.

Problem Set 4-4

4-4.P1. Consider the partial sums of $\sum_{k=1}^{\infty} z^{k-1}$.

4-4.P3. The formula implies $|\sum_{k=1}^{n} \sin kx| \le 1/|\sin \frac{x}{2}|$ when $\sin \frac{x}{2} \ne 0$.

Problem Set 4-5

4-5.P2. (b) If $\{ a_{n_k}\}$ is any subsequence, then $n_k \ge k$.

4-5.P3. For $\{a_k + b_k\}$, put the following idea in precise mathematical terms: Given any bound $M > 0$, take k large enough to ensure that, on the one hand, b_k exceeds $(\lim b_k - 1)$, and on the other hand, a_k exceeds $M - (\lim b_k - 1)$.

4-5.P4. Check $a_1, a_3, a_5, \ldots$.

4-5.P7. First argue why it is sufficient to show that a subsequential limit of $\{x_n\}$ cannot be less than $1/(\limsup_{n\to\infty} {}^1\!/_{x_n})$.

4-5.P8. By Problem 4-5.P7, the sequence cannot consist of positive terms.

Problem Set 4-6

4-6.P1. Prove by induction that $\sum_{k=0}^{n} c_k = (\sum_{k=0}^{n} a_k)(\sum_{k=0}^{n} b_k)$.

4-6.P2. Use the facts that ${}^1\!/_{(j+1)(k-j+1)} = [{}^1\!/_{(j+1)} + {}^1\!/_{(k-j+1)}](k+2)^{-1}$ and $\sum_{j=0}^{k} {}^1\!/_{(j+1)} = \sum_{j=0}^{k} {}^1\!/_{(k-j+1)}$.

Problem Set 5-1

5-1.P2. If f_A and f_B denote the uniform limit of the sequence $\{f_n\}$ on A and B respectively, first show that $c \in A \cap B \Rightarrow f_A(c) = f_B(c)$.

5-1.P3. (a) Check the value at $x = n \in [0,\infty)$.
(b) Check the pointwise limit for continuity.

5-1.P4. Use the inequality ${}^{|u|}\!/_{(1+u^2)} \le \frac{1}{2}$.

5-1.P5. Check the value at $x = 1/n \in [0,1]$.

5-1.P6. The sequence is not decreasing.

5-1.P7. (b) Use the triangle inequality to relate $f_n(x_n)$, $f(x_n)$ and $f(x)$.

5-1.P8. Show that the pointwise limit is 0. Then examine $f({}^1\!/_{\sqrt{n}})$ in the light of Problem 3-3.P3.

5-1.P9. $\cos(\pi\theta) = 1$ if θ is an even integer and 0 if θ is not an integer.

Problem Set 5-2

5-2.P2. The partial sum can be computed explicitly.

5-2.P3. Each term in the series is dominated by a fixed multiple of $1/k^2$.

5-2.P4. (b) Relate lim sup to subsequential limits.
(c) Consider Σz^k and $\Sigma (2z)^{2k}$.

5-2.P5. (a) $|I| \le 1$; so, $|c_j I(x-x_j)| \le |c_j|$.
(b) $f(x) = c_k I(x-x_k) + \Sigma_{j\neq k} c_j I(x-x_j)$.

5-2.P6. (a) For convergence, show for $x \in [0,1]$ that $g_k(x)$ is nonzero for at most one value of k. For the lack of uniformity, check out $g_k(1/(k+\frac{1}{2}))$.
(b) Use part (a) and apply Theorem 5-2.2.

5-2.P7. $k > \sqrt{(2A)}$, $0 \le x \le A \Rightarrow f(x/k^2) = x/k^2 \le A/k^2$. On $(0,\infty)$, use the fact that $f(k+\frac{1}{2}) = 1$.

5-2.P8. Ratio Test. See the example preceding Prop.4-3.6.

5-2.P9. (a) Use Leibnitz' Theorem 4-1.7.
(b) $\left|\frac{(-x)^k}{k}\right| \le x^k$.

(c) $\Sigma_{k=n+1}^{2n}\left(\frac{x^k}{k}\right) > n\left(\frac{x^{2n}}{2n}\right) = \frac{x^{2n}}{2}$.

Problem Set 5-3

5-3.P1. (b) We have $f(x)^2 \ge 1/2 \ \forall \ x \ge 0$. Consider the relation between the function and the sequence $\{b_n\}$ defined inductively as $b_1 = 1/\sqrt{2}$ and $b_{n+1} = \sqrt{[(1+b_n)/2]}$.

5-3.P2. $1/k! \le (\frac{1}{2})^{k-1}$.

5-3.P3. If p is rational, then $x+p$ is rational $\Leftrightarrow x$ is.

5-3.P4. Use trigonometric identities about such products as $\cos kx \cos jx$.

5-3.P5. Apply the Limit Comparison Test 4-3.1.

5-3.P6. (d) $0 \le x < y \le \pi \ \Rightarrow \ 0 < \frac{1}{2}(y+x) < \pi$, $0 < \frac{1}{2}(y-x) \le \frac{\pi}{2} < \pi$.

5-3.P7. (a) Compute the ratio of consecutive terms of the series for $\cos x$.

(b) $\cos 3x = \cos(2x+x)$; use established formulas.

(c) Find an equation satisfied by $u = \cos\frac{\pi}{3}$, using part (b) and fact that $\cos\pi = -1$.

(d) Contradiction proof, using parts (a) and (c) together with 5-3.P6(d).

5-3.P8. (a) Consider the series for $\cos\frac{1}{2}$.
(b) Use part (b) of Problem 5-3.P7.

5-3.P9. Apply Dirichlet's Test as in 4-4.P2.

5-3.P10. (a) Use 5-3.P6(d).
(c) Let $k = [\frac{t}{2\pi}]$, where [ ] denotes the integer part function. Part (a) is relevant.

5-3.P11. $-1 \le x_1 \le 1$. Use the Intermediate Value Theorem on cos over $[0, \pi]$.

5-3.P12. Follow the Hint for 5-2.P6.

5-3.P13. Weierstrass M-test; regarding $(0, \infty)$, note that $\sin^2(\pi(k+\frac{1}{2})) = 1$.

Problem Set 6-1

6-1.P1. Try taking $\{a, c\}$ as one of the sets in the cover.

6-1.P5. Consider the largest n for any finite subfamily.

Problem Set 6-2

6-2.P1. Imitate the idea of the Remark following Def.3-1.3.

6-2.P3. (a) Let $M = 1 + \max\{|f(c+)|, |f(c-)|, |f(c)|\}$ if c is not an endpoint; drop $|f(c-)|$ [resp. $|f(c+)|$] if c is the left [resp. right] endpoint.

(b) Either imitate the proof of Theorem 2-3.2 or consider a convergent subsequence of a sequence $\{x_n\}$ in the domain satisfying $f(x_n) \to \infty$.

6-2.P4. Rewrite a partial sum for the function as $\sum_{j=1}^{n} c_j I(x - y_j)$, where $y_1, \ldots, y_n$ are the numbers $x_1, \ldots, x_n$ arranged in increasing order. Then show that this is constant on each (y_i, y_{i+1}) and apply Prop.6-2.6.

6-2.P6. (c) Induction. (e) $x^{k+2} - c^{k+2} = x(x^{k+1} - c^{k+1}) + (x-c)c^{k+1}$. (g) Induction using (f); the hypothesis that $f \notin P_0$ is crucial at the initial step. (h) Consider the set K of all nonnegative $k < n+1$ for which there exists $g_k \in P_k$ such that $f(x) = (x-c)^m g_k(x)$, where $m > 0$ and $m + k = n + 1$. Use (f) to argue that $K \neq \varnothing$. Separate the cases $0 \in K$ and $0 \notin K$. For the latter, use (f) again.

Problem Set 6-3

6-3.P2. Props.6-2.6&6-2.7 are relevant.

6-3.P4. Use 2-5.P1(a).

6-3.P7. Modify Step 1 in the proof of the Stone-Weierstrass Theorem appropriately and then check how the rest of the proof is affected, particularly Step 4.

6-3.P8. 6-3.P7 is relevant here.

6-3.P10. (a)&(b) Apply 6-3.P7 for both. Problem 5-3.P6(c) is useful.

Problem Set 7-1

7-1.P1. (a) Use Prop.7-1.3. (b) The only isolated point is 7.

7-1.P2. Imitate the idea of the Remark following Def.3-1.3.

7-1.P4. To show that a nonzero x cannot be a limit point, consider separate the cases $x < 0, x > 1, x = 0, 0 < x < 1$.

7-1.P5. If $\lim_{t \to x} f(t) = \lambda$ exists, then for any $\varepsilon > 0$, there exists $\delta' > 0$ such that $0 < |h| < \delta'$, $x + h \in A \Rightarrow |f(x+h) - \lambda| < \varepsilon$, in particular, $0 < h < \delta'$, $x + h \in A \Rightarrow |f(x+h) - \lambda| < \varepsilon$. What else is needed for concluding that the one sided limits exist and are equal to λ?

Problem Set 7-2

7-2.P1. (a)&(b) The sequence $\{-\frac{1}{n}\}$ lies in the range of f. Consider $\{f^{-1}(-\frac{1}{n})\}$.

7-2.P3. First show that $f'(x) = 3|x|^2$ for all x and hence that $f''(x) = 6|x|$ for all x.

7-2.P4. Extend the domain of f to $[a, b+1]$ by setting $f(x) = f(b) + f'(b)(x-b)$ for $x \in (b, b+1]$. Argue why the extended function is continuous on its domain and consider $f_n(t) = n[f(t+\frac{1}{n}) - f(t)]$.

Problem Set 7-3

7-3.P2. A certain inequality satisfied by $\frac{\pi}{3}$ has to be taken into account.

7-3.P3. Separate the cases $x = 0$, $x < y < 0$ and $x, y > 0$.

Problem Set 7-4

7-4.P3. Use Cor.7-4.7 for the limit and the derivative.

7-4.P5. Suppose $e^{-1} = \frac{m}{n}$, where $m \in \mathbb{N}$, $n \in \mathbb{N}$. Then $n!e^{-1}$ is an integer. Therefore $n!$ times the alternating series for e^{-1} has an integer sum. The first $n+1$ terms in this series are obviously integers. Use the Leibnitz Test to show that the sum of the remaining terms is not an integer. [The series for e can be used instead, but then something else has to be used in place of the Leibnitz Test.]

7-4.P11. Apply the Intermediate Value Theorem to an appropriate function on $[0,1]$; keep in mind that $e > 2$ and $\pi > 3$.

7-4.P12. What is the derivative of $f(x)e^{-x}$?

Problem Set 7-5

7-5.P4. (a) Suppose, if possible, that $f(c) \geq 0$, where $c \in (a, b)$. Apply the Lagrange MVT on $[a, c]$ and on $[c, b]$.
(b) Suppose, if possible, that $f(c) \geq \max \{f(a), f(b)\}$, where $c \in (a, b)$. Apply the Lagrange MVT on $[a, c]$ and on $[c, b]$.

7-5.P5. Apply Problem 7-5.P4(a) to a suitable function.

7-5.P9. Use Prop.7-5.7.

7-5.P10. Use Lemma 7-5.1 and Prop.7-5.2.

7-5.P11. Show that g is invertible and apply the Lagrange Mean Value Theorem to $f \circ g^{-1}$.

7-5.P12. (a)&(b) Find the subintervals of $(0, \infty)$ on which $f(x) = \ln x - x^p$ is increasing or decreasing and note that $f(p^{-(1/p)}) = -\frac{1}{p}(1 + \ln p)$. (c) Prop.7-4.13 reveals what f does as $x \to \infty$.

7-5.P13. $\ln a \leq a^{1/e}$ by 7-5.P12.

7-5.P16. Check $f(u)$ and $f(2u)$. For the uniqueness part, apply Rolle's Theorem and exploit the fact that $f(x) = x(x^2 - u^2) - (v^2 + w^2)x - 2uvw$.

7-5.P21. Begin by showing that $g'(x) = \frac{1}{x}[f'(x) - \frac{f(x) - f(0)}{x - 0}]$ for $x > 0$.

7-5.P22. Apply the Lagrange Mean Value Theorem to f and the Uniform Continuity Theorem to f'.

7-5.P23. By 6-2.P6(g), there must be two consecutive roots between any two given roots. Apply 6-2.P6(h) at each of these roots and check the derivative.

7-5.P28. Consider $e^{-Mx} f(x)$.

7-5.P29. Begin by applying Lemma 7-5.1 at x_0. Then consider $\inf\{x > 0 : f(x) \leq 0\}$.

Problem Set 7-6

7-6.P2. (b) First show that $\frac{d^k}{dx^k} x^n = \left(\prod_{j=1}^{k} (n - j + 1)\right) x^{n-k}$ for $k \geq 1$. Evaluate at $x = 0$ for all k. Use these values and the binomial theorem to deduce that $f^{(k)}(0)$ is an integer for all integers $k \geq 0$. Then apply the result of part (a).

7-6.P4. (b) Apply the result of part (a) to $G(x) = f(x) - \sum_{k=0}^{n} f^{(k)}(a)(x - a)^k \frac{1}{k!} + \varepsilon(x - a)^n \frac{1}{n!}$.

7-6.P6. For the induction step, suppose k to be an integer such that $f^{(k)}(0) = k! a_k$ for every polynomial of any degree, not just n, in particular, for the polynomial f'.

7-6.P7. $f(x)-f(\xi) = \frac{(x-\xi)^n}{n!}f^{(n)}(\eta)$. Since $f^{(n)}$ is continuous at ξ, it follows that $f^{(n)}(\eta)$ will have the same sign as $f^{(n)}(\xi)$, provided the latter is nonzero and η is sufficiently close to ξ.

Problem Set 8-1

8-1.P2. Imitate the solution of 7-6.P3(b). To obtain the limit, begin by subtracting the Taylor expansion to n terms plus remainder from the Taylor expansion to $n+1$ terms plus remainder and simplifying the resulting equality.

8-1.P3. Select ε suitably and use an appropriate form of Taylor's Theorem.

8-1.P4. Take Taylor expansions to two terms plus remainder at a as well as b; evaluate $f(\frac{a+b}{2})$ using both, subtract and proceed.

8-1.P5. Take the Taylor expansion at x to two terms plus remainder, express $f'(x)$ in terms of everything else and evaluate at $x+2\sqrt{\frac{A}{B}}$.

Problem Set 8-2

8-2.P1. (b) Compare with Problem 7-5.P11.

8-2.P2. By Darboux's Theorem of 7-5.P10, g' is either positive everywhere or negative everywhere. Apply Lemma 7-5.1 and proceed.

Problem Set 8-3

8-3.P4. Note that $f_n(x) = -\sum_{k=1}^{n}\frac{(-x)^{3k-2}}{3k-2}$. Check out $f_n'(x)$ and the value of $f_n(0)$, and deduce that $\lim_{n\to\infty} f_n(x) = f(x)+\frac{1}{\sqrt{3}}\tan^{-1}\frac{1}{\sqrt{3}}$ for $0 \le x < 1$; then extend the equality to $x = 1$ by using Leibnitz' Theorem to obtain uniform convergence.

Problem Set 9-1

9-1.P9. For any partition, $M_j^f \le M_j^g + k$ in obvious notation.

9-1.P10. Use the fact that sup $\{-f(x) : x_{j-1} \le x \le x_j\} = -m_j^f$ to show that $-M_j^{|f|} \le m_j^f \le M_j^f \le M_j^{|f|}$.

9-1.P11. $M_j = f(x_{j-1})\ \forall\, j \ge 1$.

Problem Set 9-2

9-2.P2. See Problem 9-1.P5; choose your own partition.

9-2.P5. For a given $\varepsilon > 0$,, start with partitions P_1 and P_2 of $[a,c]$ and $[c,b]$ respectively such that $\underline{\int_a^c} f - L(f_1,P_1) < \frac{\varepsilon}{2}$ and $\underline{\int_c^b} f - L(f_2,P_2) < \frac{\varepsilon}{2}$. Let P be the partition of $[a,b]$ consisting of all the points of P_1 and P_2 taken together. See how $L(f,P)$ is related to $L(f_1,P_1)+L(f_2,P_2)$.

9-2.P8. Prove first that $\left|\int_\alpha^\beta \sin mx\,dx\right| = |(\cos m\alpha - \cos m\beta)/m| \le 2/m$ and that, for any partition $P : a = x_0, x_1, \ldots, x_n = b$, the inequality

$$\Sigma_{j=1}^{n} \int_{x_{j-1}}^{x_j} |f(x) - f(x_j)|\,|\sin mx|\,dx \le U(f,P) - L(f,P)$$

holds. Apply Prop.9-2.9 and proceed from the equality

$$\int_a^b f(x)\sin mx\,dx = \left|\Sigma_{j=1}^{n} \int_{x_{j-1}}^{x_j} [f(x_j) + f(x) - f(x_k)]\sin mx\,dx\right|.$$

9-2.P9. Let the function f_j on $[0,1]$ be 1 on the subinterval I_j and 0 elsewhere. Consider what the hypothesis about the intervals means for $\sum_{j=1}^{n} f_j(x)$ and look at its integral.

9-2.P11. The lower integral can be positive if, and only if some lower sum contains at least one positive term.

Problem Set 9-3

9-3.P1. Show for any $\varepsilon > 0$ that a partition P such that $U(f,P) - L(f,P) < \varepsilon$ exists.

9-3.P3. The given function is a step function on $[\alpha, 1]$ whenever $0 < \alpha < 1$. Use 9-3.P2.

Problem Set 9-4

9-4.P2. Borrow ideas from the proofs of Prop.9-2.10 and Prop.9-4.8.

9-4.P3. Use 9-1.P10.

9-4.P5. Start from the fact that $F = \int_a^b f^2 \ge 0$, $G = \int_a^b g^2 \ge 0$ and $H = \int_a^b fg$ satisfy

$\int_a^b (\lambda f - g)^2 = F\lambda^2 - 2H\lambda + G$ and $F\int_a^b (\lambda f - g)^2 = (F\lambda - H)^2 + (FG - H^2)$.

9-4.P6. If not, then continuity implies that the function must have a positive lower bound on some interval contained in (a,b).

9-4.P8. Apply Problem 7-5.P17 to $F = |f|/(\int_a^b |f|^p)^{1/p}$ and $G = |g|/(\int_a^b |g|^q)^{1/q}$.

Problem Set 9-5

9-5.P2. Modify the proof of the First Fundamental Theorem; use Theorem 7-5.6.

9-5.P5. $|f(x)| \le K\int_a^x |f| \le (\sup|f|)K(x-a)$. Use the Second Fundamental Theorem repeatedly to obtain stronger and stronger inequalities.

9-5.P6. The hypothesis leads to $Kf(x)/[c + K\int_a^x f] \le K$. An antiderivative of the left side can be identified by the First Fundamental Theorem. So the Second

Fundamental Theorem can be used for obtaining another inequality. Correlate it with the given inequality.

9-5.P7. Split $g(x)$ as the sum of integrals over $[0,x]$ and $[x,1]$. Then differentiate.

9-5.P8. (d) Check $\frac{d}{dx}[x+\ln(1-x)]$ for $0<x<1$ and evaluate $0+\ln(1-0)$.

9-5.P9. (b) Take $N=([K]+2)^2$ and consider the intervals from 0 to $1/n$ and from $1/n$ to $1/N$.

9-5.P10. Denote the minimum and maximum values of f by m nd M respectively; then $mx^2 \le f(x)x^2 \le Mx^2\ \forall\ x \in [0,1]$. Use this and the Intermediate Value theorem.

9-5.P12. For the inequality, use 9-4.P5(a), 9-4.P5(c) and 9-4.P6. The fact that the derivative of $f(x)\exp(-\lambda x^2/2)$ is $\exp(-\lambda x^2/2)(f'(x)-\lambda x f(x))$ is relevant.

9-5.P14. For each fixed $y>0$, consider the derivative of $L(xy)$ with respect to x.

9-5.P15. First obtain $F'(t)=f(t)\ge F(t)^{1/2}$, which implies $1\le F'(t)F(t)^{-1/2}$. Then use the fact that $x-1=\int_1^x 1\,dt$.

Problem Set 10-1

10-1.P3. Relate $S(fg,P,\bar{P}_1)$ to $U(f,P)-L(f,P)$ and sup $|g|$.

Problem Set 10-2

10-2.P2. (a) Follow Example 10-2.5, using $f(x)=1/x^3$.
(b) Follow Example 10-2.5, using $f(x)=1/(1+x^2)$ with $n=3$.

10-2.P5. Show that $\frac{1}{x}\left(\frac{1}{x}-\frac{2}{e^x-e^{-x}}\right)<3/2x^2$ for $x\ge 1$.

10-2.P6. Show $0<\ln(1+4/(e^x+e^{-x}))\le 4e^{-x}$.

10-2.P7. For the first one, show that $1/\ln x \ge 1/2(x-1)\ge 0$ on $(1,b]$ and apply Prop.10-2.6. Convert the second integral to the first one.

10-2.P8. Show that $x/(1+k^2x^2)\ge \tan^{-1}((k+1)x)-\tan^{-1}(kx)$, take the sum over $N\le k\le 2N-1$ and evaluate at $x=1/N\in(0,1]$.

***10-2.P9.** (c) Use the Hölder Inequality of Problem 9-4.P8.
(d) Choose p and q appropriately in part (c).

10-2.P10. Substitute $t=x+s$.

Problem Set 10-3

10-3.P1. Apply the Cauchy MVT or 7-5.P11 to a properly chosen pair of functions and then use the Second Fundamental Theorem.

10-3.P3. Dirichlet's Test.

10-3.P4. Dirichlet's Test. Distinguish the cases $p < 0$, $p = 0$, $p > 0$.

10-3.P6. (a) $f(b)-f(a) = (b-a)f'(\eta)$, where $\eta \in (a,b)$. Now, the numbers $\gamma = \min\{f'(\eta), f'(a)\}$, $\Gamma = \max\{f'(\eta), f'(a)\}$ satisfy $(b-a)\gamma \le f(b)-f(a) \le (b-a)\Gamma$ and also $a\gamma \le af'(a) \le a\Gamma$. Use these two double inequalities to derive another one with $b\gamma$ and $b\Gamma$. Look at the double inequality so derived in the light of Darboux's Theorem (7-5.P10) applied to the function bf on $[a,\eta]$. Lastly, argue why the number $\xi \in [a,\eta] \subseteq [a,b)$ obtained from Darboux's Theorem can be taken to be in the open interval (a,b).
(b) Proceed from (a) as in 7-5.P11.
(c) Proceed from (b) as in the Second MVT of Weierstrass 10-3.2.

10-3.P9. (a)&(b) For the part about bounded domains, use the Hőlder Inequality of 9-4.P8 and Prop.10-3.4.

10-3.P13. If not, then for some $\varepsilon > 0$ and some sequence $\{x_n\}$ of real numbers, we have $x_n \to \infty$ and either $f(x_n) \ge \varepsilon$ for every n or $-f(x_n) \ge \varepsilon$ for every n; suppose the former. By uniform continuity, choose $\delta > 0$ corresponding to $\frac{\varepsilon}{2}$ and consider the sum of integrals over $[x_n - \delta, x_n + \delta]$.

Problem Set 10-4

10-4.P1. f' is uniformly continuous on $[0,b]$.

10.4.P2. Use 10-4.P1 and proceed as in the proof of Theorem 10-4.1.

Problem Set 11-1

11-1.P8. Adapt the solution of 9-1.P10.

Problem Set 11-2

11-2.P1. (a) There are partitions P such that $L(f, P, \alpha) = U(f, P, \alpha)$. Find one.
(b) $\alpha(x_j) - \alpha(x_{j-1})$ is nonzero for only one j, whatever the partition.

11-2.P4. Imitate the solution of 9-2.P5.

11-2.P6. Consider a partition of the form $a, s-\frac{\delta}{2}, s, b$, where $\delta > 0$ is chosen in consonance with the given left continuity at s.

11-2.P7. Consider a partition in which the subintervals are of length less than δ, where $\delta > 0$ is chosen in consonance with the given continuity of α at c. Then c must lie in at least one of the subintervals but in at most two.

11-2.P8. For a given $\varepsilon > 0$, choose a positive $\delta < 1$ for $\frac{\varepsilon}{2}$ in accordance with $f(0+) = f(0)$ and consider the partition $-1, 0, \delta, 1$. Show that $|U(f, P, \alpha) - f(0)| \le \frac{\varepsilon}{2}$ and also $|L(f,P,\alpha)-f(0)|\le \frac{\varepsilon}{2}$. For the converse, take any partition P with 0 as one of its points, say x_{i-1}. How is $M_i - m_i$ related to $U(f,P,\alpha)-L(f,P,\alpha)$? Exploit the fact that such a P can be chosen so as to have $U(f,P,\alpha)-L(f,P,\alpha)<\varepsilon$.

Problem Set 11-3

11-3.P1. For any partition that has 1 as one of its points, there can be at most two nonzero terms in the lower or upper sum. Consider such a partition with the additional feature that the upper and lower sums are arbitrarily close to each other.

11-3.P3. Take $c\in(a,b)$ such that $\alpha(c)-\alpha(a)$ and $\int_c^b f\,d\alpha - A$ are arbitrarily small. Then take a partition of $[c,b]$ that brings both the lower and upper sum arbitrarily close to $\int_c^b f\,d\alpha$. Now consider the partition of $[a,b]$ obtained by adding the point a to P.

Problem Set 11-4

11-4.P4. For any partition that has 0 as one of its points, there can be at most two nonzero terms in the lower or upper sum. Consider such a partition with the additonal feature that the upper and lower sums are arbitrarily close to each other. Use an idea from the solution of 11-2.P7; apply Prop.11-2.11 and Remark 11-4.9.

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